TAMIL NADU VETERINARY AND ANIMAL SCIENCES UNIVERSITY

COURSE MATERIAL FOR

VETERINARY MICROBIOLOGY

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UNIT 1

GENERAL AND SYSTEMATIC VETERINARY BACTERIOLOGY

INTRODUCTION & HISTORY TO MICROBIOLOGY

History

The history of bacteriology dates back as early as to the existence of man. It originated from man's effort to get him saved from infectious diseases. For a very long time man had no idea about microbes though they were associated with his day to day life in various forms including food. It was also believed that the incidence of the disease was due to divine reasons since man had no idea of microbes at this point of time. In the earlier days leprosy was considered as a dangerous infectious disease and spread of the disease was attributed to direct contact with the affected persons. Hence, certain procedures were specified for segregation of the affected persons. At this point of time various theories were also attributed to the genesis of disease. Some of the important theories are as below

Theories

- **Theurgical theory** – It attributes the incidence of diseases to divine origin. It also relates man's involvement to sins, spirits etc. to the occurrence of diseases.
- **Miasmatic theory** – This theory was proposed by Hippocrates and further developed by his student Galen. It states that the diseases originate from earth and various natural powers like sun, water, fire etc. were associated with incidence of disease. This theory was hypothetical and philosophical.
- **Pore theory** – It is not very important.

At this same point of time some of the naturally occurring events also aroused man's curiosity like appearance of maggots on carcass and a new theory called **theory of spontaneous generation** was developed. Number of workers attributed that living organisms were generated suddenly. This theory lacked a scientific background (e.g.) generation of rats from dirty clothes. The theory's brief success was due to the lack of knowledge about sterilisation and microbes. Francisco Redi, Needham (1745) and Spallanzani (1769) countered this by heating the putrefied fluid and disproved the spontaneous generation.

Timeline

- **Fracastorius of Verna** (1546) said that a *Contagium vivum* is the cause of infectious disease. It is based on microbial origin of disease. Later, von Plenciz (1762) substantiated this theory.
- **Antoni van Leeuwenhoek** (1683) – a Dutch glassmaker by profession devised an apparatus by assembling lenses at a particular distance. This enabled him to observe small organisms, which were not seen till then by man. He observed small living organisms in body fluids and called them as ***animalcules*** meaning small animals. This finding gave a firm foundation for the later developments. Hence, he is called ***father of bacteriology***.
- **Robert Hooke** (1865) identified cells using his compound microscope. His findings were responsible for the theory that all living things are composed of cells.
- **Carl Linneus** (1735) developed the system of binomial naming of living organisms.
- **Edward Jenner** (1796) **-** Vaccination against small pox
- **Louis Pasteur** (1822-95) – No other person has contributed so significantly than Louis Pasteur for the subject than any other in earlier days. He has contributed significantly to the various aspects of the subject.
    - He has studied the fermentation process in wine manufacture and identified certain substances which are formed only by the living organisms and laid down certain guidelines for clean

production of wine which saved the French wine industry in those days. Hence he is also called ***saviour of French wine industry.***

- He also developed technique for laboratory cultivation of bacteria.
- His accidental finding on loss of virulence by *Pasteurella multocida,* causing fowl cholera and systematic work on *Bacillus anthracis* causing anthrax and rabies virus paved the way for control of infectious diseases by a process called vaccination.
- He evolved the practice of heating to 120°C under pressure for sterilisation, which even killed spores (a forerunner for sterilisation by moist heat (autoclaving) and dry heat (hot air oven - 170 °C.
- For his significant contributions, Louis Pasteur is called ***father of Microbiology***.

❖ **Tyndall** – Established a sterilisation procedure called ***Tyndallisation*** in which the fluids were heated at periodical time intervals. This method was proved effective in killing vegetative organisms and spores.

❖ **Joseph Lister** (1827-1912) – Developed anti-septic wound dressing procedure, which saved the lives of persons with, suspected wound contamination.

❖ **Robert Koch** (1843-1910) – A German physician who moved from the field of medicine to microbiology and has contributed significantly.

- His notable achievements were the two important concepts called ***Koch's Postulates*** (which deal with the establishment of aetiology of infectious diseases) and ***Koch's Phenomenon-*** (which gave firm basis for cell mediated immune response).
- He also established the procedure for preparation, fixation and staining of smears for bacterial identification by aniline dyes.
- He also developed nutrient gelatin and nutrient agar for artificial cultivation of bacteria.
- He identified bacteria causing tuberculosis (*Mycobacteium*) and cholera (*Vibrio*)
- His visit to IVRI at Mukteshwar is note worthy to remember.

❖ **Hansen** (1874) – Identified the bacteria causing leprosy in man.

❖ **Hess** (1882) developed agar for the cultivation of bacteria.

❖ **Paul Ehrlich** (1882) - Acid fast staining – staining procedure used for tuberculosis organisms

❖ **Christian Gram** (1884) – Grouped bacteria into two groups based on a staining procedure called Grams staining.

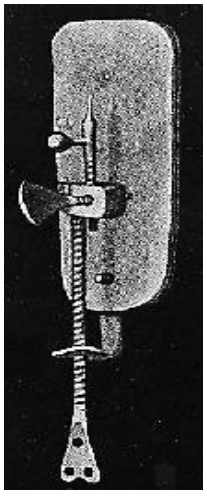
❖ **Loeffler** (1884) – Identified the bacteria causing diphtheria.

❖ **Petri** (1887) developed Petri dish that is still widely used in the cultivation of bacteria.

❖ **Dmitri Iosifovich Ivanovski** (1892) - Discovered viruses

❖ **Gilbert, W and Sanger, F** (1977) – standardized DNA Sequencing method

❖ **Kary Mullis** (1983) – Standardized Polymerase chain reaction



Golden Age

The period between 1857 and 1914 is called as Golden age of Microbiology since during this period number important discoveries were made. The two noted scientists during this period were Louis Pasteur and Robert Koch

The three golden age time periods

- I Golden age of microbiology – 1857-1914 - Louis Pasteur, Robert Koch, Paul Ehrlich, Joseph Lister

- II Golden age of microbiology – 1943-1970 –Alexander Flemming –
- III Golden age of microbiology – 1970 – till date - –Genomics, Bioterrorism, Biomediation

Definitions

Microbiology

Microbiology is the study of microorganisms and the subject covers wide variety of microorganisms that protozoa, algae, fungi, bacteria and virus

Microorganisms

Microorganisms are also referred as microbes and these are unicellular and do not form-specialised tissues and organ systems as found in higher plants and animals

The term microbiology is a broad one and covers a wide variety of organisms. It can be broadly defined as study of organisms that are microscopic in nature. The term microscopic organisms include protozoa, algae, fungi, bacteria and virus. All the members under the 'term' microbes are unicellular and do not form-specialised tissues and organ systems as found in higher plants and animals. Hence they are also called *Protists*.

Grouping of microbes

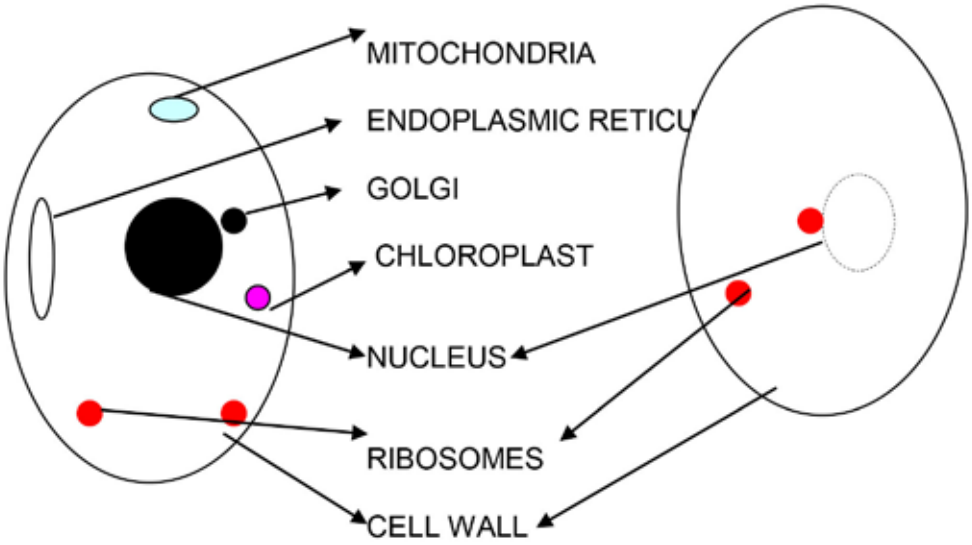
Generally the cells of all living things are classified as *prokaryotes (bacteria)* and *eukaryote* (protozoa, algae, fungi). With number of important points differentiating them. Generally two characters differentiate microbes in to different groups namely the cell wall and chlorophyll.

Sl.No	Characters	Protozoa	Algae	Fungi	Bacteria	Virus
1.	Chlorophyll	-	+	-	+ or -	-
2.	Cell wall	-	+	+	+	NA

Prokaryotes Vs Eukaryotes

Sl. No	Characters	Prokaryotic Cells	Eukaryotic Cells
1.	Nucleus surrounded by a membrane	Absent	Present
2.	Nucleolus	Absent	Present
3.	Chromosome number	1	>1
4.	Reproduction	Asexual	Sexual
5.	Mitotic nuclear divisions	Absent	Present
6.	Cytoplasmic ribosomes	70S	80S
7.	Endoplasmic reticulum	Absent	Present
8.	Mitochondria	Absent	Present
9.	Chloroplast	Absent	Present
10.	Golgi apparatus	Absent	Present
11.	Cytoplasmic membrane – presence of sterols	Absent	Present

EUKARYOTIC CELL PROKARYOTIC CELL



TAXONOMY

Taxonomy is defined as the science of classification particularly living forms. The important objective of classification is to establish relationship between organisms. It is the basic and necessary tool in the study of living organisms. The term **taxa** denotes the degree of relatedness among living organisms and provides means for phylogenetic relationship.

Taxonomy originated from Aristotle, who classified living organisms into plants and animals. Subsequently in 1859 Charles Darwin published his theory of natural selection that explained similarities among organisms and the similarities were attributed to common ancestors. Initially bacteria were placed under plants by Carl Von Nageli (1857). The term prokaryote was introduced by Edward Chatton (1937) and in 1968 a separate kingdom prokaryote was proposed by RGE Murray

Systems of classification

At present there are two systems of classification

- Five Kingdom system proposed in 1969 by Robert H. Whittaker of Cornell University
- Three domain system proposed in 1978 by Carl R. Woese.

Earlier systems of classification

- **Two kingdom system** – Proposed by Aristotle in 4BC in which living organisms were classification into plantae and animalia. This classification is based on observation (phenetics)
- **Three kingdom system** – Propsed by Ernst Haeckel in 1894 in which living organisms were classified into Plantae, animalia and protista
- **Four kingdom system** – Proposed by Herbert Copeland in 1956 in which living organisms were classified into Plantae, animalia, single cell eukaryotes (Protista), prokaryotes
- **Five-kingdom system**: It is the most accepted system of classification in which all prokaryotes are put under one kingdom and eukaryotes are grouped under four different kingdoms.
  - ➢ Prokaryotae – *Bacteria*

Other four are eukaryotic kingdoms – classified based on nutritional requirement, patterns of development, tissue differentiation, flagellar differences.

- ➢ Protista – no tissue organisation, mold, primitive algae, protozoa
- ➢ Fungi – include unicellular yeast, multicellular mold and mushrooms

- ➢ Plantae – some algae, mosses, ferns,conifers and flowering plants
- ➢ Animalia - Sponges, worms, insects, birds and animals

Modern system of classification

Three domain system: This system is based on variations by using molecular biological and biochemical techniques and variations in rRNA sequences (cladistics). In this system, there are two kingdoms for prokaryotes, whereas all eukaryotes are put under one kingdom.

- Eubacteria – All true bacteria
- Archaeobacteria: True methanogens, halophilic, thermoacidophilic
- Eukaryotes

Position of bacteria

Bacteria were initially classified under plants (Carl Von Nageli 1857), later under *Protista (*organisms that lack nucleus (Earnest Haeckel 1866)) and Prokaryote (Edward Chatton, 1937). Now Prokaryote is defined as separate kingdom (RGE Murray)

The kingdom prokaryote is divided into four division - 3 for eubacteria and 1 for archaeobacteria. The three division of eubacteria are **gracilicutes**, f**irmicutes**, **tenericutes** and **archaeobacteria.** Though there are several classes, bacteria of pathogenic importance are found only in seven classes.

Taxonomic hierarchy

The following is the hierarchy followed for classification of bacteria

- Kingdom
- Phylum of Division
- Class
- Order
- Family
- Genus
- Species

Hierarchy	Suffix	Description
Class	*-al*	A class consists of related orders
Order	*-ales*	An order contains a group of related families
Family	*-aceae*	In this category are placed closely related genera or tribes
Tribe	*-ieae*	A tribe contains closely related genera
Genus		This most important category contains closely related species
Species		Included in the same species are strains of organisms that have many characteristics in common, e.g., different strains of *Escherichia coli* will give substantially the same reactions to many biochemical tests
Subspecies		Some species may be further subdivided into subspecies on the basis of small but consistent differences, e.g., *Campylobacter fetus subsp. Intestinalis, subsp. Jejuni*
Strain		A strain consists of the descendants (clone) of a single isolate in pure culture
Serovar		A strain with distinctive antigenic properties.
Biovar		A strain with special biochemical or physiological properties

Dr. Vet

Taxonomy and diversity

Two types of classification is followed – Phenetic system and cladistic system. Phenetic system refers to grouping of system based on mutual similarity of phenotypic characteristics, which may or may not correctly match with evolutionary grouping. In this system a variety of characteristics are used. Under this system two types of classification are followed for bacteria namely Classical and Numerical.

Classical taxonomy

Classical taxonomy: Under this type of classification the following characters are normally used to classify bacteria.

- Morphological characteristics
- Differential staining
- Biochemical tests
- Serology
- Phage typing
- Amino acid sequencing
- Protein analysis
- Base composition of nucleic acid G+C content (paradox ?) – DNA fingerprints
- Nucleic acid hybridization (Southern blot)
- Flow cytometry

Numerical taxonomy

It was proposed by Robert Sneath. It is based on mathematical concepts. Bacteria are given the value 1 for possession of a character and the value o for absence of a character. The values are pooled and weighted average is given. This serves as basis for classification. In this no different levels of importance are given for assigned traits and all traits are given equal importance. It is also based on per centage of similarities ie. how many traits do organisms have in common. There are two methods by which weighted average is calculated - Matching coefficient and Similarity coefficient. In matching coefficient the sum of all the traits that are shared is divided by the total number of traits. In this calculation both positive and negative characters are included {(Common + character) + (Common – character) / Total characters used}. In similarity coefficient, only positive characters are included in calculation and characters that are absent are not considered. It is the most accepted method of finding weightage in numerical taxonomy. It is calculated as {Shared + character / (shared + character) + (+&-) + (-&+)}. Later based on weightage, bacteria are compared and grouped into clusters. The clusters are referred as **phenons**. The similarity matrix and converted in to **dendrogram** (phenogram).

Cladistics or Phylogenetic system

In Phylogenetic system grouping of organisms based on shared evolutionary heritage. It is also referred as Molecular taxonomy. In this system proteins and genes are used to determine evolutionary relationships. This system portrays and accurate picture of how and when organisms diverged from common ancestors over time

Molecular taxonomy

It is based on the ubiquity of rRNA in all living organisms as part of the ribosomes. The bacterial 16S rRNA has a length of 1650 nucleotides and its sequence has been divided into distinct areas according to their degree of variability among different taxonomic groups. The idea behind this approach is that different areas on the rRNA-molecule have a different mutation rate. Highly conserved regions can

therefore be used to group bacteria into higher taxonomic orders whereas more variable regions allow classification at lower taxonomic levels such as the genus or species level. The 16S rRNA is also referred as signature sequence.

Nomenclature

Nomenclature is refers to art of naming. For bacteria the binomial nomenclature system developed by Carl Linnaeus is followed. The names are written in Latin and hence they are written in italics. The names are assigned to bacteria by the **International Committee on Systematic Bacteriology,** which publishes Bacteriological code and International Journal of Systematic Bacteriology. The rules for assigning names to newly classified bacteria and for assigning bacteria to taxa are published in Bacteriological code and the evidence for their classification are published in International Journal of Systematic Bacteriology. Later it is also published in Bergey's Manual of (Determinative) Systematic Bacteriology, which is also referred as Bible of Bacteriology. The book is named so in memory of of taxonomist David Bergey. It was first published in 1923 now it is in 9th edition. In this book both morphological and physiological characters are used for classification. However, the present edition emphasises on phylogenetics.

Grouping of bacteria

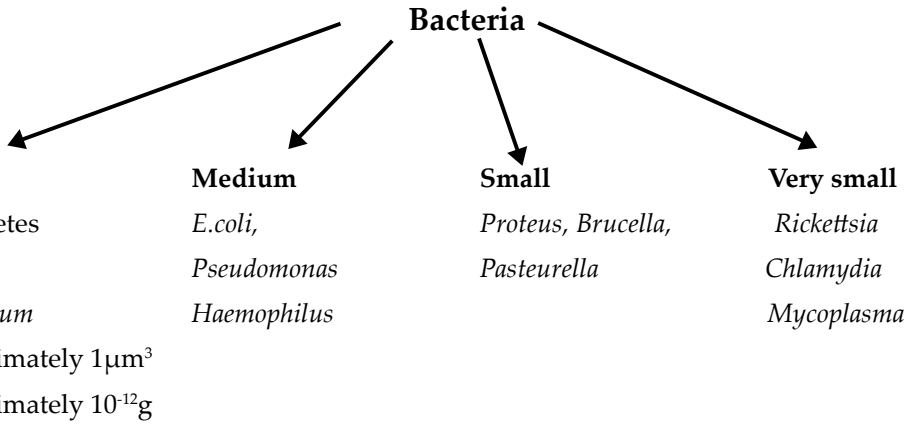
No.	Divisions	Groups	Example
1.	Gracilicutes (Gram negative eubacteria with cell wall)	Aerobic rods and cocci	*Brucella sp*
		Facultative anaerobic rods	*E.coli*
		Anaerobic cocci	*Bacteroides*
		Aerobic/microaerophilic, motile, helical vibrioid bacteria	*Campylobacter*
		Spirochetes	*Leptospira*
		Rickettsias and chalmydias	
2.	Firmicutes (Gram positive eubacteria with cell wall)	Gram positive cocci	*Streptococcus, Staphylococcus*
		Endospore formers	*Clostridium, Bacillus*
		Regular non-spore forming rods	*Listeria*
		Irregular non-sporing Gram positive rods	*Corynebacterium•Mycobacteria*
		Actinomycetes	
3.	Tenericutes (Gram negative eubacteria without cell wall)	Mycoplasma	
4.	Archaeobacteria	Methanogens	
		Halophiles	
		Thermoacidophiles	

STRUCTURE OF BACTERIA

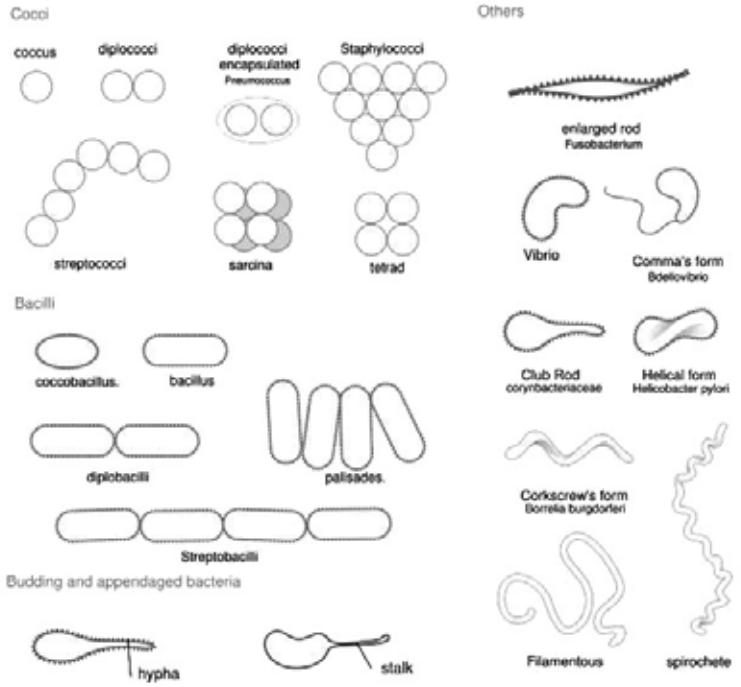
Bacterium

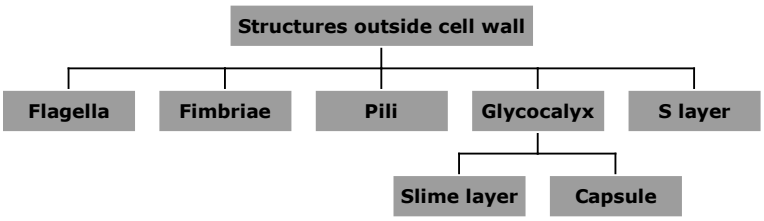
The bacterium (pleural – bacteria) is a single celled, small microscopic organism that does not have a clear nucleus and multiplies by asexual method. Prokaryotes are divided into four divisions – Gracilicutes, Firmicutes, Tenericutes and Mendosicutes. All these terms are derived from Latin words *gracilis* (thin), *firmus* (thick), *tener* (soft) and *mendosus* (faults) respectively.

Shape:	Bacteria exist in three common morphologic forms; they are bacillus (short rod), cocci (spherical) and spirochete or spirillum (curved rods). Some of the variations in bacterial shape are coccobacillary, ovoid and filamentous forms.
Arrangement	The arrangement of bacteria is based upon their dividing planes. **Cocci** *Staphylococci* appear as bunches of clusters *Streptococci* appear in chains *Pneumococci* appear as paired *Micrococci* appear as tetrads (fours) *Sarcina* appear as packets of eight **Bacilli** Regular rods, coccobacillary, chains (short and long). **Other forms -** *Corynebacteria* appear as **club shaped** *Actinomyces* and *Nocardia* appear as **filamentous branches** *Fusobacterium* appear as **spindle** form *Vibryo and Campylobacter* appear as **comma** or **S shaped** *Leptospira and Trepanoma* appear as **loosely coiled**
Size:	Varies considerably. Most rods measure between 2 to 5 μm in length by 0.5 to 1 μm in width. Spirochetes are longer – up to 20 μm and narrower 0.1 to 0.2 μm. Cocci approximately have 1μm diameter. Based on size bacteria are grouped into

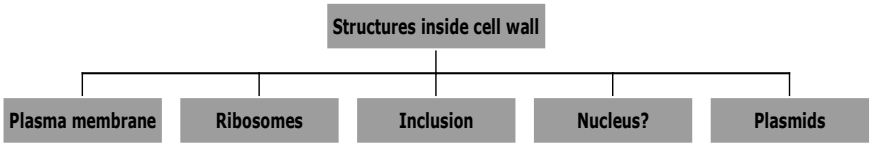


Forms of bacteria





Cell wall



Structures found outside the cell

Glycocalyx

The term glycocalyx is used to indicate material that surround the cell. The bacterial glycocalyx is a visous gelatinous polymer lying outside the cell wall. Chemically, the glycocalyx is either polysaccharide or polypeptides and some bacteria possess both on their glycocalyx. When the glycocalyx substance is firmly attached to cell wall it is referred as capsule and if it is arranged loosely attached to cell wall it is referred as slime layer. The capsule is Well organized glycocalyx, firmly attached to bacterial cell wall and not easily washed off, whereas the slime layer is Diffuse, unorganized, loosely attached glycocalyx and easily washed off. Some bacteria contain hyaluronic acid on their capsule. The important role of bacterial capsule is conferment of anti-phagocytic property and is

i. One of the important virulence factors
ii. Adhesion of bacteria to cells
iii. Supply of nutrition during adverse conditions
iv. Protection of bacteria from dehydration
v. Prevention of migration of nutrients from bacteria
vi. Formation of films on inert surface referred as biofilm.

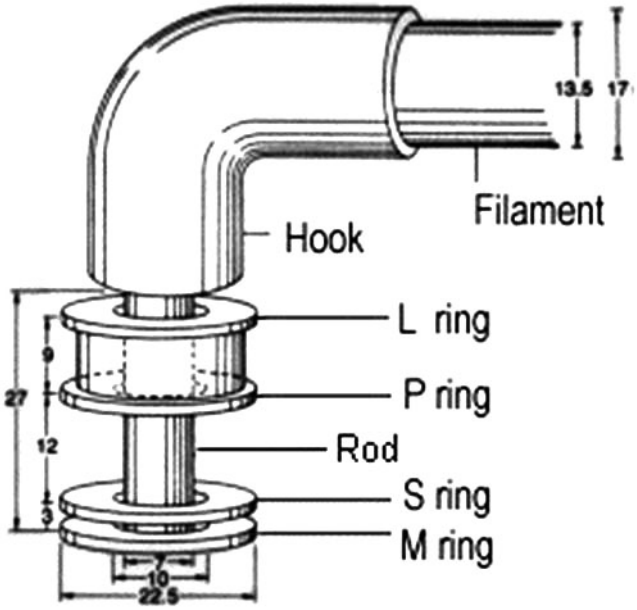
Flagella

It is an external appendage that helps the bacteria to propel. Flagellum is the organ of locomotion for bacteria. Flagella are a thin, filamentous, whip like structure of uniform thickness with a diameter of 12-19nm. Bacteria with active flagella are referred as motile bacteria.

The flagellum is composed of three parts filament, hook and basal body. The filament is made up of number of units of a protein called ***flagellin.*** It is arranged as chains over hallow. The protein of hook is different from filament protein ie. Flagellin. The diameter of a flagellum is 20nm. The basal body is composed of set of rings and is embedded in the plasma membrane that gives the flagella rotary motion (the way the shaft in electrical motor rotates) and propels the organisms. The inner most rings are referrd as M and S rings and they are located in the plasma membrane comprising of the motor apparatus. The P and L rings are the outermost rings and are located in the periplasm and the outer membrane respectively. The outer rings functions as bushings to support the rod where it is joined to the hook of the filament on the cell surface. There is a hook near the outer cell surface Inside these structures the filament is located.

The energy required motion of bacterial flagellum is provided by the proton gradient of cytoplasmic membrane. Approximately 256 protons must cross the cytoplasmic membrane to power a single flagellum. Flagellum can rotate at speeds of 1200 rpm that enables the bacteria to move at

speeds up to 100μm/second or 0.0002 mile/hour. The bacterial motion is referred as tumble, run or swim. The bacterial flagellum provides the bacterium with a mechanism of swimming towards or away from chemical stimuli that is known as **chemotaxis.** The chemicals that attract bacterium are known as ***chemoattractants*** and that repel bacterium are known as ***chemorepellents.*** Some bacteria move in response to a magnetic field and such movement is known as **magnetotaxis**. Magnetotaxis is mediated by a special inclusion structures known as **magnetosomes** that are magnetic iron oxide Fe_3O_4. Some bacteria detect and move in response to variations in light intensity and such movement is called **phototaxis**.



Classification of bacteria based on flagella

Bacteria are classified into 5 groups based on location and number of flagella

Atrichous – No flagella

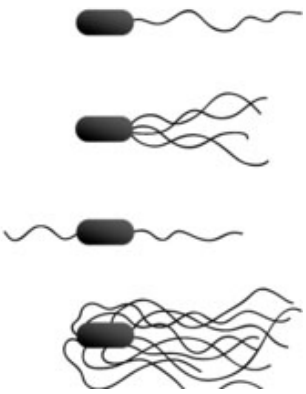
Monotrichous – One flagellum at one end

Lophotrichous – Flagella throughout the surface of cell

Peritrichous - Tuft of flagella at one end; and

AmphitrichousTuft of flagella at both ends

Either a single flagellum at one end or tuft of flagella at one end is also referred as polar flagella



Axial filament

The equivalent of flagella is certain bacteria (Spirochete) are referred as **axial filaments**. They are bundles of fibrils that arise at the ends of the cell wall beneath the outer sheath and spiral around the

cell. They are anchored at one end and the rotation of filaments propels the bacteria like the way the cork screw moves through a cork

Fimbriae

Fimbriae are short, thin, straight, hair like projections found either at the poles or throughout the entire surface. They are mainly responsible for attachment. They are not organ of motility except in few bacteria. They are one of the virulence factors in certain bacteria. The major protein component of fimbriae is referred as fibmbirn.

The type of attachment is used to distinguish between pili and fimbriae. Pili refer to attachment between mating bacterial cell and fimbriae refer to all other types of attachment.

Pili

Pili are similar in structure to fimbrae but longer and present on the bacterial cell numbers in low numbers. A bacteria may have either one or two pili per cell. They are responsible for a process called bacterial conjugation in which transfer of genetic material occur between bacteria. The pili that is responsible for conjugation is referred as sex pili (F pilus). The major component of pili is a phosphate-carbohydrate-protein complex containing a single type of peptide unit called pilin. Pili also act as receptors for bacteriophages. Apart from sex pilin non-sex pili are also observd in bacteria and they in attachment of bacteria to surfaces. In some bacteria pili are considered as one of the virulence factors.

S-layer

S-layer is a cell protein layer found close to bacterial capsule. Its exact function is not known. It acts as a permeability barrier and protects the bacteria from host defence.

Cell wall

The cell wall is a complex semi rigid but permeable structure positioned between capsule and cytoplasmic membrane. The cell wall confers shape to the bacterium. The important functions of cell wall are to protect the contents of cell from adverse conditions and to prevent the bacteria from rupturing when there is variation in osmotic conditions. It also confers rigidity and shape to the bacterial cell besides protecting from chemical and physical actions. It is also responsible for the transport of substances either by simple diffusion or by complex transport systems. The cell wall makes up approximately 20 per cent of the total dry weight of bacterium.

Chemical composition

Chemically the major component of cell wall is peptidoglycan. Peptidoglycan is composed of N acetylglucosamine and N acetylmuramic acid disaccharaide polymers and peptides consisting of amino acids L-alanine, D-alanine, and D-glutamic acid and either lysine or diaminopimelic acid. Alternating the NAG and NAM molecules are linked in rows of 10-65 sugars to form a carbohydrate backbone.

Thickness

The thickness of cell wall varies between Gram positive (100 A-800A) and Gram negative (100A) bacteria. Besides the thickness, the chemical composition between Gram positive and negative bacteria also varies.

Variation

Cell wall of Gram-positive bacteria (GPB): It is a featureless layer of approximately 20-30nm in thickness. It is made up of many layers of peptidoglycan forming a thick and rigid structure compared to GNB. The major component of cell wall of Gram-positive bacteria is **Peptidoglycan** (80-90%). In addition to peptidoglycan, the cell wall also contains polysaccharides and t*ecihoic acid*. There are two types of **teichoicacid** a lipid teichoic acid and wall teichoic acid. Tecichoic acid is responsible for

regulating the movement of cations into and out of the cells and prevents the cell from lysis besides cell at low pH. It also provides antigenic character to bacteria.

Cell wall of Gram-negative bacteria (GNB): Unlike Gram positive bacteria, the cell wall of Gram-negative bacteria are multi-layered (three layers). The three layers are outer membrane, intermediate and transparent (inner most). Of the three layers, the outer membrane is most important. It is more or less like a molecular sieve. It is a lipid bi-layer containing phospholipids, proteins, lipo-proteins and lipo-polysaccharide (LPS). The LPS is also referred as **endotoxin**, which is one of the important virulence factors of GNB. The LPS is a complex molecule consisting of a lipid – Lipid A, which is attached to a polysaccharide. The polysaccharides consist of a core polysaccharide and repeat polysaccharide. The later is also called as O-polysaccharide of O-antigens. The outer membrane also contains protein called **porin** which form micro channels through which molecules diffuse. Unlike GPB, the cell wall does not contain teichoic acid and is made up of less amount of peptidoglycan. Hence, a GNB is more susceptible for mechanical damage. The space between outer membrane and plasma membrane is called periplasmic space.

Atypical cell wall

Mycoplasma in prokaryotes does not have cell walls whereas the cell walls of aracheobacteria are composed of pseudomurin. As a result of damage to cell wall by lysozymes, the gram-positive bacteria loose the cell wall, but still are surrounded by plasma membrane and can carry out metabolism. Such structures are referred as **protoplasts.** The damage to cell wall of GNB caused by lysozyme is not as extensive as in GPB. The GNB retain certain layers of outer membrane and plasma membrane and such structures are referred as **spheroplasts.**

Structures inside the cell wall

Plasma membrane

It is also referred as cytoplasmic membrane. It is a thin structure lying inside the cell wall covering the cytoplasm of the cell. In bacteria it is composed of phospholipid molecules, which are arranged as two parallel layers referred as phospholipid bilayer. The sterols that are normally present in eukaryotic cells are absent. Among the phospholipid bilayer two different protein molecules are arranged. One type of proteins is referred as peripheral proteins, which act enzymes and also as scaffold. The other type of protein is called integral proteins and some of these molecules function as channels through which various molecules enter into the cell. This membrane is a differentially permeable barrier that allows selective restriction on molecules transported. Smaller molecules enter easily into the cell, whereas the ions are transported slowly. The molecules that are fat soluble enter very easily into the cell.

Mechanism of movement

Movement across plasma membrane is mediated by concentration gradient. Movement is divided into active and passive movements. The active process includes **group dislocation** and the passive processes are **simple diffusion, facilitated diffusion and osmosis**.

Adhesion sites

They are also referred as **Bayer junction.** In GNB (bacilli) the cell wall and cytoplasmic membrane are joined in number of areas and these areas are called adhesion sites or Bayer junctions. Through these sites the polysaccharides are transported.

Cytoplasm

The cytoplasm is enclosed by the cytoplasmic membrane and 80% of the cytoplasm is composed of water. The liquid component of the cytoplasm is referred as **cytosol.** The contents of cytoplasm include

nucleic acids (DNA and RNA), enzymes, amino acids, carbohydrates, lipids, inorganic ions, and many low molecular weight compounds. Besides these many inclusion bodies with specific functions are also observed. Cytoplasm is the site of most bacterial metabolism (both catabolism and anabolism). The helical actin-like proteins of cytoplasm contribute to cell shape. Cytoplasm secrete two types of enzymes

1. **Exoenzyme** - Breaks macromolecules into smaller molecules capable of being transported across the cytoplasmic membrane
2. **Endoenzyme** – Responsible for chemical reactions (metabolism) that occur inside bacteria. The main functions of the enzymes are cell growth, metabolism, and replication

Ribosomes

Ribosomes are the sites where proteins are synthesized. The ribosomes of bacteria are 70S composed of 30S and 50S ribosomes subunits. The ribosomes contain a protein and one (30S) or two (50S) molecules of rRNA. They are responsible for translation of genetic information carried by mRNA into proteins (protein synthesis)

Mesosomes

Mesosomes are invaginated or infolded structures of the cytoplasmic membrane and are mostly located near the area where the bacteria divides. They are more prominent in GPB than in GNB. They are also called as ***peripheral bodies*** or ***chondroids***. Though a number of functions are attributed to this structure, now the very presence of this structure in bacterial cell is disputed.

Inclusions

1. **Metachromatic granules**: They are responsible for staining property. They stain red with certain blue dyes like methylene blue. They are also referred as **volutin** granules that are rich in inorganic phosphate and used for synthesis of ATP. Eg. *Corynebacterium*
2. **Polysaccharide granules:** They are rich in glycogen and starch and stained by iodine as reddish blue and blue respectively.
3. **Lipid inclusions:** They are storage site for lipid and contain poly beta hydroxybutyric acid. These granules are stained by Sudan dyes. Eg. *Mycobacterium*
4. **Sulfur granules:** Present only in Sulphur bacteria as energy reserves. Eg. *Thiobacillus*
5. **Carboxysomes:** These bodies are seen in bacteria that utilises carbon dioxide as their sole source of carbon. Eg. Cyanobacteria
6. **Gas vacuoles:** These vacuoles are present in bacteria that live in depth of water to maintain buoyancy. They are present as hollow cylinders surrounded by a protein.

Endospores

The formation of spores by bacteria was first observed by Cohn (1875). Not all bacteria produce spores and it is a special character of some group of bacteria. Spores are produced when the bacterial cells are deprived of some factor or nutrient necessary for growth. Spore production is noticed in natural conditions and also during artificial cultivation.

Structure

Spores are highly resistant and thick walled structure. The position of spore is of some taxonomical importance. Depending upon their position in the cell they may be classified as terminal, sub-terminal or equatorial. They also give a specific shape to the bacteria like '**drumstick'** or '**barrel'** appearance. Mature spore is a multi-layered structure. The innermost part is called **spore protoplast** or **core** that contains the DNA, RNA and protein. The DNA is found in the complete chromosome and the proteins found are of ribosomal proteins and enzyme activities have also been detected. This core is surrounded

by a membrane, which in turn is surrounded by the **germ cell wall** that is rich in peptidoglycan. A peptidoglycan rich layer called **cortex** surrounds the germ cell wall. The peptidoglycan of cortex is different from that of vegetative cells in that it is rich in muramicanhydride, incomplete peptides and absence of teichoic acid. The cortex is surrounded by a membrane, which is again surrounded by **sporecoats** and **exosporium**. The spore coat is characterized by the presence of hydrophobic proteins.

Spores are dehydrated and are the contributing factors for heat resistance. Heat resistance is also provided by **dipicolinic acid** (pyridine 2-6-dicarboxylic acid). Dipicolinic acid is found only in spores. It concentrates and forms insoluble salts of Ca, which are responsible for heat resistance.

Sporulation

The process of spore formation is referred as sporulation and is divided into five stages.

1. **Stage I**: The DNA of bacterial cell assembles into an axial filament, which is followed by formation of an asymmetric double membranous septum.
2. **Stage II**: Formation of a ring like intrusion of the cell wall leading to formation of forespores.
3. **Stage III**: Engulfment of spores by cytoplasmic membrane of vegetative cell.
4. **Stage IV**: Formation of spore cortex and exosporium.
5. **State V**: Release of mature spore.

Germination

The revival of a dormant spore into vegetative cell is called germination and it takes place in three stages – activation, germination and outgrowth.

1. **Activation**: Activation is generally sluggish and is accelerated by heating at 60°C for an hour or by lowering the pH or by treating with mercaptoethanol. No morphological stages are reported during activation.
2. **Germination**: When activated spores are exposed to substances like amino acids, nucleosides or sugars, germination starts. This stage is characterized by refractility and heat resistance. This stage also involves active metabolism.
3. **Outgrowth**: It takes 50-60 minutes. First there is swelling due to excessive water intake, which is followed by synthesis of DNA.

Plasmids

In addition to bacterial chromosome, bacteria also contain one or more small, circular DNA molecules called plasmids. They are also referred as extra chromosomal DNA that replicates independent of chromosomal DNA. All bacteria possess chromosomal DNA but not plasmid. Sometimes the genetic information carried by plasmids important in that they are responsible for mating, resistance to antibiotics, production of toxins and tolerance to toxic metals. Because of their small size they are easy to manipulate and are useful tools in applied microbiology

Bacterial chromosome

The term nucleus cannot be used for bacteria since in contrast to eukaryotic cells, the genetic material is amorphous in nature and the nuclear membrane that separates nucleus from cytoplasm is absent. Hence, the genetic material inside the bacterial cell is referred as ***nuceloid, nuclear body, nuclear region or bacterial nucleus***. Since the bacterial chromosome does not have a covering membrane it is also referred as ***naked DNA***. The bacterial chromosome is circular in shape (except in *Streptomyces* where it is linear), arranged as a super coiled structure and is approximately the length of one double stranded DNA. It has got 4.7×10^6 nucleotides. The chromosomes of the eukaryotic cells are diploid - that is they have two copies of each chromosome whereas the chromosomes of bacteria are haploid

(only one chromosome) except *Rhabdobacter*. If stretched the length of bacterial DNA can be extended up to 1mm (roughly 100 times more than the length of bacteria). The weight of bacterial chromosome is approximately $5x10^{-15}$g. Replication of bacterial chromosome always precedes the binary fission of cell. The nuclear region is composed of about 80% DNA, 10% RNA (mostly nascent) and 10% protein (RNA polymerase).

DNA

The DNA (**DEOXYRIBONUCLEIC ACID)** is the macromolecule that stores the hereditary information of the cell. The DNA is a double helix structure that is composed of two polynucleotide strands that run in opposite direction. The two strands are joined together by **hydrogen bonds**. In addition to hydrogen bonding, the double helix is also stabilised by hydrophobic interactions. The basic subunits of nucleic acid are called ***nucleotides*** that are also referred as **genetic alphabets**. The nucleotides that form the DNA are called as **deoxyribo-nucleotides.** Each deoxyribo-nucleotide is made up of a nucleic acid base, deoxyribose sugar and phosphate. Four nucleic acid bases form the deoxyribose-nucleotides. They are **Adenine (A), Guanine (G), Cytosine (C) and Thymine (T)**. Of these four, chemically A and G are **purines** thatare composed of two fused ringsand C and T are **pyrimidines** that are composed of only one ring**.** Structurally both purines and pyrimidines are heterocyclic molecules with the ring having a **carbon and nitrogen** instead of carbon alone. The strands contain alternating units of sugar and phosphate in other words the phosphate attaches 3′ end of one sugar molecule with 5′ end of another molecule. This structure is also known as "**sugar-phosphate backbone'.** The bases are attached to sugar molecules forming deoxyribonucleosides and these nucleosides are joined to a phosphate group on carbon 5′ of the sugar to form deoxyribo-nucleotide subunits of DNA. The nucleotides of DNA are linked by 3′-5′ **phosphodiester** bonds. Each strand will have a free 3′OH free end and 5′-P free end. The strand which runs in 5′-3′ direction is called **leading strand** and that runs from 3′-5′ is called **lagging strand**. Adenine of one strand joins with Thymine of another strand and Guanine of one strand joins with Cytosine of another strand. This is called **base pairing** and is specific. The ends of the DNA strands are covalently bonded together forming the typical circular molecule.

The double helix has a diameter of 2nm and each turn is approximately of 3.4nm in length with ten nucleotides. The most common form of DNA helix is called **B-DNA** – the right-handed form (the helix rotates clockwise). Based on the numbers of bases per turn the right-handed form is subdivided into **A-DNA, B-DNA, D-DNA** and **E-DNA**. In some cellular conditions a left-handed form of the DNA known as **Z-DNA** is also found.

Generally the base pairing occurs between strands. Sometimes it is also noticed in the same strand due to a special kind of sequence arrangement called **palindromic sequence** meaning a sequence that read the same when read either from right to left or left to right. Palindromic sequences are also referred as **inverted repeats.** Due to this, base pairing occurs in the same strand-forming hairpin like structures known as **cruciform loops**.

The bacterial DNA is arranged in the form of a super-coiled structure inside the cell. The DNA in simple circular form is known as **relaxed DNA**. The relaxed DNA assumes the shape of either **positively super-coiled** structure or **negatively super-coiled** structure depending upon the direction of twists. Histones that are responsible for coiling of the DNA in eukaryotes are absent in bacteria. However, histone like proteins is reported in some bacteria.

RNA

The RNA (**RIBONUCLEIC ACID)** is chemically similar to DNA in that it is a macro-molecule composed of nucleotides. Further the sugar in RNA is **ribose** and a 5′3′ phosphodiester bond joins the **ribonucleotides**. The RNA molecule has got a free 3′-OH end and 5′-P end. Chemically, the ribose sugar

has got an extra –OH group, which makes the RNA less stable. RNA has got purine bases **adenine (A)** and **guanine (G)** and pyrimidine bases **Cytosine (C)** and **Uracil (U).** However, RNA is **single stranded** in contrast to DNA. RNA is **single stranded** in contrast to DNA. However, the strand folds at several places result in base pairing between A and U and G and G through hydrogen bonds result in hairpin structure.

In cells RNA occurs in three classes as a result of transcription, they are rRNA (ribosomal RNA), mRNA (messenger RNA) and tRNA (transfer RNA). The three types have different functions. While in the cell, the DNA serves as genetic material RNA acts as informational mediators between DNA and Proteins. The process by which the information stored in the DNA is used to code for the synthesis of RNA is called **transcription**. For transcription, one strand of the DNA, usually sense strand act as template and only one DNA sequence code for an mRNA.

Ribosomal RNA

Ribosomal RNA (rRNA) combines with various ribosomal proteins to produce ribosomes. It is also important in positioning tRNA in ribosomes during protein synthesis and in peptide bond formation.

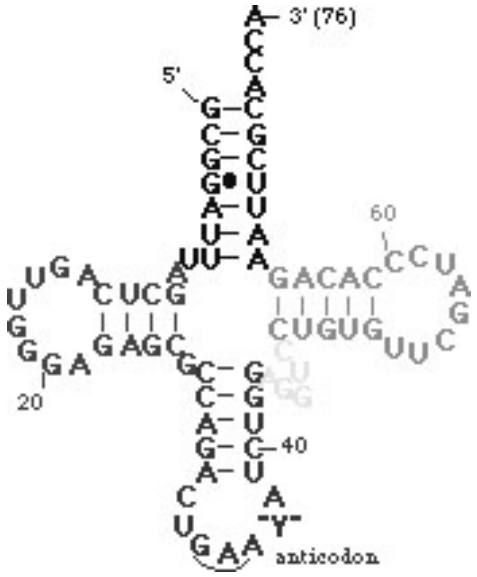
Messenger RNA

Messenger RNA (mRNA) bears the message for protein synthesis that is sequence of amino acids. The stability of mRNA molecule between prokaryotes and eukaryotes vary widely. The mRNA prokaryotes are stable for few minutes whereas eukaryotes are stable for a long time.

Transfer RNA

Transfer RNA helps in decoding the mRNA into correct amino acid sequence. It carries the specific amino acid to ribosome. The tRNA molecules have characteristic four-leaf clover structure that results as a result of internal base pairing. The tRNA molecules are characterized by presence of unusual ribonucleotides namely pseudo uridine, thymidine and dihydro uridine. There are different loops with important functions. The DHU loop or D loop, that derives its name due to presence of unusual base dihydroxy uracil binds to the enzyme involved in forming the peptide during translation. The TψC loop contains psuedouracil (ψ) A and it is responsible for binding with ribosomes. The fourth loop is called anti codon loop, which has got bases complementary to codons on mRNA. The **amino acid arm** carries the correct amino acid in relation to mRNA sequence. The 3'-OH end of tRNA molecule always ends with sequence ACC and amino acids bind here. This arm is also known as CCA loop.

Structure of tRNA



BACTERIAL GROWTH

Definition

The term bacterial growth is defined as an increase in number of cells.

Binary fission

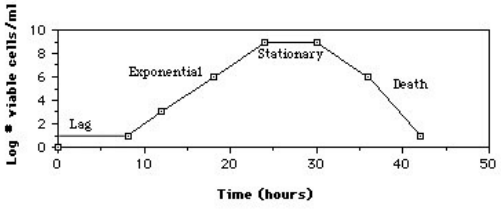
Bacterial cells increase in number by a simple process of division called **binary fission.** This process involves as many as 2000 various reactions. It is a process of polymerisation. In this process copies of the macromolecules are produced first and later they are assembled and cell structures like cell wall, cytoplasmic membrane, and chromosome are formed. This is followed by cell division. The chromosome division precedes cell division always. The first stage in binary fission is the elongation of cells to double length and formation of septum. Septum is formed as a result of inward folding of cytoplasmic membrane and cell wall. Assembly of synthesised macromolecules is the next stage. Following this the daughter cells are pinched off. Of the two copies of chromosome one each goes to daughter cells and sufficient number of copies of other cell structures is provided for each daughter cells. On good nutritional conditions a bacterial cell will divide into two in 20 minutes.

Population growth

Population growth is defined as an increase in the number of microbial cells in a population. This increase in population is measure by an increase in cell mass. **Growth rate** change in cell number or cell mass.**Generation** is the interval for the formation two cells from one cell and **generation time** is the time required for this to occur. Generation time is also referred as **doubling time.** The generation time for bacteria varies from few minutes to even days. The bacterial growth does not follow a constantly increasing slope pattern. During increase in population there is initial period of no activity followed by rapid increase in number. This stage is followed a periods of sustenance and death.

The pattern of population increase where the number of cells double during each unit time is referred as **exponential growth.** The growth curve can be divided into distinct phases like **lag phase, exponential phase, stationary phase** and **death phase.**

1. **Lag phase:** When a fresh population is introduced into a growth medium, bacterial growth does not take place immediately. The brief period of no activity is called as lag phase. This is time taken by the bacteria to adjust to the environment in terms of utilisation of nutrients. The lag phase is not seen in growth curve if the medium used in inoculums and the growth medium are same in the availability of nutrients. However, the lag phase is reported even in above conditions if the inoculum contains more number of damaged bacteria.
2. **Exponential phase:** This period is influenced by the environmental conditions like temperature, availability of nutrients etc. During this phase there is rapid increase in the number of bacteria.
3. **Stationary phase:** During this phase there is no increase in number of cells. However, the metabolic activities and some biosynthetic processes continue. This stage occurs as a result of utilisation of nutrients. The nutrients available at this stage only allow the bacteria to survive but not to multiply. Some bacteria have slow growth rate during this stage whereas in some the increase in growth rate is compensated by death (**Cryptic growth**). Expression of certain genes called *sur* (Survival genes) is essential for bacteria to survive during this stage.
4. **Death phase:** The death phase is also exponential but it is slow. It is due to lack of nutrients. During this phase the cells die by lysis.



The most important factors controlling the bacterial growth are from the environment in which the bacteria grow. The four common environmental factors that control the bacterial growth are temperature, pH, water availability and oxygen.

1. **Temperature**: The growing bacteria have a temperature range, which is divided into minimum temperature below which no growth occurs, and an optimum temperature at which the growth is rapid and maximum temperature above which no growth occur. These temperatures are also known as cardinal temperatures that are characteristic of each organism. At optimum temperature, the chemical and enzymatic reactions in cell are carried out at a faster rate and cell grows rapidly. However, at above maximum temperature the proteins and nucleic acid found in the cell are inactivated and the cell dies. The temperature requirements of bacteria vary widely. It ranges from 5°C to 113°C. Based on the temperature requirement, bacteria are classified in to five categories. They are **psychrophile** (13 °C), **mesophile** (39 °C), **thermophile** (60°C), **hyperthermophile** (88 °C) and **hyperthermophile** (105°C)**.** Most of the bacteria of veterinary importance are mesophiles.
2. **pH**: Like temperature requirement bacteria also prefer a pH range in which they grow better. The pH requirement of bacteria varies widely from 2-12. Based on the pH preference, bacteria are classified into **acidophile** (2-5), **neutrophile** (6-8) and **alkalophile** (10-11).
3. **Water availability (water activity and osmosis)**: It is also expressed as water activity (a_w), which is the ratio of the vapour pressure of the air in equilibrium with a substance or solution to the vapour pressure at the same temperature of pure water. The value of water activity ranges from 0-1. Based on this bacteria are grouped in to **halophile** (salt loving, prefer NaCl concentration between 1-30%), **osmophile**, which grow in high sugar concentration and **xerophile**, which grow in very dry conditions.
4. **Oxygen requirement**: Based on the oxygen requirement, bacteria are classified into **aerobic** that required oxygen and **anaerobic** that does not require oxygen.

BACTERIAL METABOLISM

Definition

The term **metabolism** refers to the sum of the biochemical reactions required for energy generation and the use of energy to synthesize cell material from small molecules in the environment. Hence, metabolism has an **energy-generating component**, called **catabolism**, and **an energy-consuming**, **biosynthetic component**, called **anabolism**. Catabolic reactions or sequences produce energy as **ATP**, which can be utilized in anabolic reactions to build cell material of bacteria

ATP

Adenosine triphosphate - During catabolism, energy is changed from one form to another, and keeping with the laws of thermodynamics. During catabolism, useful energy is temporarily conserved in the "high energy bond" of ATP - adenosine triphosphate. ATP is considered as the universal currency of energy exchange in biological systems. When energy is required during anabolism, it may be spent as the high-energy bond of ATP.

NAD

It is the coenzyme commonly involved in energy-producing metabolism, derived from the vitamin niacin (**Nicotinamide Adenine Dinucleotide)**. The basis for chemical transformations of energy usually involves oxidation/reduction reactions. For a biochemical to become oxidized, electrons must be removed by an oxidizing agent. The oxidizing agent is an electron acceptor that becomes reduced in the reaction. During the reaction, the oxidizing agent is converted to a reducing agent that can add its electrons to another chemical, thereby reducing it, and reoxidizing itself. The molecule that usually

functions as the electron carrier in these types of coupled oxidation-reduction reactions in biological systems is NAD and its phosphorylated derivative, **NADP**.

Energy production

There are two aspects concerned with energy production. They are oxidation - reduction and mechanisms of generation of ATP.

1. **Oxidation-reduction:** The term oxidation refers to removal electrons from an atom, which always results in production of energy. When an atom looses electron some other atom gains it, which is referred as reduction. For Eg. Molecule X looses an electron, which is accepted by molecule Y it is referred as X is oxdised Y is reduced. Oxidation and reduction are always coupled and hence it is referred as oxidation-reduction. In cellular oxidation reactions, along with an electron, proton is also removed and referred as **dehydrogenation**. Compounds such as glucose, which contain 12 hydrogen atoms are reduced continuously to produce potential energy.
2. **Generation of ATP:** The energy released during oxidation-reduction is trapped within the cell by the formation of ATP molecules. ATP molecules are produced by the addition of one phosphate group to ADP. The addition of a phosphate group is called **phosphorylation.** There are three methods by which ATP are produced in bacteria. They are
    i. **Substrate level phosphorylation (SLP):** Substrate level phosphorylation occurs during respiration and fermentation. In this method ATP is made during the conversion of an organic molecule from one form to another. Energy released during the conversion is partially conserved during the synthesis of the high energy bond of ATP. SLP also occurs in some bacteria that utilises inorganic substrates (Lithotrophy).
    ii. **Electron Transport Phosphorylation (ETP)**: This method is also called as **oxidative phosphorylation**. This method of production of energy is more common in respiration. ETP requires that electrons removed from substrates be dumped into an electron transport system (ETS) contained within a membrane. The electrons are transferred through the ETS to some final electron acceptor in the membrane (like O_2 in aerobic respiration), while their traverse through the ETS results in the extrusion of protons and the establishment of a **proton motive force** (**pmf**) across the membrane. An essential component of the membrane for synthesis of ATP is a **membrane-bound ATPase** (ATP synthetase) enzyme. The ATPase enzyme transports protons, thereby utilizing the pmf (protons) during the synthesis of ATP.
    iii. **Photophosphorylation:** This method is seen in bacteria, which contain chlrophylls. (The details will not be covered since photosynthetic bacteria are generally not of veterinary importance).

Types of energy production in bacteria

Eukaryotes produce energy (ATP) through ethanol fermentation (e.g. yeast), lactic fermentation (e.g. muscle cells, neutrophils), aerobic respiration (e.g. molds, protozoa, animals) or oxygenic photosynthesis (e.g. algae, plants). However, apart from these modes of energy-generating metabolism all following types of energy production which are virtually non-existent in eukaryotes are also observed in bacteria. They are as follows;

1. **Unique fermentations** proceeding through the Embden-Meyerhof pathway.
2. **Other fermentation pathways** such as the phosphoketolase and Entner-Doudoroff pathways.
3. **Anaerobic respiration**: respiration that uses substance other than 0_2 as a final electron acceptor
4. **Lithotrophy**: use of inorganic substances as sources of energy
5. **Photoheterotrophy**: use inorganic compounds as a carbon source during bacterial photosynthesis

6. **Anoxygenic photosynthesis**: photophosphorylation in the absence of O_2
7. **Methanogenesis**: an ancient type of archaeon metabolism that uses H_2 as an energy source and produces methane
8. **Light-driven nonphotosynthetic photophosphorylation**: unique archaeon metabolism that converts light energy into chemical energy.
9. Autotrophic procaryotes, there are three ways to fix CO_2, two of which are unknown among eukaryotes, the **CODH (acetyl CoA pathway)** and the **reverse TCA cycle**.

Enzymes

In a living cell, chemical reactions occur as a result of formation of bonds, which are formed as well as broken. A normal chemical reaction required collision of molecules for formation of bond between them. For collision to occur, certain physiological conditions like temperature and pressure inside the cell have to be either increased or decreased. However, if such alterations are made inside a cell, it may lead to death of cells. Hence, in a living cell certain substances catalyses chemical reactions by bringing two molecules close to each other. Such catalytic molecules, which are found in biological systems are referred as **enzymes.** These enzymes act on specific molecules, which are referred as **substrates.** The binding of a enzyme to a substrate is always specific. The enzymes are named by the reaction they perform followed by extension –ase.

Classes of enzymes: There are six different classes of enzymes depending on the actions they perform.

1. *Oxidoreductase* – Performs oxidation-reduction in which electrons are gained or lost. Eg. Cytochrome oxidase.
2. *Trasferase* – Helps in transfer of a functional group such as amino group. Eg. Alanine deaminase
3. *Hydrolase* – Helps in addition of water (hydrolysis) – Lipase, Sucrase
4. *Lyase* – Helps in removal of atoms without hydrolysis – Oxalate decarboxylase
5. *Isomerasse* – Helps in rearrangement of atoms within a molecule – Glucose phosphate isomerase
6. *Ligase* – Helps in joining two molecules – DNA ligase.

Components of enzymes: Only few enzymes are composed of entirely of proteins. Whereas, most of the proteins are made up of a protein component called **'apoenzyme'** and a non-protein component called **'cofactor'.** The cofactor is called **coenzyme** when it is a metallic ion or a complex molecule. The apoenzyme along with cofactor is called **haloenzyme.**

Coenzymes: They are very important components in metabolic reactions. They assist the enzyme by accepting electrons from the substrate or donates the electron required to the substrate (**electon donator and acceptor).** Two coenzymes are very important for metabolic reactions they are **nicotinamide adenine dinucleotide (NAD^+)** and **nicotinamide adenine dinucleotide phophate ($NADP^+$).** Generally the derivatives of B-complex vitamins serve as coenzymes.

Mechanism of enzyme action: Enzymes act on specific molecules, which are referred as substrate. The specific region where the enzyme binds with substrate is called **active site.** Binding of enzyme to active site in substrate lead to the formation enzyme-substrate complex. The three dimensional shape of the active site should match with that of enzyme (lock and key fashion) for formation of complex. This leads of rearrangement of atoms in the substrate molecule resulting in its breakdown or conversion to another molecule. As a result this the active site changes in substrate molecule and enzyme molecule is released. The unchanged enzyme molecule goes and combines with another unaltered substrate.Thus a small number of enzyme molecules are enough to perform a big reaction.

Factors influencing enzymatic activity: The important factors that influence the enzymatic activity are temperature, pH, substrate concentration and presence or absence of inhibitors. Temperature to a certain extent will favour the reaction and beyond threshold level it will denature the enzymes. The optimum pH for different enzymes vary widely and above or below the optimum pH the enzyme reaction will get affected. The substrate concentration should be in a such a level that the active sites are not over saturated. Inhibitors play a major role in enzyme substrate combination. There are two types of inhibitors **competitive and non-competitive.** Competitive inhibitors are those whose shape and chemical structure match that of substrate leading to competition with normal substrates. The non-competitive inhibitors not bind with enzyme's active site, but may bind on some other area resulting alteration of the shape active site. Because of this, normal substrate cannot bind with enzyme. Such type of inhibition is called as **allosteric inhibition.**

Feedback inhibition: Allosteric inhibitors are mainly responsible for feedback inhibition. In this method the end product of a complex reaction, which involves multiple enzymes, go a bind with the enzyme that mediates first reaction in the complex. As a result of such allosteric binding, substrate can no longer bind with enzyme and whole reaction is thus stopped. This is essential to shut down the metabolism, and save enzymes when they are not required.

Carbohydrate catabolism

Definition

The breakdown of carbohydrate molecules to produce energy is called carbohydrate catabolism. Among the different carbohydrates glucose is the most common energy source used by bacteria.

There are two methods by which energy is produced from glucose namely **respiration** and **fermentation.** Respiration is defined as an ATP generating process in which molecules are oxidized and the final electron acceptor is almost always an inorganic molecule. In aerobic respiration the final electron acceptor is molecular oxygen, whereas in anaerobic respiration the final electron acceptor is an inorganic molecule other than oxygen (rarely organic molecules are also used as final electron acceptor). Respiration is broadly divided into three stages namely **Glycolysis, TCA cycle (Kreb's cycle)** and **Electron transport chain and chemiosmosis.** In fermentation, after glycolysis a separate pathway is followed. The end products of respiration and fermentation are also different. In respiration water is the end product, whereas in fermentation ethanol is the end product.

Types of respiration

The method of production of energy by bacteria is further divided into aerobic and anaerobic respiration. The important difference between these two methods is in the terminal electron acceptor. In aerobic respiration, the terminal electron acceptor is molecular oxygen. Whereas in anaerobic respiration instead of molecular oxygen, the terminal electron acceptors are nitrate, sulfate or carbonate ions. When nitrate ions are utilised they are reduced to nitrous oxide and nitrogen gas, sulfate ions are reduced to hydrogen sulfide and carbonate ions are reduced to methane.

Aerobic Respiration

There are three principal stages in respiration. They are

1. **Glycolysis:** Glycolysis is oxidation of glucose into pyruvic acid with production of some ATP and energy containing NADH.
2. **TCA cycle:** In this cycle the derivative of pyruvic acid – Acetyl CoA is oxidized to production of some ATP, energy containing NADH and $FADH_2$.
3. **Electron transport chain and chemiosmosis (ETS):** In this stage, the electon carriers NADH and $FADH_2$ are contribute the electrons and energy from these reactions is used to generate considerable amount of ATP. Most of ATP produced in respiration is produced in this stage.

Glycolysis

Glycolysis (Embden-Meyerhof pathyway)

Glycolysis is defined as oxidation of glucose to pyruvic acid. The most common sequence of reaction through which glucose is split into pyruvic acid by most of the bacteria is referred as Embden-Meyerhof pathyway (EMP). The EMP consists of ten stages, which are broadly grouped into two stages namely preparatory stage and energy conserving stage.

In the preparatory stage two ATP molecules are utilized to phosphorylate and restructure glucose into two three carbon atom compounds namely dihydroxyacetone phosphate (DHAP) and glyceraldehydes 3-phosphate (GP).

In the energy conserving stage the two three carbon atom compounds are oxidized by several steps to yield two molecules of pyruvic acid. Further in this stage four molecules of ATP are produced and two molecules are NADH are also produced.

Alternatives to Glycolysis (Embden-Meyerhof pathway): There are two pathways alternative to EMP namely Pentose phosphate pathway and Entner-Doudoroff pathway.

Pentose phosphate pathway - It is also called **hexose monophosphate shunt (HMP).** It operates simultaneously with EMP. One of the important features of this pathway is that it produces pentose sugars as intermediates, which act as precursor in the synthesis of nucleic acid and certain amino acids. Unlike EMP, only one molecule of ATP is produced in this pathway. However, it is an important pathway for production of NADPH from NADP. Eg.*E.coli*

Entner-Doudoroff pathway – This pathway functions independent of EMP and HMP shunt. In other words bacteria that have enzymes for this pathway will not use EMP and HMP. In this pathway one molecule of ATP and two molecules of NADPH are produced. Eg.*Pseudomonas.*

TCA cycle

The TCA cycle is also called as Kreb's cycle or citric acid cycle. In this large amount of potential energy stored in the acetyl CoA is released step by step. Pyruvic acid obtained from glycolysis cannot enter into TCA cycle as such, it first looses one molecule of carbon dioxide (**decarboxylation**) to produce 2 carbon acetyl group, which combines with coenzyme A to form Acetyl CoA that enters into TCA cycle. Acetyl CoA condenses with the 4C-compound, oxaloacetate, to form the first stable intermediate of the TCA cycle, 6C-citric acid (citrate), a tricarboxylic acid. Citrate is isomerized to isocitrate, which is oxidized and decarboxylated forming alpha-ketoglutarate (akg). Alpha ketoglutarate dehydrogenase uses CoA and NAD to oxidize akg to succinyl CoA in a reaction analogous to the pyruvate dehydrogenase reaction above. Succinyl CoA is converted to succinate during a substrate level phosphorylation yielding high energy GTP (equivalent to ATP). This completes the decarboxylation of pyruvate forming 3 CO2. The remaining three steps in the cycle complete the oxidation of succinate and regenerate the oxaloacetate necessary to drive the cycle. During the oxidation of pyruvate to 3 CO2 by one turn of the TCA cycle, 4 NADH2, 1 FADH2 and one ATP (actually GTP) are produced.

The TCA cycle (including the steps leading into it) accounts for the complete oxidation of the substrate and provides 10 pairs of electrons (from glucose) for transit through the ETS. For every pair of electrons put into the ETS, 2 or 3 ATP may be produced, so a huge amount of ATP is produced in a respiration, compared to fermentation. Glucose is dissimilated in a pathway of glycolysis to the intermediate, pyruvate, and it is the pyruvate that is moved into the TCA cycle to become oxidized to 3 CO_2. Since 2 pyruvates are formed from one glucose, the cycle must turn twice for every molecule of glucose oxidized to 6 CO_2.

Electron transport chain

An electron transport chain consists of a sequence of carrier molecules that are capable of oxidation and reduction. There are three different types of carrier molecules – **flavoproteins (FMN)**, **cytochromes (cyt)** and **ubiquinones** (**coenzymeQ**).

An ETS consists of 7 carrier molecules, which are arranged as **FMN, Coenzyme Q, cyt b, cyt c1, cyt c, cyt a and cyt a3.** High-energy electrons as NADH from TCA cycle, are first accepted by 1st carrier molecule and passed through other carrier molecules to last carrier molecule and finally to terminal electron acceptor. The carrier molecules are reduced as a result of accepting high-energy electrons from TCA cycle (as NADH) and oxidized as a result of passing it to next carrier molecule. As electrons pass through the chain, there is stepwise release of energy, which is used to drive the chemiosmotic generation of ATP.

Chemiosmosis

Chemiosmotic generation of ATP: There are three stages in generation of ATP, which are

1. Active transport of protons (**proton pumps**) by carrier molecules FMN and coenzyme Q.
2. As a result of pumping of protons, high concentration of protons are established on one side of plasma membrane resulting in formation of **proton motive force.**
3. Protons pass from high concentration to low concentration through special channels in plasma membrane. The walls of these channels contain the enzyme adenoside triphosphatase (ATP synthase). This enzymes catalyses formation of ATP from ADP utilizing the energy. After oxidation of the last carrier molecule (cyt a3), the released electron is accepted by molecular oxygen, which is negatively charged and this picks up protons from surrounding medium to form water. Water is the end product of respiration.

Yield

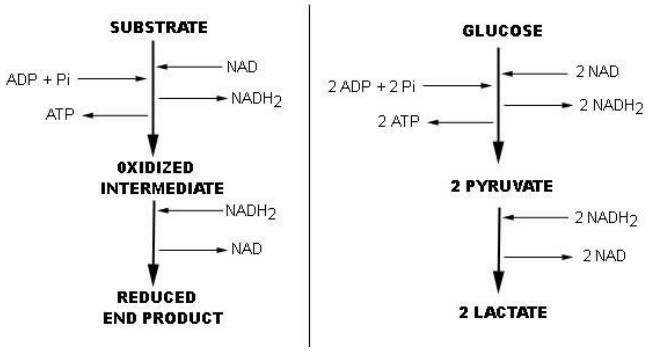
Thirty-eight molecules of ATP are produced through respiration.

$$C_6H_{12}O_6 + 6O_2 + 38ADP + 38P \longrightarrow 6CO_2 + 6H_2O + 38ATP$$

Fermentation

The second method by which energy is produced from carbohydrate is referred as fermentation. In fermentation, energy is derived from the **partial oxidation of an organic compound** using **organic intermediates as electron donors and electron acceptors**. No outside electron acceptors are involved; no membrane or electron transport system is required; all **ATP is produced by substrate level phosphorylation**. The **pathways of fermentation** are a bit more complex, usually involving several preliminary steps to prime the energy source for oxidation and substrate level phosphorylation. In prokaryotes there are three major pathways of glycolysis (the dissimilation of sugars).

Fermentation and respiration are common up to glycolysis. The end product of glycolysis ie. Pyruvic acid enters into TCA cycle as acetyl coA in case or respiration, whereas it is converted either to lactic acid or ethanol in fermentation. The end products of respiration are water and carbon-dioxide, which do not carry any energy. But in fermentation, the end products lactic acid and ethanol are high-energy compounds carrying residual energy. Further the number of ATP molecules produced is also much less in fermentation compared to respiration.



Types of Fermentation

Depending on the end product fermentation is classified as below;

1. **Homolactic Fermentation:Lactic acid** is the sole end product. These bacteria are used to ferment milk and milk products in the manufacture of yogurt, buttermilk, sour cream, cottage cheese, cheddar cheese, and most fermented dairy products.
2. **Mixed Acid Fermentations:** The end products in this pathway are a mixture of **lactate**, **acetate**, **formate**, **succinate** and **ethanol**, with the possibility of gas formation (**CO_2** and **H_2**). This pathway is commonly seen among members of E*nterobacteriaceae.*
3. **Butanediol Fermentation:** Forms mixed acids and gases as above, but, in addition, also produces **2,3 butanediol** from the condensation of 2 pyruvate molecules.
4. **Butyric acid fermentations**: In addition to butyric acid acetate, CO_2 and H_2 are also produced from sugars. Small amounts of ethanol and isopropanol may also be formed. This pathway is present in clostridia.
5. **Butanol-acetone fermentation**. Butanol and acetone are the end products.
6. **Propionic acid fermentation**. This is a bizarre fermentation carried out by the propionic acid bacteria which include Corynebacteria, *Propionibacterium* and *Bifidobacterium*. The propionic acid bacteria are used in the manufacture of Swiss cheese, which is distinguished by the distinct flavor of propionate and acetate, and holes caused by entrapment of CO_2

Lipid catabolism

The catabolism or breaking of lipids is as follows;

1. **Action of lipase on fats:** Microbes produce enzymes called lipase, which breaks fats into fatty acid and glycerol.
2. **Conversion of glycerol into DHAP:** The glycerol is converted into dihyroxyacetone phosphate (DHAP) (DHAP is one of the products of glycolysis). DHAP is then metabolized by glycolysis and TCA cycle to produce enery.

Component	Precursor
Polysaccharide capsules or inclusions	Polymers of glucose and other sugars
Cell wall peptidoglycan (NAG and NAM	Glucose phosphate
Nucleotides (DNA and RNA)	Ribose phosphatase
Triose-phosphate	Glycerol
Acetyl CoA	Lipids for membranes
Amino acids	Pyruvate, alpha-ketoglutarate and oxaloacetate
Vitamins and coenzymes	Various pathways that leave central metabolism

3. **Beta oxidation:** The fatty acids are oxidized. In the process of oxidation, carbon atoms from long chains of fatty acids are removed two at a time till Acetyl CoA is formed. Then Acetyl CoA enters into TCA cycle and metabolized.

Protein catabolism

The catabolism or breaking of proteins is as follows;

1. **Action of proteases:** Microbes produce extracellular proteases and peptidases, which break the proteins into component aminoacids. These amino acids pass through the membrane into the cell for further breakdown.
2. The amino acids enter into TCA cycle by any one of the following chemical reaction – deamination, decarboxylation or dehydrogenation.
3. **Deamination:** The amino groups in amino acids are removed and the resultant organic acid enters into TCA cycle and metabolized.
4. **Decarboxylation:** The -COOH groups in amino acids are removed and the resultant organic acid enters into TCA cycle and metabolized.

Anabolism

Anabolism is referred as biosynthesis. The pathways of central metabolism (glycolysis and TCA cycle) also provide the precursors for the biosynthesis of cell material apart from production of energy. Hence, Embden-Meyerhof pathway and the TCA cycle, which functions to provide energy in addition to chemical intermediates for the synthesis of cell material, are referred as **amphibolic pathways**. The fundamental metabolic pathways of biosynthesis are similar in all organisms, in the same way that protein synthesis or DNA structure are similar in all organisms. Some of the important components of bacterial cell and their precursors are given below.

Component	Precursor
Polysaccharide capsules or inclusions	Polymers of glucose and other sugars
Cell wall peptidoglycan (NAG and NAM	Glucose phosphate
Nucleotides (DNA and RNA)	Ribose phosphatase
Triose-phosphate	Glycerol
Acetyl CoA	Lipids for membranes
Amino acids	Pyruvate, alpha-ketoglutarate and oxaloacetate
Vitamins and coenzymes	Various pathways that leave central metabolism

Lithotrophic metabolism

Lithotrophy is the use of an inorganic compound as a source of energy. Most lithotrophic bacteria are aerobic respires that produce energy in the same manner as all aerobic respiring organisms: they remove electrons from a substrate and put them through an electron transport system that will produce ATP by electron transport phosphorylation. However, the source of electrons is from an inorganic, rather than an organic compound.

BACTERIAL GENETICS

Bacterial variation

In a bacterial population, the daughter cells not always resemble parents and some variations are noticed in certain of them. When these variations become inheritable, they gain importance. Bacterial variation could be due to either **mutation** or **recombination. Mutation** is defined as inheritable change in the base sequence of nucleic acid comprising the genome of the organism. **Genetic recombination** is the process by which genetic elements contained in two separate genomes are brought together in one unit. While the mutation bring about small amount of genetic changes, recombination bring large amount of changes. In recombination, genes, sets of genes or even the entire chromosome is shifted between organisms.

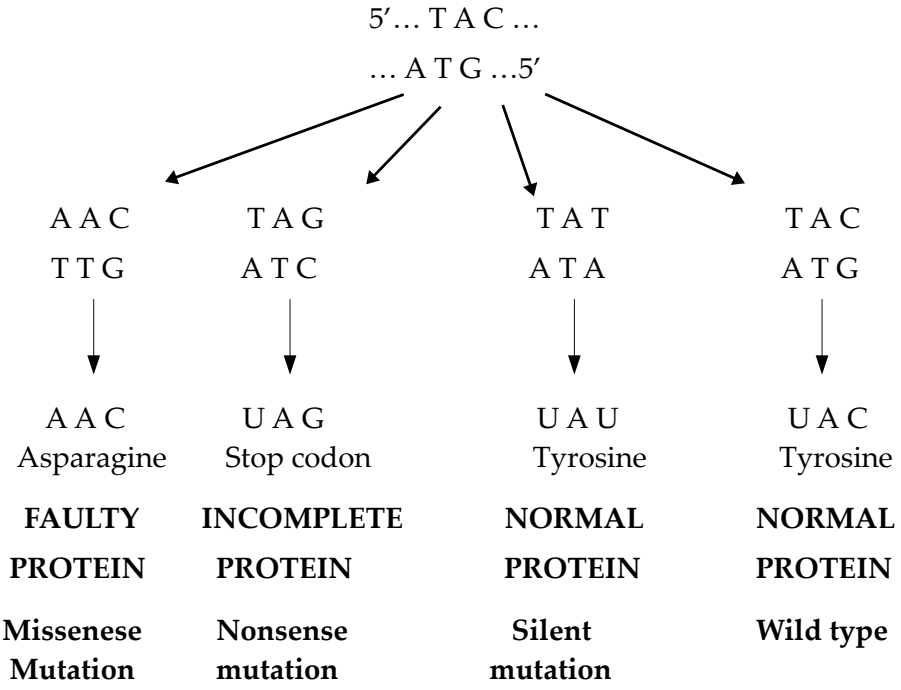
Mutation

Mutation is defines as inheritable change in base sequence of nucleic acid. A strain carrying a mutation in its genome is called as **mutant.** A mutant strain differs from its parental strain in genotype whereas the phenotype may or may not be altered. Mutations could be either **spontaneous** or **induced.** Spontaneous mutation occurs as a result of natural radiation or during replication of genome. The frequency of such errors during replication is approximately 10^{-7}tp10^{-11}. Induced mutation is mediated by number of chemicals, x-ray or radiation at laboratory condition.

Spontaneous mutation

Mutations involving one or very few base pairs:

Point mutation: Mutations involving one or very few base pairs are referred as **point mutation**. Point mutation could be either a single base pair insertion (**microinsertion**) or single base pair deletion (**micro deletion**) or even alteration. The phenotypic changes brought about by point mutation depend upon the changes in amino acid sequence as a result of point mutation. Due to degeneracy of code and wobbling, point mutation not always brings change in the phenotype. This is well illustrated by the following



Sometimes a change in a single nucleotide base results in alteration of the codon that may code for a different amino acid. As a result of this, the protein structure will alter. Such mutations are called **missense mutation**. If a mutation results in sudden appearance of stop codon that will result in translation of a faulty protein and such mutation are called **nonsense mutation**. Due to degeneracy of genetic code, change in the third base may still result in production or same amino acid and such mutation are called **silent mutation**. The term **wild type** refers to strains isolated from nature. A missense mutation can result in production of an altered enzyme that may change the character of the organism. Such alterations result variations in the exhibition of characters from wild type. For example, a wild type may grow in both 30°C and 40°C; whereas an organism with missense mutation may not grow at 40°C and will grow only at 30°C. Such mutation is also called as **conditionally lethal mutation.** It is also known as **temperature sensitive mutation**. The genetic code is always read from left to right as blocks of three. Any insertion or deletion of base pair may result in a shift in the reading frame resulting in entirely different translation. Such mutations are called **frame shift mutation**. Point mutations are revertible. If the reverting occurs as a result of change at the same site of mutation, it is called **same site**

revertant and if it occurs at different site it is called **second site revertant.** Such mutations are called **back mutation** or **reversion**. As a result of back mutation, if the same wild sequences are obtained again it is called **true reversions** and such strains are called as **truer evertants.** True reversion could be obtained as result of suppressor mutation. When the effects of one mutation are compensated by other mutation it is termed as **suppressor mutation**.

Mutations involving many base pairs: These mutations involve insertion of deletion of many numbers of bases and are termed as **insertions** or **deletions**. The number of bases inserted or deleted may range from hundreds or thousands. When so many bases are deleted that may result in loss of some character. Micro insertion occurs as a result of error in copying whereas insertion of large number of bases occurs as a result of genetic recombination. The sequence that is inserted is called as **insertion sequence.** The other types of mutations involving many base pairs are **translocation** and **inversions.** In translocation there number rearrangements occur as a result of recombination. In inversion, orientation of a particular segment is reversed with respect to surrounding DNA.

Mutagens

Mutagens are physical, chemical or biological substances that induce mutation in the genome. Some of the important chemical and mutagens are listed in the table below.

Agent	Action
Base analogues	
5-bromouracil	Incorporated like T, occasional faulty pairing with G
2-aminopurine	Incorporated like A, faulty pairing with C
Chemicals reacting with DNA	
Nitrous acid	Deaminates A and C
Hydroxylamine	Reacts with C
Alkylating agents	
Ethylmethanesulphate	Puts methyl of C, faulty pairing with T
Nitrogen mustard	
Mitomycin	Cross links DNA strands faulty excision by DNase, and inserts beween two base pairs
Nitrosoguanidine	
Intercalative dyes	
Radiation	
Ultraviolet	Pyrimidine dimer formation
Ionising radiation (X-ray)	Free radical attack on DNA, breaking chain
Biological	
Transposable element	Transposons, Mu bacteriophage

Induced mutation

Mutations can be induced at specific genes in genetic engineering such mutation are called **site directed mutation** and the process is termed as **site directed mutagenesis**.

Isolation of mutants

Mutants are identified by the phenotypic characters they exhibit. **Screening** is one of the common methods to identify the mutants. For example due to nonsense mutation an organisms might have lost of the character of pigmented colonies. This could be identified by screening large number of colonies. The variation in nutritional characters could be identified by replica plating. This identifies **auxotrophs** – organisms that obtained variations in nutritional character through mutation. The wild type from which **auxotrophs** are derived is called **prototrophs.**

Genetic recombination

The second form of bacterial variation is known as genetic recombination. In this method there is physical exchange of genetic material between two organisms. While the genetic mutation is self-mediated and involves only one organism, genetic recombination involves two organisms. The genetic exchange of homologous DNA sequences between two organisms is known as **general** or **homologous** genetic recombination. The process of genetic recombination is complex involving number of pathways.

Mechanism: The process begins with a nick in one of the DNA molecules. The enzyme DNase mediates it. Then the proteins having helicase activity separates the strand with nick from other strand. Then a single stranded DNA binding protein binds with separated DNA strand. Next the RecA protein binds with single stranded fragment forming a complex and facilitates annealing (joining) with complementary sequence of adjacent DNA, simultaneously displacing it. This process is known as **strand invasion**. This result in a DNA molecule having strands from different chromosomes and the recombined regions are known as **heteroduplex** regions. This process also involves the activities of DNA polymerase and DNA ligase.

In prokaryotes, the genetic recombination occurs as fragments of homologous DNA from a donor chromosome transferred to a recipient cell. It occurs by any one of the following processes, **transformation –** that involves donor DNA that is free in the environment, **transduction** – that involves the transfer of donor DNA mediated by bacteriophage and **conjugation** – that is mediated by conjugative plasmids and sex pili.

Transformation

Genetic transformation is a process by which the free DNA found in the environment is incorporated into the recipient cell. Free bacterial DNA is commonly found in the environment as a result of lysis of the bacteria. Due to its large size, the DNA breaks into small pieces and each piece will have number of genes. The recipient cell that takes up DNA molecule from environment is called as **competent cell** and the character is known as **competence**. The uptake of DNA molecule by competent cells is controlled by number of proteins. DNA binding protein first binds the transforming DNA to the cell surface**.** Some bacteria take dsDNA molecules whereas some bacterial take only ssDNA molecules. Inside the cell, the DNA molecule is prevented from nuclease activity by a **competence-specific protein**. Then the DNA is integrated into the bacterial genome, which is controlled by RecA protein.

Transduction

Transduction is a process in which DNA is transferred from cell to cell through viruses that infect bacteria called as **bacteriophages.** There are two types of transduction. In **generalised transduction**, host DNA derived from virtually any portion of the host genome becomes a part of mature virus particle in place of virus genome. Whereas in **specialised transduction**, only some specific region of host chromosome is integrated directly into virus genome replacing some virus genes and it occurs in temperate phages.

1. **Generalised transduction:** In this method any host gene can be transferred from donor to recipient. When a lytic phage infects a bacterial population, it causes lysis of bacterial cell. During the assembly stage of the virion, sometimes accidentally fragments of host genome are packaged (inserted) into the phages instead of virion DNA. Such bacteriophage carrying host genome instead of their own DNA are called **transducing particles.** They are also known to be **defective** since they cannot replicate on their own. When a population of bacteriophage containing transducing particles infect a bacteria, the normal particles cause lysis of the cell. Whereas the transducing particles recombine with the host genome as per the mechanism mentioned earlier. However, the number of such transducing particles in a virulent phage population is low.

2. **Specialised transduction:** This method is efficient in transfer of genetic material compared to the previous method and involves temperate phages, which replicate via lysogenic cycle. In specialised transduction, the phage DNA is integrated into host genome and forms prophage. When it remains as prophage the replication is under the control of host genome. As a result of induction, prophage separates from host genome and enters a lytic cycle. However, some times, the excision of phage DNA is faulty. Due to faulty excision, the phage DNA carries with itself some host DNA and loses some of its own DNA. As a result of this, it cannot replicate on its own and require a helper virus. When a population of phages containing defective particles (carrying host genome) and helper virus infect susceptible cells, the defective particles (carrying host genome of donor) provide the character of gene it carries to the recipient.
3. **Phage conversion:** This phenomenon is same with specialised transduction. When a temperate phage infects bacteria and becomes prophage, the lysogen becomes immune to subsequent infection by the same phage, which is considered a change in phenotype. Besides this, some unrelated phenotypic changes are also noticed in lysogenised cells. The change brought about in a cell as a result of lysogenisation by a temperate phage is called phage conversion.

Conjugation

Conjugation is a process of transfer of DNA directly from one bacterial cell to another cell by a mechanism that requires cell-to-cell contact. The DNA transferred could be either chromosomal DNA or plasmid DNA. It is a replicative process in which both the donor and recipient cell get a copy of the transferred material. Conjugation involves a **donor cell** and recipient cell. The donor cell contains a conjugative plasmid (F+) whereas the recipient cell (F-) does not. Hence, the donors are referred as **males** and recipients are called as **females.** The gene that controls the conjugation *tra* also produces a surface structure called **sex pilus.** This structure is found only in donor cell and not in recipient cell. The sex pilus of donor cell attaches with a suitable receptor found in the recipient cell forming a **conjugation bridge.** The transfer of genetic material takes place through this bridge. Once connection is established between donor and recipient cell, the sex pilus retracts and brings the cell close to each other. For DNA transfer to occur, DNA synthesis is essential. It occurs close to the conjugation bridge. The first stage in transfer is a nick at the host plasmid DNA and separation of strand and one strand is transferred to the recipient cell. While one strand passes through the conjugation bridge, new complementary DNA strand synthesis is initiated at two sites one at recipient cell and another at donor cell. This type of DNA synthesis is also referred as **rolling circle model.** Thus a recipient cell receives plasmid from donor and obtains phenotypic characters. Further, the recipient also becomes a donor now. Conjugation is not only mediated by sex pilus. In certain bacteria, the recipient cell produce a substance called **pheromones** that attracts the plasmid containing bacteria which in turn allows the donor cells to produce a substance that facilitates attachment of donor and recipient cells and plasmids are transferred. When chromosomal DNA are transferred it is referred as Hfr conjugation. In this process, in the donor cell first the plasmid integrates with the chromosomal DNA followed by transfer of one strand to F- cell. But in this process, the host loses its plasmid.

Antibiotic resistance

Antibiotic resistance is the acquired ability of an organism to resist the effects of an antibiotic to which the organism is normally susceptible. There are several mechanisms by which the organisms acquire antibiotic resistance. They are as follows;

1. The organism may lack a structure, the synthesis of which is inhibited by an antibiotic. Eg. *Mycoplasma.*
2. The organism may be impermeable to an antibiotic.

3. The organism may have the capacity to convert the antibiotic to an inactive form.
4. The organism may modify the target of the antibiotic.
5. The organism may modify its metabolic pathway to escape the action of an antibiotic.
6. Efflux (pumping out) action of some bacteria.
7. Resistance plasmid (R-plasmid) mediated.

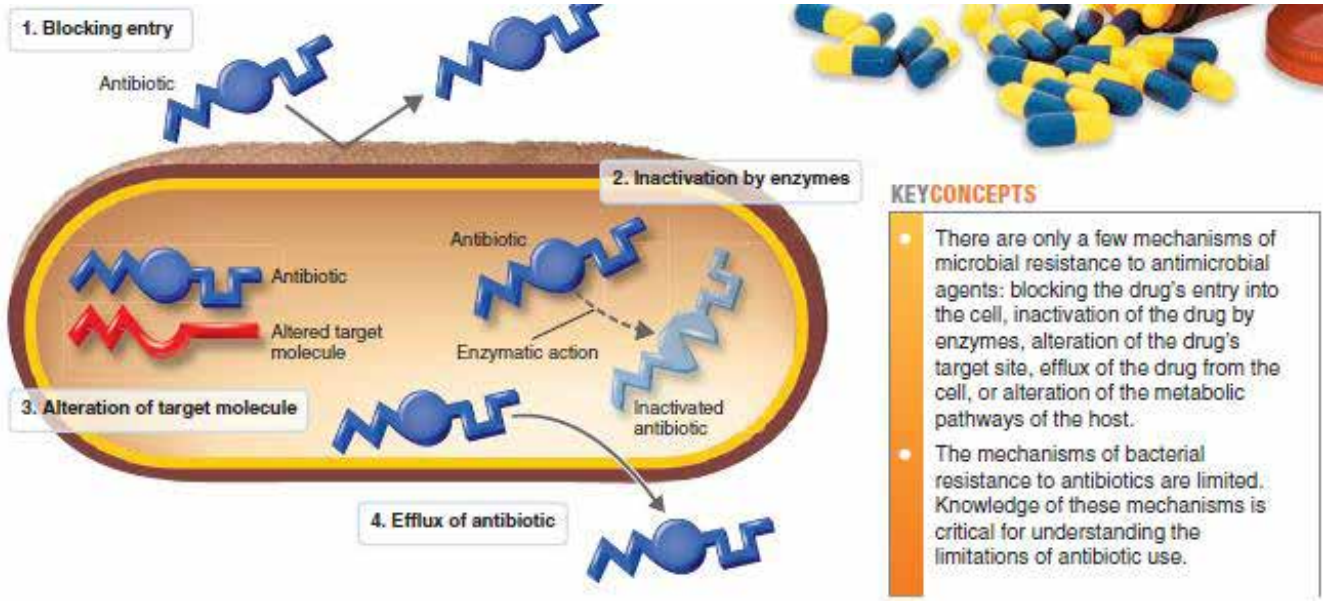
Bacteriocins

Certain microorganisms produce substances which provide them competitive advantage or means of defence in establishing themselves in natural environments. One of such substances is **bacteriocin**. They are also called as **protein antibiotics**. These substances have high specificity and extremely active at low concentrations.

There are three types of bacteriocins. They are

1. **Defective bacteriophages:** Eg. Phonics.
2. **Broad-spectrum bacteriocins:** (Megacins) chemically they are phospholipase that cause susceptible cell to leak intracellular contents by attacking the membrane phospholipids.
3. **Classic bacteriocins:** They are soluble proteins with molecular weight 50000-100000 Da. The best-studied classic bacteriocins are colicins produced by coliform bacteria. The colicins are classified depending on their receptor specificity and mode of action. Eg. E1,E2,E3, DF13 etc.

Acquired resistance that develops due to chromosomal mutation and selection is termed *vertical evolution*. Acquired resistance that develops due to acquisition of new genetic material from other resistant organisms is termed *horizontal evolution*, and may occur between strains of the same species or between different bacterial species or genera. Mechanisms of genetic exchange include conjugation, transduction, and transformation.



Development of Resistance during Treatment

Mutation: Resistant mutants may emerge during treatment. The occurrence and establishment of resistant mutants vary with different drugs. Selection of mutants is favored by under dosage, prolonged administration, and the presence of a "closed" focus of infection such as is found in an abscess.

The drugs are grouped below according to the likelihood that resistant mutants will emerge during treatment.

Frequent	:	sulfonamides, streptomycin, nalidixic acid, rifamycin.
Less *frequent*	:	erythromycin.
Infrequent	:	tetracyclines, penicillin, cephalosporins, chloramphenicol, nitrofurans.

Superinfection:

This term is frequently used to describe the infection that develops and is caused by "alien" organisms such as *Candida albicans,* as a result of the alteration or suppression of the normal flora by prolonged antibiotic administration. The term is also used to describe the replacement of the original infecting agent with a new strain of the same species that is resistant, or a resistant strain of another species.

Infectious Drug Resistance:

Transferable multiple drug resistance caused by R factors occurs most commonly in members of the *Enterobacteriaceae.* This type of resistance, i.e., the whole multiple pattern, is transferable among strains of the same species and among strains of closely related species, e.g., from *Salmonella* sp. to an *Escherichia coli.*

BACTERAEMIA, SEPTICAEMIA AND TOXAEMIA (ENDOTOXIN & EXOTOXIN)

Bacteraemia

Bacteraemia is the presence of bacteria in blood. The pathogenic organisms may gain entry into a blood capillary or venule by active or passive means from the initial site of entry. Once in blood stream, the organism can spread to various parts of the body and cause localized infections. Eg: Leptospira can reach the kidneys by means of bacteraemia. Organisms can also directly gain access to the blood by first infecting the lymphoid system.

Septicaemia

In septicaemia, the bacteria in the blood stream actively multiply and produce toxic products. One of most severe septicaemia is Anthrax in which the number of bacteria in blood may often exceed the erythrocytes in blood. Septicaemic infections often start as localized infections that later become generalized. Eg: Streptococcal pharyngitis, bubonic plague.

Toxaemia

Toxaemia is the presence of toxins in blood.

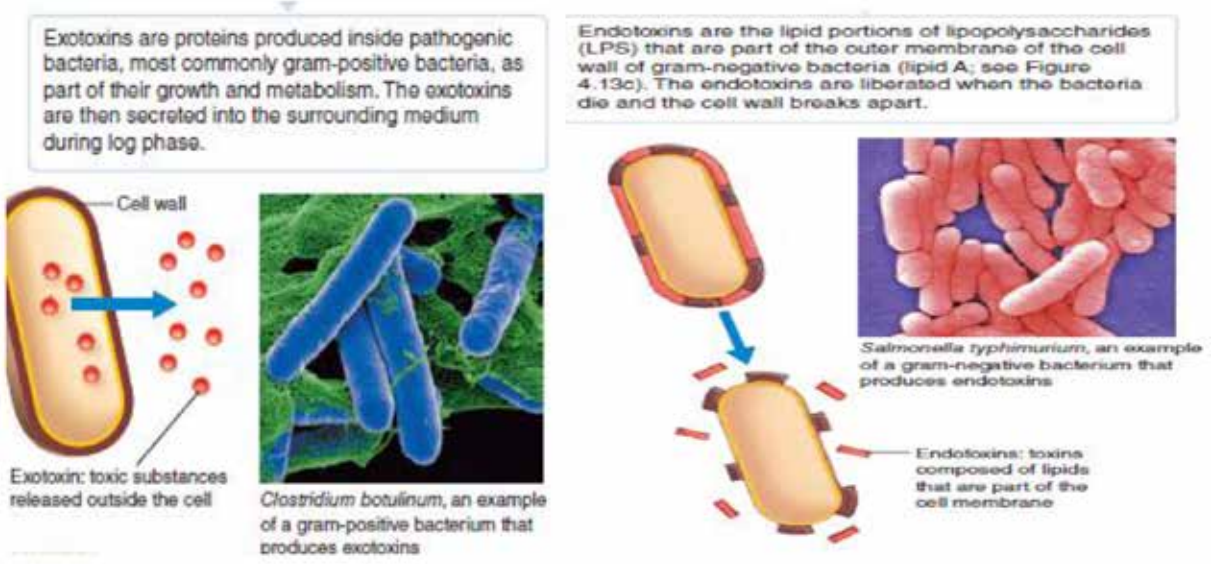
Bacteria can damage host tissues directly through the effects of exotoxins and endotoxins. Moreover, tissue damage can result indirectly from the activity of enzymes secreted bythe bacteria from the inflammatory reactions and immune responses of the host. Bacterial exotoxins and endotoxins differ in their structures and modes of action.

1). Exotoxins

Exotoxins are produced by Gram- positive / Gram-negative bacteria within the body and exert their effects either locally or systemically. Occasionally, exotoxins, such as the potent toxin of *Clostridium botulinum* are ingested in contaminated food and produce systemic effects. Some exotoxins cause cell death either by digesting lipids in cytoplasmic membranes or by insertion into the membranes, forming protein pores.

- Exotoxins are toxic proteins secreted by living microorganisms.
- Exotoxins can be divided based on site of action and the cell type affected.
- Neurotoxins – tetanus toxins, enterotoxin – *Vibrio cholera*.

- Diphtheria toxin kills several types of cells and hence termed cytotoxin.
- Some cytotoxins kill leukocytes and hence called leukocidins.
- Those which lyse the erythrocytes are called hemolysins.
- Formaldehyde treatment of exotoxins causes inactivation of the toxigenicity without affecting the immunogenicity. The formaldehyde treated toxin is called toxoid. It induces antitoxins in the body which react with toxins and neutralize them. Eg: tetanus and diphtheria toxoids.



2). Endotoxins

Endotoxins, which are lipopolysaccharides present in the outer membrane of Gram-negative bacteria, are released when the cells are lyzed. This occurs during the later growth stages of culture. All endotoxins exhibit similar pharmacological effects. They cause pyrexia, blood changes and shock. Pyrogenicity is the ability to change body temperature - fever as a response.

The salient points in comparison between the endotoxins and exotoxins are summarised below;

Feature	Exotoxins	Endotoxins
Source	Live bacteria, both Gram-positive and Gram-negative	Component of the cell wall of Gram-negative bacteria released following cell death
Composition	Proteins, usually of high molecular weight	Lipopolysaccharide complex containing lipid A, the toxic component
Sensitivity to heat	Thermolabile	Thermostable
Toxins and functions	Potent toxins, usually with specific activity; not pyrogenic. Highly antigenic; readily converted into toxoids which induce neutralizing antibodies	Toxins with moderate, non-specific generalized activity; potent pyrogens, weakly antigenic; not amenable to toxoid production. Neutralizing antibodies not associated with natural exposure
Synthesis determined	Extrachromosomally	In chromosome
Mechanism of action	Each toxin has a highly characteristic action	All act similarly
Lethal dose	Small	Large

PATHOGENESIS, VIRULENCE AND INFECTION

The following are the terms that are frequently used in the understanding of infectious diseases;

- ***Infectivity*** – It is the capacity of the organism to become established in the tissues of the host. It is the ability to penetrate the tissues, to survive in the host defenses and to multiply and disseminate in the animal.
- ***Pathogenicity*** – It is the capacity of the organism to produce disease.
- ***Virulence*** – It is the measure of degree of pathogenicity.

  For example, pathogenic strains of *Streptococcus equi* may vary in their capacity to produce disease in the horse. All may cause disease, but some may cause more severe disease than others.
- ***Toxigenicity*****:** This is the capacity of certain organisms to produce exotoxins. For example, there are both toxigenic and non-toxigenic strains of are both toxigenic and non-toxigenic strains of *Clostridium perfringens*; only the former cause disease.

PATHOGENIC PROPERTIES OF BACTERIA:

The two basic mechanisms by which bacteria cause disease include;

i. Invasion of tissues

ii. Production of toxins.

A great variation exists in the extent of invasiveness of microorganisms ranging from the highly invasive *B. anthracis*, moderately invasive *Pasteurella* sp. to the non-invasive tetanus bacilli. Invasive bacteria are further classified as intracellular or extracellular parasites.

Intracellular parasites:

a) Facultative intracellular parasite

These microorganisms are not confined to cells, but they can survive and multiply in phagocytic cells. The phagocytes may also destroy the parasite and prevent or ultimately eliminate an infection. Eg: *Brucella abortus*. When a balance is struck between the bacterium and phagocyte, the bacteria may survive in this intracellular state of relative equilibrium for months or years.

b) Obligate intracellular parasites

These microbes can propagate only within cells. Eg: Chlamydia, Rickettsia, viruses.

Extracellular parasites:

These microorganisms cannot survive for long inside the phagocytic cells and hence readily destroyed. They damage the tissues while they are outside the phagocytes and other cells. Eg: *Klebsiella* and *Pasteurella* spp.

- **Antiphagocytic capsules:**

  These surface structures are found on extracellular bacteria and protect the microorganisms from ingestion by phagocytic cells. Capsules are typically polysaccharides (Eg; *P. multocida*) or protein (Eg; *B. anthracis*) in composition.
- **Extracellular enzymes:**

  These are substances produced by some bacteria that are not toxic directly and do not produce disease.

  - Coagulase produced by S. aureus clots fibrin that protects the bacteria.
  - Hyaluronidase produced by many bacteria help in spread of organisms by breaking down hyaluronic acid in tissues.

- Collagenase produced by some strains of *C. perfringens* help in the bacterial spread by break down of collagen in tissues.
- Streptokinase and streptodornase produced by Streptococci.

- **Adherence to surfaces:**
  Most of the organisms possess surface components called adhesins or ligands that enable them to adhere specifically to host cells through suitable receptors on the surface of these cells. These adhesins include components of outer membranes, capsules and pili (fimbriae). Host cell receptors are mostly sugars like mannose.

Bacterial toxins:

Bacteria candamage host tissues directly through the effects of exotoxins and endotoxins. Toxins are of 2 types; Exotoxins and Endotoxins. Exotoxins are produced by Gram- positive and Gram-negative bacteria which cause systemic effects or cell death. Endotoxins, which are lipopolysaccharides present in the outer membrane of Gram-negative bacteria, are released when the cells are lyzed.

RESISTANCE AND SUSCEPTIBILITY OF HOST

To establish an infection, an invading microorganism must first overcome many surface barriers, such as skin, degradative enzymes, and mucus that have either direct antimicrobial activity or inhibit attachment of the microorganism to the host. Any microorganism that penetrates these barriers encounters two levels of resistance - other nonspecific resistance mechanisms and the specific immune response.

The **nonspecific immune response** is also known as **nonspecific resistance** and **innate** or **natural immunity;** it offers resistance to any microorganism or foreign material encountered by the vertebrate host. It includes general mechanisms inherited as part of the innate structure and function of each animal (such as skin, mucus, and constituitively produced antimicrobial mediators like lysozyme), and acts as a first line of defense.

In contrast, the **specific immune responses,** also known as **acquired, adaptive,** or **specific immunity,** resist a particular foreign agent. Moreover, the effectiveness of specific immune responses increases on repeated exposure to foreign agents such as viruses, bacteria, or toxins; that is to say, specific responses have "memory." Substances that are recognized as foreign and provoke immune responses are called **antigens** (contraction of the words *anti*body and *gen*erators). The antigens cause specific cells to replicate and manufacture a variety of proteins that function to protect the host. One such cell, the B cell, produces and secretes glycoproteins called antibodies. **Antibodies** bind to specific antigens and inactivate them or contribute to their elimination.

NON-SPECIFIC RESISTANCE OF HOST:

The important mechanisms of non-specific or innate host resistance are as follows;

a). ***Skin:***

- It is an effective barrier against microbes. Any cut or wound in the skin allows entry of microbes which would otherwise not have access to the body.
- Microorganisms like *B. abortus* can penetrate skin, while others can enter and cause infections of sweat and sebaceous glands, hair follicles. The glands secrete acidic, antibacterial substances that contain fatty acids. Young animals are more susceptible due to their lower concentration of fatty acids on the skin than the adult animals.
- Lysozyme which breaks down cell walls of bacteria is present on the skin. This mucolytic enzyme breaks sugars from the glycopeptides of the cell wall of many Gram-positive bacteria.

b). *Mucous membrane:*

- Mucous membranes of various parts of the body are covered by mucus. The bacteria are trapped in this and readily phagocytosed. These are then carried to the lymph nodes and specific immune responses are generated.
- Cilia in the respiratory tract move the bacteria and mucus constantly towards the oral and nasal openings.
- Both mucus and tears contain protective enzyme-lysozyme.
- Saliva, stomach acid and proteolytic enzymes have an anti-bacterial action.
- Acidic pH of vagina caused by Lactobacilli has a limiting and stabilizing influence on the vaginal flora.
- Disturbance of flora by antibiotic therapy may allow establishment and multiplication of disease-producing organisms.

c). *Phagocytosis*:

- Microorganisms entering blood, organs and tissues are engulfed and killed by the phagocytic cells namely, leucocytes and macrophages.
- Resulting phagocytic vacuole created by the engulfing of a microorganism is called a **phagosome.** The phagosome fuses with the lysosome containing lysozyme, which aids in the destruction of microorganism.
- Phagocytosis may be non-specific in the absence of antibody but specific when antibodies to the microorganism are present.
- Some microbes may survive and actually multiply within the phagocytic cells.

d). Resistance of tissues:

- Some constituents of tissues have an inhibitory effect on microorganisms. For example, interferons produced by tissues in response to viral infections have antiviral properties.
- Mostly the healthy tissues inhibit the multiplication of microbes. This resistance can be impaired by depression of inflammatory response, x-rays, corticosteroids and anti-neoplastic drugs.

e). Natural antibodies:

- These are immunoglobulins that react with a wide range of organisms and antigens to which the animal has not yet been exposed or immunised.

Eg: antibodies against blood group antigens.

f). Inflammatory response:

- Pathogenic microbes elicit inflammatory responses of varying intensity.
- These responses include increased blood flow, fever, pain, swelling fluid exudation and increased vascular permeability.
- Inflammatory mediators like serotonin, histamine, leukotrienes are involved in these responses.

g). Fever:

- This is a manifestation of any inflammatory response and often a cardinal symptom of infectious disease.
- Lipopolysaccharides released from phagocytes that have engulfed gram-negative bacteria cause macrophages to produce the cytokine interleukin-1.
- This is carried by the blood to the hypothalamus, where it stimulates the temperature control centre to release prostaglandins that increase the temperature causing fever.
- Endocrine disturbances and hypersensitivity reactions can also cause fever.

SPECIFIC MECHANISMS OF HOST RESISTANCE

Any pathogen must evade both natural and specific immune responses of the host to achieve infection. Specific mechanism of host resistance or immunity comprises of natural and acquired immunity. Acquired immunity consists of active and passive immunity.

Active immunity results from the immune response mounted by the host and consists of cell-mediated and humoral immunity. Recognition of foreign material, specificity and memory are the salient features of both types of immunity.

The major points differentiating active and passive immunity are tabulated below;

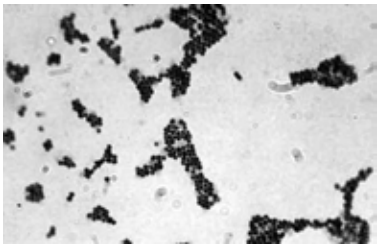
Feature	Active immunity	Passive immunity
Host involvement	Produced actively by host immune system	Received passively by host. No involvement of host immune system
Source	Induced by infection or contact with immunogens (vaccines/ allergens)	Conferred by introduction of readymade antibodies
Level of protection	Durable and effective	Transient and less effective
Time of onset	Immunity effective only after a lag period (for generation of antibodies)	Immediate onset of immunity
Immunological memory	Present	Absent
In immunodeficient hosts	Not applicable	Applicable

Staphylococcus

Staphylococcus (der. Greek, "Staphylo" = bunch of grapes"), Size 1 μ dia., Gram positive, non-motile, non-spore forming, facultative anaerobic (ferments sugars but does not produce gas). Members are pathogenic species associated with **suppurative** (pus forming) **disease** (**abscesses** - hallmark clinical manifestation is a ***boil*** (eg,. pimples)). They can also cause pneumonia, osteomyelitis, meningitis, arthritis in invasion into deeper tissues and organs. They also produce toxic shock syndrome and food poisoning in humans. Organisms of this genus are among the hardiest, most resistant to environmental stress of all of the non-spore forming organisms - resist dehydration, relatively heat resistant and tolerate many common disinfectants.

Staphylococci are often called "opportunists" because they wait for suitable conditions before invading the body. Humans carry the bacteria on their hands and arms, in their nose, and in the hairy parts of their body. There are nearly 27 species in this genus and the important animal pathogens are *S.aureus, S.pseudintermedius, S.hyicus and S.epidermidis. S.aureus* is considered as type species for this genus. Generally these organisms are found on the surface of skin as commensal and also on the mucous membrane. Though they are found on the surface of skin, they have habitat preference to head, ears and anterior nares. They are generally non-pathogenic, occasionally causing mastitis, abscesses and wound infections. Some species of staphylococci like *S.saprophyticus* cause urinary tract infection and wound infection in humans.

Morphology: Staphylococci are gram-positive cocci occurring in pairs, short chains or in clusters. The clusters resemble 'bunches of grapes'. They are aerobic or facultative anaerobic. They are non-motile, non-sporulative and fermentative (no gas production). Some strains possess a polysaccharide capsule. They are catalase positive and oxidase negative. These organisms are relatively resistant. They can withstand a temperature of 60°C for 30 minutes. They are resistant to disinfectants including phenolic compounds and can withstand a high salt concentration.



1. ***S.aureus***: principal pathogenic species and is *coagulase-positive.* It is so named because of the yellow or gold color of its colonies - due to the production of **beta carotene** pigment. This pigment is also thought to function as a virulence factor due to its ability to act as an electron-sink (conjugated diene structure) and thus scavenge reactive oxygen species (ROS) produced by inflammatory cells. Pigmented Staph are consistently more resistant to phagocytic oxygen-dependent killing mechanisms. However, some strains (particularly pathogenic strains of veterinary significance) lose the ability to produce beta carotene as evidenced by their lack of yellow/gold color.
2. ***S.epidermidis*** - gets its name because of its prevalence on human skin - it is also found on skin, teats, hair and superficial mucosa of many animals but in less numbers than seen in humans - colonies are non-pigmented and non-hemolytic and *coagulase negative*- it is an opportunistic invader of low virulence.
3. ***S.pseudintermedius***- most prevalent disease-producing species in dogs and other carnivores. Colonies are non-pigmented, *coagulase-positive*, vary in hemolyticpattern.
4. ***S.hyicus***– It was thought to be biotype (biotype 2) of *S. epidermidis*. It is the etiologic agent of Exudative Epidermitis of Swine (Greasy Pig Disease)

Cultural characteristics: These organisms grow well in ordinary laboratory media. However, the most preferred medium for cultivation is the **mannitol salt agar.** Blood agar is the preferred medium for primary isolation. Colonies appear as round, smooth, glistening and may or may not have golden yellow pigments. Colonies are roughly 4mm in diameter. They produce double zone haemolysis and haemolysis. Haemolysis is best demonstrated in bovine red blood cells.

Biochemical tests: The common biochemical tests used to differentiate species of staphylococci are Oxidase test, Coagulase test, beta haemolysis, Pigment production, utilisation of maltose and mannitol, sensitivity to antibiotic novobiocin.

Antigenic nature: The antigenic structure is complex and heterogeneous. The cell wall is composed of Teichoic acid, Peptidoglycan and Protein A. Teichoic acid is species specific and Protein A is antiphagocytic. The polysaccharide capsule is alsoantiphagocytic.

Pathogenesis: The infection may be endogenous or exogenous. Endogenous infections are more common. The main mode of spread is by direct contact and by fomites. Some species have the ability to invade the tissues, produce abscesses, pustules, various pyogenic infection, bacteraemia and septicaemia. The inflammatory response following infection results in the infiltration of neutrophils leading to pyogenic response and production of pus. Accumulation of pus results in formation of abscesses. Staphylococci produce number of extra cellular products that are responsible for the establishment of pathogenesis. Not all the extra cellular products produced are toxic. Some of them are enzymes and virulence enhancers. The enzymes are responsible for protecting the organisms from body defense mechanisms. Some of the important extra cellular products and their roles are listed below.

Sl.No	Virulence factors	Important role in pathogenesis
1	Coagulase:	Responsible for clotting of plasma and conversion of prothrombin to thrombin and fibrinogen to fibrin. Most of the staphylococci produce coagulase.

2	Enterotoxin A, B, C1, C2, C3, D and E:	These toxins are highly heat resistant (100^0C for 30 minutes). These toxins are produced by *S.aureus* in custards, raw milk, cream, ice cream, meat gravy, fish, cheese etc. Clinical symptoms in humans include nausea, vomiting and diarrhoea.
3	Haemolysin α, β, γ and δ (Haemolysins and cytolysins:	The α and β haemolysins are the potent toxins. The α toxin is responsible for the inner clear zone of haemolysis and the β toxin is responsible for outer partial zone of haemolysis. Chemically the γ toxin is sphingomylinase C. The role of other two toxins in pathogenesis is not clear.
4	Lipase:	Degrades protective fatty acids on skin and help in abscess formation.
5	Staphylokinase:	Degrades fibrin clots.
6	Leukocidin:	Kills granulocytes and macrophages. It is made up of two heat labile interacting proteins.
7	Exfoliative toxin A and B (Exfoliatin):	Causes cleavage of desmosomes in the stratum granulosum of the epidermis.
8	Toxic shock syndrome toxin	
9	Hyaluronidase:	Spreading factor.
10	Penicillinase:	
11	Lysostaphin:	
12	Protein A:	It is found on the surface of staphylococci and has got a special property of binding with Fc region of antibodies. Used in the co- agglutination test.
13	Slime:	Surface mucoid substance responsible for adhesiveness.
14	Collagenase:	
15	Acid and alkaline phosphatase:	
16	Pyrogenic exotoxin:	
17	Elastase:	
18	Catalase:	
19	Protease:	
20	Nuclease:	

Mechanism of pathogenesis:

1. Microbial adhesion: not completely understood but initial host cell recognition and attachment may involve binding to glycolipids as well as fibronectin and fibrinogen. *S. aureus* and especially coagulase-negative strains (CoNS) of *S. epidermidis* (and presumably intermedius) can also form **biofilms**. **Biofilms** are an assemblage of microscopic animals, plants and bacteria attached to a surface. They often produce a slime layer in which they live and which helps them to stick to a surface and provides a protective environment. Following the initial attachment phase, the CoNS adopt a highly characteristic mode of growth in which they begin to produce a mucoid substance or 'slime' which promotes biofilm development. The slime mainly consists of polysaccharides, with about 10- 20% proteins that presumably stabilises the biofilms by promoting bacterial cell-to-cell and cell-to-surface associations so that multi-layered cell clusters accumulate on the implant surface.
2. Penetration, invasion and disease (lesion development and tissue damage) are dependent on a number of factors such as skin trauma, lacerations, macerations, underlying disease, competing flora, immune status, etc. Thus, most often these organisms are opportunistic pathogens - not particularly invasive unless given the opportunity.

3. Multiplication and spread: Most important factors in Staphylococcal multiplication and spread following invasionare:
    a. resistance to phagocytosis (protein A and polymeric uronic acid capsules
    b. intracellular survival inphagocytes
    c. coagulase & hylauronidase production - toxin production
    d. adhesion to epithelial cells; eg. vet strains of *S. aureus* are particularly adept at adhesion to intramammary glandepithelia
4. Damage: (Suppurative (pus producing), abscesses)- local infections lead to the formation of a collection of pus (neutrophils) called an abscess - abscesses in the skin are called boils, pimples, furuncles, etc.
    a. Stages of abscess formation (bacterial and hostfactors)
        i. acute inflammation - rapid and extensive infiltration neutrophils
        ii. chemotactic factors (bacteria products and host-derived complement)
        iii. outpouring of large amounts of lysosomal enzymes from damaged neutrophils (leucocidin of Staph) causes surrounding tissue damage
        iv. inflammatory area begins to be "walled off" by a thick- walled fibrin capsule - possible participation by Staph coagulase
    b. Abscess is thus a well-defined area containing pus - from the host's point of view it has contained the invading organism - however, if the abscess happens to be in or near a vital organ or tissue it can often have serious or even life-threatening consequences.

Important Staphylococcal Infections

1. S.aureus

a. **Botyromycosis** - The infections caused by staphylococci are referred as **botryomycosis.** It is a chronic granulomatous lesion involving udder of mare, cow, sow and spermatic cord of stallion invasion into stump of spermatic cord of horses following castration

- infected cord becomes greatly enlarged and small pockets of pus are found containing granules of encapsulated organisms.

b. **Mastitis** – Characterized by suppurative lesions in cattle mammary gland ductular epithelia. The organisms enter from human hands and opportunistic invasion due to trauma or tissue damage from milking machines. The relative resistance of the organism insures continued environmental contamination and tendency for infection and re-infection. The organisms also have a great adhesive tenacity for epithelium at the tip of the teat. Mastitis varies in severity from subclinical to severe gangrene. Subclinical, chronic form is the most damaging economically because of reduced milk production or milk losses. Staph mastitis cases also occur in sheep, goats, mares, sows, cats and mink.

c. **Purulent synoviti**s in chickens and turkeys (**bumble foot**) – Characterized by lameness, swellings in the feet. The organisms enter opportunistically in birds with underlying disease or because of stress-inducing husbandry.

d. Porcine Necrotizing Staphylococcal Endometriti*s* – Characterized by abortion in swine

e. **Tick Pyemia** – Characterized by septicemia in lambs heavily infested with ticks. The ticks inject staphylococci deep into tissue. If septicemia occurs it is rapidly fatal.

f. **Exfoliative Skin Disease** – It comprises a variety of syndromes, generalized exfoliative dermatitis in newborns called as Ritter's disease, toxic epidermal necrolysis in children and occasionally in adults, bullous impetigo and Staphylococcal scarlatina. This group of diseases is collectively

called **Staphylococcal Scalded Skin Syndrome** - as in pigs the disease spreads over the entire skin surface primarily because **exfoliative toxin.**

g. **Staphylococcal Food Poisoning**: It is due to the powerful **enterotoxin.** The symptoms are seen within 1-6 hours after ingesting contaminated food and results in severe cramps, nausea, vomiting, and diarrhoea.

h. **Toxic Shock Syndrome:** It is characterized by fever, skin rash, hypotension, and dysfunction of several systems.

2. ***S. pseudintermedius***

a. **Canine Pyoderma** - organism enters opportunistically form endogenous normal colonization of nasopharynx and skin. Factors such as dry skin, trauma, ectoparasitism (demodectic mange), matted dirty hair, predispose the animal to invasions of superficial and deeper skin layers. Staphylococcal cellulitis is often a complication of mange or other ectoparasitic disease

3. ***S. hyicus***

a. It is the etiological agent of **Exudative Epidermitis of Swine** - (**Greasy Pig Disease).** It is a highly contagious disease and easily spread from pig to pig. Piglets between age grooup 1-7 weeks are most often affected. Organism often enters through skin lacerations caused by bites from other pigs competing for food. Whole skin can be covered with a moist, greasy exudate similar to "Staphylococcal Scalded Skin Syndrome" in humans. The disease starts with lesions as vesicles on hairless skin surfaces (behind the ears), as the disease progresses the skin becomes thickened and layers of epidermis peel off (in milder forms it resembles dandruff-like scaling). The affected pigs usually recover from 15-40 days after symptoms, although it can be fatal.

4. A type of enterocolitis referred as Staphylococcal enterocolitis is seen in humans after prolonged antibiotic therapy.

Diagnosis:

1. **Materials for diagnosis:** Pus, affected tissue and milk samples.

2. **Examination of culture and pus smear** by Grams staining method will reveal gram-positive organisms with typical arrangement.

3. **Isolation and identification by culture method:** These organisms grow well in ordinary laboratory media. However, the most preferred media for cultivation is the **mannitol salt agar.** Blood agar is the preferred medium for primary isolation. Colonies appear as round, smooth, glistening and may or may not have golden yellow pigments. Colonies are roughly mm in diameter. They produce double zone haemolysis and γ haemolysis. Haemolysis is best demonstrated in bovine red blood cells.

4. **Coagulase test:** To a loopful of bacterial culture in saline a drop of rabbit plasma is added over a slide. Appearance of clumps within ten minutes is considered as positive. Most of the Staphylococci produce coagulase enzyme. The appearance of the organisms in smear, cultural characteristics and result of this are generally used to confirm staphylococci.

5. Latex agglutination test:

6. Rapid miniaturised commercialkits:

7. Phage typing:

8. **Biochemical tests:** The common biochemical tests used to differentiate species of staphylococci are Oxidase test, Coagulase test, beta haemolysis, Pigment production, utilisation of maltose and mannitol, sensitivity to antibiotic novobiocin.

Treatment: Penicillin is the drug of choice. However, penicillin resistant strains are also reported.

Synthetic penicillins like methicillin, oxacillin and nafcillin, tetracyclines, bacitracin, nitrofurans and erythrocmycin are also used

EXTRA READING:

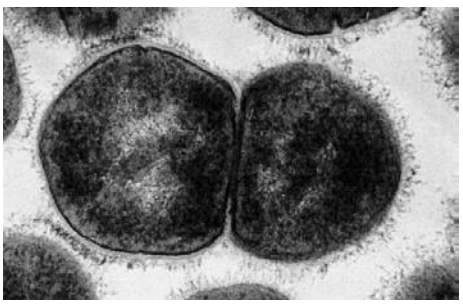
Superantigens: enterotoxins and toxic shock syndrome toxin

S.aureus secretes two types of toxin with superantigen activity, **enterotoxins**, of which there are six antigenic types (named SE-A, B, C, D, E and G), and **toxic shock syndrome toxin** (**TSST-1**). Enterotoxins cause diarrhoea and vomiting when ingested and are responsible for staphylococcal food poisoning. TSST-1 is expressed systemically and is the cause of toxic shock syndrome (TSS). When expressed systemically, enterotoxins can also cause toxic shock syndrome. In fact, enterotoxins B and C cause 50% of non-menstrual cases of TSS. TSST-1 is weakly related to enterotoxins, but it does not have emetic activity. TSST-1 is responsible for 75% of TSS, including all menstrual cases. TSS can occur as a sequel to any staphylococcal infection if an enterotoxin or TSST-1 is released systemically and the host lacks appropriate neutralizingantibodies.

Superantigens stimulate T cells non-specifically without normal antigenic recognition. Up to one in five T cells may be activated, whereas only 1 in 10,000 are stimulated during a usual antigen presentation. Cytokines are released in large amounts, causing the symptoms of TSS. Superantigens bind directly to class II major histocompatibility complexes of antigen-presenting cells outside the conventional antigen-binding grove. This complex recognizes only the Vb element of the T cell receptor. Thus any T cell with the appropriate Vb element can be stimulated, whereas normally, antigen specificity is also required inbinding.

Food poisoning: Symptoms of staphylococcal food poisoning are caused by staphylococcal enterotoxin, not by staphylococcus itself. It is a common cause of food poisoning, and the potential for outbreaks is high when food handlers with skin infections contaminate foods left at room temperature. Custards, cream-filled pastry, milk, processed meat, and fish provide media where coagulase-positive staphylococci grow and produce enterotoxin. Onset is usually abrupt. Symptoms characteristically are severe nausea and vomiting, which begin 2 to 8 h after eating food containing the toxin. Other symptoms may include abdominal cramps, diarrhoea, and occasionally, headache and fever. Because the toxin does not cause mucosal ulceration, the diarrhoea is usually non-bloody. Acid-base imbalance, prostration, and shock may ensue in severe cases. The attack is brief, often lasting < 12 h, and recovery is usually complete. Rare deaths occur as a result of fluid and metabolic stresses, especially among very young or old or chronically ill patients. Diagnosis relies on recognizing the clinical syndrome. Usually, several persons are similarly affected, constituting a point source outbreak. Diagnostic confirmation, although rarely required, entails isolating coagulase-positive staphylococci from the suspected food. Gram stain of specimens of vomitus may show staphylococci. Careful food preparation is essential for prevention. Persons with furunculosis or impetigo should not prepare food until their lesions have healed. Treatment is described under General Principles of Treatment, above. Rapid IV replacement of electrolytes and fluids often brings dramatic relief

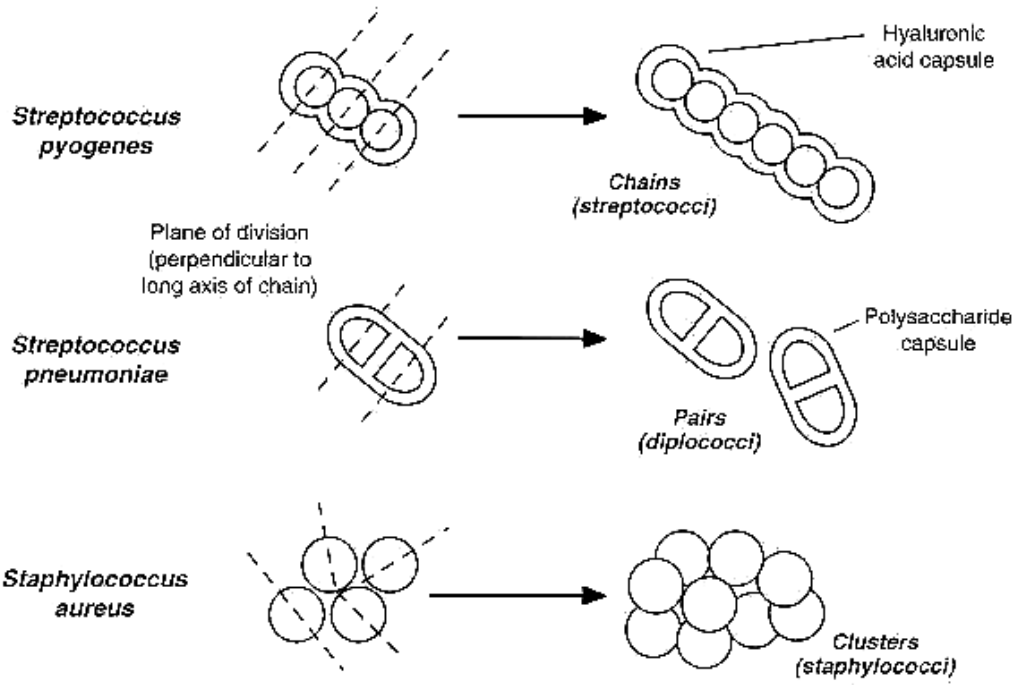
Streptococcus



The genus **Streptococcus** is comprised of a wide variety of both pathogenic and commensal **gram-positive bacteria**, which are found to inhabit a wide range of hosts, including humans, horses, pigs and cows. Within the host, streptococci are often found to colonize the mucosal surfaces of the mouth, nares and pharynx. However, in certain circumstances, they may also inhabit the skin, heart or muscle tissue. This genus contains more

than 37 species that include both pathogenic and non-pathogenic species. The pathogenic *Streptococci* are present in skin and mucous membrane of the genital, upper respiratory and digestive tracts. They are widely distributed in nature. Streptococci are also essential in industrial and dairy processes and as indicators of pollution.

Morphology: Members of the genus *Streptococcus* are gram-positive, non-motile, non-spore forming cocci occurring singly, pairs or in chains. Older cultures may lose their Gram-positive character. Most of them are facultative anaerobes or strict anaerobes. *S pyogenes* characteristically is a round- to ovoid coccus 0.6-1.0 μm in diameter. They divide in one plane and thus occur in pairs, or (especially in liquid media or clinical material)in chains of varying lengths. S *pneumoniae* appears as a 0.5-1.25 μm diplococcus, typically described as lancet-shaped but sometimes difficult to distinguish morphologically from other streptococci



Classification: Different types of classification are followed for *Streptococci.* Streptococci are classified on the basis of colony morphology, haemolysis, biochemical reactions, and (most definitively) serologic specificity.

1. Based on growth characteristics, types of haemolysis and biochemical activities they are classified in to six categories. They are pyogenic streptococci, oral streptococci, enterococci, lactic streptococci, anaerobic streptococci and other streptococci. Most of the pathogenic streptococci are placed in first category.
2. Based on haemolysis pattern they are grouped into four types. The haemolytic property is the ability of the streptococci to lyse the red blood cells in blood agar medium.
   a. α-Haemolysis: Partial haemolysis surrounded by zone of green coloration around small colonies (haemolysis with an inner zone of partial haemolysis).
   b. β-Haemolysis: Complete haemolysis. Clear zone colourless zone around colonies.
   c. γ-Haemolysis: No detectable haemolysis.
   d. δ-Haemolysis: a small zone of partial haemolysis followed by zone of complete haemolysis.

Lancefield classification: In this method streptococci are divided into different groups and each group is further divided into types. This classification of streptococci is based on serologic difference to a carbohydrate substance found on the cell wall of streptococci, which is called as component C.

Precipitation test is commonly used in this method to differentiate. Letters A, B, C etc., designates the major groups in this classification and the types under each group are assigned Arabian numbers. The typing under each group is based on serological difference to the M protein. Group A comprises of human pathogens and B, C, G comprise of animal pathogens. Group D streptococci are important etiologic agents of urinary tract infections and infections associated with biliary tract procedures, as well as cases of disseminated infection, bacteremia, and endocarditis. Group F streptococci are associated with abscess formation and purulent disease. Group R streptococci, well-documented causes of meningitis and septicemia in pigs, also pose a serious health hazard to workers in the pork industry. List of important streptococci causing infection in animals are provided below.

Sl.No	Species	Group	Infection produced
1	***S.pyogenes***	A	Streptococcal disease of human beings, bovine mastitis and lymphangitis in foals
2	***S.agalactiae***	B	Bovine mastitis. Mastitis in sheep and goat. It is an obligate pathogen. Infection spread by contaminated teat cups and milker's hands
3	***S.dysgalactiae***	C	Bovine mastitis and lamb polyarthritis
4	***S.equisimilis***	C	Associated with Strangles, wound infection, genital infection and mastitis in horses
5	***S.equi***	C	Strangles and udder infections in mares
6	***S.zooepidemicus***	C	Genital infections in mares, epididymitis in stallion, naval infection in foals, cervicitis, mastitis and metritis in cattle, arthritis and abortion swine and fatal septicaemia in chickens
7	***S.bovis***	D	Lactic acidosis and other rumen disorders ofcattle
8	***S.equinus***	D	Alimentary tract infection of horses
9	***Enterococcus faecalis S.suis***	D, R, S	Meningitis, septicaemia, pyaemia, abscess and encephalitis and abortion in pigs
10	***S.uberis***	C, D, E, P.U	Bovine mastitis
11	***S.porcinus***	E, P, U, and V	Mandibular abscess, jowl abscess or cervical lymphadenoma
12	***S.avium***	Q	Significance not known
13	***S.canis***	-	



Cultural characteristics: The organisms grow well in ordinary laboratory medium enriched with blood or serum. Blood agar and Edwards's medium are the most preferred media however; growth is luxuriant in nutrient agar also. Streptococci often have a mucoid or smooth colonial morphology. They produce small colonies of 1mm diameter, round, smooth, glistening and look like dew drops. Haemolytic pattern in blood agar as indicated above may be noticed depending upon the species of streptococci and type of blood used. The colonies produced are mucoid, matt or glossy. Virulent streptococci produce matt colonies and less virulent types produce glossy colonies.

Biochemical characters: They are catalase and oxidase negative and fermentative. Ferments sugars like trehalose, sorbitol, mannose, salicin, lactose, raffinose, inulin, esculin etc with variation. Hydrolyse esculin and sodium hippurate.

Pathogenesis: The infection may be either endogenous or exogenous. The exogenous infection is through inhalation or ingestion. Aerosols, direct contact and fomites are the important methods of spread. Carrier animals are important source of spread of infection. The streptococcal infection is characterized by production of pus (pyogenic infection). When the pyogenic bacteria invade the system they stimulate inflammatory response that is characterized by vascular dilation and exudation of plasma and neutrophils. Neutrophils engulf the bacteria (phagocytosis). Most of the bacteria are killed in the process of phagocytosis. However, some bacteria can grow in neutrophils resulting in production of toxins, which kill the phagocytic cells. The enzymes liberated by the dead phagocytic cells liquefy dead tissues and phagocytic cells. The liquefied mass is yellow and thick in consistency and called as pus. Variation in consistency of pus is due to deoxyribonucleo protein. Streptococcal is infection is generally localised and rarely it becomes septicaemic or bacteraemic. The major virulence factors of streptococci are surface M proteins and hyaluronic acid. The extra cellular products produced by streptococci are important in spread and localisation of the organisms inside the tissue. Some of the extra cellular products are also responsible for damage to the tissue.

Extracellular products: The important extra cellular products of streptococci and their role are listed below.

1. **Haemolysin**: Two types Streptolysin O and Streptolysin S. Toxic for neutrophils and macrophages.
2. **Streptokinase**: Also known as **Fibrinolysin**. It activates plsminogens to plasmin, which prevents the formation fibrin clots.
3. **DNases A, B, C** and **D**: Also known as **Streptodornase**. Assist in the production of substances required for growth.
4. **Hyaluronidase A:** Major virulence factor. Promotes the spread of organisms in tissues.
5. **Erythrogenic toxins A, B** and **C:** Responsible for rashes in scarlet fever.
6. **NADases:** Kills the phagocytic cells.
7. Proteinase
8. Lipoproteinase
9. Amylase
10. Esterase

Diagnosis:

1. **Materials for diagnosis:** Pus, joint fluid, milk, organs, blood swab and meningeal swab.
2. **Examination of culture smear** by Grams staining method and milk smear by **Newman's staining method.**
3. **Cultivation:** The media commonly used are nutrient agar, blood agar or Edward's medium. Most pyogenic bacteria produce beta haemolysis. Edward's medium contains esculin, crystal violet and thalliumacetate.
4. Biochemical reactions:
   a. Haemolytic patterns
   b. **Differentiation in fermentation of sugars** trehalose, sorbitol, mannose, salicin, lactose, raffinose, inulin, esculin etc. hydrolysis of esculin and sodium hippurate, reduction of methylene blue milk and preference of 6.5% sodium chloride for growth.

CAMP test (Christie, **A**tkinson, **M**uench and **P**eterson): It is a presumptive test for the diagnosis of *S.dysgalactiae.* This test is based on the ability of streptococcal organisms to complete partial haemolysis produced by Staphylococcus aureus.

5. **Hydrolysis of esculin:** Esculin agar is a selective and differential media that is used primarily to distinguish fecal streptococci (*Enterococcus* species) from other *Streptococcus* organisms. *Enterococcus* is the only organism that can hydrolyze esculin. The hydrolyzed esculin complexes with iron to form a dark, brown-black color in the tube.
6. **The cystine tryptic agar (CTA) sugar fermentation test** is used to identify fastidious organisms (especially *Streptococcus* species), by detecting fermentation reactions. The fermentation of carbohydrates in the media produces acid, which turns the pH indicator to a yellow color. Colonies obtained from a blood agar culture are stabbed into the media in a CTA tube. After incubation, a positive fermentation test will be yellow. A tube of CTA base media is inocculated along with the CTA sugar tube; this base media tube is used to confirm the color change of the CTA sugar tube. If the bacteria grow in the base media tube, that media will remain red.

Treatment: Penicillin is the drug of choice. However sulphamethixine, erythromycin, lincomycin and tetracycline are also effective.

IMPORTANT STREPTOCOCCAL INFECTIONS:

STRANGLES

Strangles is a highly contagious and serious infection of horses and other equids caused by the bacterium, *Streptococcus equi*. The disease is characterized by severe inflammation of the mucosa of the head and throat, with extensive swelling and often ruptures of the lymph nodes, which produces large amounts of thick, creamy pus. Strangles is caused by *Streptococcus equi* subspecies *equi*, better known as *Streptococcus equi (S. equi)*. The organism can be isolated from the nose or lymph nodes of affected animals, and is usually readily identified in the laboratory by simple sugar tests. Horses of all ages are susceptible, though strangles is most common in animals less than 5 years of age and especially in groups of weanling foals or yearlings.

Symptoms & lesions: Susceptible horses develop strangles within 3–14 days of exposure. Animals show typical signs of a generalized infectious process (depression, inappetence**,** and fever of 39°C–39.5°C). More typically of strangles, horses develop a nasal discharge (initially mucoid, rapidly thickening and purulent), a soft cough and slight but painful swelling between the mandibles, with swelling of the submandibular lymph node. Horses are often seen positioning their heads low and extended, so as to relieve the throat and lymph node pain. With the progression of the disease, abscesses develop in the submandibular (between the jaw bones) and/or retropharyngeal (at the back of the throat) lymph nodes. The lymph nodes become hard and very painful, and may obstruct breathing ("strangles"). The lymph node abscesses will burst (or can be lanced) in 7–14 days, releasing thick pus heavily contaminated with *S. equi*. The horse will usually rapidly recover once abscesses have ruptured. Although the disease process described above is classic, some horses (especially older animals) will develop a mild, short lasting disease without or with minor lymph node abscessation. This is thought to be the result of partial immunity although this may also result from infection by *S. equi* of relatively low virulence. Classic strangles is a severe infection that can be fatal, usually because of a variety of complications that occur. The main and most important fatal complications of strangles are Bastard strangles and purpura haemorrhagica. Bastard strangles, which describes the dissemination of infection to unusual sites other than the lymph nodes draining the throat. For example, abdominal or lung lymph nodes may develop abscesses and rupture, sometimes weeks or longer after the infection seems to have resolved. A brain abscess may rupture causing sudden death or a retropharyngeal lymph node abscess may burst in the throat and the pus will be inhaled into the lung. Purpura haemorrhagica, which is an immune-mediated acute inflammation of peripheral blood vessels that occurs within 4 weeks of strangles, while the animal is convalescing. It results from

the formation of immune complexes between the horse's antibodies and bacterial components. These immune complexes become trapped in capillaries where they cause inflammation, visible in the mucous membranes as pinpoint haemorrhages. These haemorrhages lead to a widespread severe edema of the head, limbs, and other parts of the body. Purpura can also be a complication of routine vaccination.

Diagnosis: Diagnosis can be confirmed by culturing pus from the nose, from abscessated lymph nodes or from the throat of clinically affected horses. Although *S. equi* isolates are thought to be genetically identical, isolates may vary in virulence and atypical isolates occur, which differ in their sugar tests from typical *S. equi*.

Treatment: The organism is amenable to Penicillin G.

Prevention and control: A killed and live vaccines are used in Western countries to control the infection. However, isolating the affected animal, cleaning the shed and proper disinfection will help getting rid of the infection.

MASTITIS: Mastitis is defined as inflammation of udders and it is the most important disease threatening dairy industry. Clinical mastitis is of greatest concern to producers since the cow is ill and the milk cannot be included in milk for sale. The clinical signs include abnormal milk, sore gland and often sick cow. Mastitis is caused by variety of organisms and *S.agalactiae* is one of the important organisms causing mastitis. This highly contagious pathogen is found in the mammary gland and spreads from cow to cow. This organism is generally subclinical and causes severe milk loss and a costly disease in a herd. However, *S.aglactiae* is easily treated with antibiotic therapy in the mammary gland during lactation or during the dry period. There are three methods to control or eliminate *S.aglactiae,* which are as follows;

1. Use a predipping and post-dipping iodine for teat sanitation.
2. Treat all quarter of all cows at dryoff.
3. Treat all quarter of cows identified with *Strepag.*

California Mastitis Test (CMT). The CMT is a good cow-side test to determine if a quarter has a high milk somatic cell count (SCC). A high SCC is a good indicator of an intramammary infection.

Procedure*:*

1. Clean teat and remove a single stripping of milk and discard.
2. Collect milk (2ml) in appropriate well: A-RF, B-RR, C-LF,D-LR
3. Add equal amount of CMT solution
4. Rotate mixture gently for 20seconds
5. Determine solution agglutination (gel formation in well)
6. Record each quarter reaction: quarter with mild to strong should be considered infected

BACILLUS

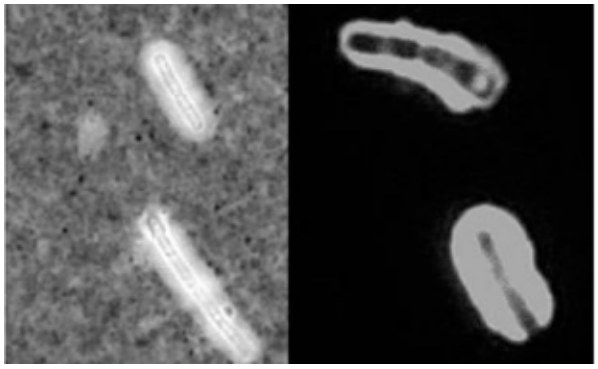
Introduction: The anthrax bacillus, ***Bacillus anthracis,*** was the first bacterium shown to be the cause of a disease. In 1877, Robert Koch grew the organism in pure culture, demonstrated its ability to form endospores, and produced experimental anthrax by injecting it into animals. This genus contains number of species, which are ubiquitous in nature and are found in air, soil, dust, water etc. They are also common contaminants of the laboratory cultures. This genus consists of four important species *B.anthracis, B.cereus, B.thuringiensis* and *B.subtilis*. The only species of pathogenic importance is *B.anthracis.* Yet another species *B.cereus* rarely produce disease in animals. It is a normal inhabitant of the soil, but it can also be regularly isolated from foods such as grains and spices. ***B. thuringiensis***

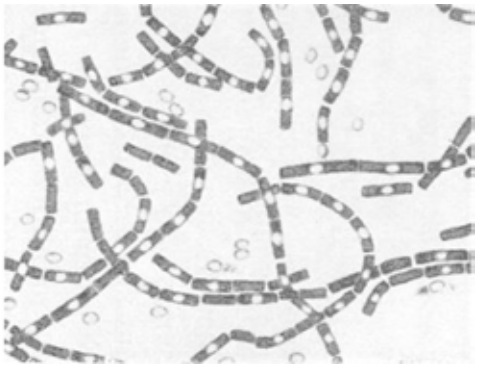
is identified by its pathogenicity for lepidopteran insects (moths and caterpillars) and by production of an intracellular **parasporal crystal** in association with spore formation. The bacteria and protein crystals are sold as "Bt" insecticide, which is used for the biological control of certain garden and crop pests.

Morphology: The species under the genus *Bacillus* are gram-positive; rod shaped organisms measuring 1-1.2µ min width x 3-5µm in length; arranged single or in short chains. They are aerobic or facultative anaerobic. They possess capsule and are sporulative. Most of the species under genus Bacillus are motile, whereas the *B.anthracis* that causes anthrax is non-motile. They are catalase positive and fermentative. B.*anthracis* has certain special morphological features, which distinguishes it from other anthracoids (anthracoids are normal PM invaders). Morphologically *B.anthracis* is rod shaped organismwith truncated ends and in blood smears they appear either as single organism or chain of one or two organisms. They sporulate with very great frequency in soils with pH of 6 and above and the spores are centrally located. The spores of the organisms are considerably resistant than the vegetative cells to the physical and chemical agents. The spores can remain viable in soil and dried cultures for a long time (many years). Boiling for 10 minutes and by dry heat at 140°C for 3 hours destroy the spores. However, freezing temperature has no effect on the spores. Only prolong action of chemical disinfectants at high concentration has killing effect on the spores. 5% phenol for 2 days, 10-20% formalin in 10 minutes and autoclaving destroys the spores completely.

Cultural characteristics: The organisms grow well in all laboratory media. However, serum and blood in the media enhances the growth. In blood agar, the colonies appear with in 24 hours and are flat, rough, grey and non-haemolytic. The colonies are referred as medusa heads or judges wig. On microscopical examination, the edges of the colonies appear as a tangled mass of curly hairs. In nutrient agar the colonies could be smooth and mucoid or rough.

The laboratory animals of choice for cultivation of the organisms are guinea pigs and mice. The affected tissues after grinding them in a mortar and pestle or blood as such may be used as an inoculum. The animals will die from 24 hours and anthrax organisms as large capsulated rods can be seen in blood smear and spleen.





Antigenic characters: *Bacillus anthracis* forms a single antigenic type of capsule consisting of a poly-D-glutamate polypeptide. All virulent strains of *B. anthracis* form this capsule. Production of capsular material is associated with the formation of a characteristic mucoid or "smooth" colony type. "Smooth" (S) to "rough" colonial variants occur, which is correlated with ability to produce the capsule. R variants are relatively avirulent. Capsule production depends on a 60 megadalton plasmid termed as pX02.

Pathogenesis: The anthrax bacilli are distributed worldwide. Animals may become infected from contaminated soil, water, bone meal, oil cake, offal etc. Carrion birds and wild animals also have a role in the spread of the infection. The important modes of infection are by ingestion, inhalation and

through wounds and scratches. In human beings blood feeding insects are also found to have a role in spread of the infection.

The virulence of *Bacillus anthracis* is attributable to three bacterial components namely;

1. Capsular material composed of poly-D-glutamate polypeptide
2. EF component ofexotoxin
3. LF component ofexotoxin

The poly-D-glutamyl capsule is non-toxic, but functions to protect the organism against the bactericidal components of serum and phagocytes, and against phagocytic engulfment. The capsule plays its most important role during the establishment of the infection, and a less significant role in the terminal phases of the disease, which are mediated by the anthrax toxin. The anthrax exotoxin is referred as anthrax toxin, which is protein and comprised of three components I (Oedema factor), II (Protective antigen) and III (lethal factor). Production of the anthrax toxin is mediated by a temperature-sensitive plasmid termed as pX01.

Factor I is the **edema factor (EF)** which is necessary for the edema producing activity of the toxin. EF is known to be an **inherent adenylate cyclase**, similar to the *Bordetella pertussis* adenylate cyclase toxin.

Factor II is the **protective antigen (PA)**, because it induces protective antitoxic antibodies in guinea pigs. PA is the **binding (B) domain** of the anthrax toxin, which has two active (A) domains, EF (above) and LF (below).

Factor III is known as the **lethal factor (LF)** because it is essential for the **lethal effects** of the anthrax toxin. Apart from their antigenicity, each of the three factors exhibits no significant biological activity in an animal. However, combinations of two or three of the toxin components yield the following results in experimental animals.

PA+LF combine to produce lethal activity EF+PA produce edema

EF+LF is inactive

PA+LF+EF produces edema and necrosis and is lethal

Both the capsule and the anthrax toxin play a role in the early stages of infection, through their direct effects on phagocytes. Virulent anthrax bacilli multiply at the site of the lesion. Phagocytes migrate to the area but the encapsulated organisms can resist phagocytic engulfment, or if engulfed, can resist killing and digestion. A short range effect of the toxin is its further impairment of phagocytic activity and its lethal effect on leukocytes, including phagocytes, at the site. After the organisms and their toxin enter the circulation lethality will result. *Bacillus anthracis* coordinates the expression of its virulence factors in response to a specific environmental signal. Anthrax toxin proteins and the antiphagocytic capsule are produced in response to growth in increased atmospheric CO_2. This CO_2 signal is thought to be of physiological significance for a pathogen, which invades mammalian host tissues.

The spores enter through the skin or mucous membrane and germinate at the site of entry and later spread throughout the system. In septicaemic forms the spread is via lymphatics and blood stream. Respiratory failure and anoxia produced by the toxin cause death.

In pigs, the infection occurs as local form affecting lymph nodes of head and neck. Large number of bacilli is excreted via the natural orifices during the terminal stage. The important virulence factors are the toxin and capsule.

Pathogenicity: The infection occurs as per-acute, acute, sub-acute, chronic or cutaneous forms. Cattle, horses, sheep, swine and human being are affected by this infection. It is also considered as a

zoonotic infection. In per-acute infection, animals suddenly die at the grazing field without exhibiting any symptoms. Mortality and morbidity rates are very high. Generally the symptoms exhibited are very few before death. The important observation during terminal stage is oozing of blood from natural orifice. Since the clotting mechanism is affected the blood will not clot. The important lesion in anthrax infection is the enlargement of spleen (Spleenomegaly). The body cavity will be filled with unclotted blood.

In pigs, the infection is usually sub-acute with pharyngitis, swelling and haemorrhage of throat and mouth. An intestinal form with gastroenteritis is also seen. In chronic infection, localisation in the tonsils and lymph nodes of head and cervical region by the organisms are frequent.

In human beings, the infection occurs as pulmonary form or as malignant carbuncle or pustule (wool sorter's disease). Intestinal anthrax is also reported. In humans, anthrax is fairly rare; the risk of infection is about 1/100,000. The most common form of the disease in humans is **cutaneous anthrax**, which is usually acquired via injured skin or mucous membranes. A minor scratch or abrasion, usually on an exposed area of the face or neck or arms, is inoculated by spores from the soil or a contaminated animal or carcass. The spores germinate, vegetative cells multiply, and a characteristic gelatinous edema develops at the site. This develops into papule within 12-36 hours after infection. The papule changes rapidly to a vesicle, then a pustule (malignant pustule), and finally into a necrotic ulcer from which infection may disseminate, giving rise to septicemia. Lymphatic swelling also occurs within seven days. In severe cases, where the blood stream is eventually invaded, the disease is frequently fatal.

Another form of the disease, **inhalation anthrax** (woolsorters' disease), results most commonly from inhalation of spore-containing dust where animal hair or hides are being handled. The disease begins abruptly with high fever and chest pain. It progresses rapidly to a systemic hemorrhagic pathology and is often fatal if treatment cannot stop the invasive aspect of the infection

Diagnosis:

Materials to be sent: As a rule, necropsy should not be conducted on animals suspected to be died of anthrax, since the organisms will sporulate on exposure to atmosphere. Blood smears and blood are ideal material of choice. Isolation under laboratory condition can be attempted with spleen.

1. **Direct examination:** Smears from tissues or blood stained by Lesihman's method will reveal anthrax organisms as individual rods with truncated edges. Other staining methods like Giemsa, Grams can also be used. While diagnosing by this method the anthracoids have to be differentiated. The following table lists some important differences between anthrax bacilli and anthracoids.

Sl.No	Characters	B.anthracis	Anthracoids
1.	Motility	Non-motile	Motile
2.	Edges of bacilli	Truncated	Round
3.	Arrangement	Single	Chain
4.	Salicin utilisation	Slow	Rapid
5.	Methylene blue reduction	Slow	Rapid
6.	Gelatin liquefaction	Slow	Rapid
7.	Haemolysis	Slight	Haemolytic
8.	Susceptibility to Penicillin	Susceptible (some)	Resistant
9.	Capsule	Present(only in virulent organisms)	Absent

2. **McFadyean's reaction:** On staining by methylene blue, the anthrax organism appears as blue colour rods in pink stained amorphous polyglutamic acid capsule. This is used as one of the confirmative tests for anthrax.

3. **Isolation and identification:** The organisms grow well in all laboratory media. However, serum and blood in the media enhances the growth. In blood agar, the colonies appear within 24 hours and are flat, rough, grey and non-haemolytic. The colonies are referred as medusa heads or judges wig. On microscopical examination, the edges of the colonies appear as a tangled mass of curly hairs. In nutrient agar the colonies could be smooth and mucoid or rough.
4. **Isolation in lab animals:** The laboratory animals of choice for cultivation of the organisms are guinea pigs and mice. The affected tissues after grinding them in a mortar and pestle or blood as such may be used as an inoculum. The animals will die from 24 hours and anthrax organisms as large capsulated rods can be seen in blood smear and spleen.
5. **Ascoli's test:** It is an immunological test based on the principles of precipitation reaction. This test is used to detect anthrax in decomposed carcasses. In an arrow tube, extract from the decomposed carcass and antibodies against anthrax bacilli are added. Development of a precipitation ring is an indication of anthrax infection.
6. **String of pearls test:** This test is based on the principle of impairment of cell wall development by penicillin. When anthrax bacilli is grown in a medium containing penicillin, due to cell wall impairment it will produce colonies which will resemble string of pearls.
7. **Phage typing:** Gamma phages that specifically lyse *B.anthracis* are used to identify anthrax bacilli.
8. **Fluorescent antibody test:** This test is used to identify anthrax organisms in tissue smears. Either direct or indirect test may be employed.

Treatment: Penicillin, tetracycline, gentamycin, streptomycin, enrofloxacin, erythromycin and chlormaphenicol are used to treat the sick animals suspected to have anthrax.

Control:

1. Different vaccines like attenuated spore vaccine, avirulent spore vaccine (Sterne's vaccine) and non-living vaccine containing protective antigen are used to provide protection to risk group. The Sterne's vaccine is used in Tamil Nadu to control anthrax infection in susceptible population
2. To prevent the spread of infection, sporulation should be controlled. Hence, necropsy of the animals susceptible to have anthrax should not be performed. The carcass should be buried in a six feet deep pit and covered with lime.
3. Animal by-products should be screened for the presence of anthrax bacilli and if they are found to contain spores they should be disposed as mentioned in the previous point.
4. Animals should not be allowed to graze on pastures suspected to have anthrax spores.

Clostridium species

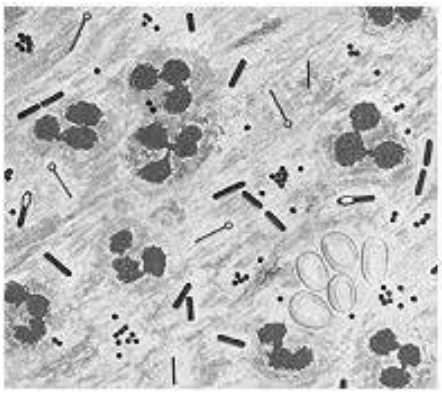
The **clostridia** are relatively large, Gram-positive, rod-shaped organisms. All species form endospores and have a strictly fermentative mode of metabolism. They produce end products such as butyric acid, acetic acid, butanol and acetone, and large amounts of gas (CO_2 and H_2) during fermentation of sugars. They also produce foul smelling compounds during the fermentation of amino acids and fatty acids. The clostridia also produce a wide variety of extracellular enzymes to degrade large biological molecules in the environment into fermentable components. Hence, the clostridia play an important role in nature in biodegradation and the carbon cycle. Most clostridia will not grow under aerobic conditions and vegetative cells are killed by exposure to O_2, but their spores are able to survive long periods of exposure to air. The clostridia live in all anaerobic habitats of nature where organic compounds are present, including soils, aquatic sediments and the intestinal tracts of animals. Most of the clostridia are saprophytes but a few are pathogenic for animals. Those that are pathogens have

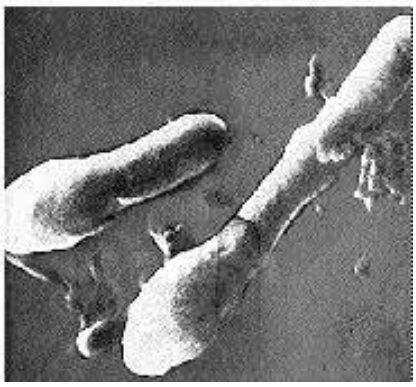
primarily a saprophytic existence in nature and are opportunistic pathogens. *Clostridium tetani* and *Clostridium botulinum* produce the most potent biological toxins.

Important species:

1. Clostridium tetani,
2. Cl.botulinum
3. Cl.novyi,
4. Cl.chauvoei,
5. Cl.septicum,
6. Cl.perfringes,
7. C.haemolyticum,

Important diseases: Tetanus, Black Quarter, Black disease, Enterotoxaemia, Bacillary haemoglobinuria, botulism (food poisoning)





General characters:

1. **Morphology:** In general, clostridial organisms are gram-positive long rod shaped organisms with rounded ends occurring either singly, short chains or in filaments. They are variable in size measuring 0.4 to 1.2μm in width and 0.3 to 8μm in length. They are motile (except *C.perfringens*) by peritrichous flagella and non-capsulative. They are known for being anaerobic (some are microaerophilic) and spore forming. The spores are located centrally, terminally or subterminally.
2. **Habitat:** Clostridial organisms are mainly found on the soil since they are frequently excreted in the faeces of the infected animal. In the soil, they survive as spores. Under favourable conditions, these spores germinate into vegetative cells and cause infection.
3. **Classification:** Clostridial organisms are classified into three groups according to habitats and natural history. Many species are **saprophytes** closely associated with soil and are reason for putrefaction of fruits and vegetables. Some are **commensals** that are seen in the digestive tract. Few species are **toxigenic** that produce toxins to host tissue. These toxigenic strain are classified into four groups depending upon the tissue they affect viz,
    a. **Histotoxic clostridia –** To a variety of tissue particularly muscle.
    b. **Hepatotoxc clostridia –** To liver
    c. **Enterotoxic clostridia –** To digestive system
    d. **Neurotxic clostridia –** ToCNS
4. **Resistance:** The spores of the organisms are considerably resistant than the vegetative cells to the physical and chemical agents as resistant as the spores of *B.anthracis*.. The spores can remain viable in soil and dried cultures for a long time (many years). Boiling for 30 minutes, dry heat at 140°C for 3 hours and autoclaving at 121°C will destroy spores. However, freezing temperature has no effect

on the spores. Only prolong action of chemical disinfectants at high concentration has killing effect on the spores - 5% phenol for 2 days and 10-20% formalin in 10 minutes.

5. **Cultural characters:** Clostridial organisms are known for their strict anaerobic growth characters. Certain species like *C.novyi* are very sensitive to oxygen, whereas *C.perfringens* grow in microaerophilic conditions. The media commonly used for cultivation are blood agar, cooked meat medium and thioglycolate broth. The colonies are 3 mm in diameter, round or slightly irregular, slightly raised, granular, transparent or translucent with filamentous margins. Special media are used to study the toxin production. When media like blood agar are used for the cultivation of clostridial organisms, the plates have to be incubated in anaerobic conditions. The anaerobic condition can be achieved in laboratory in GasPak system, McIntosh-Fildes jar or by using certain reducing substances. When fluid media are such anaerobic incubation is not necessary since the organisms will grow at the bottom of the media where oxygen tension is low.

TETANUS - Clostridium tetani Morphology and general characters:

1. ***C.tetani*** is a long, slender rod shaped organism. They are motile by peritrichous flagella and non-capsulated. The spores of the organisms are round, two times bigger than the diameter of the bacilli and are arranged terminally giving them the characteristic **drumstick** appearance.
2. These organisms sporulate faster and readily than other clostridia. The spores are destroyed at boiling temperature after 15 minutes, autoclaving and at dry temperature of 150°C after one hour.

Antigens and toxins: There are 11 strains of *C. tetani* distinguished primarily on the basis of flagellar antigens. Some strains are toxigenic and some are non-toxigenic. Tetanus toxin is one of the three most poisonous substances known, the other two being the toxins of botulism and diphtheria. The toxin is produced by growing cells and released only on cell lysis. Cells lyse naturally during germination the outgrowth of spores, as well as during vegetative growth. After inoculation of a wound with *C. tetani* spores, only a minimal amount of spore germination and vegetative cell growth are required until the toxin is produced. The toxigenic strains produce two toxic substances termed as **tetanolysin** and **tetanospasmin.** Tetanolysin is a haemolysin and cytotoxin. It is responsible for areas of haemolysis in blood agar. It is heat labile and oxygen labile. Tetanospasmin is a potent lethal neurotoxin. It is protein with a molecular weight of 150,000. The toxin is produced inside the cell and not released into the surrounding media. It is released only after the bacterial cell disintegrates. Hence, the toxin is not strictly considered as an exotoxin. **Tetanospasmin** initially binds to peripheral nerve terminals. It is transported within the axon and across synaptic junctions until it reaches the central nervous system. There it becomes rapidly **fixed to gangliosides at the presynaptic inhibitory motor nerve endings**, and is taken up into the axon by endocytosis. The effect of the toxin is to **block the release of inhibitory neurotransmitters** (glycine and gamma-amino butyric acid) across the synaptic cleft, which is required to check the nervous impulse. If nervous impulses cannot be checked by normal inhibitory mechanisms, it produces the generalized muscular spasms characteristic of tetanus. Horses and man are more susceptible to the toxin than other animals. Birds are insusceptible to the toxin. One mg of the toxin contains 100-million mouse lethal dose. Unpurified toxin is denatured at 65°C for 5 minutes, whereas dried form resists a temperature of 120°C for one hour.

Cultural characters: They are strict anaerobes and optimum temperature of growth is 37°C. However, growth is also reported at 14-44°C. In cooked meat broth good growth along with blackening of meat particles is noticed and addition of egg white to the meat broth yield better growth. In blood agar haemolytic, translucent, grey colonies of 2-4mm diameter with filamentous edges and fuzzy appearance are noticed.

Pathogenesis: Many domestic animals are affected by tetanus. However, birds are resistant to the infection. Young animals are more susceptible than older animals. The incubation period is variable

(form 6 days to several months). The organism does not have any invasive power and development of tetanus entirely depends on the favourable conditions suitable for germination of spores and production of toxin. The favourable conditions generally occur during traumatic conditions that involve reduced blood supply, necrosis and multiplication of other bacteria which results in reduced oxygen tension in the area. The toxins produced by the bacteria produce two types of infection, local and general. These two forms are more common in man than in animals. In animals general form alone is seen. The toxins move in the inter-neuronal tissue space of motor nerves and then into interstitial fluid of CNS. The main site of toxin action is the synaptic junction of the motor neurones.

Symptoms: Initial symptoms are stiffness and unwillingness to move that is followed by general stiffness of the limbs, head and tail that become stiff. There is mild twitching of muscles that later turn into muscular spasms. Rigidity of the muscles extends from limbs to trunk, with dilated nostrils, erect ears, protruding nictitating membrane and impossible mastication (due to locked jaw) are characteristic symptoms. Respiration is shallow and rapid followed by respiratory failure. The condition of the horses is referred as 'wooden horse' sincethe animals have difficulty in moving.

Lesions: No characteristic lesions are noticed.

Diagnosis:

1. Based on the **symptoms.**
2. **Micorscopical examination** of clinical specimens (wound materials) for 'drumstick bacilli'.
3. **Cultural examination** on meat broth where the colour of the meat particles turn into black and addition of egg white to the meat broth yield better growth. In blood agar haemolytic, translucent, grey colonies of 2- 4mm diameter with filamentous edges and fuzzy appearance arenoticed.

Control:

1. Tetanus toxoid should be give when wound contamination with *C.tetani* is suspected. In horses two dosed at an interval of 3-4 weeks produce good response.
2. Hygienic wound management and good animal husbandry practices.

BOTULISM – *Clostridium botulinum*

This organism is found commonly in the soil. The toxin produced by this organism is a very potent one and cause botulism in animals, birds and human beings.

Morphology: They are gram-positive rod shaped organism occurring singly, in pairs or in short chains. They are sluggishly motile by peritrichous flagella. They are non-capsulative. They produce oval shaped spores that are located centrally, sub-terminally or terminally inside the cell. Seven toxigenic types of the organism exist, each producing an immunologically distinct form of botulinum toxin. The toxins are designated A, B, C1, D, E, F, and G. Not all strains of *C. botulinum* produce the botulinum toxin. The purified toxins are protein with difference in molecular weight. The toxins are very lethal – 1.0 mg of toxin is lethal for 40 million mice. Rarely the toxins are activated inside the gut by trypsin. This toxin acts on the cholinergic nerve endings of peripheral somatic and autonomic fibres and does not have effect on brain and spinal cord.

Cultural characters: They are strict anaerobic organisms and grow better at an incubation temperature of 35°C. Growth occurs in ordinary laboratory media and the colonies are transparent and irregular. In blood agar they produce narrow zone of haemolysis. The proteolytic A, B and F cause blackening of the meat particles in the meat medium whereas the non-proteolytic C, D and E do not blacken the meat. They liquefy gelatin.

Pathogenesis: The spores of the organisms are commonly found on the soil. They are ingested by the animals and excreted without any change. Besides the organism have very poor invasive power. The important aspect of botulism is that the disease arises due to consumption of preformed toxin in the putrefied meat or decayed vegetation. *C.botulinum* does not invade the body and produce toxin inside living body. The ideal environment for the production of toxin is found in putrefied carcases, decomposed feed and decayed vegetation. Another important aspect in production of the disease is that there should be no degeneration of the toxin produced inside the putrefied materials. In cattle the infection is called as **lamsiekete** and is associated with phosphorus deficiency in cattle. In cattle, phosphorus deficiency is characterized by depraved appetite and the animals will consume pieces of carcass found in the soil and get the infection. In poultry the infection is seen commonly in ducks, chicken and pheasants. In ducks the infection is referred as **Duck sickness** or **Western duck disease**. The infection is due to consumption of decayed vegetation containing toxin found onthe marshy lands. In chickens and pheasants the infection is termed as **limberneck,**where the infection is due to consumption of green bottle fly and other insects that feed on dead animals. In brooders the infection is termed as **pseudo limberneck** and is due to pecking at the litter that may contain botulinum toxin.

Symptoms and lesions: The incubation period is 2-10 days and symptoms are characteristic of nervous symptoms. Initially there is excitability followed by in- coordination and paralysis of hind limbs. Paralysis the mouth, pharynx and neck with protrusion of tongue are characteristic of botulism. The animals die of the infection. In poultry, there is paralysis of wings, legs and neck, protrusion of nictitating membrane followed by death.

Diagnosis:

1. Based on symptoms andlesion.
2. **Detection of toxin** in intestinal contents or in feed stuff. The laboratory animals used are mice and guinea pigs
3. **Differentiation of *C.perfirngens*** with other clostridia by FAT.

Control: A toxoid used in certain countries. However, good animal husbandry practices with prevent the infection.

BIG HEAD DISEASE or BLACK DISEASE - Clostridium *novyi (C.oedematiens)*

Morphology: They are gram-positive rod shaped organism occurring singly, in pairs or in short chains. They are sluggishly motile by peritrichous flagella. They are non-capsulative. They produce oval shaped spores that are located centrally, sub-terminally or terminally inside the cell.

The organisms are divided into three types based on the antigenic structure and toxins produced.

Type A (*C.novyi*) – Gas gangrene and big head in ram Type B (*C.gigas*) – Black disease (chronic hepatitis)

Type C (*C.bubalorum*) – Osteomyelitis in buffaloes (Kranefeld's bacillus)

Cultural characters: The organisms are strict anaerobes. Primary cultures are difficult to grow and the growth is enhanced by the addition of glucose, fresh blood or brain extract. The growth is characterized by **colonial motility** (**swarming**) that is noticed after 3-4 days. Swarming is characterized by movement of daughter colonies away from parent colonies. In plates the swarming looks like arcs or spirals. Later the daughter colonies fuse with parent colonies. In horse blood agar they produce wider zone of haemolysis. In meat medium the meat particles turn into pink. In egg yolk medium, the colonies of type A and B are surrounded by opalescent zone on precipitation that is due to lecithinase. This reaction is called as **Nagler's reaction.**

Pathogenesis: The spores are commonly found on the soil and the organisms are commensals in the intestine. The big head in ram is a result of contamination of the wounds on the heads of rams.

Head wounds are common in ram as result of fighting among themselves. The alpha toxin produced by the organism is responsible for the oedema and death. Black disease occurs in areas where liver fluke infection is common. The organisms multiply the areas where the immature flukes have migrated. The inflammatory response in the tracts of immature fluke migration provide ideal environment for the organism to multiply, produce alpha toxin resulting in death due to toxaemia.

Symptoms and lesions: Big head is characterized by gas gangrenous lesion around the face and head. Affected animals will collapse and die. Black disease is an acute toxic infection with sudden death. The symptoms observed are inability to move and unsteady gait. The lesions are more prominent in liver with necrotic areas. There is extensive blood stained oedema that gives the carcass a dark colour. Exudates are seen in heart.

Diagnosis:

1. Based on symptoms and lesions.
2. **Microscopical observation** of liver impression smears
3. **Cultural characteristics -** The organisms are strict anaerobes. Primary cultures are difficult to grow and the growth is enhanced by the addition of glucose, fresh blood or brain extract. The growth is characterized by **colonial motility** (**swarming**) that is noticed after 3-4 days. Swarming is characterized by movement of daughter colonies away from parent colonies. In plates the swarming looks like arcs or spirals. Later the daughter colonies fuse with parent colonies. In horse blood agar they produce wider zone of haemolysis. In meat medium the meat particles turn into pink. In egg yolk medium, the colonies of type A and B are surrounded by opalescent zone on precipitation that is due to lecithinase. This reaction is called as **Nagler's reaction.**
4. **Detection of toxin** in intestinal contents or in feed stuff. The laboratory animals used are mice and guinea pigs.

Control: A formalised whole culture vaccine adsorbed onto aluminium hydroxide is used to control the infection.

BLACK QUARTER - *Clostridium Chauvoei*

General characters:

1. **Morphology:** They are gram-positive rod shaped pleomorphic organisms. They appear as large cigar shaped rods or citron forms. They are sluggishly motile by peritrichous flagella. They are non-capsulative. They produce oval shaped spores that are located centrally, sub-terminally or terminally inside the cell. The staining character is highly variable, sometimes only one end of the organisms will take stain and the other side remain unstained.
2. **Habitat:** They are commonly found on the pastures where the incidence is endemic. They are also found in the soil.
3. **Resistance:** The spores are resistant and are killed by steam in 40-50 minutes and by autoclaving. The spores are also killed by 3 per cent formalin for 15 minutes.
4. **Antigens and toxins:** *C.chauvoei* have one common somatic antigen and the organisms are divided into two groups based on flagellar antigens. They share spore antigen with *C.septicum.* They produce four different types of toxins alpha (neurotoxin), beta (DNAse), gamma (hyaluronidase) and delta (haemolysin).
5. **Classification:** Not very important. They share a spore antigen with *C.septicum.*

Cultural characteristics: They are strict anaerobic organisms. The growth is enhanced by the addition of liver extract of glucose. The optimum temperature for growth is 37^0C. In blood agar the colonies have irregular edges, 2-4mm in diameter and are surrounded by a zone of haemolysis. In

cooked meat medium, the colour of the meat changes into pink and a sour odour is produced. *C.chauvoei* ferments glucose, lactose, sucrose and maltose with production of acid and gas. However, they do not ferment salicin. This is a differentiating feature with *C.septicum.*

Pathogenesis: The main mode of spread is through wound contamination. The contaminated pasture contains the spores that enter into the animal through wounds. Infection through ingestion is also possible but the spores have to be activated. Activation of spores becomes possible only when the environment becomes anaerobic. Necrotic cells provide ideal environment for the organisms to multiply. Cattle, sheep and goat that are in pasture are more prone to infection than stall fed animals. Young animals are more affected than adult animals.

Symptoms: The infection may occur as per acute or acute. In per acute infection, no symptoms are usually seen and animals die suddenly in the pasture. Important symptoms include crepitant swelling on the fore or hindquarter that crackles when rubbed with fingers. It also pits on pressure. This is dueto production of large quantities of gas and oedema. The affected animal will have fever, anorexia and becomes lame. Death occurs after 24 hours.

Lesions: The muscle becomes dark red in colour, dry necrotic and filled with small gas bubbles. The lesion also emits a rancid odour. The muscle is surrounded by yellow or blood stained fluid.

Diagnosis:

1. Based on the **symptoms** and **lesions**.
2. **Microscopical examination** of the impression smears prepared from affected muscles or oedematous fluid and stained by Gram's method for *C.chauvoei* organisms and spores. The spores can also be confirmed by malachite green staining method.
3. **Cultural examination**: The clinical materials are affected muscle piece either dry of as such. In blood agar the colonies have irregular edges, 2-4mm in diameter and are surrounded by a zone of haemolysis. In cooked meat medium, the colour of the meat changes into pink and a sour odour is produced.
4. Differentiation with *C.septicum*
   - By biochemical reaction – Salicin utilization
   - Fluorescent antibody test using specific antibodies.
5. Modern methods like ELISA.
6. **Biological test:** Guinea pig is laboratory animal of choice. Suspected material is mixed with Calcium chloride and given as deep intramuscular injection to guinea pigs. Guinea pigs will develop characteristic symptoms of black quarter after 48 hours.

Treatment: Penicillin is the drug of choice. Usually it is given coupled with antiserum. Oxytetracycline and chloramphenicol can also be given.

Control: Vaccination is the best method to control the infection in endemic areas. Different vaccines are available. The commonly used vaccine is killed alum of aluminium hydroxide adjuvanted vaccine. It protects the animals for 6-24 months.

ENTEROTOXAEMIA - *Clostridium perfringens* Synonym ;*Clostridium welchii*

General characters:

1. **Morphology:** They are gram-positive rod shaped organisms occur, either singly, in pairs or in short chains. Unlike most other clostridia they are non-motile, capsulative and sporulate rarely. If sporulate the spores are located centrally, sub-terminally or terminally inside the cell. Young cultures appear as gram-positive whereas old cultures appear gram-negative.

2. **Habitat:** They are present everywhere. They are found in the soil, and intestines commonly.
3. **Resistance:** They are not as resistant as other clostridia. Boiling for five minutes kills them.
4. **Classification:** There are five important types.
   - **Type A –** Causes gas gangrene
   - **Type B –** Causes lamb dysentery
   - **Type C –** Causes struck
   - **Type D –** Causes pulpy kidney or over eating disease
   - **Type E –** Causes enterotoxaemia in calves.
5. **Antigens and toxins:** *C.perfringens* possesses somatic and flagellar antigens but they are of less importance in classification. They produce number of toxins and enzymes that are of importance in pathogenesis. Some of important toxins and enzymes are listed below.
   1. Alpha toxin – (Lecithinase C). Found abundant in type A. Responsible for Nagler reaction.
   2. Beta toxin – Found abundant in B and C. It is a lethal necrotising toxin.
   3. Gamma toxin – Found in B and C. Minor lethal toxin.
   4. Delta toxin – Found in type C and is responsible for struck in sheep. Lethal haemolytic toxin.
   5. Epsilon toxin – Found in types B and D. Major lethal necrotising toxin.
   6. Eta toxin – Found in type A. Minor lethaltoxin.
   7. Theta toxin –Haemolysin.
   8. Iota toxin – Inert major lethal necrotising toxin that is activated by lambda toxin.
   9. Kappa toxin –Collagenase.
   10. Lambda toxin – Proteinase and gelatinase
   11. Mu toxin – Hyaluronidase like enzyme.
   12. Nu toxin –DNAse

Cultural characteristics: Unlike other clostridia, *C.perfringens* are not strict anaerobic and they will grow well in ordinary laboratory media with a temperature range of 37-47C (optimum 43-47C).

1. In blood agar the colonies are small 2-3mm in diameter, smooth, evenly convex and opaque. Occasionally mucoid variants also occur. A narrow zone of complete haemolysis surrounded by wider zone of partial heamolysis is produced in blood agar.
2. The colour of the meat in cooled meat medium changes into pink and it is not digested. The growth in cooled meat medium is characterized by production of large amount of gas. After few hours of incubation gas bubbles can be seen rising from the bottom. This can be achieved when heating at 100C, followed by cooling at 37C, brings down the residual oxygen of the meat medium.
3. In media containing egg yolk or horse sera, *C.perfringens* produces a marked opalescence. This is caused by lecithinase. This reaction is referred as **Nagler reaction** or **lecithinase reaction.**
4. *C.perfringens* ferments glucose, lactose, sucrose and maltose and produce acid and gas. In litmus milk acid and clot is produced and due to large amount of gas production the clot breaks up in number of areas, this reaction is referred as **stormy clot reaction.**

Pathogenesis: Generally the infections associated with *C.perfringens* are human food poisoning and gas gangrene.

1. **Gas gangrene:** It is caused by Type A. This infection has an incubation period of 12-24 hours. Initially there is pain and oedema in the affected areas followed by accumulation of gas bubbles and extensive haemorrhage. The area, which initially appears pale later, turns into brown, dark green of black. Death follows septicaemia.
2. **Enterotoxaemia in sheep (Lamb dysentery):** Type B organisms cause this infection. The organisms are found in the intestinal tract as commensals. Infection occurs as a result of production of large amount of toxins. The toxins are produced when there is heavy consumption of milk followed by indigestion. The affected animals show signs of abdominal pain, bleat repeatedly, cease to suck, collapse and die. The faeces is blood stained. The intestine consist of intense areas of haemorrhage and ulcers. The stomach is filled with undigested milk.
3. **Enterotoxaemia in sheep (Struck):** The disease occurs during early spring and affects adult sheep. The infection is characterized by fatal toxaemia with ulceration of the intestine. Death occurs suddenly without any symptoms. The affected animals develop symptoms of rigidity, immobility and abdominal pain but no diarrhoea. The important lesions are ulceration of the intestine, degenerative changes in kidneys and fluid accumulation in abdominal and thoracic cavities.

   Enterotoxaemia in sheep (Pulpy kidney disease, Over eating disease): It is caused by type D organisms. The infection is due to rapid multiplication of the organisms in the intestine and abomasum and absorption of the toxins produced. Young animals between age groups of 6-12 months are more prone to infection. The infection occurs as result of sheep exposed to lush green pastures after monsoon and resultant over eating. Excessive concentrate feeding is also a predisposing factor. The organisms are found in the soil and they are consumed by the animals resulting in production of small amount of toxin in the intestine. These small quantities are not dangerous. Only when there is large amount of toxin production the infection sets in. The infection is characterized by sudden death of animal in the field. The symptoms exhibited are animals become dullness and weakness. There is spasmodic struggling before the animal collapse. The important lesions are the degeneration of the kidney cortex, soft and pulpy kidneys, degeneration of other organs, accumulation of straw-coloured fluid in the pericardium, endocardial haemorrhages and areas of congestion in abomasum and small intestine.
4. **Enterotoxaemia in cattle:** Less significant infection. The infection confuses with lead poisoning.

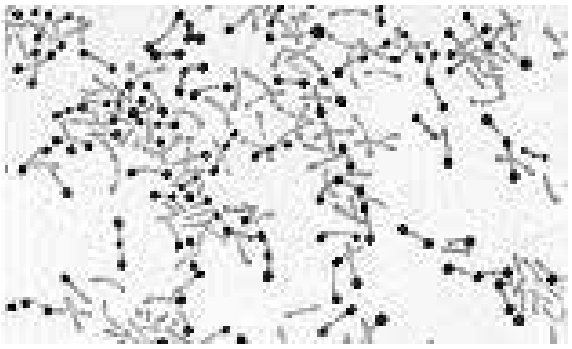
Diagnosis:

1. Based on symptoms and lesions
2. **Microscopical examination** of the smears prepared from intestinal contents reveal gram-positive organisms with characteristic morphology.
3. **Demonstration of toxin activity in mice –** Test for lethal action and toxin neutralisation test.
4. **Cultural examination – The intestinal contents** from dead animals is the ideal material. The contents are inoculated into cooked meat medium. The colour of the meat in cooled meat medium changes into pink and it is not digested. The growth in cooled meat medium is characterized by production of large amount of gas. After few hours of incubation gas bubbles can be seen rising from the bottom. This can be achieved when the residual oxygen of the meat medium is brought down by heating at 100^0C, followed by cooling at 37^0C.
5. Presence of glucose in the urine is an indication of infection.

Treatment: Most of the infections are of acute type. Hence, treatment is of little value.

Control: Alum precipitated formalin killed culture is used as vaccine that offers satisfactory immunity. Vaccination of mothers provided passive immunity to the young animals.

CORYNEBACTERIUM



Introduction: Corynebacteria are Gram-positive, aerobic, non-motile, rod-shaped bacteria related to the Actinomycetes. They do not form spores or branch as do the actinomycetes, but they have the characteristic of forming irregular shaped, club-shaped or V-shaped arrangements in normal growth. They undergo snapping movements just after cell division, which brings them into characteristic arrangements resembling Chinese letters. *Corynebacterial* cell walls contain thin spots, which leads to some Gram variability and "ballooning" that produces a "club-shaped" cell). Old cells store inorganic phosphate, which can appear as metachromatic granules when stained.

Corynebacteria belong in the family *Mycobacteriaceae* and are part of the CMN group (*Corynebacteria, Mycobacteria and Nocardia*). As a group, they produce characteristic long chain fatty acids termed mycolic acids. The genus *Corynebacterium* consists of a diverse group of bacteria including animal and plant pathogens, as well as saprophytes. Some corynebacteria are part of the normal flora of humans, finding a suitable niche in virtually every anatomic site. The best known and most widely studied species is *Corynebacterium diphtheriae*, the causal agent of the disease diphtheria. The family *Mycobacteriaceae* are Gram-positive, nonmotile, catalase-positive and have a rod like to filamentous morphology (*Corynebacteria* are often pleomorphic.

The genus *Corynebacterium* consisted of number of important species of veterinary importance, but most of them have been moved to new genus as a result of improved methods of classification. The following table lists important pathogenic species of *Corynebacterium* and their new names;

Old name	Infection	New name
C. diphtheria	Diphtheria in man	
C. equi	Suppurative bronchopenumonia	*Rhodococcus equi*
C. Pseudotuberculosis	Caseous lymphadenitis	
C. pyogenes	Caseous pneumonia in pigs Polyarthritis Chronicsuppurative pneumonia, arthritis, umbilical infections in cattle Summer mastitis and pneumoniain sheep and goat	*Trueperella pyogenes*
C. renale; C.pilosum; C.cystitidis	Pyelonephritis in cattle	*C. renale*
C. mastitidis	Mastitis in sheep and goat	

C.pseudotuberculosis

General characters:

1. **Morphology:** They are small, Gram-positive, pleomorphic rod bacteria that occur in a characteristic filamentous, angular, palisade arrangement often located intracellularly. They are non-motile, aerobic, fermentative, non-spore forming. C. pseudotuberculosis is a facultative intracellular parasite that has been implicated in the diseases caseous lymphadenitis in sheep and goats, ulcerative lymphangitis in horses and formation of abscesses in many other animal species.
2. **Location:** They are facultative intracellular.
3. **Habitat and resistance**: *C. pseudotuberculosis* is present in the environment following contamination

by infected animals. Low temperatures and damp conditions prolong survival time. *C. pseudotuberculosis* survive for prolonged periods (>6 mths) in contaminated environments and has been isolated from feed, feeders, fences, shears and sheep dips, plus dust and soil samples from around stock handling areas. The main source of infection and potential spread of the organism is via the rupture of affected lymph nodes and abscesses with the discharge of thick caseous pus containing millions of organisms into the environment.

4. **Antigens andtoxins**:

- **Cell wall surface lipid**. It allows organism to resist digestion by cellular enzymes and to persist as a facultative intracellular parasite within phagocytes. It also cytotoxic and induces caseation giving the organism pathogenic potential.
- **Heat labile phospholipase exotoxin, phospholipase D (PLD)**. It is the major virulence factor. The toxin's major effect of phospholipase relates to the establishment of infection and possible local effects. It helps in the survival and multiplication of the organism in the host. It has effects on phagocytic cells (inhibition of chemotaxis, degranulation, and lethality in neutrophils) or other effects such as compliment depletion. This exotoxin also functions as a permeability factor acting on the local blood vessels and lymphatics. The resulting leakage of plasma increases the extravascular circulation of fluid and therefore the probability of spread of infection both locally and to the regional lymph nodes via the lymphatics. This exotoxin also acts as a partial haemolysin having a limited necrosing action within caseous lesions, and contributing to the degradation of RBC membranes.

Cultural characters: *C.pseudotuberculosis* is aerobic or facultative anaerobic. Optimum growth is obtained at 37C at a pH of 7.2 to 7.4. Growth is enhanced by addition of serum to the culture media. On serum agar the organisms produce small, finely granular, translucent colonies with irregular border. After few days the colonies become opaque, concentrically ringed, dry and crumbly. Colonies have cream to orange colored. The organisms grow in Loeffler's coagulated blood serum containing potassium tellurite salt. The colonies are greyish black in colour. Uniform turbidity is produced in broth.

Pathogenesis:

The usual mode of entry of bacteria is via superficial wounds in the skin or mucous membranes followed by extension to the regional lymph nodes and beyond. Organisms also enter via the respiratory tract. Ingestion of bacteria may produce lesions in lymph nodes of head and thorax.

Caseous lymphadenitis: It is a chronic purulent disease of sheep and goats characterized by discrete chronic abscesses with palpable enlargement of the superficial lymph nodes, visceral lymph nodes and internal organs. It mainly occurs in sheep. In sheep the, infection is often preempted by wounds obtained during shearing, castration, docking, crutching etc. Other general wounds such as dog bites, scratches etc can also lead to this infection. Contamination of these wounds by bacteria in environments, or via direct contact with infected animals is of importance in transmission. The most common potential source of infection is contaminated sheep dips, with bacteria able to survive in commercial dips up to 24 hours without significant reduction in viable count. Pus from a single discharging node is able to contaminate large volumes. Sheep dipped in contaminated fluid within 2 weeks of being shorn are especially susceptible to infection due to ease of contact between bacteria and the skin. Other predisposing factors are conditions where animals are kept in close contact with infected animals in areas like yards with wooden fences, dirt floors etc., which are difficult to disinfect. In goats skin abrasions, fight wounds or traumatised buccal mucosa act as portals of infection for the organism. Formation of abscess is characteristic lesions in caseous lymphadenitis. The abscesses are most common in superficial nodes, (eg; mandibular, parotid, retropharyngeal, prescapular, subiliac,

superficial inguinal) with lung and thoracic lymph node involvement. The lesions are rarely seen in sheep in head and neck lesions and most commonly noticed at pre-femoral and pre-scapular sites, while in the goat mandibular and parotid nodes are frequently involved. Visceral lesions involving lung parenchyma, internal lymph nodes and most other organs develop from haematogenous spread from the regional lesions. Common organs involved are the liver and kidneys plus lungs and thoracic nodes. The Abscess reaches a diameter of 5-10 cm and is associated with a firm fibrous capsule. It contains thick creamy pus which becomes firmer with age leading to a firm, caseous friable dry texture. In sheep the contents are usually light green and often develop a concentric laminated onion-ring appearance due to repeated stages of necrosis and capsule formation. Deposition of calcerous granules frequently occurs at the margin of the expanding lesion. In goats abscess contents are more uniformly pasty than dry, do not have a laminated appearance and rarely calcify. When the disease affects superficial lymph nodes there is palpable enlargement of these nodes. Affected nodes may rupture forming a discharging wound. When internal lesions occur lung and thoracic nodes are frequently involved. Animals with interstitial lung lesions causing pneumonia, pleurisy or other serious internal lesions may show signs of ill thrift. In many cases there are frequently no outward clinical signs, therefore CLA is often referred to as a "hidden disease" and only becomes apparent at slaughter. The major effects of the disease are due to economic losses through trimming and condemnation of carcass, decreases in wool production, value of hides, body weights, milk production and reproductive efficiency.

Ulcerative lymphangitis: It is a contagious disease of equines characterized by inflammation of the subcutaneous lymphatic vessels with formation of nodules, abscesses and ulcers on the ventral abdomen, pectoral region, distal portions of limbs. Lymphatics become corded and infection can spread to form abscesses in internal organs. Superficial skin wounds are a major cause of infection. High incidence is related to poor management, with infection most common in situations of overcrowding when animals are picketed together on muddy standings and subject to kicks, chaffs and other injuries. Contaminated bedding and grooming utensils act as fomites. Flies and other arthropods act as mechanical vectors. The abscesses vary markedly in appearance, range in diameter from 5-30 cm, often with pitting oedema of adjacent tissues. Most abscesses are localised and tend to come to the surface where they rupture or can be lanced easily. The most common sign is swelling of the pectoral region or ventral abdominal wall. Other common signs include diffuse or localised swellings elsewhere on the body, ventral oedema, ventral midline dermatitis, fever, weight loss plus presence of draining abscesses, wounds or tracts. Lameness is also noticed and is attributed to stiffness and discomfort from oedema and abscesses adjacent to limbs. Affected animals are often noticeably depressed due to the effects of pyrexia, lameness and pain. The organism has also been cited as a cause of equine abortion, pyelonephritis and equinemastitis.

Infections of farm workers and veterinarians exposed to infected animals usually appear as lymphadenitis.

Diagnosis

1. Clinical materials: Pus or affected tissue from lymph nodes.
2. Based on symptoms and lesions.
3. Microscopy – Smears prepared from pus smear and culture smear stained by Grams staining method reveal onion ring appearance and Chinese letter shaped organisms
4. Isolation and identification: Growth is enhanced by addition of serum to the culture media. On serum agar the organisms produce small, finely granular, translucent colonies with irregular border. After few days the colonies become opaque, concentrically ringed, dry and crumbly. Colonies have cream to orange colored. The organisms grow in Loeffler's coagulated blood serum containing

potassium tellurite salt. The colonies are greyish black in colour. Uniform turbidity is produced in broth.

5. Slide agglutination test

Treatment

Treatment is not normally attempted.

Control

A formalised toxin (toxoid) is used to control the infection

Contagious bovine pyelonephritis:

Bovine cystitis is an inflammation of the urinary bladder of cattle that may ascend the ureters to cause infection of the kidneys referred as pyelonephritis. A similar condition can occur in sheep. The condition is sporadic and worldwide in distribution. It is most often seen after parturition. The most common causative agents are the *Corynebacterium renale* group of bacteria, including *C renale*, *C cystitidis*, and *C pilosum*, as well as *Escherichia coli* . The latter is most often seen in chronic cases and may be an opportunist following the corynebacterial infection. Other organisms that may be involved include, but are not limited to, *Staphylococci* and *Streptococci*. The most common causative agents, *Corynebacterium* spp and *E. coli* are ubiquitous in the environment and are common inhabitants of the vagina and prepuce. The organisms attack or colonize the mucosal lining of the bladder and ureters usually after some traumatic insult (such as parturition or abnormal deformity of the vaginal tract). The stresses of parturition, peak lactation, and a high-protein diet (which increases the pH of the urine and is therefore conducive to colonization of the attacking organisms) are all contributing factors. The symptoms include a slight fever, loss of production, colic with restlessness, tail switching, polyuria, hematuria, or pyuria. In chronic cases, the animal may show colic, diarrhoea, polyuria, polydipsia, stranguria, and anaemia. As the disease progresses, the bladder becomes thickened and inflamed. The ureters become thickened and dilated with a purulent exudate. The involved kidneys develop multiple small abscesses on the surface that may extend into the cortex and medulla. Diagnosis is based on clinical signs; a history of recent parturition; palpation of the left kidney for enlargement, loss of lobulation, and pain; vaginal inspection for detection of inflamed and enlarged ureters; microscopical examination of the urine for WBC and bacteria and urine culture to identify the organism. In early acute cases, enlarged ureters and involvement of the kidney may not be detectable on rectal palpation. Pyelonephritis can be cured by administration of Penicillin.

RHODOCOCCUS

Disease caused: Suppurative bronchopneumonia in horses

Important species: *Rhodococcus equi*

Morphology and general characters:

1. **Morphology:** This organism was earlier grouped under the genus *Corynebacterium.* They are gram-positive and weakly acid fast. They are pleomorphic organism occurring as large rods, cocco bacilli or club shaped, arranged in groups or clusters. They are non-motile and non-sporulative. In contrast to corynebacteria, these organisms do not possess metachromatic granules. Further, they possess large polysaccharide capsule. They are non-fermentative and catalase and urease positive.
2. **Habitat and resistance:** These organisms are soil borne and often found in manure. They are also present in the intestines of horses and persist for long periods in the manure and litter of stables. They are relatively heat resistant and are killed only when exposed to a temperature of 60°C for 1 hour. Sensitivity of disinfectants is variable.
3. **Antigenicity and serotypes:** In contrast to corynebacteria, members of the genus *Rhodococcus* do

not produce any toxins. They are serotyped into 10 serotypes based on their capsular (K) antigens. Serotypes 1 and 2 are more predominant.

4. **Location:** They are facultative intracellular.

Cultural characters: They are aerobic or facultative anaerobic and grow well on ordinary lab media at 37°C. In nutrient agar, they produce glistening mucoid colonies that are pale pink in colour. These organisms are known for their ability to produce pale pink to red colour pigments. No haemolysis is produced in blood agar. One of the important characteristics of these organisms is that the primary cultures take a long time to develop. However, the secondary cultures appear within 24-48 hours. Gelatin and serum are not liquefied.

Pathogenesis: The pathogenesis of the organism is not very clear. Infection arise through direct contact, inhalation or from contaminated soil, manure etc. They are pyogenic organisms and produce abscesses. The main route of infection is through respiratory tract. They are facultative intracellular organisms and have the ability to survive inside macrophages. Though no toxins are produced, two substances phospholipase and cholesterol oxidases are considered as virulence factors.

Pathogenicity: The organism causes suppurative bronchopneumonia in foals, submandibular and cervical lymphadenitis in swine and abscess in lungs and lymphnodes in many animals. It is also responsible for abortion in mares.Pneumonia occurs in foals during 2-4 months of age. Affected animals cough, become dull and listless due to pyrexia and respiratory distress. In severe cases, a purulent discharge is noticed from the nostrils. Abscesses are noticed throughout the lungs and associated lymphnodes. Besides these, suppurative lesions are noticed in uterus also.

Diagnosis:

1. Clinical materials: Pus or pieces of lung and lymph nodes from dead animals. The clinical material may be mixed with oxalic acid (to a final concentration of 2.5%) for better isolation.
2. Morphological characters – Staining properties:
3. Isolation and identification: They are aerobic or facultative anaerobic and grow well on ordinary lab media at 37°C. In nutrient agar, they produce glistening mucoid colonies that are pale pink in colour. These organisms are known for their ability to produce pale pink to red colour pigments. No haemolysis is produced in blood agar. One of the important characteristics of these organisms is that the primary cultures take a long time to develop however, the secondary cultures appear within 24-48 hours. Gelatin and serum are not liquefied.
4. Based on symptoms and lesions:
5. CAMP reaction: *R.equi* can produce only partial haemolysis or no haemolysis. When combined with *Corynebacterium pseudotuberculosis* it causes haemolysis. It also enhances the beta haemolytic action of *Staphylococcus aureus.*

Treatment: Treatment is generally ineffective. It should be started well in advance. Penicillin, lincomycin, rifampicin, neomycin, gentamycin and streptomycin combined with erythromycin are effective.

Zoonotic importance: The organisms cause pneumonia and abscess in AIDS and immuno-compromised patients.

ACTINOMYCES, TRUEPERELLA & NOCARDIA

ACTINOMYCES

Important species: *Actinomyces bovis, A.pyogenes*

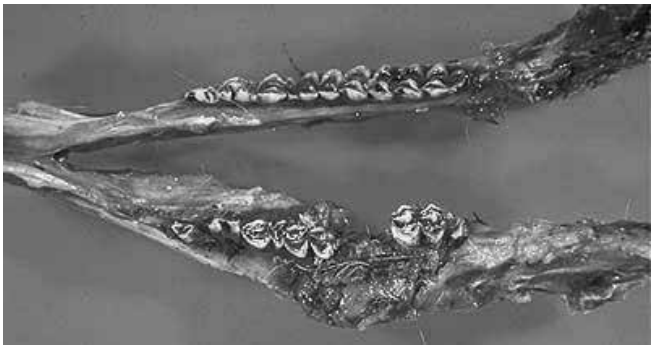
Important diseases: Lumpy jaw, Pyaemic infections

General characters:

1. **Morphology:** Members of the genus *Actinomyces, Nocardia* and *Dermatophilus* are referred as higher bacteria since they show extensive filamentation, aerial hyphae with asexual spores or conidia. The cultural and morphological characters are similar to fungi than bacteria. Members of the genus *Actinomyces* are gram-positive, non-acid-fast rod shaped organisms that show extensive branching. They are non-motile and non- spore forming. They are catalase negative and fermentative.
2. **Habitat:** These organisms are commonly found as commensal in the oral cavity of animals. They are obligate parasites in tonsils and carious teeth. These organisms have an affinity to hard tissues.
3. **Classification:** There are three important species *A.bovis* affects cattle and produce lumpy jaw. A.israeli produces the same type of infection in human. *A.bovis* has been moved from the genus *Corynebacterium* and it causes pyaemic infections.
4. **Resistance:** They are fragile organisms and are killed at a temperature of 60°C for 20 minutes.
5. **Antigens and toxins:** They are antigenically homogenous.

Cultural characters: They grow better at anaerobic environment at 5% carbon dioxide tension. The media used for culturing are agar enriched with glucose, serum or blood, Dorset egg medium, Loeffler's serum, brain heart infusion agar, thioglycollate broth, heart broth or glucose broth. The optimum temperature for growth is 37C. They are slow growing organisms and growth is noticed after 4-6 days of incubation. The colonies are 1 mm in diameter, round, flat and pale yellow. The colonies do not adhere to the medium. In fluid growth occurs as small granules in the bottom of the tube.

Pathogenesis: These organisms are commonly found as commensal in the oral cavity of animals. However, they are obligate parasites in crypts of tonsils and carious teeth. These organisms have an affinity to hard tissues. The common areas of infection are cheek muscles, lower jaw, submaxillary region and other areas of throat and head. In pigs, the organisms are found in the udder and suckling piglets get the infection through abrasions found in their mouth. In horses, the organism is associated with fistulous withers and poll evil.



Symptoms: In lumpy jaw in cattle, there is suppurating osteitis of cheek and mandibular region associated with swelling that affects chewing. In pigs, the mammary glands become soft and excrete pus. In horses, abscesses are seen in the fistula and poll region. Lumpy jaw produces lumps on the upper and lower jawbones, hence the common name 'lumpy jaw'. These lumps are immovable hard swellings of the bones, usually at the level of the central molar teeth. The swellings develop slowly and may take months to reach the size of a tennis ball. They consist of honeycombed masses of thin bone filled with yellow pus. If neglected the swellings may become very large. In advanced cases, sinuses or openings develop and discharge small amounts of sticky pus containing gritty yellow granules. Unlike the case with wooden tongue, the local lymph nodes do not become involved. Lumpy jaw may be well advanced before external signs are visible. Difficulty in breathing due to involvement of the nasal bones may be the first sign. As the disease progresses, chewing becomes more difficult and painful,

resulting in loss of condition. Occasionally, the soft tissues of the head and alimentary tract can be involved. Lesions in the alimentary tract give vague symptoms of indigestion, often with chronic bloat.

Lesions: There is suppuration followed by granulation, erosion of bone tissues and formation of new bone tissues. Pockets of pus seen in this area increase in size become fistula discharging the pus. The pus containing small granules referred as sulphur granules. Secondary infection of lung and digestive tract results as a result of swallowing the pus. Microscopically a central area of necrosis surrounded by granulation tissue that is enclosed in fibrous tissue is seen. Lumpy jaw is usually progressive. As the bony swellings continue to enlarge, gross disfiguration of the head can occur, much condition will be lost and death may result.

Diagnosis:

1. Based on **symptoms and lesions** and differential diagnosis with Actinobacillosis that affect soft tissue.
2. **Microscopical examination** of granules found in the pus. The pus granules are collected in a petridish and later transferred to a drop of 10%NaOH on a glass slide, crushed with help of a cover slip and stained by Gram's method. The organisms are seen as a central mass of filaments that show branching. The periphery of the filamentous mass is covered by a circle of club shaped bodies arranged radially with narrow ends pointing towards centre of filamentous mass (hence, this arrangement is known as ray fungus). The club shaped bodies stain Gram-negative.
3. **Isolation of the organisms by culturing**: The media used for culturing are agar enriched with glucose, serum or blood, Dorset egg medium, Loeffler's serum, brain heart infusion agar, thioglycollate broth, heart broth or glucose broth. The optimum temperature for growth is 37C. They are slow growing organisms and growth is noticed after 4-6 days of incubation. The colonies are 1 mm in diameter, round, flat and pale yellow. The colonies do not adhere to the medium. In fluid growth occurs as small granules in the bottom of the tube.

Treatment: Treatment of **lumpy jaw** is often ineffective. If the disease is detected early, it may be better to dispose of the animal while it is still in good condition. The head is usually condemned by meat inspectors, unless the lesions have spread elsewhere in the body. Repeated doses of penicillin coupled with sulphonamides and streptomycin give good results.

Actinomyces viscosus: It causes actinomycosis of dogs. The infection in dogs is characterised by granulomatous infection of skin and thoracic tissues.

Trueperella

Important species: *Trueperella pyogenes* (previous name: *Arcanobacterium pyogenes*)

T. pyogenes: It is a commensal found on the mucous membrane of nasopharynx of cattle, sheep and pigs. It is also found in the udder. It produces a exotoxin. It is an important pyogenic organism for cattle, sheep and pigs. It causes chronic abscessing mastitis in cattle (summer mastitis), suppurative pneumonia, nephritis, endometritis, pyometra, wound contamination and seminal vesiculitis in bulls.

Summer mastitis: Summer mastitis is produced by *T. pyogenes*. It is an acute illness of dry cows and heifers, which causes extensive and painful damage to the udder. The infected quarter is permanently damaged and results in early culling of the cow. Infections are more likely to occur when cows are in an environment where teats can be damaged and fly populations are high. The highest incidence is seen over the summer period (July, August and September), especially if warm and damp where the fly population is high usually close to wooded pastures. Its antibiotic treatment is difficult. In many cases the problem is not rapidly detected because the animals are in the fields and the farmer fails to

notice the condition. These bacteria may gain entrance through the wounds caused by the flies, but more often they are commensals in udder.

Symptoms: The exotoxins produced by the organisms are absorbed into the bloodstream making the cow very ill. Affected animals have a high temperature, a swollen udder and stiffness due to joint involvement and friction of the swollen quarter against the side of the leg. When the contents of the teat are drawn they may vary from cheese-like to whitish yellow, to thick reddish and dark clots and a typical rotten smell. The animal may be depressed, the udder may be cold, clammy, secretions are scanty and difficult to drain. Close examination of the teats, reveals a mass of tiny red fly bites. This damage allows the common mastitis germs, usually streptococci, staphylococci and *Peptostreptococcus indolicus* to multiply and grow in the udder.

Treatment & Prevention: In most cases when summer mastitis is detected very late. The first requirement is to remove as much as possible of the toxins in the infected quarter. This is achieved by bathing with hot water, firm massaging of the upper quarter to break pockets of pus and finally stripping or drawing every few hours. This procedure prevents the absorption of toxic materials into the animals system and clears the way for the natural healing process to commence and for intramammary tubes to gain access to the udder tissue. The infection may be so severe that the teat may have to be cut off to allow drainage. Further means to prevention of summer mastitis to the herd are application of teat dips to dry cows every few days, spaying fly repellent on the udders of suspected animals, inspect the udders of dry cows at least once a day, keep all dry cows with the milkers in open fields and bring them in with the milkers twice daily.

NOCARDIA

Importantspecies : *Nocardia asteroids, N.caviae, N.farcinica(??)*

Importantdisease : Bovine farcy,Nocardiosis

These organisms are long filamentous branching with terminal spores. They are gram-positive and some are acid-fast. They are found as saprophytes and few are pathogenic producing farcy characterised by infection of lymphatic vessels. The vessels and glands become enlarged, corded and nodular due to slow formation of caseous material. The exact pathogenesis is not clear. They are also isolated from bovine mastitis.

MYCOBACTERIUM

Introduction: *Mycobacterium tuberculosis* was the cause of the "White Plague" of the 17th and 18th centuries in Europe. During this period nearly 100 per cent of the European population was infected with tubeculosis and 25 percent of all adult deaths were caused by tuberculosis.

1. ***Mycobacterium tuberculosis*** is the etiologic agent of **tuberculosis (TB)** in humans. Humans are the only reservoir for the bacterium.
2. ***Mycobacterium bovis*** is the etiologic agent of TB in cows and rarely in humans. Both cows and humans can serve as reservoirs. Humans can also be infected by the consumption of unpasteurized milk. This route of transmission can lead to the development of **extrapulmonary TB,** exemplified in history by bone infections that led to hunched backs.
3. Other human pathogens belonging to the *Mycobacterium* genus include ***Mycobacterium avium,*** which causes a TB-like disease especially prevalent in AIDS patients, and ***Mycobacterium leprae,*** the causative agent of **leprosy**.

General characters of *M.tuberculosis, M.bovis and M.avium* :

1. **Morphology:** Members of the genus *Mycobacteria* are Gram-positive, non-branching, acid-fast,

slender and small rods. They are non-motile, non-sporulative and do not produce any toxins.They are aerobic organisms. These organisms are known as acid-fast or acid-alcohol- fast organisms since they with stand decolourisation with acid or acid-alcohol.

2. **Cell wall structure:** Cell wall constituents of mycobacteria: The cell wall constituents of mycobacteria are highly complex. They have a high concentration of lipid 20-40%, which protects the bacteria from destructive actions of disinfectants, strong acids, alkalis and antibodies. Three different types of lipids are commonly found in the cell wall. They are **mycolic acid, mycosides and glycolipids.** Mycolic acids are b-hydroxy fatty acids that are responsible for acid fastness. Acid fastness is the ability of the organisms to retain the carbol fuchsin after application of decolouriser acid-alcohol. Mycosides are substances that are responsible for the controlling the permeability of the organisms to water, enzymes, disinfectants and antibiotics. These substances are always found in association with Wax-D that in association with proteins induce delayed hypersensitivity. Various virulence factors are found to be mycosides. Glycolipids produce toxicity, granulomatous response and enhance the survival of mycobacteria inside the macrophages. Cord factor, one of the glycolipids is responsible for characteristic cultural characters.

3. **Resistance:** They are generally killed at a temperature of 60°C for 15 minutes and comparatively resistant to the lethal action of disinfectants than other organisms. They can remain viable for long periods in the darkness, putrefying material (1-4 years), faeces (150 days), sputum etc. However, they are highly susceptible to ionic detergents and to sunlight. Freezing temperature has little effect on them. Phenol (5%), Lysol (3%), cresols, formaldehyde (3-8%), alkaline gluteraldehyde (2%) and cresylic acid are fairly effective. Sodium hypochlorite at a concentration of 1:2000 or 1:1000, phenol soap mixtures and other phenol derivatives are effective disinfectants.

4. **Antigenic structure:** A strong relationship exists between different species of mycobacteria. *M.tuberculosis* and *M.bovis* are very closely related. The murine type is also very closely related to human and bovine types. Avian type share few antigens with human type. The avian type is very closely related with *M.intracellulare* (Battey bacilli) and is often indistinguishable. Hence, it is frequently referred as *M.avium-intracellulare* complex.

Classification:

1. **Classic mycobacteria** or **classic species -** they produce disease in animals and man. They are
   - *Mycobacterium tuberculosis*
   - *M.bovis*
   - *M.avium*
   - *M.paratuberculosis*
   - *M.leprae*

2. **Anonymous, unclassified** *or* **atypical Mycobacteria:** The classification of these group of organisms is based on rate of growth, colony morphology and pigment production. This classification was proposed by **Runyon (1959).** He classified atypical mycobacteria into four groups. These groups are further subdivided into number of subgroups based on biochemical reactions and cultural characteristics.

   a. **Group I. Photochromogenic:** Produce yellow pigmented colonies that appear 7 days or later after exposure to light and slow growing (*M.kansasii, M.marinum, M.asiaticum, M.simiae)*

   b. **Group II. Scotochromogenic:** Produce yellow or orange pigments in the absence of light and slow growing *M.gordonae, M.scrofulaceum, M.szulgai, M.xenopi*

   c. **Group III. Nonphotochromogenic:** Produce no or slight pigment after exposure to light and are slow growing *M.avium, M.intracellulare, M.terrae, M.ulcerans*

d. Group IV: Variable pigmentation and fast growing. Growth appears in less than seven days *M.phlei, M.smegmatis, M.fortuitum, and M.chelonei.*

Cultural characters:

1. ***M.tuberculosis and M.bovis:*** These organisms are aerobic and grow at a temperature of 37-38°C except *M.avium* that grows at 40°C and *M.piscium* that grows at 25°C. Growth does not occur in ordinary laboratory media but occurs in media that contains egg and serum after 10-14 days. The common media that contain egg and serum – **Dorset egg medium** is commonly used in cultivation of mycobacteria. The generation time is 18 hours and primary cultures are difficult to grow. The addition of glycerine to egg medium (**Lowenstein-Jensen medium)** enhances the growth of human strains whereas it has no effect or inhibit the growth of bovine strains. The relatively luxuriant growth of human strains in the presence of glycerine is termed as **eugoinc** and less luxuriant growth of bovine strains are termed as **dysgonic. Stonebrink's medium** contains inspissated egg with malachite green to inhibit the growth of contaminants and sodium pyruvate to enhance the growth of bovine strains is commonly used to cultivate bovine strains. In the fluid media these organisms form tenacious friable bacterial masses instead of homogenous suspension since they strongly adhere to each other. The fluid media that are generally used contain casein hydrolysate, bovine serum albumen, asparagin and various salts.The growth of *M.tuberculosis* is dry, crumbly and luxuriant. The colonies are yellow in colour with rough surface. Whereas the growth of *M.bovis* is dry, sparse, delicate and non-luxuriant. Chains of cells in smears made from in vitro-grown colonies often form distinctive **serpentine cords**. This observation was first made by Robert Koch who associated **cord factor** with virulent strains of the bacterium
2. ***M. avium:*** The optimal temperature for growth of avian strains is 40°C. However, growth will also take place at 42-43°C. Avian strains are fast growing and produce colonies that contain yellow or slightly pink pigments. They grow well in Dorset egg medium with enhanced growth in the presence of glycerine. The growth of *M.avium* is moist, slimy, glistening and luxuriant.

Pathogenesis: Host affected

1. ***M.bovis:*** It mainly affects cattle. Swine are readily and severely infected. Dogs, horses, sheep are also affected. Cats are susceptible and may perpetuate the infection. In cattle pulmonary tuberculosis with involvement of lymph nodes is common. In human beings the infection is mainly due to consumption of milk. Chickens are resistant. Laboratory animals like rabbits, guinea pigs and mice are susceptible to infection.
2. ***M.avium:*** It mainly affects chicken. Cattle are sensitised. Infection in horses, dogs and cats is rare. Rabbits and guinea pigs are susceptible.
3. ***M.tuberculosis:*** Mainly affects human beings and primates. Cattle are sensitised. Pigs contract the infection by eating squid from infected source. Dogs are infected, whereas cats are resistant. Chickens are rarely infected. Guinea pigs and mice are susceptible.

The infection is generally exogenous and the most important modes of infection are by ingestion and inhalation. Less frequently the organism also enters through skin and congenital (via placenta). Factors like crowding of animals in stalls, genetic factors like race and breed and resistance to tuberculosis both acquired and natural as a result of previous infection play a major role in the pathogenesis. Other factors like husbandry, hygiene, environment and ability of the organisms to withstand infection also contribute significantly in pathogenesis. *M.bovis* spread via respiratory discharges, faeces, milk, urine, semen and genital discharge. *M. tuberculosis* spread via sputum, fomites and respiratory discharges. *M.avium* spread via faeces. The manifestations of the disease are associated with the local lymphnode (mescentric in case of ingestion and tracheobronchal in case of inhalation). From the lymphnode the organisms reach thoracic duct and disseminated. At the visceral organs they produce tubercle nodules. In acute form, there is general dissemination where tubercle nodules are found throughout (miliary tuberculosis). A progressive cellular

granulomatous type of lesion with concentration of polymorphs is the first reaction, which is later, replaced by mononuclear cells with number of giant cells. These cells are separated from surrounding cells by macrophages, lymphocytes, plasma cells and fibrous tissue. As the size of the lesion increases, necrosis and caseation develops centrally that discharges caseous material containing bacilli. This caseous material is source of infection for other sites. The lesion later becomes cavity and calcification starts. Alternatively, there is also an exudative form that is characterised by exudate, which contains neutrophils, eosinophils, macrophages and fibrin. The tubercle lesions of M.bovis are commonly found in lymph nodes of lungs, intestinal wall, mescentric lymph nodes, liver and spleen and lungs. Lesions of *M.tuberculosis* are found in lungs and associated lymph nodes whereas the lesions of *M.avium* are found in digestive tract, liver, spleen and bone marrow. Lung lesions are rare in *M.avium* infection.

Symptoms: The symptoms depend upon the organs involved and the extent of lesion. In pulmonary TB, dry cough that increases in frequency accompanied by weight loss are the important symptoms. Generally symptoms of tuberculosis in visceral organs are not very specific. Tubercular mastitis is one of the common conditions in *M.bovis infection.* In poultry tuberculosis is a disease of digestive tract, the symptoms include severe emaciation, pale mucous membranes and intermittent diarrhoea.

Lesions: Tuberculous nodules are seen in lungs in case of pulmonary form and in other visceral organs and associated lymph nodes in generalised infection. The nodules contain thick caseating mass, which tend to calcify. Miliary lesions resembling seeds of millet seeds develop when there is distribution of bacilli via blood stream. In pigs the skeleton especially vertebrae and long bones are common sites for tuberculosis.

Diagnosis:

1. **Microscopical examination:** Sputa, milk, uterine discharges, pleural and peritoneal fluids, urine or faeces are the ideal material for demonstration of organisms under this method. Samples of milk, uterine discharges, and pleural and peritoneal fluids should be centrifuged and the smear should be prepared from the sediment. An initial volume of 20-25 ml is essential. The smears are stained by Ziehl-Neelsen method. The tubercle bacilli appear in clumps as slender rods stained pink in a blue background. In milk samples, the organisms are commonly seen in epitheloid cells.
2. **Based on symptoms and lesions:** The symptoms are generally inconclusive, whereas the lesions have to differentiate from other bacterial infections where caseation is common.
3. **Cultural examination:** Demonstration of acid fast tubercle bacilli in the tubercle nodule is one confirmative method for TB diagnosis. The clinical materials have to be processed in the following method before attempting for isolation.
    a. The fat has to be trimmed off and the clinical material should be treated with 10ml 4% NaOH containing phenol red,
    b. Then the tissue should be crushed in a sterile mortar and pestle with sterile sand.
    c. The sediment should be neutralised with 2N HCl for a maximum of 30 minutes.
    d. The sediment should be centrifuged at low speed for 20 minutes and decant the supernatant.
    e. The sediment should be inoculated into media like Lowenstein- Jensen medium or Stonebrink medium or egg yolk agar. The cultures should be kept for a period of atleast 8 weeks before discarding them as negative.

Alternative methods are to treat the samples with 5% oxalic acid,10%trisodium phosphate, 4% caustic soda or 20% antiformin.

Colonial morphology:

a. ***M.tuberculosis and M.bovis:*** These organisms are aerobic and grow at a temperature of 37-38°C

except *M.avium* that grows at 40°C and *M.piscium* that grows at 25°C. Growth does not occur in ordinary laboratory media but occurs in media that contains egg and serum after 10-14 days. The common media that contain egg and serum – **Dorset egg medium** is commonly used in cultivation of mycobacteria. The generation time is 18 hours and primary cultures are difficult to grow. The addition of glycerine to egg medium (**Lowenstein-Jensen medium)** enhances the growth of human strains whereas it has no effect or inhibit the growth of bovine strains. The relatively luxuriant growth of human strains in the presence of glycerine is termed as **eugoinc** and less luxuriant growth of bovine strains are termed as **dysgonic. Stonebrink's medium** contains inspissated egg with malachite green to inhibit the growth of contaminants and sodium pyruvate to enhance the growth of bovine strains is commonly used to cultivate bovine strains. In the fluid media these organisms form tenacious friable bacterial masses instead of homogenous suspension since they strongly adhere to each other. The fluid media that are generally used contain casein hydrolysate, bovine serum albumen, asparagin and various salts. The growth of *M.tuberculosis* is dry, crumbly and luxuriant. The colonies are yellow in colour with rough surface. Whereas the growth of *M.bovis* is dry, sparse, delicate and non-luxuriant.

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4. **Animal inoculation:** Guinea pigs are the laboratory animals of choice. Rabbits and chicken can also be used. Suspected materials are inoculated into the thighs of guinea pig. Two animals are inoculated for each sample. The animals are killed after 4-6 weeks and examined for typical lesions in the liver, spleen and lymphnodes. Rabbits are used to differentiate human, bovine and avian types. Rabbits generally die from generalised infection with bovine type 4-5 weeks after intravenous injection, but survive similar infection with human type, Avian type is less virulent for rabbits and produce chronic form of the disease unless the inoculum given in large dose. It does not produce macroscopically visible tubercles. However, the spleen and liver become enlarged containing tubercle bacilli. This type of infection is referred asYersin type of infection.

5. **Polymerase chainreaction**

6. **DNA probe** based diagnosis

Treatment: Treatment is not generally feasible in animals. One of the common drugs is isoniazid. In humans para-aminosalicylic acid, ethambutol combined with streptomycin constitute the triple therapy.

Control and prevention:

Tuberculin testing: In the field, tuberculosis is mainly identified by tuberculin testing. The basis of the test is that animals infected with tuberculosis develop a hypersensitivity to *M.tuberculosis*, so when an extract of the *M.tuberculosis* called tuberculin is injected intradermally into the skin, a localised hypersensitive reaction occurs at site of injection. This reaction is not produced in health animals. Two types of tuberculins are commonly used – old tuberculin (OT) and protein purified derivative (PPD). OT was prepared by evaporating a glycerinated broth culture to one-tenth of its volume. This preparation has got only little active principle besides various non-specific proteins. PPD is a pure form, where the active principle is precipitated by trichloroacetic acid and some other precipitant. Tuberculin is prepared from human, bovine and avian types. Different types of tests are used to test animals using tuberculin.

1. **Intradermal test:** In this method 0.1 ml of tuberculin is injected at caudal fold, vulvar lip or sides of neck. A firm swelling after 72 hours indicates positive reaction. Thickness of swelling 3mm is

doubtful and 4mm is positive.

2. **Comparative cervical:** Intradermal inoculation of regular and avian tuberculin and various sites on the sides of neck and results are read after 72hours.
3. **Ophthalmic:** Used mostly for primates. 0.1 ml of 1:10 dilution of bovine tuberculin is injected intradermally into the upper eyelid.

Vaccination: Though a number of vaccines were tried, the best vaccine is a live attenuated vaccine known BCG (*Bacille Calmette Guerin)*. Animals vaccinated with this vaccine give positive reaction to tuberculin testing and the immunity offered by this vaccine is also questionable. Calfhood vaccination is commonly practiced.

Control:

1. The control methods are aimed at identifying the reactors and eliminating them from the herd. For this tuberculin testing is routinely employed in the herd. Comparative tuberculin testing is used when the results of single intradermal test is questionable. Calfhood vaccination is also advisable.
2. The animal sheds containing infected material are sterilised by first washing them with water under pressure and them with disinfectant under pressure. The two common disinfectants used are cresylic compound and sodium orthophenylphenate.
3. Since it is a zoonotic disease and one of the common mode of spread in humans is by ingestion of milk from affected animals, raw milk should never be consumed and only pasteurised milk should be consumed.
4. The infection may also spread to humans from affected pet animals. Hence such animals should not be maintained in close proximity.
5. Pigs can get human type infection by consuming squid. Hence if pigs are fed with squid the source should be clean.

   During meat inspection the affected carcass should be condemned and should not be fed to carnivore animals.

PARATUBERCULOSIS

Organism – *Mycobacterium avium sub sp. paratuberculosis (M.johnei)* Johne's disease (pronounced "yo-knees") is a contagious, chronic and usually fatal infection that affects primarily the small intestine of ruminants. All ruminants are susceptible to Johne's disease. Johne's disease is caused by *Mycobacterium paratuberculosis,* a hardy bacteria related to the agents of leprosy and tuberculosis. The disease is worldwide in distribution. Johne's disease has been diagnosed in cattle, goats, sheep, deer, llamas, camels and other wild ruminants. It is also speculated that *M. paratuberculosis* is also the causative agent for Crohne's disease in humans. This disease is characterized by ulceration of the intestine causing extreme pain and debilitation. However, to date this research has not been confirmed and the organism has not been cultured from human cases.

Morphology and general characters: It is a gram-positive, rod shaped organism with rounded ends. The shape changes in artificial media into club form. It is a non-motile, non-sporulative and non-capsulated organism. It is strongly acid-fast. The cell wall constituents and resistance of *M.paratuberculosis* resemble other Mycobacteria. The organisms can remain viable for a long time in dried faeces in the pasture. They are resistant to disinfectants and to drying by the sun. Snow cover will help their survival whereas sunlight, drying or exposure to intense cold will reduce their numbers.

Strains and antigenic relationship: This organism has got a close antigenic relationship with avian tubercle bacilli. It has been found that animals with paratuberculosis often react positively to

avian tuberculin. There are three major strains of this bacterium (which is related but different from *Mycobacterium avium* subsp. *avium*, a disease of birds and rarely humans [*avium tuberculosis*]). There is a cattle strain, a sheep strain (Islandic strain) and an intermediate strain (yellow or orange pigment producing strain). While cattle are susceptible to all three strains, they are usually not infected with the sheep strain. Sheep usually get only the sheep strain but can also succumb to the intermediate strain. Goats usually have the cattle strain.

Cultural characters: They are aerobic organisms and are obligatory parasites. They are difficult to cultivate in ordinary laboratory media. However, the organisms will grow slowly in glycerine egg media containing extracts from other killed acid-fast bacteria. The extract of *M.phlei* (called Mycobactin) is commonly used in the media used for cultivation of *M.paratuberculosis*. The optimum temperature for growth is 38-39°C. However, growth is reported from 20-40°C. Initially the colonies appear as minute, greyish white, friable and irregular. Later, these colonies turn into pale yellow in colour and become large. Subcultures from primary cultures grow better.

Pathogenesis: The main mode of infection is by ingestion of the organism from contaminated environment. Young animals are more susceptible. The most common mode of infection for young animals is by ingestion of milk contaminated with faeces from animals having paratuberculosis. The infection is characterised by long incubation period (ranging from 12 months to many years). The pathogenesis of the infection is correlated with high amount of intracellular iron. Huge consumption of iron containing feed is also correlated with the severity of the lesion and shedding of bacteria. Cattle and sheep are mostly affected. Pigs are rarely affected. Mycobacteria are taken up by specialized cells (M cells) in the ileum.M cells present the micro be to macro phages and lymphocytes in Peyer's patches. The immune system reacts to the invasion by recruiting more macrophages and lymphocytes to the site of infection. Lymphocytes release a variety of cytokines, in an attempt to increase the bacterial killing power of the macrophages. Macrophages fuse forming large cells in an apparent attempt to kill the mycobacteria. Infiltration of infected tissues with millions of lymphocytes and macrophages leads to visible thickening of the intestine. Late in the infection, antibody production by the animal occurs, but this does not affect control of mycobacterium's multiplication.

Sheep become infected through eating feed contaminated with faecal material. Infected sheep on pasture will contaminate that pasture for other grazing animals. Lambs born into these contaminated environments are most susceptible to infection. Dirty udders and wool tags, barn, yards, bedding, etc are all sources of infection to lamb. The bacteria have a very thick cell wall and can survive in the environment for perhaps as long as a year. The bacteria invade the intestine and the intestinal lymph nodes where they become established and interfere with absorption of nutrients, hence weight loss occurs. This progression, however, takes two to seven years to go to completion, depending on the number of bacteria that infect the sheep and the age at which the sheep is infected. When they are first infected, sheep do not shed the bacteria. This period of incubation may be as short as a year or as long as 5 yrs but is often 2 to 3 yrs. Although Johne's disease is primarily a disease of the intestine, during the advanced stages of disease the agent may spread throughout the body. Foetuses are very susceptible to infection while still in the womb. Lambs may be born infected (in cattle, up to 60% of calves born to cows in advanced disease, were infected) and these animals may develop clinical disease at an earlier age.

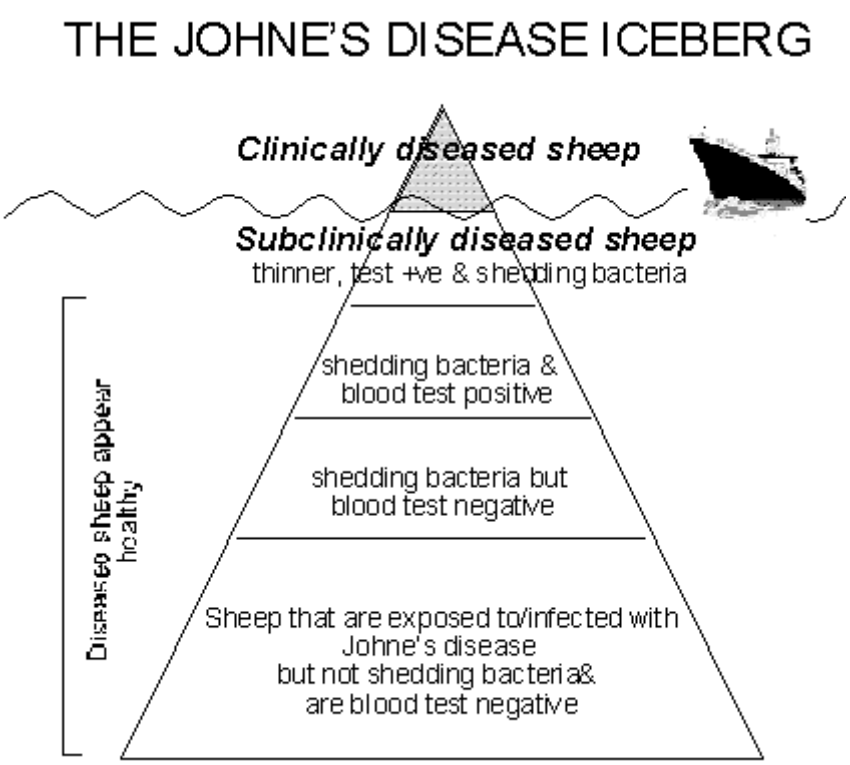
Symptoms:

Cattle: It is a slow progressive infection. Hence, signs of Johne's disease are not seen until years after initial infection. Cattle may be infected for years before they show any signs of disease. The signs of Johne's disease are intermittent bouts with diarrhoea eventually becoming chronic diarrhoea and weight loss (wasting – cachexia) despite a good appetite. Affected cattle do not generally have a fever.

Some infected animals appear unthrifty and often weak while others just have chronic diarrhoea. The signs of this disease can easily be confused with several other diseases. In the infected cow or heifer, noticeable signs commonly start within a few weeks after a stressful event like calving. In the sub-clinical stages cattle are more susceptible to other infectious diseases. These animals also may suffer from inefficient utilization of nutrients and drop off in performance especially milk production. In cattle, the bacteria have been isolated from semen from bulls with advanced Johne's disease. Some cows may develop "bottle jaw" due to a low protein edema, or appear unthrifty overall. Clinical signs of the disease occur more commonly during the end stages of the infection, typically at three to six years of age.

Sheep: Sheep can become infected at any age. When infected as a lamb or foetus, then the disease is manifested at 18 months of age. Stress such as lambing or mastitis will hasten the onset of more severe clinical disease. The ewe is depressed and pale and very thin and about 20% of cases develop diarrhoea. Diarrhoea indicates that the sheep is nearing the end and will die soon. The infection is also mistaken for other wasting type diseases like bad teeth due to incisor loss (broken mouth) or uneven molar wear and gingivitis. Slow virus infections in sheep also cause wasting. Severe parasitism, intestinal abscesses due to caseous lymphadenitis can also cause wasting. Management problems like poor nutrition and competition for feed can also cause wasting. In sheep wasting associated with or without diarrhoea is seen.

Lesions: Lesions are seen only in the intestine and associated lymph nodes (mesenteric nodes). Lesions are more pronounced in terminal part of the intestine particularly in rectum and ileo-caecal valve. The intestinal lesions are characterised by thickening of the intestinal cell wall. Due to thickening, the intestine looks like a hosepipe. The thickening is due to infiltration of mucosa by plasma cells and endothelial cells. There is neither giant cell formation nor caseation and calcification of lesions that are characteristic of tuberculosis. In the initial stages, the intestinal wall is oedematous with haemorrhagic streaks. In chronic cases, the intestinal mucosa is wrinkled and corrugated resembling brain. The raised areas are warty whereas the depressed areas are smooth. The lymph nodes are oedematous, enlarged and spongy. In sheep infected with pigmented strains, orange discoloration of the mucosal surface is common. Caseation and calcification are sometimes noticed in sheep.



Diagnosis:

1. Based on symptoms and lesions.
2. **Microscopical examination:** Rectal pinch smears stained by Ziehl- Neelsen method reveal acid-fast organisms that appear as pink in colour, single or in characteristic clumps. However, to provide a confirmative diagnosis, multiple samples have to be taken from different areas in rectum. However, normally the first diagnosis is often made at necropsy. As mentioned before, the intestinal lymph nodes become enlarged and the intestinal wall may appear thickened. Signs in sheep are more subtle than in cattle, in which the large intestine becomes quite thickened and corrugated like cardboard. Bacteria can be stained and viewed by a microscope. Sometimes the bacteria can be seen in a faecal smear, particularly in animals that have reached the diarrhoea stage. To be diagnostic however, large numbers of bacteria must be seen and follow-up tests need to be done.
3. **Smears from biopsy material of lymph node or terminal ileum** stained by Ziehl-Neelsen method reveal acid-fast organisms that appear as pink in colour, single or in characteristic clumps.
4. **Cultural examination:** It is not very successful in sheep. In cattle and goats, the bacteria can be cultured about 60% of the time when they are present in the faeces. However, the bacteria are very slow growing and take 4 to 16 weeks to grow. This means that negative results take at least 4 months and then a negative result may only mean failure to grow or that the animal is still in the early stages of disease. The situation in sheep is even worse. Most of the diagnostic laboratories in North America and elsewhere are unable to grow the bacteria with any degree of success. Sensitivity of bacterial culture in sheep even when showing severe clinical disease is only 8% (8 animal is culture positive out of 100 shedding animals). Faeces are the material of choice for culturing. Faeces have to be treated with 20 per cent antiformin for twenty minutes or with malachite green and oxalic acid before culturing. Penicillin has to be added to the media. The organisms will grow slowly in glycerine egg media containing extracts from other killed acid-fast bacteria. The extract of *M.phlei* (called Mycobactin) is commonly used in the media used for cultivation of *M.paratuberculosis*. The optimum temperature for growth is 38-39°C. However, growth is reported from 20-40°C. Initially the colonies appear as minute, greyish white, friable and irregular. Later, the colonies turn into

pale yellow in colour and become large. Subcultures from primary cultures grow better.

5. **Serological tests** like AGID, ELISA, CFT, RIA, and FAT etc can be performed to diagnose Johne's disease. These tests have very less sensitivity.
6. **DNA probes** are used to rapidly identify *M.paratuberculosis* in faeces.
7. **Johnin test:** PPD prepared from *M.paratuberculosis* is used to screen animals for paratuberculosis. But this method is not reliable.
8. **Gamma-interferon assay**: It is done in cattle. In cattle, the sensitivity is thought to be between 70 and 94% with excellent specificity (almost 100%). This test proves to be a good test for cattle but evaluation is pending for sheep.

Control:

1. Removal of animals with chronic diarrhoea and wasting.
2. Faecal culture of animals (2 years and above) in a herd, once in six months and removal of animals positive for paratuberculosis.
3. Disinfection of the suspected premises with cresylic compounds.
4. Clean animal husbandry practices that include separate rearing of calves.
5. Ploughing infected grazing land will dilute the bacteria and help to kill them.
6. AI from bulls negative in Johnin test.
7. A live vaccine using avirulent *M.paratuberculosis* strains is used for calves and sheep. This vaccine given to calves at 4 months of age reduces incidence of paratuberculosis. This vaccine given along with an adjuvant that includes mineral oil, liquid paraffin and ground pumice stone.

PASTEURELLA

Taxonomic name: *Pasteurella multocida subsp. multocida* (Lehmann and Neumann 1899) Rosenbusch and Merchant 1939. [Source: NCBI]

The species *Pasteurella multocida* includes a heterogeneous group of Gram- negative bacteria that are inhabitants of the upper respiratory tract of many vertebrate hosts including birds, cattle, swine, cats, dogs and rodents. Members of this species are responsible for a number of infections that normally are secondary to colonization of the upper respiratory tract including avian cholera (in waterfowl, chickens and turkeys), respiratory disease and hemorrhagic septicemia in ruminants (cattle, sheep, goats and buffalo), atrophic rhinitis in pigs and snuffles/septicemia in rodents (mice & rabbits). These infections are primarily transmitted by the respiratory route and are associated with crowding and other stressors. *P. multocida* is also a rare cause of infection in humans that is normally associated with dog or cat bites or scratches.

General characters:

1. **Morphology:** Members of the genus *Pasteurella* gram-negative short ovoid shaped organisms. In laboratory media they also appear as long rods. Generally the shape of the organisms is referred as cocco-bacilli. They are non-motile and non-sporulative organisms. They possess a capsule that is composed of polysaccharide and hyaluronic acid. They are facultatively anaerobic organisms, fermentative and oxidase positive. These organisms are known for their bi-polar staining property in tissue smears stained by Leishman's stain or methylene blue.
2. **Habitat**: They are found as commensals in the upper respiratory and digestive tracts of animals. They are also found on mouth, tonsils and nasopharynx.

3. **Resistance:** They are not very resistant organisms. They are killed by chemical and agents like 0.5% phenol when exposed for 15 minutes. They are also killed on exposure to sunlight for 3-4 hours and heating at 55°C for 15 minutes.
4. **Antigens and toxins:** Based on differences in capsular substances (Carter's classification), *P.multocida* classified into A, B, D, E and F. These types are further subdivided into 16 subtypes based on differences in somatic antigens. These subtypes are assigned numerals. Thus a serotype is designated by an alphabet followed by a number eg. Serotype B2 causes HS. A toxin in addition to endotoxin (LPS) is produced by B serotype.

Cultural characters: They are aerobic or facultatively anaerobic organisms and grow at a temperature of 37°C. In ordinary media, three types of colonies – mucoid, smooth (iridescent) and non-iridescent are produced after overnight incubation. The mucoid colonies are round, flat, mucoid and sticky of 2-3 mm in diameter. Organisms with abundant capsular material produce these types of colonies. Organisms with less capsular material produce the iridescent colonies whereas rough colonies are produced by organisms that have no capsular material. Blood agar is preferred medium for growth. There is darkening of the medium with no haemolysis and a specific odour. In broth cultures they produce uniform turbidity.

Biochemical characters: HS organisms produce oxidase, catalase and indole, and will reduce nitrates. They do not produce hydrogen sulphide or urease, and fail to use citrate or liquefy gelatin. Glucose and sucrose are always fermented with the production of acid only. Most strains also ferment sorbitol. Some strains ferment arabinose, xylose and maltose, whereas salicin and lactose are almost invariably not fermented

HAEMORRHAGIC SEPTICAEMIA

Haemorrhagic septicemia (HS) is an acute pasteurellosis, caused by particular serotypes of *Pasteurella multocida* and manifested by an acute and highly fatal septicemia principally in cattle and water buffaloes; the latter are thought to be more susceptible than cattle. HS occurs infrequently in swine and even less commonly in sheep and goats. It has been reported in bison, camels, elephants, horses, and donkeys, and there is evidence of its occurrence in yak. Presently, two serotypes are recognised – the Asian serotype and the African serotype. The Asian strains belong to capsular type B only while the African strains are types B and E. Epidemic HS is caused by one of two serotypes of *P multocida*, designated B:2 and E:2.

Pathogenesis: Infection is exogenous and animals acquire the infection by contact, inhalation or ingestion. It is hypothesized that animals become susceptible as a result of various stresses, eg, the inanition seen in cattle and water buffalo at the beginning of the rainy season. The heaviest losses occur during the monsoon rains in south east Asia, and it is thought that the organisms, which can survive for hours and probably days in the moist soil and water, are transmitted widely at this time. Biting arthropods rarely transmits infection. Pasteurella organisms are found in the upper respiratory and digestive tract of animals and birds as commensals. Most of the animals carry these organisms with them without exhibiting any symptoms. The organisms produce infection when they are exposed to stress conditions like extreme weather conditions, transport, immunosuppression etc. During these conditions the organisms rapidly multiply and excreted via droplets and through digestive tract. The droplets with organisms are the main source of infection for other animals. In *Pasteurella* infection, the morbidity is always higher. *P.multocida* is also a secondary invader in respiratory tract infection. The important virulence factor is the endotoxin (LPS). Besides this a thermostable, cell associated toxin that is released by dying cells also plays a role in pathogenesis.

Symptoms: The clinical syndrome of HS consists of an initial phase of temperature elevation (often unnoticed), a phase of respiratory involvement, and a terminal phase of septicaemia and recumbency leading to death. The incubation period is usually 1–3 days, and the course of the disease may range from sudden death, with no observable clinical signs, to a protracted course extending up to 5 days. Buffaloes are generally believed to be more susceptible to HS than cattle, and in this species, the disease course is shorter. In endemic areas, most deaths are confined to older calves and young adults. In non-endemic areas, massive epizootics may occur. Case fatality approaches 100% if treatment is not carried out sufficiently early (in the pyrexic stage). Three forms of HS are noticed among animals – acute, sub-acute (oedematous form) and sub-acute (pectoral form). The important symptoms in acute cases are rise in body temperature, drop in milk yield, abdominal pain, severe diarrhoea and dysentery. The respiration becomes rapid before death and the mucous membrane becomes cyanotic. Less acute cases are characterised by rise in body temperature and oedema in the head, neck and brisket region. A purulent, blood stained nasal discharge is also noticed. Occasional cases linger for several days. Recovery is rare. There appears to be no chronic form.

Lesions: The most obvious changes in affected animals are the edema, widely distributed hemorrhages, and general hyperemia. In most cases, there is an edematous swelling of the head, neck, and brisket region. Incision of the swellings reveals a clear or straw-colored serous fluid. The edema is also found in the musculature, and the subserous petechial hemorrhages, which are found throughout the animal, are particularly characteristic. Blood-tinged fluid is often found in the pericardial sac and in the thoracic and abdominal cavities. The lesions are also coupled with gastroenteritis, marbled lungs, blood stained stools and enlarged and haemorrhagic mesenteric lymph nodes Petechial hemorrhages are seen in the pharyngeal and cervical lymph nodes. Gastroenteritis is seen only occasionally and, unlike pneumonic pasteurellosis, pneumonia usually is not extensive.

Diagnosis:

- Based on symptoms and lesions.
- **Microscopical examination:** The septicaemia in HS occurs at the terminal stage of the disease. Therefore, blood samples taken from sick animals before death may not always contain *P. multocida* organisms. Also, they are not consistently present in the nasal secretions of sick animals. A blood sample or swab collected from the heart with in few hours after death is most ideal. Blood smears from affected animals are stained with Gram, Leishman's or methylene blue stains. The organisms appear as Gram-negative, bipolar-staining short bacilli. No conclusive diagnosis can be made on the basis of direct microscopic examinations alone.
- **Isolation and identification:** Blood in transport medium is the most ideal material for isolation. If the animal has been dead for a long time, a long bone, free of tissue, can also be taken. The most ideal medium for isolation of Pasteurella is casein/sucrose/yeast (CSY) agar containing 5% blood (calf blood). Freshly isolated *P. multocida* forms smooth, greyish glistening translucent colonies, approximately 1 mm in diameter, on blood agar after 24 hours' incubation at 37°C. Colonies grown on CSY agar are larger. Old cultures, particularly those grown on media devoid of blood, may produce smaller colonies.
- **Test to confirm the production of hyaluronidase:** HS-causing strains of *P. multocida* have the ability to produce the enzyme hyaluronidase. A hyaluronic-acid-producing culture is streaked across the centre of a dextrose starch agar plate. The *Pasteurella* culture to be tested for hyaluronidase production is streaked at right angles. The plates are incubated at 37°C for 18 hours. At the point of intersection, the mucoid growth of the hyaluronic acid producer will diminish into a thin line of growth, indicating the production of hyaluronidase by the test culture.

- **Immunological methods:** Several immunological tests are used for the identification of the HS-causing serotypes of *P. multocida*. These consist of a rapid slide agglutination test, an IHA test for capsular typing, the AGID test, and the counter immunoelectrophoresis test (CIEP).
- **Nucleic acid recognition methods:** PCR amplification of specific DNA sequences allows rapid detection and presumptive identification of organisms directly from either clinical specimens or from small amounts of mixed or pure bacterial cultures. Some of the common methods performed to identify Pasteurella are as below
    i. Pasteurella-multocida-specific PCR assay
    ii. Pasteurella multocida multiplex capsular PCR typing system
    iii. HS-causing type-B-specific PCR assay
    iv. Genotypic differentiation of isolates
- **Serological methods:** Serological tests for detecting antibodies are not normally used for diagnosis.
- **Animal Inoculation:** The mouse usually serves as a biological 'screen' for extraneous organisms. A small volume (0.2 ml) of eluted blood swabs or a portion of bone marrow in saline is inoculated subcutaneously or intramuscularly into mice. If viable *P. multocida* is present, the mice die 24–36 hours following inoculation, and a pure growth of *P. multocida* can be seen in blood smears.

Treatment: Various sulfonamides, tetracyclines, penicillin, and chloramphenicol (are effective if administered early. Because of the rapid course of the disease and the frequent difficulty of access to animals, antimicrobial therapy often is not practicable.

Control: The principal means of prevention is by vaccination. Three kinds of vaccine are widely used: plain bacterin, alum-type precipitated bacterin, and oil-adjuvant bacterin. The most effective bacterin is the oil-adjuvant— one dose provides protection for 9-12 months; it should be administered annually. The alum-precipitated-type bacterin is given at 6-months intervals. A live vaccine prepared from a B:3,4 serotype of deer origin is being used with reported success in southeast Asia.

FOWL CHOLERA (AVIAN CHOLERA)

Fowl cholera (avian pasteurellosis) is a commonly occurring avian disease that can affect all types of birds and is often fatal. It is contagious and usually occurs as a septicemia of sudden onset with high morbidity and mortality, but chronic and asymptomatic infections also occur.

Pathogenesis: *Pasteurella multocida*, the causal agent, is a small, gram- negative and non-motile rod. Although *P multocida* may infect a wide variety of animals, strains isolated from non-avian hosts generally do not produce fowl cholera. Strains that cause fowl cholera represent a number of immunotypes, which complicates widespread prevention by using bacterins. The organism is susceptible to ordinary disinfectants, sunlight, drying, and heat. Turkeys are more susceptible than chickens, older chickens are more susceptible than young ones, and some breeds of chickens are more susceptible than others. Chronically infected birds are considered to be a major source of infection. Dissemination of *P multocida* within a flock is primarily by excretions from mouth, nose, and conjunctiva of diseased birds that contaminate their environment.

Symptoms and lesions: These vary greatly depending on the course of disease. In acute fowl cholera, dead birds are usually the first indication of disease. Fever, depression, anorexia, mucoid discharge from the mouth, ruffled feathers, diarrhoea, and increased respiratory rate are usually seen. Many of the lesions are related to vascular disturbances. Hyperemia isespecially evident in the vessels of the abdominal viscera. Petechial and ecchymotic hemorrhages are common, particularly

in subepicardial and subserosal locations. Increased amounts of peritoneal and pericardial fluids are frequently seen. Livers may be swollen and often develop multiple, small, necrotic foci. Pneumonia is particularly common in turkeys. In **chronic fowl cholera**, signs and lesions are generally related to localized infections. Sternal bursae, wattles, joints, tendon sheaths, and footpads are often swollen because of accumulated fibrinosuppurative exudate. There may be exudative conjunctivitis and pharyngitis. Torticollis may result when the meninges, middle ear, or cranial bones are infected.

Diagnosis

1. Based on the symptoms and lesions
2. **Microscopical examination**: The cells are coccobacillary or short rod-shaped, usually 0.2–0.4 by 0.6–2.5 μm in size, stain Gram negative, and generally occur singly or in pairs. Recently isolated organisms or those found in tissue smears show bipolar staining with Wright or Giemsa stains or methylene blue, and are usually encapsulated.
3. **Isolation and identification**: Isolation of the organism from visceral organs, such as liver, bone marrow, spleen, or heart blood of birds that succumb to the acute form of the disease, and from exudative lesions of birds with the chronic form of the disease, is generally easily accomplished. *Pasteurella multocida* is a facultative anaerobic bacterium that grows best at 35–37°C. Primary isolation is usually accomplished using media such as blood agar, trypticase–soy agar or dextrose starch agar, and isolation may be improved by supplementing these media with 5% heat-inactivated serum. Colonies range from 1to3 mm in diameter after 18–24 hours of incubation. They usually are discrete, circular, convex, translucent, and butyraceous. Capsulated organisms usually produce larger colonies than those of noncapsulated organisms. Watery mucoid colonies, often observed with mammalian respiratory tract isolates, are very rare with avian isolates.
4. **Nucleic acid identification methods**: The most ideal method is the DNA fingerprinting of *P. multocida* by restriction endonuclease analysis (REA).
5. **Serological tests:** Serological tests for the presence of specific antibodies are not used for diagnosis of fowl cholera.

Treatment: Sulfonamides and antibiotics are commonly used; early treatment and adequate dosages are important. High levels of tetracycline antibiotics in the feed (0.04%) or administered parenterally may be useful.

Control: Good management practices are essential to prevention. Adjuvant bacterins are widely used and generally effective; autogenous bacterins are recommended when polyvalent bacterins are found to be ineffective. Attenuated vaccines are available for administration in drinking water to turkeys and by wing-web inoculation to chickens. These live vaccines can effectively induce immunity against different serotypes of *P multocida*. They are recommended for use in healthy flocks only.

OTHER IMPORTANT PASTEURELLAINFECTIONS

- **Rabbits**: Peracute infection is common in rabbits. In chronic infection coryza like respiratory symptoms are common.
- **Sheep**: Mastitis (blue bag) and pneumonia
- **Dogs and cats**: Septic contamination of wounds
- **Pigs**: Respiratory infection and rhinitis. Atrophic rhinitis is an infectious disease of swine characterised by purulent nasal discharge, shortening or twisting of the snout, atrophy of the turbinate (conchal) bones and reduced productivity. It may occur enzootically or more sporadically, depending on a variety of factors including herd immunity. The most severe progressive form is caused by infection with toxigenic strains of Pasteurella multocida alone or in combination with *Bordetella bronchiseptica*.

Mannheimia haemolytica (P. heamolytica): Unlike *P.multocida,* it is beta haemolytic. Causes transport of **shipping fever** in cattle. It is a pneumonic condition with rise in temperature, rapid respiration followed by death with 12-48 hours. In less acute cases coughing, debility and death are noticed.

Riemeralla anatipestifer (P.anatipestifer): It is a non-fermenting bacterium. It causes infectious serositis (New duck disease) in ducklings. Anatipestifer infection causes high mortality, weight loss and condemnation. In the acute form, listlessness, eye discharge and diarrhoea are commonly seen. Ducks show incoordination, shaking of the head and twisted neck. Birds are commonly found on their backs, paddling their legs. Typical lesions found in dead birds are infected air sacs, membranes covering the heart and liver, and meningitis. Preventive management and vaccination are effective means of control. Penicillin, enrofloxacin and sulfadimethoxine-trimetoprim (0.04-0.08% in feed) are effective in reducing mortality.

BRUCELLA

Brucellosis is caused by bacteria of the genus *Brucella* and is characterized by abortion, retained placenta, and to a lesser extent, orchitis and infection of the accessory sex glands in males. Diagnosis depends on the isolation of Brucella from aborted materials, udder secretions or from tissues removed at post- mortem. Presumptive diagnosis can be made by assessing specific cell-mediated or serological responses to *Brucella* antigens. The disease is prevalent in most countries of the world. It primarily affects cattle, buffalo, bison, pigs, sheep, goats, dogs, elk, and occasionally horses. The disease in man, sometimes referred to as undulant fever, is a serious public health problem, especially when caused by *Bmelitensis*

Classification: All members of the *Brucella* genus are closely related. There are six classical nomenspecies: *Brucella abortus, B. melitensis, B. suis, B. neotomae,ovis* and *B. canis*. The first three of these are subdivided into biovars based on cultural and serological properties. Two species *B. cetaceae* and *B. pinnipediae* have been isolated from marine organisms. *Brucella* shows close genetic relatedness to some plant pathogens and symbionts of the genera *Agrobacterium* and *Rhizobium*, as well as, animal pathogens (*Bartonella*) and opportunistic or soil bacteria (*Ochrobactrum*).

Important diseases - Brucellosis, Bang's disease, Malta fever, Undulant Fever, Fistulous withers, Poll evil

General characters:

1. **Morphology:** *Brucella* are coccobacilli or short rods measuring from 0.6 to1.5 μm long and from 0.5 to 0.7 μm wide. They are usually arranged singly, and less frequently in pairs or small groups. The organisms appear in pleomorphic forms in old cultures. *Brucella* is non-motile. They do not form spores, and do not have flagella, pili, or true capsules. *Brucella* is Gram negative and usually does not show bipolar staining. Although they are not acid-fast organisms, they withstand decolourisation of 0.5 per cent acetic acid.
2. **Resistance:** The organisms are killed by heating them to 60°C for 10 minutes and in milk readily killed by pasteurisation. They are susceptible to acidic pH, disinfectants and sunlight. They can survive for a long time foetus that is not exposed to sunlight. They can remain for a long time in cold temperatures.
3. **Antigens and Toxins:** Brucella produce two types of colonies smooth and rough that are associated with virulence, auto-agglutination etc. Virulent organisms produce smooth colonies and avirulent organisms produce rough colonies. Certain organisms produce intermediate colonies. *B.abortus, B.melitensis* and *B.suis* produce two antigens called A and M. Amount of these antigens in these species varies widely. The A and M antigens occur in ration of 20:1 in *B.abortus* and still lower

in *B.suis* where as it occurs in the ratio of 1:20 *B.melitensis.* The LPS (endotoxin) is the important virulence factor. The organisms produce noexotoxins.

Cultural characters: They are aerobic organisms. However, certain organisms like *B.abortus* and *B.suis* require 5-10% CO_2 tension. They are slow growing organisms and growth is visible after 48 hours at a temperature of 37°C. Numbers of media are used for the cultivation of organisms like nutrient agar, serum agar, liver infusion serum agar, tryptose agar, and dextrose potato and glycerol potato agar. On nutrient agar the colonies are small, round, convex, translucent with a smooth glistening surface. In broth cultures they produce fine granular deposit. Other media used include serum–dextrose agar (SDA) or glycerol dextrose agar. A nonselective, biphasic medium, known as Castañeda's medium, is used for the isolation of *Brucella* from blood and other body fluids or milk.

Biochemical characters: They utilise carbohydrates but produce very little or no acid and gas. H_2S production is variable and this character is useful to differentiate brucella species. Production of urease by various species is also variable. Inhibition of growth in the presence of different concentration of dyes like thionin, basic fuchsin, methyl violet and pyronin is also useful character in differentiation of various brucella species. In general requirement of CO_2 for growth, hydrogen sulphide production, urease production and sensitivity to various dyes are the characters that are used to differentiate different Brucella species.

BRUCELLOSIS IN CATTLE

The disease in cattle, water buffalo, and bison is mostly caused by *Brucella abortus*. However, *B suis*or *B melitensis* occasionally cause infection in cattle.

Pathogenesis: Brucella infection in a herd is through introduction of infected animals. The main sources of infection are aborted foetuses, placentae and genital discharge, milk and semen of infected bulls. Natural transmission occurs by ingestion of organisms, which are present in large numbers in aborted fetus, fetal membranes, and uterine discharges. Cattle may ingest contaminated feed and water, or lick contaminated genitals of other animals. Venereal transmission by infected bulls to susceptible cows appears to be rare. Transmission may occur by artificial insemination when *Brucella* -contaminated semen is deposited in the uterus but, reportedly, not when deposited in the midcervix. Brucellae may enter the body through mucous membranes, conjunctivae, wounds, or even intact skin. Calves are relatively insusceptible and excrete the organisms in faeces and serve as source of infection for other animals. *B.abortus* organisms are excreted in the colostrum and milk after parturition for many months up to 2 years. Mechanical vectors (eg, other animals, including man) may spread infection. The organisms pass from point of entry to the lymphatics to lymph node and multiply in thoracic duct. From there it reaches the paranchymatous organs and other tissues. They have the capacity to survive inside phagocytic cells. In the paranchmatous organs, they develop granulomatous lesions. They have an affinity for gravid uterus and foetal tissues. The affinity of brucella organisms for foetal tissue is due to **erythritol** – a saccharide alcohol that is present in placenta of cattle, sheep, goat and pigs. In male animals the organisms multiply in testicles, seminal vesicles and epididymis.

Symptoms: Abortion during last stages of pregnancy is the important symptom (seventh or eighth month of pregnancy in cattle, third or fourth month in sheep, 50th day in dogs). In cattle abortion is associated with retention of placenta. Infections may also cause stillborn or weak calves, retained placenta, and reduced milk yield. Usually, general health is not impaired in uncomplicated abortions. Seminal vesicles, ampullae, testicles, and epididymides are infected in bulls. Testicular abscesses may occur. Long-standing infections may result in arthritic joints in some cattle.

Lesions: Necrotic placentitis, swollen, hyperaemic cotyledons surrounded by brown exudates, thickened intercotyledonary spaces with leathery appearance are the common lesions. Udder may look normal but cellular infiltration; degeneration and necrosis are also seen. Orchitis and necrosis of the testicular tissues surrounded by fibrous tissue are the common lesions in male animals.

BRUCELLOSIS IN GOATS

In goats *B.melitensis* is the causative agent. The disease is prevalent in most countries where goats are a significant part of the animal industry. The signs of brucellosis in goats are similar to those in cattle. *B.melitensis* organisms have a preference for udder tissue than gravid uterus. Infection occurs primarily through ingestion of the organisms. The disease causes abortion about the fourth month of pregnancy. Arthritis, epididymitis and orchitis also occur.

BRUCELLOSIS IN SHEEP

In sheep *B ovis* is the causative agent, in which epididymitis and orchitis are observed. Occasionally, placentitis and abortion are also seen.

Pathogenesis: The disease can be transmitted among rams by direct contact. Infection in ewes occurs after mating with naturally infected rams. Contaminated pastures do not appear to be important in spread of the disease. Infection frequently persists in rams, and a high percentage shed *B ovis* intermittently.

Symptoms & Lesions: In sheep apart from abortion, lameness, mastitis, pyrexia and diarrhoea are also seen. Lesions develop rapidly. In rams, the first detectable abnormality is the marked deterioration in semen quality associated with the presence of inflammatory cells and organisms. The epididymal enlargement is unilateral or bilateral. The most prominent lesion is spermatoceles of variable size containing partially inspissated spermatic fluid. The tunics frequently become thickened and fibrous, and extensive adhesions develop between them. The testes show fibrous atrophy. In few cases the organisms are present in semen over long periods without clinically detectable lesions.

BRUCELLOSIS IN HORSES

The causative agent is *B.abortus.* The organisms produce chronic inflammatory conditions referred as fistulous withers, poll evil and joint infections. Fistulous wither and poll evil are clinically same conditions occurring in supraspinous and supra-atlantal bursae respectively. The inflammation of bursae leads to considerable thickening of the wall and the supraspinous bursa distends with a clear, straw-colored, viscid exudates and rupture later. In more chronic, advanced cases, the ligament and the dorsal vertebral spines are also affected become necrosed.

BRUCELLOSIS IN PIGS

B.suis is the causative organism for abortion in swine. In pigs' abortion, birth of weak piglets, lameness, paralysis, affection of spinal column and infertility in male and female animals are the common symptoms.

BRUCELLOSIS IN DOGS

B. canis is the causative organism in dogs. In dogs' abortion, early embryonic deaths, dermatitis of scrotum and testicular atrophy are common symptoms.

BRUCELLOSIS IN HUMANS

Infection known as **Undulant fever (*B.abortus*) or Malta fever (*B.melitensis*)** in humans occurs

as a result of coming in contact with animals or animal products that are contaminated with these bacteria. Human brucellosis is more common in countries that do not have good standardized and effective public health and domestic animal health programs. Areas currently listed as high risk are the Mediterranean Basin (Portugal, Spain, Southern France, Italy, Greece, Turkey, North Africa), South and Central America, Eastern Europe, Asia, Africa, the Caribbean, and the Middle East. Unpasteurized cheeses, sometimes called "village cheeses," from these areas may represent a particular risk for tourists. Humans are generally infected in one of three ways: eating or drinking something that is contaminated with *Brucella*, breathing in the organism (inhalation), or having the bacteria enter the body through skin wounds. Direct person-to-person spread of brucellosis is extremely rare. Mothers who are breast-feeding may transmit the infection to their infants. Sexual transmission has also been reported. Veterinarians treating the animals are in risk group along with butchers and animal attendants. The symptoms of brucellosis are similar to the flu and include fever, sweats, headaches, back pains, and physical weakness. Severe infections of the central nervous systems or lining of the heart may occur. Brucellosis can also cause long- lasting or chronic symptoms that include recurrent fevers, joint pain, and fatigue.

Public Health Significance: The World Health Organization (WHO) laboratory biosafety manual classifies *Brucella* in Risk group III. Brucellosis is readily transmissible to humans, causing acute febrile illness – undulant fever – which may progress to a more chronic form and can also produce serious complications affecting the musculo–skeletal, cardiovascular, and central nervous systems. Infection is often due to occupational exposure and is essentially acquired by the oral, respiratory, or conjunctival routes, but ingestion of dairy products constitutes the main risk to the general public. There is an occupational risk to veterinarians and farmers who handle infected animals and aborted fetuses or placentae. Brucellosis is one of the most easily acquired laboratory infections, and strict safety precautions should be observed when handling cultures and heavily infected samples, such as products of abortion. Specific recommendations have been made for the safety precautions to be observed with *Brucella*- infected materials.

Diagnosis:

1. **Based on symptoms and lesion and history of abortion** in a herd.
2. **Microscopical examination:** Smears prepared from chorionic tissues, foetal stomach contents and uterine discharges stained by modification of Macchiavello's method or Koster's method will reveal brucella organisms. *Brucella* are Gram negative and usually do not show bipolar staining. They are not truly acid-fast, but are resistant to decolorisation by weak acids and thus stain red by **the Stamp's modification of the Ziehl–Neelsen method**. This is the usual procedure for the examination of smears of organs or biological fluids that have been previously fixed with heat or ethanol, and by this method, *Brucella* organisms stain red against a blue background. A fluorochrome or peroxidase-labelled antibody conjugate based technique is also used.
3. **Bacteriological examination:** For the diagnosis of animal brucellosis by cultural examination, the choice of samples usually depends on the clinical signs observed. The important samples are aborted fetus (stomach contents, spleen and lung), fetal membranes, vaginal secretions (swabs), milk, semen and arthritis or hygroma fluids. From animal carcasses, the preferred tissues for culture are those of the reticulo-endothelial system (i.e. head, mammary and genital lymph nodes and spleen), the late pregnant or early post-parturient uterus, and the udder. When milk products are checked for contamination, they have to be enriched since the milk products contain minimal number of organisms. **Albimi agar** is the medium of choice. Other media include nutrient agar, serum agar, liver infusion serum agar, tryptose agar, and dextrose potato and glycerol potato agar. The colonies are small, round, convex, and translucent with a smooth glistening surface. In broth

cultures they produce fine granular deposit. Growth normally appears after 2–3 days, but cultures should not be discarded as negative until 8–10 days have elapsed.

4. **Nucleic acid recognition methods:** Polymerase chain reaction is used to differentiate causative organisms from vaccine organisms.

5. **Animalinoculation:**
   a. Guinea pigs are the laboratory animals of choice for brucella identification. Clinical materials are inoculated intra muscularly or sub-cutaneously. Two animals are inoculated and one animal is killed after 3 week and another after 6 weeks. Typical lesions include miliary necrotic foci in spleen, liver and lymphatic glands.
   b. **Strauss reaction**: Development of orchitis after intraperitoneal injection of infective material.

6. **Serum agglutination tests:** These tests are performed to identify antibodies against *Brucella* in animals. Two types of tests a rapid plate test called **Rose Bengal plate test (RBPT),** which is a qualitative test and **serum agglutination test (SAT),** which is a quantitative test are performed to identify animals infected with brucella. In RBPT, the brucella antigen is stained with Rose Bengal dye. 0.3 ml of antigen and 0.3 ml of antibody are mixed and development of agglutination after four minutes is positive reaction. It is a rapid method. In SAT, the serum sample is diluted and brucella antigen without any dye is added. Development of 50% agglutination at dilutions 1:40 and above is considered as positive and 1:20 is considered as suspected. The test is of useful in unvaccinated cattle and in farms where calfhood vaccination is practiced.

7. **Milk ring test (MRT) or Abortus Bang's ring (ABR) test:** This test is performed to identify brucella in herds. It is based on detection of antibodies in milk against stained antigen. Milk samples from herd are mixed with stained antigen and incubated. Development of a ring above cream layer is a positive reaction.

8. ***Whey agglutination test:***

9. **Coombstest:**

10. **ELISA:** It is a prescribed tests for international trade.

11. **Complement fixation test:** It is also a prescribed test for international trade. The CFT is a widely used and accepted confirmatory test although it is complex to perform, requiring good laboratory facilities and adequately trained staff to accurately titrate and maintain the reagents. There are numerous variations of the CFT in use, but this test is most conveniently carried out in a microtitreformat.

12. **Fluorescence polarisation assay**: It is also a prescribed test for international trade. FPA is a simple technique for measuring antigen/antibody interaction and may be performed in a laboratory setting or in the field. It is a homogeneous assay in which analytes are not separated and it is therefore very rapid. For FPA antigen is labelled with a fluorochrome and added to serum or other fluid to be tested for the presence of antibody. If antibody is present, attachment to the labelled antigen will cause its rotational rate to decrease and this decrease can be measured.

13. **Brucellin skin test:** It is used for screening unvaccinated herds. A purified (free of sLPS) and standardised antigen preparation (e.g. brucellin) is used.

Control and Prevention:

1. Vaccination of calves with strain 19 (avirulent strain) at an age of 3-6 months offer high level of immunity. One dose gives immunity up to fifth pregnancy. Now RB51 strain is used instead of strain 19 vaccine.
2. Periodical testing of herds by SAT and elimination of reactors.
3. Pasteurisation of milk and milk products.

4. Clean animal husbandry procedures.
5. Proper disposal of aborted foetuses.
6. Elimination of male animals from breeding programme

ENTEROBACTERIACEAE

What is enteric bacterium? Enteric bacterium is one that is found in the intestinal tract of warm-blooded animals in health and disease.

The **Enterobacteriaceae** are a family of bacteria, including many familiar pathogens, such as *Salmonella* and *E. coli*. They are placed among the proteobacteria. The **Proteobacteria** are a major group of bacteria. They include a wide variety of pathogens, such as *Escherichia*, *Salmonella*, *Vibrio*, *Helicobacter*, and many other notable genera. Others are free-living, and include many of the bacteria responsible for nitrogen fixation. The group is defined mainly in terms of ribosomal RNA sequences, and is named for the Greek god Proteus, who could change his shape, because of the great diversity of forms found in it.

General Characters: Members of the family ***Enterobacteriaceae*** are Gram- negative, oxidase-negative, rod-shaped bacteria measuring 0.3-1.0 x 1.0-6.0 um. They are motile by peritrichous flagella. They are non-spore forming. Some members possess a prominent capsule and some don't. Some members also have fimbriae and pili. They are facultative anaerobes. They exhibit both respiratory and fermentative metabolism. Most grow well between 22 and 35°C on media containing peptone or beef extract. They also grow on MacConkey's agar, which may be used for their selective isolation. Most grow on glucose as a sole carbon source, although some require vitamins and/or amino acids for growth.

Distribution: They are worldwide in distribution and found in water or as commensals. They live as saprophytes, as symbionts, as epiphytes or as parasites. Their host range includes animals ranging from insects to humans, as well as fruits, vegetables, grains, flowering plants, and trees. Faecal contamination is one of the important modes of spread. Most of the members produce enteric infection. They are also associated with wound infection, mastitis etc.

GENUS – ESCHERICHIA

Type Species – *Escherichia coli*

This organism was first identified by Theodore Escherich in 1885, who isolated this bateria from the faeces of neonates.

General characters:

1. **Morphology:** They are Gram-negative, oxidase-negative, rod-shaped bacteria measuring 0.3-1.0 x 1.0-6.0 um. They usually occur as single organisms or in short chains. They are motile by peritrichous flagella. They are non-spore forming and capsulative. They also have fimbirae and pili. They are normally present in intestinal tract as commensals.
2. **Antigens and toxins:** There are over 700 antigenic types (**serotypes**) recognized based on somatic (**O), flagellar (H), and capsular (K) antigens**. *E.coli* organisms are classified based on O, H and K antigens and this system of classification is called as ***Kauffmann-Knipschidt- Vahlne*** scheme. The following are the virulence factors of *E.coli.*

a. Adhesins

- CFAI/CFAII
- Type 1fimbriae
- Pfimbriae

- Sfimbriae
- Intimin (non-fimbrialadhesin)

Invasins

- hemolysisn
- siderophores and siderophore uptakesystems
- Shigella-like "invasins" for intracellular invasion andspread

Motility/chemotaxis

- Flagella

Toxins

- LTtoxin
- STtoxin
- Shiga-liketoxin
- Cytotoxins
- Endotoxin LPS)

Antiphagocytic surface properties

- capsules
- K antigens
- LPS

Defense against serum bactericidal reactions

- LPS
- Kantigens

Defense against immune responses

- capsules
- Kantigens
- LPS antigenic variation

Genetic attributes

- genetic exchange by transduction and conjugation
- transmissible plasmids
- R factors and drug resistance plasmids
- toxin and other virulence plasmids

3. **Resistance:** They are destroyed by 60^0C at 30 minutes. They are also sensitive to common disinfectants. They are also killed at freezing temperature when exposed to two hours. A variation in the heat sensitivity is observed between strains. *E.coli* is known for its resistance to antibiotics.

Cultural characters: These organisms are aerobic or facultative anaerobic and fermentative. The optimum temperature for growth is 37.5^0C. They grow readily on ordinary laboratory media. On nutrient agar, the colonies are white or yellowish white, moist, glistening, opaque and circular with an entire edge. In MacConkey agar the organisms produce pink coloured colonies since they utilize the sugar lactose. In Eosin-Methylene blue agar, the colonies appear as blackish centre with a metallic sheen. In litmus lactose agar the organisms produce red coloured colonies.

Biochemical reactions: *E.coil* produces acid and gas from glucose, lactose, mannitol and number of other sugars. It reduces nitrate and does not produce hydrogen sulphide. The biochemical reactions are commonly referred as IMViC reaction. IMViC is an acronym in which the capital letters stand for **I**ndole, **M**ethyl red, **V**oges-Proskauer, and **C**itrate utilization tests. The **IMViC tests** were developed in order to distinguish strains of *E. coli* from related species that also produced acid and gas from the fermentation of lactose.

The IMViC set of tests examines: the ability of an organism to

1. produce **I**ndole;
2. produce sufficient acid to change the color of **M**ethyl red indicator
3. produce acetoin, an intermediate in the butanediol fermentation pathway (a positive result of the **V**oges-Proskauer test);
4. the ability to grow on **C**itrate as the sole source of carbon.

Lactose fermenters are considered *E. coli* if they are positive in the first two tests and negative in the second two.

Pathogenesis

E. coli causes disease in all kinds of domestic and wild animals. As a pathogen, *E. coli* cause intestinal diseases. Five classes (virotypes) of *E. coli* that cause diarrhoeal diseases have been recognized. They are as follows;

1. Enterotoxigenic *E. coli* (ETEC)
2. Enteroinvasive *E. coli* (EIEC)
3. Enterohemorrhagic *E. coli* (EHEC)
4. Enteropathogenic *E. coli* (EPEC)
5. Eeroaggregative *E. coli* (EaggEC)

1. **Enterotoxigenic *E. coli* (ETEC):** ETEC are an important cause of diarrhoea in infants and travellers in underdeveloped countries or regions of poor sanitation. The diseases vary from minor discomfort to a severe cholera-like syndrome. ETEC also cause infection in piglets and calves. The virulence factors for ETEC adhesins are fimbriae, which are species- specific. For example, the K-88 fimbrial Ag is found on strains from piglets; K-99 Ag is found on strains from calves and lambs. Enterotoxins produced by ETEC include the LT(heat-labile) toxin and/or the ST (heat- stable) toxin. The LT enterotoxin is very similar to cholera toxin in both structure and mode of action. The ST enterotoxin is actually a family of toxins which are peptides of molecular weight about 2,000 daltons. One such toxin STa is responsible for secretion of fluid and electrolytes resulting in diarrhoea. Symptoms ETEC infections include diarrhoea without fever.
2. **Enteroinvasive *E. coli* (EIEC):** EIEC resemble *Shigella* in their pathogenic mechanisms and the kind of clinical illness they produce. EIEC penetrate and multiply within epithelial cells of the colon causing widespread cell destruction. The clinical syndrome is identical to *Shigella* dysentery and includes dysentery-like diarrhoea with fever. EIEC apparently lack fimbrial adhesins but do possess a specific adhesin that, as in *Shigella*, is thought to be an outer membrane protein. Also, like *Shigella*, EIEC are invasive organisms. They do not produce LT or ST toxin and, unlike *Shigella*, they do not produce the shigatoxin.
3. **Enteropathogenic *E. coli* (EPEC):** EPEC induce watery diarrhoea similar to ETEC, but they do not possess the same colonization factors and do not produce ST or LT toxins.
4. **Enteroaggregative *E. coli* (EaggEC):** These strains are associated with persistent diarrhoea in young children. A heat-labile is associated with this type of infection

5. **Enterohemorrhagic *E. coli* (EHEC):** EHEC is identified with serotype O157:H7, which causes a diarrhoeal syndrome where there is copious bloody discharge and no fever. Usually kidneys are affected in this type of infection

There are, three situations where the otherwise harmless *E. coli* can cause illness:

1. When the bacteria get out of the intestinal tract and into the urinary tract they can cause an infection sometimes referred to as "honeymoon cystitis" in human beings.
2. When the bacteria get out of the intestinal tract through a perforation (= a hole or tear, which could be caused by an ulcer, for example) and into the abdomen, they usually cause an infection called "peritonitis" that can be fatal.
3. Certain strains of E.coli are toxigenic (some produce a toxin very similar to that seen in dysentery) and can cause food-poisoning usually associated with eating contaminated meat (contaminated during or shortly after slaughter or during storage or display). Severity of the illness varies considerably; it can, rarely, be fatal, but is more often mild.

Calves: Calves are susceptible to both systemic colibacillosis and neonatal diarrhoea (calf scours or white scour). The infection is usually fatal if not promptly treated. Specific heat-stable enterotoxigenic *E. coli* serotypes containing K99 fimbrial adhesin are considered the causative agents. The organisms are also associated with mastitis. ETEC cause diarrhoea in very young calves, less than 3- 4 days of age (typically less than 48 hours of age). Calves are depressed, do not drink or suckle, become dehydrated, and die rapidly. Very profuse and watery diarrhoea is typical of ETEC scours. Septicaemic colibacillosis, caused by another serotype of ETEC, is an acute disease with very few diagnostic signs and is the most common cause of acute, fatal illness in neonatal calves. Depressed, weak animals initially have a fever but become hypothermic rapidly. Mortality rates are high and survivors are often affected by post-septicaemic localisation of infection in the form of arthritis, meningitis or pneumonia. Inadequate transfer of passive immunity from the dam is considered the main risk factor for colibacillosis. Multiresistant (to antibiotics) strains of *E. coli* have been identified and hence antibiotics should not be the main approach to treatment. Control is aimed as two aspects as mentioned below;

Maintenance of immune status

- Do not separate the dam until 24 hours after calving
- Provide adequate bedding to allow calf to stand without difficulty
- Ensure early feeding, assist if needed, monitor the intake as closely as possible and record it ("maximum supervision, minimum interference")
- Keep a supply of frozen colostrum in case the dam leaks colostrum before calving
- Feed fresh or fermented colostrum to all calves up to three weeks of age
- Encourage outdoor calving
- In cases where ETEC has been identified as a herd problem, immunise pregnant dams before calving as part of a targeted control plan.

Minimisation of exposure

- Provide adequate numbers of calving pens and clean and disinfect them between batches
- House calves well away from the calving pens
- Keep up high hygiene standards throughout the calving period and in calf housing
- Spread the calving period over a longer period of time (maximum 15 calves per month)
- Keep the infection out of the herd if not present, by a closed herd policy and the avoidance of visitors from other farms, unless precautions are taken.

Sheep: Infant diarrhoea in lambs is usually caused by strains of *E. coli* producing a heat-stable enterotoxin. Most of these strains also contain the K-99 fimbrial adhesin.

Pigs: *E. coli* infections in pigs are as exhibited as diarrhoea in piglets, or as edema (edema disease) preceded by mild diarrhoea. Both forms are acute and highly fatal. As in sheep and cows, the causative strains produce a heat-stable enterotoxin, but they may also produce a heat-labile enterotoxin. Swine strains usually possess a K88 fimbrial adhesin, which is antigenically distinct from K99.

Chickens: Colisepticaemia is the commonest infectious disease of farmed poultry. It is most commonly seen following upper respiratory disease (such as Infectious Bronchitis) or Mycoplasmosis. It is frequently associated with immunosuppressive diseases such as Infectious Bursal Disease Virus (Gumboro Disease) in chickens or Haemorrhagic Enteritis in turkeys, or in young birds that are immunologically immature. Infection is by the oral or inhalation routes, and via shell membranes/ yolk/navel, water, fomites, with an incubation period of 3–5 days. Poor navel healing, mucosal damage due to viral infections and immunosuppression are predisposing factors. The important symptoms are Respiratory signs like coughing, sneezing, dejection, reduced appetite, poor growth and omphalitis. The important lesions are air sacculitis. pericarditis, perihepatitis, swollen liver and spleen, peritonitis, salpingitis, omphalitis, synovitis, arthritis, enteritis, granulomata in liver and spleen (Coligranuloma – Hjarre's disease) and Cellulitis over the abdomen or in the leg. The infection can be treated with amoxycillin, tetracyclines, neomycin or gentamycin. Good hygiene in handling of hatching eggs, hatchery hygiene and good sanitation of house, feed and water will prevent infection in chickens.

Diagnosis

1. Based on the symptoms and lesions.
2. Microscopy: Culture smears will reveal Gram-negative organisms.
3. Isolation and identification of *E.coli* by culture methods and biochemical reactions (IMViC tests)
4. ***E. coli* as an Indicator of Fecal Pollution**: *E. coli* has been used as the principal indicator of fecal pollution in both tropical and temperate countries. *E. coli* comprises about 1% of the total fecal bacterial flora of humans and most warm-blooded animals. Tests to identify isolates as *E. coli* aim to differentiate them from organisms normally associated with uncontaminated water. The term "coliform" describes *E. coli*-like organisms. The coliforms are further divided into "fecal coliforms" (*E coli*) and "nonfecal coliforms" (e.g. *Klebsiella* and *Enterobacter).* As per the prevailing Federal standards of US, drinking water should not have any coliforms.
5. **Detection of *E. coli* in Food:** The International Commission on Microbiological Specifications for Foods (ICMSF, 1978) has adopted a set of standard techniques for the enumeration of *E. coli* in food products, which is also accepted by the International Standards Organization (ISO, 1984). This method employs the use of lauryl sulfate tryptose broth at 35 or 37°C as a mildly selective-enrichment medium. This is followed by growth in EC broth containing 0.15% bile salts at 45°C as a second selective step. The ability to produce indole from tryptophan (in tryptone broth) at 45°C defines the strains as *E. coli.*
6. **Detection of *E. coli* in Water:** There is no method for the detection of *E. coli* in water that is accepted throughout the world. The presence of *E. coli* in water is identified by the ability of the organisms to produce gas from lactose and produce indole from tryptophan at 44°C. A method for enumeration employs a standard multiple tube test with a modified glutamate synthetic medium at 37°C as a first selective step, followed by further cultivation in standard media at 44°C.
7. **Detection of *E. coli* in Clinical Specimens:** Direct plating of clinical materials should be attempted when the specimens are supposed to have heavy load of organisms. Media such as MacConkey agar or Eosin Methylene Blue (EMB) agar are commonly used. If the number of *E. coli* is likely to be

very low or the amount of specimen is limited, enrichment in a rich nutrient medium such as brain heart infusion broth may beused.

8. **Rapid Methods for Detecting *E. coli*** : A fluorogenic detection method has been developed based on the cleavage of methylumbelliferyl-D- glucuronide (MUG) to the free methylumbelliferyl moiety, which fluoresces a blue color after irradiation with long-wave ultraviolet radiation. Automated or semi-automated systems are also being used for the detection of *E. coli* as part of the detection methods for *Enterobacteriaceae.* Other techniques such as immunoassays and nucleic acid hybridization studies can also be used to enumerate *E. coli,* and DNA probes directed at a number of genes have also been developed.

Salmonella

Salmonellosis is a bacterial disease caused by strains of *Salmonella*. It occurs in animals and man. In both cases it is an enteric disease of varying severity, usually involving diarrhoea. In poultry *Salmonella* causes two important infections namely pullorum disease and fowl typhoid.

General characteristics:

1. **Morphology:** They are small rod shaped or pleomorphic gram-negative organisms. Most of the species are motile except S.Pullorum and S.gallinarum. They are actively motile by peritrichous flagella. Capsule formation is associated with mucoid strain. Most serotypes also possess pili. They are nonspore-forming.
2. **Habitat:** They are typical intestinal pathogens and affected animals excrete the organisms in the excreta, contaminate the environment and act as source of infection to other animals. The most important aspect as for as the habitat is concerned is the species specificity of the members. Species affecting one type host does not affect other type of hosts. S.Typhi and S.Paratyphi are essentially pathogens of human beings and do not affect animals. However, S.gallinarum affects both human and animals.
3. Classification:
   a. **Old systems:** In this system, classification of the member of the genus *Salmonella* is based on the antigenic structure, biochemical characters, host specificity, geographic distribution and DNA relatedness. As per this system, there are more than 2000 serotypes of *Salmonella (Salmonella – Arizona group)* and they are classified into 7 subgroups. Species of *Salmonella* causing infection in man and warm animals are placed in sub group 1. The other groups contain species that cause infection in cold-blooded animals and species isolated from environment. Further, on the basis of biochemical characteristics the *Salmonella* organisms are classified into three species namely S.Enteritidis, S.Cholerasuis and S.Typhi.
   b. The other method of classification proposed is placing the members of salmonella under one species. The serovars are further classified into biovars based on variations in biochemical reactions. However, the old method of assigning a name to serotype is still followed considering the simplicity.
   c. According to the latest nomenclature the genus *Salmonella* consists of only two species: *S. enterica* and *S. bongori*. *Salmonella enterica* is divided into six subspecies, which are distinguishable by certain biochemical characteristics. Strains of *Salmonella* are classified into serovars on the basis of extensive diversity of lipopolysaccharide (LPS) antigens (O) and flagellar proteinantigens (H) in accordance with the **Kauffmann–White** scheme; currently approximately 2500 serovars are recognised. This number is constantly being increased. The most common serovars that cause infections in humans and food animals belong to subspecies *enterica*. The serovars of the other subspecies are common in poikilothermic (cold- blooded) animals and in the environment. Names are retained only for subspecies *enterica* serovars. These names are no longer italicised.

The first letter is a capital letter. For example S.Pullorum causing Pullorum disease is written as *S.enterica* subspecies *enterica* serovar Pullorum.

4. **Resistance:** They are killed by a temperature of 55°C for 1 hour and 60°C for 20 minutes. They are destroyed at autoclaving. They are also susceptible to phenol, QAC, cresol and other disinfectants. However, the effect of these disinfectants is reduced in the presence of mucus. Formaldehyde vapour also destroys the organisms (it is used to fumigate egg incubators).
5. **Antigens, toxins and virulence factors:** *Salmonella* possess three types of antigens - somatic (O), flagellar (H) and somatic (Vi). The somatic antigens are designated by numbers (1,2,3 etc.). The flagellar antigens (H) are present in two phases phase 1 and phase 2. Phase 1 antigens are indicated by alphabets and after Z they are indicated by numbers as subscripts to alphabet Z like Z_{10}. Phase 2 antigens are indicated by numbers. Generally Salmonella are grouped based on O antigen and each group is subgrouped based on flagellar antigens. Vi antigen is an important virulence factor for human infection with little significance in animal infections. *Salmonella* are also classified based on O and H antigens by agglutination reactions.

Cultural characteristics: They are aerobic or facultatively anaerobic organisms and grow well at a temperature of 37°C. The growth is enhanced by the addition of blood, serum, ascitic fluid, glucose etc. The colonies are smooth, round, convex, 2-3 mm in diameter and dewdrop like. During isolation of *Salmonella* contamination of other enteric organisms are eliminated by enrichment that allows only the growth of *Salmonella* and destroys the growth of other organisms. Selenite F broth is commonly used as an enrichment media. Numbers of selective media are used for *Salmonella* isolation. The commonly used selective media are desoxycholate citrate agar (DCA agar), McConkey's agar and brilliant green agar. Salmonella isolation is generally done first in enrichment media followed by selective media. The suspected materials are first inoculated into enrichment media (Selenite F broth) and incubated for 18-24 hours at 43°C followed by streaking at selective media like brilliant green agar. The selenite F broth is reincubated again and the contents are streaked again on selective media for better isolation. In McConkey agar they produce pale colour colonies in contrast to pink colonies of *E.coli.*

Salmonella organisms on repeated laboratory cultivation produce rough colonies instead of routine smooth colonies. This is due to loss of O antigen and virulence. This variation is referred as S◎R (smooth to rough). Some strains also have transient stage (T) between S and R having intermediate characters. This is referred as S◎T◎R.

Biochemical characters: They ferment glucose, maltose, mannitol, dulcitol and dextrin and produce acid and gas. They do not ferment lactose, sucrose and salicin. They reduce nitrates to nitrites and produce hydrogen sulphide. They produce indole, VP negative, liquefy gelatin and urease negative.

Pathogenesis: The infection is generally exogenous in nature. The organisms generally produce enteric infections and affected animals excrete the organismsin the faeces. Faecal contamination of water and fomites are the important source of transmission. Vertical transmission is also common among poultry.

SALMONELLA INFECTIONS IN POULTRY

Pullorum disease (S.Pullorum**):** It is an important poultry disease. The infection spreads both vertically and horizontally. Infected adult birds lay eggs that contain S.Pullorum in the yolk. However, only few such eggs hatch and act as source of infection to other chicks in the incubator. The dried spicules of fluff and faeces of infected source contain S.Pullorum, *which* spread through the air currents in incubator. The faeces of infected contaminates the feed and water that act as a source of infection. Chick boxes, litter, utensils and chick sexers also spread the infection. Droplet infection is also reported.

OIE Classification: List B disease

Symptoms: The incubation period is usually a few days. Young chicks are more affected with a morality of up to 95%. The chicks appear sleepy, dull and huddle close to the heating source. Inappetance, increased thirst, lour chirping, respiratory distress and nervous symptoms like staggering gait, incoordination of the limbs are other symptoms. The vent of the affected chicks will be covered with adherent mass of faeces. Growth rate is slow among growers with intermittent diarrhoea. In adults decreased egg production is the most important symptoms.

Lesions: In chicks, there is congestion of spleen, yellow colouration of spleen with streaks of haemorrhage. There is also hyperaemia of liver, necrotic foci on kidney, spleen, distended ureters with urates, distended caeca and degeneration of the myocardium. Lung and heart may have white nodules, pericardium may be thickened, with yellow or fibrinous exudates. Gastro intestinal tract may have white nodules on the gizzard, caeca and large intestinal wall. In adult birds the ovaries contain misshaped ovules and detachment of ovules is also noticed. Yolk sac retention, with yolk appearing creamy or caseous are other important lesions. Caseous cores may also be seen in the caeca. Joints may also be swollen with yellow viscous fluid.

Diagnosis:

1. **Based on symptoms and lesions.**
2. **Rapid Whole blood test:** It is done with whole blood to identify carrier adult birds. It is based on agglutination principle. The rapid whole blood agglutination test can be used under field conditions for detecting both S. Pullorum and S.Gallinarum, and the reactors can be identified immediately. However, it is not reliable in turkeys as the test results in a significant proportion of false-positive results.
3. **Rapid Serum agglutination test:** The RST is performed in the same manner, except that serum is substituted for whole blood. For export test purposes an initial screening of sera by RST followed by confirmation of positives by the tube aglutination test is approved.
4. **Tube agglutination test:** Fresh serum from chickens, turkeys or other birds is used at an initial dilution of 1/25. A titre of 1/50 is usually considered to be positive.
5. **Micro agglutination test:** This resembles the tube agglutination test, but requires much smaller volumes of reactants. The test is performed in microtest plates.
6. **ELISA:** The indirect ELISA using lipopolysaccharide antigen is likely to be the most sensitive and specific serological flock test for *Salmonella*, including *S.* Gallinarum and S.Pullorum.
7. **Isolation of the causative organism by culturing.** In clinical cases direct plating on Brilliant Green, McConkey and non-selective agar can be done. Enrichment procedures usually rely on selenite broth, tetrathionate broth and Rappaport-Vassiliadis followed by plating on selective media.

Control:

1. Identification of the reactors and removing them fromflock.
2. Periodical disinfection of the incubators using formaldehyde vapour (fumigation).
3. Fumigation of egg or dipping them in bactericidal agents before incubating them.
4. Vaccines are not normally used as they interfere with serological testing and elimination of carriers

Treatment

Amoxycillin, poteniated sulponamide, tetracylines, fluoroquinolones.

Fowl typhoid *(S.gallinarum)*

In contrast to pullorum disease fowl typhoid is a disease of adult birds and growers. Vertical transmission from hen to chicks is also possible. The organism can survive in litter for a long time and the litter acts as source of infection. Broiler parents and brown-shell egg layers are especially susceptible. Chickens are most commonly affected but it also infects turkeys, game birds, guinea fowls, sparrows, parrots and canaries. Infections still occur worldwide in non-commercial poultry but are rare in most commercial systems now. Morbidity is 10–100%; mortality is increased in stressed or immunocompromised flocks and may be up to 100%. The route of infection is oral or via the navel/ yolk. Transmission may be transovarian or horizontal by faecal–oral contamination, egg eating etc, even in adults. The bacterium is fairly resistant to normal climate, surviving months, but is susceptible to normal disinfectants.

OIE Classification: List B disease

Symptoms: In chicks the symptoms are as same as pullorum disease. The mortality in adults is up to 50%. The affected birds show are listlessness, green or yellow coloured diarrhoea, purple colouration of comb and wattles, anaemia and intermittent diarrhoea.

Lesions: The most important lesions are enlarged liver that turns into dark red or reddish brown after exposure to atmosphere. Enlargement of spleen, focal necrosis on liver, spleen and myocardium are other lesions. The intestine may show catarrhal inflammation and petechial haemorrhage.

Diagnosis:

1. Based on symptoms and lesions.
2. Tests 2-6 mentioned for pullorum can be done for fowl typhoid also but the antigen used should be S.gallinarum.
3. Isolation of causative organisms by culturing.

Treatment

Amoxycillin, potentiated sulponamide, tetracylines, fluoroquinolones.

Prevention

Biosecurity and clean poultry husbandry practices will reduce the incidence. As with other salmonellae, recovered birds are resistant to the effects of infection but may remain carriers. Vaccines for fowl typhoid have been used in some areas, both live (usually based on the Houghton 9R strain) and bacterins.

S. Enteritidis and S. Typhimuriuminfections

Salmonella Enteritidis and S.Typhimurium are the predominant sero-types associated with human disease in most countries. Infections in chickens, turkeys and ducks cause problems worldwide with morbidity of 0–90% and a low to moderate mortality. The route of infection is oral; many species are intestinal carriers and infection may be carried by faeces, fomites and on egg shells. Vertical transmission may be either by shell contamination or internal transovarian contamination of yolk. Feed and feed raw material contamination is less common than for other sero-types. The bacteria are often persistent in the environment, especially in dry dusty areas, but are susceptible to disinfectants that are suitable for the particular contaminated surfaces and conditions, applied at sufficient concentrations. The infection caused by S.Typhimurium in poultry is also referred as paratyphoid.

Symptoms: The important symptoms in affected birds include dejection. Ruffled feathers, closed eyes, diarrhoea, vent pasting, loss of appetite and thirst and stunting in older birds.

Lesions: The lesions include enteritis, focal necrotic intestinal lesions, foci in liver, unabsorbed yolk, cheesy cores in caecae, pericarditis, perihepatitis and misshapen ovules in the ovaries.

Diagnosis

1. Isolation and identification. In clinical cases direct plating on Brilliant Green and McConkey agar will be useful. Enrichment media such as buffered peptone followed by selective broth or semi-solid media (e.g. Rappaport-Vassiliadis) followed by plating on two selective media will greatly increase sensitivity
2. ELISA to differentiate with S.Pullorum

Treatment

Sulphonamides, neomycin, tetracyclines, amoxycillin and fluoroquinolones

Prevention

1. Clean poultry husbandry practices.
2. Routine monitoring of breeding flocks,hatcheries
3. Vaccines – Inactivated and live vaccines for S. Enteritidis and S. Typhimurium are available.

SALMONELLA INFECTION IN CATTLE

The two species that affect cattle are S.Dublin and S.Typhimurium. The infection in cattle spreads by ingestion. Poor health conditions also predispose the animal for infection. The infection in cattle is characterised by high temperature, inappetance, drop in milk yield and diarrhoea. Pregnant animals may abort. Affected animals die in 2-3 days in acute infection. In chronic cased the animal become emaciated, dehydrated and show signs of abdominal pain. In young calves the loss of appetite, weakness and blood stained mucoid are characteristic of salmonellosis. The lesions are found in the intestine and mesenteric lymph nodes. There is extension of the gall bladder and focal necrosis of liver. Diagnosis is based on symptoms and lesions, isolation of the organisms and agglutination tests. The clinical infection can be treated with Chloromycetin, oxytetracycline,neomycin and furazolidone. Both killed and live vaccines are available as protective measure.

SALMONELLA INFECTION IN SHEEP AND GOAT

The species affecting sheep and goat are S.Abortus-ovis, S.Typhimurium and S.Dublin. The infection is characterised by abortion during last two months of pregnancy preceded by blood stained purulent vaginal discharge. Affected animals deliver weak lambs that show symptoms of diarrhoea and dysentery from day one. Surviving lambs become debilitated. In adult animals intermittent diarrhoea and fleece loss are the characteristic symptoms. There is congestion and haemorrhage in abomasum, small intestine, myocardium and kidneys.

SALMONELLA INFECTION IN PIGS

The pigs act as a reservoir for many serotypes of salmonella. The most common infection in pigs is the pig paratyphoid caused by S.Cholerae-suis. S.Typhimurium can also cause infection in pigs. Pig paratyphoid is an acute, subacute or chronic infection with high temperature, loss of appetite and appearance of bright pink colour areas on the skin. Affected animals will have profuse yellow colour diarrhoea. Survivors act as carriers for other animals.

SALMONELLA INFECTION IN DOGS AND CATS

Salmonella infection in dogs and cats are caused by number of serotypes due to consumption of

contaminated meat, milk and wild rodents. The infection is characterised by abdominal pain, vomiting, diarrhoea, dysentery and death.

PUBLIC HEALTH IMPORTANCE

The main mode of spread to humans is through animal products like meat, milk and egg. Contact with infected pet and domestic animals also act as source of infection.

Proteus

Important species : *Proteus vulgaris, P.mirabilis, P.rettgeri*

Important infection : Urinary tract infection, Enteritis and Otitis

General characters:

1. **Morphology:** They are gram-negative pleomorphic rod shaped organisms. They occur as coccal form or as filaments. They are motile by peritrichous flagella. They are non-spore forming and non-capsulative organisms. Majority of the organisms possess flagella.
2. **Habitat:** They are found as commonly in the intestine of man and animals, animal manure, sewage, soil and water, meat and eggs. They are also isolated from number of wild animals.
3. **Resistance:** They are killed by a temperature of 55°C for 1 hour and 60°C for 20 minutes. They are destroyed at autoclaving. They are also susceptible to phenol, QAC, cresol and other disinfectants. However, the effect of these disinfectants is reduced in the presence of mucus. Formaldehyde vapour also destroys the organisms (it is used to fumigate egg incubators).
4. **Classification:** Not very significant.
5. **Antigens and toxins:** The organisms possess O and H antigens.

Cultural characteristics: They are aerobic organisms grow at a temperature range of 20-40°C. However, the optimal temperature is 37°C. On the surface of solid media the organisms (colonies) rapidly spread as a thin film and this phenomenon is called **swarming.** This phenomenon occurs in waves and the culture look like rings. Swarming can be inhibited by sodium azide or media containing 6% agar. A characteristic odour from the plate is also characteristic of *Proteus*.

Biochemical characters: They are non-lactose fermenters, indole variable, citrate utilisation negative, and urease positive and produce hydrogen sulphide. Proteus organisms are positive to (phenyl pyruvic acid) PPA reaction. Proteus is the only genus that is positive for this test.

Pathogenesis: They are normal inhabitants of the intestine. Under some conditions they produce severe diarrhoea and dysentery. In dogs it produces otitis.

Diagnosis: The symptoms are not characteristic. Hence diagnosis is based on isolation of the organisms.

KLEBSIELLA

Important species : *Klebsiella pneumoniae* (**Friedlander's bacillus**);

Important disease : Mastitis, Metritis, Suppurative infection, and atrophic rhinitis

The genus *Klebsiella* belongs to the tribe *Klebsiellae*, a member of the family Enterobacteriaceae. The organisms are named after Edwin Klebs, a 19th century German microbiologist. Klebsiellae are non-motile, rod-shaped, gram-negative bacteria with a prominent polysaccharide capsule. This capsule accounts for their large appearance on samples stained with Gram stain.

General characters:

1. **Morphology:** They are gram-negative rod shaped organisms. They are non-motile and non-spore forming organisms. The characteristic feature of the organisms is the presence of a very prominent polysaccharide capsule.
2. **Habitat:** These organisms are found as saprophytes in water and soil. They are also found as commensals in the intestinal and respiratory tract of animals.
3. **Classification:** There are number of serotypes. The classification is not very important.
4. **Resistance:** They are killed at a temperature of 60^0C for 20 minutes and by number of disinfectants. They can remain for a long time in room temperature.
5. **Antigens and toxins:** The O and K antigens are important. Serotypes are based on the structural variability of the capsular polysaccharides (K antigens) and lipopolysaccharides (O antigens). There are 77 K antigens and 8 O antigens. The virulence of all serotypes appears to be similar.

Cultural characteristics: They grow well in MacConkey's agar and produce prominent mucoid pink colour colonies. Pink colouration is due to utilisation of lactose and mucoid consistency is due to polysaccharide capsule. No haemolysis is produced in blood agar.

Biochemical characters: They are VP positive, MR negative, catalase positive and hydrolyse urea.

Pathology: They are generally secondary invaders. In foals they produce pneumonia and metritis. In cattle they produce mastitis and in pigs they produce metritis and atrophic rhinitis. In poultry they cause air sac infection. The symptoms vary according to the system.

Diagnosis:

Diagnosis is based on the isolation of the organisms by culture on selective medium and biochemical tests.

YERSINIA

Important Species: *Yersinia pestis* CO92; *Yersinia pestis* KIM; *Yersinia enterocolitica; Yersinia pseudotuberculosis*

Important *disease:* Plague, Pseudotuberculosis

Yersinia spp. are responsible for disease syndromes ranging from gastroenteritis to plague. *Y. pestis* is the cause of the plague and is actually catagorized into three subtypes or biovars; Antiqua, Medievalis, and Orientalis, each associated with a major pandemic. *Y. pestis* strand KIM belongs to biovar Mediaevalis while strand CO92 is in biovar Orientalis. Biovar Mediaevalis is thought to of descended from the bacteria that caused the second pandemic (the Black Death), while biovar Orientalis bacteria are responsible for the current pandemic (modern plague).

General characters:

1. **Morphology:** They are Gram-negative, short ovoid rod shaped organisms. They have a variable motility character. *Y.pseudotuberculosis* organisms are non-motile at 37^0C but motile at 22^0C. Whereas *Y.pestis* is non-motile. They appear bipolar stained in fresh isolates. They are non-spore forming. The capsule is more prominent in tissues where as it is not prominent in cultures.
2. **Habitat:** *Y.pseudotuberculosis* is worldwide in distribution and distributed by wild rodents and pigeons and causes pseudotuberculosis in guinea pigs. *Y.pestis* is found only in certain parts of Asia, America and Africa and causes bubonic plague or pneumonic plague in man. *Rat flea transmits Y.pestis*.

3. **Resistance:** They are not very resistant organisms. They are killed by chemical and agents like 0.5% phenol when exposed for 15 minutes. They are also killed on exposure to sunlight for 3-4 hours and heating at 55°C for 15 minutes.
4. **Classification:** *Y.pseudotubeculosis* is classified based on O antigen and *Y.pestis* is classified based on O and K antigens.
5. **Antigens and Toxins:** *Y.pseudotuberculosis* has got O antigens and *Y.pestis* has got K antigens. *Y.pestis* produces a toxin that is lethal to both mice and guinea pigs.

Cultural characteristics: They are aerobic organisms that grow in a wide range of temperature. *Y.pseudotubeculosis* and *Y.pestis* grow well in MacConkey's bile salt medium (this character is used to differentiate with *Pasteurella multocida*). *Y.pestis* grows between temperature ranges of 14-37^{0}C. Their growth is enhanced by blood or serum in the media.

Pathogenesis:

1. ***Y.pseudotuberculosis:*** The main mode of spread is by ingestion of food contaminated with urine of wild rodents that suffer from clinical infection of *Y.pseudotuberculosis.* Guinea pigs, rabbits, chickens and turkeys are affected by pseudotuberculosis. The acute infection develops as sudden fatal septicaemia with high mortality. The chronic form is manifested as intermittent diarrhoea and emaciation. The infection remains localised in lymph nodes during chronic infection. The spleen and mesenteric lymph nodes are enlarged and congested. The lesions appear as necrotic caseous foci surrounded by polymorph cells but there is no calcification.
2. ***Y.pestis:*** Plague is a disease of rodents and man is considered as an accidental host. The infection is spread through bite of rat flea *Xenopsylla cheopis.* Bubonic plague that mainly affects lymphatic system is spread through rat flea whereas pulmonary plague spreads via infected droplets from affected person.
3. ***Y.enterocolitica:*** Associated with contamination of food.

Diagnosis:

Based on symptoms, lesions and isolation of organisms by culturing.

Treatment and control: Chloramphenicol, tetracyclines and streptomycin are the drugs of choice. Control is mainly by controlling wild rodent population.

CAMPYLOBACTER

Campylobacteriosis is a bacterial disease caused by *Campylobacter jejuni* or *Campylobacter coli*. *C.fetus* causes abortion and genital infections in animals. *Campylobacter jejuni* and C. *coli* are the most frequently isolated Campylobacter species in infected and diseased humans and can be transmitted from animals to humans directly from contact with animals or through consumption and handling of animal food products. Both C. jejuni and C. coli are commonly present in the intestines and being shed. The faecal contamination of meat (especially poultry meat) during processing is considered to be a major source of human food-borne disease. Both *C. jejuni* and C.*coli*, like C.*fetus*, can also be isolated from bovine and ovine aborted foetus due to translocation across the intestinal mucosa or by ascending infections

Important diseases - Infertility and abortion, winter dysentery

General characters

1. **Morphology:** Members of the genus are typically Gram-negative, S- shaped or spiral shaped bacteria (0.2–0.8 μm wide and 0.5–5 μm long). The organisms may have two or more curvatures. The shape is also referred as comma shaped, S shaped or flying seagull like appearance. They are

pleomorphic rods. They are motile by polar flagella at one or both ends, conferring a characteristic corkscrew- like motility. They are capsulative and non sporulative. They are microaerophilic organisms requiring 3- 15% $C0_2$ tension. They oxidase positive and do not attack carbohydrates.

2. **Habitat:** These organisms are commonly found on the mucous membrane of reproductive and digestive tracts. Some birds are found to transmit the infection.
3. **Classification:** The genus *Campylobacter* has been carved from *Vibryo, which* is very important genus since one of the members of the genus *V.cholera* cause cholera in human beings. The species *C.fetus* is grouped into two sub species *C.fetus* subsp. *fetus* and *C.fetus* subsp. *venerealis.* The sub species *venerealis* is further subdivided into two biotypes based on hydrogen sulphide production into *venerealis* and *intermedius. C.fetus subsp fetus* causes abortion and *C.fetus subsp venerealis* causes bovine genital infections.
4. **Resistance:** They are fragile organisms and cannot withstand drying and sunshine and are readily killed by disinfectants. They are killed when exposed to a temperature of 60°C for 5 minutes. They can survive in soil, hay and manure for 10-20 days depending upon the humidity and temperature.
5. **Antigens and toxins:** Somatic (O), capsular (K) and flagellar (H) antigens are found in *C.fetus.* There are some similarities and dissimilarities between organisms isolated from different species like cattle and sheep.

Cultural characters: They are microaerophilic organisms requiring reduced oxygen tension and 3-15 % $C0_2$ tension for growth. The optimum temperature for growth is 37°C. They are also slow growing organisms. The media commonly used for isolation are blood agar with or without thioglycollate, serum-dextrose agar, Thiol medium, Clark and Dufty medium and chocolate agar. Colonies appear after 2-6 days as pinpoint, delicate, circular and opaque. The colonies fuse together to produce a large mucoid colony. The surface of the colonies may have a frosted appearance. Organisms without capsule produce rough colonies.

Pathogenesis: *C.fetus* subsp *venerealis* cause venereal infection in cattle. The main mode of infection is through coitus and fomites. Bulls get the infection from infected cows through coitus and the affected bulls remain as main source of infection to other animals. Bulls can also transmit the infection mechanically for several hours after copulating with an infected cow. In cows, the duration of the carrier state is also variable; some clear the infection rapidly, while others can carry *C fetus* for ≥2 yr. The organisms produce both acute and chronic infections. During acute infections the inflammatory changes in the uterus cause death of fertilized ovum leading to either its expulsion (abortion) or absorption. This infection occurs 3-4 week after fertilisation. In cows with immunity abortion occurs in later stage of pregnancy (fifth month) due to inflammation of uterus and placenta. *C.fetus* subsp *venerealis* biotype intestinalis causes infection in sheep that is not venereal. The infection spread through ingestionof contaminated water and feed. Birds also spread infection. Rams have little or no role in the pathogenesis. The death of foetus is due to inflammatory changes in the uterus. The important virulence factor is the endotoxin.

Symptoms: In cattle the symptoms are more pronounced in cows than in bulls. The affected bulls remain normal without any symptoms and no abnormalities are found in the semen. In cows endometritis that causes early embryonic death, prolonged luteal phases, irregular oestrous cycles, repeat breeding abortion and infertility are the main symptoms. Abortion occurs during fifth or sixth month of pregnancy. Infertility rate is higher in heifers than in cows. In sheep abortion during end of pregnancy preceded by vaginal discharge for several days are the main symptoms. Large number of animals is generally affected in flock and the condition is referred as **abortion storm.**

Lesions: The carcass of affected foetus is oedematous with pericarditis, peritonitis and petechial haemorrhages. Areas of focal necrosis are seen in liver. The foetal membranes are oedematous with areas of necrosis on cotyledons. Placentitis occurs with hemorrhagic necrotic cotyledons and edematous or leathery intercotyledonary areas. Catarrhal inflammation of vagina and cervix is common in cows. The same lesions are noticed in ewes also. The ewes will appear normal with oedematous uterus and areas of marked hyperaemia. Metritis is also seen in ewes.

Diagnosis:

1. Based on symptoms, lesion and history of abortion in a herd.
2. **Microscopical examination** of smears prepared from the foetal stomach contents or visceral organs stained by Gram's method reveal organisms with characteristic morphology. *C.fetus* can also be identified in dark field preparations from abomasal or placental smears or in uterine discharge.
3. **Isolation of the bacteria by culturing:** Vaginal mucus and preputial washings are the material of choice. Vaginal mucus three weeks after AI or service is the most ideal material. *C.fetus* will survive only for 6-8 hr after collection, but inoculation into Clark's or similar media will allow survival for >48 hr. Special care must be taken to avoid contamination of the clinical materials with other bacteria. The media commonly used for isolation are blood agar with or without thioglycollate, serum-dextrose agar, Thiol medium, Clark and Dufty medium and chocolate agar. Colonies appear after 2-6 days as pinpoint, delicate, circular and opaque.
4. **Vaginal mucus agglutination test:** This test is performed in herd basis to detect antibodies against *C.fetus*, which are found up to 2-12 months after infection. 10% of the herd or atleast 10 cows should be sampled.

The mucus is collected in sterile tube of tampon. Antibodies in the mucus are separated by mixing it with agar and treating it with phenol-saline. An agglutination test is performed in serial dilution. A positive reaction is atleast 75% agglutination in the first two tubes.

5. Indirect HA test
6. Serum agglutination test
7. ELISA
8. **Mating test:** Infected bulls are allowed to mate virgin heifers and examination of the vaginal mucosa after 3-4 weeks. But this is seldom practical. Instead the preputial cavity is infused with buffered sterile saline, and the prepuce is massaged vigorously in the area of the fornix. The sheath washing is then examined using a fluorescent antibody test and culture.

Treatment: Treatment is usually not feasible. However, penicillin and streptomycin are used as therapeutic agents. Washing the vagina and prepuce of affected animal with streptomycin is of some use.

Control: Elimination of the infected bull and treating the semen with streptomycin are the two important aspects in control programme. Killed vaccines have found to reduce the incidence of infertility. Artificial insemination instead of natural coitus will also avoid campylobacteriosis.

OTHER SPECIES OF *CAMPYLOBACTER*

The species, particularly *C. jejuni*, *C. coli* and *C. lari*, are thermophilic, growing optimally at 42°C. Hence, in the isolation procedure the media are incubated at 42°C to prevent contamination from other bacteria.

***Campylobacter jujuni*:** It is found in the intestinal tract of domestic animals and birds as commensals. It causes infectious diarrhoea (**winter dysentery**) in cattle. Affected animals become listless, show signs of abdominal pain with dark coloured fluid consistency faeces with blood and mucus. The powerful heat labile enterotoxin produced by the organism is responsible for the diarrhoeic symptoms. It causes avian infectious hepatitis (vibryonic hepatitis) in chickens characterised by haemorrhage and necrosis of liver.

Pathogenesis: Environmental contamination is the main source of infection for poults, chicks, and ducklings. Litter can remain infective for long periods. Infected chicks and poults become colonized and can continue to excrete *C jejuni* for up to 2 months. Contaminated water can introduce infection into poultry flocks, and nonchlorinated water derived from a dam, river, or shallow well should be regarded as a possible source. Flocks can be infected by rats, wild birds, and houseflies; equipment and footwear contaminated with faeces from an infected source. Once introduced into the environment, rapid transmission within the flock occurs. C.*jejuni* is not transmitted vertically, either on the surface of eggs or by transovarial transmission.

Symptoms and lesions: Rapid onset of diarrhoea, which may persist for four days is the predominant symptom. Gross lesions include distention of the jejunum, disseminated hemorrhagic enteritis, and in some cases, focal hepatic necrosis. Microscopic lesions include edema of the mucosa of the ileum and cecum with *C jejuni* in the brush border of enterocytes. Mononuclear infiltration of the submucosa and villous atrophy occur with intraluminal accumulation of mucus, erythrocytes, and both mononuclear and polymorphonuclear cells.

Diagnosis:

1. **Isolation and identification by culture method:** Fecal specimens should be collected using swabs. Enrichment culture of specimens in semisolid motility medium facilitates isolation when small numbers of *C jejuni* are present in a sample. The materials should be cultured on selective media containing brucella agar base and bovine blood with up to seven antibiotics that inhibit overgrowth of other Enterobacteriaceae.
2. **Nucleic acid identification method:** Bacterial restriction endonuclease DNA analysis can distinguish among various *C jejuni* isolates.

Control:

Control in commercial poultry is based on strict biosecurity, decontamination of housing between successive flocks, exclusion of rodents and wild birds, and insect eradication. Chlorination of drinking water to 2 ppm and operating farms on a strict all-in/all-out basis can reduce the prevalence of infection.

Public Health Significance

C.jejuni is a major source of food-borne enteritis in consumers; contaminated, undercooked poultry is responsible for >50% of cases. In addition to contaminated poultry, nonchlorinated ground water, young diarrheic pets, and contaminated beef and pork products may also be responsible for infection of people. C.*jejuni* contamination can be reduced by improved washing of carcasses, the use of counter-flow scalding, elimination of immersion chillers, and reduction in manual handling by installation of advanced automated equipment. Chemical disinfectants, such as glutaraldehyde (0.125%) and succinic acid (3%), and organic compounds, such as lactic and acetic acids, may be used to destroy *C. jejuni* when used as a spray or an immersion treatment. Gamma irradiation at levels ranging from 1 to 3 kGy effectively eliminates *C. jejuni* from poultry carcasses and products. The only current measure to reduce the risk of *C. jejuni* infection to consumers is thorough cooking of poultry to achieve a core temperature of 74°C for 1 min. This ensures destruction of *C. jejuni*.

PSEUDOMONAS & BURKHOLDERIA

Members of the genus are found abundantly as free-living organisms in soils, fresh water and marine environments, and in many other natural habitats. They are also found in associations with plants and animals as normal flora or as agents of disease. The term "**pseudomonad**" refers to a bacterium with ecophysiological properties similar to members of the genus ***Pseudomonas***. Some of these bacteria were formerly in the genus ***Pseudomonas*** but have been moved to other genera, families, or orders. Pseudomonads are globally active in aerobic decomposition and biodegradation, and hence, they play a key role in the carbon cycle. *Pseudomonas* species are renowned for their abilities to degrade compounds, which are highly refractory to other organisms, including aliphatic and aromatic hydrocarbons, fatty acids, insecticides and other environmental pollutants. Apparently, the only organic compounds that these pseudomonads can't attack are teflon, styrofoam and one- carbon organic compounds (methane, methanol, formaldehyde, etc.). Pseudomonads are also a regular component of microbial food spoilage in the field, in the market place, and in the home.

Pseudomonas and certain other pseudomonads include species pathogenic for humans, domestic animals, and cultivated plants. *Pseudomonas* species, as well as species included in the newly- created genera *Burkholderia* and *Ralstonia* (ex-*Pseudomonas*) are among the most important bacteria that are pathogens of plants. They cause economically significant crop disease and crop loss world-wide. *Pseudomonas aeruginosa* infects both plants and animals and has evolved into one of the most common and refractory nosocomial pathogens of the post-antibioticera.

Important species : *Pseudomonas aeruginosa, Burkholderia mallei and B. pseudomallei*

Important diseases : Pyogenic infection, Melioidosis, Glanders

General characters:

1. **Morphology:** They are gram-negative rod shaped organisms (0.5-0.8 um x 1-3 um). They are actively motile by polar flagella some strains also produce lateral flagella. They are non-capsulative and non-spore forming. They are strictly aerobic. Catalase-positive, Usually oxidase-positive
2. **Habitat:** They are worldwide in distribution and occur as commensals in the intestine of animals and man. Animal pathogens are less host specific and most are considered to be opportunistic pathogens. One species, *B. mallei*, is an exception to this rule. This bacterium is the agent of glanders in horses, mules, and donkeys. *B. mallei* is not found living saprophytically, and it is considered to be the only pseudomonad with exclusive parasitism.
3. **Resistance:** They are fragile organisms and are lethal to heat and disinfectants. They are killed at a temperature of 55^0C for 1 hour.
4. **Classification:** Pseudomonads species are classified into five natural clusters based on ribosomal RNA homology, called **"RNA similarity groups"**. The *Pseudomonas spp.* are further classified into subgroups based on O antigen.

   Pseudomonas aeruginosa – RNA similarity Group I (Palleroni's Group I) *Burkholderia mallei and B. pseudomallei* – RNA similarity Group II (Palleroni's Group II)
5. **Antigens and toxins:** They have somatic (O) antigen. No exotoxins are produced.
6. **Metabolism:** *Pseudomonas* species are respiratory and never fermentative. All species respire aerobically, and some respire anaerobically with NO_3 as a final electron acceptor.
7. **Pigment production:** A common characteristic of the fluorescent pseudomonads is the production of pigments that fluoresce under short wave length (254 nm) ultraviolet light. These pigments and/ or their derivatives are known to play a role as siderophores in the iron uptake systems of the bacteria. *P. aeruginosa* is capable of producing several pigments, of which the most characteristic is

pyocyanin. This blue pigment is a diagnostic character, since no other species has been found to produce it. The other pigments are **Pyoverdin** (fluorescent pigment), **pyorubrin (**reddish pigment), and a brown pigment, **pyomelanin**. The production of pigments is readily demonstrated by culturing the bacteria in media such as King's Medium B.

Cultural characteristics: They grow at a wide temperature of 5-43^{0}C. Optimum growth is obtained at 37^{0}C in ordinary laboratory media. No organic growth factors are required for the growth of these organisms. The colonies are convex, glistening, bluish metallic sheen and have a tendency to spread. The colonies fluoresce due to production of two pigments called **pyocyanin** and **fluorescin.** The cultures have a characteristic sticky odour (grape like).

Pathogenesis:

P.aeruginosa is found worldwide and is also found in soil. They produce variety of pyogenic infection. The pus is greenish yellow in colour. It also causes bovine mastitis, urogenital infections, liver abscess, calf enteritis, pneumonia etc. In sheep it causes a condition called green wool.

GLANDERS – (Syn. Dros, Farcy, Malleus)

Glanders is an infectious disease caused by the bacterium *Burkholderia mallei.* Glanders is primarily a disease affecting horses, but it also affects donkeys and mules and can be naturally contracted by goats, dogs, and cats. Human infection occurs rarely and sporadically among laboratory workers and those in direct and prolonged contact with infected domestic animals.

Morphology of *B.mallei*: The organisms are mainly extracellular, fairly straight Gram-negative rods with rounded ends, 2–5 μm long and0.3–0.8μm wide with granular inclusions of various size, often stain irregularly. They do not have capsules or form spores. However, the presence of a capsule-like cover has been established by electron microscopy recently. This capsule is composed of neutral carbohydrates and serves to protect the cell from unfavourable environmental factors. Unlike other organisms in the *Pseudomonas* group, *Burkholderia mallei* have no flagellae and are therefore nonmotile. The causative organism *B.mallei* is closely related to *B.pseudomallei*, the cause of melioidosis, and is serologically indistinguishable. The organism is destroyed by direct sunlight and is sensitive to desiccation. It is readily killed by common disinfectants. It may survive for up to 6 weeks in infected stables.

Cultural charcters: The organisms are aerobic and facultatively anaerobic only in the presence of nitrate. Optimum temperature for growth is 37°C. It grows well, but slowly, on ordinary culture media, 48- hour incubation of cultures is recommended. Glycerol enrichment enhances the growth. After a few days on glycerol agar confluent, slightly cream-coloured growth that is smooth, moist, and viscid can be observed. With continued incubation, the growth thickens and becomes dark brown and tough. It also grows well on glycerol potato agar and in glycerol broth, on which a slimy pellicle forms. On plain nutrient agar, the growth is much less luxuriant, and growth is poor on gelatin.

Pathogenesis: Glanders occurs in horses, mules and donkeys. It also occurs in wild animals eating the meat of carcasses infected with B.*mallei.* Cattle, pig and poultry are resistant. The disease is introduced into horse populations by diseased or latently infected animals. Ingestion ofthe pathogen, present in secretions from infected animals, constitutes the major route of infection in glanders. Inhalation and entry through wounds in skin are of minor importance in the natural spread of the disease. Close proximity alone does not usually result in transmission of glanders; transmission is facilitated if the animals share feeding troughs or watering facilities or if they nuzzle each other. The primary infection is seen in lungs. However, digestive tract infection is seen when the organisms are

ingested. The susceptibility to infection increases by poor nutrition, over work and other stress factors. The initial distribution of the organisms in the system is via lymphatic system followed by blood stream. The incubation period varies.

Symptoms: The clinical forms of the disease are classified as pulmonary form, nasal form and cutaneous form (farcy). The infection occurs either as any one of the individual form or as combination of forms.

The pulmonary form is characterised by pyrexia and severe cough that produce blood stained mucus. In the **nasal form**, there is reddening of nasal mucosa and discharge from nostrils that may turn purulent and blood stained later. Ulceration of nasal mucosa and enlargement of the lymph glands are other important symptoms. **Farcy** is characterised by cutaneous or subcutaneous nodules on the limbs or flanks. Later these nodules turn into hollow ulcers releasing yellow and oily pus. The local lymphatic vessels become corded. These may be referred to as "Farcy pipes."

Lesions: Nodular lesions of glanders are found beneath the pleura of the lung. In some acute cases lobular pneumonia may be present. The nodular lesions, typically about 1 cm in diameter, consist of a gray or white core of necrotic material that may become calcified and are surrounded by a zone of hyperemia and edema. Similar lesions may be found in other viscera. Glanderous orchitis may be seen in intact males.

Nasal lesions consist of submucosal nodules surrounded by a small zone of hyperemia. These nodules may rupture, leaving exudative ulcers. As new lesions develop it is not unusual to find small nodules, ulcers, and scars side by side. Lymphadenitis of associated lymph nodes is a consistent finding. In some cases laryngeal lesions similar to the nasal lesions may be found.

Cutaneous lesions consist of cord-like thickening of subcutaneous lymphatics along which are distributed chains of nodules, some of which are ulcerated. Histologically, there is infiltration of neutrophils, fragmentation of cellular nuclei, giant cell formation and rarely calcification.

Diagnosis:

1. Based on symptoms and lesions.
2. **Mallein test (Test prescribed for International Trade):** It is an allergic test. Mallein is the active principle form *B.mallei.* Various sites are preferred for mallein test like intradermopalpebral, ophthalmic and subcultaneous.
    a. **The intradermopalpebral** (below the eyelids) test is more reliable. Swelling of the eyelid and congestion of conjunctiva after 36-48 hours indicate a positive reaction.
    b. **Ophthalmic**: A few drops of mallein are instilled into the eye at the canthus. In an infected animal, the eyelids, and sometimes the side of the face, become swollen and there may be a little discharge from the eye. The reaction may also occur to a lesser extent in the opposite eye.
    c. **Subcutaneous**: A 10 cm square skin patch in the middle of the neck is clipped and disinfected; 2.5 ml of dilute mallein are injected subcutaneously into the centre of the patch. With a positive test, the horse develops a pyrexia of 104°F (40.0°C) or over during the first 15 hours, and a firm painful swelling with raised edges develops within 24 hours at the injection site. In nonglandered horses, there is no, or minimal, transient local swelling. Doubtful reactors may be retested after 14 days using a double dose of mallein
3. **Strauss reaction:** Strauss reaction is observed when infectious material from glanders patients is injected intraperitoneally into male guinea pigs. In positive cases, the guinea pig develops localized peritonitis involving the scrotal sac. Glanderous orchitis follows with painful enlargement of the

testes. The testis becomes enlarged and painful and ultimately necrotic and is discharged through the scrotal skin.

4. **Nucleic acid identification methods:** Polymerase chain reaction (PCR) is used for the specific detection of *B. mallei* DNA. This test also helps in the differentiation between *B. mallei* and *B.pseudomallei*.
5. **Serological tests:** Agglutination, precipitation, ELISA, CFT and conglutination tests (CFT is approved for International Trade) are routinely performed.
6. **Isolation of bacteria** from pus in glycerol agar or potato agar or serum agar.
7. Differential diagnosis:
   a. **Strangles** is caused by *Streptococcus equi*. It is characterized by fever, anorexia, and depression with swollen submandibular lymph nodes and mucopurulent nasal discharge. The nasal discharge is usually bilateral, whereas it is most often unilateral in cases of glanders. Skin nodules and typical lung lesions are absent. Animals with strangles will not react to mallein testing or serological tests for glanders. *S. equi* is readily demonstrable. Strangles does not develop into a chronic, debilitating condition, and most infected horses recover within a few weeks.
   b. **Epizootic lymphangitis** (caused by *Histoplasma farciminosum*) is characterized by cutaneous nodules originating from superficial lymph vessels. In epizootic lymphangitis, conjunctivitis is a common lesion. Demonstration of the infectious agent and application of the mallein test and serological testing will help distinguish between these diseases.
   c. **Ulcerative lymphangitis** (caused by *Corynebacterium pseudotuberculosis*) is characterized by dermatitis and abscess formation predominantly in the pectoral and ventral abdominal regions. Standard diagnostic tests are again valuable in distinguishing this disease from glanders.
   d. **Melioidosis** (caused by *B.pseudomallei*) is characterized by multiple abscesses in a variety of tissues and organs. Unlike glanders, it is not specifically a disease of equids and occurs most often in sheep, goats, and swine. It is characterized by dyspnea and lameness, but a wide array of clinical signs may be elicited. Diagnosis is confirmed by isolation of the causative organism. Serological cross-reactions occur with *B.mallei*.

Treatment and Control: There is no vaccine. Prevention and control depend on early detection and elimination of affected animals, as well as complete quarantine and rigorous disinfection of the area involved. Treatment is given only in endemic areas. Antibiotics are not very effective. Combinations of sulfazine or sulfamonomethoxine with trimethoprim were found to be efficient in the prevention and treatment of experimental glanders.

Zoonotic importance: The human form of the disease is painful and frequently fatal. Laboratory workers and animal attendants are most at risk. Symptoms of glanders in people include nodular eruption on the face, legs, and arms; involvement of the nasal mucosa; and later pyemia and metastatic pneumonia. Human glanders may be confused with a variety of other diseases, including typhoid fever, tuberculosis, syphilis, erysipelas, lymphangitis, pyemia, yaws, and melioidosis. The diagnosis can be confirmed by serology and by isolation of the causative organism.

Control of Glanders in India: Glanders was considered as very serious disease and as early as in 1899 and Act was passed to control the infection in horses (includes asses and mules). The extracts are given below;

- THE GLANDERS AND FARCY ACT 1899 WITH NOTIFICATIONS AND RULES THERE UNDER (ACT No. XIII OF 1899) PASSED BY THE GOVERNOR GENERAL OF INDIA IN COUNCIL (Received the assent of the Governor General on the 20th March1899).

- It extends to the whole of British India
- When this Act has been so applied to a local area, the local Government may by notification in the local official Gazette appoint such persons as it thinks fit to be Inspectors under this Act and to exercise and perform within the whole of the local area, of such portions thereof as it may prescribe the powers conferred and the duties imposed by this Act on such officers
- Within such limits as aforesaid, the Inspector may seize Power of any horse, which he has reason, to believe to be diseased.
- On any such seizure as aforesaid, the Inspector shall cause the horse seized to be examined as soon as possible by such Veterinary Practitioner as the Local Government may appoint in this behalf.
- If the Veterinary Practitioner certifies in writing that the horse is diseased, the Inspector shall cause the same to be immediately destroyed.
- When any diseased horse has been in any building, shed or other enclosed place, or in any open lines, the Inspector may issue a notice to the owner of the building, shed place or lines or to the person in charge thereof, directing him to have the same disinfected and the internal fittings thereof or such other things found therein or near there to as the Local Government may by rule prescribe, destroyed.
- On the failure or neglect of such owner or other person as aforesaid to comply with the notice within a reasonable time, the Inspector shall cause the billing, shed place or lines to be disinfected and the fittings or other things to be destroyed, and the expense (if any) thereby incurred may be recovered from the owner or other person as if it were a fine.
- Whoever refuses or neglects to comply with any notice issued by the Inspector under section 9, or removes any horse in contravention of section 11, shall be punishable with imprisonment for a term which may extend to one month, or with fine which may extend to fifty rupees, or with both.

MORAXELLA

Important species: *Moraxellabovis*

Important disease: Bovine keratitis, Pink eye, and Bovinekerato-conjunctivitis

General characters:

1. **Morphology:** They are gram-negative rod shaped organisms arranged in pairs and hence are referred as diplobacilli. They are non-motile, non- spore forming organisms with prominent capsules.
2. **Habitat:** They are found generally found in the eye. They can even remain in the eye without causing any infection.
3. **Classification:** *M.bovis* earlier placed in the genus *Haemophius*. It has been moved out of *Haemophius* since it does not require X and V factors for growth.
4. **Resistance:** They are relatively fragile organisms and are sensitive to heat at a temperature of 60°C for 15 minutes and to number of disinfectants.

Cultural characters: It is an aerobic organism and grows at 37°C on blood agar. The colonies are small, circular, translucent and grey in colour surrounded by beta haemolysis. A granular deposit is noticed in broth culture and gelatin is liquefied.

Pathology: The organisms are found in the conjunctiva casing highly contagious kerato-conjunctivitis. The disease is transmitted from one animal to another though infective material. The infection also spreads via conjunctival exudates. There is conjunctivitis followed by keratitis and ulceration of the cornea.

Diagnosis is based on isolation of the organisms in blood agar from conjunctival swabs.

Treatment: Chloramphenicol, tetracycline and sulphonamides.

LEPTOSPIRA

The spirochetes comprise of three important genus that caused infection in man and animals. The three genus are

1. *Leptospira*
   - Leptospirosis
2. *Borrelia*
   - Lyme disease or relapsing fever
   - Fowl spirochetosis
3. *Spirillum*
   - Rat bite fever

Leptospirosis is an infectious disease of animals and man caused by a spiral- shaped organism (spirochete) of the genus *Leptospira*. The important serotypes recognized in livestock are *Leptospira pomona*, *L. canicola*, *L. icterohaemorrhagiae*, *L. grippotyphosa* and *L. hardjo*. These organisms have a wide host range, including man. However, humans are accidental hosts in whom this disseminated disease varies in severity from subclinical to fatal. In humans this infection is referred as **Weil's disease.** Among domestic animals, swine, cattle, dogs, and horses are most frequently affected. Known wildlife hosts include many of the small rodents, raccoons, foxes, opossums, skunks, deer, and moose. Because of the nature of the disease, it should not be considered a problem of the individual animal but a problem of a population, either a herd or a species within the area. The disease can be controlled by proper management and the correct use of available vaccines.

General Characters

1. **Morphology:** *Leptospira* are thin, tightly coiled obligate aerobes that are highly motile (flexuous type of motility). *Leptospira* has the general structural characteristicsthat distinguish spirochetes from other bacteria. The cell is encased in a three- to five- layer outer membrane or envelope. Beneath this outer membrane are the flexible, helical peptidoglycan layer and the cytoplasmic membrane; these encompass the cytoplasmic contents of the cell. The structures surrounded by the outer membrane are collectively called the protoplasmic cylinder. An unusual feature of the spirochetes is the location of the flagella, which lie between the outer membrane and the peptidoglycan layer. They are referred to as periplasmic flagella. The periplasmic flagella are attached to the protoplasmic cylinder subterminally at each end and extend toward the center of the cell. The number of periplasmic flagella per cell varies among the spirochetes. The leptospires have two periplasmic flagella, one originating at each end of the cell. The motility of bacteria with external flagella is impeded in viscous environments, but that of spirochetes is enhanced. The slender (0. 1 μm by 8 to 20 μm) leptospires are tightly coiled, flexible cells. In liquid media, one or both ends are usually hooked. Leptospires are too slender to be visualized with the bright-field microscope but are clearly seen by dark- field or phase microscopy. They do not stain well with aniline dyes.
2. **Habitat:** The primary reservoir hosts are wild animals such as rodents, which can shed leptospires throughout their lifetimes. Leptospires have been isolated from approximately 160 mammalian species. Most leptopsires are pathogenic whereas serotypes of *L biflexa* are not pathogenic and exist in water and soil as free-living organisms.
3. **Classification:** All the pathogenic leptospires were formerly classified as members of the species *Leptospira interrogans*, however the genus has recently been reorganised and pathogenic leptospires

are now identified in several species of *Leptospira*. The recent classification is based on nucleotide sequences. There are more than 200 distinct leptospiral serovars recognised and these are arranged in 23 serogroups. Some of the important pathogenic species are listed below; *Leptospira pomona*, *L. canicola*, *L. icterohaemorrhagiae*, *L. grippotyphosa* and *L.hardjo*

4. **Resistance:** They are fragile and do not withstand dryness or heat. They can survive in alkaline waters for upto six months.
5. **Antigens and Toxins:** They do not produce exotoxin and do not have endotoxin

Cultural characters: The leptospires are the most readily cultivated of the pathogenic spirochetes. They have relatively simple nutritional requirements; long-chain fatty acids and vitamins B1 and B12 are the only organic compounds known to be necessary for growth. When cultivated in media of pH 7.4 at 30°C, their average generation time is about 12 hours. Aeration is required for maximal growth. They can be cultivated in plates containing soft (1 percent) agar medium, in which they form primarily subsurface colonies. They are also grown in semisolid (0.1–0.2% agar) medium containing BSA and either Tween 80. Fastidious leptospires prefer a medium containing o.4–1% rabbit serum (Stuart's media; Fletchers media). Cultures are incubated at 29 +/- 1°C for at least 16 weeks, and preferably for 26 weeks.

Pathogenesis

Leptospirosis is generally contracted by the direct splashing of urine from infected or carrier animals into the eyes of susceptible animals. It can also be spread through the skin and mucous membranes from contact with water contaminated with leptospires. Transmission may also occur during breeding through residual urine in the genital tract or through infectious semen. The major sources of contamination are swine, cattle, dogs, and wildlife that have recovered from the disease and have become carriers. Cattle and swine are major sources because of the volume of urine and the extent and duration of leptospires in the urine. Human cases are acquired through direct contact with the urine of infected livestock or with contaminated soil or water. Swimming in water contaminated with the urine of infected livestock also serves as a source for human infection. The organism may survive up to six months in alkaline water.

The mucosa and broken skin are the most likely sites of entry for the pathogenic leptospires. A generalized infection ensues, but no lesion develops at the site of entry. Bacteremia occurs during the acute, leptospiremic phase of the disease. The host responds by producing antibodies that, in combination with complement, are leptospiricidal. The leptospires are rapidly eliminated from all host tissues except the brain, eyes, and kidneys. Leptospires surviving in the brain and eyes multiply slowly if at all; however, in the kidneys they multiply in the convoluted tubules and are shed in the urine (the leptospiruric phase). The leptospires may persist in the host for weeks to months; in rodents they may be shed in the urine for the lifetime of the animal. Leptospiruric urine is the vehicle of transmission of this disease. The mechanism by which leptospires cause disease remains unresolved because neither endotoxin nor exotoxins have been associated with them. The marked contrast between the extent of functional impairment in leptospirosis and the scarcity of histologic lesions suggests that most damage occurs at the subcellular level. Damage to the endothelial lining of the capillaries and subsequent interference with blood flow appear responsible for the lesions associated with leptospirosis. The most notable feature of severe leptospirosis is the progressive impairment of hepatic and renal function. Renal failure is the most common cause of death.

Symptoms

Cattle: Leptospirosis in cattle may vary in severity from a mild, inapparent infection to an acute infection that may cause death. Acute clinical disease occurs more often in young calves, but all ages

may be affected. Cattle may develop a high fever of 104° to 107°F, depression, loss of appetite, decreased milk production, and weakness. Hemoglobinuria (coffee-colored urine), anemia, icterus (jaundice), and bloody milk are also seen. In lactating cows, the udder becomes flaccid and milk flow nearly ceases. The milk is usually abnormal and may contain thick yellow or reddish clots. Reddish milk may develop with the onset of hemoglobinuria. Abortion occurs two to five weeks after initial infection with leptospires in the last three months of pregnancy.

Swine: The disease in swine is largely subclinical (no apparent signs of illness) except for abortions, which usually occur during the last two to three weeks of pregnancy. The sow aborts pigs rapidly with no apparent signs of illness. Some aborted fetuses may have been dead a short time. Piglets may be born weak and die shortly after birth.

Horses: The infected animal may have a slight temperature rise and mild loss of appetite. Leptospirosis caused by *L. pomona*is widespread among susceptible horse populations. Within 12 to 14 months after the initial infection, the eyes of many horses show evidence of uveitis (inflammation of the internal structures of the eye), a disease commonly known as periodic ophthalmia or "moon blindness." Leptospirosis is not a common cause of abortion in horses.

Sheep and goats: The signs reported are similar to those described for cattle.

Dogs: *Leptospira canicola* is the most common serotype; *L. icterohaemorrhagiae*, *L. pomona,* and *L. grippotyphosa* are also responsible for some infections. The acute infection is characterized by elevated body temperature, vomiting, muscular stiffness, weakness, and nephritis (inflammation of the kidney). In severe cases, jaundice and death may occur. Central nervous system signs may occur with or without other clinical signs, and organisms may be present in the brain tissue for extended periods. Chronic leptospirosis is primarily associated with chronic kidney degeneration. Shedding of leptospires in the urine may continue for over ayear.

Cats: *Leptospira* infection and disease are very rare in cats.

Rodents: Wild rat (*Rattus norvegicus*) populations in urban and rural areas are frequently infected with *L. icterohaemorrhagiae*. *Leptospira ballum* is found primarily in wild mice (*Mus musculus*), but is also present in some rat populations and has been present in commercially raised laboratory mice. Other serotypes occur with less frequency in rodent populations.

Diagnosis

1. **Based on symptoms and lesions:** Leptospirosis diagnosis in all animals, including livestock, companion, and wild animals, must be confirmed by laboratory tests. There are no clinical signs distinctive to this disease but certain signs are suggestive.
2. **Serological tests:** The ideal method for diagnosis of leptospirosis is the demonstration of significant levels of antibodies to leptospires in the serums of recovered animals. Plate agglutination tests or microscopic agglutination tests are normally performed.
    a. Plate agglutination test with killed antigens or the microscopic agglutination (MA) test, in which living organisms are used as antigens. Titers greater than 1:100 by MA test or 1:40 or greater by plate test against one or more serotypes are generally considered significant.
    b. ELISA:

3. Microscopic demonstration:

a. Leptospires are too slender to be visualized with the bright-field microscope but are clearly seen by dark-field or phase microscopy.

b. Leptospires do not stain satisfactorily with aniline dyes, and silver- staining techniques lack sensitivity and specificity.

c. **Immunohistochemical staining:** Leptospires can also be demonstrated by a variety of immunochemical staining techniques,e.g. immunofluorescence. This technique is in diagnosing infection in pathological material that is unsuitable for culture or where a rapid diagnosis is required. The technique is less suitable for diagnosing the chronic carrier state, where the numbers of organisms may be very low or localised.

4. **Isolation and identification of leptospires:** Leptospires are most readily isolated from blood or milk taken from animals during the acute phase, or from the urine, kidneys, spinal fluid, and brains of recovered animals.

5. **Nucleic acid identification methods:** Polymerase chain reaction (PCR)- based assays are now used in the detection of leptospires in tissues and body fluids.

Control

1. Livestock herds can be protected against leptospirosis by a combination of proper management and vaccination procedures.
2. Cattle used for breeding purposes should be vaccinated during early pregnancy to provide the greatest degree of protection during the last two-thirds of gestation.
3. The vaccine used in infected herds should be identical with the serotype causing the diseases, as there is little or no cross-protection between vaccine serotypes. The vaccine should be given to all susceptible livestock on the premises where infection has been identified.
4. Annual vaccination is recommended for dogs with a booster given if an outbreak occurs in the area.
5. Control of rodents in animal houses.

Treatment

Streptomycin, chlortetracycline, or oxytetracycline are the antibiotics of choice for treatment.

BORRELIA - Fowl spirochetosis

General characters: The causal organism, *Borrelia anserina*, is an actively motiles pirochete,~0.2-0.3 μm × 8-20μm and consists of 5-8 loosely arranged coils.

Cultural characters: in vitro cultivation is not possible. However, they can be propagated in embryonating duck or chick embryos or in young ducks or chicks.

Pathogenesis: *Argas* spp transmit the disease in different geographic areas. Generally, an infected *Argas* tick can transmit the disease at every feeding and maintains the infection throughout larval, nymphal, and adult stages. The ticks also transmit the infection transovarially, ie, the F_1 larvae are infective. Other vectors (lice, mosquitoes, some species of ticks, inanimate objects) can transmit the spirochete mechanically to a susceptible host. Ingestion of bile-stained fecal droppings containing the spirochete, as well as acts of cannibalism during spirochetemia can result in infection. After the bite of an infected tick, the incubation period is ~4-7 days.

Symptoms

Signs are highly variable, depending on the virulence of the spirochete, and are not pathognomonic. They include listlessness, depression, somnolence, moderate to marked shivering, and increased thirst. Young birds are affected more severely than older ones. During the initial stages of the disease, there

is usually a greenish yellow diarrhoea with increased urates. The course of the disease is 1-2 wk. In tick-infested geographical areas, morbidity can approach 100% and mortality >90% has been recorded.

Lesions: An enlarged spleen with petechial or ecchymotic hemorrhages is the most notable gross lesion. Occasionally, the liver may be swollen and contain focal areas of necrosis. Kidneys may be enlarged and pale. A green, catarrhal enteritis is common.

Diagnosis:

1. Diagnosis rests on demonstration of *Borrelia* in the blood, either as actively motile *Borrelia* during darkfield microscopy, or as stained spirochetes in Giemsa-stained blood smears.
2. Agar-gel diffusion and various serological tests have been described but are of questionable value

Treatment

The most widely used antibiotics are penicillin derivatives, but the streptomycins and tetracyclines are also effective.

Control

1. Control of biological vector *Argas*ticks
2. Bacterins prepared from infective blood are used as vaccines.

MYCOPLASMA

Members of the genus *Mycoplasma* are the smallest organisms lacking cell walls that are capable of self-replication and cause various diseases in humans, animals, and plants. Mycoplasmas are unique because they are neither virus nor bacteria. Mycoplasmas do not have a protective cell wall and are minimal in the area of genetic material. Due to this fact it is very hard for the hosts' immunity to fight off the disease. The animal's immune system targets disease-causing bacteria for destruction by locking on the shape and size of the bacteria's surface structures. In the case of mycoplasma, these structures keep changing making their removal difficult for immune system. *Mycoplasma* often colonizes in the respiratory tract, the bloodstream, and other mucosal surfaces. For this reason it is very hard to control due to animal's contact. Even in confinement houses animals sneeze and can spread the disease. Over 90 organisms belonging to this group have now been recognized.

Type species: *Mycoplasma mycoides*

General characteristics:

1. **Morphology:** *Mycoplasma* are the smallest and simplest prokaryotic cells capable of self-replication. They do not have cell wall hence they are highly pleomorphic (filamentous, pear shaped, spherical). They are surrounded by a lipoprotein plasma membrane. Some multiply by binary fission where as some have a reproductive cycle. They are poorly gram-negative and are stained by Giemsa method.

   The genome of mycoplasma is dsDNA and is smallest of any replicating prokaryotic cell. They have the smallest genome size and, as a result, lack many metabolic pathways and require complex media for their isolation.

2. **Habitat:** They are generally found as commensals in mucous membrane of upper respiratory, digestive, genital tracts and udder.

3. **Classification:** All mycoplasma belong to the class Mollicutes having two families Mycoplasmataceae and Acholeplasmataceae and three genuses *Mycoplasma, Acholeplasma and Ureaplasma.*

4. **Resistance:** They are killed at moist heat and by dry temperatures of 60^0C for 5 minutes. They can remain viable for long time in freeze dried cultures and frozen lung. They are sensitive to disinfectants.
5. **Antigens and toxins:** Multiple antigenic components are reported in mycoplasmas. Certain organisms produce exotoxins. *M.neurolyticum,* which causes rolling disease in mice, produces a neurotoxin.

Cultural characteristics: They are fastidious organisms and require cholesterol for their growth. They grow aerobically at a carbon dioxide tension of 10%. The media used for isolation of mycoplasma contain 20% unheated horse serum, 10% fresh yeast agar, and very minute quantity of DNA, penicillin and thallous acetate. The pH should be between 7 and 8. The optimum temperature is 36-38C. The colonies are microscopic and appear after 3-4 days. They have yellowish, opaque centre surrounded by a thin translucent rim resembling poached egg or fried egg appearance. Growth is difficult in fluid medium and a granular type growth is noticed. Seven-day-old embryonated eggs are used for cultivation of mycoplasmas. After two days the embryo will die showing extensive oedema and swelling. Cell culture systems like HeLa cells, chicken heart fibroblasts and human conjunctival cells are used for cultivation. The biochemical reactions, growth inhibition test and metabolic inhibition test are of use in diagnosis.

Pathogenesis:

Many Mycoplasmas live as commensals in the respiratory, digestive, urinary tracts and udder. Their plastic nature besides an adhesion protein helps the organisms to attach themselves to the mucous membrane. The destruction of cilia in mucous membrane predispose the cells for mycoplasma infection. The fibrinous exudates protect them from phagocytic cells and antimicrobial drugs. The accumulation of mycoplasmal metabolites is also responsible for the pathogenesis. The important virulence factors are adherence factors and toxic metabolic products. The immune response also aids in spread of the infection.

Adherence factors – The adherence proteins are one of the major virulence factors. The adherence protein called P1. The P1 Adhesin localizes at tips of the bacterial cells and binds to sialic acid residues on host epithelial cells. This protein has been identified only in *M.pneumoniae* and not identified in other Mycoplasmas. Colonization of the respiratory tract results in the cessation of ciliary movement. The normal clearance mechanisms of the respiratory tract do not function, resulting in contamination of the respiratory tract and the development of a dry cough.

Toxic Metabolic Products - The toxic metabolic products of Mycoplasmas accumulate and damage host tissues. Both hydrogen peroxide and superoxide are products of mycoplasma metabolism and are responsible for pathogenesis. The mycoplasmas also inhibit host cell catalase and increase the peroxide concentrations further facilitating the infection.

Both acute and chronic infections are produced. The important mycoplasma species and the infection produced by them are listed below

1. Mycoplasma of birds

   Avian mycoplasmas are generally transmitted by eggs. Affected birds will have respiratory distress, drop in egg production and poor growth rate.

   - *M.gallisepticum* – Chronic respiratory disease (CRD), air sac disease of chicken, turkeys and other fowl.
   - *M.synoviae:* Infectious synovitis of chickens and turkeys. Infections are seen in hock and wing joints.

- *M.meleagridis:* Air sacculitis in turkeys.
- *M.iowae:*Air sacculitis

2. Mycoplasma of cattle:
    - *M.mycoides ss. mycoides:* Causes contagious bovine pleuropnemonia (CBPP). The infection is characterised by septicaemia and suppurative lesions in lungs, pleura and pericardium. It also causes abortion and results in infertility.
3. Mycoplasma of sheep and goat:
    - *M.mycoides ss. Capri:* Causes contagious caprine pleuropneumonia (CCCP).
    - *M.agalactiae:* Contagious agalactia in sheep and goat.
4. Mycoplasma of pigs:
    - *M.hyorhinis:* Polyserositis and arthritis.
    - *M.hyosynoviae:* Arthritis.

MYCOPLASMA INFECTION IN POULTRY

MYCOPLASMAGALLISEPTICUM

Mycoplasma gallisepticum infection is commonly designated as chronic respiratory disease in chickens and infectious sinusitis in turkeys. Infection may also occur in pheasants, chukar partridges, and peafowl. The disease is worldwide. Its effects are most severe in large commercial operations during winter. MG (*Mycoplasma gallisepticum*) is one of the major health problems facing the poultry industry. The severity and appearance of disease may be influenced by concomitant infection and other pathogens and/or by predisposing factors. Predisposing factors include nutritional deficiencies and intensive management with excessive ammonia and dust. Other pathogens include the viruses of Newcastle disease and infectious bronchitis, including vaccine strains and pathogenic strains of *E. coli* and *Avibacterium paragallinarium*.

Under experimental conditions the incubation period between exposure and infection is from 6 to 21 days. Under field conditions, the incubation period is not established due to many variables, which influence the onset and the extent of disease. Field outbreaks of MG near the onset of production (between the 26th and 38th weeks of age) may suggest a long incubation period. Mycoplasma spread rapidly within a population of birds (7-14 days). The disease spread in the flock by direct contact (bird to bird) with infected carriers, airborne dust and droplets, contact with contaminated equipment, crates, feed, water, and litter, and other species of birds, wild birds and rodents. MG can also spread through the hatching eggs (transovarian) and/or hatchery. The severity and intensity of MG infection is variable. It is more severe and of longer duration in the cold months and it affects younger birds more severely than mature birds. Recovered birds may become carriers of MG.

Symptoms and lesions: Morbidity and mortality in the field outbreaksvary widely depending on the environmental and climatic conditions and type of secondary infection. The infection may vary from mild with low mortality to severe with high mortality In broilers, mortality may be high if management is poor and birds are exposed to stress and other secondary infection. Laying hens usually experience low mortality, but many of the birds have an unthrifty appearance. Very high mortality may be seen in turkeys. Signs of MG may develop slowly in the flock. Respiratory signs usually persist for weeks. The first sign is a nasal discharge followed by a foamy or bubbly condition of the eyes. Birds may show a persistent hacking cough, sneezing and sniffing, and tracheal rales. Poor physical condition and loss of weight are usually apparent. In broilers, the onset of signs is severe

and includes severe reduced growth rate and feed intake. Layers will show a drop in egg production and feed consumption. Production may continue at a lower level. The primary lesions in turkeys are air sacs and lung lesions. There is also swelling of one or both infraorbital sinuses. Important lesions include swelling of nasal passages, sinuses, trachea, and bronchi. Air sacs are often thick and opaque and may contain slimy or cheesy exudate.

Control strategy for MG infection in poultry

1. Completely depopulate the infected premises (removing all birds as well as all used feed supplies, removable equipment, and supply items that may be contaminated) to establish a "MG-free"flock.
2. Remove all of the litter and manure completely.Wash down all inside surfaces (walls, ceilings, rafters, ventilation openings, air intake ducts, etc.) with a high pressure washer using a large volume of water. The addition of a cleaning agent speeds up and improves the process.
3. Disinfect all of the interior surfaces of the building with recommended concentration of phenolic or cresylic acid disinfectant solution. Chlorine bleach solution (.025%) may be substituted for phenol or cresylic acid solutions.
4. Spray the entire floor area, the lower three to four feet of the support poles, the perimeter walls, and the area outside the entrance with 0.1% glutaraldehyde solution at one gallon per 10 square feet. Also, spray a five- to ten- foot band around the entire outside perimeter to prevent contamination from outside the building. Close up the building and make it tight without any ventilation until the next day, and then open it up and air out the formaldehyde fumes.
5. Spray the entire floor area, support poles, and side walls with carbaryl insecticide at recommended levels to control insect pests. Malathion may be added to, and applied with, the cresylic acid spray in place of the separated carbaryl applications.
6. Bring back the necessary removable equipment that has been cleaned and disinfected and bring in feed and supply items that are also clean and not contaminated with pathogens.
7. Allow the buildings to remain empty for at least two weeks after clean out.
8. Always consider the area outside the building contaminated, particularly if the previous flock had MG. Set up your traffic pattern to prevent recontaminating the building from the area outside.
9. Obtain chicks hatched from eggs from MG-free breeder flocks, hatched in a hatchery where no infected eggs are set, and delivered in "MG-free" trucks by people who have no contact with infected birds.
10. Farm workers go from younger to older birds (not vice-versa) if mutliple ages are present on the farm and cared for by the same person.
11. Have only one age group on the farm (all-in, all-out) or have barns adequately separated (about 650 feet or more) and treat each age group as a separate unit. Clean up and disinfect barns between flocks.
12. Enforce regulations on clothing and footwear worn by flock caretakers, servicemen, and equipment repairmen that move between flocks and farms. You can significantly influence the security and isolation of a healthy flock from the rest. Use "on- farm" clothing (including coveralls, boots, hats, etc.) in poultry barns.
13. Feed truck, egg pick-up, and other service vehicle drivers must stay in the truck if servicing both infected and free farms (or if that is not possible, avoid the barn entrance area).

Treatment: Most strains of *M gallisepticum* are sensitive to a number of antibiotics, such as chlortetracycline, erythromycin, oxytetracycline, spectinomycin, tiamulin, tylosin, or a fluoroquinolone such as enrofloxacin.

MYCOPLASMA IOWAE: Infection is common in turkey flocks. It is a relatively uncommon infection of chickens. *Mycoplasma iowae* is egg transmitted. Many strains of *M iowae* are lethal to turkey embryos. Affected turkey breeder flocks show no clinical signs other than reduced hatchability (usually2-5%). In many flocks, the hatchability returns to normal after 1-2 months. Most embryos die during the later stages of incubation. Dead turkey embryos are edematous, congested, and stunted; they may have clubbed down. Poults infected *in ovo* or at 1 day of age may develop various skeletal deformities such as rotated tibia, deviated toes, chondrodystrophy, or erosion of the articular cartilage of the hock joint. Feathers may also be poorly developed.

MYCOPLASMA MELEAGRIDIS: It is a widespread, egg-transmitted disease of turkeys found worldwide. The primary lesion is airsacculitis. *Mycoplasma meleagridis* is thought to be a specific pathogen for turkeys, and the organism is commonly found in the respiratory and reproductive tract.

MYCOPLASMA SYNOVEAE: The organisms cause airsacculitis and synovitis. It is also a egg transmitted infection.

MYCOPLASMA INFECTIONS IN CATTLE

CONTAGIOUS BOVINE PLEUROPNEUMONIA(CBPP)

CBPP is a disease of bovines. The causative agent of contagious bovine pleuropneumonia (CBPP) is *Mycoplasma mycoides* subspecies *mycoides* SC (small colony). CBPP is a respiratory illness characterized by the presence of sero-fibrinous, interstitial pneumonia, interlobular oedema and hepatization giving a marbled appearance of the lung and capsulated lesions termed sequestra in the lungs of affected cattle. The occurrence of subacute, symptomless infections and chronic carriers after the clinical phase of the disease creates major problems in the control of this disease. Infection is spread to a susceptible animal by the inhalation of expired breath laden with M. m. mycoides SC from an infected animal.

Symptoms: There is considerable variation in the degree of symptoms seen in cattle affected with CBPP ranging from the hyperacute through acute to chronic and sub-clinical forms. Respiratory distress and coughing, evident on stimulation of resting animals, are the main signs of CBPP.

Acute Form: Animals show dullness, anorexia, and irregular rumination with moderate fever, and may show signs of respiratory distress. Coughing is usually persistent and is slight or dry. Sometimes body temperature rises from 40 to 42^0C and the animal prostrates with difficulty of movement.

Hyperacute Form: The clinical signs observed in the hyperacute form are greatly accelerated. The pathological signs are usually characteristic with marked pleural adhesion accompanied by exudative pericarditis. Affected animals may die within a week of exhibiting classical respiratory signs.

Subacute/Chronic FormsL In the subacute form, symptoms may be limited to a slight cough, only noticeable when the animal is exercised.

Lesions: The pathological lesions of this disease are confined to the thoracic cavity and lungs, and lesions are usually unilateral. The thoracic cavity of affected animals may contain many litres of clear yellowish brown fluid containing some pieces of fibrin. Caseous fibrinous deposits are observed on the parietal and visceral surfaces of the lungs. The interlobular septa of the affected lung show distension with amber- coloured fluid surrounding the distended lymphatics. This fluid separates the lung lobules, which vary in colour with red, grey and yellow hepatization being evident indicating different stages of inflammatory lesions. Consolidation of the lungs with typical marbled appearance, sometimes accompanied by adhesion of the parietal and visceral surfaces is also characteristic.In

addition to respiratory disease, affected calves may present exudative peritonitis, arthritis, bursitis and fibrinous arthritis of the carpal and tarsal joints.

MYCOPLASMA INFECTIONS IN SHEEP AND GOAT

CONTAGIOUS AGALACTIA OIE LISTING – LIST B DISEASE

Contagious agalactia is a disease of sheep and goats that is characterised by mastitis, arthritis and keratoconjunctivitis. It is mainly caused by *Mycoplasma agalactiae*. Of late *M. capricolum* subsp. *capricolum* (*Mcc*) and *M. mycoides* subsp. *mycoides* LC (*Mmm*LC; LC = large colonies) have also been isolated in many countries from sheep and goats with mastitis and arthritis.

Symptoms: Following an incubation period of 60 days from natural exposure, body temperature rises to a range from 41 to 42C. The period of fever corresponds to the time of septicaemia. During early stages, infected animals become depressed and anorectic, and some may die. As it progresses, the disease usually develops signs of keratitis and arthritis and, in female sheep, mastitis and abortion. The acidity of the milk from infected udders changes from a normal pH 6.8 to pH 7.8, becomes yellow in colour and, on standing, separates into a light green supernatant layer and a grumous sediment. Gradually, the udder atrophies and the milk yield diminish. Joints, especially carpal and tarsal, become swollen, painful, and lame. In some animals, a joint may rupture and discharge exudate. One or both eyes may become infected, leading to impaired vision and blindness. Depending on the stage of disease development, the infected eye may show congestion and swelling, punctiform keratitis with yellow opacity of the cornea, or rupture of the cornea. Ewes in the third trimester of gestation may abort either dead or living infected lambs and later develop vaginal discharges.

Lesions: The infected udder is grossly atrophic in either one or both halves of the organ. Microscopically, the chronic inflammatory reaction in the stroma shows increased fibrosis and a reduced number of glandular acini. Infected joint capsules are oedematous, and the synovium may contain clumps of fibrin. Articular surfaces maybe eroded and occasionally ankylosed. In early stages of keratitis, the cornea is oedematous and infiltrated with leukocytes. In advanced stages, abundant purulent exudate infiltrates both cornea and the ciliary body.

CONTAGIOUS CAPRINE PLEUROPNEUMONIA (CCPP): CCPP is an infectious disease which affects only goats. The causative agent of contagious caprine pleuropneumonia (CCPP) is *Mycoplasma capricolum* subsp. *capripneumoniae* (Mccp). In natural infections, the organisms are acquired by susceptible goats by inhalation of contaminated droplets from infected goats. Due to the high sensitivity of mycoplasmas to the external environment, close contact is essential between infected and naive animals for transmission to take place, and, overcrowding and confinement favours close contact and circulation of mycoplasmas. Stress factors due to malnutrition and movement over long distances can predispose the animal to disease.

Symptoms: The primary clinical signs are cough with animals tending to lie down or lag behind the flock. Affected animals continue to graze for some time but eventually become anorectic; breathing becomes laboured with painful grunting and a rise in temperature up to 41^0C. Gradually, the respiratory symptoms become prominent, respiration is accelerated and painful, and is followed by violent coughing. In the terminal stages, the animals are unable to move. They stand with their legs abducted, the neck is stiff and extended downward, saliva continuously drips from their mouth and their nose is obstructed by mucopurulent discharge. The tongue protrudes and the animals bleat distressingly. The organism is not reported to affect organ systems other than the respiratory tract of goats. In endemic areas subacute and chronic cases are common and the symptoms are milder, dominated by intermittent coughing.

Lesions: The gross pathological lesions are localized exclusively to lung and pleura and are often unilateral. Affected lungs can be totally hepatized, and have a port wine colour. A lung section shows a fine granular texture with various colours, but usually without any thickening of the interlobular septa. There is often an abundant pleural exudate and conspicuous pleuritis. The pleural exudates can solidify and form a gelatinous covering sometimes over the whole lung. In acute cases, the pleural cavity contains an excess of straw-coloured fluid with fibrin flocculations. In chronic cases there is a black discolouration of the lung tissue and sequestration of the necrotic lung areas. Adhesions between the lung and the pleura are very common and often very thick.

Diagnosis:

1. **Symptoms and lesions** are very conclusive.
2. **Isolation and identification by culturing:** The medial used for isolation of mycoplasma contain 20% unheated horse serum, 10% fresh yeast agar, and very minute quantity of DNA, penicillin and thallous acetate. The pH should be between 7 and 8. The optimum temperature is 36-38^{0}C. The colonies are microscopic and appear after 3-4 days. They have yellowish, opaque centre surrounded by a thin translucent rim resembling poached egg appearance. Growth is difficult in fluid medium and a granular type growth is noticed. Seven-day-old embryonated eggs are used for cultivation of mycoplasmas. After two days the embryo will die showing extensive oedema and swelling. Cell culture systems like HeLa cells, chicken heart fibroblasts and human conjunctival cells are used for cultivation.
3. Nucleic acid identification method
4. Agglutination reaction using specific antisera.
5. Complement fixation test (OIE accepted test).
6. ELISA.
7. Direct and indirect FAT.
8. Growth inhibition test using specific antisera.
9. Biochemical reaction.

Treatment and control: screening the flock supplying egg and eliminating the flock can control avian mycoplasma. Treatment of eggs with antiseptics prevents the occurrence of Mycoplasma in poultry. Live and killed vaccines are available for cattle infection.

RICKETTSIA

The study about *Rickettsia* and *Chlamydia* organisms is called as rickettsiology. Rickettsia and Chlamydia are group of organisms that are placed in between the bacteria and viruses. They have some properties that are found in both bacteria and viruses. However, the majority of the properties of rickettsia are found in bacteria and they are closely related to bacteria. Rickettsias are so named to perpetuate the memory of **Dr.H.D.Ricketts**, who worked on typhus (important disease caused by *Rickettsia* organisms) and died of the same disease after accidentally contracting it. The rickettsiae are a diverse collection of obligately intracellular Gram-negative bacteria found in ticks, lice, fleas, mites, chiggers, and mammals. They include the genera *Rickettsiae, Ehrlichia, Orientia*, and *Coxiella*. This genus consists of two antigenically defined groups: spotted fever group and typhus group, which are related; scrub typhus rickettsiae differ in lacking lipopolysaccharide, peptidoglycan, and a slime layer, and belong in the separate, although related, genus *Orientia*. Rickettsias are coccoid or rod shaped organisms in the size range of 0.3-0.7 m wide and 1-2 m long. Except one organism all the remaining rickettsia are obligate parasites (*Rochalimaea Quintana* – trench fever). They cannot be cultivated in

the artificial laboratory media. The following table summarises some of the important properties of rickettsia, Chlamydia, viruses and bacteria

Properties	*Rickettsia*	*Chlamydia*	Viruses	Bacteria
Structural				
1) Nucleic acid	RNA and DNA	RNA and DNA	RNA or DNA	RNA and DNA
2) Ribosomes	Present	Present	Absent	Present
3) Cell wall	Muramic acid, Diaminopimelic acid (DPA) preset	Muramic acid, DPA absent	No cell wall	Muramic acid
4) Structural integrity during multiplication	Maintained	Maintained	Lost	Maintained
5) Visibility under light microscopy	Visible	Visible	Not visible	Visible
Metabolic capabilities				
1) Macromolecular synthesis	Carried out	Carried out	Only with use of host machinery	Carried out
2) ATP generating system	Present	Absent	Absent	Present
3) Sensitivity to antibiotics	Sensitive	Sensitive (except penicillin)	Not sensitive	Sensitive
4) Sensitivity to interferon	Not sensitive	Sensitive	Sensitive	--
Replication				
1) Mode of replication	Binary fission	Binary fission and budding	Dependence on host cell machinery	Binary fission
Cultivation				
1) Growth in artificial media	Not possible	Not possible	Not possible	Possible
2) Intracellular replication	Possible	Possible	Possible	Variable

GENERAL PROPERTIES OF RICKETTSIA

DEFINITION: The rickettsiae are bacteria that are obligate intracellular parasites. They are considered a separate group of bacteria because they have the common feature of being spread by arthropod vectors (lice, fleas, mites and ticks). They are true bacteria as they have metabolic enzymes and cell walls, use O_2, and are susceptible to antibiotics, although they require living cells for growth. Most rickettsia are maintained in nature by a cycle involving an animal reservoir and an insect vector (usually an arthropod) that infects animals. Most species are found only in the cytoplasm of host cells, but those, which cause spotted fevers multiply in nuclei as well as in cytoplasm. In the laboratory, they may be cultivated in living tissues such as embryonated chicken eggs or vertebrate cell cultures.

GENERAL CHARACTERS:

1. **Morphology:** The cells are extremely small (0.25 u in diameter) rod- shaped, coccoid and often pleomorphic microorganisms, which have typical bacterial cell walls, no flagella (except for *Rickettsia prowazekii*), are gram-negative and multiply via binary fission only inside host cells. They occur singly, in pairs, or in strands**.** The structure of the typical rickettsia is very similar to that of Gram-negative bacteria. The typical envelope consists of three major layers: an innermost cytoplasmic membrane, a thin electron dense rigid cell wall and an outer layer. The outer layer resembles typical membranes in its chemical composition and its tri-laminar appearance. The cell wall is chemically similar to that of Gram-negative bacteria in that it contains diamino pimelic acid and lacks teichoic acid. Intracytoplasmic invaginations of the plasma membrane (mesosomes) and ribosomes are also seen. There are no discrete nuclear structures. The flagellum of *R. prowazekii* is similar to that of other bacteria. The rickettsiae have many of the metabolic capabilities of bacteria, but require an exogenous supply of cofactors to express these capabilities. The response to exogenous cofactors implies an unusually permeable cytoplasmic membrane.

   Electron microscopic studies of the rickettsial cell show its close resemblance with bacterial cell. The cell wall contains muramic acid and diaminopimelic acid. The nucleolar material contains both RNA and DNA. The DNA is of double stranded form with GC content varying from 29-33% in different species of rickettsia.

Species	Cellular Location	Delevopmental Cycle	Axenic Cultivation	Lipopolysaccharide	Major Outer Membrane Protiens (kDa) Antigens	Percent G+C Content
Rickettsia rickettsii	Endothelial and smooth muscle cytosol and nucleus	No	No	Yes	190, 135	33
Rickettsia prowazekii	Endothelial cytosol	No	No	Yes	135	29
Orientia (*Rickettsia*) *tsutsugamushi*	Cytosol, rarely nucleus	No	No	No	70, 54-56, 46-47	28.5-30
Ehrlichia chaffeensis	Cytoplasmic endosomal membrane-bound vacuole	No	No	Unknown	30, 29, 28	Unknown
Coxiella burnetii	Macrophage phagolysosome	Apparently	No	Yes	28	43
Bartonella quintana	Extracellular, attached to louse gut epithelium	No	Yes	Yes	100, 75, 60, 35, 17	39-41

2. **Replication:** Rickettsia normally multiply by transverse binary fission with a doubling time of about 8 hours. For binary fission to take place both rickettsial cell and host cell should be in active state. First the rickettsial cell enters into the host cell, multiply inside it and when sufficient numbers or daughter cells are produced they come out of the host cell by rupturing it. Under poor nutritional conditions, the rickettsia cease dividing and grow into long filamentous forms, which subsequently undergo rapid and multiple division intothe typical short rod forms when fresh nutrientis added. Immediately after division, therickettsia engages in extensive movements through the cytoplasm of the cell. *C. burnetii* differs from other rickettsia in that it is enclosed in a persistent vacuole during growth and division. Six to ten daughter cells will form within a host cell before the cell ruptures and releases them.
3. **Classification:** The family Rickettsiaceae is taxonomically divided as per the following flow chart. The important differentiating features of three important genus are given below:
   a. *Rickettsia* - obligate intracellular parasites, which do not multiply within vacuoles and do not parasitise white blood cells.
   b. *Ehrlichia* - obligate intracellular parasites, which do not multiply within vacuoles but do parasitize white blood cells.

c. *Coxiella* -obligate intracellular parasite, which grows preferentially in vacuoles of host cells.

Genus *Rochalimaea: Bartonella (Rochalimaea) quintana*, the agent of trench fever, was formerly considered as a rickettsial agent and now moved to the family Bartonellaceae. Earlier it was considered as theonly rickettsia that can be cultivated extracellularly. ***Bartonella henselae* – cat scratch fever**

4. **Resistance:** The resistance of different members of the family Rickettsiaceae varies widely. Generally they are fragile organisms and cannot survive for a long time outside a cell. Certain rickettsial organisms like *Coxiella burnetti* that causes Q fever are very resistant to physical damages since they produce an endospore form. They can withstand even pasteurisation.
5. **Transmission:** Rickettsias do not survive outside the host cell for a long time and hence they must be transported between animals by arthropod vectors. When and arthropod bites a rickettsia infected patients, the organisms enters into the arthropod where they penetrate into epithelial cells of gastrointestinal tract, multiply there and are released in through the faeces. Other methods of transmission are also possible.

CULTIVATION: They are difficult to cultivate and require a living system for the growth. They are cultivated in embryonated eggs of 6-8 days old by yolk sac route. The antigen is separated from yolk membrane. Lab animals like guinea pigs, rats and feral animals are also used. Most of the organisms are potential pathogens to human beings. Hence, attempts to isolate the organism from clinical cases or routine laboratory cultivation are done in laboratories with containment facilities.

PATHOGENESIS: Cattle, sheep, goat, and dog cats are the important hosts affected by rickettsial organisms. The main mode of spread is through ticks. Ticks act as biological vectors. Other modes of spread are ingestion and droplet infection. Rickettsia generally multiplies in the gut epithelium of ticks and in vascular endothelial cells of animals. In their arthropod vectors, the rickettsia multiply in the epithelium of the intestinal tract; they are excreted in the faeces, but occasionally gain access to the arthropods salivary glands. They are transmitted to man, via the arthropod saliva, through a bite. In their mammalian host, they are found principally in the endothelium of the small blood vessels, particularly in those of the brain, skin and heart. Hyperplasia of endothelial cells and localized thrombus formation lead to obstruction of blood flow, with escape of RBC's into the surrounding tissue.

Inflammatory cells also accumulate about affected segments of blood vessels. This angiitis appears to account for some of the more prominent clinical manifestations, such as petechial rash, stupor and terminal shock. Death is ascribed to damage of endothelial cells, resulting in leakage of plasma, decrease in blood volume, and shock. It is also assumed that the observed clinical manifestations of a rickettsial infection are due to production of an endotoxin, although this endotoxin is quite different in physiological effects from that produced by members of the Enterobacteriaceae. No evidence for immunopathological damage has been obtained. Both humoral and cell mediated immunity are important in recovery from infection. Antibody-opsonized Rickettsia are phagocytosed and killed by macrophages and delayed type hypersensitivity develops following rickettsial infections.

DIAGNOSIS:

1. Based on the symptoms and lesions.
2. Microscopical identification of the organisms in tissue or blood. Although the organisms are gram-negative, they only weakly take the counter stain, safranin. Hence, special staining procedures are commonly used. The special staining procedures used are
   - **a. Macchiavello stain**:-organisms are bright red against the blue background of the tissue.
   - **b. Castaneda stain**:-blue organisms against a red background.
   - **c. Giemsa stain**:-bluish purple organisms.

3. **Weil-Felix reaction:** It is an agglutination test. It is normally done in human beings using *Proteus* organisms. In this test, the titre of the agglutinins in the patient's serum against the *Proteus* strains OX-19, OX-2 and OX-K are determined. These *Proteus* strains have no etiological role in rickettsial infections, but appear to share antigens in common with certain rickettsia. These antigens are alkali stable polysaccharide haptens, which are distinct from the group-specific and type-specific antigens. In interpreting the results, it must be kept in mind that *Proteus* infections are fairly common (especially in the urinary tract) and that they, too, may evoke antibodies to the *Proteus*-OX strains. This test is usually positive seven days after the initial infection.
4. **Complement fixation test:** It is a better test than Weil-Felix reaction.
5. **Immuno-fluorescense assay:** It is a better test than Weil-Felix reaction and complement fixation test.
6. **Nucleic acid identification methods**

TREATMENT: The drugs of choice for the treatment of rickettsial diseases are chloramphenicol and tetracycline. The sulfonamides stimulate rickettsial growth and thus are contraindicated in the treatment of these diseases.

CONTROL:

1. Vector control is the best method.
2. Proper disposal of the dead animals, and foetal membranes also prevent the zoonotic rickettsial infections.
3. Raw milk should never be consumed.
4. Vaccination is not normally practiced in animals.
5. Most of the rickettsia infections are found in few geographical locations. Hence, animal movement from endemic areas should be restricted.

Q FEVER

Causative organism : *Coxiella burnetii*

Synonyms : Abattoir fever, Query fever, Burnet's rickettsiosis

INTRODUCTION: Query *(Q)* fever is a zoonosis that occurs in most countries. Humans acquire infection from animal reservoirs, especially from domestic ruminants. Q fever is a highly infectious disease, which is due to the proliferation of *Coxiella burnetii*, a small and pleomorphic bacterium. Q fever was first described in Australia in 1937. Cattle, goats, sheep and other animals are affected by the infection. Infection in man usually occurs as a result of inhalation of aerosols containing the organism.

GENERAL CHARACTERS:

1. **Morphology:** *Coxiella burnetii* is a small, obligate and pleomorphic bacterium measuring 0.3–1.5 μm long x 0.2–0.4 μm wide. It is acid– alcohol resistant and can be stained by several methods, which include Stamp, modified Ziehl–Neelsen, Gimenez, Giemsa and modified Koster techniques. They are also weakly gram-negative. However, they stain gram-positive with alcoholic iodine as mordent. Unlike rickettsiae, *C. burnetii* produces a small, dense, highly resistant spore-like form that is highly stable in the environment. This particular character is important for transmission of the infection. This ability has been attributed to the existence of *C. burnetii* developmental cycle variants: large-cell variants (LCV), small-cell variants (SCV), and small dense cells (SDC). The SDC and SCV represent the forms of the bacteria likely to survive extracellularly as infectious particles. The organism has 11 chromosomal genes and 1plasmid.

2. **Classification:** *C.burnetii* was earlier under the family Rickettsiae, but now as per the molecular taxonomy (based on 16S ribosomes), they are now under a separate family –*Coxiellaceae*
3. **Habitat:** *C.burnetii* is found in all most all animals. The reservoirs for this organism include wild and domestic mammals, birds, and arthropod ssuch as ticks. Cattle, goats, and sheep are considered the primary reservoirs from which human contamination occurs.
4. **Antigens:** *C. burnetii* displays antigenic variations similar to the smooth- rough variation in the family *Enterobacteriaceae*. The organisms exhibit two phases. Phase variation is related mainly to mutational variation in the lipopolysaccharide (LPS). Phase I is the natural phase found in infected animals, arthropods, or humans. Organisms as Phase I is highly infectious and corresponds to smooth LPS. In contrast, phase II is not very infectious and is obtained only in laboratories after serial passages in cell cultures or embryonated egg cultures. It corresponds to rough LPS.
5. **Resistance:** The organism is very stable in environment. It is resistant to drying, chemicals, and many disinfectants. *C. burnetii* is very resistant to killing in nature and may survive for several weeks in areas where animals have been present; the organism may also be spread by the wind.

OIE Classification: List B infection (Multispecies)

Human risk: *C. burnetii* is extremely hazardous to humans, and laboratory infections are common. Because of its low infectious dose (even a single bacterium can cause infection), resistance in the environment, and aerosol route of transmission, *C. burnetii* is considered a potential agent of **bioterrorism**. The OIE has classified *C.burnetii* as risk group 3 agent.Live culture or contaminated material from infected animals must only be handled in facilities that meet the requirements for Containment Group 3 pathogens.

CULTURAL CHARACTERS: It cannot be grown in non-living media. Lab animals like guinea pigs are better source for cultivation. In guinea pigs the organisms are cultivated through intraperitoneal route. In male guinea pigs development of orchitis (called as Strauss reaction) is rare. They are also cultivated in embryonated eggs by yolk sac route. *C. burnetii* infection leads to death of the embryo within 14 days. However, only Phase II bacteria are recovered from embryonated eggs. *C. burnetii* can also be grown in vitro in a number of cell types, including mouse macrophage-like cells, fibroblast cells and Vero cells

PATHOGENESIS: Cattle, sheep, and goats are the primary reservoirs of *C. burnetii*. Infection has been noted in a wide variety of other animals, including other species of livestock and in domesticated pets. Cats and dogs may represent reservoirs of *C. burnetii*. *Coxiella burnetii* does not usually cause clinical disease in these animals, although abortion in goats and sheep has been linked to *C. burnetii* infection. Organisms are excreted in milk, urine, and feces of infected animals. Most importantly, during birth the organisms are shed in high numbers within the amniotic fluids and the placenta. The organisms are resistant to heat, drying, and many common disinfectants. These features enable the bacteria to survive for long periods in the environment.

Infection of humans usually occurs by inhalation of these organisms from air that contains airborne barn yard dust contaminated by dried placental material, birth fluids, and excreta of infected herd animals. Humans are often very susceptible to the disease, and very few organisms may be required to cause infection. Ingestion of contaminated milk is also a less common mode of transmission. Other modes of transmission to humans, including tick bites and human to human transmission are rare. Ticks are not considered essential in the natural cycle of *C. burnetii* infection in livestock and domestic animals. However, ticks play a significant role in the transmission of coxiellosis among the wild vertebrates, especially in rodents, and wild birds. Over 40 tick species are naturally infected with *C. burnetii*, including *Rhipicephalus sanguineus* found in dogs. *C. burnetii* infection has been reported less

frequently in a number of other domestic or wild mammals, including horses, rabbits, swine, camels, water buffalo, rats, and mice. Birds may also be infected, and *C. burnetii* has been isolated from pigeons, chickens, ducks, geese, and turkeys. Humans may acquire Q fever from infected domestic poultry by consumption of raw eggs or inhalation of infected fomites.

SYMPTOMS: Animals do not usually experience symptomatic C. *burnetii* infections. In the acute phase, the presence of C. *burnetii* may be demonstrated in the blood, lungs, spleen, and liver. Most animals remain totally asymptomatic, including a lack of fever. C. *burnetii* infection often becomes chronic, with persistent shedding of C. burnetii in feces and urine. Animals do not develop chronic endocarditis as is observed in humans. The female uterus and mammary glands are primary sites of chronic C. *burnetii* infection. Thus, the shedding of C.*burnetii* into the environment occurs mainly during parturition. Birth products, mainly the placenta, are heavily contaminated with C. *burnetii*. C.*burnetii* can also be recovered from milk. The only pathological manifestations that have been associated with chronic C.*burnetii* infection in animals are abortion, mainly in sheep and goats and lower birth weight and infertility in cattle.

In human beings, variety of symptoms ranging from inapparent or mild to severe infections occurs. Acute infection in humans presents as a febrile, self-limiting disease marked by symptoms including fever, headache and myalgia. Some of the common symptoms are a sudden onset, chills, headache, muscle cramps, weakness and malaise, severe sweats, hepatitis, meningitis pneumonitis and other respiratory symptoms. The infection may lead to complications in pregnant women. A serious complication of chronic Q fever is endocarditis, generally involving the aortic heart valves, less commonly the mitral valve.

DIAGNOSIS:

1. Q fever may be confused with Brucellosis and Leptospirosis. Hence, it should be differentiated from these two infections.
2. Most common method of diagnosis is detection of specific antibodies in serum
3. Isolation of organisms by culture method is hazardous, therefore not routinely carried out. It also requires specially equipped laboratory with bio-containment facilities. As isolation of the organism is hazardous, the most common method of diagnosis is through the detection of specific antibody in serum, with ELISA considered the most sensitive method.
4. ELISA: It is considered most sensitive method of diagnosis (IgG studies). It correlates with conventional CFT. Diagnosis can be made with single serum specimen. It can measure response to different classes of antibody.
5. Complement fixation test (CFT)- Involves a >2 fold increase in paired sera. Usually phase II antigens are used. Slow production of antibodies and static titres are problems of thistest.
6. Immunofluorescence assay (IFA): It is an accepted method of diagnosis. It is more sensitive than CFT with antibodies appearing as early as 5 days. It can measure individual antibody classes to different phases
7. Nucleic acid identification methods like PCR.

PREVENTION:

- Vaccination of abattoir workers and others in hazardous occupations
- Pasteurisation of milk from cows, goats and sheep to inactivate organism

TREATMENT

Tetracyclines administered after patient is afebrile.

Significance for Bioterrorism: *Coxiella burnetii* is a highly infectious agent that is rather resistant to heat and drying. It can become airborne and inhaled by humans. A single *C. burnetii* organism may

cause disease in a susceptible person. This agent could be developed for use in biological warfare and is considered a potential terrorist threat.

HEART WATER or COWDRIOSIS

Causative organism : *Ehrlichia ruminantium(*formerly *Cowdria ruminantium)*

Synonym : Veld poisoning

Definition: *Heartwater* (*also known as cowdriosis*) is an infectious, non- contagious rickettsial disease of ruminants caused by *Ehrlichia ruminantium* (*formerly Cowdria ruminantium*) and transmitted by *Amblyomma ticks.* The disease affects domestic and wild ruminants, including cattle, sheep, goats, antelope, and buffalo. Non-ruminant hosts of *E. ruminantium*, such as guinea-fowl, leopard tortoises, and scrub hare, may also be important in the maintenance of the organism in nature because they are all known carriers of the agent. Heart water is usually an acute disease and is commonly fatal within a week of onset of clinical signs. The name "heartwater" is derived from the hydropericardium, which is commonly seen with this disease. The disease is widespread in most of Africa and on several islands in the West Indies. With increased trade and movement of animals in today's global market, heartwater may present a significant threat to the domestic livestock industry in any country

General characters of *E. ruminantium:*

1. **Morphology:** It is a very small (0.4 to 1.0μ), pleomorphic, obligate intra cellular parasite. They stain Gram-negative and stain blue or purplishblue in Giemsa in smears prepared from affected organs. They have an affinity for multiplication in the vascular endothelial cells. They appear as colonies inside the endothelial cells. In such colonies, the organisms appear as pleomorphic rods to horse shoe shaped structures. It usually occurs in clumps of from less than five to several thousand organisms within the cytoplasm of infected capillary endothelial cells, especially in the brain. They remain firmly attached to RBC and it is difficult to remove the organisms by washing in buffered saline.
2. **Resistance:** These organisms are very fragile organisms and cannot survive for longer time at room temperature in blood. Because of its fragility, the organism must be stored in dry ice or liquid nitrogen to preserve its infectivity. The organisms remain viable for two years at -80^0C in spleen and blood.
3. **Habitat:** They are obligate intra cellular parasites and are found either in the affected animals or the *Ambylomma* ticks that are biological vectors.
4. **Antigens:** There are many strains of *E. ruminantium,* which vary in virulence. All strains are apparently pathogenic for sheep and goats and at least one strain is nonpathogenic for cattle.
5. **OIE listing:** List B disease (Multispecies)
6. **Human risk:** Humans are not susceptible to *E. ruminantium*.

Cultivation: The organisms are very difficult to grow and cannot be cultivated in embryonated eggs or in tissue cultures. Of late endothelial cells are used in the isolation of *E.ruminantium.* Ferrets are the only laboratory animals of choice. The animals used for routine cultivation under laboratory cultivation are the sheep. In mice, the organisms can remain upto 90 days but they do not multiply.

Pathogenesis: The important mode of spread is by tick bites. Heart water is transmitted only by ticks of the genus *Amblyomma*, with the tropical bont tick as one of the most important vectors. This tick is widely distributed throughout Africa and number of other countries. The life cycle of *Amblyomma* ticks may take from 5 months to 4 years to complete. Thus, the infection may persist in the environment, inside the tick, for a long time. The immature stages of the tick will feed on a wide variety of livestock,

wild ungulates, ground birds, small mammals, reptiles, and amphibians. Cattle egrets have also been implicated in much of the recent spread of heart water.

The incubation period is usually 12 days in sheep and goats and 12-18 days in cattle. After infection of the host with *E. ruminantium*, from the sites of tick bite the organisms are taken to various organs through blood circulation. Initial replication of the organism takes place in reticuloendothelial cells and macrophages in the regional lymph nodes. From here the organisms are disseminated via the blood stream to invade endothelial cells of blood vessels of various organs where further multiplication occurs. The organisms multiply in vascular endothelial cells and produce a toxin. This toxin increases the permeability of the capillary walls that causes oedema of the lungs and accumulation of fluid in pericardium (*this gives the name heartwater to the infection)*. Endothelial cell parasitization coincides with the onset of fever. There is an increased vascular permeability allowing the seepage of plasma proteins, which result in transudation through the serous membranes with resultant tissue oedema and effusion into body cavities. This causes the drastic fall in blood volume before death. Oedema of the brain is responsible for the nervous signs, hydropericardium contributes to cardiac dysfunction during the terminal stages of the disease and progressive pulmonary oedema and hydrothorax result in asphyxiation.

Symptoms: Heartwater occurs in four different clinical forms - peracute, acute, subacute and subclinical as determined by variations in susceptibility of the hosts and the virulence of various strains of the HW agent. The peracute form is rare and is seen in exotic breeds of cattle, sheep, and goats introduced to an HW enzootic area. Heavily pregnant cows are especially prone to develop the peracute disease. In this form sudden death occurs, usually preceded only by a fever, severe respiratory distress, and terminal convulsions. Severe diarrhea may be seen in some breeds of cattle. The acute form of heartwater is the most commonly observed form of the disease. A sudden high fever (107 degrees F) is followed by loss of appetite, depression, and respiratory problems. Animals may initially have an increased respiratory rate, followed within a few days by severe respiratory distress. Nervous disorders often follow the respiratory signs and can include a variety of abnormal behaviours such as protrusion of tongue, excessive chewing movements, twitching of eyelids, incoordination, circling, star-gazing, head tilt, overly rigid posture, and walking with a high-stepping gait. These nervous signs usually last up to 48 hours, followed by the animal's death. In some cases, the nervous signs may not be noticed prior to death. The nervous signs increase in severity, and the animal goes down in convulsions. Galloping movements and opisthotonos are commonly seen before death. Hyperesthesia is often observed in the terminal stages of the disease, as is nystagmus and frothing at the mouth. Diarrhea is occasionally seen, especially in younger animals. The acute disease is usually fatal within a week of the onset of signs. The subacute infection is characterized by prolonged fever, coughing (a result of lung edema), and mild incoordination; recovery or death occurs in 1 to 2 weeks. A mild or subclinical form of the disease, known as "heartwater fever," is seen in partially immune cattle or sheep, in calves less than 3 weeks old, in antelope, and in some indigenous breeds of sheep and cattle with high natural resistance to the disease. The only clinical sign in this form of the disease is a transient febrile response.

Lesions: The gross lesions in cattle, sheep, and goats are similar. Hydropericardium is the most important lesions of this infection. The accumulation of straw-colored to reddish fluid in the pericardium is more consistently observed in sheep and goats than in cattle. Ascites, hydrothorax, mediastinal edema, and edema of the lungs, all resulting from increased vascular permeability with consequent transudation, are frequently encountered. Subendocardial petechial hemorrhages are also seen, and submucosal and subserosal hemorrhages may occur elsewhere in the body. Degeneration of

the myocardium and liver parenchyma, splenomegaly, edema of lymph nodes, nephrosis, and catarrhal and hemorrhagic abomasitis and enteritis are all commonly encountered. Meningeal congestion and edema are often present. Brain congestion may occur, but brain lesions can be remarkably few when one considers the severity of the nervous signs observed in this disease.

Diagnosis:

1. **Based on the symptoms and lesions**. The symptoms should be correlated with the presence of *Ambylomma* ticks in the region.
2. Microscopy:
    a. Demonstration of the colonies or the organisms inside the cytoplasm of the vascular endothelial cells. In the colonies the organisms appear as pleomorphic rods to horseshoe shaped structures.
    b. Demonstration of the organisms: The HW organism stains purplish- blue with Giemsa stain and can be seen by microscopic examination of brain smears prepared from cerebrum, cerebellum, hippocampus, or other well-vascularized portion of the brain. These portions are macerated between two microscope slides. The resultant pulp is then drawn across a slide with varying pressure, which results in "ridges and valleys" on the slide. The slide is then air-dried, fixed with methanol, and stained with Giemsa. Under low magnification, the capillaries will be found extending from the "thick" areas of the slide. Examination of the capillary endothelial cells under oil immersion will reveal the blue to reddish-purple clumps of organisms. A rapid method for obtaining brain tissue for examination is to drive a large nail through the unopened skull and make a smear from the tissue adhering to the nail. The HW organisms can also be observed in smears prepared from the intima of large blood vessels or in stained sections of kidney glomeruli and lymphnodes.
    c. Transmission electron microscopy is also used to demonstrate *Ehrlichia* organisms in the endothelial cell's cytoplasm
3. **Isolation of the organisms by culturing:** *Ehrlichia* can be isolated from the blood of reacting animals by cultivation on ruminant endothelial cells. Endothelial cells from umbilical cord, aorta, or the pulmonary artery of different ruminant species (cattle, goat, sheep) are used most often for isolation. No standard cell line has yet been designated forisolation. However, the best method to isolate the organism in susceptible ruminants.
4. **Nucleic acid identification methods**: Of late PCR and DNA probes are used to identify infected animals.
5. **Serological tests**: Indirect fluorescent assay and cELISA are performed to identify antibodies against these organisms in serum.
6. **Differential diagnosis**: The observed nervous system abnormalities suggest other diseases (such as rabies, tetanus, meningitis, or encephalitis) or toxic poisoning. Other diseases to be eliminated are anthrax, piroplasmosis, east coast fever, listeriosis, louping ill etc. A definitive diagnosis of heartwater is made by microscopic examination and observation of the causative rickettsia in a brain tissue smear.

Treatment: Administration of chlortetracycline and oxytetracycline intravenously is very effective.

Control: Eradiation of *Ambylomma* sp. ticks is the best method of control. Since the young animals are not normally affected inoculation of young animals with infected blood protects them from infection during adult stage.

SALMON POISONING

Causative organism : ***Neorickettsia heminthoeca***

Synonym : Salmon disease of dogs

Definition: It is a non-contagious, infectious, fatal acute infection of dogs characterised by extreme thirst, severe depression and frequent vomiting. Salmon poisoning is characterized by fever, lymph node enlargement and severe hemorrhagic enteritis, which may lead to the animal's death if left untreated. The infection is spread by the metacercariae of the fluke *Nanophyetus salmincola.* The fluke is a parasite of fish and the infection is more commonly seen in areas where the dogs eat raw fishes. The infection is more commonly seen in the northwestern region of American continent. The infection was first reported in 1924. However, the aetiology was confirmed in 1947. It is one of very few rickettsial infections that are spread by trematode vector.

General characters of *N. helminthoeca*:

1. **Morphology:** They are very small pleomorphic organisms (coccoid, rod or crescent shaped) often arranged in clusters. They stain gram-negative but stain blue or purple by Giemsa and red with Macchiavello's stain. In cells the organisms are appear as LCL bodies of *Chlamydia.* In dogs, they are found inside the reticuloendothelial cells of the lymph nodes, intestinal lymphoid follicles, tonsils, thymus and spleen. Rarely they are also seen inside macrophages.
2. **Habitat:** The organisms are normally found in the metacercariae of the fluke *Nanophyetus salmincola* that affects fish. Dogs get the infection when they consume raw fish.

Cultivation: Very difficult and very scarce information available on this.

Pathogenesis: This microorganism is found in salmon, steelhead, trout, Pacific giant salamanders and fresh water fish found in and around the Pacific Ocean from Northern California to Seattle. The geographical limitations are likely caused by the limited habitats of infected snails. The infection is commonly seen in dogs, foxes and other carnivores that feed on raw fishes.

The fluke *Nanophyetus salmincola* is host to *Neorickettsia helminthoeca.* The organism develops in snails (*Oxytrema plicifer*), infects and develops into cysts in fish, is ingested by dogs where they infect the intestinal tract. The dogs excrete eggs in their stool, the organism re-enters the water, infects snails and the cycle begins again.

SNAIL ==> FISH ===> DOG ===> SNAIL

The egg matures (3 months) and hatches in water. The miracidium finds and penetrates a snail. Redia develops in the snail. Cercariae develop in the redia, and when mature emerge from the snail. The cercariae swim around until they find a trout or salmon whose skin they penetrate and enscyst as metacercaria in various tissues. When the definitive host eats the fish the metacercaria excyst and develops to the adult stage in the host's small intestine. Eggs are laid in the small intestine and pass out with the faeces.

Symptoms: The onset of symptoms is usually sudden, usually 5-7 days after ingestion but can be delayed up to a month. Symptoms last for 7-10 days and can be **fatal in a majority (upto90%)** of untreated dogs. The dog's temperature can peak (104-107F) suddenly and then return to normal or even below normal. The fever may persist for 37 days. There can be severe and bloody diarrhoea, dehydration, vomition, extreme thirst, severe weight loss and complete loss of appetite. Some animals show serous nasal discharge and conjunctivitis. Symptoms can look like parvo or distemper. Mortality is as high as upto 90%. Death is generally caused by the toll the symptoms take on the dog's body. These include electrolyte imbalance, anaemia and dehydration.

Lesions: There is hyperplasia of carcase, visceral lymph nodes with necrosis. The tonsils, thymus and spleen may be enlarged. Haemorrhagic enteritis is the pathognomonic lesion.

Diagnosis:

1. Diagnosis is made by finding fluke eggs in the stool.
2. Demonstration of the organisms in the buffy coat of infected dogs.
3. Based on symptoms and lesions.
4. Microscopical examination of smear prepared from affected animals and stained by Romanowsky stains.

Treatment: Treatments include hydration and nutrition, blood transfusions as well as antibiotics (sulphonamides or any other broad spectrum antibiotics) and related medications.

Control: No vaccines are available. Dogs that have been infected and recovered can develop immunity.

CANINE ROCKY MOUNTAIN SPOTTED FEVER

Causative organism: *Rickettsia rickettsii*

Rocky Mountain Spotted Fever (RMSF) is an important zoonotic disease that may cause clinical signs in both dogs and humans. It is caused by the organism *Rickettsia rickettsii*, a small Gram-negative obligate intracellular parasite from the family *Rickettsiaceae.* Although it has been long identified as a human pathogen, RMSF was not recognized in dogs until the 1970's.

General characters:

1. **Morphology:** *R. rickettsiiis* a very small bacterium that lives inside the cells of its hosts. These bacteria range in size from 0.2 x 0.5 micrometers to 0.3 x 2.0 micrometers. They are difficult to see in tissues by using routine histologic stains and generally require the use of special staining methods. *R.rickettsii* is a weakly Gram negative. *R. rickettsii* are observed using Giemsa, Machiavello, and Gimenez staining and by the use of direct fluorescent antibody staining techniques. *R rickettsii* has ribosomes and indistinct strands of DNA in an amorphous cytosol surrounded by a plasma membrane. In addition, an indistinct microcapsular layer is present on the outer surface of the cell wall. An electron-lucent zone separates this layer from the host cytosol. This zone is thought to represent a slime layer which may be important in pathogenicity.

Pathogenesis: *R. rickettsii* is transmitted to the dog through the bite of an infected tick. Hard-bodied ticks are the vectors of RMSF. The most common vectors are *Dermacentor variabili* and *D.andersoni. Amblyomma americanum* and *Rhipicephalus sanguineous* also have been reported to carry the organisms. The ticks acquire the organism by feeding on parasitemic small mammals such as chipmunks and squirrels. Larger mammals rarely achieve the degree of parasitemia necessary to transmit the disease to a feeding tick. These small mammals function as reservoirs for the organism. Dogs and humans are incidental hosts and are the only reservoirs that display clinical signs of disease. The larvae and nymphs of the *Dermacentor* species typically feed on small mammals, while the adult ticks prefer larger mammalian hosts. Therefore, it is the larval and nymphal stages that are most often infected with *R. rickettsii* during feeding. Once infected, ticks spread RMSF among their own population through transfer of bodily fluids during mating, or transovarially spread the organism from the pregnant female to her offspring. Transovarial infection is the primary means by which *R. rickettsii* propagates in nature. Once infected, ticks then transmit RMSF to vertebrates (including dogs and humans) through their saliva when taking blood meals. The incubation period is 2-14 days. Then the organism enters into the circulation and then invades endothelial cells of the venules and capillaries and begins replicating,

causing a vasculitis. The organism prevents apoptosis (programmed death) of the vascular endothelial cell, allowing adequate time for the organism to effectively replicate and spread. Vasculitis lead to edema, hemorrhage, shock, and vascular collapse. Organs like brain, skin, heart and kidneys are affected most often. Vascular leakage also triggers activation of the platelets and coagulation system.

Symptoms & Lesions: The first symptom is development of a red rash approximately 12 days after the initial tick bite. However, it is not very common among dogs. High fever (102.6-104.9º F) approximately 5 days after the tick bite, is the more common clinical finding. Petechiae and ecchymotic hemorrhages associated with destruction of platelets in response to vasculitis are also seen on exposed mucosal surfaces in the dog. Vasculitis also may cause edema in the extremities including the scrotum, prepuce, and ears of affected dogs. Other signs are joint swelling, myalgia, dyspnea, and neurological abnormalities due to meningoencephalitis. Vestibular ataxia is the most common neurologic affliction. Ocular lesions include conjunctival hyperemia, hyphema, retinal hemorrhage, and anterior uveitis. These ophthalmic lesions tend to be mild and usually occur bilaterally.

Diagnosis:

1. Based on symptoms and lesions in correlation with tick infestation.
2. Blood picture–
    - Thrombocytopenia, with platelet counts ranging from 23,000 to 220,000/µl.
    - Moderate leukocytosis
    - Normocytic, normochromicanemia
3. Biochemical abnormalities may include elevated glucose and cholesterol concentrations as well as increased activity of alkaline phosphatase (ALP) and alanine aminotransferase (ALT).
4. Serologic tests: ELISA, IFA and CFT are routinely performed to identify RMSF. A four-fold increase or decrease in IgG antibody titers to *R. rickettsii* along with appropriate clinical signs of RMSF is considered diagnostic for the disease
5. Biopsy of skin
6. FAT on frozen tissue samples
7. Nucleic acid identification methods –PCR

Treatment: Response to treatment in most dogs with RMSF, treated with tetracycline (22 mg/kg, TID), is rapid. Most dogs without neurologic deficits improve within 24 to 48 hours. Chloramphenicol (15 mg/kg, TID) has been used in pups less than 12 weeks old because of the risk of staining their dental enamel with tetracycline. Dogs with neurologic deficits may have more prolonged recovery periods.

Prevention: Since no vaccine is available for RMSF, the control is aimed at control of ticks.

CANINE EHRLICHIOSIS

Causative organism: *Ehrlichia canis*

Synonyms: canine rickettsiosis, canine hemorrhagic fever, tracker dog disease, canine ticktyphus,Nairobi bleeding disorder, canine agranulocytic ehrlichiosis and tropical canine pancytopenia

Definition: A number of *Ehrlichia* species can infect dogs and their affinity for hematopoietic cells may result in leukopenia and thrombocytopenia. Monocytic ehrlichioses in dogs (CME – canine monocytic ehrlichiosis) is caused primarily by *E. canis*. *E. chaffeensis*, *E. ewingii*, *E. phagocytophila*, *E. equi*

and *E. platys cause Granulocytic ehrlichiosis* (CGE – Canine granulocytic ehrlichiosis). *Ehrlichia canis* was the first ehrlichial organism to be discovered and is the type species of the genus *Ehrlichia. E. canis* has worldwide distribution and is the cause of canine agranulocytic ehrlichiosis.. Canine ehrlichiosis is a disease of dogs and wild canids (e.g., wolves) and the infection is found worldwide. This organism is transmitted by the brown dog tick, *Rhipicephalus sanguineus* under natural conditions. *E. canis* primarily infect the host's white blood cells, especially the mononuclear phagocytic cells. The disease is then disseminated to other organ systems including the bone marrow and results in a pancytopaenia (loss of a number of blood cell lines).

General characters:

1. **Morphology:** They are very small pleomorphic organisms (coccoid or rod or cocco-bacillary) often arranged in clusters or colonies. They are generally referred as morulae. These colonies are small (initial bodies) and contain varying numbers of granules. They stain Gram-negative. The initial bodies stain as slate-grey or light blue and the granules stain deep purple in Giemsa method. In dogs, they are found inside the reticuloendothelial cells of the lymph nodes and spleen. They are also seen inside macrophages and other granulocytes.
2. **Habitat:** They are worldwide in distribution and the infection is mostly found in Asian countries. The organisms appear in clusters known as morulae in the cytoplasm of infected cells. The cells that are most commonly affected are the white blood cells. Ehrlichial organisms are classified as agranulocytic or granulocytic based on the cells they infect.
3. **Resistance:** They are very fragile organisms.
4. **Classification:** has been divided into three genogroups based on 16S rRNA. Genogroup I incorporates three species i.e. *E. canis*, *E. chaffeensis*, and *E. ewingii*. Genogroup II includes *E.phagocytophila* and *E. equi.* Genogroup III covers two species *E. sennetsu* and *E.risticii*.

Cultivation: These organisms are very difficult to cultivate in embryonated eggs or in cell cultures. However, they can be cultured *in vitro* in mammalian-derived cell line (DH82) and in an invertebrate cell line (IDE8).

Pathogenesis: The brown dog tick, *Rhipicephalus sanguineus, transmits Ehrlichia*. The tick *Dermacenter variabilis* also transmits it. The immature form of the tick feeds on an animal infected with *Ehrlichia*. When these immature forms or a mature form of the tick feeds on another animal, the *Ehrlichia* is passed on to that animal. The *Ehrlichia* can remain alive in the developing tick for up to 5 months. Because the brown dog tick transmits the disease, it can occur wherever brown dog ticks are found.

Symptoms and Lesions:

Canine Monocytic Ehrlichiosis: CME is characterized by three stages. The first, acute stage, beginning after 8–20 days following transmission by infected tick, lasts 2–4 weeks. The acute phase may be manifested by fever (103-106^{0}F), depression, dyspnoea, anorexia, and slight weight loss. The laboratory findings are thrombocytopenia, leucopoenia, mild anaemia, and hypergammaglobulinemia are observed. Dogs may have bleeding tendencies, mainly petechiae and echymoses of the skin and mucous membranes, and occasionally, epistaxis may also occur. Ocular signs include corneal opacity, anterior uveitis, hyphema, tortuous retinal vessels and focal chorioretinal lesions consisting of central pigmented spots with surrounding areas of hyper-reflectivity. Subretinal haemorrhages, resulting in retinal detachment may occur and lead to blindness. Other clinical signs may include vomiting, serous to purulent oculonasal discharge, lameness and ataxia.

The second phase is subclinical and follows the acute phase and may last 40–120 days or even years, in which dogs can remain persistently infected for years without clinical signs but with mild thrombocytopenia. In the sub clinical phase the animal may show only slight anaemia. During this

phase the dog either eexcretes the *Ehrlichia* from the body or the infection may progress to the chronic phase.

The ultimately stage is chronic,characterized by haemorrhages, epitasis and edema, with clinical signs. The results of laboratory study resemble the first phase of the disease. The course of this phase may often be complicated by superinfections by other micro- organisms. The chronic phase generally develops 1-4 months after the tick bite and can be either mild or severe. The common clinical signs of the chronic disease are weakness, depression, anorexia, chronic weight loss and emaciation, pale mucous membranes, fever and peripheral oedema, especially of the hind limbs and the scrotum. Platelet-related bleeding, such as petechiae and echymoses of the skin and mucous membranes and epistaxis are common findings. Secondary bacterial and protozoal infections, interstitial pneumonia, renal failure, and arthritis may occur during chronic severe disease. Some reproductive disorders have also been associated with chronic CME including; prolonged bleeding during oestrus, inability to conceive, abortion and neonatal death. Polymyositis has also been associated with CME.

Neurological signs may occur during the acute and chronic disease. These include signs of meningoencephalitis, e.g. arched back, severe neck or back pain, paraparesis or tetraparesis, ataxia, cranial nerve deficits and convulsions. Neurological signs may be attributed to haemorrhages, vasculitis and extensive plasma cell infiltration and perivascular cuffing of the meninges. Dogs infected with *E. canis* remain infected for their entire lives, even if they received antibiotic treatment with doxycycline.

Blood tests show that one or all of the different blood cell types are decreased. One cell type, the lymphocyte may increase and be abnormal in appearance. This can sometimes be confused with certain types of leukaemia. If a dog becomes chronically infected, the disease can keep coming back, especially during periods of stress. Decrease in the number of platelets (platelets help the blood clot) in the blood is the most common laboratory finding in all phases of the disease. Changes in the protein levels in the blood are common. The most common protein, albumin, is decreased and other types of protein called "globulins" are increased.

Since one tick could be infected with and transmit more than one disease (e.g., haemobortenellosis or babesiosis), it is not all that uncommon to see a dog infected with more than one of these diseases at a time which generally causes more severe symptoms.

Canine granulocytic ehrlichiosis: Two clinically distinct disease syndromes, including chronic, moderate to severe anaemia and polyarthritis, are associated with CGE. Clinical signs are nonspecific and include fever, lethargy, anorexia, vomiting, and diarrhoea. The most frequent laboratory abnormalities are normocytic, normochromic nonregenerative anaemia, moderate thrombocytopenia with large platelets, lymphopenia, and eosinopenia. In beagles, inoculated experimentally with granulocytic *Ehrlichia* organism, after an incubation period of 4–11 days, the most prominent clinical signs were high fever for 2–5 days, and depression

Diagnosis:

1. Based on the symptoms and lesions.
2. Examination of blood smear: A small drop of blood is spread over a microscope slide, stained and examined under the microscope. The organism can be seen as morulae and only be found in the blood stream for about 3 days during the acute phase of the disease. So this method of diagnosis could miss some cases of the disease.
3. Immunofluorescet antibody test (IFA): IFA is a highly accurate blood test that detects antibodies against *E.canis* in dogs. As the disease progresses, the antibody level will rise significantly. Often two tests will be done 2 weeks apart and the results compared. Dogs with an active infection will show a significant rise in the amount of antibody present.

4. ELISA: This test also determines the amount of antibodies present.
5. Nucleic acid identification methods: Polymerase chain reaction.
6. Western blotting.

Treatment: The antibiotics tetracycline or doxycycline is used. Treatment is for 2-3 weeks. Some dogs will need blood transfusions or intravenous fluids depending on the severity of the disease. The drug imidocarb dipropionate is sometimes used in conjunction with the antibiotics. Some of the damage caused by *Ehrlichia* may be due to the dog's own immune response to the organism. For this reason, high doses of corticosteroids (e.g., prednisolone) are often given during the early phase of the disease.

Control: Tick control is the main way to prevent ehrlichiosis. Products that repel and kill ticks, tick collars containing the active ingredient amitraz (Preventic collars) are also used. There is no vaccine for ehrlichiosis.

Zoonotic importance:

The common symptoms in man include fever, chills, headache, and muscle aches. Other less common symptoms include nausea, loss of appetite, weight loss, abdominal pain, cough, diarrhoea and change in mental status. However, man does not get infected directly from a dog, but through a tick bite. A different tick than the brown dog tick may spread human ehrlichiosis. Research suggests the Lone Star tick may be involved. Also, the *Ehrlichia* species most often implicated in human infections is *E. chaffeensis*.

ANAPLASMOSIS

Bovine Anaplasmosis

Causative organism: *Anaplasma marginale; A.marginale subsp.centrale*

Bovine anaplasmosis is an arthropod-borne hemolytic disease of cattle that is caused by the rickettsia *Anaplasma marginale* (*Rickettsiales*: *Anaplasmataceae*). It is a tick-borne disease and is endemic in tropical and subtropical areas of the world. The disease causes considerable economic loss to the dairy industry. The organism parasitises red blood cells, causing their destruction and producing emaciation, anaemia, jaundice, and, occasionally death. Analyses of 16S rRNA, groESL, and surface proteins have resulted in the recent reclassification of the order *Rickettsiales.* The genus *Anaplasma* now also includes *A. bovis*, *A. platys*, and *A. phagocytophilum*, which were previously known as *Ehrlichia bovis*, *E. platys*, and the *E. phagocytophila* respectively. *Anaplasma marginale* are called so, since they are found in the periphery of the erythrocytes. One more organism of importance in cattle *A.centrale* is less pathogenic and found slightly interior of the erythrocytes.

GENERAL CHARACTERS:

1. **Morphology:** *A. marginale* appear as dense, rounded, intraerythrocytic bodies approximately 0.3–1.0 μm in diameter with most situated on or near the margin of the erythrocyte. *Anaplasma centrale* is similar in appearance, but most of the organisms are situated away from the margin of the erythrocyte. The organisms appear as very minute roundish body at the periphery of erythrocytes (A). Within these cells the membrane-bound inclusions (also called initial bodies) contain four to eight organisms (B). The small genome of *A. marginale* is circular, and the size is estimated at 1.2 to 1.6 Mb. The organism also has six major surface proteins.
2. **Classification:** The organisms in the order *Rickettsiales* were recently reclassified based on biological characteristics and genetic analyses of 16S rRNA genes and surface protein genes. There are four distinct genus under the family *Anaplasmataceae*: - *Anaplasma*, *Ehrlichia, Wolbachia* and *Neorickettsia* Organisms classified within the family *Rickettsiaceae* (genera *Rickettsia* and *Orientia*) are all obligate intracellular bacteria that grow freely within the cytoplasm of eukaryotic cells. While organisms

placed in the family *Anaplasmataceae* are also obligate intracellular organisms, they are found exclusively within membrane-bound vacuoles in the host cell cytoplasm. All organisms in the family *Anaplasmataceae* multiply in both vertebrates and invertebrates (primarily ticks and trematodes). The genus *Anaplasma*, includes three species that infect ruminants: *A. marginale* (the typespecies). *A. marginale* subsp. *centrale* (referred *A. centrale*), and *A. ovis.* The genus *Anaplasma* also includes *A. phagocytophilum* (formerly *Ehrlichia equi*, *E. phagocytophila)*, *A. bovis* (formerly *E. bovis*), and *A. platys* (formerly *E. platys*).

3. **Habitat:** The organisms are found in cattle. The cattle tick *Boophilus microplus* act as biological vector.
4. **Resistance:** As other rickettsial organisms
5. **OIE Listing:** List B disease (Bovine)
6. **Human risk:** No human risk

CULTURAL CHARACTERS: Very difficult to cultivate and susceptible animals are normally used.

PATHOGENESIS: Cattle are the most important host affected. Sheep, goats and certain wild animals are also affected. Ticks are the natural vectors and a range of tick species has been shown to be capable of transmitting infection. In the absence of ticks, biting flies can maintain the disease though these are less efficient vectors. Fourteen different ticks transmit *A. marginale* experimentally. The important tick species are *Argas persicus, Ornithodoros lahorensis, Boophilus annulatus, B. decoloratus, B. microplus, Dermacentor albipictus, D. andersoni, D. occidentalis, D. variabilis, Hyalomma excavatum, Ixodes ricinus, Rhipicephalus bursa, R. sanguineus* and *R. simus*, However, the cattle tick, *Boophilus microplus* and *Rhipicephalus sanguineus* are the most important vectors. Various other biting arthropods have also been found to act as mechanical vectors in the transmission of *Anaplasma.* Mechanical transmission frequently occurs via blood- contaminated fomites, including needles, dehorning saws, nose tongs, tattooing instruments, ear-tagging devices, and castration instruments. In addition to mechanical and biological transmission, *A. marginale* can be transmitted fromcow to calf transplacentally duringgestation. Tick transmission can occur from stage to stage (transstadial) or within a stage (intrastadial). Transovarial transmission from one tick generation to the next does not occur. After the tick feeds on an infected animal, infected erythrocytes are ingested by ticks. The first site of infection of *A. marginale* in ticks is the gut cells. When the ticks feed a second time, many tick tissues become infected, including salivary gland cells, from where the rickettsia is transmitted back to cattle. Two forms of *A. marginale*, reticulated and dense forms, are found in infected tick cells. Reticulated forms appear first and are the vegetative stage that divides by binary fission. The reticulated form changes into the dense form, which is the infective form and can survive extracellularly. The organism can then be transmitted to succeeding generations of ticks. By this method, the infectious organism can survive outside of infected cattle for long periods of time. When future generations of infected ticks feed on susceptible cattle, the disease is transmitted.

SYMPTOMS: There is considerable variability in the severity of anaplasmosis with severity generally increasing with the age of the animal. Affected cattle have steadily increasing temperatures (up to 105F), anaemia, weakness and respiratory distress particularly after exercise, depression and anorexia become more obvious as the disease progresses, jaundice and frequently a marked loss of condition and urine is often brown due to the presence of bile pigments. Severely affected animals may die.

LESIONS: Gross pathology findings are due to anaemia and resulting anoxia. These include pale and often jaundiced tissues, watery blood and an enlarged spleen with a soft reddish-brown pulp. The liver is also enlarged, yellow-brown and often mottled.

DIAGNOSIS:

1. Based on symptoms and lesion
2. **Direct Microscopy:** The clinical sample from live animal includes thin blood smears and blood collected into an anticoagulant. From dead animals air-dried thin smears from the liver, kidney, heart and lungs should be collected. Both blood and organ smears are fixed in absolute methanol for 1 minute and stained in 10% Giemsa stain for 30 minutes. In Giemsa-stained smears, *A. marginale* appear as dense, rounded and deeply stained intraerythrocytic bodies, approximately 0.3–1.0 μm in diameter. Most of these bodies are located on or near the margin of the erythrocyte. This feature distinguishes *A. marginale* from *A. centrale,* as in the latter most of the organisms have a more central location in the erythrocyte. Fluorescent antibody staining may be used as an alternative staining technique for detecting *Anaplasma* in smears taken at post- mortem.
3. **Isolation of the organism:** The organisms are identified by inoculating blood from suspected cattle into spleenectomized calves, in which the infection is confirmed by examination of blood smear.
4. **Nucleic acid identification methods:** PCR technique and a RNA probe is used to confirm the infection. Restriction enzyme analysis, Southern blotting and DNA sequencing are the other methods of diagnosis.
5. **Serological tests:** Serological tests include competitive ELISA, card agglutination test (CAT), CFT, indirect ELISA, dot ELISA and IFA.

TREATMENT: The anaplasmosis organism is susceptible to oxytetracycline or chlortetracycline (antibiotics. Cattle showing clinical signs usually respond to high dosages of oxytetracycline given by injection. When diagnosed in an individual in the herd, it is advisable to immediately treat the entire herd with tetracylines to prevent spread of the infection and possible development of other cases of disease

CONTROL: Tick control by acaracide dipping is widely used in endemic areas. Acaracides used for this purpose include various synthetic pyrethroids, amitraz, and some organophosphates. Dipping may be done as frequently as every 4-6 weeks in heavily infested areas. Anaplasmosis vaccines are readily available and are highly effective. The most commonly used vaccine is a live *Anaplasma centrale* vaccine used either singly or in combination with *Babesia bovis* vaccine. Vaccination is particularly important for susceptible cattle entering endemic areas. An anti-tick vaccine is also commercially available in Australia. This vaccine is of limited use, but can be used as part of an integrated program for the control of ticks.

BOVINE PETECHIAL FEVER

Synonyms: Bovine infectious petechial fever, transmissible petechial fever, Ondiri disease, Nairobi quarantine disease

Etiology: *Anaplasma phagocytophilum (E. phagocytophilia; E. ondiri)*

It is an infectious non-contagious disease of cattle that is characterised by fever, oedema of the conjunctiva, petechiae of the mucous membrane and in very severe cases the affected animal collapse and dies.

General properties:

1. **Morphology:** It is caused by *E.ondiri*. *E.ondiri* is an obligate intracellular organism, which cannot survive outside the cell. The organisms occur in polymorphonucelar phagocytic cells like neutrophils and eosinophils and monocytes. They are arranged either singly or as colonies. They are very small

pleomorphic organisms with small variations in size. The initial bodies stain as slate-grey or light blue and the granules stain deep purple in Giemsa method. The particles are arranged as discrete units in the periphery of the colonies and the appearance is attributed to '**broken dinner plate'**.

2. **Habitat:** The organisms are normally seen in phagocytic cells.
3. **Resistance:** They are fragile organisms and can withstand 37^0C for only two days and retain its infectivity at refrigeration temperature for only four hours.

Cultivation: The organisms are difficult to cultivate in embryonated eggs, tissue cultures or laboratory animals like mice and guinea pig. Cattle and very few wild animals are the only host susceptible for this organism. Heparinised blood is the main source for infecting cattle in which the organisms are found in the leucocyte fraction.

Pathogenesis: Cattle are the main host and certain wild ruminants, sheep and goat are less severely affected by this infection. Blood sucking arthropods and ticks spread the infection between animals. The ticks *Rhipicephalus hurtis, R.kochi* and *Ixodus* are involved in transmission of the infection. The wild ruminants also act as reservoirs of infection. The incubation period is 7-14 days.

Symptoms: Infected animals have high fever with sharp decline in milk yield in lactating cows. On the second day of infection petechial haemorrhages appear on the gums, under the tongue, conjunctiva and mucosa of vulva and vagina. Persistent and foetid smelling diarrhoea may remain for 1-10 days. As a result of swelling and eversion of the conjunctival sac and accumulation of blood in the lower part of aqueous humor, the appearance of the eye is referred as '**poached-egg eye'** appearance. Affected pregnant animals may abort. The mortality rate is very high.

Lesions: There is persistent haemorrhage, oedema and hyperplasia throughout the body. Petechial haemorrhages are seen on the gums, under the tongue, conjunctiva and mucosa of vulva and vagina. Different types of haemorrhages on the inner and outer surface of heart, epicardial and endocardial surfaces are characteristic of this infection. Haemorrhages may also be present in various other organs and glands. Oedema of the intermuscular tissue is the frequent cause of death.

Diagnosis:

1. Based on symptoms and lesions.
2. Microscopical examination of blood film and identification of causative organism.
3. The infection should be differentiated from other acute infection of cattle like anthrax, Rift valley fever etc.

Treatment: No satisfactory treatment is available. Tetracyclines may be effective.

Control: Control is aimed at elimination of arthropod vectors and prevention of wild ruminants in grazing areas. No vaccines are available.

TICK BORNE FEVER

Etiology : *Cytocetes phagocytophilia (E. phagocytophilia, A. phagocytophilum)* (for sheep); *C. bovis* (cattle)

Definition: It is an infectious, non-contagious febrile disease of sheep and occasionally cattle. This infection is spread by ticks.

General characters:

1. **Morphology:** They are very small pleomorphic obligate intracellular organisms that are seen in phagocytic cells. They appear as pleomorphic intracytoplasmic bodies. They are usually stained

by Macchiavello's or modified Ziehl-Neelsen method. In contrast to other organisms they don't retain the basic fuchsin and take the counter stain methylene blue or malachite green. Various developmental forms like small spherical granules, large round or oval bodies and large morulae are seen.

2. **Habitat:** The organisms are commonly seen in neutrophils, eosinophils, basophils and monocytes.
3. **Resistance:** They are less fragile organisms than heart water and Ondiri infection agents.

Cultivation: The organisms are difficult to cultivate in embryonated eggs and tissue culture. The organisms can be cultivated only in sheep or in cattle.

Pathogenesis: Sheep and cattle are the main hosts affected by the infection. The infection is spread by *Ixodes ricinus* ticks. The incubation period is 4-8 days.

Symptoms: In sheep the infection is characterised by high fever (40-42^{0}C) and is associated with presence of parasites in phagocytic cells. The fever curve follows a plateau type with fever persisting for 6-22 days. The infection makes the sheep more prone for other infectious diseases. In cattle the disease relatively insignificant and seen in cattle that have come to a tick infested area from tick free area. In severe cases abortion may result.

Lesions: There are no characteristic lesions.

Diagnosis:

1. Microscopical examination of blood film and demonstration of the organisms in phagocytic cells.
2. Production of the infection in sheep by infecting them with blood from susceptible animals. Instead of cattle splenectomised guinea-pigs may also be used.

Treatment & Control: Tetracycline is the best drug. Eradication of ticks will prevent the occurrence of infection.

HAEMOBARTONELLOSIS

Causativeorganism : *Haemobartonella felis*
Diseasecaused : Feline infectious anaemia

Feline Infectious Anaemia (FIA) is caused *Haemobartonella felis,* which lives on the surface of red blood cells. The resulting structural damage can cause anaemia if the red blood cells are destroyed. The cat's own immune system may also cause death of red blood cells as it tries to kill the parasite attached to them. Clinical signs usually reflect the underlying anaemia. Cats, which have been infected with FIA, may remain carriers of the parasite for life. FIA causes anaemia, which may be accompanied by a fever in the early stages of infection. Methods of spread of *H. felis* infection are not fully known. Very young kittens can be infected, implying that infection can be vertically spread from the queen. As mentioned above, fighting and fleas have been implicated in transmission of infection between cats. Saliva and urine are not thought to be able to transmit the disease and non-infected and infected cats have been housed together without cross infection. Ingestion (such as with cat bites) and injection of infected blood (such as with a blood transfusion) can both result in infection. Clinical signs of anaemia include tiredness, depression, a reduced appetite and pale gums. Weight loss can occur. Such clinical signs can be seen with a variety of diseases that result in anaemia, and are not specific for FIA. Other clinical signs may include enlargement of the spleen and lymph nodes. It is believed that different strains of *H. felis* exist which can cause a range of clinical signs from mild non- significant disease to severe anaemia. The anaemia seen with *H. felis* infection is usually regenerative in type. This means

that the cat is able to respond to the anaemia by producing new red blood cells, which are visible in the circulation. Some infected cats are not anaemic because they are asymptomatic carrier cats, or are infected with a strain of *H. felis*, which causes only mild disease. Therefore the finding of parasites in the circulation does not always mean that the cat has clinically significant FIA disease. Doxycycline has been most commonly used and is given for two to three weeks. Corticosteroids may also be used, in conjunction with antibiotics, to suppress the immune-mediated destruction of red blood cells. In cats with severe anaemia, blood transfusions may be required. Supportive cares to encourage the cat to eat, and rehydration therapy in dehydrated cases, are also important.

CHLAMYDIOSIS – PSITTACOSIS, ORNITHOSIS

Importantspecies : *Chlamydophila psittacii; Chlamydophila abortus*

Synonym : **Parrot fever**

Introduction: Chlamydiosis refers to infection caused by Chlamydial organisms. There are two important infections caused by Chlaymdial organisms in animals.

1. Psittacosis is an infection caused by a microorganism called *Chlamydophila psittaci (*formerly referred *Chlamydia psittaci*).
2. Enzootic abortion in ewes caused by *Chlamydophila abortus (*formerly *Chlamydia ovis*).

Classification: The family *Chlamydiaceae* is recently reclassified into two genera and nine species based on sequence analysis of its 16S and 23S rRNA genes (molecular taxonomy). The two genera, *Chlamydia* and *Chlamydophila*, correlate with the former species *Chlamydia trachomatis* and *C. psittaci*.

The genus *Chlamydia* includes *C. trachomatis* (human), *C. suis*(swine), and *C. muridarum* (mouse, hamster).The genus *Chlamydophila* includes *C. psittaci (*avian), *C. felis* (cats), *C. abortus* (sheep, goats, cattle), *C. caviae* (guinea pigs), and the former species *C. pecorum* (sheep, cattle) and *C. pneumonia* (human).

C.trachomatis is a human pathogen and is also called as TRIC agent (Trachoma and inclusion conjunctivitis).

General characters:

1. **Morphology**: They are Gram-negative, obligate intracellular organisms. They stain as rickettsial organisms by Machiavello's staining method. They are comparatively larger than rickettsia and viewed easily under light microscope. The organisms contain twoparticles elementary body and initial body. The elementary body(300 nm) contains an electron dense core surrounded by three layers and a clear zone separates them. The elementary bodies aresmallerthan initial bodies. The initial body is bigger (700-1200 nm) than elementary body and has no core. The elementary body is highly infectious extracellular replicative form whereas the initial body is infectious intracellular replicative form.
2. **Resistance:** They are sensitive to heat and inactivated at 60^0C for 10 minutes. They can remain viable for a long time under–70^0C.
3. **Other properties:** They produce haemagglutination of mouse and hamster red blood cells.
4. **Replication:** The elementary bodies gain entry into the cell and they convert themselves into initial body. The elementary body (300 nm) contains an electron dense core surrounded by three layers and a clear zone separates them. The elementary bodies are smaller than initial bodies.The initial body is bigger (700-1200nm) than elementary body and has no core. The elementary body is highly

infectious extracellular replicative form whereas the initial body is infectious intracellular replicative form. The initial bodies multiply into smaller bodies by break down and remain as aggregates in the cytoplasm. A single initial body may fragment into five smaller bodies. Such aggregates cause lysis of the surrounding cytoplasm but the nucleus remains unchanged. Hence the infected cell continues to divide and multiply. Ultimately the particles are released by lysis of the cell.

5. **Antigens:** There are two main antigens a heat stable complement-fixing antigen that is group specific and heat labile species-specific antigen.
6. **OIE Listing:** Psittacosis – List B (Avian infection); Enzootic abortion – List B (Ovine infection)
7. **Human risk:** Potentially hazardous organisms for human beings.

Cultivation: They are readily cultivable in embryonated eggs by yolk sac route. Mice are the laboratory animals of choice.

AVIAN CHLAMYDIOSIS

Psittacosis (Avian chlamydiosis) is also known as parrot fever and ornithosis and is caused by the bacterium *Chlamydophila psittaci*. The disease in birds was originally called psittacosis, but the term ornithosis was introduced later to differentiate the disease in domestic and wild fowl from the disease in psittacine birds. The two syndromes are currently considered to be same.

Pathogenesis: The types of birds affected include parrots, parakeets, pigeons, turkeys, ducks etc. *C. psittaci* is excreted in the feces and nasal discharges of infected birds. The organism is environmentally labile but can remain infectious for several months if protected by organic debris (e.g. litter and feces). The two routes of transmission of psittacosis are respiratory and oral. Respiratory transmission includes the inhalation of infected particles of faecal, ocular, nasal, and respiratory discharges, and feather dust. Oral transmission includes the ingestion of food and water contaminated with *Chlamydophila* bearing faeces. Parents that are carriers can infect their nestling via the regurgitated food they feed the babies. Some infected birds can appear healthy and shed the organism intermittently. Shedding can be activated by stress factors, including relocation, shipping, crowding, chilling, and breeding

Symptoms: Symptoms of psittacosis are variable. They depend upon the strain of *Ch. psittaci* with which the bird in infected, the bird's immune system status, species, age, and the presence of other concurrent infections. Mild outbreaks of psittacosis may go unnoticed because there will be very few symptoms. Alternatively there may be very mild respiratory symptoms and diarrhoea. Symptoms are usually related to respiratory and digestive system involvement. During the acute phase the symptoms include respiratory problems (shortness of breath, noisy breathing, "runny nose," sinus infection), diarrhoea, polyuria (excess urine), lethargy, dehydration, ruffled feathers, loss of appetite and yellowish, greyish, or lime green urates. Subacute or chronic psittacosis may show the symptoms like, tremors, unusual head positions, convulsive movements, opisthotonos (neurologic disease in which the top of the head is bent over and approaches the back) and partial or complete paralysis of the legs. Other symptoms noticed are unusual tameness, lack of normal molt, poor condition in beak and nails, sneezing, swollen, infected eyelids and wasting of breast muscles.

Lesions: The important lesion is the wasting of pectoral muscle, focal liver necrosis, and enlargement of spleen, pericarditis, thickening of air sac with adhesions to the sternum, mucous plugs in digestive tract and erythematous lesions on the skin over body and legs.

Diagnosis:

1. Based on the symptoms and lesions.

2. Identification of the organisms by histochemical staining of smears of exudate and faeces, and impression smears of tissues. Giemsa, Gimenez, Ziehl–Neelsen and Macchiavello's stains are commonly used to detect chlamydiae in impression smears of liver and spleen.
3. Isolation of the organisms: Cell cultures and embryonated eggs are used isolate *C. psittaci.* The cell lines commonly used are buffalo green monkey (BGM), McCoy, HeLa, African green monkey kidney (Vero) and L cells. Chicken embryos are also used for the primary isolation of *Chlamydophila*. The standard procedure is to inject up to 0.5 ml of inoculum into the yolk sac of a specific pathogen free 6–7-day-old embryo. The eggs are incubated in a humid atmosphere at 39°C, as multiplication of chlamydia is greatly increased at the higher temperature. Replication of the organism usually causes the death of the embryo within 3–10 days. *C. psittaci* infection causes vascular congestion of the yolk sac membranes.
4. Nucleic acid identification method -PCR
5. Serologicaltests
    a. Complement fixation test: This test is performed during the infection and convalescent phase. A fourfold or more increase in antibody titre is an indication of infection.
    b. Indirect complement fixation test.
    c. ELISA

Treatment: Penicillin, aureomycin, oxytetracyliines and doxycycline are commonly used for treatment.

Control: The outcome of treatment varies, depending upon the individual bird's species, age, immune status, length of illness before treatment was sought, the virulence of the strain with which it is infected, mode of treatment, and its response to that treatment. Precautions must also be taken to protect human caretakers. It is recommended that you take the following:

1. Isolate incoming (new) birds for thirty to forty five days. (Longer is better)
2. Test suspicious birds (those with loose droppings, weight loss, or respiratory problems.)
3. Treat infected birds with Doxycycline for 45 days.
4. Thoroughly clean and disinfect cages, surroundings, and equipment used for a psittacosis bird. Quaternary ammonium disinfectants have proved very effective against these bacteria. (i.e. A-33, Barquat, Cetylcide, Floquat, Hitor, Merquat, Omega, Parvosol, Quintacide, Roccal, Zephiran. [*Avian Viruses, Function and Control*] As well as Roccal-D, Betadine and Environ-One-Stroke)
5. Keep circulation of feather dust to a minimum.
6. Droppings from an infected bird should be soaked with disinfectant and placed in a sealed plastic bag prior to disposal.
7. Contact with infected birds by humans should be kept to an absolute minimum. Strict quarantine techniques should be used.
8. Any Flu-like symptoms in human caretakers should be monitored and a physician should be contacted. Just as with birds, human psittacosis is treatable, but can develop into a serious problem without proper treatment.

Zoonotic importance: The infection spreads from birds to bird keepers. It may also be found in farmers and slaughterhouse workers who process turkeys and ducks. Psittacosis is usually spread by inhaling dust from dried bird droppings and by handling infected birds in slaughter houses. Some birds infected with psittacosis may appear healthy, but can still spread the infection to other birds or humans. Human-to-human spread is very rare. Psittacosis can cause fever, headache, chills,

muscle aches, and sometimes pneumonia with a relatively non-productive cough. The period between exposure and the beginning of symptoms may range from 7 to 28 days but is usually 10 days. Infection does not provide permanent immunity from this disease. Antibiotics such as tetracycline are usually prescribed. Erythromycin is sometimes used in young children. The disease is usually mild in humans, it may be severe, and can uncommonly result in death especially in untreated older people.

OVINE CHLAMYDIOSIS

Ovine chlamydiosis is also called as chalmydial abortion of ewes and is caused by *Chlamydophila abortus* (formerly *Chlamydia psittaci* immunotype-1). Goats, cattle and deer are also affected to a lesser extent. It is a zoonosis and the organism must be handled with biosafety precautions. Pregnant women are particularly susceptible. In sheep, abortion in late pregnancy with expulsion of necrotic fetal membranes is the important symptoms. The infection is more commonly seen in flocks that are closely congregated during the parturient period. Infected animals show no clinical illness prior to abortion. Pathogenesis commences around day 90 of gestation when rapid fetal growth occurs. At this point of time chlamydial invasion of placentomes produces a progressively diffuse inflammatory response, thrombotic vasculitis and tissue necrosis. Milder changes occur in the fetal liver and lung and, in cases in which placental damage is severe, there may be evidence of hypoxic brain damage. Abortion results from a combination of impairment of materno-fetal nutrient and gaseous exchange, disruption of hormonal regulation of pregnancy and induced cytokine aggression.

The infections should be differentiated from infectious causes of abortion such as brucellosis, coxiellosis or other bacterial pathogens like *Campylobacter*, *Listeria* and *Salmonella*.

Infected ewes shed vast numbers of infective *C. abortus* at the time of abortion or parturition, particularly in the placenta and uterine discharges. Human infection may be acquired from such sources or from carelessly handled laboratory cultures of the organism, with effects that range from subclinical infection to acute influenza-like illness. Appropriate precautions should be taken when handling cultures and potentially infected tissues. Human placentitis and abortion caused by *C. abortus* of ovine origin indicate that pregnant women are at special risk and should not be exposed to sources of infection

OTHER INFECTIONS:

Chlamydial organisms also cause catarrhal pneumonia in sheep and goats, respiratory disorders in calves, fibrinous pericarditis in swine and numerous other respiratory infections in animals.

HIGHER BACTERIA - *DERMATOPHILUS*

Dermatophilosis (also known as streptothrichosis) is an exudative, pustular dermatitis affecting cattle, sheep, horses, goats, dogs, cats, many wild mammals, reptiles and, occasionally, humans. The severe disease in ruminants is promoted by immunomodulatory effects induced by infestation with the tick, *Amblyomma variegatum*.

Dermatophilosis in animals is caused by *Dermatophilus congolensis.* Various infections caused by this organism include Streptothricosis (rain scald) in cattle, lumpy wool disease of sheep and strawberry of foot rot of sheep.

General characteristics:

1. **Morphology:** The organism is Gram positive and its morphology is more readily appreciated in smears stained with Giemsa. It hastwocharacteristic morphologic forms—filamentous hyphae and motile zoospores. The hyphae are characterized by branching filaments (1-5 μm in diameter) that

ultimately fragment by both transverse and longitudinal septation into packets of coccoid cells. The organisms appear as rows of eight cells due to division pattern. The coccoidcellsmature into flagellated ovoid zoospores (0.6-1 μm in diameter). It is also referred as higher bacteria since it shows extensive filamentation, aerial hyphae with asexual spores or conidia. They produce zoospores that are motile have four or six flagella.

2. **Habitat:** These organisms are pathogenic and are predominantly seen in backarea.
3. **Classification:** *Eubacteriae* is a class of unicellular or mycelial organisms with rigid cell walls, flagella and form conidia or endospores. This class is further divided into unicellular *Eubacteriales* and mycelial *Actinomycetes.* Under this order, the family *Dermatophilaceae* is proposed with *Dermatophilus congolensis* as type species.
4. **OIE Classification:** List B disease (Bovine disease)
5. **Human risk:** The infection can spread from affected animals to human beings.

Cultural characters: It is difficult to cultivate these organisms under laboratory conditions. They require a temperature of 37^0C and 10% CO_2. Generally shreds from underside of the scab are put in distilled water and they are kept in candle jar at a temperature of 30-35^0C and 10% CO_2 for 2 hours. This step will allow the liberation of zoospores. A small loopful of zoospores are streaked over blood agar and incubated overnight at 37^0C. The colonies are very small and can been with hand lense. The colonies appear as tiny adherent waxy and doomed. They colonies may be yellow or orange in colour. The incomple tehaemolysis is completed in association with *Corynebacterium equi or Streptococcus agalactiae.* In glucose or serum broth, small fluffy spherical colonies form the bottom of the medium or attach to the walls of the container. Smears prepared from fluid media will show organisms with characteristic morphology.

Biochemical reactions: They produce acid without gas, do not reduce nitrates and methylene blue and produce negative reaction for indole, methyl red and Voges-Proskaur tests.

Pathogenesis:

It is mainly infection of the epidermis, and is seen worldwide but more prevalent in the tropics. The infection is wrongly called mycotic dermatitis. The lesions are characterized by exudative dermatitis with scab formation. *Dermatophilus congolensis* has a wide host range. Among domestic animals, cattle, sheep, and goats are affected most frequently; horses occasionally; and pigs, dogs, and cats rarely. It is commonly called cutaneous streptothrichosis in cattle, goats, and horses; in sheep, it is termed lumpy wool when the wooled areas of the body are affected. Deaths occasionally occur, particularly in calves and lambs, because of generalized disease with or without secondary bacterial infection and secondary fly or screwworm infestation. The incidence of the infection has got close relationship with heavy rainfall, tick bites and scratches caused by thorns. During heavy rainfall season the coat of the animal is drenched and the keratin layer is spongy and allows easy entry of organisms through sites of tick bite, cut wounds, scratches etc. Blood sucking flies are play a role in the pathogenesis of the infection. The organisms can also remain in dry skin and hide for a long time. The severe disease in ruminants is promoted by immunomodulatory effects induced by infestation with adults of the tick species *Amblyomma variegatum.* The primary economic consequences are damaged hides in cattle, wool loss in sheep, and lameness and loss of performance in horses when severely affected around the pastern area.

Symptoms and lesions

1. **Streptothricosis (cattle):** Cattle are mostly affected and infection is more common in countries with tropical climate (high temperature and high rainfall). The backregions particularly hump is an important site affected by these bacteria. It affects an inverted triangular area of each side

of back immediately beneath the hump. Other parts affected include face, neck, dewlap, flank, udder, scrotum, escutcheons, axilla and groins. Eruptive eczema followed by thickening of the skin and formation of exuding pimples is the common lesion observed. The serous exudates form a crust at the base of hairs. In cattle, the lesions can be observed in three stages:

- hairs matted together as "paintbrush" lesions,
- crust or scab formation as the initial lesions coalesce,
- accumulations of cutaneous keratinized material forming "wart-like" lesions that are 0.5-2 cm in diameter.

The entire skin is covered with white powdery crusts, which fall during handling. Secondary bacterial infection results in extensive cellulites and death of affected animal.

2. **Streptothricosis (Horses):** Lesions on horses are seen in saddle region and similar to those of cattle, developing with matted hair and "paintbrush" lesions leading to crust or scab formation with yellow-green pus present under larger scabs. Matting and scab formation is uncommon; loss of hair with a fine "paintbrush" effect can be extensive. Persistent wetting in yards, stables, or at pasture leads to lower limb infection. White legs and the white-skinned areas of the lips and nose are more severely affected. Generalized disease is also associated with prolonged wet weather. Outbreaks occur on farms with previously affected horses.
3. **Lumpy wool disease:** The lesions are confined to wool-covered areas of skin especially the lumbar and flank region. The lesions are also occasionally seen in hairy parts of face, ears, scrotum and legs. The incidence of the infection is associated with mild wet weather. The infection starts are papule formation and exudation resulting in matting of the wool. The wool may appear yellow or golden coloured. Scab formation is also noticed at the base of wool fibres. The affected wool is graded less resulting in economic loss.

Diagnosis:

1. **Examination of smears from scabs**: Impression smears are made from the moist, concave undersurfaces of freshly removed scabs. Alternatively, scabs can be soaked overnight in sterile water or saline to sufficiently moisten them so that the undersurface of the scab can be used to make effective impression smears by firmly pressing this surface on to a microscope slide. Smears are then air-dried, fixed by heating or immersion in methanol for 5 minutes, and stained. The organism stains well in dilute carbol fuchsin or methylene blue stain, but Gram's stain or, preferably, a 1 in 10 dilution of Giemsa stain for 30 minutes, gives better differentiation in thick smears, the darkly stained *D. congolensis* contrasting with the paler or pink counterstained background of keratinocytes and neutrophils

2. Cultural examination:

a. **Haalstra method**: Small pieces of scab are placed in a bijou bottle containing 1 ml of sterile distilled water and allowed to stand at room temperature for 3–4 hours. The open bottle is then placedfor15 minutes in a candle jar. Samples of the surface liquid are removed with a bacteriological loop and cultured. The method depends on the release from the scab of the motile cocci of *D. congolensis* and their chemotropic attraction towards the carbon- dioxide-rich atmosphere of the candle jar. A small loopful of zoospores are streaked over blood agar and incubated overnight at 37C. Growth is accelerated under microaerophilic conditions; rough, usually haemolytic, greyish-yellow colonies, about 1 mm in diameter, are seen pitting the medium after 24 hours. Incubation in air produces similar pinpoint colonies at 24 hours that grow to about 1 mm at 48 hours. The rough colonies are formed by the branching filaments, but continued growth in air stimulates the production of the cocci, which are commonly yellow in colour. Colonies take on a smooth, often yellowish, appearance. The cocci are normally vigorously motile when taken from young cultures. The colonies are very small and can been with hand lense.

b. The other method include macerating the scab material with dry glass slide and after removing the coarse particles the finer ones are scattered over the surface of media. The colonies appear as tiny adherent waxy and doomed. They colonies may be yellow or orange in colour.

c. The incomplete haemolysis is completed in association with *Rhodococcus equi or Streptococcus agalactiae.*

d. In glucose or serum broth, small fluffy spherical colonies form the bottom of the medium or attach to the walls of the container. Smears prepared from fluid media will show organisms with characteristic morphology.

3. Fluorescent Antibody Test on smears to identify *D.congolensis.*

4. Based on symptoms and lesions.

5. Based on biochemical characters after confirming by culturing.

6. Serological tests are not currently used in the diagnosis of infection. However, ELISA is used in certain laboratories.

Treatment: Treatment is not very effective and recovery is spontaneous. Arsenicals, Copper sulphate and quaternary ammonium compounds can be used as eternal therapeutic agents. Penicillin, streptomycin and intravenous iodide therapy are also effective.

Strawberry foot rot: The signs of infection vary from mild reddening (inflammation) of the interdigital skin (skin between the digits or toes) to complete separation of the horn of the hoof. Infection commences when bacteria lodge on the interdigital skin causing inflammation; the skin-horn junction then begins to erode and the horn starts to lift. From this point, the bacteria move under the horn causing separation of horn around the heel, sole, toe and eventually to the outer wall. Infected feet may also have a characteristic foul smell. Removal of scabs reveals a raw surface with numerous bleeding points resembling strawberry fruit. Accurate diagnosis is essential, as early outbreaks of **footrot** can be difficult to distinguish from other diseases. Many feet must be examined. Examining only one or two sheep can be misleading and may result in unnecessary or incorrect treatment.

To describe **footrot** at its various stages, a scoring system has been developed called **Modified Egerton Scoring System**

Score	Description
0	Normal foot - No lesion (sign ofdisease)
1	A limited mild interdigital dermatitis (scald)
2	More extensive interdigital dermatitis.
3	Severe interdigital dermatitis and under-running of the horn of the heel and sole.
4	Severe interdigital dermatitis and under-running of the horn of the heel and sole but with the under-running extending to the walls of the hoof.
5	Under-running of the hard horn of the outside walls of hoof.

UNIT 2

VETERINARY MYCOLOGY

Introduction

Fungi are eukaryotic, **unicellular** or **multicellular,** chemoorganotrophic organisms that are **aerobic**, **non-motile** and **devoid** of **chlorophylls**. They are of two types, yeasts and moulds & mushroom. Yeasts are unicellular, while moulds and mushroom are multicellular. Some fungi are dimorphic and exist in two forms. In the host they have yeast like unicellular form. When growing saprophytically in soil or on laboratory medium they have mould form.Majority of the fungi exist as **saprophytes** in nature whereas few fungi exist as **parasitic** forms. The saprophytic fungi produce number of proteolytic, lipolytic and glycolytic enzymes that help in digestion of rotten materials. However, some mutually beneficial relationship is also reported between fungi and animals or fungi and plants.

Morphology

The basic unit of a fungus is the fungal cell. The fungal cells are **eukaryotic** and are larger than bacterial cells. They possess a clear, prominent nucleus surrounded by a nuclear membrane. The nuclear membrane separates the nucleus from cytoplasm. The nucleus also consists of a nucleolus. The cytoplasmic organelles like ribosomes, mitochondria, and endoplasmic reticulum are present except chloroplast.

The fungal cells vary in size ranging from 1-5 μm in breadth and 5-10 μm or more in length. The shape of the cell is commonly egg shaped. Some are elongated and some spherical. Yeast cells do not have flagella or organelle for locomotion. In moulds, the thallus consists of the mycelium and the spores. The mycelium is a complex of many filaments called hyphae.

Hyphae:

The hypha is 5-10 μm in diameter. It is made up of an outer tube like wall enclosing a cavity called the lumen. The lumen is filled with protoplasm. Elongation of the hypha takes place at the apex. The tip region extends as the cytoskeleton delivers materials for enzymatic dissolution of the old and construction of the new wall at the apex. Hyphae can be termed as;

- Nonseptate or coenocytic- they do not have septa.
- Septate with cells having single nuclei.
- Septate with multinucleate cells. Each cell has more than one nucleus in each compartment.

The tip of the growing hypha contains two types of vesicles called macro and micro vesicles. The macro vesicles are more than 100nm in size and micro vesicles are less than 100 nm in size. These vesicles are tightly packed and form a structure called **Spitzenkorper.** These vesicles contain enzymes that are involved in transport. The hyphae of some fungal cells are surrounded by small fibre like structures similar to bacterial fimbriae.

The hyphae of some fungi are decorated with crystals of calcium oxalate, which is assumed to have got a role in strengthening the hyphal wall. Some aerial hyphae produce stalk like structure called **conidiophores** or **sporangiophores,** giving rise to asexual/ sexual spores.

The hyphae of some species of fungi form thick strands called **mycelial cords** or **rhizomorphs.** The hypha of biotrophic fungi (*biotrophic fungi refers to obligate parasitic fungi*) forms special hyphal branches that invaginate the host cell plasma membrane without killing the host cell. These structures referred as **haustoria**are involved in uptake of nutrients from host cell.

Mycelium:

The hyphae branch in all direction and form a mat like structure that is referred as **mycelium.** These mycelia make up the body of the fungus. However, mycelium is absent in single cell fungi like

yeast. Mycelia are of two types – vegetative and reproductive. The vegetative mycelium is made up of surface hyphae and aerial hyphae. The mycelium that grows upward from the surface is referred as **aerial mycelium.** Some hyphae in the vegetative mycelium penetrate into the medium to obtain nutrients. Soluble nutrients are absorbed through the wall. Insoluble nutrients are first digested by the secreted enzymes. Reproductive mycelia usually extend from the medium into air and form the spores.

STRUCTURE OF FUNGAL CELL:

The fungi of medical importance possess structures like capsule, cell wall and cytoplasmic membrane. However, some characters like absence of structures like stems, roots leaves etc., and storage of **glycogen** instead of starch as **primary carbohydrates** differentiate them from plants.

Cell wall

The cell wall is the important structure in fungus. It determines the shape and the process of fungal morphogenesis. The cell wall is made up of mostly carbohydrates, protein and glycoprotein. The carbohydrate components are polysaccharides **chitin, glucan, mannan**and **cellulose.** These polysaccharides form a thatch work consisting of polysaccharide chains called **microfibrils.** Proteins and glycoproteins connect the polysaccharide. Many varieties of fungi share the same polysaccharides and hence have the same surface antigens. But many unique antigenic determinants resulting from the different branching patterns of the polysaccharide are also found within a certain group which are useful for classification.

Capsule

Some fungi have an external coating of slime or a more compact capsule. They are composed of amorphous polysaccharides that may cause the cells to adhere and clump together. The capsule may be antigenic and antiphagocytic in some fungi.

Cytoplasmic membrane

It is similar in structure and composition to the cell membranes of higher eukaryotes. The fungal membrane is bilayered, contains mainly ergosterol and zymosterol. Mammalian cell membrane has cholesterol.

Nucleus

Fungal cells possess a well-defined and proper nucleus surrounded by a nuclear membrane. The nuclear membrane consists of a pair of membranes separated by an inter-membrane space. The inner membrane is a sac whereas the outer membrane is continuous with cytoplasmic membrane. There are number of small holes in these membranes that are referred as **nuclear pores**. These pores facilitate the transport of molecules in and out of the nucleus.

The nucleus also contains another important structure called **nucleolus.** This structure is rich in RNA and is the site of RNA synthesis. The genetic material of fungal cells – the DNA is found in the chromosome, attached with a protein called **histones** and this protein is also responsible for staining property (basic staining). The **histones** are placed in a particular fashion in the double helix structure of the DNA at particular intervals and this forms a structure called **nucleosome**. Nucleosome aggregate and form a fibrous material called **chromatin.** This chromatin is compacted by folding and looping to form an intact chromosome. Fungus that reproduce by sexual method have diploid chromosome.

Organelles

Organelles are discrete structures found in the cell surrounded by a membrane. Apart from nucleus, fungal cells have one important organelle, the **mitochondrion.** It is involved in respiration and synthesis of ATP. The mitochondrial membranes are less rigid since they do not contain cholesterol.

The outer membrane contains number of pores that permit the passage of ATP from mitochondria to cytoplasm. The inner membrane contains number of folds and these folds are referred as **cristae.** The matrix contains number of enzymes that are required for respiration and ATP synthesis.

- **Ribosomes:** The ribosomes of fungal cells are of 80S. These are the sites of protein synthesis. They are also found on the surface of endoplasmic reticulum(ER). The ER having ribosomes is referred as rough ER.
- **Golgi apparatus:** It is involved in secretion of enzymes and other enzymes.

Growth and Nutrition

Fungi grow in a wide **temperature range** between **10** and **40°C**. Some fungi can grow at temperature of 50°C or even in freezing temperatures. Mostly fungi prefer an acidic pH for their growth. The optimum pH range is between 4-7. However, some fungi can grow even in salt water and some can grow in high concentration of solutes.

Fungi can withstand certain extreme conditions than other organisms. Moulds can grow in medium containing high concentration of sugar that inhibits most bacteria. Fungi also can tolerate more acidic conditions than other microbes. Majority of the fungi are **aerobic** with few exceptions. Some yeasts are facultative. They can grow both under aerobic and anaerobic conditions. The optimum temperature for saprophytic fungi is 22-30°C. For pathogenic fungi it is 30-37°C. Some fungi can even grow at 0°C in cold storage.

Fungi are heterotrophic. They cannot use inorganic compounds like carbon dioxide as their sole carbon source. Carbon must come from an organic source such as glucose. Some fungi can use inorganic compounds of nitrogen but all fungi can use organic nitrogen. The fungi exhibit **absorptive nutrition**. The fungi release digestive enzymes into the environment that breaks the indigestible molecules like carbohydrates, proteins and lipids into digestible and soluble small molecules that are easily absorbed into fungal cells. It should be noted here that fungi cannot carry out metabolism without some free water. Fungi can utilise any carbon source as food.

Cultivation

Most fungi can grow aerobically in the usual bacteriological media at temperatures ranging from 20-30°C. Since many fungi grow more slowly compared to bacteria, in media which support growth of bacteria and fungi, fungi may be over grown by bacterial contaminants in a mixed culture. Hence for isolation of fungi the medium which favours their growth but not suited for bacterial growth is preferred. Acidic media having high sugar concentration are tolerated by moulds but not bacteria.

One of the best known and oldest medium for fungal cultivation is Sabouraud's dextrose agar. This medium contains maltose and peptone as main constituents. The recent modification used contains glucose and any one of several specified peptones. It is widely used for isolation of moulds and yeasts. It is useful for growing pathogenic fungi from infected body fluids and exudates. The partial selective action of this medium is due to the high sugar concentration and low pH.

Most parts of the mould are capable of growth. Inoculation of a small fragment of mycelium on the medium is enough to start new mould colony. A transfer needle is used for inoculation. The needle used for moulds is stiffer and has a flattened tip for cutting the mycelium. The ordinary inoculating loop used for bacterial cultures is sufficient for inoculation of yeasts.

Classification and Reproduction of Fungi

Classification of fungi

Taxonomy is defined as the science of classification and classification means orderly arrangement of organisms. **Nomenclature** refers to assigning names to the taxonomic groups. Classification is based on number of characters including morphology.

In the traditional three kingdoms system of classification (prokaryotes, plants and animals), the fungi were placed under the plant kingdom since they have many characters that are common with plants eg. Prominent cell wall. This system of classification is changed to five kingdoms system in which two new kingdoms protista and fungi were added in addition to the three kingdoms mentioned earlier. This method of classification is based on a common ancestor and all its descendents. The common ancestor and all its descendents are referred as **monophyletic groups.** The latest method of classification followed for fungi is based on such monophyletic groups on hypothesized evolutionary relationships and the system of classification is called as **phylogenetic classification.**

The organisms broadly classified under fungi are grouped into three monophyletic kingdoms ***Fungi, Stramenopila***and ***Protista.*** The kingdom Fungi is classified into four phyla -*Chytridiomycota, Zygomycota, Ascomycota* and *Basidiomycota.* The kingdom *Stramenopila*is classified into *Oomycota, Hyphochytriomycota* and *Labyrinthulomycota.* The kingdom protista has number of phyla. However, the fungi are classified under four phyla *Plasmodiophoromycota, Dictyosteliomycota, Acrasiomycota*and *Myxomycota.*

Classification of fungi is mainly based on the characteristics of the sexual spores and fruiting bodies present during the sexual stage of reproduction. The complete or perfect life cycles of many fungi are not known yet. Hence, the imperfectly described fungi must be classified on basis other than the characteristics of the sexual stage. The morphology of the asexual spores and the thalli become important when sexual traits are not found out. So the imperfect higher fungi are provisionally placed in a special class called -class *Deuteromycetes*.

Taxonomy of fungi follows the Committee on International Rules on Botanical Nomenclature.

- Division: -*mycota*
- Subdivision: -*mycotina*
- Class: -*mycetes*
- Subclass: -*mycetidae*
- Order: -*ales*
- Family: -*aceae*
- Genera and species have no standard endings.

Kingdom: Protista

Order :Saproleginales (saprophytes)

Saprolegniaparasitica

Order: Leptomitales (saprophytes)

Order :Peronosporales (parasites)

Family: Pythiaceae, Peronosporaceae, Albuginaceae

Kingdom: Fungi

Phylum: *Chytridiomycota*

(Chytrids, formerly known as Phycomycetes, oldest group consisting of aquatic fungi, saprophytes/ parasites)

Order: *Chytridiales*

Order: *Spizellomycetales*

Order: *Blastocladiales*

Order: *Monoblepharidales*

Order: *Neocallimastigales*

Phylum: *Zygomycota*

(Bread molds, cause diseases in animals; saprophytic/parasitic/ mutualistic,

Nonseptate, asexual and sexual reproduction by zygosporangia and zygospores)

Order: *Mucorales* (Asexual reproduction by sporangia , bread mould

Rhizopus, Absidia,Mucor)

Family: *Mucoraceae*

Family: *Pilobolaceae*

Family: *Entomorphthorales* (Asexual reproduction by conidia)

Order: *Endogonales and Glomales*

Phylum: *Basidiomycota*

(Club fungi, cause diseases in plants; Septate mycelium, sexual reproduction by basidium)

Order: *Agaricales*

Order: *Ustilaginomycetes*

Order: *Uredinales, Septobasidiales, Sporidiales*

Phylum: *Ascomycota*

(Sac fungi, cause diseases in plants and animals; Septate mycelium, sexual reproduction by sac-like structure called ascus)

Order: *Taphrinales*

Order: *Saccharomycetale*s

Order: *Euascomycetes*

***Deuteromycota* (Fungi imperfecti)**

This is not a recognised taxonomic group of fungi. It includes most fungi of medical and veterinary importance, including *Candida*, Microsporum and Trichophyton.

Reproduction of fungi

Asexual reproduction

In asexual /somatic or vegetative reproduction, there are no sex cells or organs and it does not involve union of nuclei. In unicellular fungi, the asexual reproduction is just division of cell whereas in multicellular fungi, it could be fragmentation of hypha and each hypha grows into new individuals. Some of the common methods of asexual reproduction found in fungi are;

1. Fragmentation of soma, each fragment growing into separate individuals

2. Fission of somatic cells into daughter cells
3. Budding of somatic cells or spores, each bud producing a new individual
4. Production of mitotic spores, each spore germinating to form a germ tube that grows into mycelium

In some fungi, the hypha breaks into their component cells and each cell functions as spore and germinates. Such spores are called **arthrospores** or **thallic conidia.** When these spores are covered with thick walls, they are referred as **chlamydospores.**

The asexual spores vary greatly in morphology. The colour may vary from colourless to green, yellow, orange, red and brown and black. The shape may vary from spherical to oval. Fungal asexual spores produced in a sac like structure called **sporangia** and the spores are called **sporangiospores.**

Sporangiospores that are non-motile are called as **aplanospores** and motile spores are called as **zoospores.** Zoospores are motile by a whip like structure called flagellum. Fungal asexual spores produced at the tips or sides of hyphae in various ways are called as **conidia.** Conidia are developed by abstractions, division or budding of the hyphae. The stalks like branch of mycelium in which conidia are developed are called as **conidiophores.**

Sexual reproduction

Sexual reproduction occurs by fusion of the compatible nuclei of two parent cells. The sexual reproduction involves three stages.

1. In the first stage, two protoplasts bring the nuclei together within the same cell by a process called **plasmogamy**.
2. In the second stage, there is fusion of two nuclei that are brought closer. The fusion of these two nuclei is called as **karyogamy**. Such paired nuclei are referred as **dikaryon**.
3. In the third stage, there is meiosis that reduces the number of chromosomes to haploid.

Certain species of fungi possess both male and female sexual organs in the same thallus and such fungi are referred as **hermaphroditic** or **monoecious**. Whereas in some fungi the male and female sexual organs are found in separate thallus and such fungi are referred as **dioecious**. The sex organs of fungi are called **gametangia.** The male sex organs are called as **antheridium** and the female sex organ is called as **oogonium** or **ascogonium.**

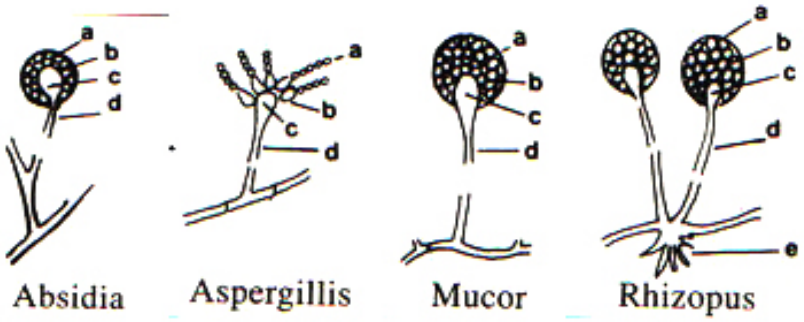
The different methods of sexual reproduction are

- gamete copulation – fusion of gametes one or both of which are motile.
- gamete – gametangial copulation – gametangia come into contact but do not fuse. The male nucleus migrates through a pore or fertilization tube in to the female gametangium.
- gametangial copulation – two gametangia or their protoplasts fuse and give rise to a zygote that develops into a resting spore.
- somatic copulation – fusion of somatic or vegetative cells.
- spermatization – union of a special male structure called spermatium with a female receptive structure. The spermatium empties its contents into the latter during plasmogamy.

Sexual spores occur less frequently and in small numbers than asexual spores. The different sexual spores are;

- **Ascospores** – single celled spores produced in a sac called **ascus**. Usually eight ascospores are present in each ascus.
- **Basidiospores** – single celled spores are borne on a club-shaped structure called a **basidium**.

- **Zygospores** – large thick walled spores formed when the tips of two sexually compatible hyphae fuse
- **Oospores** – formed in a special female structure called **oogonium**. Fertilization of eggs or oospheres by male gametes formed in an antheridium gives rise to oospores. One or more oospores are present in each oogonium.



Absidia (a) Sporangium, (b) Sporangiospores, (c) Columella, (d) Sporangiophore.

Aspergillus (a) Conida, (b) Sterigmata, (c) Vesicle, (d) Conidiospore

Mucor- same basic structure as Absidia.

Rhizopus (a) Sporangium, (b) Sporangiospores, (c) Columella, (d) Sporangiophore, (e) Rhizoids

DERMATOPHYTES (DEMATOMYCOSIS)

Dermatophytosis otherwise known as ringworm is a cutaneous infection caused by a variety of different genera of fungi called the *"dermatophytes."* The term dermatophytes refer to "plants that live on the skin". The name ringworm comes from the characteristic circular red lesions that occur commonly in humans. However, ringworm lesions in animals are much less circumscribed, even presenting as a diffuse folliculitis, a thickened, gray colored area of alopecia; or, a non-specific ulcerated lesion due to licking by the animal. The pathogenic ringworm fungi are divided into 2 major groups. The so-called *"anthrophilic"* dermatophytes are those for which human beings are the primary reservoir. These fungi are only very rarely isolated from domestic animals. The *"zoophilic"* dermatophytes are those for which domestic and wild animals are the major reservoirs. In animals, dermatophytes consist of three important genera *Microsporum, Trichophyton* and *Epidermophyton.* In animals there are 3 specific fungi responsible for causing the infection

- *Microsporum canis* - The source of this species of Ringworm is almost always a cat, especially long-haired cats.
- *Microsporum gypseum*- This species of Ringworm is usually from dogs and cats that dig into contaminated soil.
- *Trichophytonmentagrophytes*- This species infects dogs and cats when they are exposed to rodents or the burrows they live in.

General characters:

1. **Morphology:** All these genera possess a thread like, branching septate hypha forming the mycelium. The type of spores formed varies in parasitic phase and in artificial cultures. In parasitic phase, the septate hyphae forms number of asexual spores called arthrospore along the shaft of hair arranged as mosaic or in rows. These arhthrospores are called **endothrix** if they are arranged inside the hair shaft and called **ectothrix** if they are arranged outside the hair shaft. In artificial cultures they produce macroconidia and microconidia. **Microconidia**are small asexual spores of various shapes and sizes that occur alongside the hyphae. When they are arranged alongside

the hyphae it is called *en thyrse* and when arranged in clusters or in long chain it is called *en grappe.* **Macroconidia** is also called **fuseaux** that are larger imperfect spores. Macroconidia are usually elongated, multiseptate, fusiform or spindle shaped structures with thick walls. The shape of macroconidia is of importance in diagnosing dermatophytes. The structure and shape of macroconidia of three important dermatophytes are given below. Beside the macroconidia and microconidia, the hyphae formed are also of diagnostic importance. They produce racquet hyphae, pectinate hyphae, **spiral hyphae**, nodular organs and chlamydospores. **Racquet hyphae** are composed of individual hyphal cells that are elongated at one end resembling tennis racquets. **Pectinate hyphae** possess irregular projections at one end of the hyphae resembling teeth of a comb or with irregular projections along one side of the hyphal elements that resemble stag-horns that is also called **favic chandeliers**.

a. ***Microsporum:*** Elongated, multiseptate, fusiform or spindle shaped structures with thick walls which are sometimes wrinkled and wart-like.

b. ***Trichophyton:*** Long, thin, multiseptate, smooth walled and cigar shaped.

c. ***Epidermophyton:*** Oval or pear shaped with few septae.

2. **Classification:** The ringworm group of fungi is classified into three types as already mentioned.
   a. **Anthropophilics:** These are pathogenic only for man. Sometimes thy may cause infection in animals. Eg. *M. furrugineum.*
   b. **Zoophilics:** These are pathogenic only for animals. Sometimes they may cause infection in man. Eg. *M.canis,T.gallinae, T.equinum* and *T.mentagrophytes.*
   c. **Geophilics:** They are free-living dermatophytes in soil and cause infection in man and animals only in certain conditions.
3. **Habitat and distribution:** They are found throughout the world basically as saprophytes.
4. **Risk to human beings:** Zoonotic infection. Human beings get the infection from affected pet animals.

Cultural characters: The commonly used media for cultivation are Sabouraud's dextrose agar or malt agar with chloramphenicol and cyclohexamide. These two substances suppress bacterial growth and saprophytic moulds. Addition of potassium tellurite will also reduce the bacterial growth. The clinical material are embedded in the agar and incubated between 22-28^{0}C for two to three weeks. The cultures can be examined every 3-4 days for growth. After growth is noticed, a small amount of growth may be transferred to a glass slide stained by lactophenol cotton blue will reveal the structure of hyphae and macro and microconidia. The type of growth and colour are also of importance. *Trichophyton* will produce orange colouration of the plate behind white cottony mycelial growth whereas *Epidermophyton* will produce brown colour.

Pathogenesis:

Direct contact with infected animals is the common method of spread of the fungi. The infection can also spread from spore contaminated veterinary and farm equipment that have not been properly cleaned. Spores germinate and attack the shafts of the hair and the surface layers of the skin. Dermatophytes have an affinity for keratin-bearing tissues like skin, hair, feathers, nails and horns. They produce enzyme called keratinase that is capable of breaking keratin and keratin is used as source of nutriment. They do not involve the tissues beneath keratin layers or any other organs. The infection is restricted to stratum corneum and adjoining skin. The dermatophytes that affect hair parasitise superficial layers of newly keratinised cells above the root bulb. Exudate oozes from the damaged skin and mixes with debris from skin and hair, thereby forming a crusty scab. The scab is grey-white and noticeably higher than the surrounding skin. Infection spreads from the center outwards and results in

the circular lesion1 to 1.5 inches in diameter. Adjacent lesions may overlap and create larger infected areas. Lesions are most frequent on the head, tail and neck, but they may be found over the entire body in severe cases. Scabs may fall from older lesions and leave a hairless area in the center. Lesions appear as circular, scaly areas of alopecia with or without crust formation. The infection is also characterized by inflammatory and delayed hypersensitivity responses.

Symptoms and lesions:

1. **Cattle:** The important species-affecting cattle are *T. verrucosum var discoides,* and *T.mentagrophytes.* The hair in the affected area falls out with large discrete, circular confluent raised scaly lesions appear on the head, neck and less frequently the back, flanks, escutcheon and limbs.
2. **Horses:** The important species is *T.equinum.* Oval or irregular, focal inflamed oedematous lesions with loss of hair and formation of soft crust. The withers, saddle and girth regions are mostly affected.
3. **Sheep and goat:** *T.verrucosum*is the common species. Generally the infection is rare in sheep and goat. The encrusted lesions are noticed in ears, horns, nose, tail, back and chest.
4. **Pigs:** *T.mentagrophytes* is the common species. The less important species are *T.verrucosum* and *M.nanum.* The lesions appear as circular reddened areas on neck and trunk. These areas are covered with brown or reddish brown crusts.
5. **Dogs:** *M.canis, M.gypseum* and *T.mentagrophytes* are the common species that affect dogs. The lesions are more commonly seen in head, base of the ear, abdomen and scrotal area. Encrusted scaly lesions are commonly seen. The skin lesions that appear with Ringworm are variable, and do not necessarily form a ring. There will be hair loss, usually in small patches at first. As time goes on the patches may disappear or appear at other locations on the skin. There might be scratching due to itchiness.
6. **Cats:** *M.canis* is the most common species that affects cats.The lesions are well pronounced in long haired breeds. Typical skin lesions are discrete, roughly circular, areas of hair loss, particularly on the head, earsor extremities of the paws. The hairs surrounding affected areas appear broken. The affected skin is often scaly and look inflamed.
7. **Poultry:** *T.gallinae* is the most common species. The infection is also called as favus. The lesions appear as thick white crusts on the comb, wattles and other featherless areas. The infection may also involve the base of the feathers called circular favus. In severe infection the entire head is covered with crusts, loss of feathers, emaciation and death.

Diagnosis:

1. Based on **symptoms** and **lesions**.
2. Examination under **Wood'slight**:In this method the affected hairs areviewed directly by UV rays. In Wood's lamp, the light from a mercury vapour lamp is filtered through sodium-barium-silicate glass containing nickel oxide. The arthrospores that are found externally (ectothrix) on the shaft of the hairs produce bright greenish fluorescence in dark room under Wood's lamp. However, the endothrix do not produce any fluorescence.
3. **Processing of hairs,nails and skin scrapings** and examined in 10 or 20% KOH under a cover slip for arthrospores and hyphae. Before examination the materials should be warmed in KOH for 10 minutes and cooled for 30 minutes. Addition of 36% DMSO will enhance the clarity of observation. A drop of lactophenol cotton blue stain will also increase clarity.
4. **Cultural methods**: Sabouraud's dextrose agar or malt agar with chloramphenicol and cyclohexamide is commonly used. The clinical material are embedded in the agar and incubated between 22-28^{0}C for two to three weeks. The cultures can be examined every 3-4 days for growth. After growth is noticed, a small amount of growth may be transferred to a glass slide stained by lactophenol cotton

blue will reveal the structure of hyphae and macro and microconidia. The type of growth and colour are also of importance. *Trichophyton* will produce orange colouration of the plate behind white cottony mycelial growth whereas *Epidermophyton* will produce brown colour. Block slide culture and agar sausage method can also be used.

Treatment:

1. Oral administration of griseofulvin is useful in treatment at a dose rate of 20-40mg/kg bodyweight.
2. Topical application of salicylic acid, benzoic acid, sulphur containing compounds like ichthammol, tars, paraffin etc.

Dermatophytosis in humans:

Dermatophytosis in humans appear as circular patches of thickened, inflamed skin or hair loss with scaling. These may be itchy. Lesions may occur anywhere on the skin or scalp. Ringworm in humans usually responds well to treatment.

ZYGOMYCETES

Importantgenus	:	*Absidia – A. corymbifera,A.ramousa*
Mucor	:	*M.pusillus, M.spinosus, M.recimosus*
Rhizops	:	*R.suinus, R.microsporus, R.equinus*
Important disease	:	Zygomycosis

General characters:

1. **Morphology:** Zygomycetes are characterized by non-septate hyphae, asexual spores (sporangiospores) produced in thin walled sporangium and sexual zygospores and oospores.
2. **Classification:** *Mucoraceae*is one of the important families under the order zygomyctes that are characterized by zygospores, sporangia bearing clearly defined columella that includes *Mucor, Rhizopus and Absidia.* The important difference between them is *Mucor* does not possess root like rhizoids in the pseudomycelium, *Rhizopus* have rhizoids opposite to the base of aerial sporangiophores and in *Absidia* the rhizoids are found between the sporangiophores. The other order oomycetes include the genus *Saprolegina* that are characterized by formation of sexual spores (oospores) within an oosphere and elongated club shaped (clavate) sporangia opening by a single pore at the tip or side.
3. **Habitat:** They are saprophytes.
4. **Reproduction:** They reproduce by both sexual and asexual methods.
   a. **Sexual method:** Perfect spores are produced when the tips of two suitable adjacent hyphae come together and form short side branches called suspensors. Gametangium forms at this area. This gametangium enlarges and secretes a thick horny wall, which forms the resting spores or zygospores. Oospores are formed when male portion of hypha fuses with female portion. The structure that contains oospores is termed as oosphere.
   b. **Asexual method:** ***Zygomycetes*** produce long non-septate modified aerial filaments called sporangiophores which terminate in large thin walled structure called sporangium which contains asexual spores – the sporangiospores. Two types of sporangiospores are seen - the non-motile multinucleated aplanospores and motile zoospores.

Pathogenesis: The natural hosts are man and may species of domestic animals. Susceptible animals get the infection by ingesting or inhaling the spores. Debilitating conditions, over usage of corticosteroids, diabetes mellitus and leukaemia are some of the predisposing factors. The organs mostly affected are

the lymph nodes, alimentary tract, respiratory tract etc. where they produce granulomatous lesions with ulcerative characters. In guinea pigs a pseudo tuberculosis type infection is produced. The genus *Rhizopus* is also associated with placental infection in cattle resulting in abortion. In poultry and sheep the infection is mostly limited to respiratory system. In dogs all the three-genus cause otitis. Placental infection is also noticed in swine.

Diagnosis: Generally the infection goes unnoticed withoutproper exhibition of symptoms. Only during post mortem examination, histopathological examination of granulomatous lesions will reveal coenocytic hyphal elements. In aborted animals, histopathological examination of placental membranes and foetal stomach contents will reveal coenocytic hyphae.

Treatment: Not effective. Intravenous iodide therapy may be followed.

ASCOMYCETES (ASCOMYCOTA)

Important genus - *Aspergillus*

Important diseases - Brooder pneumonia, bovine mycotic abortion

Morphology:

The members of the genus Ascomycetes are characterized by presence of a hyphae that is septate. The hyphae are well developed, profusely branched, and septate and their cells are multinucleate. They multiply by through sexual and asexual spores. They possess fruiting bodies called **ascocarp (scletoium).** This ascocarp contains number of sac like structure called **ascus**that is filled with sexual spores – **ascospores.** The structure of asexual spores is also significant. The hyphal cell that branches to give conidiophore is called foot cell. The conidiophores are long erect structures that terminate in bulbous structures called **vesicle.** The development from vesicle is of two types **biseriate and monoseriate.** In biseriate, a layer of cells called **metulae** cover the vesicle and from this metulae **conidiogenous** cells or **phialides** arise. In uniseriate the structure metulae are absent.

Reproduction:

1. **Asexual:** The conidiogenous cells mature and form conidia in basipetal chains. The conidium development is blastic. Each conidium develops as spherical wall protrusions from phialide tips. The phialide nuclei undergo mitotic division and one of the nuclei enters into developing conidium. A septum then forms between conidium and phialide cell and separatesboth. The next conidium develops beneath this and pushes the already developed conidium above. Due to physical connection between each conidium, they remain in chain for sometime.
2. **Sexual:** The sex organs antheridia and ascogonia are produced close to each other on somatic hyphae. Both are multinucleate and elongate structures often coil around each other. Pairing of nuclei in ascogonium takes place irrespective of the entry of antheridia nuclei. However, if antheridial nuclei enter into ascogonia pairing takes place between antheridial and ascogonial nuclei and produce number of ascogenous hyphae that branch within the developing ascocarp. The asci are formed at the tips of such hyphae at different levels.

Habitat:

Most of the members of this group are saprophytes and non-pathogenic. They are the common laboratory contaminants. Few genera like *Aspergillus, Claviceps* contain pathogenic species. They cause respiratory infection or food poisoning.

Cultural characters:

Members of this group are aerobic and grow well in ordinary laboratory media at an incubation temperature of 25-37^{0}C. Blood agar and Sabouraud's agar, potato dextrose agar are commonly used

to study these fungi. The colonies develop very fast that appear first as white, fluffy, low filamentous growth later turning into granular and spreads across the media. The colour of the colonies changes into green. This green colouration is due to conidia.

Pathogenesis:

There are more than 200 species of the genus *Aspergillus* and it is difficult to differentiate between them. Hence the infections produced by this genus are collectively called as **Aspergillosis.** The spores of the genus are commonly foundin dust, soil, rotting plants and animal matters. The spores are abundantly found on hay, straw and grain. The settling of spores on these feed occurs during storage. When these spores are inhaled or ingested they settle in lungs or other organs and produce infection. Apart from the lung infection that is very common, *Aspergillus* is also associated with lesions on skin, external ear, nasal sinuses and occasionally bones and meninges. They also cause placental infection in cattle resulting in abortion. In horses, it causes **equine guttural pouch mycosis.** Of all the infections produced by *Aspergillus,* avian aspergillosis (brooder pneumonia) caused by *Aspergillus fumigatus* is most important.

Symptoms and lesions:

1. **Avian aspergillosis (Brooder pneumonia):** The affected chicks die suddenly within 24-48 hours without any symptoms. The symptoms if present are generally anorexia, listlessness, high temperature (pyrexia), increased respirations, diarrhoea and convulsions. The clinical symptoms and post mortem lesions are indistinguishable from pullorum disease. In post mortem the affected birds show number of small, discrete, military type, yellow white granulomata scattered throughout the lung. Small firm nodules are seen in air sacs. The air sacs consist of masses of densely packed hyphae and mycelium. In chronic cases, the fungus grows in the mucous membranes of bronchi and air sac filling them with dense dark green fluffy mass of hyphae and spores. Some times caseous exudation is also noticed.
2. **Bovine mycotic abortion:** Along with *Mucor, Absidia, Rhizopus* and *Penicillium, Aspergillus fumigatus* also cause abortion in cattle. This infection is characterized by placentitis. There is gross thickening and focal necrosis of affected cotyledons. Abortion usually occurs during second half of pregnancy. Ingestion and inhalation of spores through contaminated feed are the important modes of spread of infection. Infection generally occurs during winter months due to confinement. Affected animals do not show any symptoms. The fungus does not remain in uterus after infection.

Diagnosis:

1. Diagnosis of aspergillosis based on symptoms and lesions alone is very difficult. Demonstration of fungal hyphae by lactophenol cotton blue in affected tissue is a easy and reliable method. However it is difficult to arrive at the species based on either cultural or microscopical methods.
2. Diagnostic tests like complement fixation test and AGID are done to identify the fungal antigens or antibodies in the serum

Other miscellaneous infections:

1. Equine guttural pouch mycosis: This infection is characterized by epistaxis or pharyngeal paralysis.
2. Green mould of eggs.
3. Small encapsulated lesion of cattle.

FUNGI IMPERFECTI

This group consists of fungi that do not have sexual stage of development. This groups consists of fungi ranging from yeast and yeast like organisms to dimorphic fungi and dermatophytes. Some of

the common pathogenic organisms are *Cryptococcus sp., Candida sp.* (yeast and yeast like organisms), *Blastomyces, Histoplasma, Sporotrichum* and *Coccidiodes*(Dimorphic fungi) and *Microsporum, Trichophyton, Epidermophyton* (Dermatophytes).

Yeast and yeast like organisms:

Morphology: Yeasts are a large group of organisms with three groups based on their mode or replication. Generally yeasts are unicellular that reproduce vegetatively by **budding.** Generally the yeast cell is haploid except the zygote that is diploid. Certain yeasts produce a series of buds in succession that remain joined for a period of time forming a pseudo-mycelium. However, some yeast does produce true mycelium. The hypha of yeasts is septate with single or many pores in the septal wall (perforate septum). The septal wall grows centripetally and almost closes the pores forming a closure line. They also have an electron dense structure called **Woronian bodies** found in the hypha near the septal wall.

Reproduction: Budding and formation of conidia and arthrospore are the usual methods of asexual reproduction. Formation of bud is initiated at the time of duplication of spindle pole body. An area of enzymatic wall softening within a newly developed chitinous ring allows a portion of the cell contents along with newly synthesised wall to come out through the ring. At this stage a plate like structure called **abscission plate** begins to form at the isthmus between the two cells. Cell wall materials are deposited on either side of the plate and thus the mother and daughter cells are separated. A bud from the mother usually leaves a scar on the surface of the mother cell that resembles like a crater. The sexual reproduction of yeasts is through production of ascospores. These ascospores are found inside ascus, which in turn are located inside the ascocarp. The ascospores otherwise called as ascus are multishaped. The process of formation of new yeast from ascus is called as ascosporogenesis.

Classification:

1. Certain yeasts however, form spores and the spore that develops from mother cell is called **blastospore.** Some species of yeast produce pseduo- mycelium or mycelium that forms **arthro spore** and rarely into **chlamydospore.** However, certain yeasts reproduce sexually by forming **ascospore**.
2. Certain yeasts produce aerial spores called **ballistospores** that forcibly project into the air.
3. The third group consists of yeasts that do not produce ascospores or ballistospores and reproduce exclusively by vegetative process. This group of fungi is called as fungi imperfecti. The family *Cryptococcaceae* that contains many pathogenic types of yeast comes under this group. The further classification of this family into genera is based on the morphology of asexual spore, presence or absence of pseudo-mycelium, fermentation of sugars etc.

Habitat: The yeasts in general are saprophytes.

Cryptococcosis

This infection is otherwise called as European blastomycosis or torulosis. The causative organism is *Cryptococcus neoformans.*

a. ***Morphology:*** The fungi appear as spherical, thick walled, single budding yeast surrounded by a wide gelatinous transparent capsule.

b. ***Habitat:*** Generally they are found in the soil, fruit, milk etc. The main source of this fungus is pigeon droppings. In pigeons this fungi is found in the intestine but does not multiply due to higher body temperature of pigeons.

c. ***Cultural characters:*** They grow well in Sabouraud's dextrose agar, malt agar, blood agar and other laboratory media. The young cultures appear as white or cream-coloured later turning in to pale brown coloured. This fungus lacks the ability to ferment sugars and is ureapositive.

d. ***Pathogenesis:*** It is a non-contagious, chronic, fatal systemic infection that involves lungs, skin or other parts of the body. This fungus has got marked predilection for brain and meninges. The hosts affected by this infection are cattle, pigs, dogs, man, monkey, dog etc. The infection initially starts as lung infection followed by infection of other organs particularly brain and meninges. The fungus spreads via blood stream. The infection may take a long time even years to set in. In horses, it causes myxoma like lung infection with respiratory distress and nasal discharge. In cattle, it causes mastitis. The udder becomes swollen and hard coupled with enlargement of regional lymph nodes and drop in milk yield. In dogs, it causes CNS infection resulting in incoordination, hyperaesthesia and nasal discharge. Granulomatous lesions are also noticed in lungs, various organs and areas around ears, face and feat. Extension of infection from nasal cavity to optic nerve results in blindness.

e. Diagnosis:

i. Direct microscopical examination on specimens stained by Giemsa stain or India ink reveals spherical, thick walled, single budding yeast surrounded by a wide gelatinous transparent capsule.

ii. The fungus can also be viewed by immunofluorescense.

iii. Isolation of fungi by culturing: This fungus grows well in Sabouraud's dextrose agar, malt agar, blood agar and other laboratory media. The young cultures appear as white or cream-coloured later turning in to pale brown coloured.

iv. Biochemical reaction: This fungus do not sugars and is urea positive.

v. Animal inoculation: This is the only fungus under the family *Cryptococcaceae* that can grow in mice. Mice are infected by intra-peritoneal or intra-cerebral route. Fungus can be seen as budding cells in abdominal tissues, lungs and brain. This fungus also produces fatal infection in pigeons and chick embryos by intra-cerebral or intra-venous route.

f. ***Treatment:*** Generally the treatment is ineffective. However, amphotericin B is found to be effective.

Candida

The genus *Candida* is also known as *Monilia.* It is classified under the order *Moniliales.* The important species under this genus are *C.albicans, C.parapsilosis, C.tropicalis, C.krusei* and *C.stellatoidea.*

1. **Habitat:** The *Candida* organisms are found as saprophytes in mucous membranes of digestive, uro-genital and respiratory systems.

2. **Morphology**: The budding cells appear as gram-positive. They are characterized by formation of pseudohyphae. The yeast cells repeatedly divide and form a chain like structure called pseudohyphae. Each cell serves as spore. The species *C.albicans* is characterized by the production of thick walled chlamydospores (resting spores).

3. **Cultural characters:** The *Candida* species readily grow in bacteriological media at 25-37°C producing smooth, creamy bacterial like surface colonies. The pseudohyphae formed extended into the medium. This can be well demonstrated in agar slope culture. These colonies become sub- surfaced and furrowed after prolonged incubation. The commonly used media are malt agar or blood agar or Raulin's media with or without antibiotics. The resting spores – chlamydospores are well formed in corn- meal agar. The organisms produce severe kidney, liver and spleen lesions in rabbits and

mice by intravenous route. This organisms ferment glucose, sucrose, maltose and galactose and produce acid and gas.

4. **Pathogenesis:** The infection caused by *Candida* organisms are called and **candidiasis** or **moniliasis.** *C.albicans*is the most important species. The infection caused by *C.albicans* may be acute, subacute or chronic. The organisms are generally found as saprophytes and are non-infectious. However, it produces infection in hosts when the host is having malignant diseases or nutritional deficiency. *Candida* infection is common in individuals undergoing prolonged antibiotic therapy. *Candida* infection occurs as cutaneous and on mucous membranes. The infection in mucous membranes is referred as **thrush.** In human beings, the mucous membranes oral cavity and vagina are commonly affected. Oral thrush occurs in babies and patients at the terminal stages of wasting diseases. The infection in poultry is called as avian **candidiasis** or **moniliasis** or **thrush**. Candidiasis in poultry occurs in young birds and acute infection is characterized by rapid death and high mortality. Characteristic lesions develop in the inner lining or crop. In acute cases, the lesions appear as tiny, discrete, yellow-whine or grey-white pustules loosely adherent to the mucous membrane. In chronic infections, the crop wall is thickened and covered by a corrugated false membrane of yellowish-grey necrotic material resembling Turkish towelling. Necrotic exudate may be seen in the gizzard and intestines with miliary abscesses in lung and liver. *C.albicans* causes stomatitis and gastric ulcers in pigs, mastitis and abortion in cattle and otitis in dogs.

5. **Diagnosis:**
   - **a.** Based on symptoms and lesions.
   - b. Microscopical examination culture stained by Giemsa or Gram's stain.
   - c. Isolation of *C.albicans* by culturing.
   - d. Isolation in rabbits or mice.

6. **Treatment:** No satisfactory treatment is available. However, nystatin in drinking water, amphotericin B and iodides may give some results.

Malassezia pachydermatis

Malassezia species, commensals on the skin of animals and humans, are aerobic, non-fermentative, urease-positive yeasts which grow at 35°C to 37°C. One species, *Malassezia pachydermatis* (formerly *Pityrosporum canis)* is of veterinary importance. The cells of *M. pachydermatis,* which are bottle-shaped, thick walled and up to *6.5* μm in length, reproduce by monopolar budding on a broad base. Multiple budding may occur from the same site on a mother cell. After repeated budding, a distinct collarette forms at this site.Pseudohyphae may be produced infrequently in tissues.

Usual habit

Malassezia pachydermatis can be found on the skin of mammals and birds, particularly in areas rich in sebaceous glands. The anal region, external ear canal, lips and inter- digital skin of dogs are frequently colonized by this yeast.

Identification of *Malassezia pachydermatis*

- The unique budding pattern is demonstrable in microscopic preparations stained with methylene blue.
- ***Malassezia pachydermatis*** is the only member of the genus which grows on Sabouraud dextrose agar without lipid supplementation. Colonies, which are dull, opaque and cream-coloured, have a smooth surface.

Pathogenesis and pathogenicity

*Malassezia pachydermatis*is associated with two clinical conditions, otitis externa and dermatitis, usually in dogs. Colonization and growth of the organism in these locations may be associated with immunosuppression and other predisposing factors. When the yeast cells are present in high numbers, they apparently induce excessive sebaceous secretion, a feature of seborrhoeic dermatitis.In otitis extema, the production of proteolytic enzymes by *M. pachydermatis* results in damage to the mucosa of the ear canal. Excessive production and retention of wax, a consequence of ceruminous gland hypersecretion, combined with the activity of M. pachydermatis and other microorganisms contribute to inflammatory changes. Inflammatory exudate and necrotic debris accumulate in the canal.

Diagnostic procedures

- Involvement of *M. pachydermatis* should be considered in otitis externa and in canine seborrhoeic dermatitis. Exudate from affected ear canals should be submitted for laboratorv examination. In severe dermatitis, biopsy of skin may be considered.
- Characteristic yeast cells are demonstrable in exudates stained with methylene blue.
- *M. pachydermatis* can be cultured aerobically at 37°C for 3 to 4 days on Sabouraud dextrose agar containing chloramphenicol.

Identification criteria for isolates:

- Colonial appearance
- Growth without lipid supplementation (consistent with *M. pachydermatis*)
- Characteristic microscopic appearance

- In otitis extema, blood agar and MacConkey agar plates should be inoculated with exudate to isolate bacterial pathogens aetiologically associated with *M. pachydermatis*.

Clinical infections

M. pachydermatis has recently been implicated in canine seborrhoeic dermatitis and also in skin infections secondary to epidermal dysplasia, a genetic disorder of West Highland terriers**.** This yeast is one of many organisms which may contribute to *otitis externa* in dogs. The condition occurs less frequently in cats.

Canine seborrhoeic dermatitis

Factors which predispose to canine seborrhoeic dermatitis include hypersensitivity disorders, keratinization defects, immunosuppression and persistently moist skin folds. Lesions tend to occur more frequently and with greater severity in skin folds. Pruritis and erythema are accompanied by a foul-smelling, greasy exudate with matting of hair. Concurrent bilateral *otitis externa* may be present. Treatment with miconazole-chlorhexidine shampoo, or a combination of topical and oral ketoconazole may be effective.

Canine *otitis externa*

Otitis externa is characterized by a dark pungent discharge from the ear canal and intense pruritis with head shaking, scratching and rubbing of the ears. Damage to the pinna may manifest as a haematoma. The mucosa of the ear canal is painful and swollen. The aetiology of this condition is complex. Poor ear conformation, wax retention and immuno-suppression are among the factors which may predispose dogs to the disease. *Malassezia pachydermatis*, which is present in low numbers in the ear canal of clinically normal dogs, may proliferate in *otitis externa*. Predisposing causes should be investigated and eliminated or treated. The fungal and bacterial pathogens causing the inflammatory response should be identified by microscopic examination and culture of aural exudate. Antibiotic

sensitivity testing should be carried out on the bacterial isolates prior to initiating therapy. Proprietary ear drops, containing drugs effective against the bacteria and fungi usually involved, and also against *Otodectescynotis*, may be beneficial. In chronic cases, surgical intervention may be required.

PATHOGENIC DIMORPHIC FUNGI

Dimorphic fungi are characterized by two forms that they exhibit during their lifetime. At 37°C they exhibit yeast like growth and at 20°C they exhibit mycelial growth. The parasitic phase is characterized by absence of mycelial growth and in tissues they appear like yeast and multiply by budding. Generally they appear as yeast like cells in infected tissues and mycelial colonies in artificial media. Most of parasitic fungi that cause infections in human beings and animals belong to this category. There are four important groups under this category. They are *Histoplasma, Blastomyces, Coccidiodes* and *Sporotrichum.*

A. Histoplasmosis:

Histoplasmosis is the general term given for infection caused by species of the genus *Histoplasma.* The two important species of *Histoplasma* genus are *H.capsulatum* and *H.farciminosum. H.capsulatum* causes fatal granulomatous infection of lung and lymphatic system mostly in man and canines. *H.farcimonosum* causes epizootic lymphangitis or pseudo glanders or Japanese farcy in equines, which is a chronic infection of lymph glands and lymph nodes.

H. capsulatum: This species is worldwide in distribution and occurs sporadically in many countries. It causes fatal progressive granulomatous disease of reticuloendothelial system or a mild asymptomatic infection of the lung. The infection is non-contagious and spreads through inhalation of the infected dust particles. The species of animals affected are cattle, horses, cats, dogs and rodents. Generally the infection is asymptomatic and diagnosed after post- mortem. In dogs the infection is characterized by chronic cough and dysentery.

Morphology: The yeast bodies appear as small, capsulated oval cells that are usually found in macrophages. Budding is not a common feature of these fungi and rarely they are found outside the cell. In artificial cultures, the mycelial growth consists of fine cottony aerial hyphae bearing lateral conidia. The chlamydospores are thick walled bearing spiny projections from outside.

Diagnosis

1. Identification of non-budding yeast like cells in the tissues of reticuloendothelial system
2. Demonstration of conversion to mycelial phase at 22oC in Sabouraud's dextrose agar and reconversion of mycelial phase to yeast like colonies at glucose blood agar. The mycelial colonies appear as white coloured that later changes into tan coloured. Presence of aerial hyphae, lateral conidia and horny chalamydospores are or diagnostic importance. The yeast like colonies appears as wrinkled and creamy. Budding which is not seen in tissues during parasitic phase is common in artificial media.
3. Demonstration of yeast phase in mice through intraperitoneal inoculation.
4. Demonstration of yeast phase in HeLa cells.
5. An intradermal histoplamin test is done in dogs to identity canine histoplasmosis.
6. ***Treatment:*** Not successful. Amphotericin B at 0.1% intravenously may provide some relief.

H. farciminosum

This infection (Epizootic lymphangitis) is more common in the horse breeding areas of Asia, Africa and Europe. It is chronic infection characterized by granulomata and suppurating ulcers on the skin

with associated lymph glands and lymphatics. It is also known as Pseudo glanders, and Japanese farcy. Horses, mules and donkeys are the normal host. However, man, pigs and camels are also affected by the infection. The important modes of spread are through direct contact, through contaminated utensils, saddlery, grooming utensils (fomites) or through flies particularly biting flies. Ingestion and inhalation are other important modes of transport.

The infection is characterized by pus containing lesions on the exposed surface of skin, associated lymphatic vessels particularly limbs and neck. The infected lymphatics become dilated at intervals forming lines of hard abscesses. This is called cording. Cording is more common in limbs and neck. Later, this abscesses rupture-releasing blood stained pus. The granulating ulcers have a concave, bright red, glistening surface. A scab forms over the ulcer, which later sloughs off leaving a scar. The mucous membranes of lungs, tracheal and nasal cavity are affected. There is also a watery discharge from the nostrils and eyes. Conjunctivitis and papules on the conjunctiva and nictitating membrane are the other signs.

Diagnosis:

1. Demonstration of thin walled spherical, oval or pear shaped yeast like bodies in pus form unopened abscess or from nasal or conjunctival discharges.
2. Isolation of the causative organism in Sabouraud's agar or Harley digest agar containing 10% horse serum (pH 7.4). The colonies appear as tiny grey flakes after 2-8 weeks of incubation.
3. Demonstration of antibodies against the infection by complement fixation test (CFT).
4. Allergic intradermal test.
5. Differential diagnosis with glanders: Mallein test, isolation of causative organisms and their cultural characters will differentiate between glanders and psudoglanders.

Treatment:

1. Disinfection and cauterisation of the open lesion and removal of affected lymphatic cords.
2. Intravenous iodide therapy

Control:

1. Isolation and slaughter of the affected animal.
2. Use of hygienic saddlery, grooming equipment, other utensils etc.
3. Installation of fly proof stalls.
4. Avoiding saddle galls and other injuries to horses.

Blastomycosis

Blastomycosis is a chronic non-contagious infection of dogs and horses caused by *Blastomyces dermatitidis.* It is characterized by formation of suppurative and granulomatous lesion in any part of the body. The fungus multiplies in lungs, skins and bones.

Morphology: The yeast like cells appear as thick walled with an attached blastospore. They are found inside the phagocytic cells. The mycelial phase consists of septate aerial hyphae with large number of spherical or pear shaped conidia and many smooth walled chlamydospores.

Pathogenesis: Canine blastomycosis is a chronic, debilitating and highly fatal disease associated extensive pulmonary lesions. Nodules are also commonly seen in dogs. The chest cavity is filled with nodules and the lungs are highly congested. Bronchial and mediastinal lymph nodes show enlargement. In horses, the infection is characterized by series of long standing abscesses in the anus and vulva associated with debility. In cattle the infection is characterized by mastitis.

Diagnosis:

1. Based on symptoms and lesions.
2. Isolation of fungi and demonstration of yeast phase with wrinkled, creamy or waxy colonies on blood agar at 37°C. The mycelial phase in Sabouraud's agar at 22°C exhibit fluffy white mycelia that later turns into tan or brown. The mycelial phase consists of septate aerial hyphae with large number of spherical or pear shaped conidia and many smooth walled chlamydospores.
3. Radiographic examination.

Treatment: Not very effective. Iodide therapy or intravenous amphotericin B provides some relief.

Coccidioidomycosis (*Coccidiodes immites)*

It is an extremely infectious but non-contagious disease of man, monkeys, horses, cattle, dogs, sheep, rodents etc. The infection occurs as an acute, benign and self-limiting disease of pulmonary system. It may also occur as chronic infection involving cutaneous, visceral and osseous tissues.

Morphology: The fungus appears in several phases in infected tissues. They appear as spherical thick walled bodies called **spherules** that contain large number of irregularly shaped endospores. These endospores are released into the surroundings. Hyphal elements develop from the wall of spherules. The hyphal segments also break and forms arthrospore.

Habitat: These fungi are generally found in the soil and inhalation of the dust particles is the important source of spread among hosts.

Cultural characters: The media commonly used are Sabouraud's dextrose agar and Littman Oxgall agar. The growth initially appears as white surface colonies, moist and membranous. Later, these surface colonies turn into cottony mycelial growth. Though the fungus is a dimorphic fungus, no differences are found between colony characteristics at 22^0C and 37^0C. The arthrospore formation is very conspicuous in artificial culture. Hence, the cultures should be handled very carefully.

Pathogenesis: The organisms are commonly found in the soil and inhalation of the dust particles is the most important route of spread. The arthrospores are also introduced through the skin as a result of injury. The incidence is more during dry and dusty months. The infection remains sub clinical with automatic recovery. In dogs the infection remains for a period between 2 and 5 months and is extensive than man and cattle. The infection is characterized by granulomata of lungs, pleura, heart, spleen, liver, kidneys, bones and joints. However, granulomatous lesions are mostly restricted to bronchial and mediastinal lymph nodes and occasionally lungs.

Diagnosis:

1. Based on symptoms and lesions.
2. Examination of sputa, biopsy materials as wet mounts stained by lactophenol cotton blue reveals the spherules containing endospores and broken hyphal elements (arthrospore).
3. Isolation of the fungi is Sabouraud's dextrose agar and Littman oxgall agar. The growth initially appears as white surface colonies, moist and membranous. Later, these surface colonies turn into cottony mycelial growth.
4. Hypersensitivity test (Coccid Odin skin test): 0.1 ml of the extract of fungus (coccid Odin) when given intradermally produce erythematous lesion at the site of injection in patients having coccidiodes infection after 20-48 hours.
5. Demonstration of complement fixing antibodies through complement fixing antibodies.

6. Other serological test like agar gel immuno diffusion (AGID) and immunofluorescense.
7. Inoculation of the suspected material into mice through intraperitoneal injection produces the specific granulomatous lesions in mice after 1-2 weeks.

Treatment: Intravenous amphoterecin B therapy is useful.

Sporotrichosis (*Sporotrichum schenekii*)

It is a non-contagious granulomatous infection of skin, lymphatics and less frequently the internal organs of man and animals.

Morphology: They are Gram-positive, spindle or cigar shaped yeast like bodies. They possess septate hyphae bearing clusters of pear-shaped condia radiating from tips of unbranched conidiophores. The yeast phase is prominent at 37^0C and mycelial phase is prominent at 22^0C.

Habitat: These fungi are saprophytes that grow on decaying plants, wood and in the soil.

Cultural characters: These fungi can be cultivated in Sabouraud's dextrose agar, blood agar and brain heart infusion agar. The colonies at 37^0C are moist, flat and leathery cream coloured colonies containing budding yeast cells. At 22-28^0C cottony mycelial growth is observed.

Pathogenesis: These fungi are saprophytes that grow on decaying plants, wood and in the soil. The infection is more common among gardeners and agricultural workers as a result of entry of the fungi through cut wounds at the extremities caused by the splinters and thorns of plants. The infection is also found in dogs, horses, donkeys, mules, cattle, camels and poultry. The infection is characterized by small nodule at the site of injury that gradually increases in size becomes indurate and breaks to form inactive ulcers. Since the infection spreads via lymphatics, the lymphatics become thickened and cord like. Multiple nodules or abscesses develop throughout the course of lymphatic ducts, which may rupture and ulcerate. The nodules formed initially may also turn into a hairless area releasing exudates and form a scab. In horses, the condition resembles pseudo glanders. In dogs the infection also involves liver and lungs.

Diagnosis:

1. Based on symptoms and lesions.
2. Isolation of the organism in blood agar, Sabouraud's dextrose agar or in brain heart infusion agar. The colonies at 37^0C are moist, flat and leathery cream coloured colonies containing budding yeast cells. At 22- 28^0C cottony myceial growth is observed.
3. Inoculation into mice intraperitoneally produces peritonitis and intratesticularly produces orchitis. The oval or cigar shaped yeast bodies can be demonstrated in macrophages.

Treatment: Iodide therapy produces effective results. Oral administration of potassium iodide and intravenous administration of sodium iodide produce effective results.

Rhinosporidiosis (*Rhinosporidium seeberi*)

It is a chronic non-fatal mycotic infection of man and animals. It is primarily disease of fishes with man and animals are accidental hosts. The infection is characterized by formation of polyps in the mucous membranes of eye, nose, skin, ears, larynx and genital organs. The polyps are pedunculate or sessile, pink in colour with white spots of sporangia that may contain endospores. They are soft to touch and bleed readily. The sporangia are much larger than that or *Coccidiodes spp.* The organisms are

difficult to cultivate in the laboratory and exact mechanism of pathogenesis is also not known. In man, antimony compounds together with surgical excision or cautery is successful.

MYCOTOXICOSIS

DEFINITION: Certain strains of fungi (moulds) growing in feed or feed ingredients can produce toxins that, when eaten by man or animals, can cause a very lethal disease called mycotoxicosis. The fungal metabolites that cause mycotoxicosis are called as mycotoxins. These toxins can accumulate in maturing corn, cereals, soybeans, sorghum, peanuts, and other food and feed crops in the field and in grain during transportation. The toxins may occur in storage under conditions favourable for the growth of the toxin-producing fungus or fungi.

AFLATOXICOSIS

The most important mycotoxicosis is aflatoxicosis. *Aspergillus flavus* is a common mould that grows on many substances, and grows especially well on grain and nuts. The other important fungus is *A.parasiticus.* The toxin produced by these fungi causes aflatoxicosis in animals and poultry. The condition occurs worldwide. All species of animals are susceptible, especially younger poultry, swine and calves; adult animals are more resistant to the acute form of the disease. The infection was first called as Turkey X disease since it was first noticed in young turkeys. The aflatoxins include four closely related metabolites of *A. flavus* known as B_1, B_2, G and G_2. The B_1 toxin is the most toxic and is of greatest concern to the poultry industry. Aflatoxins M_1 and M_2 are found in milk from animals fed aflatoxin-contaminated feeds. Infection is most common after the kernels have been damaged by insects, birds, mites, hail, early frost, heat and drought stress, windstorms, and other unfavourable weather.All animal species are susceptible to aflatoxicosis as mentioned earlier, although sensitivity varies considerably from species to species. For example, birds, fish, dogs, and swine appear to be more susceptible than mature cattle.

1. In **poultry**, the aflatoxins under certain conditions cause death, reduced growth, reduced egg production, reduced hatchability, signs associated with "physiological stress" and impaired ability to develop immunity to infectious agents. Besides fatty liver and kidney disorders, leg and bone problems can develop as well as outbreaks of coccidiosis. Aflatoxins may cause vaccines to fail, increase the birds' susceptibility to disease, and result in suppression of the natural immunity to infection. The birds become susceptible to infection by bacteria such as *Salmonella* and to various viruses and other infectious agents commonly found around the farmyard, feedlot, or poultry house. Decreased blood clotting results in a greater downgrading and condemnation of the birds because of massive bleeding and bruises. Less carcass pigmentation is exhibited and egg yolks are paler. The hatchability of eggs can drop, and reduced production may be noted as well as smaller eggs with shell problems. Growth is restricted and mortality increases, especially during the growing period.
2. In **animals** there is decreased feed consumption, poor feed conversion, stunting, and decreased flesh growth. Decreased productivity may be accompanied by damage to the liver, haemorrhaging into the muscles or body cavities, and suppression of natural immunity to parasites and pathogens always present in the environment. Once the damage has been done, the animals will not fully recover, even if returned to a toxin-free ration.
3. In **human beings** it causes “Farmer's lung" - a disease that affects grain handlers and is frequently associated with skin irritation, fever, wheezing, breathlessness, cough, and ulcers. Farmer's lung is thought to be caused by an allergic reaction to fungal spores and other material in grain dust.

In both animals and birds, the liver is the principal organ affected; high doses of aflatoxins cause severe cellular necrosis. Prolonged low-level exposure can result in liver enlargement and slow growth

DIAGNOSIS OF AFLATOXIN:

a. Visual inspection of the grain that may locate lots **presumed** to be contaminated with aflatoxin (black light test)

b. Rapid screening procedures to determine the presence or absence of aflatoxin that include the fluorometric iodine rapid screening and minicolumn tests;

c. Laboratory procedures to find out the **actual** amounts of toxin present (thin-layer chromatography, gas-liquid chromatography, high-pressure liquid chromatography, fluorometric iodine, or ELISA tests).

d. Traditional method of identification of fluid cultures with UV light and ascertain by the fluorescence emitted.

DEOXYNIVALENOL (VOMITOXIN)

This toxin is produced by the fungus *Fusarium graminearum.* Corn is the most important feed ingredient that contains this toxin. Pigs are mainly affected by this toxin. Clinical signs and lesions in affected swine included feed refusal, a few instances of vomiting, lack of weight gain, poor feed efficiency, failure of mature sows to return to oestrus, reduced efficiency, high mortality of nursing pigs, intestinal tract inflammation, and acute diarrhoea in young pigs. Autopsies of young pigs revealed haemorrhaging into the abdominal cavities and pale, friable livers. The infection is not very common in cattle and poultry.

FUMONISIN - BLIND STAGGERS IN HORSES

Blind staggers (equine leucoencephalomalacia) occasionally occurs in horses, mules, or donkeys foraging corn left standing in the field after harvest or fed grain or screenings heavily infected with *F. moniliforme*. The toxins fumonisin B_1 and B_2 are produced only by certain strains of *F. moniliforme*. *F. moniliforme*is common even in food-grade corn and is often abundant in ground feeds and in silage. Research on the fumonisin toxins began only recently, and current thought is that concentrations of more than 5 to 10 ppm are necessary for mycotoxicosis in horses and more than 10 to 20 ppm for swine. As with other mycotoxins, various strains of this fungus vary greatly in their toxin producing ability.

OCHRATOXIN, CITRININ, AND PENICILLIC ACID (PA) (NEPHROTOXINS)

Ochratoxin A, produced primarily by members of the *Aspergillus ochraceus* group and a number of species of *Penicillium*, especially *P. viridicatum* have been found in some samples of food and feed grains. Frequently, citrinin or PA is produced by these same fungi simultaneously. In the field, however, injury from ochratoxin poisoning has occurred chiefly (or only) in poultry and swine. Listlessness, huddling, diarrhoea, tremors, and other neural abnormalities are often encountered in broiler poultry. Ochratoxin damage to the kidneys of swine is called "porcine nephropathy". Regular consumption of a ration containing several hundred ppb of ochratoxin results in poor feed conversion, reduced growth rate, and general unthriftiness, accompanied by reduced immunity to infection by bacteria and viruses. Other prominent features of ochratoxin poisoning are increased water consumption and increased urine production because of kidney damage. The increased urine production in pigs results in the floor of the pig pen being constantly wet and needing to be cleaned daily.

ERGOTISM

Ergot toxicity, caused by the fungus *Claviceps purpurea*, differs from other mycotoxicoses, since it results from the consumption of considerable amounts of fungal tissue.In other mycotoxicoses the toxins are secreted into plant tissues in which the fungus is growing, and very little fungal material is consumed. The ergot fungus infects the flowers of cereals and many grasses when flowering occurs during predominantly cool, moist weather Infected florets show characteristic black, spur-like sclerotia that replace the seed. The sclerotia or ergot bodies contain a variety of ergopeptine and clavine alkaloids that, when consumed regularly in small amounts, result in a complex of signs collectively called ergotism. Symptoms of ergotism are poor hair condition, gangrene or loss of extremities, and poor performance. Additional symptoms include persistent diarrhoea, agalactia and abortion in cattle

FUSARIUM TOXICITY - TIBIAL DYSCHONDROPLASIA IN POULTRY

Tibial dyschondroplasia (TDP) is a common and economically important bone deformation in growing broiler chickens and turkeys. The lesion appears in a cone of cartilage extending distally from the proximal tibiotarsalphysis. The most likely cause of this deformation is a toxin called fusarochromanone produced by *Fusarium equiseti*. It also kills chick embryos in fertilized eggs.

OTHER INFECTIONS

INFECTION	IMPORTANCE
Aspergillotoxicosis	It causes hyperkeratosis in cattle. *Aspergillus chevalieri and A.clavatus* are the causative fungi. It is basically a respiratory infection. There is also gross thickening of skin due to hyperkeratinisation
Malt sprout poisoning	A. ***oryzae* and *A.clavatus***
Facial eczema	***Pithomyces charterum***
Stachybotryotoxicosis	*Stachybotrys alternans* – sooty fungus

CONTROL:

1. Harvest at maturity and as soon as the moisture content allows minimum grain damage.
2. Adjust the harvesting equipment for minimum seed or kernel damage and maximum cleaning.
3. Dry all grain to at least 15-percent moisture as rapidly as possible. Cool the grain after drying and maintain dry storage conditions.
4. Thoroughly clean the grain and all bins before storage to remove dirt, dust, and other foreign matter, crop debris, chaff, and cracked or broken seeds and kernels.
5. Where feasible, choose varieties of grain that are resistant to insects, diseases, and mechanical damage.

MYCETOMA (Madura foot or fungal tumor)

Mycetoma is first described in Madurai district of Tamil Nadu in 1842. Mycetoma is a chronic granulomatous subcutaneous infection caused by fungi (eumycetoma) and/or microaerophilic actinomycetes (Actinomycetoma). This infection results in a granulomatous inflammatory response in the deep dermis and subcutaneous tissue which can extend to the underlying bone. Mycetoma is characterized by the formation of grains containing aggregates of the causative organisms that may be discharged on to the skin surface through multiply sinuses.

Etiology: *Maduralla mycetomatis, Exophiala, Curvularia, Pseudallescheriaboydii, Acremonium, Fusarium, Aspergillus and Leptosphaeria species.*

Distribution: Endemic in India, Africa, South and Central America (tropical and subtropical area) but rare in temperate region. Mycetoma Research Centre is in Sudan.

Host: Human; rarely in cattle, horses dog and cat.

Pathogenesis: Disease is common in farm workers and it afflicts men between 20 and 40 years old. It typically affects the lower extremities (feet); other sites include lower legs, hands, arms, head, neck, chest and shoulder. The disease results from the traumatic implantation of the aetiological agent and start as a small hard painless nodule which overtime begins to soften on the surface and ulcerate to discharge a viscous, purulent fluid containing grains. The inflammation slowly spreads to adjacent tissue including bone causing considerable deformity. Sinuses continue to discharge sero-sanguinous fluid containing the granules which vary in size, colour and degree of hardness depending on the etiologic species. These grains are the hallmark of mycetoma.

Laboratory diagnosis:

Clinical material: tissue biopsy or excised sinus, serosanguinous fluid containing the granules.

Direct Microscopy: 10% KOH or Parker ink or calcofluor white mount for serosanguinous fluid containing granules. White to yellow or black pigmented grains which are small (0.5 – 3 mm) irregularly shaped and or various colours. Microscopically, the granules consists of broad (2-5 μm) intertwined hyphae and swollen cells (15 μm or more) at periphery. Actinomycetes filaments are only 0.5-1 μm wide.

Tissue sections – Haematoxylin-Eosin staining, PAS or Geocott's methenamine silver staining.

Culture: SDA with or without antimicrobial agent. Some mycetomal fungi are fast growing but some are slow growing organism and require an incubation period of up to 6 weeks at room temperature.

Mycotic abortion

Etiology

Aspergillus fumigatus is the prime cause of mycotic abortion. Other species associated with abortion are,

Aspergillus flavus	*Mucor pusillus*	*Candida parapslosis*
Aspergillus nidulans	*Mucor racemosus*	*Candida tropicalis*
Aspergillus niger	*Rhizopus stolonifer*	*Geotrichum candidum*
Aspergillus versicolor	*Rhizopus oryzae*	*Torulopsissp*
Absidiacorymbifera	*Candida albicans*	*Trichosporon cutaneum*
Absidia ramose	*Candida guillermondii*	*Fusariumsp*
Mortierella wolfii	*Candida krusei*	

Pathogenesis

Aspergillus and *Mucor* moulds are ubiquitous in nature and are usually saprophytic. Occasionally they may localize in the body producing serious systemic diseases. It is considered that infection occurs by inhalation into lungs or by ingestion, and then carried to the placenta in the blood stream from lesions in the respiratory tract or ulcers of mycotic ruminitis, or other lesions of the digestive tract. This results in a slowly developing fungal Placentitis and interferes with the nutrition of the foetus resulting

in fetal death and abortion after a number of weeks or months. It is possible that the incubation period depends on the infecting dose and may be less than 2 months.

Epidemiology

The incidence of bovine mycotic abortion is of widespread occurrence varying from 0.5%-16% of all bovine abortions. The infection originally derived from the spores of fungi in the air or dairy shed, mouldy hay, straw and feed stuffs. The highest incidence of mycotic abortion has been reported in winter months following wet summer and rainy reason. Conditions of high humidity and temperature encourage the growth of moulds when hay is stored and this constitutes the source of infection for cattle.

Symptoms

Most of the mycotic infections of the genital organ are result in abortion, though the chronic form may end in low grade infertility manifesting a purulent vaginitis, cervicitis and endometritis. Abortion generally occurs at 6-7 months of gestation. The aborted fetus shows discrete, raised ringworm type lesion on the skin of head and back. The placenta is found thickened, hemorrhagic, edematous and necrotic.

Mycology

Fungi have been recovered from the placenta, amniotic fluid, fetal stomach contents and fetal skin lesions. Very rarely, isolations have been made from the other organs of the fetus e.g. lungs. The best isolation method from fetal stomach contents is to smear them, using a glass spreader onto a malt agar plate containing penicillin and streptomycin and to incubate at 37°C. Colonies frequently appear within 24 hours depending on the species of fungus involved but the plates should be kept at least for one week before being discarded. Isolation from the placenta presents some difficulty, for the material has invariably been exposed to the air, and more often than not, been dropped on the ground. From the fetal stomach contents the fungal growth is usually pure but there may be high bacterial contamination in fetuses aborted more than 24 hours before culture.

Diagnosis

The diagnosis is confirmed by the isolation of the pathogens from the stomach contents of the aborted fetus and placenta, demonstration of fungal hyphae in the fetal tissues and detection of *A.fumigatus* precipitins in the sera of dams. Vasculitis, associated with hyphal invasion, is demonstrable in sections of affected cotyledons.

Treatment

No clinical symptoms have been observed in the dam before or after abortion and no treatment is given to affected animals. Cow which abort fungus infected fetuses may calve normally on the next occasion. The spontaneous recovery of the dam is assumed to follow the regeneration of the uterine wall. The housing of animals in warm, dry and well ventilated sheds and avoidance of mouldy feeds for the pregnant animals may help to reduce the number of cases of fungal abortions in milch animals. Treatment of hay with fungicides during hay making in order to reduce subsequent mould growth might also help.

Mycotic mastitis

Mastitis of fungal origin is of great economic concern to dairy animals as it directly affects milk production and is also of significance of public health aspect. Of late, indiscriminate and excessive use of antibiotics without proper investigation of etiological agents involved has made mastitis cases more prone to fungal infections.

Etiology

In India, fungi isolated from the mastitic milk of cattle and buffalo include *Candida* albican, *Candida stellatoidea, C.parapsilosis, C.krusei, C.guilermondii, Cryptococcus neoformans, Crytpococcus ater, Rhodotorula sp, Trichosporon sp, Torulopsis sp, Rhizopus sp and Mucos sp .*

Epidemiology

Mycotic mastitis is prevalent throughout the world and is common in cattle and buffaloes but rare in sheep, goats, equines and swine. Mycotic mastitis occasionally occurs in epizootic form. The frequency of occurrence of fungal mastitis depends upon the ability of the fungus to set up infection in the mammary gland, resist the environmental influence including cleaning and disinfection procedures and its resistance to antifungal antibiotics.

The transmission mechanism depends on animal factors and source of infection. The animal factors include susceptibility of the animals, stage of lactation (in the first phase of lactation the occurrence of the disease is more); age of the animal (older animals are more susceptible), status of the inherited resistance related to the shape of the teat and anatomical structure of the teat canal and finally immunological status of the animal body.

Source of infection

1. "Brewers grain" has a high feeding value but an excellent source for yeast growth and most probably become quickly contaminated.
2. The pigeon droppings in a dairy unit usually contain a large number of *Cryptococcus neoformans* organisms; thus the dust from pigeon excreta or soil acts as a source of infection.
3. Contamination of dairy utensil with infecting yeasts. There is a tendency of some of the yeasts to form pellicle is liquids and this pellicle may cling to dairy utensils and under conditions of inadequate sterilization may be transferred to the hands of the milkers and to the udders of animals subsequently.
4. Spores of many of the filamentous fungi are present in the air of cow sheds and thus may lade to the production of disease in the milch animals of that shed.
5. Contaminated infusion materials transmit the pathogens easily.

Pathogenesis and clinical symptoms

After infection the pathogens establish in the mammary epithelium. Gradually they spread to the surrounding tissues. A severe tissue reaction takes place subsequently and the affected tissues suffer from pathological changes according to the type of pathogens involved. The yeasts are best adapted to grow in the mammary ducts and acini by virtue of their ability to break up into small vegetative producing units.

Aspergillus penetrates the wall of teat cistern and teat canal and extends to the surrounding tissues developing densely fibrosed granulomatous to the surrounding tissues. Sometimes they develop necrotizing pyogenic lesions.

When the infections of *Candida* have been superimposed a chronic fibrotic thickening of the underlying tissues and septum of the mammary gland is seen. The infection sometimes causes suppurative lesions and small abscesses.

In cryptococcal mastitis, there will be granulomatous lesions. Acute and chronic type of mastitis is caused and the entire network of the udder except the fibrous framework of the gland is destroyed. There will be severe cellular reaction and supra mammary lymph node becomes swollen.

The symptoms of mycotic mastitis develop gradually and are most variable. Milk yield is reduced severely or sometimes permanently stopped. Yellow clots in watery fluid and sometimes greyish or yellowish floccules are found; mucilaginous secretion is found in cryptococcal infection. Sometimes there is mucoid grey white secretion. There is acute swelling of the affected quarter, supramammary lymph node or internal iliac lymph nodes. Sometimes, there is subcutaneous oedema of the perineum. Multiple abscesses may be found in the affected quarters. The animal does not respond to the usual antibiotic treatment.

Diagnosis

1. Microscopic examination of fungi: The suspected milk samples may be centrifuged at 10000 rpm at 4oC for 10 minutes. Smears from sediments should be prepared. The slides can be stained with either 0.5% gentian violet stain or modified Grocott's silver methanamine method or lactophenol cotton blue stain. The stained smear is taken for microscopic examination for fungi.
2. Cultural isolation of fungi: A loopful of suspected sample of milk should be inoculated on Sabouraud's dextrose agar slants containing chloramphenicol (0.05 mg/ml). This slant should be incubated at 37°C. If the culture shows growth, it should be taken as positive for fungi. Then identification should be done by staining with lactophenol cotton blue stain.

Identification of Yeast or yeast like fungi

Microscopic examination of the pure culture is done. Presence of budding cells reveals the presence of species of *Candida, Cryptococcus* or true yeast (*Saccharomyces*).

If mycelium is present, look for the presence of arthrospores and blastospores. Presence of arthrospores alone is indicative of *Geotrichum candidum* and if both arthrospores and blastospores are present, the unknown may be *Trichosporon* species.

The isolate which do not produce mycelia and chlamydospores on corn meal agar or Chlamydospore agar are considered to belong to *Cryptococcus sp* or true yeasts i.e. *Saccharomyces* while the presence of round chlamydospores at the terminal ends of pseudomycelium is characteristic of *Candida albicans*.

If unknown culture does not form chlamydospores but morphologically resembles *Candida albicans*, hold the plate for 2-3 days and re-examine. Absence of chlamydospores but has the morphology of *Candida albicans*, report as *Candida* species.

If no mycelium has been formed and the cells are capsulated, urease positive, mention it as *Cryptococcus sp*.

Identification of moulds

The moulds if repeatedly isolated from the mastitic case are characterized on the following basis.

1. Examination of colonies: The colonies on SDA are examined for their various character e.g. size, form, elevation, texture (powdery, granular, velvety or cottony), surface, margin, density, pigmentation on surface and on reverse and rate of growth,
2. Morphology characters: A small portion of the colony is removed and placed on a clean glass slide with a drop of lactophenol cotton blue. This is teased apart gently with the help of teasing needles and covered with a coverslip. This is then examined under high and low powers of microscope with reduced light the following characters are seen for the identification of the mould.
    i. The mycelium is seen for the presence of septa. If the mycelium is coenocytic, the mould belongs to *Zygomycetes* and if the mycelium is septate, it may belong to *Ascomycetes* or *basidiomycetes* or *Deutromycetes*.

ii. Fructification is seen. Sporangium is present or not. If sporangium is absent, look for the type of oospores and zygospore. If all these are absent, then look for ascospores, basidiospores or conidia

iii. Also look for the presence of rhizoids

Identification of *Aspergillus* species

The hyphae are well developed, profusely branched, septate and hyaline. Their cells are multinucleate. The conidiophores are long, erect hyphae, each terminating in a bulbous head – the vesicle. Large numbers of sterigmata are produced over the entire surface of vesicle. Two layers of sterigmata may be produced according to the species.

Identification of *Rhizopus* species

The mycelium is coenocytic and sporangia are spherical with columella. Sporangiophores are branched in *Mucor* while these are unbranched in *Rhizopus*. Rhizoids are arising from stolon only in *Rhizopus* while these are absent in *Mucor*.

Treatment

Fungal mastitis does not respond to antibacterial antibiotics. Cryptococcal mastitis presents a therapeutic challenge. A better therapeutic regimen should be achieved with antifungal preparations. The treatment with iodide gives better curative effect. The mastitis due to *Candida sp* can be treated with nystatin, miconazole, amphotericin-B etc.

Control measures include cleanliness of dairy utensils and strict hygienic conditions of milking parlor. Milker's hand should be cleaned before milking and affected quarter should be milked last. Milking should be at regular intervals. Feeding of feeds with fungal growth should be avoided. No treatment of any antibacterial antibiotics should be carried out without sensitivity test.

UNIT 3

MICROBIAL BIOTECHNOLOGY

Recombinant DNA technology (Gene cloning)

Gene cloning involves separation of specific gene or DNA fragments from a donor cell, attaching it to small carrier molecule called vector and then replicating this recombinant vector into a host cell.

Steps involved in gene cloning

1. Isolation of donor DNA fragment or gene
2. Selection of suitable vector
3. Incorporation of donor DNA fragment into the vector
4. Transformation of recombinant vector into a suitable host cell
5. Isolation of recombinant host cell

1. Isolation of donor DNA fragment or gene

- At first a donor DNA fragment should be isolated. There are two methods for isolation of desired gene or DNA fragment.
- Using restriction endonuclease enzyme: the enzyme restriction endonuclease is a key enzyme in molecular gene cloning. It has specific restriction site for its action. The enzyme RE generates a DNA fragment either with blunt end or with sticky end
- Using reverse transcriptase enzyme: reverse transcriptase enzyme Synthesizes complementary DNA strand of the desired gene using its mRNA.

2. Selection of suitable cloning vector:

- When donor DNA fragment is incorporated into a host cell, it will not replicate because the isolated gene do not have the capacity to replicate itself. So before introduction of donor fragment into host, a suitable vector should be selected.
- Cloning vector is the DNA molecule capable of self-replication inside the host cell. The main function of cloning vector is to replicates the inserted DNA fragment inside the host cell.
- Examples of cloning vectors: Plasmid, BAC, YAC, Λ-bacteriophage, expression vectors etc.

Characteristics of cloning vectors

1. It must be self-replicating inside host cell
2. It must possess restriction site for RE enzymes
3. Introduction of donor DNA fragment must not interfere with replication property of the vector
4. It must possess some marker gene such that it can be used for later identification of recombinant cell.

3. Incorporation of donor DNA fragment with Plasmid vector:

- The plasmid vector is cut open by the same RE enzyme used for isolation of donor DNA fragment
- The mixture of donor DNA fragment and plasmid vector are mixed together.
- In the presence of DNA ligase, base pairing of donor DNA fragment and plasmid vector occurs forming recombinant vector in the mixture

4. Transformation of recombinant vector into suitable host:

- The recombinant vector is transformed into suitable host cell. ie bacterial cell
- Some bacteria are naturally transformable, they take up the recombinant vector automatically. For examples: *Bacillus, Haemophillus, Helicobacter pylori,* are naturally competent

- Some other bacteria are not naturally competent, in those bacteria recombinant vector are incorporated by artificial method such as Ca++ ion treatment, electroporation etc

5. Isolation of recombinant cell:

- The recombinant host cell is then grown in culture media but the culture may contains colonies both recombinant cell and non-recombinant cell.
- For isolation of recombinant cell from non-recombinant cell, marker gene of plasmid vector is employed.
- For examples, PBR322 plasmid vector contains different marker gene (Ampicillin resistant gene and Tetracycline resistant gene. When *pst1* RE is used it knock out Ampicillin resistant gene from the plasmid, so that the recombinant cell become sensitive to Ampicillin.

Cloning vector

Cloning vector is a small DNA molecule capable of self-replication inside the host cell. Cloning vector is used for replicating donor DNA fragment within host cell.

Characteristics of a cloning vectors

1. it must be small in size
2. It must be self-replicating inside host cell
3. It must possess restriction site for Restriction Endonuclease enzymes
4. Introduction of donor DNA fragment must not interfere with replication property of the vector
5. It must possess some marker gene such that it can be used for later identification of recombinant cell
6. it must possess multiple cloning site

Types of cloning vectors used in gene cloning:

1. Plasmid (PBR322):
   - Example: PBR322
   - It is isolated from E.coli
   - Size: 4361 bp
   - Cloning limit: 0.1-10 kb
   - Marker gene: Ampicillin and Tetracycline resistant gene
   - Restriction site for various restriction endonucleases
2. Bacterial artificial chromosome (BAC):
   - Example: pUvBBAC
   - It is artificially synthesized plasmid
   - size: 11827 bp
   - It is modification of bacterial F-plasmid
   - Cloning limit: 35-300 kb
   - Marker gene: chloramphenicol resistant gene and lactose metabolizing gene (LacZ)
3. Yeast artificial chromosome (YAC):
   - Example: pYAC3
   - It is an artificial chromosome having yeast centromere isolated from *Saccharomyces cerevisiae* and ligated to bacterial plasmid

- size: 11400 bp
- It has telomere sequence
- Marker: similar as for identification of yeast cell
- Cloning limit: 100-1000 kb

4. Λ-Bacteriophage:
   - Example: lambda genome
   - It is a phage (Virus) genome
   - size: 48502 bp
   - 1/3rd of the bacteriophage genome is non-essential, so that it can be cut, removed and replaced by donor DNA fragment during cloning
   - It can recombinant only 4-5 Kbp of donor DNA fragment
5. Expression vector:
   - Example: Eukaryotic expression vector pSG5
   - size: 4100 bp
   - An eukaryotic vector modified such a way that it can be expressed in prokaryotic cell known as expression vector
   - it allows RNA polymerase to transcribe gene
6. Cosmid:
   - Example: super COS1
   - size: 7900 bp
   - It has combined feature of both phase and plasmid
   - Cloning limit: 30-50 kb
7. Human artificial chromosome (HAC):
   - It is artificially synthesized chromosome used to transfer human gene
   - Cloning limit; No limit, it can carry large segment of DNA

Expression vector

An **expression vector**, otherwise known as an **expression construct**, is usually a plasmid or virus designed for gene expression in cells. The vector is used to introduce a specific gene into a target cell, and can commandeer the cell's mechanism for protein synthesis to produce the proteinencoded by the gene. Expression vectors are the basic tools in biotechnology for the production of proteins.

The vector is engineered to contain regulatory sequences that act as enhancer and promoter regions and lead to efficient transcription of the gene carried on the expression vector.[1] The goal of a well-designed expression vector is the efficient production of protein, and this may be achieved by the production of significant amount of stable messenger RNA, which can then be translated into protein. The expression of a protein may be tightly controlled, and the protein is only produced in significant quantity when necessary through the use of an inducer, in some systems however the protein may be expressed constitutively. *Escherichia coli* is commonly used as the host for protein production, but other cell types may also be used. An example of the use of expression vector is the production of insulin, which is used for medical treatments of diabetes.

Elements of expression vectors

An expression vector has features that any vector may have, such as an origin of replication, a selectable marker, and a suitable site for the insertion of a gene such as the multiple cloning site. The

cloned gene may be transferred from a specialized cloning vector to an expression vector, although it is possible to clone directly into an expression vector. The cloning process is normally performed in *Escherichia coli*, and vectors used for protein production in organisms other than *E.coli* may have, in addition to a suitable origin of replication for its propagation in *E. coli*, elements that allow them to be maintained in another organism, and these vectors are called shuttle vectors.

Elements for expression

An expression vector must have elements necessary for gene expression. These may include a promoter, the correct translation initiation sequence such as a ribosomal binding site and start codon, a termination codon, and a transcription termination sequence. There are differences in the machinery for protein synthesis between prokaryotes and eukaryotes, therefore the expression vectors must have the elements for expression that is appropriate for the chosen host. For example, prokaryotes expression vectors would have a Shine-Dalgarno sequence at its translation initiation site for the binding of ribosomes, while eukaryotes expression vectors would contain the Kozak consensus sequence.

The promoter initiates the transcription and is therefore the point of control for the expression of the cloned gene. The promoters used in expression vector are normally inducible, meaning that protein synthesis is only initiated when required by the introduction of an inducer such as IPTG. Gene expression however may also be constitutive (i.e. protein is constantly expressed) in some expression vectors. Low level of constitutive protein synthesis may occur even in expression vectors with tightly controlled promoters.

Protein tags

After the expression of the gene product, it is usually necessary to purify the expressed protein; however, separating the protein of interest from the great majority of proteins of the host cell can be a protracted process. To make this purification process easier, a purification tag may be added to the cloned gene. This tag could be histidine (His) tag, other marker peptides, or a fusion partners such as glutathione S-transferase or maltose-binding protein. Some of these fusion partners may also help to increase the solubility of some expressed proteins. Other fusion proteins such as green fluorescent protein may act as a reporter gene for the identification of successful cloned genes.

Others

The expression vector is transformed or transfected into the host cell for protein synthesis. Some expression vectors may have elements for transformation or the insertion of DNA into the host chromosome, for example the *vir* genes for plant transformation, and integrase sites for chromosomal insertion.

Some vectors may include targeting sequence that may target the expressed protein to a specific location such as the periplasmic space of bacteria.

Expression/Production systems

Different organisms may be used to express a gene's target protein, and the expression vector used will therefore have elements specific for use in the particular organism. The most commonly used organism for protein production is the bacterium *Escherichia coli*. However, not all proteins can be successfully expressed in *E. coli*, or be expressed with the correct form of post-translational modifications such as glycosylations, and other systems may therefore be used.

Bacterial expression vector

An example of a bacterial expression vector is the pGEX-3x plasmid

The expression host of choice for the expression of many proteins is *Escherichia coli* as the production of heterologous protein in *E. coli* is relatively simple and convenient, as well as being rapid and cheap. A large number of *E. coli* expression plasmids are also available suitable for a wide variety of needs. Other bacteria used for protein production include *Bacillus subtilis*.

Most heterologous proteins are expressed in the cytoplasm of *E. coli*. However, not all proteins formed may be soluble in the cytoplasm, and incorrectly folded proteins formed in cytoplasm can form insoluble aggregates called inclusion bodies. Such insoluble proteins will require refolding, which can be an involved process and may not necessarily produce high yield. Proteins which have disulphide bonds are often not able to fold correctly due to the reducing environment in the cytoplasm which prevents such bond formation, and a possible solution is to target the protein to the periplasmic space by the use of an N-terminal signal sequence. Another possibility is to manipulate the redox environment of the cytoplasm. Other more sophisticated systems are also being developed; such systems may allow for the expression of proteins previously thought impossible in *E. coli*, such as glycosylated proteins.

The promoters used for these vector are usually based on the promoter of the *lac* operon or the T7 promoter, and they are normally regulated by the *lac*operator. These promoters may also be hybrids of different promoters, for example, the Tac-Promoter is a hybrid of *trp* and *lac* promoters. Note that most commonly used *lac* or *lac*-derived promoters are based on the *lac*UV5 mutant which is insensitive to catabolite repression. This mutant allows for expression of protein under the control of the *lac* promoter when the growth medium contains glucose since glucose would inhibit gene expression if wild-type *lac* promoter is used. Presence of glucose nevertheless may still be used to reduce background expression through residual inhibition in some systems.

Examples of *E. coli* expression vectors are the pGEX series of vectors where glutathione S-transferase is used as a fusion partner and gene expression is under the control of the tac promoter, and the pET series of vectors which uses a T7 promoter.

It is possible to simultaneously express two or more different proteins in *E. coli* using different plasmids. However, when 2 or more plasmids are used, each plasmid needs to use a different antibiotic selection as well as a different origin of replication, otherwise the plasmids may not be stably maintained. Many commonly used plasmids are based on the ColE1 replicon and are therefore incompatible with each other; in order for a ColE1-based plasmid to coexist with another in the same cell, the other would need to be of a different replicon, e.g. a p15A replicon-based plasmid such as the pACYC series of plasmids. Another approach would be to use a single two-cistron vector or design the coding sequences in tandem as a bi- or poly-cistronic construct.

Yeast

Yeast commonly used for protein production is *Pichia pastoris*. Examples of yeast expression vector in *Pichia* are the pPIC series of vectors, and these vectors use the AOX1 promoter which is inducible with methanol. The plasmids may contain elements for insertion of foreign DNA into the yeast genome and signal sequence for the secretion of expressed protein. Proteins with disulphide bonds and glycosylation can be efficiently produced in yeast. Another yeast used for protein production is *Kluyveromyces lactis* and the gene is expressed, driven by a variant of the strong lactase LAC4 promoter.

Saccharomyces cerevisiae is particularly widely used for gene expression studies in yeast, for example in yeast two-hybrid system for the study of protein-protein interaction. The vectors used in yeast two-hybrid system contain fusion partners for two cloned genes that allow the transcription of a reporter gene when there is interaction between the two proteins expressed from the cloned genes.

Baculovirus

Baculovirus, a rod-shaped virus which infects insect cells, is used as the expression vector in this system. Insect cell lines derived from Lepidopterans (moths and butterflies), such as *Spodoptera frugiperda*, are used as host. A cell line derived from the cabbage looper is of particular interest, as it has been developed to grow fast and without the expensive serum normally needed to boost cell growth. The shuttle vector is called bacmid, and gene expression is under the control of a strong promoter pPolh. Baculovirus has also been used with mammalian cell lines in the BacMam system.

Baculovirus is normally used for production of glycoproteins, although the glycosylations may be different from those found in vertebrates. In general, it is safer to use than mammalian virus as it has a limited host range and does not infect vertebrates without modifications.

Plant

Many plant expression vectors are based on the Ti plasmid of *Agrobacterium tumefaciens*. In these expression vectors, DNA to be inserted into plant is cloned into the T-DNA, a stretch of DNA flanked by a 25-bp direct repeat sequence at either end, and which can integrate into the plant genome. The T-DNA also contains the selectable marker. The *Agrobacterium* provides a mechanism for transformation, integration of into the plant genome, and the promoters for its *vir* genes may also be used for the cloned genes. Concerns over the transfer of bacterial or viral genetic material into the plant however have led to the development of vectors called intragenic vectors whereby functional equivalents of plant genome are used so that there is no transfer of genetic material from an alien species into the plant.

Plant viruses may be used as vectors since the *Agrobacterium* method does not work for all plants. Examples of plant virus used are the tobacco mosaic virus (TMV), potato virus X, and cowpea mosaic virus. The protein may be expressed as a fusion to the coat protein of the virus and is displayed on the surface of assembled viral particles, or as an unfused protein that accumulates within the plant. Expression in plant using plant vectors is often constitutive, and a commonly used constitutive promoter in plant expression vectors is the cauliflower mosaic virus (CaMV) 35S promoter.

Mammalian

Mammalian expression vectors offer considerable advantages for the expression of mammalian proteins over bacterial expression systems - proper folding, post-translational modifications, and relevant enzymatic activity. It may also be more desirable than other eukaryotic non-mammalian systems whereby the proteins expressed may not contain the correct glycosylations. It is of particular use in producing membrane-associating proteins that require chaperones for proper folding and stability as well as containing numerous post-translational modifications. The downside, however, is the low yield of product in comparison to prokaryotic vectors as well as the costly nature of the techniques involved. Its complicated technology, and potential contamination with animal viruses of mammalian cell expression have also placed a constraint on its use in large-scale industrial production.

Cultured mammalian cell lines such as the Chinese hamster ovary (CHO), COS, including human cell lines such as HEK and HeLa may be used to produce protein. Vectors are transfected into the cells and the DNA may be integrated into the genome by homologous recombination in the case of stable transfection, or the cells may be transiently transfected. Examples of mammalian expression vectors include the adenoviral vectors, the pSV and the pCMV series of plasmid vectors, vaccinia and retroviral vectors, as well as baculovirus. The promoters for cytomegalovirus (CMV) and SV40 are commonly used in mammalian expression vectors to drive gene expression. Non-viral promoter, such as the elongation factor (EF)-1 promoter, is also known.

Cell-free systems

E. coli cell lysate containing the cellular components required for transcription and translation are used in this *in vitro* method of protein production. The advantage of such system is that protein may be produced much faster than those produced *in vivo* since it does not require time to culture the cells, but it is also more expensive. Vectors used for *E. coli* expression can be used in this system although specifically designed vectors for this system are also available. Eukaryotic cell extracts may also be used in other cell-free systems, for example, the wheat germ cell-free expression systems. Mammalian cell-free systems have also been produced.

Applications

Laboratory use

Expression vector in an expression host is now the usual method used in laboratories to produce proteins for research. Most proteins are produced in *E. coli*, but for glycosylated proteins and those with disulphide bonds, yeast, baculovirus and mammalian systems may be used.

Production of peptide and protein pharmaceuticals

Most protein pharmaceuticals are now produced through recombinant DNA technology using expression vectors. These peptide and protein pharmaceuticals may be hormones, vaccines, antibiotics, antibodies, and enzymes. The first human recombinant protein used for disease management, insulin, was introduced in 1982. Biotechnology allows these peptide and protein pharmaceuticals, some of which were previously rare or difficult to obtain, to be produced in large quantity. It also reduces the risks of contaminants such as host viruses, toxins and prions. Examples from the past include prion contamination in growth hormone extracted from pituitary glands harvested from human cadavers, which caused Creutzfeldt–Jakob disease in patients receiving treatment for dwarfism, and viral contaminants in clotting factor VIII isolated from human blood that resulted in the transmission of viral diseases such as hepatitis and AIDS. Such risk is reduced or removed completely when the proteins are produced in non-human host cells.

Transgenic plant and animals

In recent years, expression vectors have been used to introduce specific genes into plants and animals to produce transgenic organisms, for example in agriculture it is used to produce transgenic plants. Expression vectors have been used to introduce a vitamin A precursor, beta-carotene, into rice plants. This product is called golden rice. This process has also been used to introduce a gene into plants that produces an insecticide, called Bacillus thuringiensis toxin or Bt toxin which reduces the need for farmers to apply insecticides since it is produced by the modified organism. In addition expression vectors are used to extend the ripeness of tomatoes by altering the plant so that it produces less of the chemical that causes the tomatoes to rot. There have been controversies over using expression vectors to modify crops due to the fact that there might be unknown health risks, possibilities of companies patenting certain genetically modified food crops, and ethical concerns. Nevertheless, this technique is still being used and heavily researched.

Transgenic animals have also been produced to study animal biochemical processes and human diseases, or used to produce pharmaceuticals and other proteins. They may also be engineered to have advantageous or useful traits. Green fluorescent protein is sometimes used as tags which results

in animal that can fluoresce, and this have been exploited commercially to produce the fluorescent GloFish.

Gene therapy

Gene therapy is a promising treatment for a number of diseases where a "normal" gene carried by the vector is inserted into the genome, to replace an "abnormal" gene or supplement the expression of particular gene. Viral vectors are generally used but other nonviral methods of delivery are being developed. The treatment is still a risky option due to the viral vector used which can cause ill-effects, for example giving rise to insertional mutation that can result in cancer. However, there have been promising results

Transformation

Transformation is the uptake of DNA from the environment by a bacterial cell. This phenomenon occurs in nature between bacteria of the same species. Scientists adapted transformation for propagation of plasmid DNA, protein production, and other applications. In research, transformation introduces recombinant plasmid DNA into competent bacterial cells that take up extracellular DNA from the environment. Some bacterial species are naturally competent under certain environmental conditions, but competence is artificially induced in a laboratory setting.

Transfection

Transfection is the forced introduction of small molecules such as DNA, RNA, or antibodies into eukaryotic cells. Just to make life confusing, 'transfection' also refers to the introduction of bacteriophage into bacterial cells. But for the purpose of this article, we will focus on eukaryotic cells.

Common transfection methods include calcium phosphate, cationic polymers (such as PEI), magnetic beads, electroporation and commercial lipid-based reagents such as Lipofectamine and Fugene.

Transduction

Transduction is not a biology-specific term. The common definition of transduction is 'leading through or across.' The term has been used to describe DNA transfer between bacterial cells by bacteriophage. However, in current research, transduction also refers to the infection of mammalian cells with a viral vector.

'Transduction' was first used to describe bacterial gene transfer back in the 1950s, decades before the creation of viral vectors. It's possible that the word has been co-opted to prevent confusion, as 'transfection' doesn't distinguish between the introduction of nucleic acid or viral particles into a cell.

'Transduction' is mostly used to describe the introduction of recombinant viral vector particles into target cells, while 'infection' refers to natural infections of humans or animals with wild-type viruses. There are no universally accepted differences between these two terms. They are mostly used as a convenient way to differentiate between two biological mechanisms.

Lipid delivery using cationic lipids

- ➢ DNA, siRNA, or oligonucleotides and transfection reagent (commercially available) are diluted in separate tubes.
- ➢ DNA and transfection reagent combined to form complexes.

- ➢ Positive charge on cationic lipid binds to phosphate backbone on nucleic acid.
- ➢ Complexes added to cells. Positive charge on cationic lipid helps bind complex to membrane.
- ➢ Complexes enter the cell via endocytosis.
- ➢ Assay for gene expression or silencing.

Advantages

- Commercially available with reproducible results
- Applicable to a broad range of cell lines
- High efficiency and expression performance
- Fast and easy protocols which do not require media changes

Disadvantages

- Optimization may be necessary—some cell lines are sensitive to cationic lipids
- Some cell lines are not readily transfected with cationic lipids

Viral delivery

- ➢ Generate recombinant virus via gene cloning, cell culture, virus isolation, and clone characterization.
- ➢ Prepare and purify high titer viral vectors.
- ➢ Infect cells (containing viral receptor) at appropriate multiplicity of infection (MOI).
- ➢ Remove virus and/or add fresh medium.
- ➢ Assay for gene expression or silencing.

Advantages

- Works well for difficult-to-transfect cell types

Disadvantages

- Technically challenging and time consuming to generate recombinant viruses
- Cell lines to transfect must contain viral receptors
- Safety considerations

Delivery by electroporation

- ➢ Prepare cells by suspending in electroporation buffer.
- ➢ Pulse cells electrically in the presence of specialized buffer and DNA.
- ➢ Electrical pulse creates a potential difference across the membrane and induces temporary pores in the cell membrane for DNA entry
- ➢ Cells returned to growth conditions.
- ➢ Assay for gene expression or silencing.

Advantages

- Can work well for difficult-to-transfect cell types

Disadvantages

- Requires instrument
- Optimization of electrical pulse and field strength parameters required
- Significantly more manipulation of cells required
- High toxicity levels may be observed
- Can irreversibly damage the membrane and lyse the cells

Delivery by other chemical methods, i. e. calcium phosphate precipitation, DEAE-dextran, polybrene

- ➢ Prepare solutions. Mix calcium chloride (in phosphate buffer) and DNA.
- ➢ Allow precipitation of extremely small, insoluble particles containing condensed DNA
- ➢ $CaPO_4$: Add complexes to cell culture and incubate. Complexes adhere to cell membranes, and enter into the cytoplasm by endocytosis.
- ➢ DEAE-Dextran and polybrene: Complexes are delivered by osmotic shock using DMSO or glycerol. Positive-charged polymers complex with negatively charged DNA molecules to enable binding to the cell surface.
- ➢ Remove complexes from cells, wash, and add fresh culture medium.
- ➢ Assay for gene expression or silencing.

Advantages

- Reagents costs are inexpensive

Disadvantages

- Requires careful preparation of reagents—$CaPO_4$ solutions are pH sensitive
- DEAE-dextran is limited to transient transfections
- Cell toxicity
- Reproducibility can be problematic

BIOINFORMATICS

Bioinformatics is the field of science in which biology, computer science, and information technology merge to form a single discipline. It is the emerging field that deals with the application of computers to the collection, organization, analysis, manipulation, presentation, and sharing of biologic data to solve biological problems on the molecular level. According to Frank Tekaia, bioinformatics is the mathematical, statistical and computing methods that aim to solve biological problems using DNA and amino acid sequences and related information.

The term *bioinformatics* was coined by Paulien Hogeweg in 1979 for the study of informatic processes in biotic systems. The National Center for Biotechnology Information (NCBI, 2001) defines bioinformatics as: *"Bioinformatics is the field of science in which biology, computer science, and information technology merge into a single discipline. There are three important sub-disciplines within bioinformatics: the development of new algorithms and statistics with which to assess relationships among members of large data sets; the analysis and interpretation of various types of data including nucleotide and amino acid sequences, protein domains, and protein structures; and the development and implementation of tools that enable efficient access and management of different types of information."*

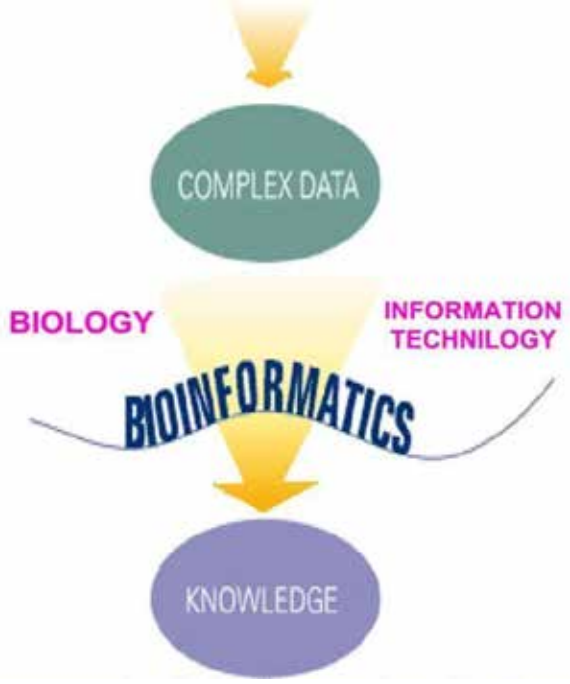


Fig 1. Concepts of Bioinformatics

Bioinformatics is a scientific discipline that has emerged in response to accelerating demand for a flexible and intelligent means of storing, managing and querying large and complex biological data sets. The ultimate aim of bioinformatics is to enable the discovery of new biological insights as well as to create a global perspective from which unifying principles in biology can be discerned. Over the past few decades rapid developments in genomic and other molecular research technologies and developments in information technologies have combined to produce a tremendous amount of information related to molecular biology. At the beginning of the genomic revolution, the main concern of bioinformatics was the creation and maintenance of a database to store biological information such as nucleotide and amino acid sequences. Development of this type of database involved not only design issues but the development of an interface whereby researchers could both access existing data as well as submit new or revised data (e.g. to the NCBI, http://www.ncbi.nlm.nih.gov/). More recently, emphasis has shifted towards the analysis of large data sets, particularly those stored in different formats in different databases. Ultimately, all of this information must be combined to form a comprehensive picture of normal cellular activities so that researchers may study how these activities are altered in different disease states. Therefore, the field of bioinformatics has evolved such that the most pressing task now involves the analysis and interpretation of various types of data, including nucleotide and amino acid sequences, protein domains, and protein structures.

Origin & History of Bioinformatics

Over a century ago, bioinformatics history started with an Austrian monk named Gregor Mendel. He is known as the–Father of Genetics". Hecross-fertilized different colors of the same species of flowers. He kept careful records of the colors of flowers that he cross-fertilized and the color(s) of flowers they produced. Mendel illustrated that the inheritance of traits could be more easily explained if it was controlled by factors passed down from generation to generation.

After this discovery of Mendel, bioinformatics and genetic record keeping have come a long way. The understanding of genetics has advanced remarkably in the last thirty years. In 1972, Paul Berg made the first recombinant DNA molecule using ligase. In that same year, Stanley Cohen, Annie Chang and Herbert Boyer produced the first recombinant DNA organism. In 1973, two important things happened in the field of genomics:

1. Joseph Sambrook led a team that refined DNA electrophoresis using agarose gel, and
2. Herbert Boyer and Stanely Cohen invented DNA cloning. By 1977, a method for sequencing DNA was discovered and the first genetic engineering company, Genetech was founded.

During 1981, 579 human genes had been mapped and mapping by *in situ* hybridization had become a standard method. Marvin Carruthers and Leory Hood made a huge leap in bioinformatics when they invented a method for automated DNA sequencing. In 1988, the Human Genome Organization (HUGO) was founded. This is an international organization of scientists involved in Human Genome Project. In 1989, the first complete genome map was published of the bacteria *Haemophilus influenza*.

The following year, the Human Genome Project was started. In 1991, a total of 1879 human genes had been mapped. In 1993, Genethon, a human genome research center in France produced roduced a physical map of the human genome. Three years later, Genethon published the final version of the Human Genetic Map which concluded the end of the first phase of the Human Genome Project.

Bioinformatics was fuelled by the need to create huge databases, such as GenBank and EMBL and DNA Database of Japan to store and compare the DNA sequence data erupting from the human genome and other genome sequencing projects. Today, bioinformatics embraces protein structure analysis, gene and protein functional information, data from patients, pre-clinical and clinical trials, and the metabolic pathways of numerous species.

3. Importance

The greatest challenge facing the molecular biology community today is to make sense of the wealth of data that has been produced by the genome sequencing projects. Cells have a central core called nucleus, which is storehouse of an important molecule known as DNA. They are packaged in units known as chromosomes. They are together known as the genome. Genes are specific regions of the genomes (about 1%) spread throughout the genome, sometimes contiguous, many times non-contiguous. RNAs similarly contains informations, their major purpose is to copy information from DNA selectively and to bring it out of the nucleus for its use. Proteins are made of amino acids, which are twenty in count (researchers are debating on increasing this count, as couple of new ones are claimed to be identified).

The gene regions of the DNA in the nucleus of the cell is copied (transcribed) into the RNA and RNA travels to protein production sites and is translated into proteins is the Central Dogma of Molecular Biology. Portions of DNA Sequence are transcribed into RNA. The first step of a cell is to copy a particular portion of its DNA nucleotide sequence (i.e. gene) which is shown in Fig 2 and Fig 3.

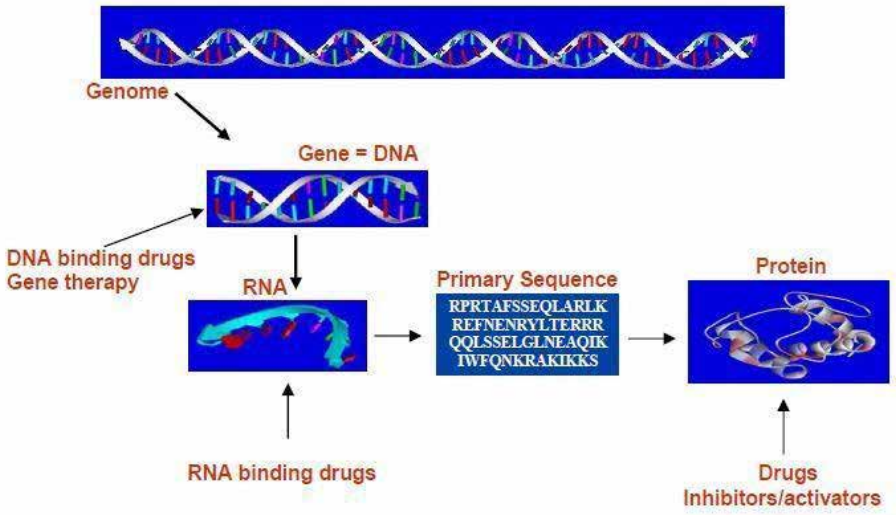


Fig 2. Biological Systems

Bioinformatics, being an interface between modern biology and informatics it involves discovery, development and implementation of computational algorithms and software tools that facilitate an understanding of the biological processes (Fig 3.) with the goal to serve primarily agriculture and healthcare sectors with several spin offs. In a developing country like India, bioinformatics has a key role to play in areas like agriculture where it can be used for increasing the nutritional content, increasing the volume of the agricultural produce and implanting disease resistance etc. In the pharmaceutical sector, it can be used to reduce the time and cost involved in drug discovery process particularly for third world diseases, to custom design drugs and to develop personalized medicine.

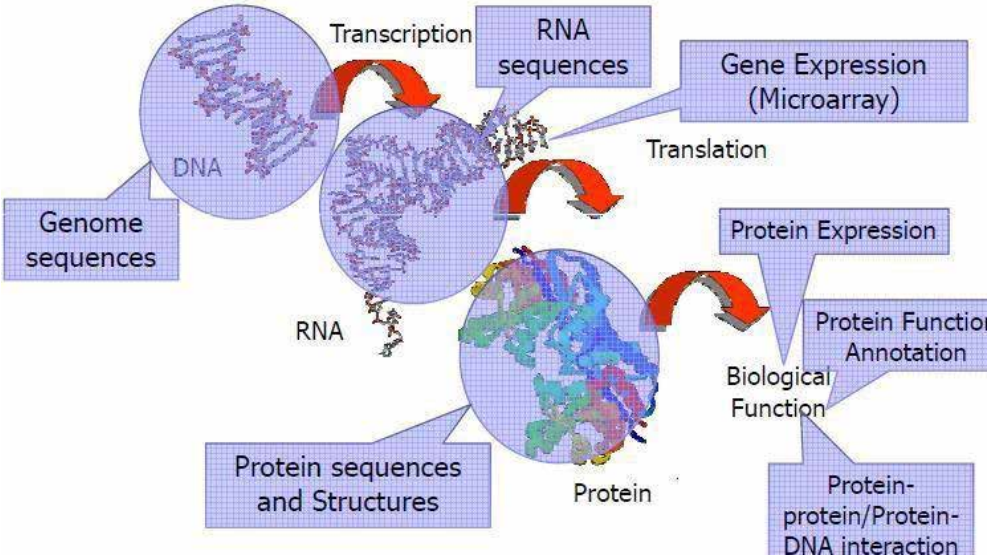


Fig 3. Biological Processes

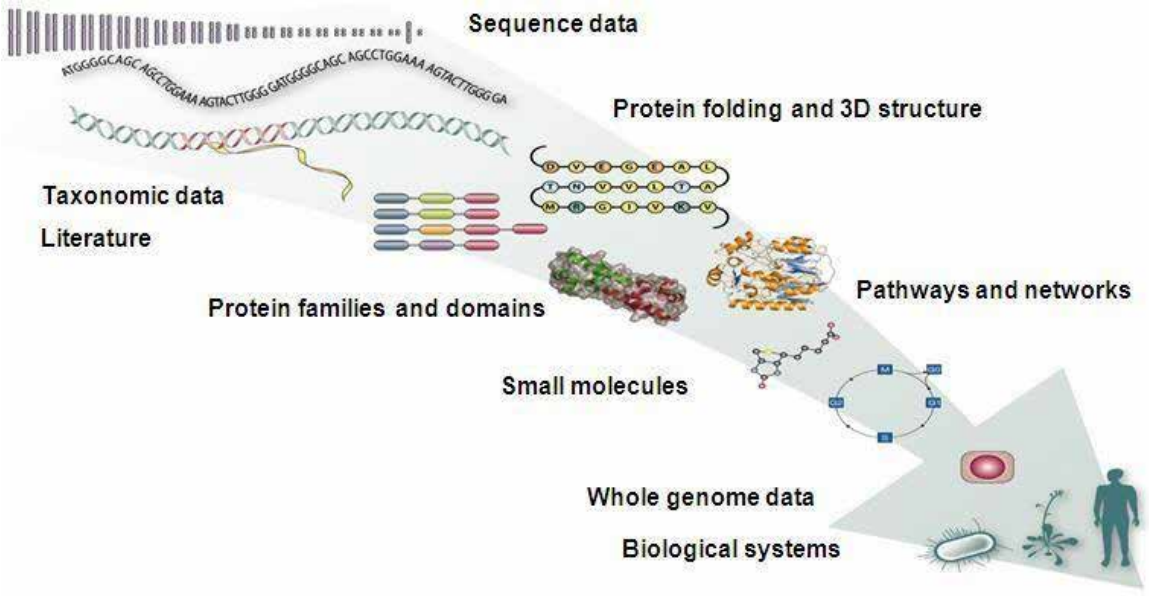


Fig 4. Information on Sequence Data

Traditionally, molecular biology research was carried out entirely at the experimental laboratory bench but the huge increase in the scale of data being produced in this genomic era has seen a need to incorporate computers into this research process. Sequence generation, its subsequent storage, interpretation and analysis are entirely computer dependent tasks. However, the molecular biology of an organism is a very complex issue with research being carried out at molecular level. The first

challenge facing the bioinformatics community today is the intelligent and efficient storage of this massive data. Moreover, it is essential to provide easy and reliable access to this data. The data itself is meaningless before analysis and it is impossible for even a trained biologist to begin to interpret it manually. Therefore, automated computer tools must be developed to allow the extraction of meaningful biological information. There are three central biological processes around which bioinformatics tools must be developed:

- DNA sequence which determines protein sequence
- Protein sequence which determines protein structure
- Protein structure which determines protein function

Concepts of Bioinformatics



Fig. 5. Hypothesis-generating bioinformatics

4. Difference between Bioinformatics and Computational Biology

Both Bioinformatics and Computational Biology are Computers and Biology. Biologists who specialize in use of computational tools and systems to answer problems of biology are bioinformaticians. Computer scientists, mathematicians, statisticians, and engineers who specialize in developing theories, algorithms and techniques for such tools and systems are computational biologists. The actual process of analyzing and interpreting data is referred to as computational biology. Important sub- disciplines within bioinformatics and computational biology include:

- the development and implementation of tools that enable efficient access to, and use and management of, various types of information.
- the development of new algorithms (mathematical formulas) and statistics with which to assess relationships among members of large data sets, such as methods to locate a gene within a sequence, predict protein structure and/or function, and cluster protein sequences into families of related sequences

Bioinformatics has become a mainstay of genomics, proteomics, and all other *.omics (such as phenomics) and many information technology companies have entered the business or are considering entering the business, creating an IT (information technology) and BT (biotechnology) convergence. **A bioinformaticist** is an expert who not only knows how to use bioinformatics tools, but also knows how to write interfaces for effective use of the tools. **A bioinformatician**, on the other hand, is a trained individual who only knows to use bioinformatics tools without a deeper understanding.

5. Biological databases

Biological databases are huge data bases of mostly sequence data pouring in from many genome sequencing projects going on all over the world. They are an important tool in assisting scientists to understand and explain a host of biological phenomena from the structure of biomolecules and their interaction, to the whole metabolism of organisms to understanding the evolution of species. This knowledge helps facilitate to fight against diseases, assists in the development of medications and in discovering basic relationships amongst species in the history of life.

The information about DNA, proteins and the function of proteins must be stored in an intelligent fashion, so that scientists can solve problems quickly and easily using all available information. Therefore, the information is stored in *databanks*, many of which are accessible to everyone on the internet. A few examples are a databank containing protein structures (the PDB or Protein Data Bank), a databank containing protein sequences and their function (Swiss-Prot), a databank with information about enzymes and their function (ENZYME), and a databank with nucleotide sequences of all genes sequenced up to date (EMBL). Due to the current state of technology, there are large differences between the sizes of databanks. EMBL, the nucleotides database contains many more sequences than the number of protein structures registered in the PDB. The reason for this is that it is a lot simpler to sequence a gene, than to find out which protein is encoded by this gene and what its function is. Also it is more difficult to determine the structure of the protein.

Using databanks, one can perform all kinds of comparisons and search queries. If, for example, you know a protein which causes a disease in humans, your might look into a databank to see if a similar protein has previously been described and what this protein does in the human body.

Using known information will make it easier and quicker to develop a drug against the disease or a test to detect the disorder in an early stage.

The Biological data can be broadly classified as:

Biological Databases	*Information they contain*
1. Bibliographic databases	Literature
2. Taxonomic databases	Classification
3. Nucleicacid databases	DNA information
4. Genomic databases	Gene level information
5. Protein databases	Protein information
6. Protein families, domains and functional sites	Classification of proteins and identifying domains
7. Enzymes/metabolic pathways	Metabolic pathways

There are many different types of database but for routine sequence analysis, the following are initially the most important

1. Primary databases: Contain sequence data such as nucleic acid or protein. Example of primary databases include:

Protein Databases	*Nucleic Acid Databases*
SWISS-PROT	• EMBL
TREMBL	• Genbank
PIR	• DDBJ

2. Secondary databases: These are also known as pattern databases contain results from the analysis of the sequences in the primary databases. Example of secondary databases include: PROSITE, Pfam, BLOCKS,PRINTS.

6. Introduction to NCBI and Entrez

The web-site of National Center for Biotechnology Information (NCBI) is one of the world's premier website for biomedical and bioinformatics research (http://www.ncbi.nlm.nih.gov/). Based within the National Library of Medicine at the National Institutes of Health, USA, the NCBI hosts many databases used by biomedical and research professionals. The services include PubMed (the bibliographic database); GenBank (the nucleotide sequence database); and the BLAST algorithm for sequence comparison, among many others. It is established in 1988 as a national resource for molecular biology information. NCBI creates public databases, conducts research in computational biology, develops software tools for analyzing genome data, and disseminates biomedical information all for the better understanding of molecular processes affecting human health and disease.

Every database has a unique identifier. Each entry in a database must have a unique identifier EMBL Identifier (ID), GENBANK Accession Number (AC). This database stores information along with the sequence. Each piece of information is written on it's own line, with a code defining the line. For example, DE (description); OS (organism species); AC (accession number). Relevant biological information is usually described in the feature table (FT).

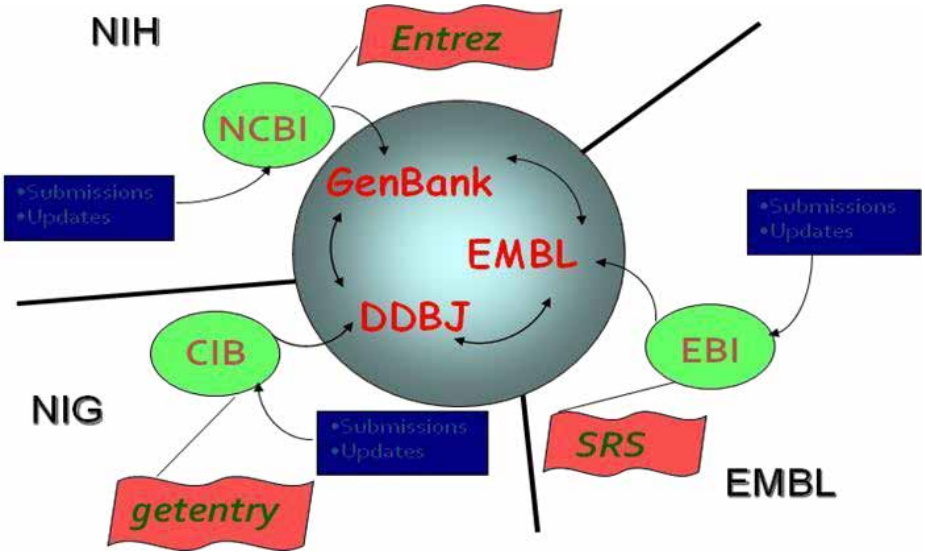


Fig 6. International Sequence Database Collaboration

7. The Entrez Search and RetrievalSystem

Entrez is the text-based search and retrieval system used at NCBI for all of the major databases, including PubMed, Nucleotide and Protein Sequences, Protein Structures, Complete Genomes, Taxonomy, OMIM, and many others. Entrez is at once an indexing and retrieval system, a collection of data from many sources, and an organizing principle for biomedical information. These general concepts are the focus of this section (Fig 7.).

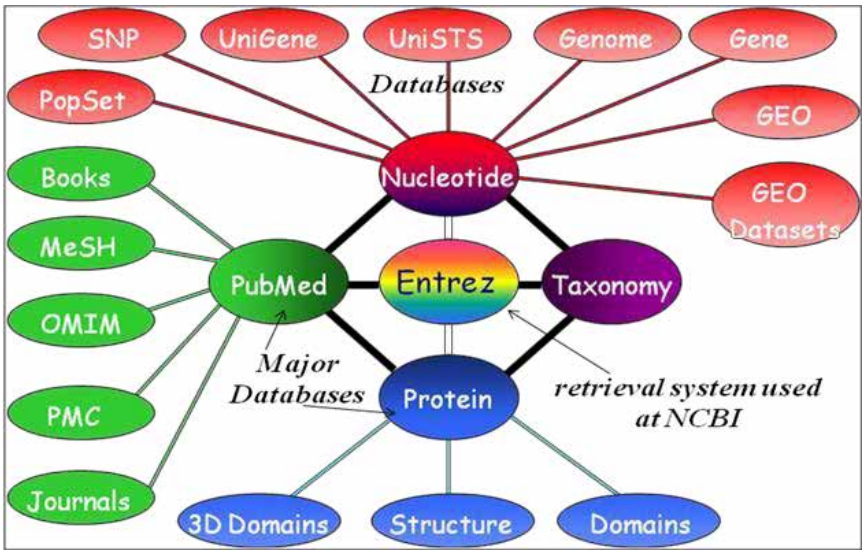


Fig 7. NCBI - RDBMS

8. The Nucleotide Sequence database

The GenBank sequence database is an annotated collection of all publicly available nucleotide sequences and their protein translations. This database is produced at National Center for Biotechnology Information (NCBI) as part of an international collaboration with the European Molecular Biology Laboratory (EMBL) as given in Fig. 8, data library from the European Bioinformatics Institute (EBI) and the DNA Data Bank of Japan (DDBJ) given in Fig. 9. GenBank and its collaborators receive sequences produced in laboratories throughout the world from more than 100,000 distinct organisms. GenBank continues to grow at an exponential rate, doubling every 10 months. Release 134, produced in February 2003, and contained over 29.3 billion nucleotide bases in more than 23.0 million sequences. GenBank is built by direct submissions from individual laboratories, as well as from bulk submissions from large-scale sequencing centers.

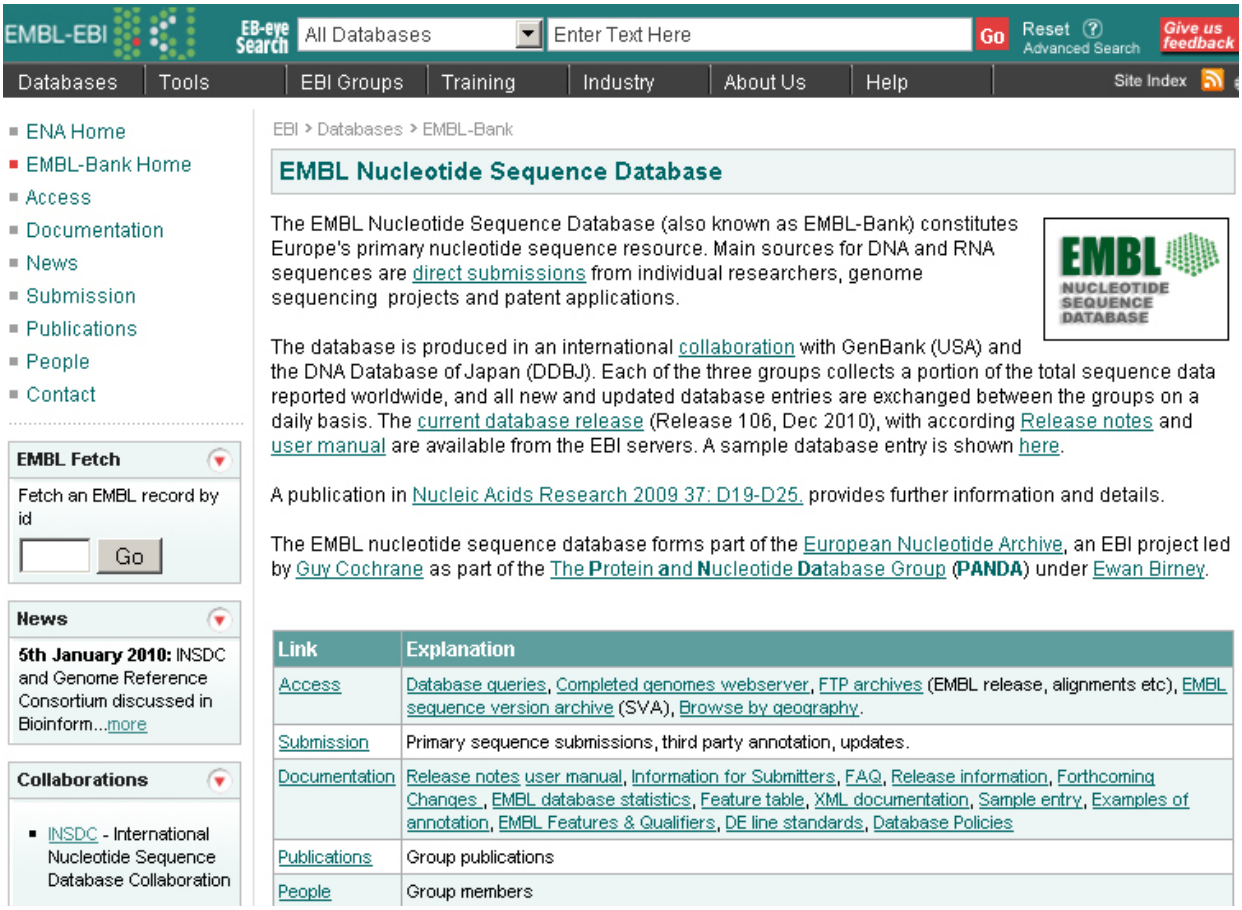


Fig. 8 EMBL Nucleotide Sequence Database



Fig. 9. DNA Data Bank of Japan

9. The Bibliographic Database

PubMed is a database developed by the NCBI. The database was designed to provide access to citations (with abstracts) from biomedical journals. Subsequently, a linking feature was added to provide access to full-text journal articles at Web sites of participating publishers, as well as to other related Web resources. PubMed is the bibliographic component of the NCBI's Entrez retrieval system.

MEDLINE is NLM's premier bibliographic database covering the fields of medicine, nursing, dentistry, veterinary medicine, and the preclinical sciences. Journal articles are indexed for MEDLINE, and their citations are searchable, using NLM's controlled vocabulary, MeSH (Medical Subject Headings). MEDLINE contains all citations published in Index Medicus, and corresponds in part to the International Nursing Index and the Index to Dental Literature.

10. Macromolecular Structure Databases

The resources provided by NCBI for studying the three-dimensional (3D) structures of proteins center around two databases: the Molecular Modeling Database (MMDB), which provides structural information about individual proteins; and the Conserved Domain Database (CDD), which provides a directory of sequence and structure alignments representing conserved functional domains within proteins(CDs). Together, these two databases allow scientists to retrieve and view structures, find structurally similar proteins to a protein of interest, and identify conserved functional sites.

11. Computer Programming in Bioinformatics: JAVA in Bioinformatics

The geographical scattered research centres all around the globe ranging from private to academic settings, and a range of hardware and OSs are being used, Java is emerging as a key player in bioinformatics. Physiome Sciences' computer-based biological simulation technologies and Bioinformatics Solutions' PatternHunter are two examples of the growing adoption of Java in bioinformatics.

12. Perl in Bioinformatics

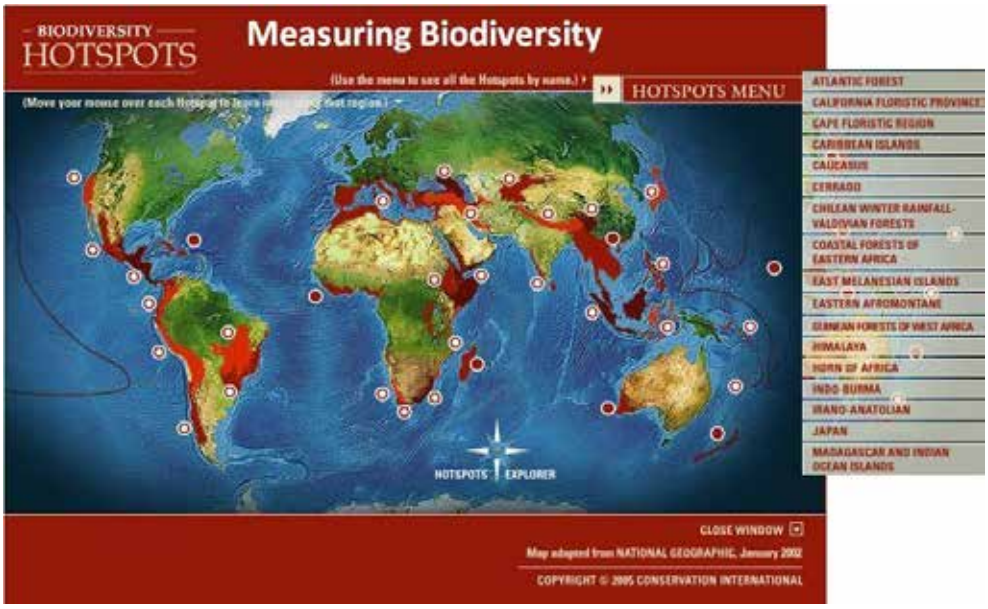
String manipulation, regular expression matching, file parsing, data format interconversionetc are the common text-processing tasks performed in bioinformatics. Perl excels in such tasks and is being used by many developers. Yet, there are no standard modules designed in Perl specifically for the field of bioinformatics. However, developers normally designed several of their own individual modules for any specific purpose, which have become quite popular and are coordinated by the BioPerl project.

13. Measuring biodiversity

Biodiversity Databases are used to collect the species names, descriptions, distributions, genetic information, status & size of populations, habitat needs, and how each organism interacts with other species etc. Computer simulations models are useful to study population dynamics, or calculate the cumulative genetic health of a breeding pool (in agriculture) or endangered population (in conservation). Entire DNA sequences or genomes of endangered species can be preserved, allowing the results of Nature's genetic experiment to be remembered *in silico*.

In these days of growing human population and habitat destruction, knowledge of centers of high biodiversity is critical for rational conservation decisions to be made. The major problem area is that this information is largely unavailable to the decision makers. It is ironic that most of these data are in the great museums, which are located in the cool temperate parts of the world whereas; most of the organisms are in the warm humid parts of the world. The data that exist are paper based. Descriptions by collectors and curators, herbarium sheets, diagrams and photographs, and of course, pickled and preserved specimens with their labels. If a researcher wishes to consult these data he/she has to travel to the museum in question. For people who need a breadth of information to make decisions, this is obviously not an option. There are two areas in biology where enormous amounts of information are generated. One is in molecular biology which deals with base sequences in DNA and amino acid sequences in proteins, and the other is the biodiversity information crisis. Mathematics and computers are being used to tackle these problems with procedures which come under the label of Bioinformatics.

Fig 10. Biodiversity Hotspots regions



14. Sequence analysis and alignment

The most well-known application of bioinformatics is sequence analysis. In sequence analysis, DNA sequences of various organisms are stored in databases for easy retrieval and comparison. The well-reported Human Genome Project (Fig. 11) is an example of sequence analysis bioinformatics. Using massive computers and various methods of collecting sequences, the entire human genome was sequenced and stored within a structured database. DNA sequences used for bioinformatics can be collected in a number of ways. One method is to go through a genome and search out individual sequences to record and store. Another method is to compare all fragments for finding whole sequences by overlapping the redundant segments. The latter method, known as shotgun sequencing, is currently the most popular because of its ease and speed. By comparing known sequences of a genome to specific mutations, much information can be assembled about undesirable mutations such as cancers. With the completed mapping of the human genome, bioinformatics has become very important in the research of cancers in the hope of an eventual cure. Computers are also used to collect and store broader data about species. The Species 2000 project, for example, aims to collect a large amount of information about every species of plant, fungus, and animal on the earth. This information can then be used for a number of applications, including tracking changes in populations and biomes.

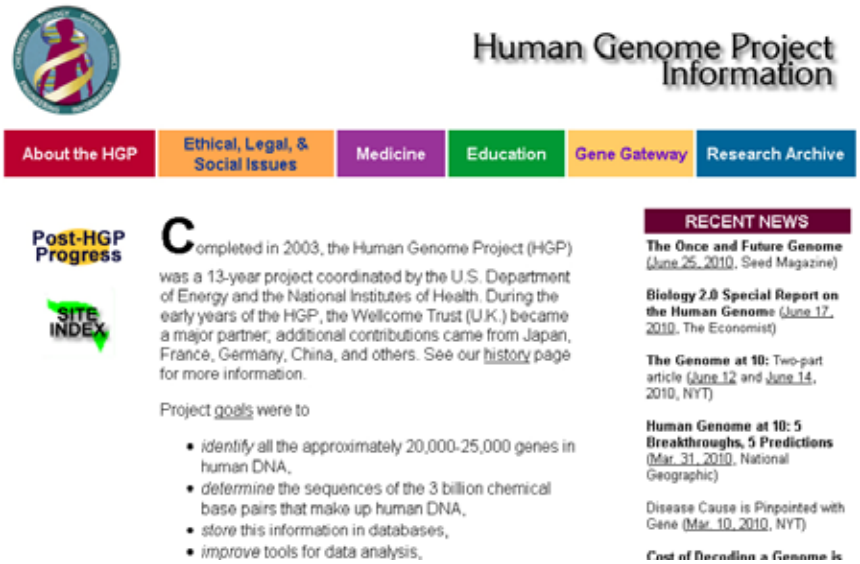


Fig. 11. Human Genome Project

With the growing amount of data, earlier it was impractical to analyze DNA sequences manually. Nowadays, many tools and techniques are available provide the sequence comparisons (sequence alignment) and analyze the alignment product to understand the biology. For example, BLAST is used to search the genomes of thousands of organisms, containing billions of nucleotides. BLAST is software which can do this using dynamic programming, as fast as google searches for your keywords, considering the length of query words of bio-sequences.

Sequence Alignment: The sequence alignment can be categorized into two groups i.e. global and local alignment

Global Alignment

Input: two sequences S_1, S_2 over the same alphabet Output: two sequences S'_1, S'_2 of equal length (S'_1, S'_2 are S_1, S_2 with possibly additional gaps)

Example:

u S_1=GCGCATGGATTGAGCGA

u S_2=TGCGCCATTGATGACC

u A possible alignment:

S'_1= -GCGC-ATGGATTGAGCGA

S'_2= TGCGCCATTGAT-GACC—

Local Alignment

Goal: Find the pair of substrings in two input sequences which have the highest similarity Input: two sequences S_1, S_2 over the same alphabet

Output: two sequences S'_1, S'_2 of equal length

(S'_1, S'_2 are substrings of S_1, S_2 with possibly additional gaps)

Example:

S_1=GCGCATGGATTGAGCGA S_2=TGCGCCATTGATGACC

A possible alignment:

S'_1= ATTGA-G

S'_2= ATTGATG

FASTA: In bioinformatics, FASTA format is a text-based format for representing either nucleotide sequences or peptide sequences, in which base pairs or amino acids are represented using single-letter codes. The format also allows for sequence names and comments to precede the sequences. The FASTA format may be used to represent either single sequences or many sequences in a single file. A series of single sequences, concatenated, constitute a multisequence file. A sequence in FASTA format is represented as a series of lines, which should be no longer than 120 characters and usually do not exceed 80 characters. This probably was because to allow for preallocation of fixed line sizes in software: at the time, most users relied on DEC VT (or compatible) terminals which could display 80 or 132 characters per line. Most people would prefer normally the bigger font in 80- character modes and so it became the recommended fashion to use 80 characters or less (often 70) in FASTA lines. The first line in a FASTA file starts either with a ">" (greater-than) symbol or a ";" (semicolon) and was taken as a comment. Subsequent lines starting with a semicolon would be ignored by software. Since the only comment used was the first, it quickly became used to hold a summary description of the sequence, often starting with a unique library accession number, and with time it has become common place use

to always use ">" for the first line and to not use ";" comments (which would otherwise be ignored).

>gi|5524211|gb|AAD44166.1| cytochrome b [*Elephas maximus maximus*]

LCLYTHIGRNIYYGSYLYSETWNTGIMLLLITMATAFMGYVLPWGQMSFWGATVITNLFSA
IPYIGTNLVEWIWGGFSVDKATLNRFFAFHFILPFTMVALAGVHLTFLHETGSNNPLGLTSDS
DKIPFHPYYTIKDFLGLLILILLLLLLALLSPDMLGDPDNHMPADPLNTPLHIKPEWFLFAYAI
LRSVPNKLGGVLALFLSIVILGLMPFLHTSKHRSMMLRPLSQALFWTLTMDLLTLTWIGSQP
VEYPYTIIGQMASILYFSIILAFLPIAGXIENY

15. Prediction of protein structure

Proteins play crucial functional roles in all biological processes: enzymatic catalysis, signaling messengers, structural elements. Function depends on unique 3-D structure. It is easy to obtain protein sequences but difficult to determine structure. Protein structure prediction is another important application of bioinformatics. The amino acid sequence of a protein, the so-called primary structure, can be easily determined from the sequence on the gene that codes for it. In the vast majority of cases, this primary structure uniquely determines a structure in its native environment. Knowledge of this structure is vital in understanding the function of the protein. For lack of better terms, structural information is usually classified as one of *secondary*, *tertiary* and *quaternary* structure. Protein structure prediction is the prediction of the three-dimensional structure of a protein from its amino acid sequence i.e, the prediction of its tertiary structure from its primary structure. Protein structures are being determined with increasing speed. Consequently, automated and fast bioinformatics tools are required for exploring structure–function relationships in large numbers of proteins. These are necessary both when the function has been characterized experimentally and when it must be predicted.

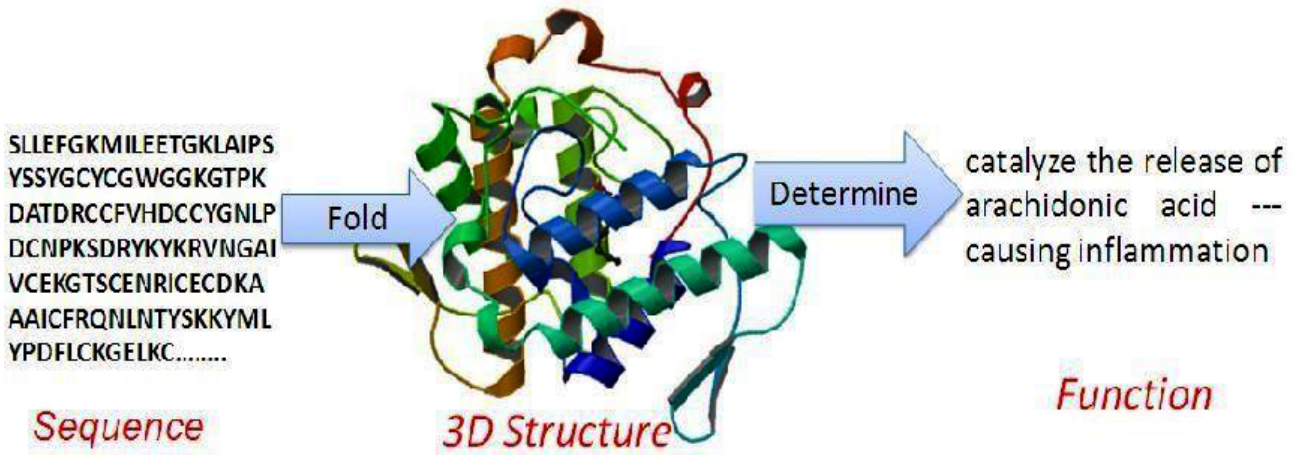


Fig. 12. Protein Structure Prediction

In the genomic branch of bioinformatics, homology is used to predict the function of a gene: if the sequence of gene *A*, whose function is known, is homologous to the sequence of gene *B*, whose function is unknown, one could infer that B may share A's function. In the structural branch of bioinformatics, homology is used to determine which parts of a protein are important in structure formation and interaction with other proteins. In a technique called homology modeling, this information is used to predict the structure of a protein once the structure of a homologous protein is known. One example of this is the similar protein homology between hemoglobin in humans and the hemoglobin in legumes (leghemoglobin). Both serve the same purpose of transporting oxygen in the organism. Though both of these proteins have completely different amino acid sequences, their protein structures are virtually identical, which reflects their near identical purposes.

16. Molecular docking

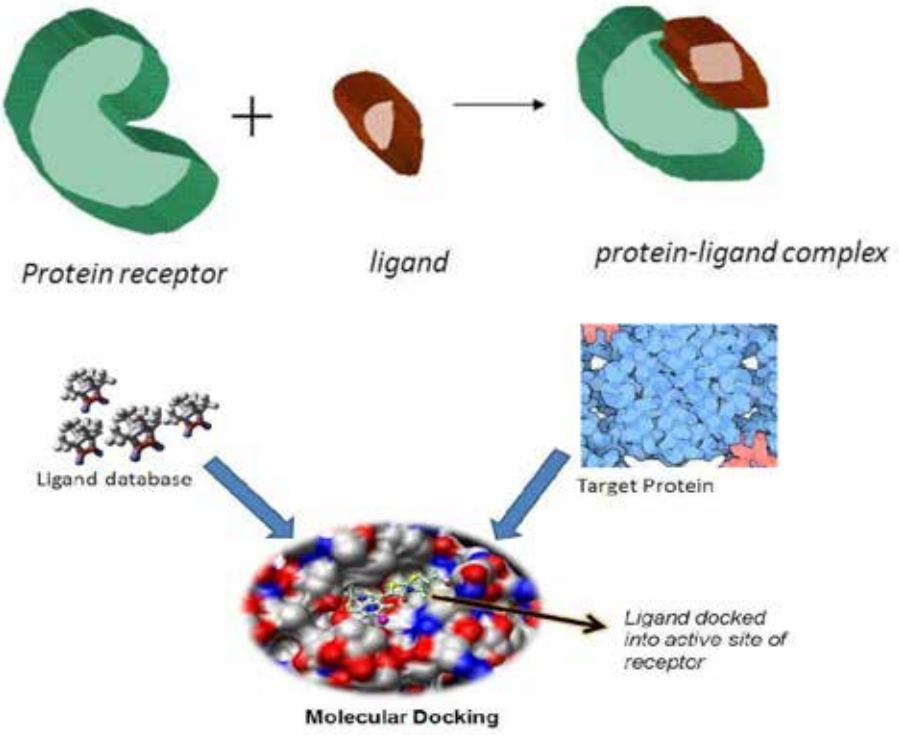


Fig. 13 Protein-ligand Docking

In the last two decades, tens of thousands of protein three-dimensional structures have been determined by X-ray crystallography and Protein nuclear magnetic resonance spectroscopy (protein NMR). One central question for the biological scientist is whether it is practical to predict possible protein-protein interactions only based on these 3D shapes, without doing protein-protein interaction experiments. A variety of methods have been developed to tackle the Protein-protein docking problem, though it seems that there is still much work to be done in this field. We are interested in information about our DNA, proteins and the function of proteins. Genes and proteins can be sequenced, so the sequence of bases in genes or amino acids in proteins can be determined. This information must be stored in an intelligent fashion, so that scientists can solve problems quickly and easily using all available information. Therefore, the information is stored in *databanks*, many of which are accessible to everyone on the internet. A few examples are a databank containing protein structures (the PDB or Protein Data Bank), a databank containing protein sequences and their function (Swiss-Prot), a databank with information about enzymes and their function (ENZYME), and a databank with nucleotide sequences of all genes sequenced up to date (EMBL).

17. Bioinformatics in Agriculture

The most critical tasks in bioinformatics involves the finding of genes in the DNA sequences of various organisms, developing methods to predict the structure and function of newly discovered proteins and structural RNA sequences, clustering protein sequences into families of related sequences, development of protein models, aligning similar proteins and generating phylogenetic trees to examine evolutionary relationships. The sequencing of the genomes of microbes, plants and animals should have enormous benefits for the agricultural community. Computational analysis of these sequence data generated by genome sequencing, proteomics and array-based technologies is critically important. Bioinformatics tools can be used to search for the genes within these genomes and to elucidate their functions.

The sequencing of the genomes of plants and animals should have enormous benefits for the agricultural community. Bioinformatic tools can be used to search for the genes within these genomes and to elucidate their functions. This specific genetic knowledge could then be used to produce stronger, more drought, disease and insect resistant crops and improve the quality of livestock making them healthier, more disease resistant and more productive.

18. Bioinformatics in India

Studies of IDC points out that India will be a potential star in bioscience field in the coming years after considering the factors like bio-diversity, human resources, infrastructure facilities and governments initiatives.

Bioinformatics has emerged out of the inputs from several different areas such as biology, biochemistry, biophysics, molecular biology, biostatics, and computer science. Specially designed algorithms and organized databases is the core of all informatics operations. The requirements for such an activity make heavy and high level demands on both the hardware and software capabilities. This sector is the quickest growing field in the country. The vertical growth is because of the linkages between IT and biotechnology, spurred by the human genome project. The promising start- ups are already there in Bangalore, Hyderabad, Pune, Chennai, and Delhi. There are over 200 companies functioning in these places. IT majors such as Intel, IBM, Wipro are getting into this segment spurred by the promises in technological developments.

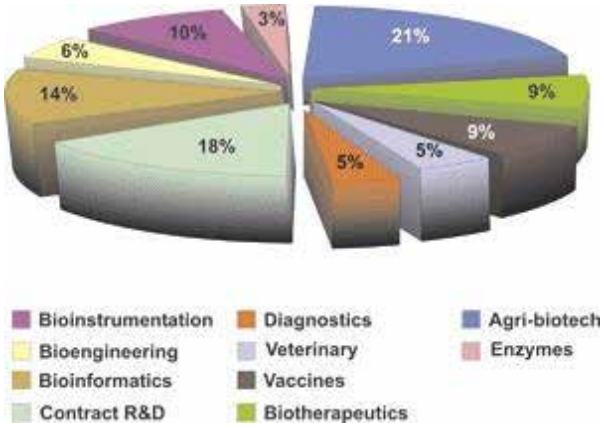
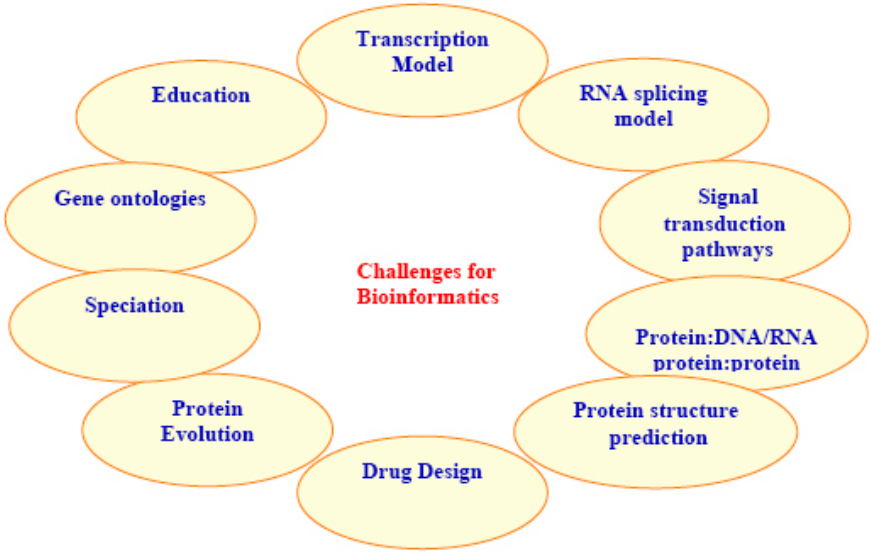


Fig. 14. Applications and Challenges in Bioinformatics



Gene Sanctuaries and Gene Banks

Gene Sanctuaries:

The genetic diversity is sometimes conserved under natural habitat. In other words, areas of great genetic diversity are protected from human interference. These protected areas in natural habitat are referred to as gene sanctuaries. Gene sanctuary is generally established in the centres of diversity or microcenter. Gene sanctuary is Also Known As Natural Park or biosphere reserve. India has setup its first gene sanctuary in the Garo Hills of Assam for wild relatives of citrus. Efforts are also being made to setup gene sanctuaries for banana, sugarcane, rice and Mango. In Ethiopia gene sanctuaries for conservation of wild relatives of coffee was setup in 1984. Gene sanctuaries have two main advantages. Firstly, it protects the loss of genetic diversity caused by human intervention. Secondly, it allows natural selection and evolution to operate. There are two main drawbacks of gene sanctuary. Firstly, entire variability of a crop species can not conserved. Secondly, its maintenance and establishment is a difficult task. It is a very good method of *in situ* conservation.

Gene Banks:

Gene bank refers to a place or organisation where germplasm can be conserved in living state. Gene banks are also known as germplasm banks. The germplasm is stored in the form of seeds, pollen or in vitro cultures, or in the case of a field gene bank, as plants growing in the field. Gene banks are mainly of two types, viz: 1) Seed gene banks, and 2) Field gene banks. These are briefly discussed below:

1. Seed Gene Bank:

A place where germplasm is conserved in the form of seeds is called seed gene bank. Seeds are very convenient for storage because they occupy smaller space than whole plants. However, seeds of all crops can not be stored at low temperature in the seed banks. The germplasm of only orthodox species (whose seed can be dried to low moisture content without losing variability) can be conserved in seed banks. In the seed banks, there are three types of conservation, viz. 1) Short term, 2) Medium term, 3) Long term. Base collections are conserved for long term (50 years or more) at 18 or 20 ^{0}C. Active collections are stored for short term (3-5 years) at 5-10 ^{0}C. The main advantages of gene banks are as follows:

1. Large number of germplasm samples or entire variability can be conserved in a very small space.
2. In seed banks, handling of germplasm is easy.
3. Germplasm is conserved under pathogen and insect free environment.

There are some disadvantages of germplasm conservation in the seed banks. These are listed below:

1. Seeds of recalcitrant species can not be stored in seed banks.
2. Failure of power supply may lead to loss of viability and there by loss of germplasm.
3. It requires periodical evaluation of seed viability. After some time multiplication is essential to get new or fresh seeds of storage.

2. Field Gene Banks:

Field gene banks also called plant gene banks are areas of land in which germplasm collections of growing plants are assembled. This is also *ex situ* conservation of germplasm. Those plant species that have recalcitrant seeds or do not produce seeds readily are conserved in field gene bank. In field gene banks, germplasm is maintained in the form of plants as a permanent living collection. Field gene banks are often established to maintain working collections of living plants for experimental purposes.

They are used as source of germplasm for species such as coconut, rubber, mango, cassava, yam, and cocoa. Field gene banks have been established in many countries for different crops.

Field Gene Banks in Some Countries:

Name of Country	Crop Species for which Field Gene Bank is Established
Malaysia	Oil palm has been conserved on 500 hectares.
Indonesia	Earmarked 1000 hectares area for coconut and other perennial crops.
Philippines	South East Asian germplasm of banana has been conserved.
India	Global collection of coconut has been conserved in Andaman and Nicobar.

Field gene banks have some advantages and disadvantages which are discussed below: There are two main advantages.

1. It provides opportunities for continuous evaluation for various economic characters.
2. It can be directly utilized in the breeding programmes.

There are three main demerits of field gene banks as given below:

1. Field gene banks can not cover the entire genetic diversity of a species. It can cover only a fraction of the full range of diversity of a species.
2. The germplasm in field gene banks is exposed to pathogen and insects and some times is damaged by natural disasters such as bushfires, cyclones, floods , etc.
3. Maintenance of germplasm in the field gene banks is costly affair.

Meristem Gene Banks:

Germplasm of asexually propagated species can be conserved in the form of meristem. This method is widely used for conservation and propagation of horticultural species. In vitro method can be used in two ways, first, for storage of tissue under slow growth conditions. Second, for long term conservation of germplasm by cryopreservation. In cryopreservation, the tissues are stored at very low temperatures i.e. at -196 ^{0}C in liquid nitrogen. At these temperatures, all biological processes virtually come to a stop.

Based on status of Research Institutes, gene banks are again of two types, viz. 1) National gene banks, and 2) International or global gene banks . National gene banks are maintained by each country and global gene banks are located in International Crop Research Institutes/ Centres. In India, gene banks are maintained by concerned crop research institute of ICAR. National Bureau of Plant Genetic Resources, New Delhi is also maintaining germplasm of various field crops.

Types of gene banks

Seed bank

A seed bank preserves dried seeds by storing them at a very low temperature. Spores and pteridophytes are conserved in seed banks, but other seedless plants, such as tuber crops cannot be preserved this way. The largest seed bank in world is the Millennium Seed Bank housed at the Wellcome Trust Millennium Building (WTMB), located in the grounds of Wakehurst Place in West Sussex, near London.

Tissue bank

In this technique, buds, protocorm and meristematic cells are preserved through particular light and temperature arrangements in a nutrient medium. This technique is used to preserve seedless plants and plants that reproduce sexually.

Cryobank

In this technique, a seed or embryo is preserved at very low temperatures. It is usually preserved in liquid nitrogen at -196 °C. This is helpful for the conservation of species facing extinction. Cryobanks are utilized for the cryoconservation of animal genetic resources Storage of pollen This is a method in which pollen grains are stored. We can make plants which are facing extinction in the present world. Using this technique, we can make plants with one set of chromosomes. The pollen is stored in liquid nitrogen. This method is useful for crossbreeding.

Field gene bank

This is a method of planting plants for the conservation of genes. For this purpose, an ecosystem is created artificially. Through this method one can compare the differences among plants of different species and can study them in detail. It needs more land, adequate soil, weather, etc. Germ plasma of important crops are conserved through this method. 42,000 varieties of rice are conserved in the Central Rice Research Institute in Orissa.

Principle of PCR

The polymerase chain reaction (PCR) is a laboratory technique for DNA replication that allows a "target" DNA sequence to be selectively amplified. PCR can use the smallest sample of the DNA to be cloned and amplify it to millions of copies in just a few hours. Discovered in 1985 by Kerry Mullis, PCR has become both and essential and routine tool in most biological laboratories.

The PCR involves the primer mediated enzymatic amplification of DNA. PCR is based on using the ability of DNA polymerase to synthesize new strand of DNA complementary to the offered template strand. Primer is needed because DNA polymerase can add a nucleotide only onto a preexisting 3'-OH group to add the first nucleotide. DNA polymerase then elongate its 3 end by adding more nucleotides to generate an extended region of double stranded DNA.

Components of PCR

The PCR reaction requires the following components:

- DNA Template : The double stranded DNA (dsDNA) of interest, separated from the sample.
- DNA Polymerase : Usually a thermostableTaq polymerase that does not rapidly denature at high temperatures (98°), and can function at a temperature optimum of about 70°C.
- Oligonucleotide primers : Short pieces of single stranded DNA (often 20-30 base pairs) which are complementary to the 3' ends of the sense and anti-sense strands of the target sequence.
- Deoxynucleotide triphosphates : Single units of the bases A, T, G, and C (dATP, dTTP, dGTP, dCTP) provide the energy for polymerization and the building blocks for DNA synthesis.
- Buffer system : Includes magnesium and potassium to provide the optimal conditions for DNA denaturation and renaturation; also important for polymerase activity, stability and fidelity.

Procedure of PCR

All the PCR components are mixed together and are taken through series of 3 major cyclic reactions conducted in an automated, self-contained thermocycler machine.

Denaturation : This step involves heating the reaction mixture to 94°C for 15-30 seconds. During this, the double stranded DNA is denatured to single strands due to breakage in weak hydrogen bonds.

Annealing : The reaction temperature is rapidly lowered to 54-60°C for 20-40 seconds. This allows the primers to bind (anneal) to their complementary sequence in the template DNA.

Elongation : Also known at extension, this step usually occurs at 72-80°C (most commonly 72°C). In this step, the polymerase enzyme sequentially adds bases to the 3′ each primer, extending the DNA sequence in the 5′ to 3′ direction. Under optimal conditions, DNA polymerase will add about 1,000 bp/ minute.

With one cycle, a single segment of double-stranded DNA template is amplified into two separate pieces of double-stranded DNA. These two pieces are then available for amplification in the next cycle. As the cycles are repeated, more and more copies are generated and the number of copies of the template is increased exponentially.

Applications of PCR

Some common applications of PCR in various fields can be explained in following categories.

Medical Applications:

1. Genetic testing for presence of genetic disease mutations. Eg: hemoglobinopathies, cystic fibrosis, other inborn errors of metabolism
2. Detection of disease causing genes in suspected parents who act as carriers.
3. Study of alteration to oncogenes may help in customization of therapy
4. Can also be used as part of a sensitive test for tissue typing, vital to organ transplantation genotyping of embryo
5. Helps to monitor the gene in gene therapy

Infectious disease Applications:

1. Analyzing clinical specimens for the presence of infectious agents, including HIV, hepatitis, malaria, tuberulosis etc.
2. Detection of new virulent subtypes of organism that is responsible for epidemics.

Forensic Applications: Can be used as a tool in genetic fingerprinting. This technology can identify any one person from millions of others in case of : crime scence, rule out suspects during police investigation, paternity testing even in case of avaibility of very small amount of specimens (stains of blood, semen, hair etc)

Research and Molecular Genetics:

1. In genomic studies: PCR helps to compare the genomes of two organisms and identify the difference between them.
2. In phylogenetic analysis. Minute quantities of DNA from any source such a fossilized material, hair, bones, mummified tissues.
3. In study of gene expression analysis, PCR based mutagenesis
4. In Human genome project for aim to complete mapping and understanding of all genes of human beings.

Factors affecting PCR

i. Primer

- PCR reaction needs two primer, a forward and a reverse primer
- Primer are synthesized oligonucleotide usually ranging from 15-30 bases long
- Primers are designed such that at 3′ end they do not have more than two bases complementary to each other as this results in PRIMER-DIMER formation
- The G+C contents is in the range of 40-60%
- The melting temperature (Tm) of both forward and reverse primer is usually the same.

- Low concentration of primer results in poor yield while high concentration may results in non specific amplification. Hence optimal concentration of primer is needed ie 0.1-1μM.

ii. Amount of Template DNA

- Optimal amount of template DNA usually in nano gram. Higher concentration inhibit or results in non specific amplification.
- Taq DNA polymerase:
- Taq DNA polymerase is 94 KD thermostable DNA polymerase isolated from *Thermus aquaticus*.
- Optimal temperature for activity of Taq polymerase is 72° but it can tolerate high temperature and do not affects by denaturating temperature of 94°C.
- Taq DNA polymerase have both 5′-3′ polymerase activity and 5′-3′ exonuclease activity. But it lacks 3′-5′ exonuclease activity (proof reading activity).

Types of PCR

In addition to the amplification of a target DNA sequence by the typical PCR procedures already described, several specialised types of PCR have been developed for specific applications.

1. Standard PCR:

- Nested PCR
- Random amplified polymorphic DNA
- Long PCR
- Restriction fragment length polymorphism (RFLP)
- Amplified fragment length polymorphism (AFLP)
- Multiplex PCR
- Single cell PCR
- Fast cycling PCR
- In situ PCR
- High fidelity PCR
- Asymmetric PCR
- Repetitive sequence based PCR
- Overlap extension PCR
- Assemble PCR
- Mini primer PCR
- Solid phase PCR
- Touch Down PCR

2. Reverse transcriptase Polymerase chain reaction (RT-PCT): for RNA

- One step RT-PCR
- Two step RT-PCR

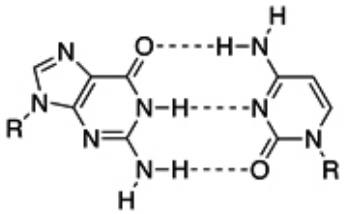
3. Real time PCR: for DNA or RNA

- Dye binding to ds DNA
- Fluorescent probes

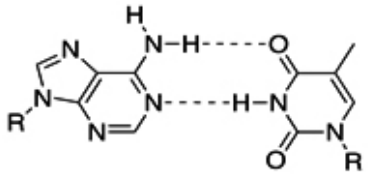
Nucleic acid hybridization

Introduction:

Nucleic acid hybridization is a basic technique in molecular biology which takes advantage of the ability of individual single-stranded nucleic acid molecules to form double-stranded molecules. According to Watson-Crick base pairing, adenine binds with thymine and guanine binds with cytosine by hydrogen bonding.



A GC base pair demonstrating three intermolecular hydrogen bonds



An AT base pair demonstrating two intermolecular hydrogen bonds

Southern hybridization:

The basic principle behind the southern hybridization is the nucleic acid hybridization. Southern hybridization commonly known as southern blot is a technique employed for detection of a specific DNA sequence in DNA samples that are complementary to a given RNA or DNA sequence. It was the first blotting technique to be devised, named after its pioneer E.M Southern, a British biologist. Southern blotting involves separation of restricted DNA fragments by electrophoresis and then transferred to a nitrocellulose or a nylon membrane, followed by detection of the fragment using probe hybridization.

Separated by electrophoresis is transferred from gel to a membrane which in turn is used as a substrate for hybridization analysis employing labeled DNA or RNA probes specific to target fragments in the blotted DNA. Southern hybridization helps to detect specific fragment against a background of many other restriction fragments. Southern blotting is a technique which is used to confirm the identity of a cloned fragment or for recognition of a sub-fragment of interest from within the cloned DNA, or a genomic DNA. Southern blotting is a prerequisite to techniques such as restriction fragment length polymorphism (RFLP) analysis.

Procedure:

1. The high-molecular-weight DNA strands are fractioned using restriction enzymes.
2. The DNA fragments are separated based on size by agarose gel electrophoresis.
3. The gel with the restricted fragments is then laid on a filter paper wick which serves as a connection between the membrane and the high salt buffer.
4. The nitrocellulose membrane is placed on top of the gel and a tower of filter papers is used to cover it and these are kept in place with a weight. The capillary action drives the buffer soaking through the filter paper wick, through the gel and the membrane and into the paper towels. Along with the buffer passing through the gel the DNA fragments are also carried with it into the membrane and they bind to the membrane. This causes an effective transfer of fragments (up to 15 kb in length taking around 18hours or overnight.

   For DNA fragments larger than 15 kb, before blotting an acid such as diluted HCl is used to treat the gel that depurinates the DNA fragments causing breakage of DNA into smaller pieces, resulting in more efficient transfer from the gel to membrane.

Now a days blotting is also done by applying electric field. This **electro blotting** technique depends upon current and transfer buffer solution to nucleic acids onto a membrane. Following electrophoresis, a standard tank or semi-dry blotting transfer system is set up. A stack is put together in the following order from cathode to anode: sponge, three sheets of filter paper soaked in transfer buffer gel, PVDF or nitrocellulose membrane, three sheets of filter paper soaked in transfer buffer and then again sponge. Importantly the membrane should be located between the gel and the positively-charged anode, as the current and sample will be moving in that direction. Once the stack is prepared, it is placed in the transfer system, and suitable current is applied for a specific period of time according to the materials being used.

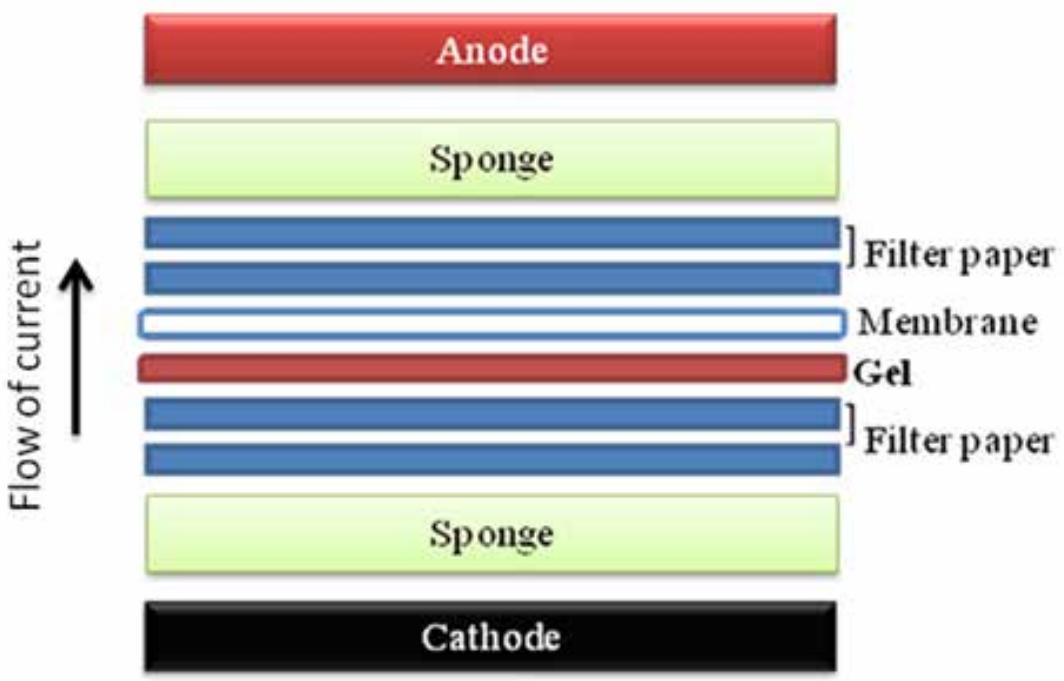


Fig: Set up for Electro-blotting system

5. For using alkaline transfer methods, the DNA gel is placed into an alkaline solution (like that of sodium hydroxide) causing denaturation of the double-stranded DNA. Denaturation in an alkaline environment enhances the binding between the negatively charged DNA and the positively charged membrane, causing separation to single DNA strands for further hybridization to the probe, alongside destroying any residual RNA that may persist in DNA. The membrane is washed with buffer to remove unbound DNA fragments.
6. The membrane which contains the transferred fragments is heated in presence or absence of vacuum at 80°C for 2 hours or exposed to ultraviolet radiation (nylon membrane) for permanent attachment of the transferred DNA to the membrane.
7. The obtained membrane is then hybridized with a probe (a DNA fragment with a specific sequence whose presence in the target DNA is to be determined).
8. Labeling of the probe DNA is done for easy detection, usually radioactivity is incorporated or the molecule is tagged with a fluorescent or chromogenic dye. The hybridization probe may be made of RNA, instead of DNA in some cases where the target is RNA specific.
9. Washing of the excess probe from the membrane is done by using saline sodium citrate(SSC buffer) after the hybridization step and the hybridization pattern is studied on an X-ray film by autoradiography (for a radioactive or fluorescent probe), or via color development on membrane if a chromogenic detection method is employed.

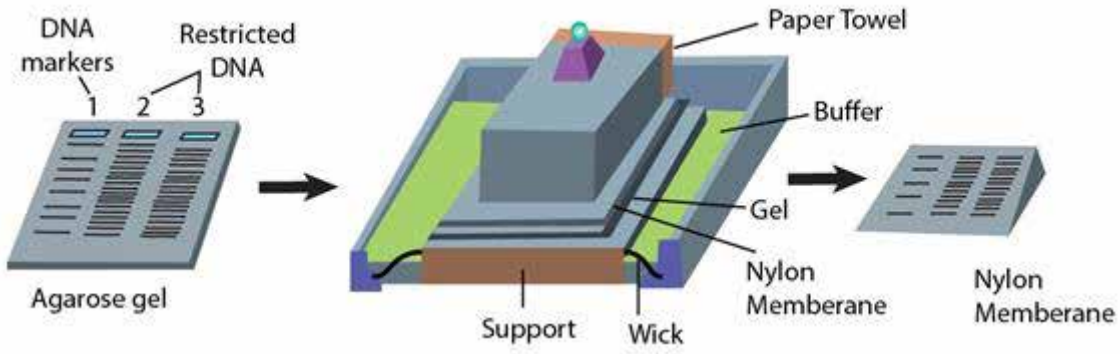


Fig: Analysis of Southern Blot

Hybridization of the probe to a specific DNA fragment on the membrane indicates the presence of a complementary fragment in the DNA sequence. Southern hybridization performed by digestion of genomic DNA using a restriction enzyme digestion, helps in determining the number of sequences (or gene copies) in the genome. For a probe hybridizing to a single DNA segment that has not been cut by the restriction enzyme, a single band is observed and on the other hand multiple bands will likely be observed when the hybridization occurs between the probe and several highly similar target sequence (Due to sequence duplication). Alterations in the hybridization conditions like enhancing the hybridization temperature or decreasing salt concentration, helps in altering specificity and hybridization of the probe to sequences that are less than 100% similar.

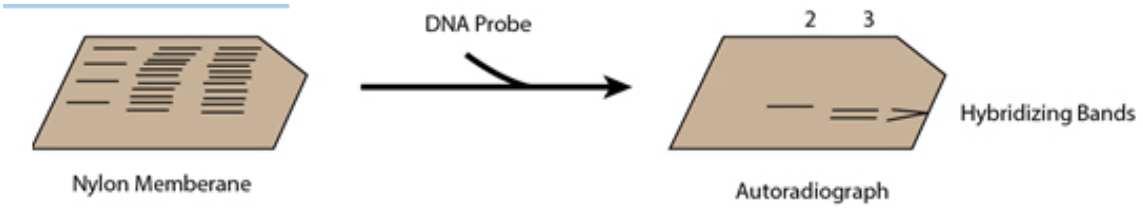


Fig: Southern hybridization analysis

Applications:

Southern blotting has been exploited for various applications which include:

a) Clone identification: One of the most common applications of Southern blotting is identification and cloning of a specific gene of interest. Southern blotting is carried out for identification of one or more restriction fragments that contain the gene of interest in genomic DNA.. After cloning and tentative identification of the desired recombinant by employing colony or plaque hybridization, southern blotting is further is used to confirm the clone identification and possibly to locate a shorter restriction fragment, containing the sequence of interest.

b) Restriction fragment length polymorphism Analysis: Another major application of Southern hybridization is restriction fragment length polymorphism (RFLP) mapping, which is crucial in construction of genome maps.

Northern hybridization:

Northern blotting was developed by James Alwine, George Stark and David Kemp (1977). Northern blotting drives its name because of its similarity to the first blotting technique, which is Southern blotting, named after the biologist Edwin Southern. The major difference is that RNA being analyzed rather than DNA in the northern blot.

Expression of a particular gene can be detected by estimating the corresponding mRNA by Northern

blotting. Northern blotting is a technique where RNA fragments are separated by electrophoresis and immobilized on a paper sheet.Identification of a specific RNA is then done by hybridization using a labeled nucleic acid probe. It helps to study gene expression by detection of RNA (or isolated mRNA) in a sample.

In Northern blotting, probes formed of nucleic acids with a sequence which is complementary to the sequence or to a part of the RNA of interest. The probe can be DNA, RNA or chemically synthesized oligonucleotides of minimum 25 complementary bases to the target sequence.

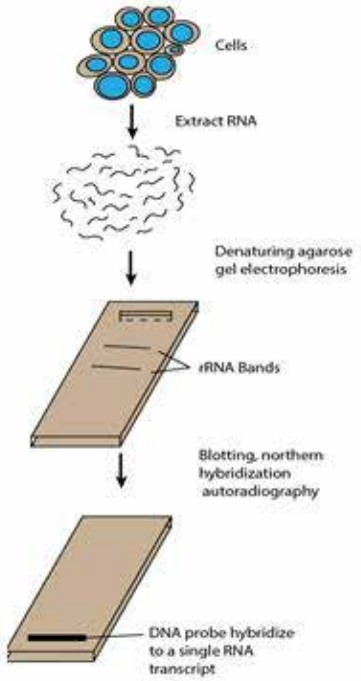


Fig: Steps of Northern Hybridization

Procedure:

The northern blotting involves the following steps:

1. Total RNA is extracted from a homogenized tissue sample or cells. Further eukaryotic mRNA can then be isolated by using of oligo (dT) cellulose chromatography to isolate only those RNAs by making use of a poly A tail.
2. The isolated RNA is then separated by gel electrophoresis.
3. The RNA samples separated on the basis of size are transferred to a nylon membrane employing a capillary or vacuum based system for blotting.

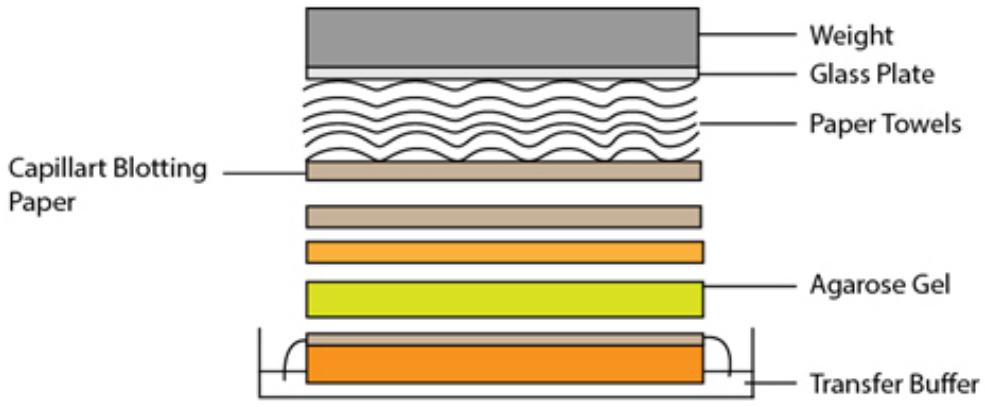


Fig: Setup for Northern blotting

4. Similar to Southern blotting, the membrane filter is revealed to a labeled DNA probe that is complementary to the gene of interest and binds.
5. The labeled filter is then subjected to autoradiography for detection.

The net amount of a specific RNA in a sample can be estimated by using Northern blot. This technique is widely used for comparing the amounts of a particular mRNA in cells under different conditions. The separation of RNA samples is often done on agarose gels containing formaldehyde as a denaturing agent as it limits the RNA to form secondary structure.

Analysis of Northern Blot:

RNA extract is electrophoresed in an agarose gel, using a denaturing electrophoresis buffer (containing formaldehyde) to ensure that the RNAs do not form inter- or intra-molecular base pairs, as base pairing would affect the rate at which the molecules migrate through the gel. After electrophoresis, the gel is blotted onto a nylon or nitrocellulose membrane, and hybridized with a labeled probe. If the probe is a cloned gene, the band that appears in the autoradiograph is the transcript of that gene. The size of the transcript can be determined from its position within the gel, and if RNA from different tissues is run in different lanes of the gel, then the possibility of differentially expressed gene can be examined.

Applications:

Northern blotting helps in studying gene expression pattern of various tissues, organs, developmental stages, pathogen infection, and also over the course of treatment. It has been employed to study overexpression of oncogenes and down-regulation of tumor-suppressor genes in cancerous cells on comparison with healthy tissue, and also for gene expression of immune-rejection of transplanted organ.

The examination of the patterns of gene expressions obtained under given conditions can help determine the function of that gene.

Northern blotting is also used for the analysis of alternate spliced products of same gene or repetitive sequence motif by investigating the various sized RNA of the gene. This is done when only probe type with variation in one location is used to bind to the target RNA molecule.

Variations in size of a gene product may also help to identify deletions or errors in transcript processing, by altering the probe target that can be used along the known sequence and make it possible to determine the missing region of the RNA.

Colony Hybridization:

It is a rapid method of isolating a colony containing a plasmid harboring a particular sequence or a gene from a mixed population. The colonies to be screened are first replica-plated on to a nitrocellulose filter disc that has been placed on the surface of an agar plate prior to inoculation. Master plate is retained for reference set of colonies. The filter bearing the colonies is removed and treated with alkali so that the bacterial colonies are lysed and the DNA they contain is denatured. The filter is then treated with proteinase K to remove protein and leave denatured DNA bound to the nitrocellulose. The DNA is fixed firmly by baking the filter at 80°C. A labeled probe is hybridized to this DNA which is monitored by autoradiography. A colony whose DNA print gives a positive auto radiographic result on X-ray film can then be picked from the reference plate.

Colony hybridization can be used to screen plasmid or cosmid based libraries.

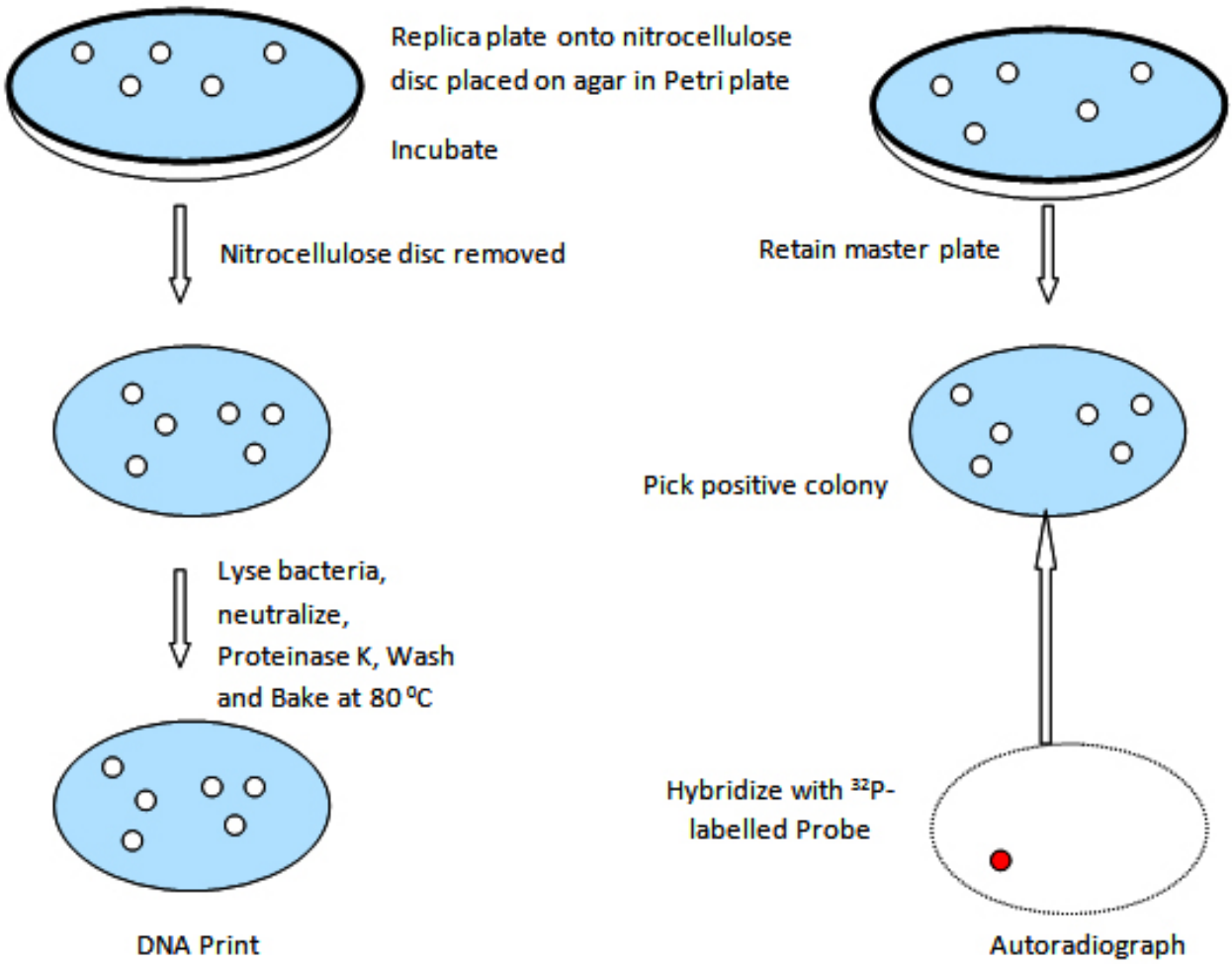


Fig: Colony hybridization

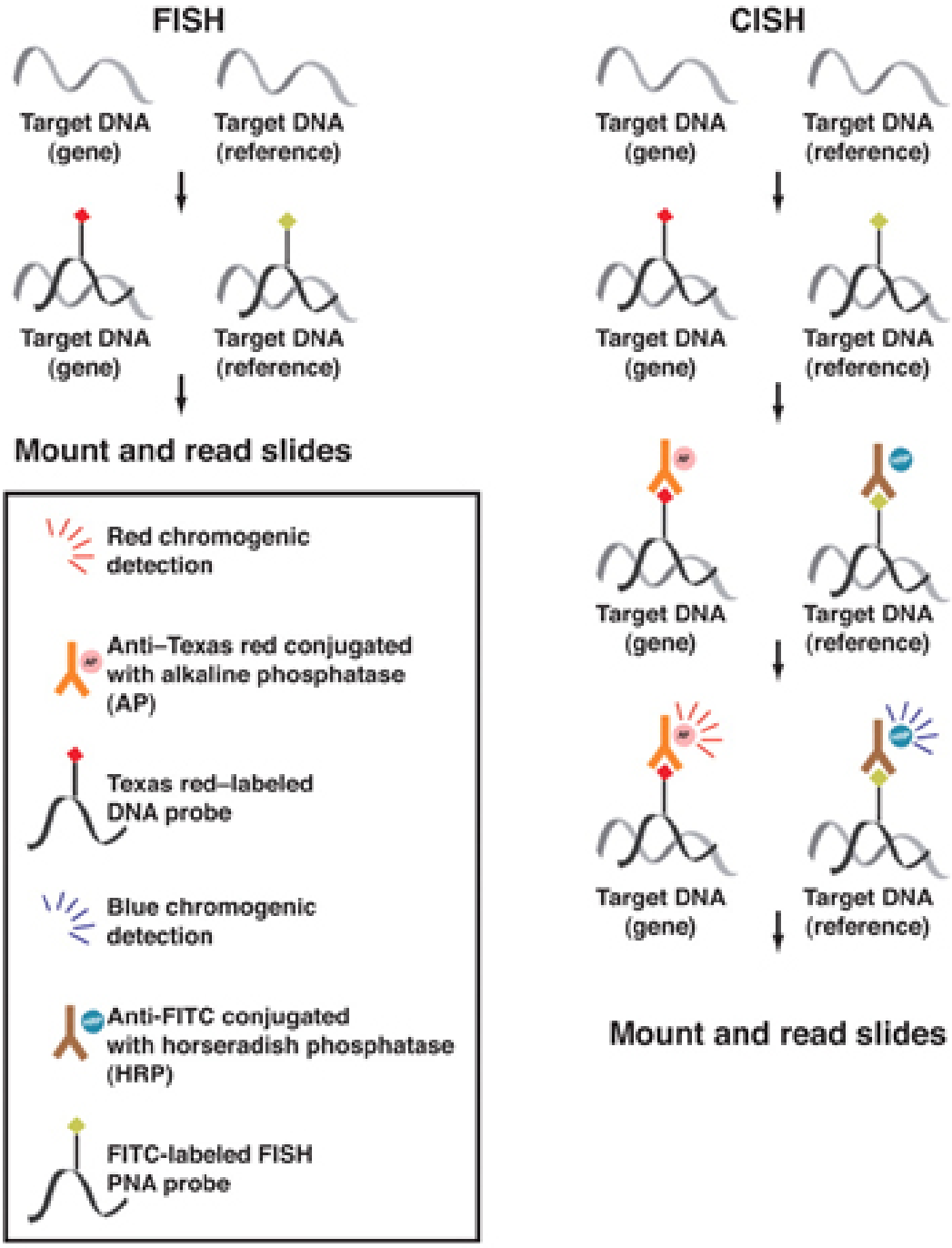
***In Situ* Hybridization (ISH)**

It is a technique that employs a labeled complementary nucleotide strand (i.e. probe) for localizing specific DNA or RNA sequence targets within fixed tissues and cells (i.e *in situ*). Probes used for hybridization can be double-stranded DNA probes, single-stranded DNA probes, RNA probes, synthetic oligonucleotides. There are two ways available to detect DNA or RNA targets

Chromogenic (**CISH**) **in situ** hybridization and Fluorescence *in situ* hybridization (**FISH**).

Chromogenic in situ hybridization (CISH)

It uses the labeling reactions involving alkaline phosphatase or peroxidase reactions to visualize the sample using bright-field microscopy. It is primarily used in molecular pathology diagnostics. CISH can also be employed for samples like fixed cells or tissues, blood or bone marrow smears and metaphase chromosome spreads.



Fluorescence *in situ* hybridization (FISH)

FISH is a cytogenetic technique that uses fluorescent probes that bind to complementary targets and sample is visualized using epi-fluorescence or confocal microscopy. Using differently labeled probes, we can visualize several targets in a single sample. It is used for spatial-temporal patterns of gene expression and resolving genetic elements in chromosomal preparations.

Cells or tissues to be analyzed are fixed and permeabilized with Proteinase K to allow target accessibility. Probe is constructed and tagged using non-radioactive labels like biotin, digoxigenin or fluorescent dye (FISH). Probe must be large enough to hybridize specifically with its target. Probe is applied to fixed sample and incubated to several hours to allow hybridization. Washing is done to remove non-specific or unbound probe removal. Results are then visualized using either bright-field or confocal microscopy.

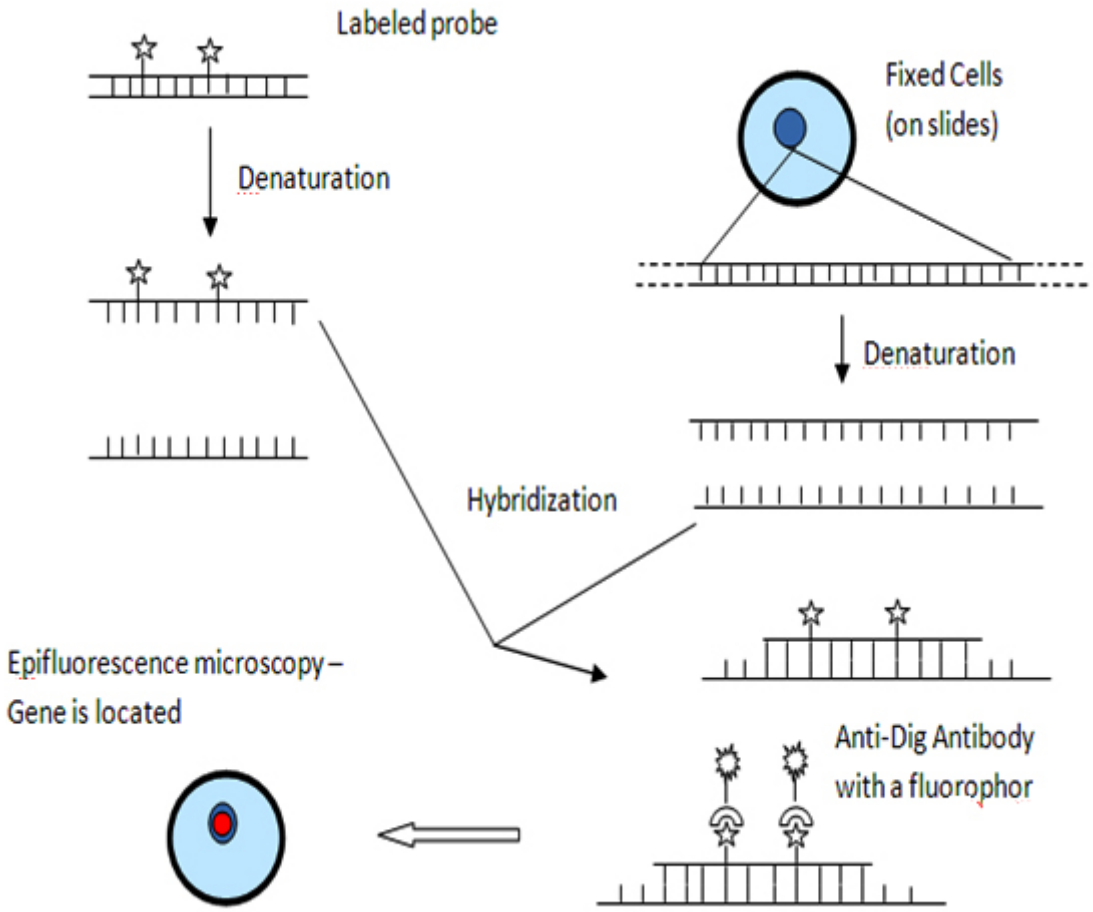


Fig: Fluorescent *in-situ* hybridization

Dot Blot and Slot Blot Hybridization

These two techniques represents the simplification of Southern and Western blots saving the time involved in procedures of chromatography, electrophoresis, restriction digestion and blotting of DNA or proteins from the gel to membrane. Here nucleic acid mixture is directly applied (blotted) on to the nylon or nitrocellulose membrane where hybridization between probe and target takes place, denatured to single-stranded form and baked at 80°C to bind DNA target to membrane. In dot-blot, target is blotted as circular blots whereas in slot-blots, it is in the form of rectangular blots. Due to this, slot-blot offers greater precision in observing different hybridization signals. After blotting, membrane is allowed to dry and non-specific sites are blocked by soaking in blocking buffer containing BSA. It is then followed by hybridization of labeled probe for detection of specific sequences or gene.

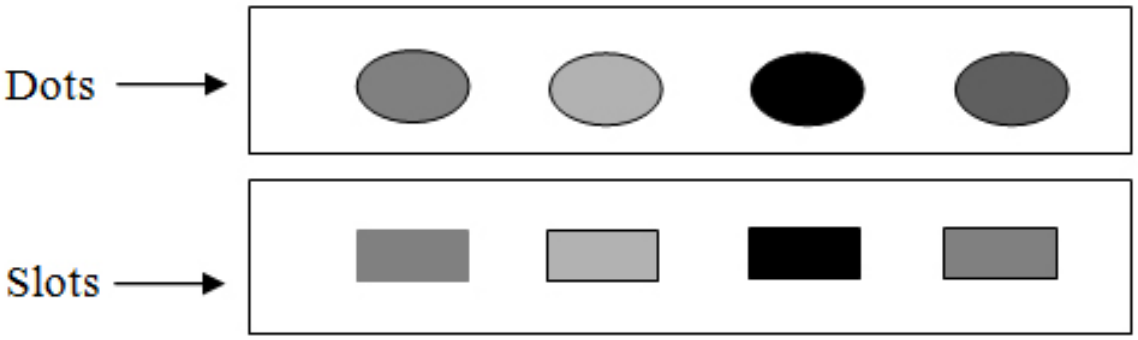


Fig: Dots and slots in dot and slot blot hybridization

These procedures can only detect presence and absence of particular sequence or gene. It cannot distinguish between two molecules of different sizes as they appear as single dot on membrane. It also has application in detecting alleles that differ in single nucleotide with the help of allele-specific oligonucleotides.

Genomic DNA and cDNA Libraries

A DNA library is a collection of DNA fragments that have been cloned into vectors so that researchers can identify and isolate the DNA fragments that interest them for further study. There are basically two kinds of libraries: genomic DNA and cDNA libraries. Genomic DNA libraries contain large fragments of DNA in either bacteriophages or bacterial or P1-derived artificial chromosomes (BACs and. PACs). cDNA libraries are made with cloned, reverse-transcribed mRNA, and therefore lack DNA sequences corresponding to genomic regions that are not expressed, such as introns and 5′ and 3′ noncoding regions. cDNA libraries generally contain much smaller fragments than genomic DNA libraries, and are usually cloned into plasmid vectors.

Preparation of Genomic Library-A genomic library represents complete genome in multiple clones containing small DNA fragments. Depending upon organism and size of genome, this library is either prepared in a bacterial vector or in yeast artificial chromosome (YAC). An outline of the construction of genomic library has following steps:

1. Isolation of genomic DNA
2. Generation of suitable size DNAfragments
3. Cloning in suitable vector system (depending on size)
4. Transformation in suitable host.

Principle of cDNA Library:

In the case of cDNA libraries we produce DNA copies of the RNA sequences (usually the mRNA) of an organism and clone them. It is called a cDNA library because all the DNA in this library is complementary to mRNA and are produced by the reverse transcription of the latter.

Much of eukaryotic DNA consists of repetitive sequences that are not transcribed into mRNA and the sequences are not represented in a cDNA library. It must be noted that prokaryotes and lower eukaryotes do not contain introns, and preparation of cDNA is generally unnecessary for these organisms. Hence, cDNA libraries are produced only from higher eukaryotes.

Vectors used in the Construction of cDNA Library:

Both the bacterial and bacteriophage DNA are used as vectors in the construction of cDNA library.

The following table give a detailed information:

Table 6.2: Vectors used in the construction of cDNA library

Vectors	Insert size	Remarks
λ-phages	Up to 20-30kb (for replacement vectors) and 10-15kb (for insertion vectors)	• Maximum size of mRNA is about 8kb. Hence the capicity of DNA insert is not a major conern • Insertion vector system is usually employed • Useful for study of individual genes and their putative functions • Efficient packaging system, easy for gene transfer into *E. coli*, more representative than plasmid libraries, subcloning and subsequent DNA manipulation process are less convenient than plasmid systems
Bacterial plasmids	Up to 10-15kb	• Relatively easy to transform *E. coli* cells although may not be efficient as the λ-phage system for large scale gene transfer • Less representative than λ-phage libraries, subcloning and subsequent DNA manipulation processes are more convenient than the λ-pahge systems.

Procedure in the Construction of cDNA Library:

The steps involved in the construction of a cDNA library are as follows:

1. Extraction of mRNA from the eukaryotic Cell:

Firstly, the mRNA is obtained and purified from the rest of the RNAs. Several methods exist for purifying RNA such as trizol extraction and column purification. Column purification is done by using oligomericdT nucleotide coated resins where only the mRNA having the poly-A tail will bind.

The rest of the RNAs are eluted out. The mRNA is eluted by using eluting buffer and some heat to separate the mRNA strands from oligo-dT.

2. Construction of cDNA from the Extracted mRNA:

There are different strategies for the construction of a cDNA. These are discussed as follows:

(a) The RNase Method:

The principle of this method is that a complementary DNA strand is synthesized using reverse transcriptase to make an RNA: DNA duplex. The RNA strand is then nicked and replaced by DNA. In this method the first step is to anneal a chemically synthesized oligo-dT primer to the 3′ polyA-tail of the RNA.

The primer is typically 10-15 residues long, and it primes (by providing a free 3′ end) the synthesis of the first DNA strand in the presence of reverse transcriptase and deoxyribonucleotides. This leaves an RNA: DNA duplex.

The next step is to replace the RNA strand with a DNA strand. This is done by using RNase H enzyme which removes the RNA from RNA: DNA duplex. The DNA strand thus left behind is then considered as the template and the second DNA strand is synthesized by the action of DNA polymerase II.

(b) The Self-Priming method:

This involved the use of an oligo-dT primer annealing at the polyadenylate tail of the mRNA to prime first DNA strand synthesis against the mRNA. This cDNA thus formed has the tendency to transiently fold back on itself, forming a hairpin loop. This results in the self-priming of the second strand.

After the synthesis of the second DNA strand, this loop must be cleaved with a single-strand-specific nuclease, e.g., SI nuclease, to allow insertion into the cloning vector. This method has a serious disadvantage. The cleavage with SI nuclease results in the loss of a certain amount of sequence at the 5′ end of the clone.

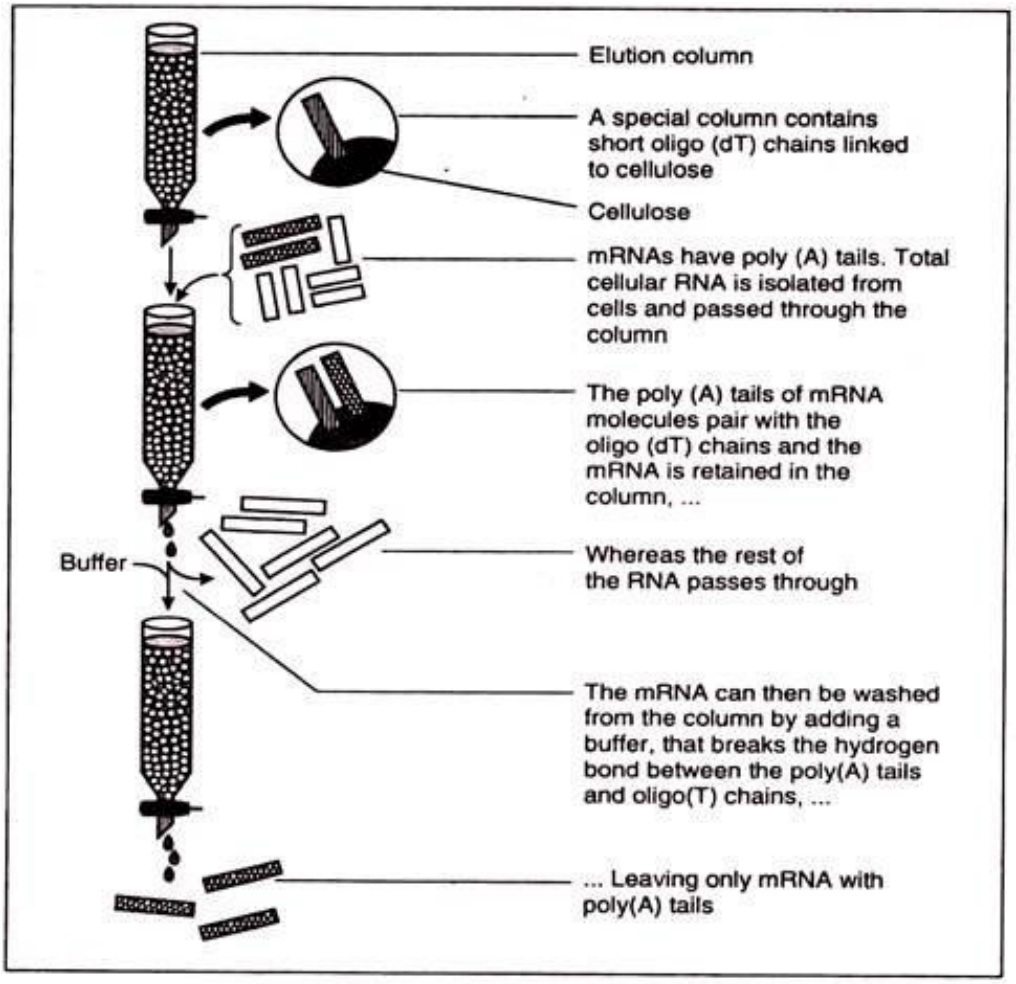


Fig. 6.4: Extraction of mRNA from the eukaryotic cell

(c) Land *et al.* Strategy:

After first-strand synthesis, which is primed with an oligo- dT primer as usual, the cDNA is tailed with a string of cytidine residues using the enzyme terminal transferase. This artificial oligo-dC tail is then used as an annealing site for a synthetic oligo-dG primer, allowing synthesis of the second strand.

(d) Homopolymer Tailing:

This approach uses the enzyme terminal transferase, which can polymerize nucleotides onto the 3′-hydroxyl of both DNA and RNA molecules. We carry out the synthesis of the first DNA strand essentially as before, to produce an RNA: DNA hybrid.

We then use terminal transferase and a single deoxyribonucleotide to add tails of that nucleotide to the 3′ ends of both RNA and DNA strands. The result of this is that the DNA strand now has a known sequence at its 3′ end Typically, dCTP or dATP are used.

A complementary oligomer (synthesized chemically) can now be annealed and used as a primer to direct second strand synthesis. This oligomer (and also the one used for first strand synthesis) may additionally incorporate a restriction site, to help in cloning the resulting double- stranded cDNA.

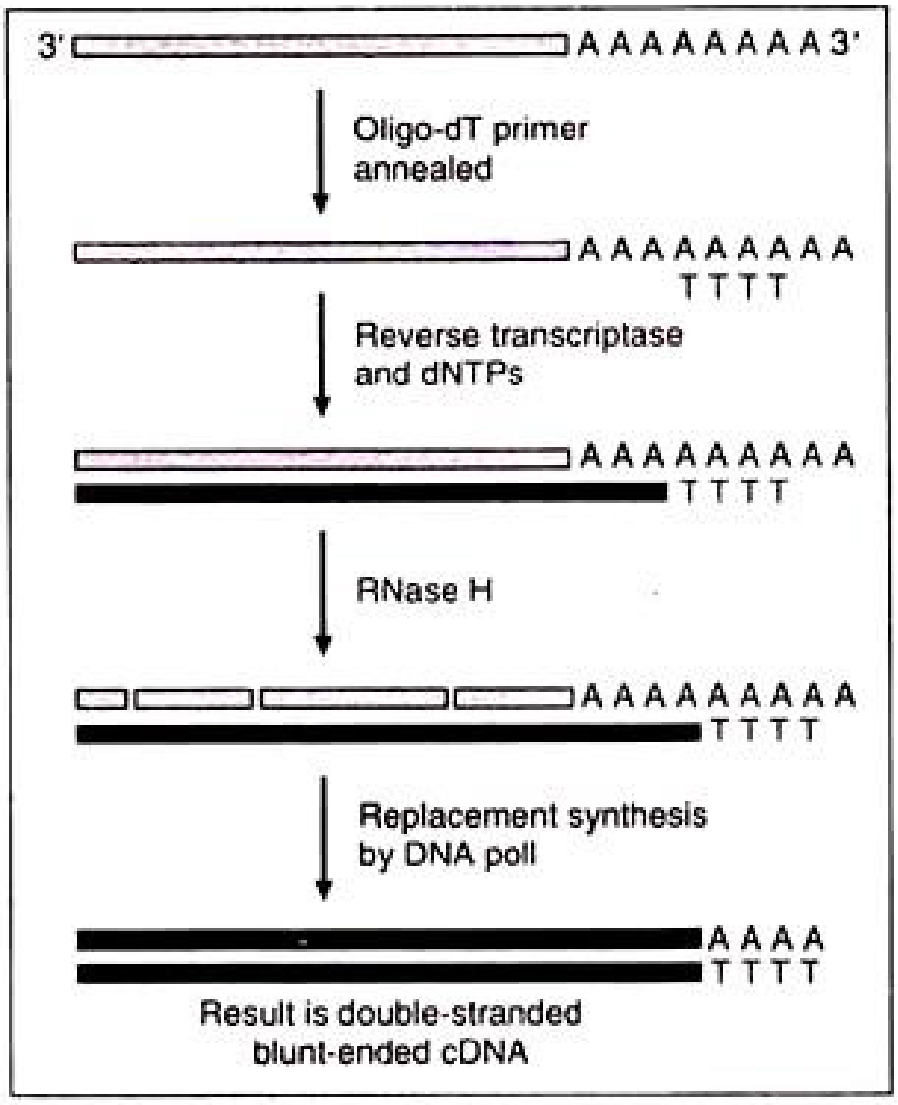


Fig. 6.5: The RNase H method of c-DNA synthesis

(e) Rapid Amplification of cDNA Ends (RACE):

It is sometimes the case that we wish to clone a particular cDNA for which we already have some sequence data, but with particular emphasis on the integrity of the 5′ or 3′ ends. RACE techniques (Rapid Amplification of cDNA Ends) are available for this. The RACE methods are divided into 3′RACE and 5′RACE, according to which end of the cDNA we are interested in.

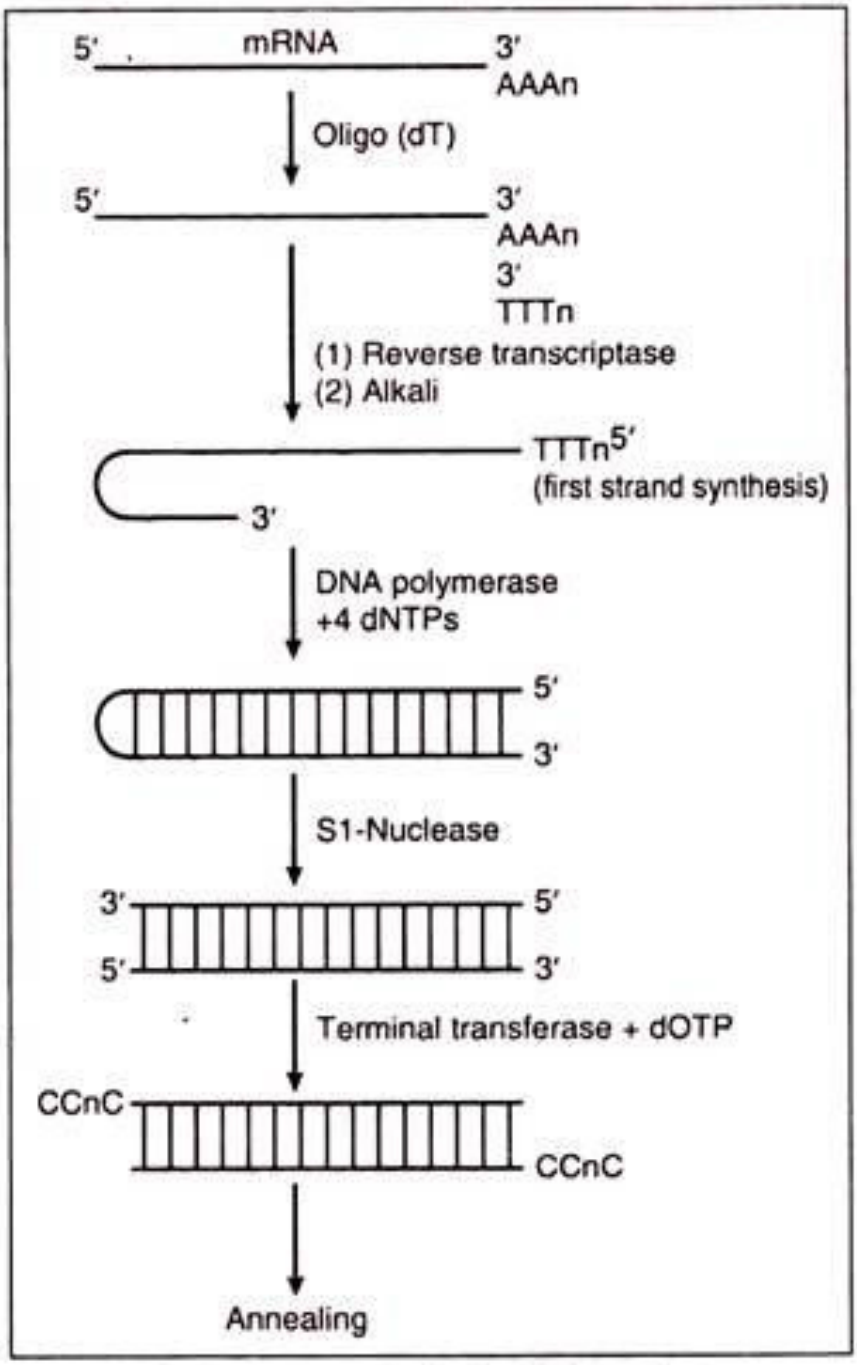


Fig. 6.6: Self-priming method of c-DNA synthesis

(a) 3'RACE:

In this type of RACE, reverse transcriptase synthesis of a first DNA strand is carried out using a modified oligo-dT primer. This primer comprises a stretch of unique adaptor sequence followed by an oligo-dT stretch. The first strand synthesis is followed by a second strand synthesis using a primer internal to the coding sequence of interest.

This is followed by PCR using

(i) The same internal primer and

(ii) The adaptor sequence (i.e., omitting the oligo-dT). Although in theory it should be possible to use a simple oligo- dT primer throughout instead of the adaptor-oligo-dT and adaptor combination, the low melting temperature for an oligo-dT primer may interfere with the subsequent rounds of PCR.

(b) 5′RACE:

In this type of RACE first cDNA strand is synthesized with re- verse transcriptase and a primer from within the coding sequence. Unincorporated primer is removed and the cDNA strands are tailed with oligo-dA. A second cDNA strand is then synthesized with an adaptor-oligo-dT primer.

The resulting double-stranded molecules are then subject to PCR using

(i) A primer nested within the coding region and

(ii) The adaptor sequence. A nested primer is used in the final PCR to improve specificity. The adaptor sequence is used in the PCR because of the low melting temperature of a simple oligo-dT primer, as in 3′RACE above. A number of kits for RACE are commercially available.

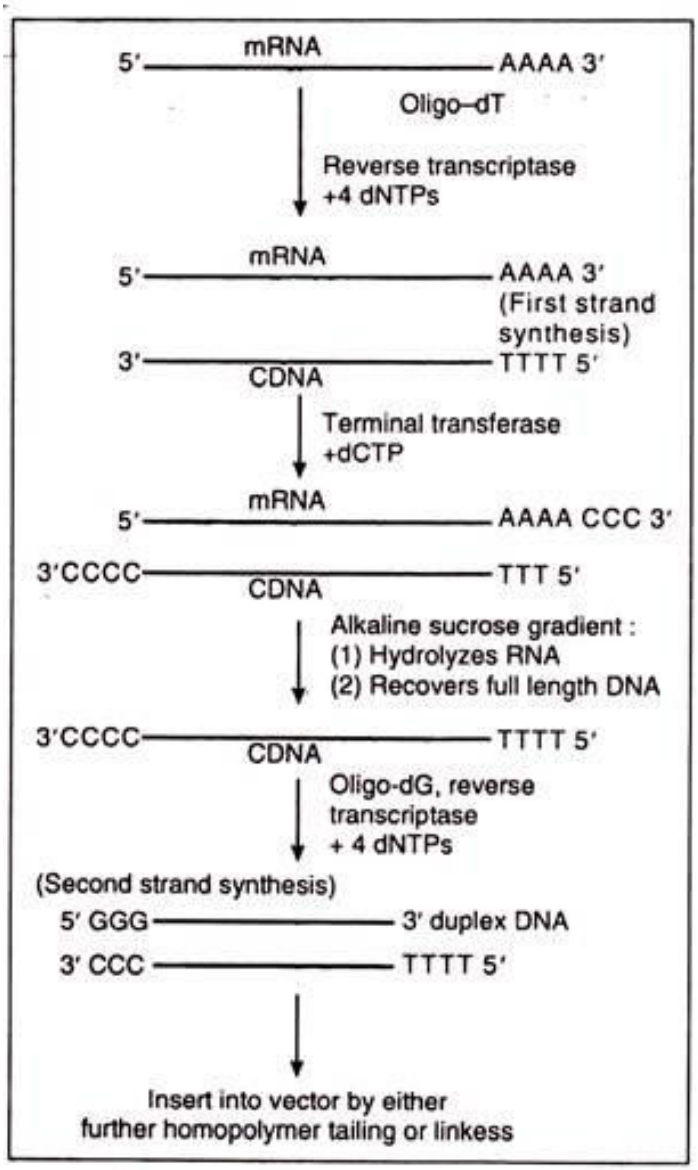


Fig. 6.7: Land et al. strategy

3. Cloning the c-DNA:

(a) Linkers:

The RNaseH and homopolymer tailing methods ultimately generate a collection of double-stranded, blunt-ended cDNA molecules. They must now be attached to the vector molecules. This could be done by blunt-ended ligation, or by the addition of linkers, digestion with the relevant enzyme and ligation into vector.

(b) Incorporation of Restriction Sites:

It is possible to adapt the homopolymer tailing method by using primers that are modified to incorporate restriction. In the diagram shown next page, the oligo-dT primer is modified to contain a restriction site (in the figure, a Sail site GTCGAC).

The 3′ end of the newly synthesized first cDNA strand is tailed with C's. An oligo-dG primer, again preceded by a Sail site within a short double-stranded region of the oligonucleotide, is then used for second-strand synthesis.

Note that this method requires the use of an oligonucleotide containing a double-stranded region. Such oligonucleotides are made by synthesizing the two strands separately and then allowing them to anneal to one another.

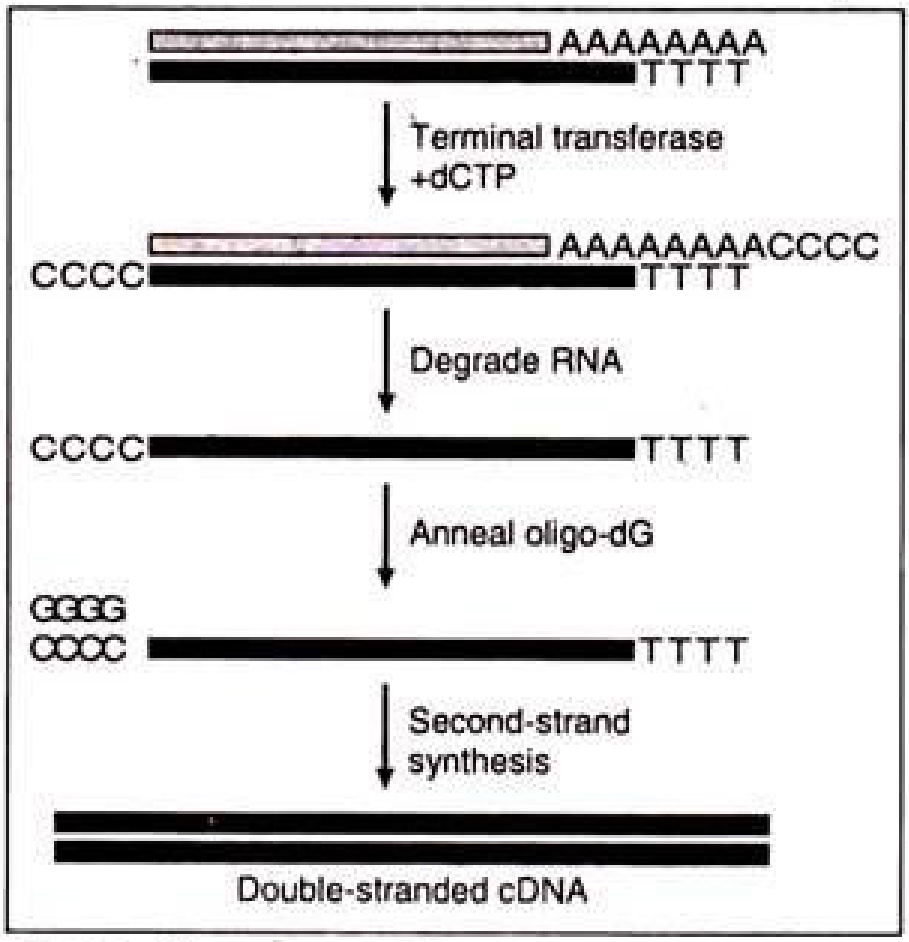


Fig. 6.8: Homopolymer tailing

(c) Homopolymer Tailing of cDNA:

Another option is to use terminal transferase again. Treatment of the blunt-ended double-stranded cDNA with terminal transferase and dCTP leads to the polymerization of several C residues (typically 20 or so) to the 3′ hydroxyl at each end.

Treatment of the vector with terminal transferase and dGTP leads to the incorporation of several G residues onto the ends of the vector. (Alternatively, dATP and dTTP can be used.) The vector and cDNA can now anneal, and the base-paired region is often so extensive that treatment with DNA ligase is unnecessary.

In fact, there may be gaps rather than nicks at the vector insert boundaries, but these are repaired by physiological processes once the recombinant molecules have been introduced into a host.

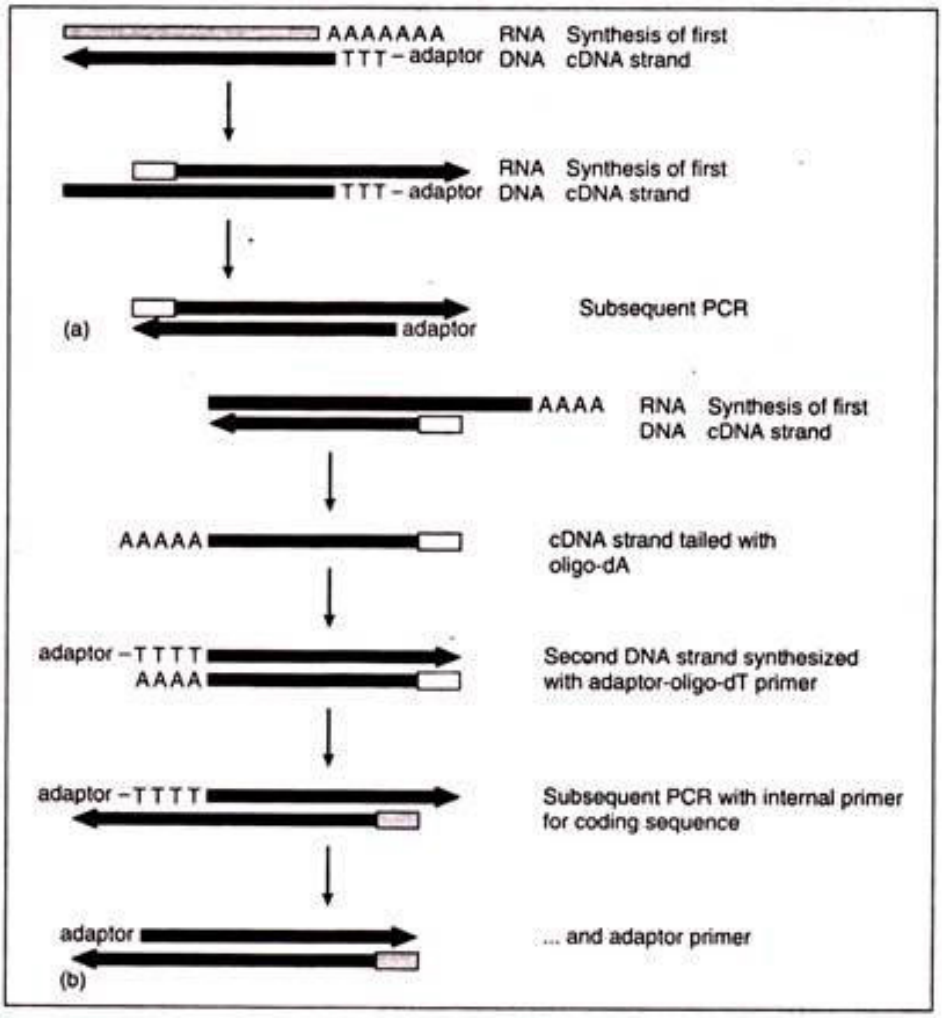


Fig. 6.9: RACE. (a) 3'RACE. The first primer is the oligo-dT-adaptor molecule. The second primer (open box) is internal to the coding sequence of interest. This is used in conjunction with the adaptor primer (rather than the oligo-dT-adaptor primer) in subsequent PCR. (b) 5'RACE. Synthesis of the first cDNA strand uses a primer within the coding region (open box). The first cDNA strand is tailed with oligo-dA. A second DNA strand is synthesized with an adaptor-oligo-dT primer. This is followed by PCR with (i) a primer nested within the coding sequence (shaded box) and (ii) the adaptor.

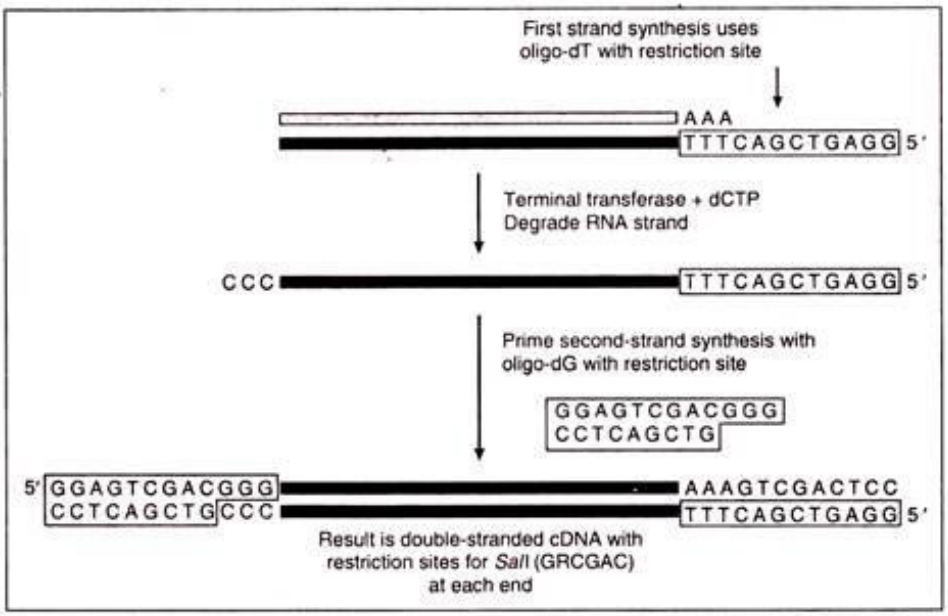


Fig. 6.10: Modification of homopolymer tailing, incorporating restriction sites

Advantages of cDNA Library:

A cDNA library has two additional advantages. First, it is enriched with fragments from actively transcribed genes. Second, introns do not interrupt the cloned sequences; introns would pose a problem when the goal is to produce a eukaryotic protein in bacteria, because most bacteria have no means of removing the introns.

Disadvantages of cDNA Library:

The disadvantage of a cDNA library is that it contains only sequences that are present in mature mRNA. Introns and any other sequences that are altered after transcription are not present; sequences, such as promoters and enhancers, that are not transcribed into RNA also are not present in a cDNA library.

It is also important to note that the cDNA library represents only those gene sequences expressed in the tissue from which the RNA was isolated. Furthermore, the frequency of a particular DNA sequence in a cDNA library depends on the abundance of the corresponding mRNA in the given tissue. In contrast, almost all genes are present at the same frequency in a genomic DNA library.

Applications of cDNA Library:

Following are the applications of cDNA libraries:

1. Discovery of novel genes.
2. Cloning of full-length cDNA molecules for in vitro study of gene function.
3. Study of the repertoire of mRNAs expressed in different cells or tissues.
4. Study of alternative splicing in different cells or tissues.

DNA FINGERPRINTING

Alec Jeffreys (1984) invented the DNA fingerprinting technique at Leicester University, United Kingdom. Dr. V.K. Kashyap and Dr. Lalji Singh started the DNA fingerprinting technology in India at CCMB (Centre for Cell and Molecular Biology) Hyderabad.

The DNA profiling of each individual is unique because of the diverse in polymorphic regions present in genome of every individual. These polymorphic regions used for identification are the non-coding regions of the genome. The polymorphic regions of the DNA do not code for proteins and which make-up 95% of our genetic DNA. Hence these regions are therefore called the —junk DNA‖. Although these —junk DNA‖ regions do not code for proteins, they are involved in regulating gene expression, they help in reading of other genes that code for protein and are a large portion of the chromosome structure. The junk DNA regions are made-up of length polymorphisms, which show variations in the physical length of the DNA molecule. In DNA profile the length of the polymorphisms in the non-coding areas is measured as it varies with each individual. These polymorphisms are identical repeat sequences that are present in non- coding DNA region. At specific loci on the chromosome the number of tandem repeats varies between individuals. There will be a certain number of repeats for any specific loci on the chromosome. Depending on the size of the repeat, the repeat regions are classified into two groups. Short tandem repeats (STRs) contain 2-5 base pair repeats and variable number of tandem repeats (VNTRs) have repeats of 9-80 base pairs.

We inherit a copy of chromosome, one from father and one from mother for each of the 23 pairs of chromosomes, which indicate that we carry two copies of each VNTR locus, just like we have two copies of genes donated by our parents. At a particular VNTR site if you have the same number of sequence repeats, you are called homozygous at that site; if you have a different number of repeats, you are said to be heterozygous‖. VNTR sequences from different loci can be combined to create DNA fingerprint. Resulting patter of each individual is theoretically unique.

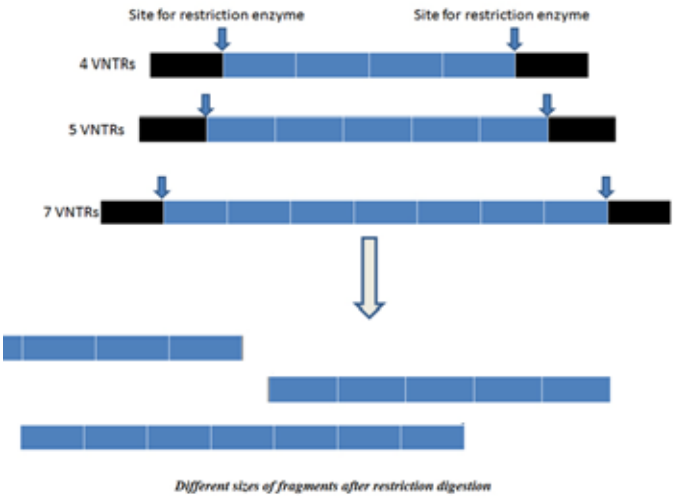
The diagnosis of DNA is a laboratory process which requires a number of sequential steps. There are many techniques used by various laboratories in analyzing DNA. However, the two most popular techniques used are Restriction fragment length polymorphism (RFLP), and polymerase chain reaction (PCR) using short tandem repeats (STRs). DNA extraction and purification is the first step involved in both the techniques. Techniques for DNA isolation depend on the type of sample used for isolation. Generally the sample may be treated with detergent that degrades protein and not affecting the DNA. To break the proteins and other cellular material, enzymes can be used. DNA can be separated using organic solvents form organic and non-organic material. By using the above methods the DNA is separated from thesample.

The two types of DNA fingerprinting tests: RFLP and PCR/STR

Restriction fragment length polymorphism (RFLP) and polymerase chain reaction (PCR) amplification of short tandem repeats (STRs) are two main DNA tests widely used for DNA fingerprinting. Other diagnostic methods exist, but they lack accuracy and precision.

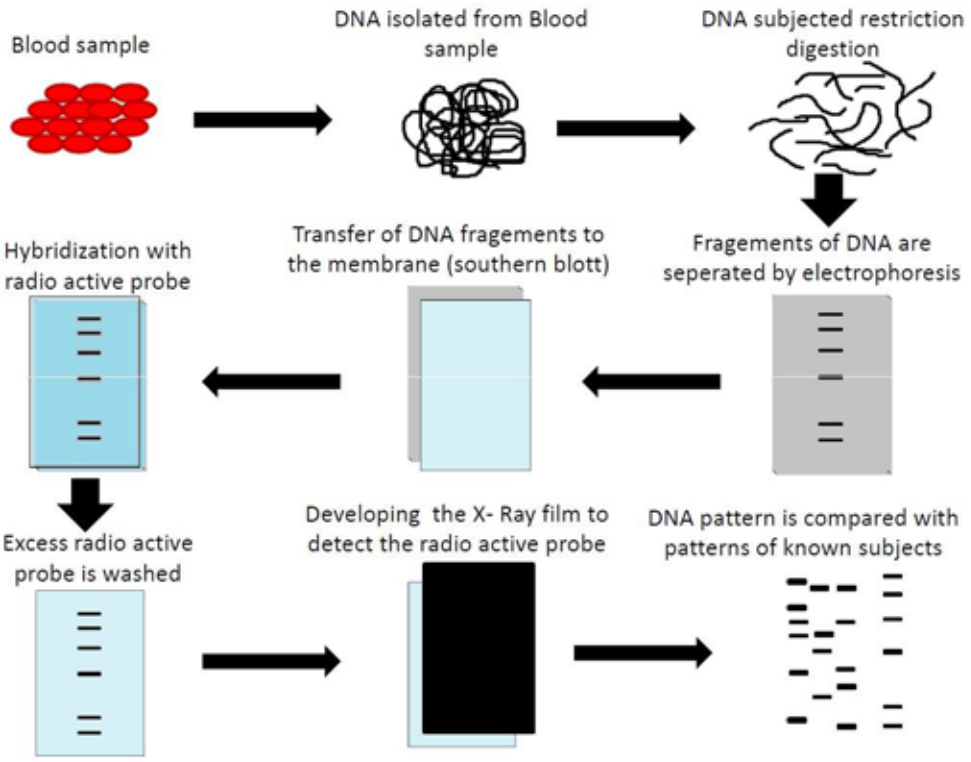
Restriction fragment length polymorphism (RFLP):

The RFLP is considered to be more accurate than the PCR, mainly because the size of the sample used more, use of a fresh DNA sample, and no amplification contamination. The RFLP, however, require longer time period to complete the analysis and is costly. The first step in this process is to isolate the DNA from the sample material to be tested. As mentioned, the sample size for RFLP test must be large enough to get the proper result. Once the required size of the sample is available, the DNA is isolated from the sample and is subjected to restriction digestion usingrestrictionenzymes. ThedigestedDNAsampleisthenseparatedbyagarosegel electrophoresis, in which the DNA is separated based on the size. The next step is transfer of separated DNA from gel slab onto the nitrocellulose membrane to hybridize with a labeled probe that is specific for one VNTR region (radio activity labeled complimentary sequence for VNTR region nucleotide sequence). This technique of transferring and hybridizing DNA onto nitrocellulose membrane is known as southern blotting, a most widely used DNA detection technique by molecular biologists. After the hybridization with the radioactive probes, the X- ray film is developed form the southern blotting and only the areas where the radioactive probe binds will show up on the film. Now these bands when compared with the other known samples, will give the final result of the DNA fingerprinting.



Identification of band using VNTR probe (Radio activity labeled nucleotide sequence complimentary to VNTR)

Note: **VNTR sequences from different loci can be combined to create DNA fingerprint. Resulting patter of each individual is theoretically unique.**



Restriction fragment length polymorphism (RFLP): a diagrammatic representation of sequential steps involved in DNA finger printing using RFLP.

Polymerase chain reaction (PCR) amplification of short tandem repeats (STRs)

Karymullis in the year 1991 developed a technique to amplify a DNA. PCR generates the repeated copies of a specific area of the DNA fragment. These areas are the alleles and are specific sequences of base pairs. They are the target regions with variable length. On either side of the target regions there are –flanker‖ sequences which are non-variable regions. On chromosomes –flanker‖ occurs at same location for all individuals. Thousands ofcopies of a particular variable region are amplified by PCR which forms the basis of this detection. STR with a known repeat sequence is amplified and separated using gel-electrophoresis. The distance migrated by the STR is examined. For the amplification of STRs using PCR, a short synthetic DNA, called primers are specially designed to attach to a highly conserved common non- variable region of DNA that flanks the variable region of the DNA. By comparing the STR sequence size amplified by PCR with the other known samples, will give the final result of the DNA fingerprinting.

Applications of DNA Fingerprinting:

1. **In Forensics Science**: DNA Fingerprinting and Forensics Forensic science can be defined as the intersection of law and science. The DNA profile of each individual is highly specific. The chances of two people having the exact DNA profile are 30,000 million to 1 (except for identical twins). Biological materials used for DNA profiling are: Blood, Hair, Saliva, Semen, Body tissue cells etc.

2. DNA isolated from the evidence sample can be compared through VNTR (Variable number of tandem repeats) prototype. It is useful in solving crimes like murder and rape. For example; the sex scandal of President Clinton with Monica Lewinsky, Double murders of O.J. Simpson in 1995. Colin Pitchfork was the first criminal caught based on DNA fingerprinting evidence. He was arrested in 1986 for the rape and murder of two girls and was sentenced in 1988. O.J. Simpson was cleared of a double murder charge in 1994, which relied heavily on DNA evidence.

3. **Paternity and Maternity Determination:** A Person accedes to his or her VNTRs from his or her parents. Parent-child VNTR prototype analysis has been used to solve disputed cases. This information can also be used in inheritance cases, immigration cases. For Example: In 2002 Elizabeth Hurley used DNA profiling to prove that Steve Bing was the father of her child Damien.

4. **Personal Identification**: The concept of using DNA fingerprints as a sort of genetic bar code to pinpoint individuals has already been discussed above.

5. **Diagnosis of Inherited Disorders:** It is also useful in diagnosing inherited disorders in both prenatal and newborn babies. These disorders may include cystic fibrosis, hemophilia, Huntington's disease, familial Alzheimer's, sickle cell anemia, thalassemia, and many others.

6. **Development of Cures for Inherited Disorders:** By studying the DNA fingerprints of relatives who have a history of some particular disorder, DNA prototypes associated with the disease can be ascertained. The Hong Kong Baptist University was able to use DNA fingerprinting to identify the Chinese medicine—Lingzhi in 2000.

7. **Breeding Program:** Breeders conventionally use the phenotype to evaluate the genotype of a plant or an animal. For example, homozygous dominant genotype AABB is always desirable. As it is difficult to make out homozygous or heterozygous dominance from appearance, the DNA fingerprinting allows a fastidious and precise determination of genotype. Offspring from the discerning mating of superior animals are expected to inherit desirable characters like strong cardiopulmonary capacity and speed. It is basically useful in breeding race horses and hunting dogs.

(iii) Human Lineage:

DNA from various probables is being studied to find out human lineage,

(v) Forensics:

DNA finger printing is very useful in the detection of crime and legal pursuits. DNA fingerprinting has proved that Dhanu, the human bomb, was the real murderer of Shri Rajiv Gandhi, the former Prime Minister of India,

(vi) Sociology:

It can identify racial groups, their origin, historical migration and invasions. Genography is the study of migratory history of human species.

DNA SEQUENCING

The term DNA sequencing refers to methods for determining the order of the nucleotides bases adenine,guanine,cytosine and thymine in a molecule of DNA. The first DNA sequence were obtained by academic researchers,using laboratories methods based on 2- dimensional chromatography in the early 1970s. By the development of dye based sequencing method with automated analysis,DNA sequencing has become easier and faster. The knowledge of DNA sequences of genes and other parts of the genome of organisms has become indispensable for basic research studying biological processes, as well as in applied fields such as diagnostic or forensic research.

DNA is the information store that ultimately dictates the structure of every gene product, delineates every part of the organisms. The order of the bases along DNA contains the complete set of instructions that make up the genetic inheritance.

DNA can be sequenced by a chemical procedure that breaks a terminally labelled DNA molecule partially at each repetition of a base. The length of the labelled fragments then identify the position of that base. We describe reactions that cleave DNA preferentially at guanines,atadenines,at cytosine and thymines equally, and at cytosine alone. When the product of these four reactions are resolved by size,by electrophoresis on a polyacrylamide gel, the DNA sequences can be read from the pattern of radioactive bands. The technique will permit sequencing of atleast 100 bases from the point of labelling. The purine specific reagent is dimethyl sulphate; and the pyrimidine specific reagent is hydrazine.

In 1973 , Gilbert and Maxam reported the sequence of 24 base pairs using a method known as wandering- spot analysis.

The chain termination method developed by Sanger and coworkers in 1975 owing to its relative easy and reliability.

In 1975 the first complete DNA genome to be sequenced is that of bacteriophage ϕX174.

By knowing the DNA sequence, the cause of the various diseases can be known. We can determine the sequence responsible for various disease and can be treated with the help of Gene therapy.

DNA sequencing is very significant in research and forensic science. The main objective of DNA sequence generation method is to evaluate the sequencing with very high accuracy and reliability.

There are some common automated DNA sequencing problems :-

1. Failure of the DNA sequence reaction.
2. Mixed signal in the trace (multiple peaks).

3. Short read lengths and poor quality data.
4. Excessive free dye peaks "dye blobs" in the trace.
5. Primer dimer formation in sequence reaction
6. DNA polymerase slippage on the template mononucleotide regions.

So, we should have to do the sequencing in such a manner to avoid or minimize these problems. DNA sequencing can solve a lot of problems and perform a lot of work for human wellfare A sequencing can be done by different methods :

1. Maxam – Gilbert sequencing
2. Chain-termination methods
3. Dye-terminator sequencing
4. Automation and sample preperation
5. Large scale sequencing strategies
6. New sequencing methods.

2. Maxam-Gilbert sequencing

In 1976-1977, Allan Maxam and Walter Gilbert developed a DNA sequencing method based on chemical modification of DNA and subsequent cleavage at specific bases.

The method requires radioactive labelling at one end and purification of the DNA fragment to be sequenced. Chemical treatment generates breaks at a small proportions of one or two of the four nucleotide based in each of four reactions (G,A+G, C, C+T). Thus a series of labelled fragments is generated,from the radiolabelled end to the first 'cut' site in each molecule. The fragments in the four reactions are arranged side by side ingel electrophoresis for size separation. To visualize the fragments,the gel is exposed to X-ray film for autoradiography, yielding a series of dark bands each corresponding to a radiolabelled DNA fragment, from which the sequence may be inferred.

3. Chain-terminationmethod

The chain terminator method is more efficient and uses fewer toxic chemicals and lower amount of radioactivity than the method of Maxam and Gilbert.

The key principle of the Sanger method was the use of dideoxynucleotide triphosphates (ddNTPs) as DNA chain terminators.

The chain termination method requires a single-stranded DNA template, a DNA primer, a DNA polymerase, radioactively or fluorescently labelled nucleotides, and modified nucleotides that terminate DNA strand elongation. The DNA sample is divided into four separate sequencing reactions, containing all four of the standard deoxynucleotides (dATP, dGTP, dCTP, dTTP) and the DNA polymerase. To each reaction is added only one of the four dideoxynucleotide (ddATP, ddGTP, ddCTP, ddTTP) which are the chain terminating nucleotides, lacking a 3'-OH group required for the formation of a phosphodiester bond between two nucleotides, thus terminating DNA strand extension and resulting in DNA fragments of varying length.

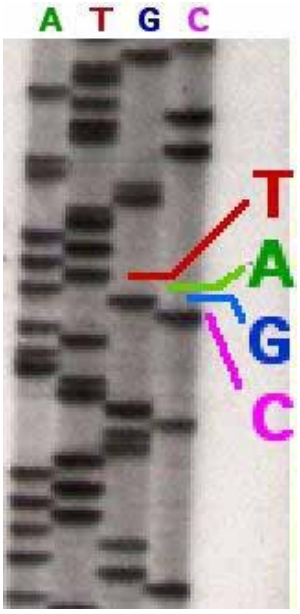


Fig. Part of a radioactively labelled sequencing gel

The newly synthesized and labelled DNA fragments are heat denatured , and separated by size by gel electrophoresis on a denaturing polyacrylamide-urea gel with each of the four reactions run in one of the four individual lanes (lanes A, T, G,C), the DNA bands are then visualized by autoradiography or UV light,and the DNA sequence can be directly read off the X-ray film or gel image. A dark band in a lane indicates a DNA fragment that is result of chain termination after incorporation of a dideoxynucleotide (ddATP, ddGTP, ddCTP, or ddTTP). The relative position of the different bands among the four lanes are then used to read (from bottom to top) the DNA sequence.

The technical variations of chain termination sequencing include tagging with nucleotides containing radioactive phosphorus for labelling, or using a primer labelled at the 5′ end with a fluorescent dye. Dye- primer sequencing facilitates reading in an optical system for faster and more economical analysis and automation.

Chain termination methods have greatly simplified DNA sequencing. Limitations include non-specific binding of the primer to the DNA, affecting accurate read-out of the DNA sequence, and DNA secondary structures affecting the fidelity of the sequence.

3.1 Dye-terminator sequencing

Dye-terminator sequencing utilizes labelling of the chain terminator ddNTPs,which permits sequencing in a single reaction,rather than four reactions as in the labelled- primer method. In dye-terminator sequencing ,each of the four dideoxynucleotide chain terminators is labelled with fluorescent dyes,each of which with different wavelengths of fluorescence and emission. Owing to its greater expediency and speed,dye terminator sequencing is now the mainstay in automated sequencing. Its limitation include dye effects due to differences in the incorporation of the dye-labelled chain terminators into the DNA fragment,resulting in unequal peak heights and shapes in the electronic DNA sequence trace chromatogram.

The common challenges of DNA sequencing include poor quality in the first 15-40 bases of the sequence and deteriorating quality of sequencing traces after 700-900 bases.

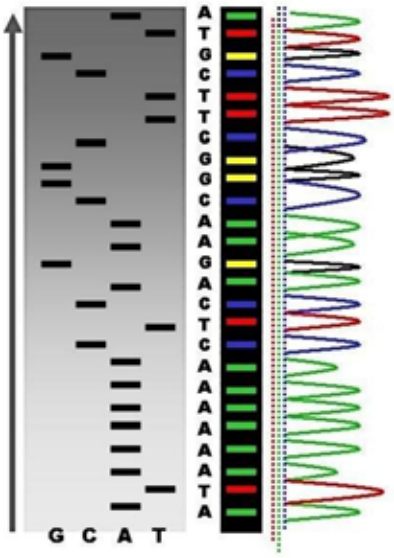


Fig. Sequence ladder by radioactive sequencing compared to fluorescent peaks

3.2 Automation and sample preparation

Automated DNA sequencing instruments (DNA sequencers) can sequence upto 384 DNA samples in a single batch (run) in up to 24 runs a day. DNA sequencers carry out capillary electrophoresis for size seperation,detection and recording of dye fluorescence,and data output as fluorescent peak trace chromatograms.

A number of commercial and non-commercial software packages can trim low-quality DNA traces automatically. These programmes score the quality of each peak and remove low- quality base peaks (generally located at the ends of the sequence).

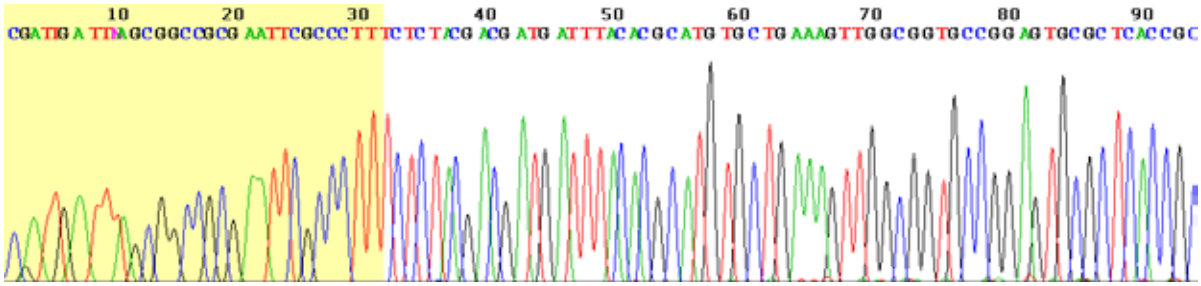


Fig. View of the start of an example dye-terminator read

4. Large-scale sequencing strategies

Current methods can directly sequence only relative short (300-1000 nucleotides long) DNA fragments in a single reaction. The main obstacle to sequencing DNA fragments above this size limit is insufficient power of separation for resolving large DNA fragments that differ in length by only one nucleotide.

Large scale sequencing aims at sequencing very long DNA pieces, such as whole chromosomes. It consist of cutting (with restriction enzymes) or shearing (with mechanical forces) large DNA fragments into shorter DNA fragments. The fragmented DNA is cloned into a DNA vector, and amplified in *E.coli.* Short DNA fragments purified from individual bacterial colonies are individually sequenced and assembled electronically into one long, contiguous sequence. This method does not require any pre- existing information about the sequence of the DNA and is reffered to as de novo sequencing. Gaps in the assembled sequence may be filled by primer walking. The different strategies have different

tradeoffs in speed and accuracy.

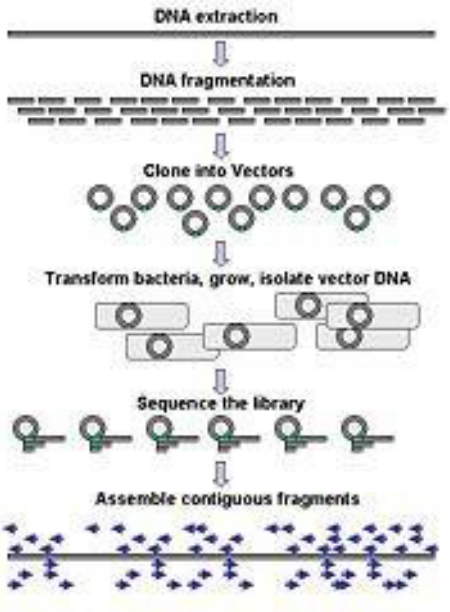


Fig. Genomic DNA is fragmented into random pieces and cloned as a bacterial library. DNA from individual bacterial clones is sequenced and the sequence is assembled by using overlapping regions.

2. New sequencing methods

The high demand for low-cost sequencing has driven the development of high- throughput sequencing technologies that parallelize the sequencing process,producing thousands or millions of sequences at once. High-throughput sequencing technologies are intended to lower the cost of DNA sequencing .

Molecular detection method are not sensitive enough for single molecule sequencing, so most approaches use an in vitro cloning step to amplify individual DNA molecules.

In microfluidic Sanger sequencing the entire thermocycling amplification of DNA fragments as well as their separation by electrophoresis is done on a single chip (appoximately 100cm in diameter) thus reducing the reagent usage as well as cost. In some instances researchers have shown that they can increase the throughput of conventional sequencing through the use of microchips.

3. High throughput sequencing

The high demand for low-cost sequencing has driven the development of high-throughput sequencing technologies that parallelize the sequencing process, producing thousands or millions of sequences at once. High- throughput sequencing technologies are intended to lower the cost of DNA sequencing beyond what is possible with standard dye-terminator methods.

3.1 Lynx therapeutics' massively parallel signature sequencing (MPSS)

The first of the "next-generation" sequencing technologies, MPSS was developed in the 1990s at Lynx Therapeutics, a company founded in 1992 by Sydney Brenner and Sam Eletr. MPSS is an ultra high throughput sequencing technology. When applied to expression profile, it reveal almost every transcript in the sample and provide its accurate expression level. MPSS was a bead-based method that used a complex approach of adapter ligation followed by adapter decoding, reading the sequence in increments of four nucleotides; this method made it susceptible to sequence-specific bias or loss of specific sequences. However, the essential properties of the MPSS output were typical of later "next-gen" data types, including hundreds of thousands of short DNA sequences. In the case of MPSS, these were typically used for sequencing cDNA for measurements of gene expression levels. Lynx Therapeutics merged with Solexa in 2004, and this company was later purchased by Illumina.

3.2 Polony sequencing

It is an inexpensive but highly accurate multiplex sequencing technique that can be used to read millions of immobilized DNA sequences in parallel. This techniques was first developed by Dr. George Church in Harvard Medical college. It combined an *in vitro* paired-tag library with emulsion PCR, an automated microscope, and ligation-based sequencing chemistry to sequence an *E. coli* genome at an accuracy of > 99.9999% and a cost approximately 1/10 that of Sanger sequencing.

3.3 Pyrosequencing

A parallelized version of pyrosequencing was developed by 454 Life Sciences, which has since been acquired by Roche Diagnostics. The method amplifies DNA inside water droplets in an oil solution (emulsion PCR), with each droplet containing a single DNA template attached to a single primer-coated bead that then forms a clonal colony. The sequencing machine contains many picolitre-volume wells each containing a single bead and sequencing enzymes. Pyrosequencing uses luciferase to generate light for detection of the individual nucleotides added to the nascent DNA, and the combined data are usedto generate sequence read-outs.This technology provides intermediate read length and price per base compared to Sanger sequencing on one end and Solexa and SOLiD on the other.

3.4 Illumina (Solexa) sequencing

Solexa developed a sequencing technology based on dye terminators. In this, DNA molecule are first attached to primers on a slide and amplified, this is known as bridge amplification. Unlike pyrosequencing, the DNA can only be extended one neucleotode at a time. A camera takes images of the fluorescently labeled nucleotides, then the dye along with the terminal 3' blocker is chemically removed from the DNA, allowing the next cycle.

3.5 SOLiD sequencing

The technology for sequencing used in ABISolid sequencing is oligonucleotide ligation and detection. In this, a pool of all possible oligonucleotides of fixed length arelabelled according to the sequenced position. This sequencing results to the sequences of quantities and lengths comparable to illumine sequencing.

3.6 DNA nanoball sequencing

It is high throughput sequencing technology that is used to determine the entire genomic sequence of an organisms. The method uses rolling circle replication to amplify fragments of genomic DNA molecules. This DNA sequencing allows large number of DNA nanoballs to be sequenced per run and at low reagent cost compared to other next generation sequencing platforms. However, only short sequences of DNA are determined from each DNA nanoball which makes mapping the short reads to a reference genome difficult. This technology has been used for multiple genome sequencing projects and is scheduled to be used for more.

3.7 Helioscope(TM) single molecule sequencing

Helioscope sequencing uses DNA fragments with added polyA tail adapters, which are attached to the flow cell surface. The next steps involve extension-based sequencing with cyclic washes of the flow cell with fluorescently labeled nucleotides. The reads are performed by the Helioscope sequencer. The reads are short, up to 55 bases per run, but recent improvemend of the methodology allowes more accurate reads of homopolymers and RNA sequencing.

3.8 Single molecule SMRT(TM) sequencing

SMRT sequencing is based on the sequencing by synthesis approach. The DNA is synthesisd in so calles zero-mode wave-guides (ZMWs) - small well-like containers with the capturing tools located at the bottom of the well. The sequencing is performed with use of unmodified polymerase and fluorescently labelled nucleotides flowing freely in the solution. The wells are constructed in a way that only the fluorescence occurring by the bottom of the well is detected. The fluorescent label

is detached from the nucleotide at its incorporation into the DNA strand, leaving an unmodified DNA strand. The SMTR technology allows detection of nucleotide modifications. This happens through the observation of polymerase kinetics. This approach allows reads of 1000 nucleotides.

3.9 Single molecule real time (RNAP) sequencing

This method is based on RNA polymerase (RNAP), which is attached to a polystyrene bead, with distal end of sequenced DNA is attached to another bead, with both beads being placed in optical traps. RNAP motion during transcription brings the beads in closer and their relative distance changes, which can then be recorded at a single nucleotide resolution. The sequence is deduced based on the four readouts with lowered concentrations of each of the four nucleotide types.

4. Other sequencing technologies

Sequencing by hybridization is a non-enzymatic method that uses a DNA microarray. A single pool of DNA whose sequence is to be determined is fluorescently labelled and hybridized to an array containing known sequences. Strong hybridization signals from a given spot on the array identifies its sequence in the DNA being sequenced. Mass spectrometry may be used to determine mass differences between DNA fragments produced in chain-termination reactions.

Some important applications of DNA sequencing are :

1. To analyse any protein structure and function we must have the knowledge of its primary structure i.e its DNA sequence.
2. With its study we can understand the function of a specific sequence and the sequence responsible for any disease.
3. With the help of comparative DNA sequence study we can detect any mutation.
4. Kinship study.
5. DNA fingerprinting.
6. By knowing the whole genome sequence, Human genome project get completed.

The main problem with sequencing is its intactness. If we perform the sequencing of same sample with different methods the result may be different so we should have to do it in such a manner that atleast 40-50% sequence must be same of similar sample.

Intellectual Property Rights (IPRs)

Intellectual Property Rights (IPRs) are legal rights that protect creations and/or inventions resulting from intellectual activity in the industrial, scientific, literary or artistic fields. The most common IPRs include patents, copyrights, marks and trade secrets.

Intellectual property is divided into two categories:

Industrial Property includes patents for inventions, trademarks, industrial designs and geographical indications.

Copyright covers literary works (such as novels, poems and plays), films, music, artistic works (e.g., drawings, paintings, photographs and sculptures) and architectural design. Rights related to copyright include those of performing artists in their performances, producers of phonograms in their recordings, and broadcasters in their radio and television programs.

Patent

A patent is an exclusive rightgranted for an invention –a product or process that providesa new way of doing something,or that offers a new technicalsolution to a problem.A patent provides patent ownerswith protection for their inventions.Protection is granted for a limitedperiod, generally 20 years.

Trademarks

A trademark is a distinctive signthat identifies certain goods orservices produced or provided byan individual or a company. Itsorigin dates back to ancient timeswhen craftsmen reproduced theirsignatures, or "marks", on theirartistic works or products of afunctional or practical nature. Over the years, these marks haveevolved into today's system oftrademark registration andprotection. The system helpsconsumers to identify andpurchase a product or servicebased on whether its specificcharacteristics and quality – asindicated by its unique trademark– meet their needs.

Industrial design

An industrial design refers to theornamental or aesthetic aspects ofan article. A design may consist of three-dimensional features, such asthe shape or surface of an article,or two-dimensional features, suchas patterns, lines or color.Industrial designs are applied to awide variety of industrial productsand handicrafts: from technicaland medical instruments towatches, jewelry and other luxuryitems; from house wares andelectrical appliances to vehiclesand architectural structures; fromtextile designs to leisure goods.

Geographical indication

A geographical indication is a signused on goods that have a specificgeographical origin and possessqualities or a reputation dueto that place of origin. Mostcommonly, a geographicalindication consists of the nameof the place of origin of thegoods. Agricultural productstypically have qualities that derivefrom their place of production andare influenced by specific localgeographical factors, such asclimate and soil. Whether a signfunctions as a geographicalindication is a matter of nationallaw and consumer perception.

Geographical indications maybe used for a wide variety ofagricultural products, such as, forexample, "Tuscany" for olive oilproduced in a specific area of Italy,or "Roquefort" for cheeseproduced in that region of France.The use of geographical indicationsis not limited to agriculturalproducts. They may also highlightspecific qualities of a product thatare due to human factors found inthe product's place of origin, suchas specific manufacturing skillsand traditions. The place of originmay be a village or town, a regionor a country. An example of thelatter is "Switzerland" or "Swiss",perceived as a geographicalindication in many countries forproducts made in Switzerland and,in particular, for watches.

Difference between a geographical indication and a trademark

A trademark is a sign used by a company to distinguish its goods and services from those produced by others. It gives its owner the right to prevent others from using the trademark. A geographicalindication guarantees to consumersthat a product was produced ina certain place and has certaincharacteristics that are due to thatplace of production. It may beused by all producers who makeproducts that share certain qualitiesin the place designated by ageographical indication.

"Generic" geographical indication

If the name of a place is used to designate a particular type of product, rather than to indicate its place of origin, the term no longer functions as a geographical indication. For example, "Dijon mustard", a kind of mustard that originated many years ago in the French town of Dijon, has, over time, come to denote mustard of that kind made in many places. Hence, "Dijon mustard" is now a generic indication and refers to a type of product, rather than a place.

World Intellectual Property Organization (WIPO)

Established in 1970, the World Intellectual Property Organization (WIPO) is an international organization dedicated to helping ensure that the rights of creators and owners of intellectual property

are protected worldwide, and that inventors and authors are therefore recognized and rewarded for their ingenuity. It is present in Geneva, Switzerland.

The patent cooperation treaty (PCT) is a multilateral treaty entered into force in 1978. Through PCT, an inventor of a member country contracting state of PCT can simultaneously obtain priority for his/her invention in all or any of the member countries, without having to file a separate application in the countries of interest, by designating them in the PCT application. All activities related to PCT are coordinated by the world intellectual property organization (WIPO) situated in Geneva.

India

The Patent Office is headquartered in Kolkata with branches in Delhi, Chennai, and Mumbai. The Trade Marks Registry, headquartered at Mumbai has branches in Ahmedabad, Chennai, Delhi, and Kolkata. The Design Office is located in Kolkata and the GI Registry is in Chennai. Separate facilities house the International Searching Authority (ISA) / International Preliminary Examining Authority (IPEA) in Delhi; an Intellectual Property Office Archives center has been setup at Ahmedabad.

TYPES OF IPR IN INDIA

PATENT:

Definition and significance:

- A patent is granted for an invention which is a new product or process involving an inventive step and capable of industrial application. “New invention" means the subject matter has not fallen in public domain or that it does not form part of the state of the art; Inventive step is the feature(s) of the invention that involves technical advance as compared to the existing knowledge or having economic significance or both and that makes the invention not obvious to a person skilled in the art. Capable of Industrial application means that the invention is capable of being made or used in an industry.

DESIGN:

Definition and significance:

- Definition and significance: a design refers only to the features of shape, configuration, pattern, ornamentation, composition of colour or line or a combination thereof, applied to any article, whether two or three dimensional or in both forms by any industrial process or means which, in the finished article, appeal to and are judged solely by the eye.

TRADE MARK:

Definition and significance:

- A trademark means a mark capable of being represented graphically and which is capable of distinguishing the goods or services of one undertaking from those of other undertakings.
- A trademark can be a sign, words, letters, numbers, drawings, pictures, emblem, colours or combination of colours, shape of goods, graphic representation or packaging or sound or any combination of the above as applied to goods or services.

GEOGRAPHICAL INDICATIONS:

Definition and significance:

- A geographical indication identifies agricultural or natural or manufactured goods as originating or manufactured in the territory of a country or region or locality in that territory, where a given quality, reputation or other characteristic of such goods is essentially attributable to its geographical origin and, in case where such goods are manufactured goods, one of the activities of either the production or processing or preparation of the goods concerned takes place in such territory, region, or locality, as the case may be.

COPYRIGHT:

Definition and significance:

- Copyright is a right given by the law to creators of literary, dramatic, musical and artistic works and producers of cinematograph films and sound recordings. In fact, it is a bundle of rights including, inter alia, rights of reproduction, communication to the public, adaptation and translation of the work.

SEMICONDUCTOR INTEGRATED CIRCUITS LAYOUT-DESIGN:

Definition and significance:

- The aim of the Semiconductor Integrated Circuits Layout-Design Act 2000 is to provide protection of Intellectual Property Right (IPR) in the area of Semiconductor Integrated Circuit Layout-Designs and for matters connected therewith or incidental thereto.

PLANT VARIETY PROTECTION:

Definition and significance:

Protection granted for plant varieties, the rights of farmers and plant breeders and to encourage the development of new varieties of plants.

Biotechnology

Biotechnology isn't something new - selective breeding to create more useful varieties of animals and plants is a form of biotechnology that human beings have used for thousands of years.

Biotechnology includes any use of science or technology to alter the characteristics of a particular breed or animal.Biotechnology can be good or bad for animals - and it may also produce an answer to the ethical problems of experimenting on animals.Transgenic animals raise a particularly difficult problem.

Human problems

Newspaper articles about the ethical problems of genetically engineered animals are usually concerned about the danger these animals may pose to human beings (usually to human health), rather than any implications for the animals themselves.

Animal rights

Genetic engineering and selective breeding appear to violate animal rights, because they involve manipulating animals for human ends as if the animals were nothing more than human property, rather than treating the animals as being of value in themselves.

Recent action to allow animals to be patented reinforces the idea of animals as human property, rather than beings in their own right.

Animal welfare

- Biotechnology can be good for animals. Selective breeding and genetic engineering can benefit animals in many ways:
- Improving resistance to disease
- Breeding to remove characteristics that cause injury
    - eg selecting cattle without horns

But biotechnology can also be bad for animals - the good effects for the breeder can offset by painful side-effects for the animals:

- Modern pigs have been bred to grow extra fast - some breeds now grow too fast for their hearts, causing discomfort when animals are too active
- Broiler chickens are bred to grow fast - some now grow too fast for their legs

Regulating genetic engineering

Profitability is one of the major drivers of both selective breeding and genetic engineering.

If animal welfare is not to be compromised, research must be restricted by a counter-balancing ethical principle that prevents altering animals in a way that was bad for the animal.

One writer, Bernard Rollin, suggests that a suitable rule to regulate genetic engineering would be this:

Genetically engineered animals should be no worse off than the parent stock would be if they were not so engineered.

This principle can easily be adapted to cover selective breeding.

Biotechnology and experimental animals

It's been suggested that genetic engineering may solve all the ethical problems of laboratory experiments on animals. The goal is to create a genetically engineered mammal that lacks sentience, but is otherwise identical to normal experimental animals.

Such an animal could not suffer whatever was done to it, so there should be no ethical difficulty in performing experiments on it.

Ethical problems:

- This argument seems convincing, but do you feel comfortable about it?
- Is there any ethical objection to creating genetically engineered human beings without sentience, and experimenting on them?

Transgenic animals

Transgenic animals

Transgenic animals are animals that have been deliberately bred for research and that contain elements of two different species - they are creatures that blur the barrier between species.These animals are often deliberately created with genetic defects, and these defects may well cause the animal to have a bad quality of life. A mouse has been created, for example, that has been genetically modified to develop cancer.

Ethical issues of transgenic animals

Transgenic animals raise several particular moral issues (quite apart from any damage they might do to the environment):

- Are animals that combine species an unethical alteration of the natural order of the universe?
- Is it unethical to modify an animal's genetic make-up for a specific purpose, without knowing in advance if there will be any side-effects that will cause suffering to the animal?
- Does 'creating' animals by genetic engineering amount to treat the animals entirely as commodities?
- Is it unethical to create 'diseased' animals that are very likely to suffer?
    - ❖ Suffering may last for a long time in these animals as researchers want to conduct long-term investigations into the development of diseases

Religious views of transgenic animals

Against transgenic animals:

- God laid down the structure of creation and any tampering with it is sinful.
- Manipulating DNA is manipulating 'life itself' - and this is tampering with something that God did not intend humanity to meddle with.

In favour of transgenic animals:

- As human beings have been given 'dominion' over the animals, they are entitled to tamper with them.
- Palaeontology shows that the structure of creation has changed over time as some species became extinct and new ones came into being. They say that this shows that there is nothing fixed about the structure of creation.

Transgenic animals and religious food laws

Transgenic animals pose problems for religions that restrict the foods that their believers can eat, since they may produce animals that appear to be one species, but contain some elements of a forbidden species.

UNIT 4

VETERINARY IMMUNOLOGY AND SEROLOGY

IMMUNOLOGY

DEFINITION:

Immunology is defined as study about structure and functioning of immune system. It also deals with the study of host responses to the introduction of foreign substances into the tissues and the methods by which the body tries to eliminate them and protect itself from further invasion by them. The term **immunity** is derived from the Latin word *immunitas* that refers to the exemption from various civic duties and legal prosecution offered to Roman senators during their tenures in office. However, historically the term immunity refers to **protection**, particularly from infectious diseases. The cells and molecules responsible for immunity constitute the **immune system** and their collective and coordinated response to the introduction of foreign substances is defined as **immune response.**

BRIEF HISTORY:

The history of immunology is as old as human civilization and in the olden days man has always tried to protect himself from infectious diseases. In the olden days certain important diseases like bubonic plague, small pox etc. produced severe loss of human lives and attempts were made to protect human beings against these diseases. It was the **Indians** and **Chinese**, who proved to the world that persons once cured of the dreadful disease small pox did not get the infection again. So they deliberately infected the children with scab materials or made the children to inhale powder made from dried scab of infected persons and induced infection. This practice became established in Western Europe in 1718 when **Lady Mary Wortley Montagu (1718)** performed this technique on children. This method is also called **variolation (***Variola* in Latin means small pox). In that process, though certain percentage of mortality was produced large proportion of population was protected. **Thucydides (B.C 430)** reported about immunity to an infection that he called plague (not bubonic plague) and historians give him the credit of first report about protection. However, it was **Edward Jenner (1798),** who with methodical approach put the first step in development of immunology. He observed that milk- maids who had recovered from cowpox never contracted small pox during their life. Hence, he injected material collected from cowpox pustules in to human beings and none of them get the infection. He coined the term vaccination for this and *vaccines* in Latin means of or from cows.

A century later, Louis Pasteur who was credited with the formulation of germ theory did extensive work on fowl cholera, anthrax and rabies and developed vaccines against them. His work on development of vaccine against fowl cholera was accidental. Pasteur had a virulent culture of fowl cholera organisms (*Pasteurella multocida*) that had the capacity to cause death of chickens and he left it on the shelf and went for vacation. He came after couple of weeks and infected a group of birds with that culture. To his surprise the virulent culture did not produce death of chickens. Pasteur being starved of funds used the same group of chickens and infected them with fresh virulent *P.multocida* organisms. Again the chickens were not killed. However, the fresh virulent culture produced death in another group of birds. Thus Pasteur found out that some organisms lose their virulence when stored for a long time however, they retain their ability to induce protection. Using the same principle he developed vaccines for anthrax and rabies. His vaccines were of attenuated vaccines type. He attenuated anthrax organisms by growing them at high temperature and rabies virus by drying the affected rabbit brain. Since then number of workers have worked on the different concepts and developed vaccines like **Salmon** who developed killed vaccine for *Salmonella sp.* and Van Behring and Kitasato who found that filtrates of tetanus bacilli can protect animals against tetanus. They also proved that serum from a person affected by diphtheria can protect another person from diphtheria infection. During Pasteur's days, the knowledge on the cells involved in the immune system or on antibodies was limited. It was **Paul Ehrlich (1900)** proposed a theory about certain receptors found on surface of cells that are specific

for antigens and these receptors when liberated were called as antibodies. In the same way, it was **Elie Metchnikoff (1893)** who first observed the involvement certain special type of cells in removing foreign substances entered into the cells (phagocytosis) and put the basis for cellular immunity.

IMMUNITY – INTRODUCTION AND TYPES

The term immunity that otherwise means protection or to protect comprise of two different mechanisms **innate (natural or native) immunity** and **adaptive (specific) immunity.** Both specific and innate immunities act closely in removal of antigens

INNATE IMMUNITY:

The innate immunity is also called as **natural defence.** The innate immunity is composed of three mechanisms namely the **physical barriers** (skin, hairs, tears etc.), **phagocytosis** and **inflammation.** It has limited capacity to differentiate among antigens and is stereotypic. Innate immunity non-specifically clears antigens using preformed components without immunologic memory. Further it is also dependant on species, age, sex etc. Some of the important roles of natural barriers that contribute for innate immunity to a host are listed below.

Skin	-	Prevents penetration of pathogens
Body fluids	-	Prevents entry of pathogens
Tears	-	Contain lysozymes that prevent the entry of pathogens
Ciliated epithelium	-	Acts as a filter against invaded pathogen Mucus
	-	Prevents entry of pathogen
Acidic pH of stomach	-	Kills organisms

Phagocytosis involves the engulfment and destruction of pathogens and particulate matter by specialized cells that fall in two categories **polymorphonuclear (neutrophils and eosinophils)** and **mononuclear (macrophages and monocytes).** These phagocytes are also called **first line of defence**. The term **inflammation** that means 'setting on fire' is the third component of innate immunity and is initiated when phagocytosis fails to control the infection. Inflammation collectively involves a sequence of vascular events, which serve as a defence mechanism. The events are clotting mechanism activation, increased blood flow, increased capillary permeability and increased influx of phagocytic cells. Innate immunity provides early defence against microbes and plays important role in the induction of specific immune responses. The important difference between innate and adaptive immunity are listed below.

Feature	Innate	Adaptive
Specificity for antigens	Low	High
Diversity	Limited	Large
Specialization	Stereotypic	Highly specialized
Memory	No	Yes
Physical and chemical barriers	Skin, mucosal epithelia, anti-microbial chemicals	Cutaneous and mucosal immune systems. Secretes antibodies
Blood proteins	Complement	Antibodies
Cells	Phagocytic cells Natural killer cells	Lymphocytes

ADAPTIVE (SPECIFIC IMMUNITY):

The adaptive immunity is extremely specific against a particular antigen. It specifically recognizes foreign antigens and selectively removes them. The six important characters of this immunity are **specificity, inducibility, diversity, memory, distinguishing from self and non-self** and **self-limiting.** The inflammatory processes that are associated with many infections triggers specific immune responses. The classical example for this is the production of a cytokine by antigen-ingested macrophages that stimulate specific T lymphocyte against that antigen. The specific immunity increases the protective mechanisms of innate immunity and makes it efficient in removing foreign antigens. Further the specific immunity compensates the stereotyped mechanism of innate immunity with its high specificity. Another most important aspect of specific immunity is its ability to remember each antigen that has entered the body. This **immunologic memory** helps in mounting a faster and efficient removal when the same antigen enters subsequently. Specific immunity can also be called as **acquired immunity.** Hosts get this immunity as a result of exposure to an infection of some intervention (vaccination).

This specific immunity can also be classified into **humoral** and **cell mediated immunity.**

Humoral immunity is mediated by antigen specific blood glycoproteins called **antibodies.** These antibodies are secreted by plasma cells that are modified B-lymphocytes. This immunity is very effective against antigens that are found in the circulation. The antigen could be bacteria, exotoxin or virus. This immunity can be transferred to susceptible individual by transferring the cell free portion of blood either plasma or serum.

Cell mediated immunity is mediated by antigen specific cells that are thymus derived called T-lymphocytes. There are two populations of T cells – T helper cells (T_H cells) and T cytotoxic cells (T_C cells). This immunity protects the host against intracellular antigens like viruses, cancer antigens etc. This immunity is also responsible for graft rejection in transplants.

Humoral and cell mediated immunity can each be divided into **active** and **passive immunity.** Specific immune responses are stimulated when a host is exposed to and antigen. This immunity is called active immunity, which is acquired gradually, lasts longer and is highly protective. It also stimulates immunological memory. Passive immunity is obtained either by transfer of serum or T cell. The recipient of such transfer becomes immune immediately. However, the immunity is short lived with moderate protection. Besides this, there is no immunological memory associated with passive immunity.

The differences between humoral and cell mediated immunity are summarized below.

	Humoral immunity	**Cell mediated immunity**
Antigen	Extracellular antigens	Intracellular antigens
Responding lymphocytes	B lymphocytes	T lymphocytes
Effector mechanism	Antibody mediated elimination	Lysis of infected cell
Transferred by	Serum	T lymphocytes

THE IMMUNE SYSTEM – CELLS AND TISSUES

The immune system is composed of different types of cells. These cells are generally found in the blood, lymph and local secretions. They are also scattered in various organs except the central nervous system. Few organs in the body are made up of a collection of these cells and such organs are called as **lymphoid organs.**

The immune system is composed of two types of cells – lymphoid and non-lymphoid cells. The lymphoid cells are known as **lymphocytes** and there are three classes of lymphocytes – B-lymphocytes, T-lymphocytes and natural killer cells (NK cells). The non-lymphoid cells are also called as **accessory cells.** The important accessory cells of immune system are the mononuclear phagocytic cells (macrophage) and polymorpho-nuclear cells (PMNL). The common PMNL are neutrophils and eosinophils. Besides these, there are number of other non-lymphoid cells that have either a direct or indirect role in the immune response. The cells of immune system communicate by small non-specific molecules called **cytokines.** The cytokines produced by lymphocytes are called **lymphokines**and that produced by macrophages and monocytes are called **monokines.** The **interleukins** act as messenger between leucocytes.

PROPERTIES OF CELLS OF IMMUNE SYSTEM:

The ability of the immune system to optimally perform its protective function is entirely depending on several properties of its constituent cells. Some of the important properties of the cells of immune system are listed below.

1. The immune system is composed of specialized tissues termed as **peripheral lymphoid organs** that aggregate the antigens that entered into the body through various portals of entry. These peripheral lymphoid organs also attract lymphocytes and other cells that are required to initiate specific immune response.
2. To achieve maximum effects of specific immunity the effector and memory lymphocytes circulate and are also retained at the sites of antigen entry.
3. A strong interaction exists between lymphocytes and other non-lymphoid effector cells to achieve maximum immune response.
4. Multiple amplification loops magnify the effects of stimulating the few lymphocytes, which are specific for a particular antigen.

LYMPHOCYTES:

Morphology: The lymphocytes are small, round cells found in blood and lymphoid organs such as thymus, lymph nodes and spleen. The diameter of each lymphocyte ranges from 7-15 μm. They have a large and round nucleus that takes the haematoxylin intensely. They possess a thin rim of cytoplasm containing mitochondria, free ribosomes and a small Golgi apparatus. The surface of few lymphocytes are covered by small projections whereas, the surface is smooth for some lymphocytes. A class of lymphocytes called natural killer cells (NK cells) possesses extensive cytoplasm and number of cytoplasmic granules.

Classes of lymphocytes: Though lymphocytes are morphologically same, there are distinct subsets that differ functionally and in their protein products. The three important classes are B-lymphocytes, T-lymphocytes and NK cells.

a. The B-lymphocytes (B cells) are called thus, since they were first shown to mature in an organ in birds called bursa or Faricius. However, in mammals they mature in bone marrow. B cells are the only cells in immune system capable of producing antibodies.

b. The T-lymphocytes (T cells) arise from bone marrow and matures in thymus. The T cells are divided into distinct functional groups called helper T cells and cytotoxic T cells. The main role of T cells is to regulate immune response against all protein antigens and to produce specific immunity against intracellular antigens. Besides these two sub classes of T cells there is also one more sub class called T suppressor cells that suppress the immune response. The origin

and role of T suppressor cells are still controversial. The surface membrane proteins of T cells mainly identify the sub classes. The surface membrane proteins are also called as receptors or markers of lymphocytes. The receptors of lymphocytes are identified by CD nomenclature. The T helper cells are identified by presence of CD4 receptor (old name T4 receptor) and cytotoxic T cells are identified by presence of CD8 receptor.

c. The third class of lymphocytes NK cells lack surface markers. These cells are also referred as null cells. They have a large cytoplasm filled with cytoplasmic granules. They lyse the virus infected and tumour cells without any antigenic stimulation.

d. The main **identifying features** of T and B cells are listed below.

Sl. No	Property	B cells	T cells
1.	Site of development	Bone marrow, bursa, Peyer's patches	Thymus
2.	Distribution	Lymph node cortex, splenic follicles	Spleen periarticularsheath
3.	Circulate	No	Yes
4.	Antigen receptors	BCR	TCR
5.	Important surface antigens	Immunoglobulins	CD2, CD3, CD4, CD8
6.	Antigens recognized	Free foreign proteins	Processed foreign proteins on MHC
7.	Tolerance induction	Difficult	Easy
8.	Progeny cells	Plasma cells, memory cells	Helper T cells, cytotoxiccells
9.	Secreted protein	Immunoglobulins	Cytokines
10.	Phenotypic markers	Fc receptor, Class II MHC, CD19, CD21	Helper T cell $CD3^+$, $CD4^+$, CD^- Cytotoxic cell $CD3^+$, CD4-, CD^+

Development of lymphocytes:

Both T and B cells arise from bone marrow in mammals. The multipotent stem cells differentiates into lymphoid stem and myeloid and erythroid cell lines. From the lymphoid stem cell pre B and Pre T cell arise. The pre B cells develop into B cells and pre T cells develop into T cells. Upon antigens stimulation, the B cells differentiate into memory B cells and plasma cells where as the T cells differentiate into cytotoxic T cells, helper T cells and memory T cells.

Structure and surface markers of B cells (B Lymphocytes):

B cells are derived from bone marrow stem cells and develop within the bone marrow. Each B cell contains genetic information to produce antibody of unique antigen specificity that it expresses as membrane receptors. Mature B cells do not produce antibody but it differentiates into two daughter cells upon antigenic stimulation – plasma cells and memory cells and the plasma cells are the only cells in the body to produce antibodies. The antibody producing plasma cells are found in the lymphoid organs and at the site of immune response. They are characterized by morphology with eccentric nuclei, abundant cytoplasm with extensive rough endoplasmic reticulum and large Golgi apparatus. The memory cells are special cells that escape the programmed cell death (called apoptosis) by expressing a special gene called *bcl2*. The B cells on their surface express number of receptors. The important surface marker of B cell is termed as BCR. These BCRs are nothing but immunoglobulin molecules attached to the B cells. The soluble BCRs are called as antibodies. These receptors include antibodies of all isotypes particularly IgD and IgM. Besides these, B cells also have receptors for products of complement system (CD35, CD21), interleukins (CD122, CDw124, CD125, CD25), differentiation markers (CD5, CD9, CD10, CD19, CD20), Fc receptors (CD16, CD32 - that binds with Fc portion [stem portion] of the

antibodies), receptor for interaction with T helper cell (CD40) and class I and class II MHC molecules that are essential for presentation of antigens to T cells. Certain molecules on the surface of B cells help in adherence of cells of the immune system, regulate the signaling between them and control their movement in the system. These molecules are called as adherence molecules. These molecules belong to integrin, selectin or immunoglobulin superfamily.

Structure and surface markers of T cell (T Lymphocytes):

The T cells originate from thymus where they migrate as pre T cells and mature. The T cells are essential for full expression of specific immunity. The T cells can recognize only protein antigen but the antigen has to broke down to small peptide fragments and presented by MHC molecules. This kind of antigen recognition through MHC is called as MHC restriction. The T cells have multiple functions. They can acts as regulatory cells of immune system, mediators of immune system through production of cytokines or as effector cell in removal of antigen. Like B cells T cells also have number of surface markers. The important surface marker of T cells is the T cell receptor (TCR). Each T cell will have 10,000 to 20,000 TCRs or their surface. But interestingly all the TCRs on a single T cell are identical and can recognize only one epitope. The detailed structure of TCR will be dealt separately during antigen presentation. The T cells are divided into four subtypes based on the presence or absence of certain CD receptors. The two important CD receptors of T cells are CD4 and CD8. The helper T cells have CD4 receptors and cytotoxic T cells have CD8 receptors. The vice-versa is not possible. The delayed type hypersensitivity cells called T_D cells are cytotoxic in action but have CD4 receptors. They are responsible for delayed type hypersensitivity reaction. Some of the common T cell receptors are Fc receptors (CD16, CD32), complement receptor (CD35), interleukin receptor (CD25), transport receptor (CD71) and signal transduction receptor (CD3, CD45). In a T cell, the TCR, CD3 and CD4 or CD8 are placed close to each other and have greater role stimulating specific immunity. The exact mechanism will be dealt in coming chapters. Considering the significance of roles of different subtypes of T cells, the helper T cells are referred as commanders of immune system and cytotoxic T cells are called as combat troops of immune system.

Structure and surface markers of NK cells:

They are also referred as lymphoid null cells. They lack the markers characteristic of T cells; do not rearrange immunoglobulin or TCR genes. They are considered as primitive cytotoxic T cells. The NK cells mediate a phenomenon called antibody dependent cell mediated cytotoxicity (ADCC) that removed the antigen coated with immunoglobulins. On their surface they have CD16 and CD56 receptors. They do not have CD4, CD8, CD3 or TCR. These cells are characterized by presence of large number of lysosomes. They are considered as surveillance cells and kill intra cellular antigens non-specifically. They remove tumour antigens and virus infected cells.

MONONUCLEAR PHAGOCYTIC SYSTEM –MACROPHAGES

One of the four important cells that constitute the immune system is the macrophage. The other three being B cells, T cells and NK cells. The phagocytic cells can be divided into two distinct groups based on the morphology of nucleus into mononuclear (macrophages and their precursor monocytes) and polymorphonuclear (neutrophils and eosinophils) cells. Of the two groups, the macrophages are considered as powerful phagocytic cells and are referred as **big eaters** or **garbage collectors.**

Morphology:

They are round cells about 15μm in diameter with abundant cytoplasm. They may also assume wide variety of shapes in response to their environment. At the centre of cytoplasm they have round or bean shaped or indented nucleus. The perinuclear cytoplasm contains mitochondria, large number of

lysosomes, some rough endoplasmic reticulum and a Golgi apparatus. These cells cannot synthesize or secrete proteins. The peripheral cytoplasm is in continuous movement forming veil like ruffles.

Location of macrophages: Macrophages mature in the bone marrow and are found in the blood stream and many organs. Different names are given for macrophages found in various organs. They are listed below.

a. Blood stream - Monocytes
b. Connectivetissue - histiocytes
c. Liver - Kupfer cells
d. Brain - Glial cells
e. Lung alveoli - Alveolar macrophages
f. Lung capillaries - Pulmomary intravascular macrophages

Development of macrophages: Macrophages develop from a myelomonocytic stem cell through monoblast, promonocyte and monocyte till they reach structurally developed final stage called macrophage. The development of monocyte from promonocyte is stimulated by few factors called colony-stimulating factors. Before becoming macrophages they remain in the blood stream as monocyte for a period of time.

Characteristics of macrophages: Macrophages have six important characteristics that determine its role in the immune response. The characteristics are as follows.

a. They are mononuclear cells.
b. They show peroxidase and estrase activity.
c. They bear specific receptors for antibody and complement.
d. They have strong phagocytic properties
e. They are stimulated into active state by stimulation.
f. They have varied secretory property.
g. They can adhere to glass or plastic surface whereas T cells and B cells cannot adhere.

Surface markers of macrophages: The macrophages are characterized by presence of Fc receptors like (CD16, CD32, CD64), complement receptors (C11b, CD35), interleukin receptor (CD25), transport receptor (CD71) and Class I and II MHC molecules. The most important CD marker of macrophages is CD68, which is otherwise called as **macrosialin.**

Functions of macrophages:

- **Phagocytosis:** The primary function of macrophage is **phagocytosis**. They are potent phagocytic cells and can kill any foreign substance like microbe, carbon particles etc. They are also capable of doing repeated phagocytosis, which is not common among other phagocytic cells. The actual mechanism of macrophage- mediated phagocytosis is by **receptor-mediated endocytosis** followed by lysosomal enzyme degradation.
- **Antigen presentation:** Macrophages break the antigen into different epitopes and present them to helper T cells via MHC II molecules on their surface. Helper T cells will accept epitopes only when presented by MHC II on macrophages. This step is essential for stimulating specific immune response. Hence, they are known as antigen presenting cells (APC).

- **Secretory property:** They are strong secretory cells that secrete over 100 important substances. These substances play important role in lymphocyte differentiation, effector capability etc.

- **Activation:** Certain agents stimulate macrophages and convert them from accessory cells to effector cells. This process is called activation of macrophages and the macrophages are referred as activated macrophages. In normal conditions, the macrophages remain in circulation as monocytes or resting macrophages. When they reach the site of inflammatory response they possess enhanced phagocytic property with expression of Fc and complement receptors. At this stage the macrophages are referred as **inflammatory macrophages.** These inflammatory macrophages are further stimulated by certain substances like interferon into **activated macrophages.** The activated macrophages are bigger in size, increased mobility, and increased expression of Fc and complement receptors, increased lysosomal activity, increased MHC II expression and increased ability to kill bacteria. In certain chronic infections or in case of foreign substances remaining in body for a long time, a large number of macrophages accumulate at the site around the material and resemble epithelial cells. These accumulated cells are called as **epitheloid cells.** These cells may fuse to form multinucleated giant cells. Giant cell formation is seen when the particle to be ingested too big for a single cell.

- **Regulation of immune response:** The exact regulatory mechanisms are not clear. But macrophages through monokines control T cell proliferation.

- **Wound healing:** Apart from phagocytosis, macrophages also help in wound healing.Macrophages secrete proteases that help in break down of connective tissue and regulate the production of collagenase by fibroblasts and help in formation of blood vessels. Thus it helps in wound healing.

GRANULOCYTES

Granulocytes are group of blood leukocytes that are characterized by presence of number of cytoplasmic granule. Granulocytes are composed of three distinct populations of cells called **neutrophils, eosinophils**and **basophils.** This classification is based on the ability of the cytoplasmic granules to take up stains and the cell's function. The granules of neutrophils do not take any stain, eosinophils take acidic dye like eosin and basophils take basic dye haematoxylin.

These blood leukocytes are called **inflammatory cells** since they play an important role in inflammatory processes, innate immunity and in removal of dead tissues. Besides these functions, they also have effector functions in specific immune response.

The granulocytes neutrophils and eosinophils constitute the polymorphonuclear phagocytic system. They are called so since the nucleus is multi-lobulated. The granulocytes originate from myeloid stem cells, which in turn originate from pluripotent stem cells.

NEUTROPHILS:

This is the major cell type of granulocytes. They are formed in bone marrow and arrive at blood stream within 12 hours. They have a very short life span of just few days. The percentage of neutrophils in blood circulation among animals varies widely. It is 60-70% in carnivores, 20-30% in ruminants and 50% in horses. Their presence in circulation indicates a mild infection.

Morphology: They are round cells with a diameter of 12μm. The cytoplasm is granular with a central sausage or segmented nucleus. The cytoplasmic granules are of two types primary and secondary granules. The primary granules are rich in bactericidal enzymes such as myeloperoxidase and lysozyme etc. and secondary granules contain lysozyme, collagenase and iron binding protein

lactoferrin. The cytoplasm also contains a small Golgi apparatus, few mitochondrion and very few ribosomes or rough endoplasmic reticulum. The neutrophils are characterized by inability of the cells to divide and perform protein synthesis.

Surface receptors: The neutrophils have cell adhesion molecules (CD11a/CD18, CD11c/CD18), complement receptors (CD35, CD11b/CD18) and Fc receptor (CD32) on their surface. The cell adhesion molecules or neutrophils are called integrins. There are three type of integrins. Integrins are mixture of two receptors CD11 and CD18.

Functions: The main function of neutrophils is **phagocytosis**. They also have effector function in specific immune response. Neutrophils can phagocytose only bacteria. Unlike macrophage they cannot perform repeated phagocytosis. A neutrophil can do only one phagocytosis. At the end of phagocytosis, neutrophils are also killed. However, they secrete certain substances that in turn attract macrophages to the site. Neutrophils also have receptors for Fc portion of antibodies (CD32) and for complement protein C3b (CD35) on their surface.Hence, an antigen neutralised by and antibody or antigen covered with C3b can be taken by neutrophils. Thus the antigen antibody complex formed is removed from this system. This process is called **opsonisation** and antibodies and C3b are referred as **opsonin**.

EOSINOPHILS:

This is the second major type of granulocytes. It is called so since the granules in the cytoplasm take eosin dye and appears pink. They are formed in the bone marrow and then they reach spleen where they mature. After maturation, they leave for blood stream where they remain for a shorter time (half-life 30 minutes). Subsequently they migrate in to tissues where they remain relatively for a longer period (half-life 12 days). In contrast to neutrophils, the percentage of eosinophils is only 2% for dogs and 10% for cattle.

***Morphology*:** They are bigger than neutrophils and contain a bi-lobed nucleus. The cytoplasm consists of two types of granules - small primary granule and crystalloid granules. The small primary granules contain **arylsulfatase, eosinophil peroxidase** and **acid phosphatase.** The crystalloid granules contain four major proteins – major basic protein (MBP), eosinophilic cationic protein (ECP), eosinophil peroxidase (EPO) and eosinophil derived neurotoxin (EDN). In a crystalloid granule, the MBP is arranged as core and remaining proteins are arranged as matrix proteins. The cell membrane is characterized by the presence of large amount of lysophospholipase. The cytoplasm also contains a small Golgi apparatus, few mitochondrion and very few ribosomes or rough endoplasmic reticulum.

***Surface receptors*:** Eosinophils possess receptors for Fc portion of antibodies (CD16, CD32) and complement proteins (CD35, CD23). They also possess receptors for IgE.

Functions: The important function is phagocytosis. Like neutrophils, eosinophilscanalso phagocytose smaller particles. But they are known for their ability to destroy parasites extracellularly. The eosniophils release the granules into the surrounding fluid in tissues and the contents of granules destroy parasites.

BASOPHILS: They are less important type of granulocytes. They are called so since the granules in the cytoplasm take basophilic dyes like haematoxylin. They constitute only 0.5% of the blood cells. They are not normally found in extravascular tissues. They migrate into the tissue upon stimulation by the lymphocytes. The granules are rich in vasoactive amines like histamine and serotonin. They also possess surface receptors for IgE. Hence, inside the tissue, they stimulate inflammatory response along with mast cells and eosinophils.

DENDRITIC CELLS:

They are accessory cells and play important roles in the induction of immune responses. Dendritic cells are population of specialized mononuclear cells that are positioned at the possible places of antigen entry into the body. They are poorly phagocytic cells. However, their main role is to present the antigen to lymphocytes for its effective removal from the system.

Morphology: The cells are characterized by lobulated nuclei, clear cytoplasm and filamentous cytoplasmic processes called dendrites. The cytoplasm is filled with a special granule (cytoplasmic organelle) called **Birbeck granules.** There are two different types of dendritic cells with different properties and functions. They are **interdigitating dendtritic cells** and **follicular dendritic cells.**

Interdigitating dendritic cells: The interdigitating dendritic cells are the true dendtritic cells. They are found in high numbers in the T cells rich areas of lymph nodes and spleen. These cells are scattered throughout the skin epidermis and called as **Langerhans cells.** Langerhans cells are rich in Birbeckgranuleswhose function is not known. These cells originate from precursor bone marrow cells and are closely related to mononuclear phagocytes.

Follicular dendritic cells: These cells are found in the lymph nodes, spleen and mucosa associated lymphoid tissue (MALT). They are not derived from precursors in bone marrow and are entirely different from interdigitating dendritic cells.

Functions:

- ***Interdigitating dendritic cells:*** They are the most efficient antigen presenting cells (APC). The Langerhans cells pickup antigen that enter via the skin and transport these antigens to draining lymph nodes where immune responses are initiated.
- ***Follicular dendritic cells:*** These cells trap the antigen complexes with antibody or complement and present these antigens to B cells for recognition.

THE IMMUNE SYSTEM – ORGANS OF IMMUNE SYSTEM

The lymphoid tissues that constitute the immune system can be classified into two groups – **primary lymphoid organs or generative organs** and **peripheral organs or secondary lymphoid organs.** In generative organs, the lymphocytes first express antigen receptors and attain phenotypic and functional maturity. These organs are the maturation sites for immune cells in the absence of antigen. In peripheral organs the responses to foreign antigens are initiated and developed. These organs are the maturation sites for antigen driven immune cells. The important generative organs for the mammals are the bone marrow where all types of lymphocytes originate and mature and thymus where T cells mature and reach a stage of functional competence. In birds, apart from bone marrow they got another generative organ the bursa of Fabricius – the site where B cells mature. The bursa of Fabricius equivalent in mammals is the bone marrow. The important peripheral or secondary lymphoid organs are the lymph node, spleen; mucosa associated lymphoid tissues (MALT) and cutaneous immune system.

The important roles of generative and peripheral lymphoid organs are summarized below.

1. Bone marrow (mammals) and Bursa of Fabricius (birds) – B cell factories
2. Thymus and intestinal epithelium – T cell factories
3. Lymph nodes – Junctional filters in lymphatic system
4. Spleen – Filter in circulatory system
5. MALT – localized antibodies at major sites of pathogen entry

INVADERS OF THE BODY – ANTIGEN / IMMUNOGEN

In normal life, lot of foreign molecules enters into the body that are broken down and absorbed. But the body mounts immune response only against certain substances and **antigen** is the general term used for them. Generally the immune response is mounted against microorganisms. A microorganism is said to be **pathogenic** when it can cause infection and the ability to cause infection is termed as **virulence**.

DEFINITION: The term antigen is Greek means *anti-against gen-gignomai to generate*. **Antigens** are substances, when introduced into the host binds with specific antibodies or T lymphocytes that are reactive and specific against the antigen. Previously the terms antigen and immunogen were used together and same meaning was given. At present, a clear line of demarcation has been drawn between antigen and immunogen.

The term **immunogen** refers to substances, which can initiate lymphocyte activation necessary for an antibody response and generates immune response. The ability of a substance to induce humoral or cell mediated immune response is called as **immunogenicity.** The ability of a substance to react with the specific antibodies or activated T cells that it induces is called **antigenicity**.

CARDINAL FEATURES OF AN IMMUNOGEN: To be defined as an immunogen, a substance should have the following characters. They are

1. High molecular weight
2. Chemical complexity
3. Solubility (biodegradability)
4. Structural stability
5. Foreignness

1. **High molecular weight**: Usually macromolecules are immunogenic and larger the molecules, the more immunogenic it is likely to be. Molecules with a molecular weight of 5000 or greater are good immunogens. However, there are certain molecules with lesser molecular weight, but still good immunogens. The threshold level for a good immunogen is 1kD. Though macromolecules are good immunogens, the building blocks of macromolecules amino acids, fatty acids, and monosaccharides) are not immunogenic.

2. **Chemical complexity:** Large chemical substances like nylon, Teflon etc. that lack chemical complexity are not immunogenic. In the same way nucleic acids that are just polymers of repeating monomers are also not immunogenic. For proteins, immunogenicity increases with diversity of amino acids. If a protein contains just only one type of amino acid, it is less immunogenic than a protein with number of different amino acids.

3. **Bio-degradability:** The cells of immune system must react with immunogens before mounting immune response. Hence, the immunogens are broken in specialized cells into epitopes and immune response is mounted against these epitopes. These epitopes are very small. Thus non-biodegradable substances like metal pins are not immunogenic.

4. **Structural stability:** To mount an immune response the cells of immune system should recognize the shape of epitopes. Hence, structurally unstable substances like flagellin are poorly immunogenic.

5. **Foreignness:** A good immunogen should be a non-self-molecule. The more the non-self it is, the better is its immunogenicity. The degree of non-self is less between genetically closely related individuals.

ANTIGENIC DETERMINANTS (EPITOPES):

The antigen combining sites of antibodies or T cells are smaller than whole antigen and hence it is impossible for a whole antigen to bind entirely with antibodies or T cells. The antibodies or T cells bind with certain surface structures found on the antigen. These surface structures stimulate the immune system and they are referred as **antigenic determinants** or **epitopes.** Epitopes are the immunologically active portions an antigen that can react with antibodies or T cell receptors (TCR). These epitopes have different shapes and even a simple microbe may have thousands of such epitopes.Hence, in immune response against a particular antigen the serum sample will have antibodies against number of epitopes. But the immune response against all epitopes is not same. Certain epitopes elicit large amount of antibody response. Such epitopes are called as **immuno-dominant epitopes.**

Over the surface of an antigen a particular epitope may be present more than once and if an antigen contain only one type of epitopes it is called as **monovalent** and if it contains more than one type of epitopes it is called as **polyvalent**. In certain antigens the epitopes are arranged spatially so that binding of an antibody to one may interfere with binding of another antibody to different epitope. Such epitopes are referred as **non-overlapping epitopes.** Where as in some antigen one epitope will interfere with binding of antibody to another epitope and such epitopes are called as **overlapping epitopes.** Sometimes binding of an antibody to an epitope will result in conformational changes that may in turn help in binding of another antibody to different epitope. This is called **allosteric effect**. The epitopes can be further classified into linear epitopes, conformational epitopes and neoantigenic epitopes depending upon the arrangement and post synthesis modifications of the building blocks.

HAPTENS: Haptens are small antigens that cannot elicit antibody response individually. But they become immunogenic when coupled with larger molecules. These larger molecules that make the haptens immunogenic are called as **carrier molecules.** Haptens can react with pre formed antibodies. In other words haptens are antigenic but not immunogenic. Hence, haptens aredefined as molecules that are not immunogenic by themselves but they can react with preformed antibodies of the right specificity.

Haptens are classified according to the number of antigenic determinants they have. If a hapten has only epitope it is classifiedas simple hapten and if it has got two or more different epitopes it is classified as complex hapten.

TYPES OF ANTIGEN: The antigens could be classified into two major types – microbial and non-microbial antigens.

Microbial antigens: The microbial antigens could be classified into bacterial, viral and other microbial antigens.

a. **Bacterial antigens:** Bacteria possess different types of antigen mainly based on its morphological features. Some of the common bacterial antigens are somatic antigen (O), Capsular antigen (K), flagella antigen (H) and fimbrial or pili antigen. The major constituent of the cell wall of the Gram-positive bacteria is the peptidoglycan (PG), whereas it is lipopolysaccharide (LPS) in Gram-negative bacteria. In Gram-negative bacteria the antigenicity is associated with polysaccharide compound (an oligosaccharide attached to lipid A). In the same way the capsule of bacteria is also rich in polysaccharides, which are good antigens. Apart from these antigens that are associated with morphology, bacteria also possess other types of antigens. Some of the important antigens include **porins**, **heat shock proteins (HSP)** and **exotoxins**. The porins are proteins that form pores on the cell wall of gram-negative bacteria. HSP are produced in large quantities by bacteria undergoing stress. Exotoxins are highly immunogenic and stimulate the production of antibodies.

The antibodies against exotoxins are called **antitoxins.** When these exotoxins are precipitated by mild protein denaturing agents such as formaldehyde, the exotoxin loses its pathogenicity but retains its immunogenicity. Such precipitated toxins are called as **toxoids.**

b. **Viral antigens:** The proteins of the outer coat (capsid) of virus are good antigens. These capsid proteins elicit good antibody response when they are present in the circulation. However, when the virus is inside the cell and produce its proteins inside the cell, the virus-infected cell are identified and removed. Such types of virus proteins that are produced inside the cell are called **endogenous antigens.**

c. **Other microbial antigens:** The other micro and macro parasites likefungi, protozoa and worms also are composed of different types of antigens composed of major building blocks like lipids, carbohydrates and proteins. They also elicit immune response in the body. However, the intensity of immune response and degree of protection offered by this response is questionable. This could be due to various reasons from the development (various stages in life cycle) to certain mechanisms exhibited by them to evade immune response.

NON-MICROBIAL ANTIGENS: The non-microbial antigens could be classified into cell surface antigens, auto antigens and miscellaneous.

- **Cell surface antigens:** The surface of most of the cells is covered with different antigens. When these antigens are given to heterogonous host an immune response is mounted. Some of the important cell surface antigens are blood group antigens, and CD receptors of the leukocytes.
- **Autoantigens:** Sometimes rarely immune response is elicited against normal body components. Such substances are called as **autoantigens** and such immune response is called as autoimmunity. Some of the normal components of body against which immune response is elicited are basement membrane, myelin, mitochondrial proteins, nuclear proteins, hormones etc.
- **Miscellaneous:** Non-microbial antigens like dust particle, certain type of food particle. pollen grains, venom etc. can also elicit immune response in certain individuals. These substances produce antibodies on entering the body.

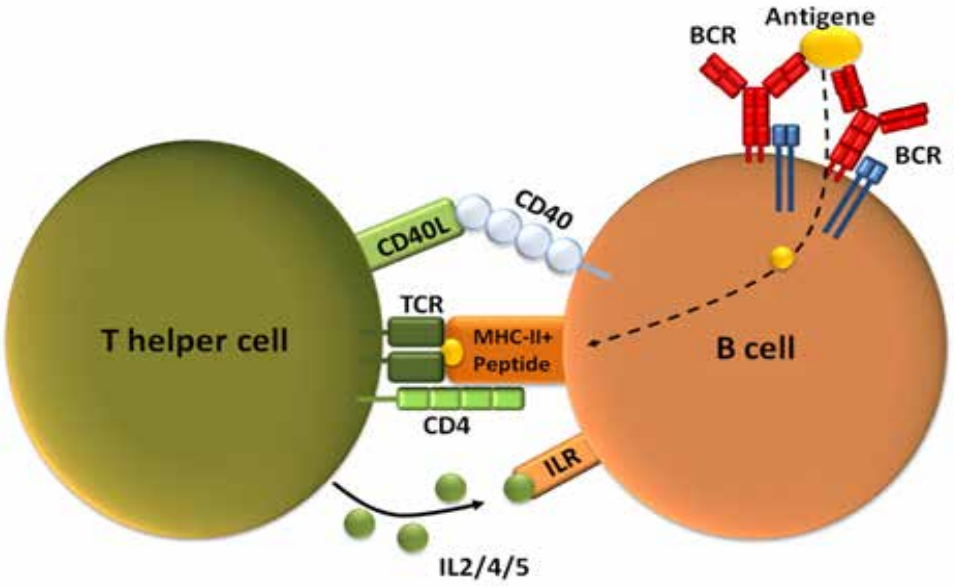
HETEROPHILE ANTIGENS: Heterophile antigens are immunologically related groups of antigens that occur in the cells of some bacterial species and also in some species of animals. This could be due to the presence of identical or similar epitopes found on them. As a result of this, antibodies directed against one antigen may react unexpectedly with an unrelated antigen. Heterophile antigens are also called as **Forssman antigen** and the antibodies directed against them are known as heterophile antibodies. This heterophile antigens form the basis for **cross-reaction**. Cross-reaction could be due to two reasons. They are

- Presence of identical epitopes between unrelated individuals. Eg some of the common examples of heterophile antigens are presence of same epitopes among certain bacteria and glycoproteins of red blood cells of mammals, *Brucella abortus* and *Yersinia enterocolitica,* feline infectious peritonitis virus and transmissible gastroenteritis virus of pigs etc.
- Structural similarity between epitopes of unrelated individuals.

T cell dependent antigens

T cell-dependent antigens are antigens that don't have the capacity of direct stimulation of B cells in the production of antibodies without the assistance of T cells. This helps in the production of

cytokines. Cytokines can be either interferons, interleukins or growth factors. Cytokines involve in the activation, differentiation, and proliferation of B cells.



T Cell Dependent B Cell activation

T cell-dependent antigens are proteins. Many antigen determinants are present in T cell-dependent antigens.

T cell independent antigens

T cell-independent antigens are a type of antigens which has the ability to induce direct stimulation of B cells in the production of antibodies without the assistance of T cells. T cell-independent antigens are polymeric antigens such as polysaccharides. The responses induced to T cell independent antigens are different from a response that is induced to a typical antigen. They possess the same antigenic determinants with many repetitions, and this is a characteristic feature of T cell independent antigens.

Many types of these antigens possess the ability to activate B cell clones which are specific to the antigens. This process is known as polyclonal activation. These antigens are further subdivided into two categories; Type I and Type II. The subdivision occurs according to the ability of Type I and Type II cells to polyclonally activate B cells. **Type I T cell independent antigens** are considered as polyclonal activators while **Type II T cell independent antigens** are not such activators. Type I antigens possess an essential B cell activating activity which induces the direct proliferation of and differentiation of B lymphocytes which occur without the stimulation of B cells. These antigens function independently of their BCR specificity. The activation of B cells occurs via Toll-like receptors that are present on the surface of B cells once the BCR stimulation is completed.

Direct activation of B Cells by antigens

Type II antigens consist of repetitive structures known as epitopes. These cells lack the activity of B cell activation. Type II antigens activate only mature B cells. They energize the immature B cells which prevent the involvement of immature B cells in any response of the immune system. These antigens are considered to be resistance to degradations and therefore tend to persist for longer for longer periods of times carrying out the specific functions of the immune system.

Difference between T cell dependent and independent antigens

T Cell Dependent Antigen vs T Cell Independent Antigen	
T cell-dependent antigens are the antigens which cannot stimulate the direct activation of B cells in the production of antibodies without the assistance of T cells.	T cell-independent antigens are the antigens which have the ability to induce direct stimulation of B cells in the production of antibodies without the assistance of T cells.
Chemical Nature	
T cell dependent antigens are proteins.	T cell-independent antigens are polysaccharides; polymeric antigens which also can be either glycolipids or nucleic acids.
Secondary Isotypes	
IgG, IgE, and IgA are the secondary isotypes of T cell dependent	IgG and IgA are the secondary isotypes of T cell independent antigens.

Mitogen

A mitogen is a chemical substance that encourages a cell to commence cell division, triggering mitosis. A mitogen is usually some form of a protein. Mitogenesis is the induction (triggering) of mitosis, typically via a mitogen. Mitogens trigger signal transduction pathways in which mitogen-activated protein kinase (MAPK) is involved, leading to mitosis.

Lymphocytes can enter mitosis when they are activated by mitogens or antigens. B cells specifically can divide when they encounter an antigen matching their immunoglobulin. T cells undergo mitosis when stimulated by mitogens to produce small lymphocytes that are then responsible for the production of lymphokines, which are substances that modify the host organism to improve its immunity. B cells, on the other hand, divide to produce plasma cells when stimulated by mitogens, which then produce immunoglobulins, or antibodies.[7] Mitogens are often used to stimulate lymphocytes and thereby assess immune function. Lipopolysaccharidetoxin from gram-negative bacteria is thymus-independent. They may directly activate B cells, regardless of their antigenicspecificity. Plasma cells are terminally differentiated and, therefore, cannot undergo mitosis. Memory B cells can proliferate to produce more memory cells or plasma B cells.

Adjuvants

Adjuvants are *substances used in combination with a specific antigen that produced a more robust immune response than the antigen alone.* An immunologic adjuvant is defined as any substance that acts to accelerate, prolong, or enhance antigen-specific immune responses when used in combination with specific vaccine antigens. **Adjuvants** (taken from the Latin, "adjuvare," meaning "to help") are designed to improve poorly immunogenic vaccines.

Adjuvants affect the immune response in various ways:

- To increase the immunogenicity of weak antigens
- To enhance speed and duration of immune response
- To stimulate and modulate humoral responses, including antibody isotype
- To stimulate cell-mediated immunity
- To improve induction of mucosal immunity
- Enhance immune responses in immunologically immature patients, particularly infants
- To decrease the dose of antigen required; reducing costs and eliminating inconvenient requirements for booster shots

Many molecules have been considered for use as an adjuvant, including mineral compounds (e.g. Alum), water-in-oil or oil-in-water emulsions (e.g. Freund's adjuvant), as well as natural and synthetic toxins derived from bacteria (e.g. cholera toxin, CT and lymphotoxin, LT). Based on their mechanism of action, adjuvants have been categorised into two broad groups; the **particulate vaccine-delivery systems** that target antigen to **antigen presenting cells** (**APCs**) and the **immunostimulatory adjuvants** that directly activate such cells through specific receptors e.g. **toll-like receptors** (**TLRs**) resulting in inflammatory responses that amplify the innate immune response. The ultimate aim is to activate the innate immune system to respond more rapidly to infection and for the adaptive immune response to become more specific.

ANTIBODIES

DEFINITION: Antibodies are recognition proteins found in the gamma fraction of serum and other body fluids of vertebrates that react specifically with the antigens that induce their formation. They belong to a family of globular proteins called **immunoglobulins.** Immunoglobulins are afamily of globular proteins that comprise antibody molecules and molecules having patterns of molecular structure in common with antibodies. The term immunoglobulin can be used to refer to any antibody like molecule, regardless of its antigen binding specificity. Chemically the antibody molecules are glycoproteins.

STRUCTURE: The structure of antibody molecules is best illustrated by four-chain model. Antibody molecules are composed of two identical heavy chains and two light chains. These chains are polypeptides made up of amino acids that are arranged in a sequence. Disulphide bonds bind these chains. A single disulphide bond connects light chain with heavy chain and two disulphide bonds connect heavy chains. An antibody molecule has got two ends an amino terminal and carboxy terminal. The structure can be best explained as primary,secondary, tertiary and quaternary structures.

Primary structure: The primary structure is the linear arrangement of amino acids in the polypeptide chain. A heavy chain is composed of approximately 440 amino acids and light chain has 220 amino acids. The sequence of amino acids from 1-107 in heavy and light chain varies between antibody molecules and this region is called as **variable region (V). W**hereas the amino acid sequence from 108-440 does not vary and this region is called as **constant region (C).** The arrangement of amino acid sequence in constant region also follows a particular pattern of similarity. Based on the similarity in arrangement the constant region is divided into **domains.** Each domain has got approximately 100-110 amino acids. In an antibody molecule the variable region of heavy and light chains have got only one domain named and **variable light (V_L) and variable heavy (V_H).** The constant region of light chain has got one domain called **constant light** (C_L). However, the constant region of heavy chain has got three domains. The domains are indicated by number coupled with type of heavy chains they have like gamma, mu, alpha, delta or epsilon. Each domain has got a particular function and structural significance. Variability of the amino acid sequence in the V regions is not random but precisely organised. It is localised only in certain areas and these areas are referred as **hypervariable regions** or **complementarity determining regions (CDR).** Each CDR is about ten amino acids in length and variable regions of both heavy and light chains have got three CDRs. The location of CDRs in VH and VL regions are as follows.

a. VH CDR Position

i. CDR1 - 31-35

ii. CDR2 - 50-65

iii. CDR3 - 96-102

b. VL CDR position

i. CDR1 - 21-34

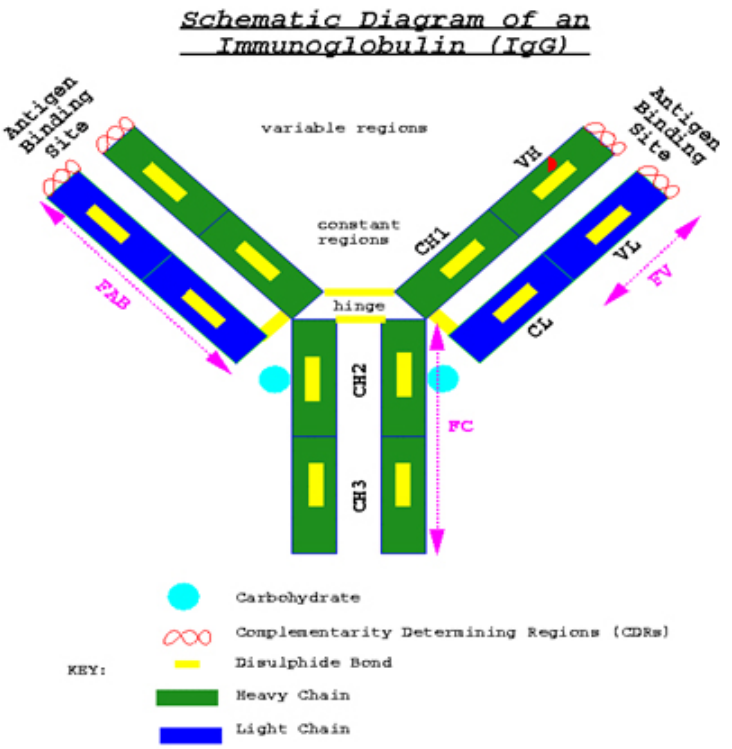
ii. CDR2 - 50-56

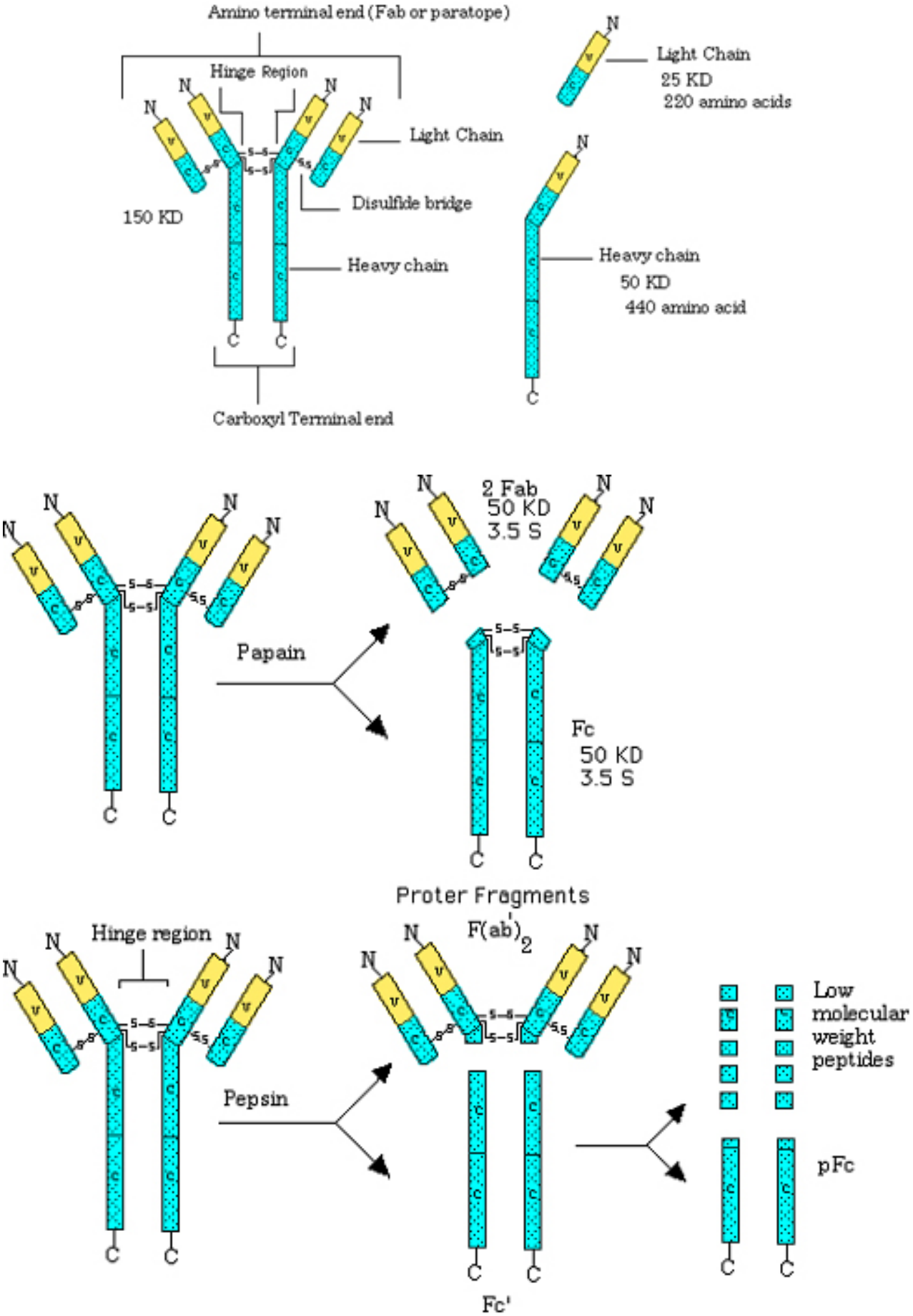
iii. CDR3 - 89-97

The intervening sequences between CDRs have constant arrangement of amino acids (conserved) and are called as **framework regions (FR).** The CDRs determine the antigen- combining site, shape of the antibody and decides the antibody specificity whereas the FR provides stability to antigen combining site. Antibodies also have segmental flexibility. The flexibility of an antibody molecule is due to a region called **hinge region** that is rich in proline and cysteine residues. This region helps in flexing the arms of antibody upto 60-180°.

Antibody molecules can be classified into five types depending upon the type of heavy chains they have. Some structural variations are found between them that will be dealt later. Some of the types like IgM have an extradomain in the heavy chain and this domain is called as **tail piece.** This tailpiece helps the antibody molecules to join together and form multimeric structures. The multimeric structures are stabilised by a polypeptide called **J chain.** Generally carbohydrates like glycan are found attached to 297th amino acid (Asparagine).

The four-chain model structure was made further clear by the digestion of antibody molecules with enzymes pepsin and papain. The enzyme papain cleaves the antibody molecule into three fragments of approximately equal molecular weight. Of them, two fragments can bind with antigens and are referred as **fragment antigen - binding** (Fab). The remaining fragment can be crystallisable and is referred as **fragment crystallisable** (Fc). The Fc fragment does not have the property of binding with antigen and has got only antibody effector functions. The Fc portion can give rise to further fragment called Fc'. The enzyme pepsin provides an entirely different digestion pattern. Pepsin digestion provides a bivalent Fab and number of small fragments of Fc region. This enzyme digestion patterns are well illustrated in the figure provided below





Secondary structure: It is the relationship between amino acids located some distance a part in the primary structure. Their interaction gives rise to periodic structural motifs in protein molecule.

Tertiary structure: It describes the spatial relationship of amino acids that are separated by great distances in the primary sequence. This refers to cysteine residues that form the intra chain disulphide bridges. Between each cysteine residue in V region 65-75 amino acids are positioned and in C region 52-59 amino acids are positioned.

Quaternary structure: It reflects the interaction between distinct polypeptide subunits of a multidomain protein. In an antibody molecule it reflects the biological activity of antibody molecule. The interaction of V_H and V_L domain yield correct confirmation for antigen combining site.

CLASSES OF ANTIBODIES

INTRODUCTION:

ISOTYPES: Generally there are five classes of antibodies. This classification is based on variations in the antigenic determinant of the heavy chain. Thus the major classes of antibodies are also known as **isotypes**. The antigenic determinants that determine the different isotypes are called as **isotypic determinants.** The Isotypic determinants in the heavy chain are restricted to constant region and termed in Greek letters as alpha (α), delta (δ), epsilon (ε), gamma (γ) and mu (μ). Thus the IgA has got alpha heavy chain, IgD has got delta heavy chain, IgE has goat epsilon heavy chain, IgG has got gamma heavy chain and IgM has got u heavy chain. Besides these, there are some other heavy chain antigenic determinants that determine the difference within an isotype and they are called as **subtypes** or **antigenic subclasses.** There are four different types of γheavy chains - γ1, γ2, γ3 and γ4 and hence four subclasses of IgG exist – IgG1, IgG2, IgG3 and IgG4. In the same way there are two subclasses of IgA – IgA1 and IgA2 depending upon the α1 and α2 heavy chains.

Antigenic determinants are also found on light chains and the light chains are classified into lambda (λ) and kappa (κ) based on them. The antigenic determinants of light chains are restricted to constant region. The λ chain is further sub classified into λ1, λ2, λ3 and λ6. The antigenic determinant of light chain is not useful to identify major classes of antibodies since λ and κ chains are found all major types.

ALLOTYPES: Only some individuals within a given species carry Allotypic determinants. These antibodies are called allotypes. Allotypic determinants have been identified for gamma, alpha and kappa chains.

IDIOTPYE: Idiotypic determinants (**idiotopes**) are found on the V region near the antigen-binding site of the antibody molecule. These determinants classify antibodies into **idiotypes.** Thesedeterminants are common to antibodies having specificity for foreign antigenic determinants. It reflects the antibody-combining site's structure and usually depends on the arrangement of heavy and light chains.

ANTIBODY CLASSES:

There are five classes of antibodies. They are IgG, IgM, IgA, IgE and IgD.

1. **IgM**: It is the biggest antibody molecule with a molecular weight of 970kD and a sedimentation coefficiency of 19S. It is a pentameric structure comprising of five four- chain units. The heavy chain is of mu type and has got 4 constant and 1 variable domains. The light chain has got one constant and one variable domain. The pentameric structure is stabilised by disulphide bond

between Cμ3 domains and by the presence of Joining chain (Jchain). A single J chain di-sulphide linked close to the C terminus of the μ chain. The cysteine amino acid located at the 575th position is involved in bonding of J chain between units. The hinge region is not present in IgM. IgM is decavalent and has got 10 antigen combining sites. As a result of this it binds the antigen avidly. It is also a good opsonin and is involved in fixing the complement. It is the first antibody to class to appear in primary immune response. Because of the size its presence is restricted only to blood and it is of not much use in body fluids or tissue secretions.

2. **IgG:** It is the main class of antibody in the serum. It has got a molecular weight of 146- 160D and a sedimentation coefficiency of 7S. It is a monomeric structure comprising of one four-chain unit. The heavy chain is of gamma type and has got 3 constant and 1 variable domains. The light chain has got one constant and one variable domain. The hinge region is present in IgG. IgG is bivalent and has got two antigen combining sites. It isalsoagoodopsonin.Though this class of antibodies appear late in primary immune response, they are responsible for secondary or anamnestic response. Apart from blood they are also found in body fluids, tissue secretions and colostrums. They also provide protection to foetus since it is the only antibody class that can pass through placenta. Number and location of disulphide bridges and minor variations in amino acid sequences for biological activity differentiate the subclasses of IgG.

3. **IgA:** It occurs in two forms. In serum it occurs as monomer and in local secretions it occurs as dimmer. As a monomer it has got a molecular weight of 160kD and a sedimentation coefficiency of 7S and as a dimer it has got a molecular weight of 385kD and a sedimentation coefficiency of 11S. The dimeric form is also known a secretory IgA (sIgA). The sIgA is characterized by presence of secretory component, which is a glycoprotein with a molecular weight of 70kD. The heavy chain is of alpha type and has got 3 constant and 1 variable domains. The light chain has got one constant and one variable domain. The hinge region and J chain are present in IgA. There are two subclasses of IgA – IgA1 and IgA2 that are differentiated by their distribution and by arrangement of disulphide bonds. IgA1 is predominantly found in serum where and IgA1 and IgA2 are present roughly in equal composition in sIgA. As monomer it has got two antigen combining sites and as dimer it has got four antigen combining sites. IgA is a component of secondary or anamnestic response. However, it cannot pass through placenta and hence is not involved in passive immunity. The microorganisms in the gut are the major stimulators or this type of antibody.

4. **IgE:** It is present very small amount in normal individuals and its presence in large amount is an indication of allergic condition. It is present as a monomer with a molecular weight of 184kD and a sedimentation coefficiency of 8S. The heavy chain is of epsilon type and has got 4 constant and 1 variable domains. The light chain has got one constant and one variable domain. It is bivalent and got two antigens combining sites. The Fc portion is characteristic in that it can combine with mast cell resulting in its degranulation. The degranulation of mast cells is responsible for allergic reaction including anaphylaxis. Hence, IgE is called reaginic antibody. It is present in blood and tissue spaces. It cannot pass through placenta.

5. **IgD:** It is present is very low concentration in serum. It is not very important type and not found in all species. The heavy chain is of delta type and has got 3 constant and 1 variable domains. The light chain has got one constant and one variable domain. It is present as a monomer with a

molecular weight of 70D and a sedimentation coefficiency of 7S. It is bivalent and got two antigens combining sites. IgD is characterized by long exposed hinge region. It is normally found on the surface of B cells and it mainly acts as a receptor forantigen.

Monoclonal antibodies and Hybridoma technology

Monoclonal antibody (MAb) is a single type of antibody that is directed against a specific antigenic determinant (epitope).

Culturing of normal B-lymphocytes is difficult and hence, the synthesis of MAb in earlier days was short-lived and very limited.

It is interesting that immortal monoclonal antibody producing cells do exist in nature. They are found in the patients suffering from a disease called multiple myeloma (a cancer of B- lymphocytes). It was in 1975. George Kohler and Cesar Milstein (Nobel Prize, 1984) achieved large scale production of MAbs. They could successfully hybridize antibody—producing B-lymphocytes with myeloma cells *in vitro* and create a hybridoma.The result is that the artificially immortalized B-lymphocytes can multiply indefinitely *in vitro* and produce MAbs. The hybridoma cells possess the growth and multiplying properties of myeloma cells but secrete antibody of B-lymphocytes. The production of monoclonal antibodies by the hybrid cells is referred to as hybridoma technology.

Principle for Creation of Hybridoma Cells:

The myeloma cells used in hybridoma technology must not be capable of synthesizing their own antibodies. The selection of hybridoma cells is based on inhibiting the nucleotide (consequently the DNA) synthesizing machinery. The mammalian cells can synthesize nucleotides by two pathways—*de novo* synthesis and salvage pathway.

Pathways for the Synthesis of Nucleotides

The de novo synthesis of nucleotides requires tetrahydrofolate which is formed from dihydrofolate. The formation of tetrahydrofolate (and therefore nucleotides) can be blocked by the inhibitor aminopterin. The salvage pathway involves the direct conversion of purines and pyrimidine's into the corresponding nucleotides. Hypoxanthine guanine phosphoribosyltransferase (HGPRT) is a key enzyme in the salvage pathway of purines.

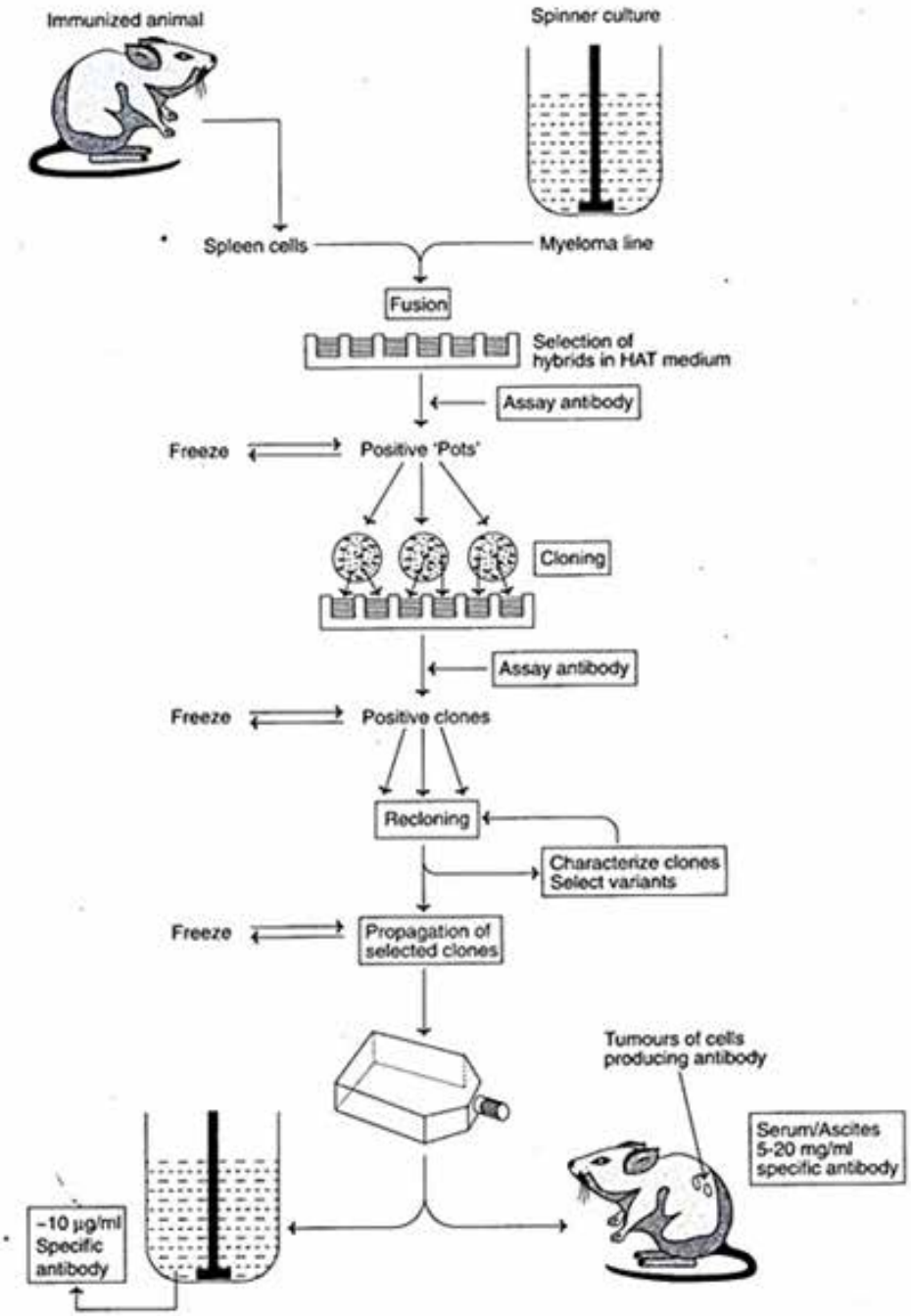
It converts hypoxanthine and guanine respectively to inosine monophosphate and guanosine monophosphate. Thymidine kinase (TK), involved in the salvage pathway of pyrimidine's converts thymidine to thymidine monophosphate (TMP). Any mutation in either one of the enzymes (HGPRT or TK) blocks the salvage pathway.

When cells deficient (mutated cells) in HGPRT are grown in a medium containing hypoxanthine aminopterin and Thymidine (HAT medium), they cannot survive due to inhibition of de novo synthesis of purine nucleotides (Note : Salvage pathway is not operative due to lack of HGPRT). Thus, cells lacking HGPRT, grown in HAT medium die.

The hybridoma cells possess the ability of myeloma cells to grow in vitro with a functional HGPRT gene obtained from lymphocytes (with which myeloma cells are fused). Thus, only the hybridoma cells can proliferate in HAT medium, and this procedure is successfully used for their selection.

Production of Monoclonal Antibodies:

The establishment of hybridomas and production of MAbs involves the following steps.



Basic Protocol for the Derivation of Monoclonal Antibodies

1. Immunization
2. Cell fusion
3. Selection of hybridomas
4. Screening the products
5. Cloning and propagation
6. Characterization and storage.

1. Immunization:

The very first step in hybridoma technology is to immunize an animal (usually a mouse), with appropriate antigen. The antigen, along with an adjuvant like Freund's complete or incomplete adjuvant is injected subcutaneously (adjuvants are non-specific potentiators of specific immune responses). The injections at multiple sites are repeated several times.This enables increased stimulation of B-lymphocytes which are responding to the antigen. Three days prior to killing of the animal, a final dose of antigen is intravenously administered. The immune-stimulated cells for synthesis of antibodies have grown maximally by this approach. The concentration of the desired antibodies is assayed in the serum of the animal at frequent intervals during the course of immunization.When the serum concentration of the antibodies is optimal, the animal is sacrificed. The spleen is aseptically removed and disrupted by mechanical or enzymatic methods to release the cells. The lymphocytes of the spleen are separated from the rest of the cells by density gradient centrifugation.

2. Cell Fusion:

The thoroughly washed lymphocytes are mixed with HGPRT defective myeloma cells. The mixture of cells is exposed to polyethylene glycol (PEG) for a short period (a few minutes), since it is toxic. PEG is removed by washing and the cells are kept in a fresh medium. These cells are composed of a mixture of hybridomas (fused cells), free myeloma cells and free lymphocytes.

3. Selection of Hybridomas:

When the cells are cultured in HAT medium (the principle described above), only the hybridoma cells grow, while the rest will slowly disappear. This happens in 7-10 days of culture. Selection of a single antibody producing hybrid cells is very important. This is possible if the hybridomas are isolated and grown individually. The suspension of hybridoma cells is so diluted that the individual aliquots contain on an average one cell each. These cells, when grown in a regular culture medium, produce the desired antibody.

4. Screening the Products:

The hybridomas must be screened for the secretion of the antibody of desired specificity. The culture medium from each hybridoma culture is periodically tested for the desired antibody specificity. The two techniques namely ELISA and RIA are commonly used for this purpose.In both the assays, the antibody binds to the specific antigen (usually coated to plastic plates) and the unbound antibody and other components of the medium can be washed off. Thus, the hybridoma cells producing the desired antibody can be identified by screening. The antibody secreted by the hybrid cells is referred to as monoclonal antibody.

5. Cloning and Propagation:

The single hybrid cells producing the desired antibody are isolated and cloned. Two techniques are commonly employed for cloning hybrid cells-limiting dilution method and soft agar method.

Limiting dilution method:

In this procedure, the suspension of hybridoma cells is serially diluted and the aliquots of each dilution are put into micro culture wells. The dilutions are so made that each aliquot in a well contains only a single hybrid cell. This ensures that the antibody produced is monoclonal.

Soft agar method:

In this technique, the hybridoma cells are cultured in soft agar. It is possible to simultaneously grow many cells in semisolid medium to form colonies. These colonies will be monoclonal in nature. In actual practice, both the above techniques are combined and used for maximal production of MAbs.

6. Characterization and Storage:

The monoclonal antibody has to be subjected to biochemical and biophysical characterization for the desired specificity. It is also important to elucidate the MAb for the immunoglobulin class or sub-class, the epitope for which it is specific and the number of binding sites it possesses.The stability of the cell lines and the MAbs are important. The cells (and MAbs) must be characterized for their ability to withstand freezing, and thawing. The desired cell lines are frozen in liquid nitrogen at several stages of cloning and culture.

ANTIGEN PROCESSING AND PRESENTATION – MHC MOLECULES

INTRODUCTION: The protection offered by physical barriers and innate immune responses are insufficient to counter the effects of an infection. Hence, specific immunity is the ultimate response in countering an infection. The specific immunity, though develops late compared to innate immune response, it offers a sustained action. Besides, this the event is also recorded (immunological memory) and subsequent infection by the same antigen is countered immediately. Development of specific immunity requires interaction of number of cells like macrophages, dendtritic cells, T cells and B cells. It also involves number of chemical mediators called chemokines. The specific immune response is generally not mounted against whole antigen. Itis generally against epitopes. Hence, a whole antigen has to be broken into epitopes and these epitopes have to be presented to antibody producing cells. This entire process is called as antigen processing and antigen presentation.

In the effective presentation of epitopes for production of either antibodies or cytotoxic T cells, two important molecules play very important role. These molecules present epitopes to T cells. These molecules are called as **Major histocompatibility molecules (MHC molecules)**. There are two types or MHC molecules – MHC Class I molecules and MHC Class II molecules. Presence of these molecules on APCs is controlled by highly polymorphic genes called **Major histocompatibility complex (MHC).** The helper T cells and cytotoxic T cells that play an important role in production of antibodies and cytotoxic T cells respectively will accept epitopes when they are presented to them for recognition by MHC molecules. The T cells generally do not recognize free flowing antigen or any other form of antigen. Hence, MHC molecules are essential for specific immunity.

STRUCTURE OF MHC MOLECULES: The MHC genes and their products are grouped into 2 classes on the basis of their chemical structure and biological properties. The two MHC proteins have a similar secondary and tertiary structure with subtle functional differences.

MHC Class I molecules:

MHC I molecules are generally found on the surface of all nucleated cells. It is composed of two polypeptide chains alpha (α) and beta (β). The alpha chain is a heavy chain (45 kD) and beta chain is a light chain called **ß2- microglobulin** (12 kD). The alpha chain is attached to the cell whereas the beta chain is not attached to the cell and is bound non-covalently with the alpha chain. There are four discrete domains in a MHC I molecules

a. External polypeptide binding domain
b. Immunoglobulin like domain
c. Transmembrane domain
d. Cytoplasmic domain

The external polypeptide-binding domain is made up of 180 amino acids with two homologous segments called as alpha 1 and alpha 2 segments. This two segments form a cleft like structure with

approximate dimension of (25Å x 10 Å x 11 Å). This cleft can take a polypeptide of 9-11 amino acid (epitope) and presents to T cells. The immunoglobulin like domain is made up of 90 amino acids and have disulphide bonds bound a loop. The amino acids in this region are highly conserved and no variations are found between different MHC I molecules. This domain is also named as alpha 3 domains and is responsible for binding with CD8 receptor. The remaining transmembrane domain and cytoplasmic domain help in anchoring the MHC I molecules to cell surface.

MHC II MOLECULES: They are composed of two polypeptide chains (alpha and ß) of similar size (34 and 30 kD). The alpha chain is slightly larger than the beta chain. As in MHC I molecule the MHC II molecules also contains four discrete domains

a. External polypeptide binding domain

b. Immunoglobulin like domain

c. Transmembrane domain

d. Cytoplasmic domain

The external polypeptide binding site of the MHC II molecule is made up 90 amino acids and is formed by interaction of part of alpha and beta chains (alpha 1 and beta 1). The immunoglobulin like domain is made up of 90 amino acids and have disulphide bonds bound a loop. This domain is formed by part of alpha and beta chains (alpha 2 and beta2). The amino acids in this region are highly conserved and no variations are found between different MHC II molecules and are responsible for binding with CD4 receptor. The remaining transmembrane domain and cytoplasmic domain help in anchoring the MHC II molecules to cell surface.

EXPRESSION OF MHC I MOLECULE:

MHC class I molecules are widely expressed among all nucleated cells, though the level varies between different cell types. Only certain cells involved in immune responses constitutively express MHC class II molecules, though they can be induced on a wider variety of cells.

Cell Type	MHC I	MHC II
T cells	+++	Varies, inducible in some species
B cells	+++	++
Macrophages	+++	+
Dendritic cells	+++ x10	+++ x10
Granulocytes	++	-
Endothelium	++	- (inducible)
Hepatocytes	+	-
Neurons	-	-

BIOLOGICAL ROLE OF THE MHC MOLECULES:

The products of the MHC play a fundamental role in regulating immune responses. T cells must recognize antigen as a complex with MHC molecules. This requires antigen to be processed by unfolding and proteolytic digestion before it complexes with the MHC molecule. Once formed the complex of antigenic peptide and MHC are generally very stable (half-life ~ 24hrs). Thus the biological role of MHC proteins is to bind small peptides and to "present" these at the cell surface for the inspection of T cell antigen receptors.

ANTIGEN PROCESSING AND PRESENTATION BY MHC MOLECULES

INTRODUCTION:

T cells are the most important cells in production of specific immune response. There are two separate populations of T cells called helper T cells and cytotoxic T cells. The former accepts the protein epitopes presented by MHC II molecules while the latter accepts the protein epitopes presented by MHC I. MHC II molecules are found mostly in the APC whereas the MHC I molecules are found in all nucleated cells. MHC II molecules have the capacity to present epitopes processed from exogenous antigens whereas MHC I molecules present the epitopes processed from endogenous antigens.

PROCESSING OF EXOGENOUS ANTIGENS:

The processing of exogenous antigens and their subsequent presentation to helper T cells are summarized below. Most of the exogenous antigens are bacterial antigens.

1. **Internalisation of antigen:** For antigen processing to take place first the antigens should enter into APC. This usually takes place by phagocytosis, pinocytosis or endocytosis. Macrophages on their surface exhibit receptors for Fc portion of antibody. Hence oposonised bacteria can easily be taken into the cell. Besides this macrophages also exhibit receptors that bind with mannose residues of the polysaccharide on bacterial cell wall.
2. The internalised antigen becomes **localised** inside the APC in a membrane bound vesicular structure called endosome.
3. The **proteolytic processing** of proteins takes place either in endosomes or in lysosomes. Both endosomes and lysosomes have an acidic pH that is pre requisite for antigen processing. Proteolytic processing requires proteases enzymes and some of the common protease enzymes found in endosomes and lysosomes are cathespin and leupeptin. These enzymes break the peptide antigen into small peptides of 10-30 amino acids long that are capable of binding with clefts of MHC II molecules.
4. The alpha and beta chains of **MHC II molecules are synthesized and associate with each other** in endoplasmic reticulum. Certain proteins like **calnexin** that are not encoded by MHC help in proper assembly and transport of the MHC molecule. Another protein called **invariant chain** prevents the nascent or any other unfolded peptide in ER from binding to MHC II molecules. This invariant chain also directs the MHC II molecule to endosomes or lysosomes where internalised proteins are broken into small peptides. This transport occurs in vesicle like structure. It is called MHC class II compartment or MIIC in macrophages and Class II vesicle or CIIV in B cells. Inside the MIIC or CIIC the invariant chain is removed. Removal of invariant chain is essential for processed peptides of endogenous antigens to bind with cleft. This removal is facilitated by proteases. Initially these enzymes leave a small fragment of invariant chain called Class II associated invariant chain polypeptide (**CLIP**) in the cleft. But this is also removed later. Only after removal of CLIP, the MHC II molecule is ready for binding with peptides of exogenous antigens.
5. Each MHC II molecule takes a peptide of 10-30 amino acids length. Once, a MHC II molecules bind with peptides they are taken in a vesicle that fuses with plasma membrane and thus the peptides are displayed to T helper cell for binding.

PROCESSING OF ENDOGENOUS ANTIGENS:

Endogenous antigens are type of antigens that are generated inside the cells. Some of the common examples of endogenous antigens are virus proteins and cancer proteins. The antigen processing is taken care by all nucleated cells. In other words all nucleated cells act APC for endogenous antigens. The MHC molecules involved in presentation is MHC Class I molecules. Large differences are observed between exogenous and endogenous antigen processing and presentation.

1. The peptides (epitopes) that bind to MHC I molecules are proteolytically generated inside the cytoplasm. The major mechanism for the generation of peptides from endogenous antigens is proteolysis is special complexes inside the cell called **proteosomes.** These proteosomes are large multiprotein complex with a broad range of proteolytic activity. Two different types of proteosomes are found inside the cell. This classification is based on the size. One proteosome with 700 kD size appears as cylinder composed of stacked array of four inner and four outer rings with each ring composed of seven distinct subunits. The subunits of inner rings are the catalytic sites for proteolysis. The second type of proteosome is of 1500 kD in size and it is composed of 700 kD structure along with additional subunits that regulate proteolytic activity.
2. The endogenous antigens that are found inside the cell are taken to this proteosomes by a special polypeptide called **ubiquitin.** Several ubiquitins are found attached to a single endogenous antigen just like string of pearls. This process is called as **polyubiquitination.**
3. The endogenous antigens are broken into small polypeptides of 5-10 amino acids inside the proteosome complexes. This breaking resembles the act of meat grinding in meat grinder. The small polypeptides thus formed are modified structurally suitable for fixing into the cleft of MHC I molecule. Now the small peptides are ready for binding with MHC I molecules are for presentation to cytotoxic T cells.
4. The two chains of MHC I molecules are synthesized separately at the rough ER and the move to smooth ER where they are assembled into a dimer structure. This structure formation is facilitated and stabilised by calnexin, BiP and TAP proteins.
5. Then the MHC I molecules move to proteosomes, bind with small peptides and move through Golgi to the surface of cell by exocytic vesicles. At the surface of the cells they are recognized by cytotoxic cells.

T LYMPHOCYTE ANTIGEN RECOGNITION AND ACTIVATION

INTRODUCTION:

Development of specific immunity requires recognition of both endogenous and exogenous antigens by cytotoxic T cells and helper T cells respectively. Recognition by T cells results in production of either antibodies or cytotoxic T cells. The T cells can recognize epitopes only when they are presented to them by MHC molecules. They fail to recognize epitopes that are presented in some other manner or found free flowing in circulation. Thus production of specific immune response requires interaction of T cells, B cells and host of other factors.

STRUCTURE OF T CELL RECEPTOR (TCR):

The TCR is the most important receptor found on the surface of T cells. A single T cell may have more than 20,000 TCRs. But all TCRs in a single T cell are of same type. Hence, a single T cell can recognize only one type of epitope. The TCR is glycoprotein structure and a single TCR is composed

of two separate polypeptide chains that are joined together by disulphide bonds. The chains are either alpha and beta or gamma and delta.

Depending upon the chains they possess there are two types of TCR. They are TCR$\alpha\beta$and TCR$\chi\delta$. The molecular weights of different chains are Alpha – 43-49 kD, Beta – 38-44 kD, Gamma – 40-55 kD and Delta – 38-42 kD. Both chains are glycosylated. The structure of the TCR is similar to the Fab portion of an Ig. Each chain has got two loops of approximately seventy amino acids and the ends of the loops are joined by disulphide bonds. There are four domains in TCRs. They are variable, constant, transmembrane and cytoplasmic domains.

1. The variable domain has got approximately 100 amino acids and as in Ig molecules CDRs and frame work regions are found. This is the region that comes into contact with the epitope. The antigen (epitope) binding site of TCR is composed of paired CDRs that form a groove like structure. The shape of the epitope determines to which TCR it should bind.
2. The constant domain is composed of 150 amino acids and as in Ig molecules, the arrangement of amino acids do not differ between TCRs.
3. The transmembrane domain is embedded in the plasma membrane and above this is s a hinge region that helps in the flexibility of the TCR molecules.
4. The cytoplasmic domain firmly anchors the TCRs to the cell.

PRODUCTION OF ANTIBODIES

The T cell antigen recognition and activation results in several biological responses that form the foundation for humoral and cell mediated immune responses. The important aspects of T cell response to antigen are

1. Proliferation of T cells based on clonal expansion that generates sufficient antigen specific T cells to counter foreign antigens.
2. Differentiation of T cells to effector cells that perform various functions. These effector cells are characterized by their unique surface receptors.
3. Development of memory T cells that make a record of the event.

Simple binding of an antigen/epitope to a T cell does not stimulate the production of antibodies or cytotoxic T cells (CTL). It includes various steps involving signal transduction and chemical mediators. Though TCR is the most important receptor in antigen recognition it is not the only receptor that is responsible for entire action. The TCR on the surface of T cells is always found in combination with number of CD receptors like CD3, CD4 and CD8 and referred as **TCR complex**. Each CD receptor is structurally and functionally different from others.

1. **CD3**: It is complosed of four polypeptide chain (1 gamma, 1 theta and 2 epsilon chains). One gamma and epsilon chain is found on one side of TCR and one theta and one epsilon chain is found on other side of TCR. However, the epsilon chains are found next to alpha and beta chains of TCR.
2. **CD4**: It is a single chain glycoprotein and as an Ig heavy chain has got four domains and a amino and carboxy ends. The domain at amino end is V like whereas the remaining three are neither C like nor V like.
3. **CD8**: It is made up of two chains alpha and beta. As an Ig molecule it has got V like domain, transmembrance domain and a cytoplasmic tail.

Zeta chains: Apart from CD receptors, a pair of polypeptide chains called zeta chains is also associated with TCR complex. These zeta chains are joined with each other and are positioned next to theta chain of CD3 receptor. It has got a role in signal transduction.

Co-stimulators: For antibody production to take place apart from TCR complex two set of signals are essential. These signals are nothing but cytokines. One set of signal comes from APC and other set is as a result of binding between APC and T cell. The cytokines from APC are Interleukin-1 (IL-1) and Interleukin-12 (IL-12). Production of these IL-1 and IL-12 is triggered by different stimuli including phagocytosis. IL-1 stimulates the division of T cells. The role of IL-12 is not clear.

Cell adherence: In addition to TCR, CD receptors, zeta chains and co-stimulators a strong bond between T cells and APC are essential for them to interact. This binding is controlled by certain specific CD receptors. Some of them are also called as integrins.

ROLE OF THE T-HELPER (Th) CELL: The special role of the T-helper (Th-cell) in production of antibodies is described below:

1. A macrophage engulfs a virus or bacteria & breaks down their proteins. Antigenic fragments of these proteins are presented on the surface of the macrophage.
2. The **few Th-cells**, which contain receptors on their surface that recognizes a particular presented foreign antigen, **interacts with** that unique-foreign antigen. This interaction triggers a series of events that activates the Th-cell.
3. The activated Th-cells are stimulated to proliferate, producing a population of this class of Th-cells (clonal selection). The population of activated Th-cells interact physically (via ligand/ receptor interactions) with ***only*** those rare B-cells that make antibody that recognizes the **same** antigen molecules that have activated the Th-cell. During this interaction the Th- & B-cells recognize each other by their common recognition of the unique antigen. The B-cells do not require the macrophage-processed antigen as they react with the whole antigen molecule (e.g. the virus, bacteria, pollen etc.)
4. The B- and Th-cell interaction stimulates the Th-cells to produce chemicals – communication molecules (**CYTOKINES**) that, in turn, stimulate the appropriate B-cells to proliferate (clonal selection) and to **differentiate** into Ab-producing plasma cells that produce the Ab that bind the antigen that the Th-cell originally reacted to.

Therefore, the Th helper cell acts as a master control cell of the immune system. It is **required** for both the humoral and cellular immune systems to function. When Th cells ***are not present*** the host's fate is sealed because the correct B-cells will not proliferate and the correct antibody will not be produced with the result that death ensues.

THEORIES OF ANTIBODY PRODUCTION

Some of the important such old theories are provided below.

1. **Ehrlich's theory:** Paul Ehrlich proposed this theory in 1900. He hypothesised that the outer surface of certain cells in the boy has receptors that can combine with foreign proteins and toxins. When these foreign proteins or toxins enter into the body they combine with such receptors and are eliminated from the body. As a result of loss or receptors, the cells produce them at increased rate and the excess receptors are detached from the cell and distributed throughout the body. He believed that these receptors were antibodies.

2. **Haurowitz theory (Instructive theory):** Haurowitz and his colleagues proposed this theory in 1930. Their theory suggested that, the first step in production of antibody is the entry of antigen into the antibody producing cells into an intracellular template. As a result of formation of such intracellular template, the cells produce globulin molecules that are specific for the antigen. This theory did not last long since no satisfactory explanation were offered about the mechanism by which the cells convert themselves to produce specific globulins and the long lasting production of antibodies.

3. **Jerne's theory:** His theory suggested that, globulin molecules carry the receptors for any foreign antigen that enters the body but not for exposed tissue antigens belonging to the host. Upon entry into the body, the foreign antigen combines with specific globulin molecule possessing specific receptors and the cell takes this combination capable of forming antibody. These cells then produce replicas of globulin molecule receptors that are liberated as antibodies.

4. **Selective theory:** This theory mainly revolves around recognition of self and immunotolerance. Burnet and Fenner proposed this theory. They hypothesised that antigens of body have self-markers that are characteristic for each antigen. When the antibody forming mechanism comes into contact with these self-markers units during foetal life recognition units are formed so that in later life antibodies are not produced against host antigen. Thus tolerance develops against host's own tissue antigens.

5. **Clonal selection theory:** Burnet proposed this. This is considered as one of the key concepts in immunology. This theory suggests that each individual possesses a repertoire of lymphocytes and each can react with one epitope. When stimulated by an antigen, only that cell alone rapidly divides and a clone of cells develop from that. These clones are generated at random and responses to self-epitopes are destroyed during foetal life.

COMPLEMENT

The complement system is a complex system of more than 34 heat labile serum proteins that interact in a cascade; many of the early components are serine proteases, which activate each other sequentially. Most of the complement proteins are enzymes that exist in serum as inactive precursors called **zymogens,** while the remaining reside on cell surfaces. The complement system is one of the major mechanisms where humoral immunity is converted into an effective defense against infection. The complement system attack extracellular forms of pathogens (e.g. certain bacteria, especially Gram-negative bacteria and enveloped viruses). Complement can be activated by two methods - spontaneously on certain pathogens as well as after immune complex formation. When the pathogen becomes coated with complement proteins that enhances its removal by phagocytic cells or results in lysis of the bacterium. Besides these, the complement reaction also generates a variety of small, diffusible products that are released and induce localized vasodilation and attract phagocytic cells chemotactically, leading to an inflammatory reaction.

Components of complement system:

The complement system is activated in three different pathways. The three pathways of complement activation are called the **classical, the alternative, and the mannan binding lectin (MBL) pathway.** Though the complement system consists of more than 34 proteins, the major protein is called as C3. All the three pathways are directed at the single most important step in activation, the cleavage of C3. All the three pathways result in a common final pathway called as the terminal pathway, or the membrane attack complex (MAC). Some of the important components of complement system are provided below.

Components of the Classical Pathway		
Native component	**Active component(s)**	**Function(s)**
C1(q,r,s)	C1q	Binds to antibody that has bound antigen, activates C1r.
	C1r	Cleaves C1s to activate protease function.
	C1s	Cleaves C2 and C4.
C2	C2a	Unknown.
	C2b	Active enzyme of classical pathway; cleaves C3 and C5.
C3	C3a	Mediates inflammation; anaphylatoxin.
	C3b	Binds C5 for cleavage by C2b. Binds cell surfaces for opsonization and activation of alternate pathway.
C4	C4a	Mediates inflammation.
	C4b	Binds C2 for cleavage by C1s. Binds cell surfaces for opsonization.
Components of the Alternate Pathway		
Native component	**Active component(s)**	**Function(s)**
C3	C3a	Mediates inflammation; anaphylatoxin.
	C3b	Binds cell surfaces for opsonization and activation of alternate pathway.
Factor B	B	Binds membrane bound C3b. Cleaved by Factor D.
	Ba	Unknown.
	Bb	Cleaved form stabilized by P produces C3 convertase.
Factor D	D	Cleaves Factor B when bound to C3b.
Properdin	P	Binds and stabilizes membrane bound C3bBb.
Components of the Membrane-Attack Complex		
Native component	**Active component(s)**	**Function(s)**
C5	C5a	Mediates inflammation; anaphylatoxin, chemotaxin.
	C5b	Initiates assembly of the membrane-attack complex (MAC).
C6	C6	Binds C5b, forms acceptor for C7.
C7	C7	Binds C5b6, inserts into membrane, forms acceptor for C8.
C8	C8	Binds C5b67, initiates C9 polymerization.
C9	C9n	Polymerizes around C5b678 to form channel that causes cell lysis.

The activity of complement system is referred as **complement cascade** meaning sequential action of these proteins in making the products and complexes that carry out the functions of complement. There are **early** and **late** components of the complement cascade that act at different points in the cascade. The early components and C3 are **proenzymes** that are activated sequentially by limited proteolytic cleavage. Cleavage of each proenzyme in the cascade sequence "activates" that component so that it can cleave the next proenzyme in the cascade, which then becomes active, etc. Since one enzyme can activate many molecules, the early components of complement form an **amplifying**

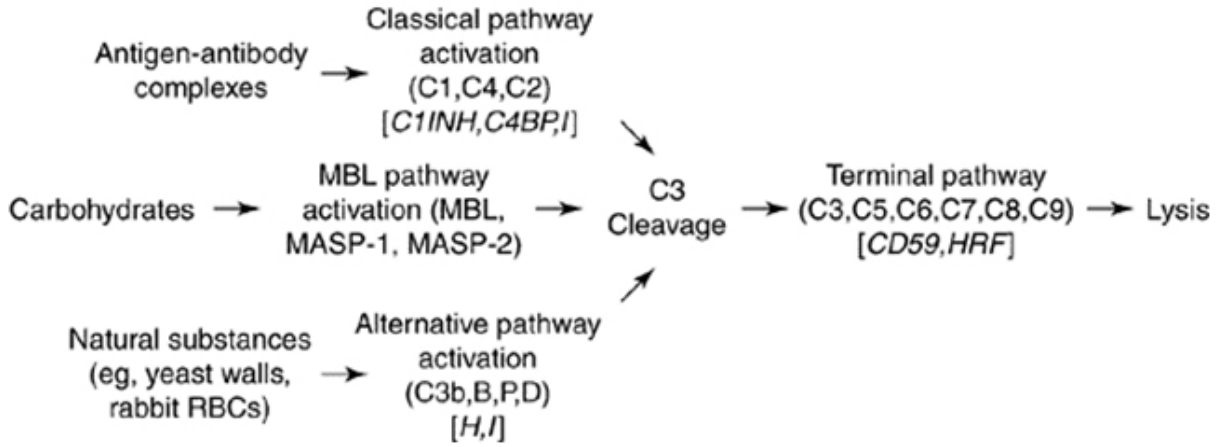
proteolytic cascade, with each proteolytic step involving more molecules.

Nomenclature:

A particular pattern of naming is followed for the components. The classical pathway components are labelled with a C and a number (eg, C1, C3). Because of the sequence in which they were identified, the first four components are numbered C1, C4, C2, and C3. Alternative pathway components are lettered (eg, B, P, D). Some components are called factors (eg, factor B, factor D). Activated components or complexes have a bar over them to indicate activation (eg, C1, C1r, C3b,Bb). Cleavage fragments are designated with a small letter after the component (eg, C3a and C3b are fragments of C3). Inactive C3b is designated iC3b. Polypeptide chains of complement proteins are designated with a Greek letter after the component (eg, C3 and β are the and β chains of C3). Cell membrane receptors for C3 are abbreviated CR1, CR2, CR3, and CR4.

Pathways of complement system:

The three pathways of complement activation are Classical pathway, Alternative pathway and Mannan binding lectin (MBL) pathway.All these pathways end in a common final pathway called the terminal pathway, or the membrane attack complex (MAC). The MAC is responsible for the lysis of extracellular pathogen by forming pores in microorganism's membrane.



Classical pathway

Classical pathway of complement activation is initiated by the binding of *IgG (IgG1, IgG2 or IgG3) or IgM*to antigen. After antibody binds to antigen, conformational changes occur in constant region of the antibody, allowing C1q to bind. (C1q binds to a single IgM molecule or aggregated IgG.). C1 is a pentameric molecular complex consisting of a single C1q, two C1r and two C1s peptides. The classical pathway can also be activated by mechanisms independent of Ab. Heparin (a polyanionic anticoagulant) and protamine (a polycation that is used to block heparin), when present in equimolar concentrations, can activate the classical pathway. Various other polyanions (eg, DNA and RNA) are thought to be able to react directly with C1q to activate the classical pathway.

Steps in classical pathway:

1. Binding of C1 to antibody causes a conformation change in C1q. This results in the activation C1r that then activates C1s.
2. *Activated C1s cleaves C4 into two fragments,* C4a (soluble, weak anaphylotoxin) and C4b (bound to cell surface, opsonin).
3. *C4b binds C2.* C2 is cleaved by activated C1s to form C2b (remains bound to C4b) and C2a (soluble).

4. **C4b2b is the classical pathway C3 convertase.** *C3 convertase (C4b2b) cleaves C3 into C3a (soluble, anaphylotoxin) and C3b.* C3b either binds to membrane or remains bound to C3 convertase (C4b2b) resulting in formation of C5 convertase (C4b2b3b).
5. C5 convertase goes on to form membrane attack complex.

Alternate pathway

Alternative pathway is an antibody independent pathway of complement activation. There are low levels of soluble C3b in serum. If this should come in contact with a foreign surface, such as a bacteria cell membrane, C3b with covalently bind and initiate the alternative pathway ofcomplementactivation.
The alternative pathway can also amplify the effect of the classical pathway. The classical pathway C3 convertase generates not only C5 convertase but also membrane bound C3b (not complexed to C3 convertase, C4bC2b). The membrane bound C3b generated by the classical pathway also initiates the alternative pathway.

Steps in alternative pathway:

1. Surface bound C3b binds Factor B to give C3bB (membrane bound).This is a substrate for Factor D which cleaves Factor B to release a small fragment, Ba (soluble) and leaves C3bBb (membrane bound).
2. C3bBb is labile but is stabilized by binding of properdin (P) to the complex (C3bBbP). C3bBbP is the alternative pathway C3 convertase.
3. C3 convertase cleaves C3 into C3a (soluble, anaphylotoxin) and C3b. C3b binds to membrane (opsonin) or remains associated with C3 convertase. The binding of C3b to C3 convertase results in formation of C5 convertase (C3bBb3b).
4. C5 convertase goes on to form membrane attack complex.

Mannan binding lectin (MBL) pathway

C-reactive protein is capable of leading to classical pathway activation without the presence of Abs. C1 bypass pathways have also been described, which do not use components of the classical pathway but result in C3 cleavage. One of these has been characterized as the MBL pathway. The mannan binding lectin (MBL) pathway depends on innate recognition of foreign substances (ie, carbohydrates) for activation. This pathway has structural and functional similarities to the classical pathway. MBL is similar to C1q, and MASP-1 and MASP-2 appear to be similar to C1r and C1s in the classical pathway, respectively. Thus, MASP-2 may cleave C4 and lead to formation of MBL pathway-derived C3 convertase.

Terminal pathway:

The final phase of complement activation is the formation of the membrane attack complex (MAC). MAC is formed by non-enzymatic assembly of the terminal complement components (C5b, C6, C7, C8,C9).

Steps in formation of MAC:

1. C5 convertase (classical or alternative) cleaves C5 into C5a (soluble, anaphylotoxin) and C5b (membrane bound).
2. C5b binds C6, forming C5b6 and this binds C7 to form C5b67. C7 inserts into the membrane. Up to this point, the complement components have bound to surface of membrane.

3. C8 binds to this complex (C5b678), also inserting into membrane.
4. There is a stepwise addition of up to fourteen C9 molecules (C5b-9), resulting in the formation of a pore in the membrane. This "hole" results in lysis of cell.

Functions of complement system: The functions of complement include:

1. Make bacteria more susceptible to phagocytosis
2. Directly lyse some bacteria and foreign cells
3. Produce chemotactic substances
4. Increases vascular permeability
5. Cause smooth muscle contraction and promote mast cell degranulation

HYPERSENSITIVITY

DEFINITION:

A normal immune system reacts when the body identifies a foreign material. This foreign material, which invokes the immune response, is called an antigen. The body can react to the antigen by producing glycoprotein molecules called antibodies, which bind the antigen. The combination of the antibody bound or attached to the antigen is called an immune complex. A hypersensitive immune system is one that over reacts to a stimulus. Besides, antibodies, various cells like mast cells can also be activated, which produce chemicals such as histamines that can affect the body. In hypersensitivity, the body produces too much of antibody, the wrong kind of antibody, a large number of antigen-antibody complexes, or antibody to proteins that aren't really foreign. In addition, an excessive number of cells may be activated to produce histamine and other chemicals.

CLASSIFICATION: There are four major types of hypersensitivity. They are

1. Type I (Immediate) Hypersensitivity
2. Type II (Antibody mediated) Hypersensitivity
3. Type III (Immune complex mediated) Hypersensitivity
4. Type IV (Delayed) Hypersensitivity

TYPE I (IMMEDIATE) HYPERSENSITIVITY:

The reaction of the immune system in Type I reaction is immediate and severe. The symptoms are mostly due to cells over-reacting and releasing very large amounts of chemicals like histamine and other vaso-active amines. The Type I reactions are generally termed as **allergy.** Different types of Type I reactions are commonly seen. Some of the important reactions are

1. **Anaphylaxis** is a rare, life-threatening, immediate allergic reaction to something ingested or injected. If untreated, it results in shock, respiratory and cardiac failure, and death. Stinging insects, antibiotics, vaccines, certain hormones and medications, and foods can cause anaphylaxis in susceptible animals. The most common symptoms are the sudden onset of diarrhoea, vomiting, shock, seizures, coma and death. The animals' gums will be very pale, and the limbs will feel cold. The heart rate is generally very fast but the pulse is weak. There is no facial swelling.
2. **Fleabite allergy** or **fleabite dermatitis** is a seasonal allergy that is worse during peak flea times. The symptoms of fleabite allergies appear to worsen in the summer and fall. Animals that have flea allergies will bite at the base of their tail and scratch frequently. In addition fleas or flea dirt

(faces) can be found on the dog the majority of the time. Besides fleas, the faeces of fleas also act as allergen. Severely affected dogs may itch over their entire bodies, have generalized hair loss, and red inflamed skin. Hot spots are often a result of flea bite allergies.

3. **Urticaria**and **angiooedema** are hypersensitivity (or allergic) reactions to drugs, chemicals, something eaten, or even to sunlight. In general, both urticaria and angiooedema are not life threatening and will go away by themselves. In urticaria, small are as of swelling occur within the skin. Often the hair will standup erect over these swellings. Sometimes itching is noticed in these areas. Angiooedema is characterized by swelling of the face, especially the muzzle and around the eyes. Sometimes the swelling is so severe; the animal cannot open its eyes. Angiooedema often results in itching. Angiooedema and urticaria generally develop within 20 minutes of being exposed to the allergen (substance to which the animal is allergic). In general, both urticaria and angiooedema are not life threatening and will go away by themselves. Rarely the swelling in angiooedema can affect the throat and make breathing difficult.

4. Type I reaction to things that are inhaled such as pollens and cat dander is called as **atopy**, **flea allergy dermatitis** and other antigens which evoke symptoms of an allergy within minutes of when the person or animal was exposed.

MECHANISM:

Type I reactions are mainly mediated by IgE antibodies. For Type I reactions to occur allergens and IgE are essential. Allergens could be either proteins or chemicals attachedwith proteins. IgE are produced by resting B cells that switches from IgD/IgM to IgE. This reaction is stimulated by IL4, IL5, IL10 and IL13. IgE can bind with number of cells through their Fc portion. But the most important cells that are responsible for Type I reactions are mast cells, eosniophils and basophiles. Of these three cells mast cells are the most important cells. There are two receptors are binding ofFcportion of IgE to these cells. One is a high affinity binding receptor and other is a low affinity binding receptor. Mast cells are large round cells distributed throughout the body in connective tissue. They are characterized by cytoplasm packed with large granules. The nucleus is bean shaped and is usually masked by granules. There are two populations of mast cells that vary in structure and functions. The usual allergic reactions are due to histamines and other amines that are present in the granules. These amines are released into the circulation as a result of degranulation. Degranulation of granules does not take place with mere binding of IgE to mast cells. After binding of IgE to mast cells, a specific allergen should join with IgE and should cross- link one or two IgEs. This activates the mast cells. The activated mast cells produce three important classes of mediators biogenic amines (histamine), lipid mediators (prostaglandin D_2, leukotrines C_4, D_4 and E_4, platelet activating factors) and cytokines (TNF, IL4, IL5). The biogenic amines and lipid mediators produce vascular leakage, vasodilation and bronchoconstriction and the cytokines mediate the late phase reactions.

DIAGNOSIS:

1. Based on the symptoms, their relation to the environment and to seasonal and situational variations, their clinical course, and the family history of similar problems should yield sufficient information.

2. **Nonspecific tests:** These tests are aimed in identifying the increased number of eosinophils, histamines and other vaso-active amines in theblood.

3. **Specific tests:** Specific tests are used to confirm sensitivity to a particular allergen or allergens. **Skin tests** are the most convenient way to confirm specific sensitivity. They should be selective and based on clues provided by the history. Test solutions are made from extracts of inhaled,

ingested, or injected materials (eg, wind-borne tree, grass, and weed pollens; house dust mites; animal danders and sera; insect venoms; foods; and penicillin and its derivatives). Until recently, few allergen extracts were standardized and their potency was highly variable. Many commonly used extracts are nowstandardized.

a. **Prick (puncture) test** is usually performed first. A drop of a dilute allergenic extract is placed on the skin, which is then pricked or punctured through the extract.

b. **Intradermal test,** just enough dilute sterile extract is injected (using a 0.5- or 1- mL syringe and a #27 short-bevel needle) to produce a 1- or 2-mm bleb. Each set of skin tests should include the diluent alone as a negative controlandhistamine (10 mg/mL of the base for the prick test or 0.1 mg/mL for the intradermal test) as a positive control. A skin test is considered positive if it produces a **wheal and flare** reaction in 15 min with a wheal diameteratleast 5 mm larger than the control.

c. **Radioallergosorbent test** (RAST) may be performed when direct skin testing is impossible because of generalized dermatitis. A RAST detects the presence of allergen-specific serum IgE. A known allergen, in the form of an insoluble polymer-allergen conjugate, is mixed with the serum to be tested. Any IgE in the serum that is specific for the allergen will attach to the conjugate. The quantity of allergen-specific IgE in the patient's circulation is determined by adding ^{125}I- labeled anti-IgE antibody and measuring the amount of radioactivity taken up by theconjugate.

d. **WBC histamine release,** an in vitro test, detects allergen-specific IgE on sensitized basophils by measuring allergen-induced histamine release from the patient'sWBCs.

e. **Provocative challenge** may be performed when a positive skin test raises a question about the role of the particular allergen in the production of symptoms. The allergen may be applied to the eyes, nose, orlungs.

TYPE II (ANTIBODY MEDIATED) HYPERSENSITIVITY:

The Type II hypersensitivity is manifested as cell or tissue destruction that is mediated by the presence of antibodies directed against cell or tissue antigens. Most of what we think of as autoimmune or immune-mediated diseases falls into this category of hypersensitivity reaction. Antigens may be either endogenous or exogenous. Endogenous antigens are "self" antigens such as erythrocytes, platelets, leukocytes, basement membranes, lens and spermatozoa, to name a few. Exogenous antigens are those that are adsorbed or attached to cell membranes that because of their close association with membranes result in cell or tissue destruction when an immune response is mounted against them. The primary immunoglobulin classes involved are IgG and/or IgM.

MECHANISM:

There are two mechanisms by which Type II reactions manifested. They are cytotoxicity and antibody-dependent cellular cytotoxicity (ADCC). In the first mechanism the primary humoral substance involved is complement, the activation of which can result in direct membrane damage and cell lysis without effector cell participation. Complement can also enhance effector cell function through opsonisation of targets by C3b and accelerating the events of cell cytotoxicity. The second mechanism involves several classes of lymphocytes such as K (killer) cells, NK (natural killer) cells, Cytotoxic T cells and myeloid cells (including mononuclear phagocytes, neutrophils, eosinophils). All these cells have Fc receptors for binding to antibody coated cells or substances.

IMPORTANT DISEASE MANIFESTATIONS:

- **Auto-haemolytic anaemia of dogs** in which there is decline in the number of erythrocytes and platelets.
- **Goodpasture's syndrome**, which is due to reaction of antibody with glomerular and alveolar basement membrane endothelium.
- **Other diseases** include Coombs'-positive haemolytic anaemia, antibody-induced thrombocytopaenic purpura, leukopaenia, pemphigus, pemphigoid and pernicious anaemia.

DIAGNOSIS:

1. **Coombs Test (direct antiglobulin test)** and **γ-globulin tests** detect antibody and complement on RBCs, respectively. These tests use rabbit antisera, one to immunoglobulin (Ig) and the other to complement. When these reagents are mixed with RBCs coated with Ig or complement, agglutination occurs. Antibodies eluted from these cells show both a specificity for RBC blood group antigens and an ability to fix complement, thus showing that they are true autoantibodies and account for the complement present on the RBCs in the direct γ-globulin test.
2. Indirect antiglobulin test
3. **Fluorescence microscopy**, which is most commonly used to detect Ig or complement in tissue (by the direct technique) and also can be used to determine the specificity of a circulating antibody
4. **Antireceptor tests** to detect antibodies against surface receptors.

TYPE III (IMMUNE COMPLEX MEDIATED) HYPERSENSITIVITY:

Antigen-antibody complexes that form in the circulating or interstitial fluids deposit in tissues where they can activate complement and other inflammatory mediators, leading to tissue damage. The antigens that elicit this type of hypersensitivity reactions may be either exogenous or endogenous. The immunoglobulins capable of fixing complement are usually involved in type III reactions. These are primarily IgG and IgM. However, IgA may be involved in a clinically important immune complex disease in horses, purpura haemorrhagica. The major cells involved in Type III reactions are those of the acute inflammatory response, namely mast cells, basophils, neutrophils, platelets, and macrophages.

MECHANISM:

Complement is the primary mediator of type III reactions. The antigen-antibody complexes activate complement. Activated complement components initiate immune complex deposition, tissue damage, and increased vascular permeability, stimulate neutrophil chemotaxis and accumulation, cause platelet aggregation, and stimulate mast cell degranulation. Other mediators include vasoactive amines (from mast cell degranulation), kinins, and leukocyte products (that produce local tissue damage). The antigen-antibody-complement complexes (immune complexes) become lodged in vessel walls and incite an acute inflammatory response.

IMPORTANT DISEASE MANIFESTATIONS:

1. **Serum sickness** is a generalised infection that occurs when an animal forms antibodies to xenogeneic immunoglobulins (Ig from other species of animals) administered during passive immunisation. This method includes the use of horse sera to treat snake-bites and murine monoclonal antibodies to treat cancer and suppress graft rejections. The resultant immune complexes become localized in the kidneys and joints where they cause glomerulonephritis and arthritis by activating complement.
2. **Arthus reaction** is a local reaction and is observed when boosters are administered to animals that have already possessed high antibody titres to vaccine molecules. The reaction takes place at a local

site in and around the walls of small blood vessels; it is most frequently demonstrated in the skin. An Arthus reaction produces an area of oedema (and perhaps haemorrhage) of 50 mm or more with a poorly defined edge. The reaction peaks 5-12 hours after subcutaneous or intradermal antigen injection. The affected area contains antigen, antibody, complement, neutrophils, mononuclear phagocytes, plasma cells and platelets; the latter may clump, leading to vascular occlusion and necrosis.

3. **In Persistent infections,** the combined effects of a low-grade persistent infection and a low- affinity antibody response lead to chronic immune complex formation and deposition in tissues, especially the kidneys. Diseases with this etiology include leprosy, malaria, dengue, viral hepatitis and staphylococcal infective endocarditis.
4. **Inhaled antigens from moulds, plants and animals** are found complexed to IgG within the alveoli of the lungs. Complement fixation leads to cell accumulation, inflammation and fibrosis. Two forms of the disease are **farmer's lung**, where there are circulating antibodies to actinomycete fungi found in mouldy hay, and **pigeon fancier's lung**, where precipitating antibodies form to the fungus *Micropolyspora faeni*.

TYPE IV (DELAYED) HYPERSENSITIVITY:

Type IV hypersensitivity is also known as **delayed- typed hypersensitivity (DTH)** or **cell- mediated hypersensitivity**. Antibody plays no role in Type IV reaction and is mediated by inflammatory Th 1 cells that have been sensitised by a previous encounter with an antigen. The Th 1 cells act by recruiting other cell types (especially additional T cells and macrophages) to the site of the reaction. Antigens that elicit type IV reactions are those that cannot be removed by the acute inflammatory response. This can include organisms such as viruses, fungi, protozoa, parasites, certain bacteria; chemicals; or transplanted (foreign) tissues.

MECHANISM:

Following initial exposure to an antigen, a specific pool of memory T cells remains. Re-exposure to the antigen causes rapid proliferation of the specific T cells at the site of antegen entry. T cells mediate type IV reactions through production of lymphokines, and the generation of cytotoxic T cells. Macrophages also play an important role here since they are necessary to present processed antigen (in association with MHC II) to T cells in order to stimulate the T cell proliferation. Besides, macrophages have numerous inflammatory functions that contribute significantly to Type IV reaction. In certain type IV reactions, basophils are the predominant cell type present (cutaneous basophil hypersensitivity), and eosinophils can be a component in a few type IV reactions. The major humoral mediators in type IV reactions are cytokines (interleukins and other cytokines) from both lymphocytes and macrophages. These play a major role in the regulation of the cell response and recruitment of further inflammatory cells to the affected focus.

There are three important stages in the development of Type IV reactions. They are sensitisation, elicitation and suppression of inflammatory reaction.

1. **Sensitisation:** Haptens adsorbed to body proteins are processed by special cells leave the epidermis and migrate as veiled cells via the lymphatics to the paracortical areas of regional lymph nodes. There they present processed hapten-protein conjugates to **Th1** lymphocytes, producing a population of memory T cells. The entire sensitisation process takes about 10-14 days.
2. **Elicitation:** Upon subsequent contact with the allergen, Langerhans' cells present the allergen to memory TH1 cells. The activated T cells release a variety of factors that recruit and activate

macrophages to the affected skin. The most important chemokines produced by the T cells are **migration inhibition factor (MIF)** and **macrophage chemotactic factor (MCF)**. The T cells also produce **IFN**, which activates **keratinocytes** to produce a number of proinflammatory cytokines. TNF and increase adhesion molecule expression and are directly cytotoxic for some tissue cells. The number of cells infiltrating the epidermis peaks at 48-72 hours. Most infiltrating lymphocytes bear CD4, but a few express CD8.

3. **Suppression of inflammatory reaction:** Suppression is mediated by prostaglandin E (from macrophages and keratinocytes) and TGF (from keratinocytes and T lymphocytes), cytokines that interfere with the production or activities of IL-1 and IL-2. In addition, ultraviolet light tends to inactivate the Langerhans' cells, and the allergen undergoes natural degradation.

DIFFERENT MANIFESTATIONS OF TYPE IV REACTIONS: There are three different forms Type IV reactions. They are contact hypersensitivity, tuberculin type hypersensitivity and granulomatous hypersensitivity.

1. **Contact hypersensitivity**: is primarily an epidermal reaction characterized by eczema at the site of contact with an allergen. The reaction is usually maximal 48-72 hours after contact. Common allergens that cause contact hypersensitivity include pentadecylcatechol (found in poison ivy), nickel, chromate, and chemicals found in rubber. These substances are generally haptens that would not themselves be antigenic; however, they penetrate the epidermis and become conjugated to normal body proteins and become antigenic.

2. **Tuberculin type hypersensitivity**: Tuberculin reactions are characterized by an area of **firm red swelling** of the skin that is maximal at **48-72** hours after challenge. The infiltrate of the tuberculin reaction is composed of monocytes (80-90%), CD4 T cells, CD8 T cells, and cells similar to Langerhans' cells. Macrophages are probably the main antigen- presenting cells. The lesion usually resolves within 5-7 days but if there is persistence of antigen, it may develop into a granulomatous reaction.

3. **Granulomatous hypersensitivity** is clinically the most important form of type IV hypersensitivity. The maximal reaction time for the development of a granuloma is **21-28 days.** The pathological changes result from the inability of the macrophage to destroy phagocytosed pathogens or to degrade large inert antigens. Where prolonged antigenic stimulation exists, activated macrophages may differentiate into **epithelioid cells** and then into **multinucleate giant cells** in granulomatous tissue.

SUMMARY OF HYPERSENSITIVITY REACTIONS

Properties	Type I	Type II	Type III	Type IV
Immune reactant	IgE	IgG, IgM	IgG, IgM	T lymphocytes
Antigens	Heterologous (nonself)	Autologous cells	Auto or heterologous	Auto or heterologous
Complement	No	Yes or no	Yes	No
Cells involved in initiation	Mast cells & basophils	None	None	APC, T H 1 cells
Inflammatory cells attracted or activated	Eosinophils	Neutrophils	Neutrophils	Macrophages, keratinocytes, Tc

Mechanism	Ag + IgE + mast cells = mediator release	Ig + tissue Ag = complement activation, lysis, opsonisation	Ig-Ag deposits = C', mast cell, and neutrophil activation	Cell-mediated cytotoxicity, mediator release, macrophage activation
Target tissue	Vascular endothelium, bronchial smooth muscle	Blood or tissue cells	Vascular endothelium and epithelial cells	Modified self, infected cells, allografts
Mediators	Early: histamine, ECF, PAF, heparin, enzymes Late: SRS-A, leukotrienes, thromboxanes, prostaglandins	C3a, C5a, chemotaxis, neutrophil proteases	C3a, C5a, chemotaxis, neutrophil proteases	MCF, MIF, IFN γ , TNF-α, GM-CSF
Primary results	Vascular permeability, transudation, erythema, oedema, wheal, bronchospasm	Cell lysis, phagocytosis, inflammation, hypocom-plementemia	Vasculitis, vasoconstriction, thrombi, necrosis, hypocom-plementemia	Erythema, induration, granuloma formation
Treatment	Avoidance, antihistamines, epinephrine, catecholamines, methylxanthine, cromolyn, hyposensitization	Immunosuppressants, corticosteroids, cyclophosphamide, azathioprine, plasmapheresis	Immuno suppressants, corticosteroids, cyclophosphamide, azathioprine	Anti-inflammatory corticosteroids, surgery
Examples	Drugs (e.g. penicillin), hayfever, insect venom, foods, vaccines	Drug-induced nephritis, rheumatic fever, erythro blastosis fetalis, pemphigus complex	Arthus reaction, serum sickness, SLE, RA , glomerulonephritis, vasculitis	RA, type I diabetes, contact dermatitis, allograft rejection, poison ivy, TB
Beneficial effects	Parasite immunity	Kills pyogenic and Gram+ bacteria by lysis, opsonisation; toxin & virus neutralization	Neutrophil and PMN activation	Immunity to M. tuberculosis, viruses, other intracellular pathogens

VACCINES AND METHODS OF VACCINATION

HISTORY:

The term vaccine is derived from the Latin word *vacca*meaning cow. Edward Jenner developed the first vaccine for small pox. He used the scab material of cowpox lesions to immunise individuals against small pox. Since then numbers of vaccines have been developed. Some of the important contributors in vaccinology include Louis Pasteur, Salmon etc.

CLASSIFICATION: Vaccines could be broadly divided into live and inactivated vaccines. There are different types of vaccines under these categories.

1. **Live vaccines:** In this category, the live organisms are given as protective agent. Live vaccines are given in different forms. Some of the important forms in which the live vaccines are given are listed below. The live vaccines produce good immunity. Besides this, the immunity produced is long lasting since memory cells are produced. In the phase of outbreak, live vaccines (avirulent strains) can offer some protection since they induce the production of interferon. They do not require adjuvants and do not stimulate hypersensitivity reactions. The main problems associated with this method are reversion of virulence in attenuated vaccines and unwanted post-vaccinal reactions. Storage of live vaccine requires proper storage conditions like low temperature (cold chain maintenance).
    - **Attenuated vaccines:** In this type, the living organisms given as vaccines are immunogenic but not virulent. The virulent organisms are made to loose their virulence by passaging them repeatedly in a heterologous host or a cell culture system for a long time. The process of reduction in virulence is called as attenuation. E.g Tissue culture rinderpest vaccine.
    - **Less virulent organisms:** Certain organisms like Newcastle disease virus has number of strains that differ in virulence but not in immunogenicity. For such diseases, the less virulent strains are given as vaccines. E.g the lentogenic LaSota and F strain vaccines for Newcastle disease. However, this type of vaccine will not hold good for organisms whose strains are immunologically distinct E.g infectious bronchitis and foot and mouth disease.
    - **Avirulent organisms:** Certain strains of Newcastle disease virus like the Australian V4 strain are pathogenic but immunogenic. Such strains can be given as vaccines. In the same way a pathogenic *Brucella abortus* strain 19 is given as vaccine for brucellosis.
    - **Genetically related organisms:** Few organisms have some similarity in the immunogenic epitopes. For example, Marek's virus is pathogenic for chickens. It has got some similarity to Herpes virus turkey (HVT), which is apathogenic for chickens. Because of the similarities between Marek's virus and HVT, when HVT is given to chickens the antibodies produced against HVT neutralise Marek's virus. Hence, HVT is given as vaccine for Marek'sdisease.
    - **Living virulent organisms:** This method is no longer practiced. It is same as giving the powder from dried scabs of small pox to normal individual who are at risk. It is discontinued due to the risks associated with this method.
2. **Inactivated vaccines:** These vaccines are also referred as killed vaccines. The living organisms are inactivated usually by a protein-denaturing agent. Formalin and Beta propiolactone are the common inactivating agents. The inactivated organisms lose their virulence but retain the immunogenicity. The inactivated vaccines are given as such or mixed with adjuvants. An adjuvant is a substance, when given with an immunogenic component augments the immune response.

Adjuvants provide a depot action. The most common adjuvants are aluminium hydroxide and mineral oil. Of the two, the approved adjuvant is aluminium hydroxide. The inactivated vaccine is usually free from the side effects and the immunity offered is good in primed animals. It is also devoid of unwanted post-vaccinal immune response. Storage of vaccine is easy.

3. **Genetically engineered vaccines:** This category of vaccines includes organisms whose genome is altered in such a way that they offer good protection. There are three important categories named as Category I that includes inactivated recombinant organisms or purified antigens derived from recombinant organisms, category II that includes live organisms in which the pathogenic genes are deleted and category III that includes live expression vectors containing heterologous immunogenic genes.

   - **Recombinant vaccines (Category I):** In this method the genes coding for the immunogenic proteins are identified and cut. The excised gene is then inserted into plasmids (cloning vectors) and later the plasmids are allowed to express in expression vectors like *E.coli.* The product of the gene the immunogenic proteins are separated and used as vaccines. Such recombinant vaccines are available for FMD, feline leukaemia etc.

   - **Deletion mutant vaccines (Category II):** In this category the genes that are responsible for pathogenicity are removed from the organisms and only essential genes like genes required for replication and immunogenicity are left. Such genetically modified organisms offer good protection but cannot produce an infection. E.g Vaccine for pseudorabies.

   - **Live recombinant organisms (Category III):** In this method, the vaccinia virus is used as vector. The genome of Vaccinia virus is very big. In the genome certain genes that are not essential for important functions of virus are removed and instead of that immunogenic genes of some other organisms are inserted. Such genetically modified vaccinia virus is used as vaccine.

4. **Naked DNA:** In this method the naked DNA of an organism is injected directly into the cell. The products of the genes are processed and presented to T cells for immune response.
5. **Synthetic peptide vaccines:** The immunogenic proteins can be synthesized in laboratory if the gene sequences are known. Such synthesized proteins can be used as vaccines.
6. **Anti-idiotype vaccines:** The binding site of immunogenic molecule (idiotype) has complementary structure to that of inducing epitope. Antibodies to this idiotype will have the same three-dimensional structure of inducing epitope. Hence, the anti-idiotype antibodies can be used as vaccines.

IMMUNOASSAYS – SEROLOFICAL TESTS

DEFINITION:

The assays that are used to measure the immune responses are called immunoassays. An immunoassay involves antigen and antibody as basic reagent besides other reagents that are of minor significance. In an immunoassay the ability of a specific antibody to bind with its antigen (epitope) is assessed. All the immunoassays are broadly classified into three types.

1. **Primary binding tests** – In these tests the binding of antigen to an antibody is measured directly. These tests are very specific and sensitive butrequire sophisticated equipments and involve cumbersome procedures. E.g. RIA, ELISA, IFA etc.

2. **Secondary binding tests** – In these tests, the results of antigen-antibody interaction (agglutination, precipitation, fixation of complement) *in-vitro* are measured. These tests are simpler to perform but are less sensitive and specific. E.g. HI, AGID, CFT etc.

3. **Tertiary binding tests** – These tests are *in-vivo* tests and require a living system. In these tests, the effects of actual protective effects of antibodies are measured in living system. These tests involve time consuming lengthier procedures but are specific and sensitive. E.g. Neutralisation assay

The ability of an assay to detect only the target and not any other is referred as **specificity** of the test. The **sensitivity** of an assay refers to ability of the test system to detect very minute amount of the target. An assay is considered as ideal, when it has got high specificity and sensitivity. At the same time all the assays are prone for certain types of errors. The error associated with an assay is broadly classified into two types – **random** and **systemic errors**. Random error is due to wrong dilution of reagents, variation between different pipetting, certain procedural mistakes etc. Random error affects the specificity and sensitivity of the result. Systemic errors are also called as **bias.** This is due to a mistake that is done throughout the assay. This may move the results of assay either to the right or to the left. Both errors are not desirable in an assay.

PRIMARY BINDING TESTS:

A primary assay involves antigen and antibody and in these reactions the antigen-antibody complex are measured directly. Attaching a reacting group like enzymes, radioisotopes, fluorescent dyes to either antigen or antibody, generally does identification of this complex. As mentioned earlier, these assays are specific and sensitive but involved costly equipments and lengthier procedures. Some of the commonly performed primary binding assays are

- Radioimmunoassay
- Enzyme immunoassay
- Fluorescent immunoassay

1. **Radioimmunoassay (RIA):** This assay involves identification of antigen-antibody complex by attaching a radioisotope to antigen. This test is mainly done to identify antigen. However, it can also be modified to identify antibodies also. Of the various isotopes that are used, the isotope of iodine ^{125}I is preferred for its half-life, relative risk- freeness to the handlers and their ability to bind with proteins. The basic principle of RIA is the competition between antigen bound to isotope called tracer and free (unbound) antigen for antibodies. In such competitions the antigen that is not bound with any isotope always bind better than the tracer. Hence, in a positive reaction the gamma counts will be less and gamma counts will be more in negative reaction. Of all the different primary immunoassays, RIA is considered as highly specific and sensitive. RIA is considered as a limited reagent test since only minimum amount of tracer and free antigen are used. However, in a modification of RIA called immuno radiometric assay (IRMA) excess reagents are used to increase the specificity and sensitivity. When RIA is done to identify, IgE is allergic animals it is called as radio allergosorbant test (RAST). Of late, RIA methods are losing their importance due to health risks and problems in disposing the radio-wastes.

2. **Enzyme immunoassays (EIA):** In this method, an enzyme is attached to either antibody or antigen to detect their opposite. When substrate is added to the reaction, if the enzymes are present it results in production of colour. The intensity of the colour indicates the amount of enzyme present that in turn indicates the amount of antigen or antibody present. Some of the commonly used enzymes in EIA are horseradish peroxidase (HRP), alkaline phosphatase (AP) and β-galactosidase. The substrates used are ABTS and DAB. The colour reaction is usually measured in a spectrophotometer (called ELISA reader) as optical density (OD) values. Some of the important EIA are enzyme linked immunosorbant assay (ELISA), immunoperoxidase test (IPT) etc.

a. **ELISA**: ELISA can be modified to identify either antigen or antibody. ELISA can be divided into direct and indirect ELISA. In direct ELISA, the antigen or antibody is attached with enzyme. Whereas in indirect ELISA, antiglobulins attached with enzymes are used. Various types of indirect ELISA are available that include blocking ELISA, competitive ELISA (as RIA), Antigen capture ELISA etc. ELISA is usually performed in polysterene plates. Now a modification using nitrocellulose is also available. This type of ELISA is called as dot ELISA and it does not require an ELISA reader. Now instead of enzymes a more sensitive system involving avidin and biotin is used. Such ELISA is called as AB-ELISA.

b. **IPT**: Basically ELISA is done to identify antigen or antibody in serum or in body fluids. IPT is done to identify antigen in tissue preparations. It is considered as one of the immunocytochemistry protocol. Both direct and indirect IPT are available.

c. **Western blotting:** It is one of the latest methods. It consists of three stages. Stage 1 involves electrophoresis of an antigen mixture in a gel that separates each fragment. Stage 2 involves transfer of these fragments from gel to nitrocellulose membranes. Stage 3 involves identification of specific fragment using antibodies against them. Antiglobulins attached to enzymes or radioisotopes are used to identify antibodies attached to protein fragment.

3. **Fluorsescent immunoassays (IFA)**: The principle of this assays are as same as RIA and ELISA. Instead of enzymes or radioisotopes, fluorescent dyes like fluorescent isothiocyanide (FITC) or rhodamine is used. As in ELISA direct and indirect IFA are available.

4. **Other primary assays**: The other assays include reacting groups like luciferin, feritin, colloidal gold involving EM (immnoelectronmicroscopy) are also available.

SECONDARY BINDING TESTS:

Antigen-antibody combination generally results in a reaction. For example, when a particulate antigen combines with antibody it results in agglutination and when a soluble antigen combines with antibody that results in precipitation. In the same way, an antigen-antibody complex can also stimulate classical complement pathway. In secondary binding tests, the antigen-antibody complex is not directly measured. Instead, the reactions that occur as a result of complex formation are measured. The intensity of the reaction reflects the amount of antigen or antibody.

Some of the common secondary binding assays isimmunodiffusion tests, immunoelectrophoresis, agglutination, passive agglutination, haemagglutination and haemagglutination inhibition, passive HA and complement fixation test. These tests are generally less sensitive than primary binding tests. The specificity and sensitivity of these tests are also lesser than primary binding tests.

1. **PRECIPITATION ASSAYS:** when a soluble antigen is mixed with antibodies, the mixture becomes cloudy within a few minutes, then flocculent and within an hour or so a precipitate is formed. The amount of precipitate formed follows a particular pattern. Large amount of precipitates are produced when the antigen and antibody are in optimal proportions. The optimal proportion of antigen and antibody that produces maximum precipitate is referred as zone of optimal proportion (zone of equivalence). The amount of precipitate produced is low in zones of excess antigen or antibody. This theory of zone of optimal proportion does not hold good for horses due to IgG (T) subclass. The amount of precipitate formed is more in horses at narrow concentration of antigen.

   - **Immunodiffusion:** In this method, antigen and antibody are loaded in wells with a diameter of 5mm with interwell distance of 1cm in an agar medium. Both antigen and antibody diffuse radially and forms a precipitate at zone of optimal concentration.

- **Radial immunodiffusion:** In this method, the antibodies are mixed with the gel and varying concentrations of antigen are loaded in different well. Precipitate forms as a circle around the well in with varying diameter depending upon the concentration. A standard curve is plotted using concentration of antigen and diameter of precipitate circle. Using this standard curve, the unknown concentration of antigen can beidentified.
- **Immunoelelectrophoresis:** In the previous methods mentioned it is difficult to resolve the components in a complex mixture. This method is used to resolve the components in a complex mixture. It involves the elelctrophoresis of antigen mixture in agar gel in one direction. Antibodies are loaded in a trough cut in the gel. When the diffusion antibodies meet their specific antigen precipitin arcs are produced. Thus different fractions in a complex mixture can be resolved,
- **Counter immunoelectrophoresis:**This is a modification of immunodiffusion. The antigen antibody combination and the resultant precipitate formation are hastened by passing mild current. Antibodies are known for their cathodic migration in an electric field. Whereas the antigen migrate to anode. When they meet they produce precipitate that is referred as precipitin arc. This is a rapid method of identification of antigen or antibody.
- **Rocket electrophoresis:**This method is a modification of radial immunodiffusion. It is done as radial immunodiffusion except that a mild current is passed in this method. Hence the precipitation arcs appear as rocket shaped.

2. **AGGLUTINATION ASSAYS:** When a particulate (insoluble antigen) is mixed with specific antibodies, it produces clumping that is termed as agglutination. This is due to the structure of antibodies. The antibodies are bivalent in nature and as a result of this it cross-links antigen and produce clumping of antigen. For agglutination to effectively take place the antigen and antibody should be in optimal concentration. If the concentration of antibody is in excess it will not produce agglutination. This phenomenon is called as **prozone reaction.** Apart from excess antibodies non-agglutinating antibodies (incomplete antibodies) can also cause prozone reaction.
   - **Passive agglutination:** Agglutination is a much more sensitive method than precipitation. Hence, the soluble antigens are converted into particulate antigens by attaching an inert particle like latex (latex agglutination test), erythrocytes (passive haemagglutination test) etc.
   - **Haemagglutination and HI:** Certain viruses like Newcastle disease virus can bind with certain receptors found on the surface of erythrocytes of certain species of animals and birds. This property of virus is referred as haemagglutination. Specific antibodies against the virus inhibit this property. Hence, the degree of inhibition of this property by the antibody reflects the amount of antibodies in serum. Hence, this test is normally done to identify and quantify antibodies against haemagglutinating viruses.
   - **Complement fixation test :** Complement is a protein that is found in the serum. This protein is capable of undergoing a series of cleavage and the products of this cleavage stimulate a cascade of reaction that ultimately lyses the antigen antibody complex. The antigen-antibody complex stimulates classical pathway. Hence, this property is used to find out the antigen antibody complex. This test involves two systems a test system and indicator system. The test system involves the known antigen and test serum sample. If the serum contains specific antibodies against the antigen antigen-antibody complex will be formed. The formation of this complex is detected by the indicator system. To the test system complement is added. The complement will cause lysis if complex is formed. If no complex is formed complement will

remain in free form. Then a known indicator system containing sheep RBC and antibodies against sheep RBC are added. If free complement is present, that will cause lysis of sheep RBC+ sheep RBC Ab complex. Hence in negative reaction lysis is observed. In positive reactions, since the complement is utilised by test antigen and antibody, no free C′ is available to cause lysis of indicator system. Hence, RBC button formation is noticed in positive reaction.

TERTIARY BINDING ASSAYS: These assays involve living systems. In these assays the ability of antibodies to neutralize the effect of antigen in living system are measured. Such neutralising activities may be death or living system, reduction in cytopathic effects etc. Due to variations between units of living only the 50% infectivity of lethality is calculated. This 50% infectivity or lethality refers to the ability of the antibodies to neutralise exactly 50% of the infectivity of antigen or 50% lethality of antigen. It is also referred as neutralisation titres. Some of the common units of titre are LD_{50}, $EID_{50}50$, $TCID_{50}$ etc. This titre is calculated by certain statistical evaluations. Some of the common methods are **Reed and Muench** method and **Karber's** method.

UNIT 5

VETERINARY VIROLOGY

GENERAL PROPERTIES OF VIRUSES

HISTORY:

Generally the information regarding viruses are only a century old. Most of the information regarding viruses were obtained after the development of electron microscope (EM) and the techniques of negative staining in EM and x-ray crystallography. However, without any idea of the nature of virus infection, certain information were found in old literatures. Few classical examples are as follows;

- The Greek poet Homer characterizes one of his characters Hector as rabid in *TheIliad.*
- The Mesopotamian law before 1000 B.C describes the dog owner's responsibility pertaining to rabid dogs.
- In Egyptian hieroglyphs, one of the characteristic squeal of polio virus infection – withered leg has been depicted.
- The spread of small pox, which was common in Ganges river basin into continental Europe has been mentioned in old records of 5 century B.C.
- Other common viral infections prevalent in those days include yellow fever, influenza and mumps.
- The control measures of small pox (variolation) that include irrational approaches of Indians and Chinese, which was fine-tuned by Lady Mary Wortley Montague in Asia Minor during 18th century.
- In 1790, Edward Jenner standardized the protection method against small pox using cowpox virus, which was termed as vaccination.
- In 1885, Louis Pasteur developed an inactivated vaccine against rabies, which was found successful in post exposure treatment of rabies.
- **First report of viruses:** In 1892, Russian scientis **Dimitrii Ivanowsky** attributed the causative agent of tobacco mosaic disease to an agent that is filterable in porcelain filters. **Martinus Beijerinck** also made the same observation seven year later. He termed the causative agent of tobacco mosaic disease as C*ontagium vivium fluidum* with clear idea about its distinct physical properties. This term evolved to **virus,**which in Latin means poison.
- In 1898, Friedrich **Loeffler and Paul Frosch** attributed the causative agent of foot and mouth disease to an agent that is filterable
- In 1915 and 1917 **Frederick Twort and Felix d'Herelle** respectively described viruses affecting bacteria, which were described as bacteriophages.
- In 1901, the **Reeds Commission** gave details of yellow fever virus, which was the first human virus to beidentified.

DEFINITION:

Viruses are defined as small ultra-microscopic obligate parasites, which have either DNA or RNA as genetic material. They replicate using host cell machinery and produce progeny virion by *de novo* synthesis. The spread of virus infection from one host to another is through progeny virions.

The term **virion** refers to whole acellular viral particles. A virion possesses nucleic acid contained in protective coat called capsid. The complete infective virus particle is also referred as virion.

TYPES OF VIRUSES: The viruses known are broadly grouped under animal viruses, plant viruses, insect viruses and bacterial. The correct classification of viruses will be dealt separately.

STRUCTURE OF VIRUSES: Viruses are extremely small particles comprises of just proteins and nucleic acid. The largest viruses are about 300nm in size, whereas the smallest known viruses are about

30nm. Viruses are comprised of two important components a protein **capsid** covering the **nucleic acid**. Some viruses also possess a lipid **envelope** but the envelope is not a virus- mediated structure.

Outer coat or capsid: The outer structure of virus is referred as capsid. It is made up of proteins. It is a protein shell that encloses the nucleic acid. It is built of **structure units**, which are the smallest functional equivalent building units of the capsid. **Capsomeres** are morphological units seen on the surface of particles and represent clusters of structural units. Capsomeres are also referred as the basic structural subunit of capsid. The capsid together with its enclosed nucleic acid is called the **nucleocapsid**. The nucleocapsid may be enclosed in an **envelope,** which may contain material of host cell as well as viral origin. The important function of the capsid of virus particle is to protect the fragile nucleic acid genome from physical, chemical and enzymatic damages. The protein subunits in a virus capsid are **multiply redundant**, i.e. present in many copies per particle. The viral nucleocapsid have the following three different morphological forms, which is also referred as symmetry. There are basic symmetries – cubical and helical.

Cubical symmetry: It is also referred as icosahedral symmetry. In this arrangement, the nucleic acids are arranged inside a shell, which is in the shape of an icosahedran. Icosahedran is a geometrical figure with 12 vertices (corners) and 20 identical facets (faces). Each facet is an equilateral triangle. There are six 5-fold axes of symmetry passing through the vertices, ten 3-fold axes extending through each face and fifteen 2-fold axes passing through the edges of an icosahedron. Icosahedral symmetry requires definite numbers of structure units to complete a shell. A virus with icosahedral symmetry (5:3:2 symmetry) requires a multiple of 60 subunits to cover the surface completely.

Helical symmetry: In this arrangement, the nucleic acid and capsomeres are helically coiled together. The length of the helical viral nucleocapsid is determined by the length of the nucleic acid. In this symmetry the identical protein subunits are arranged in the form of a circle to form a disc shaped appearance. The cylinder or helical structure is formed as a result of stacking of multiple discs, with the virus genome coated by the protein shell or contained in the hollow centre of the cylinder.

Complex symmetry: It is also referred as undefined symmetry. This arrangement does not fit into either helical or cubical symmetries. It has the feature of both cubical and helical symmetries.

VIRAL NUCLEIC ACID: The genetic material of viruses are made up of only one type of nucleic acid ie. either DNA or RNA but never composed of both. The nucleic acid both DNA and RNA may be present either as single stranded or double stranded. The nucleic acid (RNA) may also present either as a single molecule or in segments. The DNA may present either as linear or circular molecule. The RNA may present either as positive sense or negative sense. The molecular weight ranges from 2 million to 200 million. In some viruses the nucleic acids are just packed inside the capsid, whereas in some viruses, they are integrated with the nucleocapsid.

ENVELOPE: The capsid of some viruses is surrounded by a membrane called envelope. Viral envelopes are derived from cellular membranes of host, which are acquired during the release of virus from host cell. The envelope also contains proteins that are referred as matrix proteins and are specified by viral genes. The envelope proteins appear as spikes and are also referred as spike proteins. The spike proteins are responsible for attachment of virus to cells.

The viral structures are studied by electron microscopic techniques like negative staining, freeze etching and shadowing. The method that is more commonly used in X ray crystallography. Of late nuclear magnetic resonance (NMR) imaging is used to study structural details.

CLASSIFICATION OF VIRUSES

The classification of viruses is difficult due to its simple structure and presence of only few components.

Classical system of classification: In 1962, viruses were classified under classical system of classification by Lwoff, Horne and Tournier under hierarchical system consisting of phylum, class, order, family, genus and species. In this system the viruses were grouped based on the properties they share among themselves than the properties of the cells or organisms they infect. The other important features included in classical system are nature of nucleic acid, symmetry of the capsid, presence or absence of envelope and dimensions of virion.

Baltimore classification system: This system was proposed by David Baltimore. This classification is based on genetic system of each virus and the obligatory relationship between viral genome and its mRNA. In Baltimore classification, viruses are grouped under six different groups. They are as follows;
Group I – Double stranded DNA viruses

Group II – Single stranded DNA viruses

Group III – Double stranded RNA viruses

Group IV – Positive sense RNA viruses

Group V – Negative sense RNA viruses

Group VI – Reverse transcribing RNA viruses

Unclassified viruses.

ICTV classification: As for classification of bacteria, a separate committee called **International Committee on Taxonomy of Viruses** decides the grouping and regrouping of viruses. In this system certain aspects of classical system of classification like grouping under family, genera and Baltimore system of classification are used. The committee has published its latest report on classification (7th Report). At present there are more than 1,550 virus species belonging to 3 orders, 56 families, 9 subfamilies and 233 genera are recognized in this report. Only the viruses that affect vertebrates are included in the following table. Baculo virus is also included since it is commonly used in genetic engineering as vectors.

Group I: dsDNA Viruses			
Family (Subfamily)	Genus	**Type Species**	**Hosts**
Adenoviridae	*Mastadenovirus*	*Human adenovirus C*	Vertebrates
	Aviadenovirus	Fowl adenovirus A	Vertebrates
Asfarviridae	*Asfivirus*	*African swine fever virus*	Vertebrates
Herpesviridae: Alphaherpesvirinae	*Simplexvirus*	*Human herpesvirus 1*	Vertebrates
	Varicellovirus	Human herpesvirus 3	Vertebrates
Betaherpesvirinae	*Cytomegalovirus*	*Human herpesvirus 5*	Vertebrates
	Muromegalovirus	*Murine herpesvirus 1*	Vertebrates
	Roseolovirus	*Human herpesvirus 6*	Vertebrates
Gammaherpesvirinae	*Lymphocryptovirus*	*Human herpesvirus 4*	Vertebrates
	Rhadinovirus	*Simian herpesvirus 2*	Vertebrates

Iridoviridae	Ranavirus	Frog virus 3	Vertebrates
	Lymphocystivirus	*Lymphocystis disease virus 1*	Vertebrates
Polyomaviridae	*Polyomavirus*	*Simian virus 40*	Vertebrates
Papillomaviridae	Papillomavirus	Cottontail rabbit papillomavirus	Vertebrates
Poxviridae:	*Orthopoxvirus*	*Vaccinia virus*	Vertebrates
Chordopoxvirinae	*Parapoxvirus*	*Orf virus*	Vertebrates
	Avipoxvirus	Fowlpox virus	Vertebrates
	Capripoxvirus	*Sheeppox virus*	Vertebrates
	Leporipoxvirus	*Myxoma virus*	Vertebrates
	Suipoxvirus	Swinepox virus	Vertebrates
	Molluscipoxvirus	Molluscum contagiosum virus	Vertebrates
	Yatapoxvirus	Yaba monkey tumor virus	Vertebrates
Baculoviridae	Nucleopolyhedrovirus	*Autographa californica nucleopolyhedrovirus*	Invertebrates
Group II: ssDNA Viruses			
Circoviridae	Circovirus	Porcine circovirus	Vertebrates
	Gyrovirus	Chicken anemia virus	Vertebrates
Parvoviridae:	Parvovirus	Mice minute virus	Vertebrates
Parvovirinae	*Erythrovirus*	B19 virus	Vertebrates
	Dependovirus	Adeno-associated virus 2	Vertebrates
Group III: dsRNA Viruses			
Birnaviridae	Aquabirnavirus	Infectious pancreatic necrosis virus	Vertebrates
	Avibirnavirus	*Infectious bursal disease virus*	Vertebrates
Reoviridae	Orthoreovirus	*Mammalian orthoreovirus*	Vertebrates
	Orbivirus	*Bluetongue virus*	Vertebrates
	Rotavirus	*Rotavirus A*	Vertebrates
	Coltivirus	*Colorado tick fever virus*	Vertebrates
	Aquareovirus	*Golden shiner virus*	Vertebrates
Group IV: (+)sense RNA Viruses			
Order *Nidovirales* - "Nested" Viruses			
Arteriviridae	*Arterivirus*	*Equine arteritis virus*	Vertebrates
Coronaviridae	*Coronavirus*	*Infectious bronchitis virus*	Vertebrates
	Torovirus	*Equine totovirus*	Vertebrates
Astroviridae	*Astrovirus*	*Human astrovirus 1*	Vertebrates

Caliciviridae	*Lagovirus*	*Rabbit haemorrhagic disease virus*	Vertebrates
	Norwalk-like viruses	*Norwalk virus*	Vertebrates
	Sapporo-like viruses	*Sapporo virus*	Vertebrates
	Vesivirus	*Swine vesicular exanthema virus*	Vertebrates
Flaviviridae	*Flavivirus*	*Yellow fever virus*	Vertebrates
	Pestivirus	*Bovine diarrhea virus 1*	Vertebrates
	Hepacivirus	*Hepatitis C virus*	Vertebrates
Picornaviridae	*Enterovirus*	*Poliovirus*	Vertebrates
	Rhinovirus	*Human rhinovirus A*	Vertebrates
	Hepatovirus	*Hepatitis A virus*	Vertebrates
	Cardiovirus	*Encephalomyocarditis virus*	Vertebrates
	Aphthovirus	*Foot-and-mouth disease virus O*	Vertebrates
	Parechovirus	*Human parechovirus*	Vertebrates
	Erbovirus	*Equine rhinitis B virus*	Vertebrates
	Kobuvirus	*Aichi virus*	Vertebrates
	Teschovirus	*Porcine teschovirus*	Vertebrates
Togaviridae	*Alphavirus*	*Sindbis virus*	Vertebrates
	Rubivirus	*Rubella virus*	Vertebrates
Group V: (-)sense RNA Viruses			
Order Mononegavirales			
Bornaviridae	*Bornavirus*	*Borna disease virus*	Vertebrates
Filoviridae	Marburg-like viruses	*Marburg virus*	Vertebrates
	Ebola-like viruses	*Ebola virus*	Vertebrates
Paramyxoviridae: Paramyxovirinae	*Respirovirus*	*Sendai virus*	Vertebrates
	Morbillivirus	*Measles virus*	Vertebrates
	Rubulavirus	*Mumps virus*	Vertebrates
Pneumovirinae	*Pneumovirus*	*Human respiratory syncytial virus*	Vertebrates
	Metapneumovirus	*Turkey rhinotracheitis virus*	Vertebrates
Rhabdoviridae	*Vesiculovirus*	*Vesicular stomatitis Indiana virus*	Vertebrates
	Lyssavirus	*Rabies virus*	Vertebrates
	Ephemerovirus	*Bovine ephemeral fever virus*	Vertebrates
	Novirhabdovirus	*Infectious haematopoetic necrosis virus*	Vertebrates
Arenaviridae	*Arenavirus*	*Lymphocytic choriomeningitis virus*	Vertebrates

Bunyaviridae	Bunyavirus	*Bunyamwera virus*	Vertebrates
	Hantavirus	*Hantaan virus*	Vertebrates
	Nairovirus	*Nairobi sheep disease virus*	Vertebrates
	Phlebovirus	*Sandfly fever Sicilian virus*	Vertebrates
Orthomyxoviridae	*Influenza A virus*	*Influenza A virus*	Vertebrates
	Influenza B virus	*Influenza B virus*	Vertebrates
	Influenza C virus	*Influenza C virus*	Vertebrates
	Thogotovirus	*Thogoto virus*	Vertebrates
Group VI: RNA Reverse Transcribing Viruses			
Retroviridae	*Alpharetrovirus*	*Avian leukosis virus*	Vertebrates
	Betaretrovirus	*Mouse mammary tumor virus*	Vertebrates
	Gammaretrovirus	*Murine leukemia virus*	Vertebrates
	Deltaretrovirus	*Bovine leukemia virus*	Vertebrates
	Epsilonretrovirus	*Walley dermal sarcoma virus*	Vertebrates
	Lentivirus	*Human immunodeficiency virus 1*	Vertebrates
	Spumavirus	*Human spumavirus*	Vertebrates
Group VII: DNA Reverse Transcribing Viruses			
Hepadnaviridae	*Orthohepadnavirus*	*Hepatitis B virus*	Vertebrates
	Avihepadnavirus	*Duck hepatitis B virus*	Vertebrates
Subviral Agents: Satellites and Prions			
Prions			Vertebrates

REPLICATION OF VIRUSES

Introduction:

Viruses are obligate intracellular parasites. Hence, they need the help of a host cell for their replication. All viruses have to penetrate, replicate and come out of a cell. The exact way by which different viruses do this varies widely. Infection of a cell by virus may result in any one of the following types of infection.

Productive infection: The cells allow viruses to replicate and the progeny virions are released from the infected cell.

Abortive infection: The cells do not allow viruses to replicate and as a result of this daughter virions are not produced.

Restrictive infection: The cells allow minimal replication of viruses and as a result only few daughters alone are produced. However, the viral genome persists and can lead to serious consequence for the host. Eg. Epstein Barr and herpessimplex

Stages of virus replication

Virus replication can be divided into following phases and each phase may have number of stages:

Phase – I Initiation:

- Attachment
- Penetration

- Uncoating

This stage is characterized by introduction of genetic material of the virus into the cell

Phase – II Replication:

- Genome synthesis
- RNA production
- Protein synthesis

This stage is characterized by the genome size, composition and organization of viruses. There is no single pattern of replication. But all make proteins with 3 sets of functions to:

- Ensure replication of the genome
- Package the genome into virus particles
- Alter the metabolism of the infected cell so that viruses are produced.

Phase – III Assembly, Release, Maturation:

Phase I – Initiation

Attachment: Virus attaches to the cell surface. Attachment is via ionic interactions. Viral attachment proteins referred as **ligands** are present on the surface of viruses, which recognizes specific receptors on the cell surface. The ligands in viruses are usually the fibres and spikes in the virus structures. The receptors on cells are protein or carbohydrate or lipid components of the cell surface. The most common cellular receptors are glycoproteins. Some of the important cell receptors for viruses are CD4, ICAM etc. Cells without the appropriate receptors are not susceptible to the virus. Hence, a virus cannot produce infection in a host, which does not contain receptors for virus attachment. The joining of ligand to receptor on cell is also facilitated by co-receptors (Eg. gp120 of HIV) In some cases attachment leads to irreversible changes in the structure of the virion.

Penetration: It is a process by which a virus enters into the cell. It is an energy dependent reaction and occurs quickly. It occurs as fusion, endocytosis or translocation. It is different for enveloped and non-enveloped viruses.

Enveloped viruses – There are two methods for enveloped viruses.

Entry by fusing with the plasma membrane – Some enveloped viruses fuse directly with the plasma membrane. Thus, the internal components of the virion are immediately delivered to the cytoplasm of the cell.

Entry via endosomes at the cell surface – Some enveloped viruses require an acid pH for fusion to occur and are unable to fuse directly with the plasma membrane. These viruses are taken up by invagination of clathrin coated pits into endosomes. As the endosomes become acidified, the fusion activity of the virus proteins becomes activated by the fall in pH and the virion membrane fuses with the endosome membrane. This results in delivery of the internal components of the virus to the cytoplasm of the cell. This endocytosis is also called **viropexis**.

Non-enveloped viruses - Non-enveloped viruses may cross the plasma membrane directly or may be taken up via clathrin-coated pits into endosomes. They then cross the endosomal membrane.

Uncoating: This is the general term applied to events after penetration, which allows the virus to express its genome. For successful viral infection, nucleic acid has to be sufficiently uncoated. The lysosomal enzymes play a major role in uncoating.

Phase II - Replication of viral nucleic acid and protein synthesis

Once uncoating has taken place, synthesis of viral nucleic acid starts. This occurs as three different stages with differences between different families of the viruses.

1. Early transcription and translation – The proteins derived from this stage is mostly the enzymes required for virus replication.
2. Replication of Nucleic acid
3. Late transcription and translation – The proteins produced this stage are structural proteins.

The site of production of nucleic acid also varies between viruses. Most of the DNA viruses except Pox and Herpes replicate in nucleus. All RNA viruses replicate in cytoplasm except orthomyxo, borna and retro, which for certain stages of replication get into the nucleus of the cell.

RNA virus replication

RNA viruses have a relatively small genome. Because of this, RNA viruses code for only a few proteins. These normally include the enzyme Polymerase, which can copy RNA into a complementary nucleic acid and a viral attachment protein.

Problems associated with RNA virus replication

1. Prokaryotic and eukaryotic cells do not carry RNA dependant RNA polymerase enzyme, which is needed for nascent RNA strand synthesis from a RNA template. Hence, the negative strand RNA viruses carry this enzyme and positive strand RNA viruses have is as agene.
2. Eucaryotic host cell translation protein synthesis machinery in general uses monocistronic mRNAs and so there is a problem in making more than one type of protein from a single mRNA. RNA viruses have several solutions to this problem:

- The virus makes multiple monocistronic mRNAs
- The virus makes primary transcripts, which are processed by the host splicing machinery to give more than one monocistronic RNA
- The viral mRNA acts as a monocistronic transcript. A large polypeptide (called a polyprotein) is made which is then cleaved into separate proteins - Thus, one initial translation product is processed to give rise to multiple proteins. This happens, for example, in picornaviruses.

The RNA virus replication is described here as different strategies

RNA VIRUSES THAT DO NOT COPY THEIR RNA INTO DNA:

As described earlier, these viruses need an RNA-dependent RNA- polymerase to replicate their RNA.

a. **Plus-strand** RNA viruses - In these viruses, the virion (genomic) RNA is the same sense as mRNA and so functions as mRNA. This mRNA can be translated immediately upon infection of the host cell. Eg. FMD virus (picornavirus); togaviruses; flaviviruses

b. **Negative-strand** RNA viruses - The virion RNA is negative sense (complementary to mRNA) and must therefore be copied into the complementary plus-sense mRNA before proteins can be made. Eg. Newcastle disease virus (paramyxoviruses) rabies virus(rhabdovirus)

c. **Double-strand** RNA viruses - The virion (genomic) RNA is double stranded and so cannot function as mRNA. These viruses also carry the enzyme RNA polymerase as negative strand RNAviruses.

PLUS STRAND RNA VIRUSES

The plus strand viruses are also known as positive strand viruses and in this category of viruses the genomic RNA can act as mRNA. Eg. Picornaviruses; togaviruses; flaviviruses

Transcription and Translation: Poliovirus virion RNA functions as an mRNA and it is translated into a single polypeptide (polyprotein) immediately after entering into a cell. These polyprotein is later cleaved into number of small proteins. The cleavages are carried out by virus- coded proteases. The products of cleavage include RNA polymerase (replicase) needed for RNA synthesis, structural components of the virion and proteases.

RNA replication: RNA replication is the process by, which new copies of genome-length RNAs are made. The products of cleavage of protein support the replication of genomic RNA. The Viral RNA polymerase copies plus-sense genomic RNA into complementary minus-sense RNA. The new minus sense strands serve as template for new plus sense strands. The new plus strand RNA serve as a template for more minus strands.

NEGATIVE STRAND RNA VIRUSES – NON-SEGMENTED

The negative strand viruses those viruses wherein the genomic RNA cannot act as mRNA and mRNA has to be synthesized using genomic RNA as template. Hence, these viruses always carry the enzyme RNA dependent RNA polymerase. Eg. Rhabdoviruses, Paramyxoviruses **Transcription and translation: T**he RNA polymerase enzyme copy virion RNA into a mRNA. Messenger RNAs are translated on host ribosomes and viral proteins are made. There is no distinction between early and late functions.

RNA replication: RNA replication occurs in the cytoplasm and is carried out by the viral RNA polymerase. The new positive strand (mRNA) is copied into full length minus strand. New negative strands may be used as templates for the synthesis of more full length plus strands(mRNA).

SEGMENTED NEGATIVE STRAND VIRUSES

The important virus is the Influenza virus. They have negative-sense, single stranded segmented genome. Virions contain RNA polymerase packaged within the virus particle.

Transcription and translation: mRNA synthesis and replication of viral RNA occurs in the nucleus. This is very unusual for an RNA virus. Influenza virus has an unusual mechanism for acquiring a methylated capped 5' end to its mRNAs referred as **cap snatching** in which, **t**he virion mRNA go to nucleus to obtain the cap from host mRNA. The viral RNA polymerase (transcriptase) copies the template into complementary plus sense mRNA. mRNAs are translated in the cytoplasm.

RNA replication: RNA replication occurs in the nucleus using a virus- coded enzyme. A full length, exact complementary copy of virion RNA is made. Full length plus strand RNA is then used as a template for full- length minus strand synthesis. New minus strands can be used as templates for replication, mRNA synthesis, or packaged.

DOUBLE STRANDED RNA VIRUSES

The members under this category are peculiar in that the RNA exists in the form of double strands

Transcription and translation: Double stranded RNA does not function as an mRNA and so the initial step is to make mRNA (transcription). The mRNAs are made by virion-packaged RNA polymerase.

RNA VIRUSES, WHICH COPY THEIR RNA INTO DNA – REVERSE TRANSCRIPTION:

These are the retroviruses. In this case, their virion RNA, although plus-sense, is not released in the cytoplasm, and so does not function as mRNA immediately upon infection. Instead, it serves as a template for reverse transcriptase and is immediately copied into DNA. Reverse transcriptase is not available in the cell, and so these viruses need to code for this enzyme and package it in virions.

DNA VIRUS REPLICATION

Unlike RNA viruses, the DNA virus replication is a straight-forward one since the mechanism available for viral DNA replication is readily available in the cell. DNA viruses are able to use the host cell's replication machinery to transcribe their genome into mRNA immediately. DNA viruses generally replicate within the host cell nucleus. These viruses include the papovaviruses, papillomaviruses, polyomaviruses, adenoviruses and herpesviruses. Poxviruses are the exception. Members of the family *Poxviridae* replicate in the cytoplasm of the host cell and also carry their own RNA polymerase (transcriptase). A replicating DNA virus has to satisfy the following;

- The virus needs to make mRNAs that can be translated into protein by the host cell translation machinery.
- The virus needs to replicate its genome.
- Host enzymes for mRNA synthesis and DNA replication are available in nucleus hence, it needs to enter the nucleus.

Transcription and translation: The gene expression is divided into early and late phases. The early genes encode enzymes and regulatory proteins needed to start viral replication processes and late genes encode structural proteins, proteins needed for assembly of the mature virus.

DNA Replication

Semi-conservative method: DNA replication uses host cell DNA polymerase. Replication is bi-directional. (There are two replication forks per circular DNA genome and replication involves leading/ lagging strands, Okazaki fragments, DNA ligase, etc.). This process of DNA replication is very similar to that which occurs in the host cell (Eg. Papova viruses).

Strand displacement method: There are no Okazaki fragments, both strands are synthesized in a continuous fashion.

Phase III

Assembly: This stage involves the assembly of all the components necessary for the formation of the mature virion at a particular site in the cell. During this process, the basic structure of the virus is formed.

The site of assembly varies for different viruses, e.g:

- Picornaviruses, Poxviruses, Reoviruses - In the cytoplasm.
- Adenoviruses, Papovaviruses, Parvoviruses - In the nucleus.
- Retroviruses - On the inner surface of the cell membrane.

Release: For lytic viruses (most non-enveloped viruses), release is a simple process - the cell breaks open and releases the virus.

Enveloped viruses acquire the lipid membrane as the virus buds out through the cell membrane. Virion envelope proteins are picked up during this process as the virus is extruded. Budding may or may not kill the cell

Maturation: At this stage of the lifecycle normally the virus becomes infectious. Usually it involves structural changes in the particle, often resulting from specific cleavage of capsid proteins to form the mature products, which frequently leads to a conformational change in the capsid, or the condensation of nucleoproteins with the genome. For some viruses, assembly and maturation are inseparable, whereas for others, maturation may occur after the virus particle has left the cell.

ADENOVIRIDAE

Adenoviruses (Greek *adenos,* gland) were first isolated from explant cultures of human adenoids.

Key points

- ✓ Non-enveloped, double-stranded DNA viruses, Icosahedral symmetry. Fibres project from each of the twelve vertices of the capsid
- ✓ Replicate in nuclei, forming intranuclear inclusion bodies. Newly assembled virions form crystalline aggregates, demonstrable as intranuclear basophilic inclusions in stained tissue sections.
- ✓ Moderately stable in the environment. They can withstand freezing, mild acid and lipid solvents. Infectivity is abolished by heating at 56°C for more than 10 minutes.

Two genera:

- *Aviadenovirus,* avian adenoviruses

- *Mastadenovirus,* mammalian adenoviruses

Infectious canine hepatitis

This worldwide, generalized viral disease of dogs, caused by **Canine Adeno virus -1 (CAV-I),** principally affects the liver and vascular endothelium. Infectious canine hepatitis has become relatively uncommon because of the widespread use of effective vaccines. Although dogs are the most commonly affected species, foxes, wolves, coyotes, skunks and bears are also susceptible.

Transmission can occur following ingestion of urine, faeces or saliva from infected animals. The immune response usually eliminates virus from host tissues by 14 days after initial infection. However, virus may persist in the kidneys and, in some instances may be excreted in urine for more than six months.

Pathogenesis

Following ingestion, CAV-1 localizes in the tonsils and Peyer's patches (Fig. 56.3). As viraemia develops, replication in vascular endothelium results in rapid distribution of virus throughout the body. Virus replication also occurs in the parenchymal cells of the liver and kidneys. Clinical recovery in most dogs coincides with the production of neutralizing antibodies at about ten days after infection.

Lesions: Glomerulonephritis, corneal oedema and anterior uveitis, attributable to immune complex deposition, may develop in some infected animals.

Clinical signs

- ✓ The incubation period is up to seven days. Dogs of all ages are susceptible and subclinical infection is common. Clinical disease is most frequently encountered in young dogs. The mortality rate ranges from 10 to 30% in mature dogs and up to 100% in young pups.
- ✓ In peracute disease, death occurs so rapidly that poisoning may be suspected.
- ✓ In acute disease, affected dogs present with fever, depression, anorexia, increased thirst, vomiting and diarrhoea. Abdominal palpation may elicit pain and, although hepatomegaly may be detected, jaundice is uncommon. **Corneal opacity, either unilateral or bilateral ('blue eye'), which may occur within weeks of clinical recovery in about 20% of affected animals, usually resolves spontaneously.**
- ✓ Recovered animals have life-long immunity.

Diagnosis

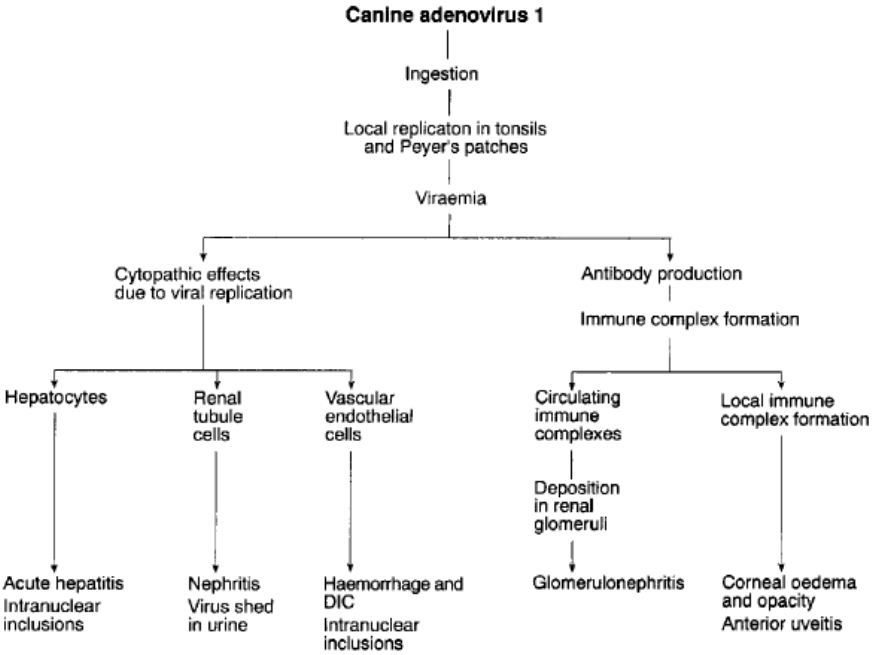
Suitable specimens for virus isolation in canine kidney cells include oropharyngeal swabs, blood, urine and faeces from affected animals sampled during the febrile stage of the disease. Samples of

spleen, lymph nodes and kidney obtained at postmortem are also suitable for virus isolation. Because of high levels of arginase activity, liver samples are unsuitable as this enzyme inhibits viral replication in tissue culture.

- ✓ A history of fever, sudden collapse and abdominal pain in young, unvaccinated dogs may suggest infectious canine hepatitis.
- ✓ Haematological examination may reveal a marked reduction in neutrophils and lymphocytes during the febrile stage. Clotting time may be prolonged.
- ✓ Basophilic intranuclear inclusion bodies in hepatocytes, Kupffer cells and endothelial cells are confirmatory.
- ✓ Viral antigen can be demonstrated by immunofluorescence in cryostat sections of liver.
- ✓ A PCR method for detecting viral DNA in clinical specimens has been described
- ✓ A rising antibody titre, using virus neutralization or haemagglutination inhibition tests, is indicative of active CAV-I infection.

Treatment and control

- Supportive treatment may allow time for hepatocellular regeneration. Fluids should be administered intravenously to counteract dehydration and control shock. In severely ill animals, blood transfusions may be required to prevent or control haemorrhage.
- Inactivated and modified live vaccines are available. Occasionally, vaccination with modified live CAV-1 vaccines results in mild nephropathy with shedding of virus in urine and, in some instances, corneal opacity. These side effects do not occur with modified live CAV-2 vaccines which stimulate effective long lasting immunity to CAV-1. Pups with unknown levels of maternally-derived antibody should be vaccinated twice, at 8 to 10 weeks and at 12 to 14 weeks of age. Booster injections, either annually o r at intervals of two years, are recommended.
- Inactivated CAV-1 vaccines, which do not induce obvious side effects, require boosters at more frequent intervals in order to maintain adequate antibody levels.



Inclusion body hepatitis / Hydropericarium syndrome / Angara disease / Lichi's disease

- Although inclusion body hepatitis (IBH) mainly affects broilers, it sometimes occurs in pullets. The cause of IBH is implicated with fowl adenovirus serotypes 4.
- In affected flocks, a sudden increase in mortality is a feature of the disease and it may reach 30% if there is immunosuppression due to infectious bursa1 disease or chicken anaemia virus infection or aflatoxin.
- Affected birds' exhibit pericardial effusion, pulmonary edema, and hepatomegaly, and have enlarged kidneys.
- Lesions include an enlarged liver with scattered haemorrhages and necrosis, intramuscular haemorrhage and anaemia. Intranuclear inclusions in hepatocytes are prominent.
- Diagnosis is based on characteristic hepatic lesions. Because apparently healthy birds excrete adenoviruses and may be serologically positive, the significance of positive serological tests is questionable.

Egg drop syndrome

- ➢ Egg drop syndrome caused by an adenovirus prevalent in ducks and probably introduced into chickens through a contaminated vaccine, was first described in 1976.
- ➢ In chicken flocks without prior experience of these viruses, the first clinical signs of infection are loss of color in pigmented eggs and soft-shelled, thin-shelled, and shellless eggs. Thin-shelled eggs may have a rough or even sandpaper-like surface.
- ➢ In flocks in which there is antibody, the disease is characterized by a drop in egg production or by failure to reach peak production
- ➢ Laying hens up to 36 weeks of age are those most commonly affected. Inflammatory lesions are found in the oviduct, particularly the pouch shell gland.
- ➢ Intranuclear inclusions may be present in the epithelial cells of this gland.
- ➢ Samples of oviduct, including material from the pouch shell gland, are suitable specimens for virus isolation in avian cell lines, especially duck kidney or fibroblast cell lines.
- ➢ As the virus agglutinates avian red cells, haemagglutination inhibition is the preferred method for serological screening of flocks.
- ➢ Control of egg drop syndrome relies on the use of inactivated vaccines before laying begins. Appropriate hygiene combined with disinfection can be used to limit the spread of infection.
- ➢ Because of the high risk of cross-infection, chickens and ducks should be housed separately.

HERPESVIRIDAE

The name, herpesvirus (Greek ***herpein,*** to creep), refers to the sequential appearance and local extension of lesions in human infection. A layer of amorphous material, the tegument, lies between the envelope and the capsid.

Key points

- ➢ Enveloped DNAviruses with icosahedral symmetry
- ➢ Replicate in the nucleus
- ➢ Labile in the environment and sensitive to detergents and lipid solvents
- ➢ Three subfamilies of veterinary importance: ***Alphaherpesvirinae, Betaherpesvirinae, Gammaherpesvirinae***
- ➢ Cause diseases of the respiratory, reproductive and nervous systems; may cause cell transformation in some species
- ➢ Latency is a common outcome of infection with these viruses

Herpesviruses enter cells by fusing with the plasma membrane. Replication occurs in the cell nucleus. The envelope, which derives from the nuclear membrane of the host cell, incorporates at least eight viral encoded glycoproteins. Enveloped virions accumulate in the endoplasmic reticulum prior to final processing of the glycoproteins in the Golgi apparatus and release by exocytosis. Active infection results in cell death. Intranuclear inclusions are characteristic of herpesvirus infections. Extension of viral infection occurs through points of cell contact without exposure of virus to neutralizing antibodies in blood or interstitial fluids. Protective antibody responses are usually directed against the envelope glycoproteins.

1. The family is divided into three subfamilies: ***Alphaherpesvirinae, Betaherpesvirinae*** and ***Gammaherpesvirinae.*** The subfamilies comprise nine genera. A tenth unnamed genus Ictalurid herpes-like viruses contains herpesviruses of fish and is currently unassigned to a subfamily.
2. Alphaherpesviruses replicate and spread rapidly, destroying host cells and often establishing latent infections in **sensory ganglia**.
3. Betaherpesviruses, which replicate and spread slowly, cause infected cells to enlarge, hence their common name cytomegaloviruses. They may become latent in **secretory glands and lymphoreticular cells.**
4. Gammaherpesviruses, which infect **T or B lymphocytes**, can produce latent infections in these cells. When lymphocytes become infected, there is minimal expression of viral antigen. Some gammaherpesvirus species also replicate in epithelial and fibroblastic cells causing cytolysis. A number of gammaherpesviruses are implicated in neoplastic transformation of lymphocytes.

Clinical infections

✓ Herpesviruses establish life-long infections with periodic reactivation resulting in bouts of clinical disease. Shedding of virus may be periodic or continuous. During latency, the episomal viral genome becomes circular and gene expression is limited.

✓ Reactivation of infection is associated with various stress factors including transportation, adverse weather conditions, overcrowding and intercurrent infection.

✓ Natural infections with particular herpesviruses are usually restricted to defined host species. Because these viruses are highly adapted to their natural hosts, infections may be inapparent or mild. However, in very young or immunosuppressed animals, infection can be life-threatening.

✓ Some herpesviruses such as Marek's disease virus and Epstein-Barr virus are implicated in neoplastic transformation of cells.

Infectious bovine rhinotracheitis and pustular vulvovaginitis/ balanoposthitis

Etiology

- Bovine herpes virus -1 (BHV-1) belongs to genus ***Varicellovirus*** under subfamily ***Alphaherpesvirinae***
- Bovine herpesvirus 1 (BHV-I), which infects domestic in cattle, deer and in other and range cattle, is associated with several clinical conditions including infectious bovine rhinotracheitis, infectious pustular vulvovaginitis, balanoposthitis, conjunctivitis and generalized disease in newborn calves. Infection with BHV-1 is an important cause of losses in livestock worldwide.
- A single antigenic type of BHV-1 which contains subtypes 1.1, 1.2a and 1.2b, has been recognized using restriction enzyme analysis. Subtype 1.1, implicated in respiratory disease, is included in most vaccines.
- Subtypes 1.2a and 1.2b cause mild respiratory disease and are implicated in the infectious halanoposthitis/infectious pustular vulvovaginitis (IBPIIPV) syndrome. Subtypes 1.1 and 1.2a rarely cause abortion; the less virulent subtype 1.2h has not been associated with abortion,

Pathogenesis and pathology

- The virus which causes infectious bovine rhinotracheitis is usually acquired through aerosols. Replication occurs in the mucous membranes of the upper respiratory tract and large amounts of virus are shed in nasal secretions. Virus also enters local nerve cell endings and is transported intra-axonally to the trigeminal ganglion where it remains latent. In most instances, infection is contained within two weeks by a strong immune response. However, tissue necrosis may facilitate secondary bacterial infection with severe systemic effects and, possibly, death. Rarely, a lymphocyte-associated viraemia in pregnant cows may produce foetal infection and abortion. Necrotic foci may be present in various organs of aborted foetuses, particularly in the liver.
- Following genital infection with subtypes 1.2a and 1.2b, virus replicates in the mucosa of the vagina or prepuce, and latent infection may become established in the sacral ganglia. Focal necrotic lesions on genital mucosae may eventually coalesce to form large ulcers. An intense inflammatory reaction can develop in the reproduc- tive tract with secondary bacterial infection leading to endometritis. Viraemia is not a feature of genital infection with BHV-1 subtypes and infected pregnant cows rarely abort.

Clinical signs

- In outbreaks of disease, either the respiratory or the genital form usually predominates. The incubation period is up to four days. The severity of the clinical signs in the respiratory form of the disease is largely determined by the extent of secondary bacterial infection.
- Affected animals develop a high temperature and nasal discharge accompanied by anorexia. The nares are inflamed ('red nose'), and conjunctivitis, lacrimal discharge and corneal opacity are often present.
- In uncomplicated infections, animals recover after about a week. If bacterial infection becomes established, animals develop dyspnoea, coughing and open-mouth breathing. Death may ensue.
- In severe outbreaks in feedlot cattle, morbidity may approach 100% and mortality may he as high as 10%.
- Cows with infectious pustular vulvovaginitis exhibit vaginal discharge and frequent urination. Animals usually recover within two weeks. However, secondary bacterial infection may result in metritis, temporary infertility and purulent vaginal discharge persisting for several weeks.
- Infected bulls have lesions on penile and preputial mucosae. A fatal generalized disease with fever, oculonasal discharge, respiratory distress, diarrhoea, incoordination and convulsions has been described in young calves.

Diagnosis

- Swabs collected from the nares and genitalia of several affected animals during the early acute phase of the disease are suitable for virus isolation. Because the virus is fragile, specimens for transport to the laboratory should be placed in viral transport medium and kept refrigerated.
- The presence of characteristic gross and microscopic lesions in ahorted foetuses is suggestive of infection with BHV-1.
- The virus produces a rapid cytopathic effect in bovine cell lines.
- Smears from nasal or genital swabs and frozen sections of tissues from aborted foetuses can be used for the rapid demonstration of viral antigen using immunofluorescence. Viral antigen can also be detected using ELISA.
- The polymerase chain reaction has been adapted for detection of BHV-I DNA in appropriate samples.

- The evidence of a rising antibody titre in paired serum samples using virus neutralization or ELISA is indicative of active infection. As part of a surveillance programme, bulk milk samples can be tested for antibodies using ELISA.

Control

Inactivated, subunit and modified live vaccines are available for the control of BHV-1. Vaccination reduces the severity of clinical signs but may not prevent infection. Modified live vaccines may cause abortion and should not be administered to pregnant animals. Vaccine strains, which are temperature-sensitive, do not replicate at temperatures above 37°C and should be administered by the intranasal route.

Appropriate serological tests permit differentiation of vaccinated animals from those infected with field virus. Marker vaccines, based either on genetically engineered viruses lacking one or more genes encoding surface glycoproteins or on subunit preparations containing one or more glycoproteins, are available. A serological test, usually a blocking ELISA, is used to detect antibodies against a glycoprotein not present in the marker vaccine. The availability of marker vaccines enables either eradication or reduction of infection in national herds to proceed economically.

Malignant catarrhal fever

Etiology

Malignant catarrhal fever (MCF) is caused by two related but distinct viruses, alcelaphine herpesvirus 1 (ALV-I) and ovine herpesvirus 2 (OHV-2). Both belongs to genus ***MacavirusRhadinovirus***, subfamily ***Gammaherpesvirinae***

As the wildebeest is the natural host of ALV-1, infection with this virus is confined to Africa. Sheep are the natural hosts of OHV-2 and infection occurs worldwide in sheep and goats. In these species infection is common and subclinical.

Epidemiology

✓ Alcelaphine herpesvirus 1 is transmitted vertically and horizontally in wildebeest populations. There is evidence that latency occurs in lymphoid cells. Some wildebeest calves are infected transplacentally but most acquire infection shortly after birth through nasal secretions from their mothers or from other calves. Viraemia persists in young wildebeest for the first few months of life. This facilitates shedding of large quantities of virus in nasal and ocular secretions. In-contact cattle may become infected.

✓ The pattern of virus shedding is thought to be similar in sheep especially at lambing, when transmission to cattle and farmed deer is most likely to occur. Infection in cattle and deer is thought to be acquired through contact with young lambs. Cattle and deer are considered to be 'end-hosts' because they do not appear to transmit virus.

Pathogenesis

The pathogenesis of MCF is poorly understood. It is presumed that virus enters the body through the upper respiratory tract. A cell-associated viraemia occurs. However, virus is virtually absent from the sites of lesions and it is thought that tissue changes in MCF have an immunopathological basis. Cell-mediated reactions have been implicated in lesion development. It has been pro-posed that stimulation of T lymphocytes through overproduction of IL-2 by deregulated natural killer (NK) cells contributes to lesion development.

Clinical signs

The incubation period, although variable, generally lasts three to four weeks. The most common clinical presentation is characterized by sudden onset of fever, oculonasal discharge, enlarged lymph

nodes, conjunc- tivitis, corneal opacity and erosive mucosal lesions in the upper respiratory tract. Profuse mucopurulent nasal discharge leads to encrustation of the muzzle. Some animals display neurological signs including muscle tremors, incoordination and head pressing. An intestinal form of the disease presents with diarrhoea or dysentery.

The course of this usually fatal disease is up to seven days. Some animals may linger for weeks or months and may recover. In peracute disease, particularly in deer, death may occur without premonitory signs.

Diagnosis

- Diagnosis is based on clinical presentation along with extensive vasculitis characterized histologically by fibrinoid degeneration and marked lymphoid infiltration. Ulceration of surface epithelia is a prominent feature of MCF.
- Viral DNA can be detected in circulating leukocytes using PCR.
- Although a competitive inhibition ELISA has been developed for the detection of serum antibodies to ALV-1 and OHV-2, it is not as reliable as PCR or histopathological examination. This ELISA test can be used to determine the prevalence of antibodies in sheep populations.
- Although ALV-1 can be isolated from the buffy coat of animals with wildebeest-associated MCF, ovine herpesvirus 2 has not yet been isolated.

Control

As no vaccine is available, control depends on the separation of susceptible species from reservoir hosts. Identification and elimination of sheep harbouring OHV-2 using the PCR assay in order to establish a virus-free flock may be worthwhile in defined circumstances.

Aujeszky's disease / Pseudorabies

Etiology:

- Procine herpes virus – 1 (Aujeszky's disease virus (ADV), belongs to genus ***Varicellovirus*** under subfamily ***Alphaherpesvirinae.***
- A single serotype of the virus is recognized. The pig, in which subclinical and latent infections can occur, is the natural host of the virus. Other domestic animals are susceptible and infection in these incidental hosts is usually fatal.

Epidemiology

✓ Infection is endemic in the pig populations of most countries. The virus is shed in oronasal secretions, milk and semen. Transmission usually occurs by nose-to-nose contact or by aerosols. Wind-borne transmission over distances of a few kilometres has been recorded. Transplacental transmission occurs and aborted foetuses are a source of virus.

✓ Although the virus is not stable in the environment, it may remain infectious for a few days under suitable conditions.

✓ Sheep, which are highly susceptible, may acquire infection following direct contact with pigs or when sharing the same airspace. Scavenging carnivores may become infected after eating pig meat; cats are particularly susceptible to infection. Because the incubation period and clinical course in these incidental hosts are short, the opportunity for transmission of virus from these animals is limited.

Pathogenesis

❖ Following infection, the virus replicates in the epithelium of the nasopharynx and tonsils.

❖ Virus spreads from these primary sites to regional lymph nodes and to the CNS along axons of the cranial nerves.

- ❖ Virulent strains of ADV produce a brief viraemia and become widely distributed around the body, particularly in the respiratory tract.
- ❖ Virus replication which occurs in alveolar macrophages interferes with their phagocytic function.
- ❖ Transplacental transfer results in generalized infection of foetuses.
- ❖ Latency occurs in a high percentage of infected animals with virus localized in the trigeminal ganglia and tonsils.

Clinical signs

- ➢ The age and susceptibility of infected pigs and the virulence of the infecting strain influence the severity of the clinical signs. Young pigs are most severely affected; mortality may approach 100% in suckling piglets. In neonatal piglets, the incubation period may be as short as 36 hours compared with five days in older pigs.
- ➢ Neurological signs including incoordination, tremors, paddling and convulsions predominate in young pigs. Affected animals usually die within two days. Mortality is much lower in weaned pigs although neurological and respiratory signs are often present.
- ➢ Fever, weight loss, sneezing, coughing, nasal discharge and dyspnoea may be evident in fatteners. Neurological signs are uncommon in these older animals and they usually recover within a week.
- ➢ Infection in sows early in pregnancy usually results in resorption of foetuses and return to oestrus. Later in pregnancy, infection frequently produces abortion; full- term piglets may be still-born, or weak. In herds with endemic ADV infection, neonatal animals are protected by matemally-derived antibody.
- ➢ Disease in other domestic animals occurs sporadically and is characterized by neurological signs resembling those of rabies. Intense pruritis ('mad itch') leading to self-mutilation is a feature of the disease, particularly in ruminants. The clinical course is short with most affected animals dying within a few days.

Diagnosis

Specimens of brain, spleen and lung from acutely affected animals are suitable for virus isolation. Nasal swabs, collected from live animals, may also be used. In the event of a delay between collection and tissue culture inoculation, samples should be refrigerated. Diagnosis is based on

- ➢ The history
- ➢ Clinical signs
- ➢ Lesions

Laboratory confirmation is based on

- ➢ Virus isolation,
- ➢ Detection of viral antigen - Cryostat sections of tonsil or brain are suitable for detection of viral antigen by immunofluorescence.
- ➢ Serology - virus neutralization, ELISA and latex agglutination tests are available for detecting ADV antibodies. In young pigs, maternally-derived antibodies may be present up to the age of four months. Differential ELISA methods have been introduced for the detection of antibodies to the surface glycoproteins gC, gE and gG. These assays are designed to differentiate animals infected with field virus from those vaccinated with a gene-deleted mutant lacking a surface glycoprotein.
- ➢ Histopathological findings.

Control

- ➢ Modified live vaccine
- ➢ Inactivated vaccine

- ➢ Marker vaccine – Gene deleted vaccines are available. Vaccines have been produced in which the thymidine kinase (TK) gene has been deleted together with a gene encoding one of the non- essential surface glycoproteins. Because endogenous levels of TK are low in neurons, this gene-deleted virus cannot replicate in neurons and as a consequence is significantly reduced in virulence. However, such vaccinal strains may fail to prevent latent infection caused by field virus in the trigeminal ganglia. In addition, the possibility of recombination of field virus with a gene-deleted virus and the subsequent generation of a virulent ADV has been suggested.
- ➢ Eradication of Aujeszky's disease can be achieved by depopulation, by test and removal, or by segregation of litters. The availability of effective marker vaccines makes eradication feasible and economically worthwhile.

Equine rhinopneumonitis and equine herpes-virus abortion

Etiology

- ➢ Equine herpesvirus -1 & Equine herpesvirus 4 belongs to genus ***Varicellovirus*** under subfamily ***Alphaherpesvirinae.***
- ➢ Equine herpesvirus 1 (EHV-1) and equine herpesvirus **4** (EHV-4), which are endemic in horse populations world-wide, are responsible for outbreaks of respiratory disease in young horses and for abortion.
- ➢ Infection with EHV-I is associated with respiratory disease, abortion, fatal generalized disease in neonatal foals and encephalomyelitis. Although infection with EHV-4 is primarily associated with respiratory disease, sporadic abortions have also been attributed to this virus.

Epidemiology

- ❖ Transmission usually occurs by the respiratory route following contact with infected nasal secretions, aborted foetuses, placentae or uterine fluids. Both EHV-1 and EHV-4 can occur as latent infections.
- ❖ It appears that foals become infected with EHV-1 from their dams or from other lactating mares in the group. Foal-to-foal spread may occur both before and after weaning.
- ❖ Mares infected with EHV-1 are potential sources of infection even without aborting. When reactivation of infection in a latent carrier mare occurs in a stud farm, exposure of non-immune, in-contact pregnant mares may lead to an abortion storm.

Pathogenesis

- ➢ These viruses replicate initially in the upper respiratory tract and regional lymph nodes with spread, in some cases, to the lower respiratory tract and lungs.
- ➢ Infections with EHV-4 appear to be restricted to the respiratory tract and viraemia is uncommon. In contrast, local replication of EHV-1 may be followed by a cell- associated viraemia which can result in abortion or neurological disease.
- ➢ The virus can spread directly from infected leukocytes to contiguous cells, thus avoiding neutralization by circulating antibody.
- ➢ Equine herpesvirus 1 has a predilection for vascular endothelium. Vasculitis and thrombosis in the placenta along with transplacental infection of the foetus result in abortion.
- ➢ Vasculitis and thrombosis, associated with EHV-1 infection, may be present in the CNS especially in the spinal cord. Neurological changes appear to be related to infection with particular strains of EHV-1.

Clinical signs

- ❖ Respiratory disease caused by EHV-4 occurs in foals over two months of age, in weanlings and in

yearlings. Following an incubation period of two to ten days, there are signs of fever, pharyngitis and serous nasal discharge. Secondary bacterial infection is common, giving rise to mucopurulent nasal discharge, coughing and, in some cases, bronchopneumonia. In the absence of serious secondary infection, recovery usually occurs within two weeks.

- Respiratory disease associated with EHV-1 is clinically indistinguishable from that caused by EHV-4. Outbreaks of disease caused by EHV-1 are less common. Immunity, following primary respiratory tract infection, lasts only a few months and is restricted to infection with antigenically similar viruses.
- Multiple infections result in significant cross-protection against the heterologous herpesvirus.
- Mares which abort following infection with EHV-1 rarely show premonitory signs. Abortion occurs several weeks or months after exposure, usually during the last four months of gestation. Such infected mares rarely abort during subsequent pregnancies and their fertility is unaffected. Infection close to term may result in the birth of an infected foal which usually dies due to interstitial pneumonia and viral damage in other tissues, sometimes complicated by secondary bacterial infection.
- Although neurological signs associated with EHV-1 infection are relatively uncommon, they may present in several horses during an outbreak of abortion or respiratory disease on a farm. The signs range from slight incoordination to paralysis, recumbency and death.

Diagnosis

1. Virus isolation and identification are used routinely for the laboratory confirmation of herpesvirus infection in horses. Nasopharyngeal swabs should be collected during the early stages of respiratory infections and dispatched in suitable transport medium to the laboratory.
2. Viral antigen may be demonstrated in cryostat sections of lung, liver and spleen from aborted foetuses using immunofluorescence.
3. The characteristic gross and microscopic lesions, particularly intranuclear inclusions, may be sufficient for confirmation of herpesvims abortion. Vasculitis is commonly found in the myeloencephalopathy caused by herpesvirus.
4. The polymerase chain reaction has been adapted for the detection of viral DNA in clinical specimens.
5. Demonstration of a fourfold rise in antibody titre in paired serum samples is useful for confirmation of a recent outbreak. Most serological tests do not distinguish infection with EHV-I from infection with EHV-4 because of antigenic cross-reactivity. Recent ELISA-type assays employing monoclonal antibody or recombinant glycoprotein G antigens can discriminate between the two viruses.

Control

- Effective management practices and vaccination are essential for control.
- Animals returning from sales, races or other events should be segregated for up to four weeks.
- On large stud farms, horses should be kept in small, physically-separated groups. It is essential that pregnant mares be segregated in a stress-free environment.
- Following an outbreak of disease, affected animals should be isolated. Premises should be disinfected and movement should be restricted until animals on the premises have been clear of the disease for at least one month.
- Modified live and inactivated virus vaccines are commercially available. Vaccination with live vaccines is not permitted in some countries. Many vaccine preparations contain both EHV-1 and EHV-4. As vaccination is not considered to be fully protective, frequent boosters are recommended. Vaccination appears to reduce the severity of clinical signs and to decrease the likelihood of abortion.

Infectious laryngotracheitis

Etiology

➢ This highly contagious respiratory disease of chickens and sometimes pheasants is caused by **gallid herpesvirus 1 (GaHV-1)**, belongs to genus **Iltovirus** under subfamily **Alphaherpesvirinae**.

➢ Although strains of GaHV-1 vary in virulence, they are antigenically homogeneous. In areas of intensive poultry production, the disease is usually well-controlled by a combination of vaccination and biosecurity. However, the virus tends to persist in small enterprises and specialized chicken flocks as an endemic infection.

Epidemiology

Infection is acquired through aerosols, especially in intensively reared birds. Latency occurs in the trigeminal ganglia and carrier birds may shed virus intermittently after periods of stress such as the onset of laying or when groups of birds are mixed. Indirect transmission from one production unit to another may occur through contaminated fomites.

Pathogenesis

Following inhalation, virus replicates locally in the upper respiratory tract. Spread along sensory nerves results in localization in the trigeminal ganglia.

Clinical signs

- The incubation period is up to 12 days. The epidemic form of the disease is caused by virulent strains of GaHV-1.
- Chickens of allages are susceptible, but disease is most common in those aged 4–18 months.
- After an incubation period of 6–12 days,mild coughing and sneezing are followed by nasal and ocular discharge, dyspnea, loud gasping and coughing, anddepression. In severe cases, the neck is raised and the head extended during inspiration—"pump handle respiration."Head shaking with coughing is characteristic, and may be associated with expectoration of bloody mucus and frank blood that appear on the beak, face, and feathers.
- Morbidity approaches 100%; the mortality for virulent strains may be 50–70% and that for strains of low virulence about 20%.
- Strains of low virulence are associated with conjunctivitis, ocular discharge, swollen infraorbital and nasal sinuses, and decreased egg production. The mild enzootic form is most common in modern poultry production, and the severe epizooticform is uncommon.
- There is severe laryngotracheitis in affected birds, characterized by necrosis, hemorrhage, ulceration, and the formation of diphtheritic membranes. Extensive diphtheritic membrane formation can plug the airway at the tracheal bifurcation, resulting in death from asphyxia, which has led to the use of the term "fowl diphtheria." The virus probably persists as a latent infection, and has been recovered from tracheal explant cultures more than 3 months after infection.

Diagnosis

1. In severe outbreaks of ILT, the clinical signs and postmortem findings may be sufficiently characteristic for diagnosis. In outbreaks of the mild form of the disease, laboratory confirmation is necessary.
2. Virus isolation can be carried out on the chorioallantoic membrane of embryonated eggs or in avian cell cultures. Rapid methods of diagnosis include demonstration of herpesvirus particles in tracheal samples by electron microscopy and detection of viral antigen in smears or frozen sections by immunofluorescence.

3. Viral antigen can be detected in tracheal samples by ELlSA or AGID.
4. Antibodies to GaHV-I can be demonstrated by virus neutralization, ELSA or AGID.

Control

Although it spreads rapidly through a flock, new clinical cases may occur over a period of 2–8 weeks; thus it spreads somewhat more slowly than acute respiratory diseases such as Newcastle disease, influenza, and infectious bronchitis. It is feasible to establish and maintain flocks free of infectious laryngotracheitis, and where management systems allow, this practice is increasingly adopted, particularly in the broiler industry where birds are harvested at 5–9 weeks of age and where "all-in–all-out" management is possible.

However, for breeding and egg production flocks, vaccination is still widely practiced, using attenuated virus vaccine.This protects birds against disease, but not against infection with virulent virus or the development of a latent carrier status for either the virulent or the vaccine viruses.

Outbreaksof acute disease have occurred in broilers as a result of reversion to virulence of vaccine virus strains.

- ➢ Flock management systems and vaccination protocols form the basis of control methods.
- ➢ In broiler flocks, short production cycles and all-in all-out management systems ensure that they remain free of the disease.
- ➢ In layer flocks, vaccination is usually carried out using live vaccines administered by aerosols or in drinking water. Vaccination protects against clinical disease but is ineffective against infection by field virus and the establishment of latency.

Marek's disease

- ❖ This contagious lymphoproliferative disease of chickens is caused by **gallid herpesvirus 2** (Marek's disease virus), belongs to the genus ***Mardivirus,*** subfamily ***Alphaherpesvirinae*** which is cell-associated and oncogenic. The disease, which is of major economic significance in the poultry industry, occurs worldwide.
- ❖ Herpesviruses of chickens and turkeys in the genus Marek's disease-like viruses can be divided into three serotypes or species. Serotype 1 (Gallid herpesvirus 2) includes all pathogenic strains and the attenuated variants derived from these strains; serotype 2 (Gallid herpesvirus **3)** contains avirulent and non-oncogenic strains; the third serotype, meleagrid herpesvirus 1, is an avirulent herpesvirus of turkeys. Serotype 1 strains can be categorized as mildly virulent, virulent and highly virulent.

Epidemiology

- ➢ Productive replication with release of infective virus occurs only in the epithelium of the feather follicle. Cell- free virus is released from the follicles along with desquamated cells. This dander can remain infective for several months in dust and litter in poultry houses.
- ➢ Infected birds remain carriers for life and their chicks, which are protected initially by maternally-derived antibody, acquire infection within a few weeks, usually by the respiratory route.
- ➢ In addition to the virulence of the infecting strain of herpesvirus, host factors which contribute to the severity of the disease include the sex, age at the time of infection and genotype. Two relevant genetic loci have been identified. One, associated with the major histocompatibility complex allele, influences the immune response to Marek's disease, and the other influences the susceptibility of T lymphocytes to transformation. Female birds are more susceptible to the disease than male birds; the reason for this difference is unknown. Resistance to the development of disease increases with age.

- Transportation, vaccination, handling and beak trimming are stress factors which increase susceptibility to disease.
- **Pathogenesis and pathology**
- Following inhalation, virus replicates locally before transfer, probably within macrophages, to the major lymphoid organs where it causes cytolysis, primarily of B cells. Latent infection occurs in T cells activated by the cytolytic process involving B cells.
- A persistent cell- associated viraemia results in dissemination of the virus throughout the body. Infection of epithelial cells in feather follicles occurs about two weeks after infection. Cytolysis of these epithelial cells results in shedding of virus particles into the environment.
- Genetically susceptible chickens are prone to tumour development and immuno-suppression, associated with apoptosis of T cells and thymocytes along with downregulation of CD8 molecule expression. Lymphomatous lesions may become evident in these birds from two weeks to several months after infection.
- Transformation of T cells probably relates to oncogenes in certain serotype 1 strains. Multiple copies of the virus genome are found in transformed cells, both as episomal DNA and as a form integrated into host cell DNA. Several genomic regions with potential roles in transformation have been identified including ***meq*** (which resembles the ***JunIFos*** family of oncogenic transcription factor proteins), a basic-leucine zipper gene.
- Defective immune surveillance mechanisms can allow transformed cells to form lymphoid tumours. The peripheral nerves are often affected, exhibiting proliferative (type A), inflammatory (type B) or minor infiltrative (type C) changes. Demyelination, which is found in type A and type B lesions, causes paralysis.
- The acute form of Marek's disease is characterized by diffuse infiltration of many internal organs with neoplastic lymphoid cells. A surface antigen, expressed on transformed lymphocytes, formerly called Marek's disease tumour-associated antigen (MATSA) is now considered to be merely a marker for activated T cells.

Features useful in differentiating Marek's disease from lymphoid Leukosis

Feature	**Marek's disease**	**Lymphoid leukosis**
Age	Six weeks or older	Not less than 16 weeks
Symptoms	Frequently paralysis	Non specific
Incidence	Frequently above 5 percent in unvaccinated flocks	Rarely above 5 percent
Macroscopic lesions		
Neural enlargement Bursa of Fabricius Tumours in skin, muscle and proventriculus	Frequent Diffuse enlargement or atrophy May be present	Absent Nodular tumours Usually absent
Microscopic lesions		
Neural involvement Liver tumours Spleen Bursa of Fabricius	Yes Often perivascular Diffuse Interfollicular tumour and/or atrophy of follicles	No Focal or diffuse Often focal Intrafollicular tumour
Central nervous system	Yes	No

Lymphoid proliferation in skin and feather follicles	Yes	No
Cytology of tumours	Pleomorphic cells including lymphoblasts, small, medium and large lymphocytes and reticulum cells, rarely can be only lymphoblasts	Lymphoblasts
Category of neoplastic lymphoid cell	T-cell	B-cell

Clinical signs

- Lymphoproliferative syndromes are most frequent with Marek's disease, lymphoma being most common, with involvement of several visceral organs and, usually, asymmetric paralysis of one or both legs or wings. Incoordination is a common early sign: one leg is held forward and the other backward when the bird is stationary, because of unilateral paresis or paralysis.
- Wing dropping and lowering of the head and neck are common. If the vagus nerve is involved, there may be dilation of the crop and gasping. Marek's disease lymphoma sometimes may occur without neurological signs, and present only as depression and comatose state, with visceral lymphomas.

Acute Marek's diseaseor fowl paralysis occurs inexplosive outbreaks in young chickens, in which a large proportion of birds in a flock show depression followed, after a few days, by ataxia and paralysis of some birds. Significant mortality occurs without localizing neurologic signs. Visceral lymphomas are typically absent in affected birds, but nerve lesions are prominent.

Ocular lymphomatosisis a rare syndrome that leads to graying of the iris of one or both eyes as a result of infiltration of transformed lymphocytes; the pupil is irregular and eccentric, and there is partial or total blindness. Mortality is rare.

Cutaneous Marek's diseaseis recognized readily after plucking, when round, nodular lesions up to 1 cm in diameter occur, particularly at feather follicles of young birds.The non-feathered area of the legs may have a distinct red coloration, and Marek's disease is therefore sometimes called "red leg syndrome."

- Other syndromes include lymphodegenerative syndromes,transient paralysis from transient brain edema, and atherosclerosis, all of which are rare and produce low to no mortality.
- Most chickens have antibody to Marek's disease virus(gallid herpesvirus 2) by the time they are mature; infection persists and virus is released in dander from the feather follicles. Congenital infection does not occur, and chicks are refractory for the first few weeks of life because of the protected and complex process that leads to lymphoma. Birds typically are infected by the inhalation of virus in the dust. Epizootics of Marek's disease usually involve sexually immature birds 2–5 months old; a high mortality (about 80%) soon peaks and then declines sharply.

Diagnosis

1. The diagnosis of Marek's disease is based on clinical signs and pathological findings. Paralysis of legs and wings in conjunction with thickening of the peripheral nerves is typical of Marek's disease. Nerve involvement is not always evident in adult birds. In these circumstances, differentiation from lymphoid leukosis is particularly important.
2. Differentiation from lymphoid leukosis is based on the age of affected birds, the incidence of clinical cases and the histopathological findings.

3. Virus can be isolated from the buffy coat of blood samples from infected birds. Chicken kidney cells or duck embryo fibroblasts can be used for isolation of the virus.
4. Viral antigen can be detected in preparations of skin or feather tips, using a radial precipitin test. Serum antibodies to GaHV-2 may be demonstrated using AGID, ELISA, immunofluorescence or virus neutralization.
5. Primers, which can distinguish attenuated and wild-type strains, have been developed for PCR assays.
6. In the absence of characteristic clinical signs, infection with gallid herpesvirus 2 is not indicative of Marek's disease in a flock.

Control

✓ The use of appropriate management strategies, genetically resistant stock and vaccination has reduced losses from Marek's disease.

✓ Disinfection, all-in all-out policies, and rearing young chicks away from older birds for the first two or three months of life reduce exposure to infection, decreasing the likelihood of serious disease. A range of modified live vaccines containing the three avian herpesvirus serotypes are commercially available. Although a single dose of virus injected into day-old chicks provides good lifelong protection, it does not prevent superinfection with virulent field viruses.

✓ A synergistic protective effect is obtained by using two or three strains in a vaccine. Consequently meleagrid herpesvirus 1 is commonly incorporated into bivalent or trivalent vaccines.

✓ Novel vaccines, based on recombinant DNA technology are being developed because of a reduction in the efficacy of conventional vaccines due to emergence of virulent mutants. Relatively recently the introduction of automated in ***ovo*** vaccination at the eighteenth day of incubation has replaced conventional vaccination methods in large commercial units

Duck Plague (Duck viral enteritis)

Etiology: Anatid herpes virus-1 under unassigned genus of Alphaherpesvirinae in Herpesviridae

❖ Duck viral enteritis, historically called duck plague, occurs worldwide among domestic and wild ducks, geese, swans, and other waterfowl.

❖ Migratory waterfowl contribute tospread within and between continents

❖ Ingestion of contaminated water is believed to be the major mode of transmission, but the virus may also be transmitted by contact. The incubation period is 3–7 days.

❖ There is anorexia, listlessness, nasal discharge, ruffled dull feathers, adherent eyelids, photophobia, extreme thirst, ataxia leading to recumbency with outstretched wings and with head extended forward, tremors, watery diarrhea, and soiled vents.

❖ Most ducks that develop clinical signs die, and sick wild ducks often conceal themselves and die in vegetation at the water's edge.

❖ Clinical findings are suggestive of duck viral enteritis, and the diagnosis may be confirmed by the finding of herpesvirus inclusion bodies in tissues of affected birds, coupled with positive immunohistochemical staining for viral antigen.

❖ Virus can be isolated in 1-day-old Muscovy or white Peking ducks, or by inoculation of chorioallantoic membranes of 9–14-day-old embryonating duck eggs.

❖ Duck viral enteritis must be differentiated from hepatitis caused by picornavirus or astrovirus infections, and from Newcastle disease and influenza.

Asfarviridae

Key points

- Enveloped DNA virus
- Icosahedral symmetry
- Replicates in cytoplasm of host cells and in soft ticks of the *Ornithodorus* species
- Causes African swine fever
- African swine fever virus is stable in the environment over a wide range of temperature (4°C to 20°C) and pH values.
- The virus may persist for months in meat.

African swine fever

African swine fever (ASF) is an economically important viral disease of pigs, characterized by fever, haemorrhages in many tissues and a high mortality rate. It is endemic in sub-Saharan Africa and Sardinia. Outbreaks have occurred in Belgium, Italy, Malta, Brazil, Cuba, Haiti and the Dominican Republic.

Epidemiology

Domestic and wild pigs are the only species susceptible to infection. In Africa, ASFV is maintained in a sylvatic cycle involving soft ticks of the genus ***Ornithodorus***and inapparent infection of warthogs and bushpigs (Fig. 58.2). Adult warthogs with persistent inapparent infection rarely develop viraemia. In contrast, young warthogs develop viraemia and are a major source of virus for soft ticks. Replication of virus occurs in the **ticks and both transovarial and trans-stadia1 transmission** have been described. Soft ticks feed for short periods on hosts before dropping off and sheltering in crevices in walls or cracks in the ground. The presence of ticks in a particular region makes the eradication of ASF difficult.

Virulent strains of ASFV, producing high mortality in infected animals, are widely distributed in Africa. Many isolates from other parts of the world are less virulent and mortality rates are usually below 50%.

- Following infection of domestic pigs with virulent virus, body fluids and tissues contain large quantities of virus until death or recovery occurs. Ingestion of uncooked meat from infected warthogs or domestic pigs is a major method of transmission. Spread can also occur by direct contact usually through oral or nasal secretions. Occasionally, animals become infected by contact with blood shed as a result of fighting. Indirect transmission can occur through contaminated transport vehicles, fomites and footwear. Feeding uncooked swill is an important mechanism of spread of ASF internationally, with outbreaks often starting in herds close to airports and harbours.
- Pigs which have recovered from clinical disease may remain infected for long periods. Carrier pigs are considered to be important sources of virus dissemination. Although recovered pigs are clinically unaffected by challenge with genotypically related ASFV, the challenge virus may replicate and spread to other pigs.

Pathogenesis and pathology

➢ Infection in domestic pigs is usually acquired via the oronasal route. The virus replicates initially in pharyngeal mucosa, tonsils and the regional lymph nodes. Infection then spreads by the bloodstream to other lymph nodes, bone marrow, spleen, lung, liver and kidneys. Secondary replication in these sites results in prolonged viraemia.

➢ Although the virus replicates primarily in cells of the **lymphoreticular system, it can also infect megakaryocytes, endothelial cells, renal epithelial cells and hepatocytes**.

- Lesions include splenic enlargement, swollen haemorrhagic gastrohepatic and renal lymph nodes, subcapsular petechiation in the kidneys, petechial and ecchymotic haemorrhages on serosal surfaces, oedema of the lungs and hydrothorax. The widespread haemorrhages result from disseminated intravascular coagulation, endothelial damage and destruction of megakaryocytes.
- Leukopenia is marked. Because the virus does not appear to replicate in T and B lymphocytes, it has been suggested that the lymphopenia follows apoptosis of lymphocytes and necrosis of lymphoid organs.
- Lesions in chronic disease include pneumonia, fibrinous pleuritis and pericarditis, pleural adhesions and hyperplasia of lymphoreticular tissues.

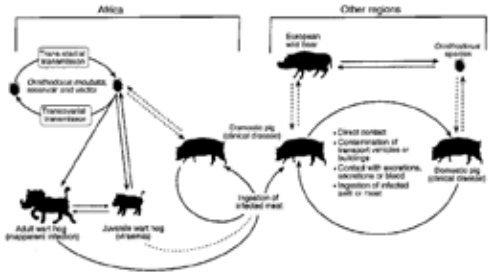


Figure 58.2 The maintenance and transmission of African swine fever virus in wild and domestic pig populations and in tick vectors.

Clinical signs

- The clinical signs of ASF, which range from inapparent to peracute, relate to the challenge dose and virulence of the virus and to the route of infection. The incubation period, which may extend from four to 19 days, is typically five to seven days in acute cases.
- Animals with peracute disease die suddenly without premonitory clinical signs. Fever, inappetence, depression and recumbency are features of acute disease. Cutaneous hyperaemia and, in some cases, haemorrhages may be evident. Other signs include dyspnoea, conjunctivitis, diarrhoea, bleeding from the nose and rectum, and abortion. The mortality rate is high.
- Subacute disease has a course of three to four weeks. Clinical signs include pneumonia, swollen joints, emaciation, depression and inappetence. Mortality rates, which are variable, depend on the age and general health of infected pigs.
- Animals may recover clinically or may develop a chronic form of the disease, which usually occurs in regions where ASFV is endemic. The immune mechanisms responsible for recovery and protection from ASFV are poorly understood. Neutralizing antibodies are not demonstrable in the sera of recovered animals. Cell-mediated immunity is considered to be an important component of the immune response.

Diagnosis

- Laboratory confirmation of ASF is mandatory because the clinical signs and lesions, which occur in some other important pig diseases such as classical swine fever, erysipelas and septicaemic salmonellosis, are similar.
- Suitable samples for laboratory examination include blood, serum, tonsil, spleen and lymph nodes.
- The most convenient and frequently used tests for detection of ASFV are direct immunofluorescence and haemadsorption.
- Direct immunofluorescence, which is fast and economical, can be carried out on impression smears or cryostat sections. However, the sensitivity of the test is only 40% in pigs with subacute or chronic ASF because of the blocking action of bound antibody in antigen- antibody complexes.
- Most field strains of ASFV induce haemadsobtion. Pig erythrocytes adhere to the surface of infected monocytes and macrophages, forming a characteristic rosette. The haemadsorption test

can be carried out by employing leukocytes from blood samples of pigs under investigation or by inoculating primary blood leukocyte cultures with blood or homogenized tissue from suspect pigs.

- ➢ A challenge experiment, involving the inoculation of suspect material into pigs vaccinated against classical swine fever and into unvaccinated pigs, may be used to differentiate the two diseases.
- ➢ The polymerase chain reaction can be used to detect DNA from ASFV in tissues unsuitable for virus isolation or antigen detection.
- ➢ Antibodies persist for long periods in recovered animals. Serological testing may be the only means of detecting animals infected with strains of low virulence. Techniques for detecting antibodies to ASF include ELISA, immunoblotting, indirect immunofluorescence, complement fixation and radioimmunoassay.

Control

- A successful vaccine is not yet available.
- Inactivated vaccines do not induce protection. Although live attenuated vaccines induce protection against challenge with homologous virus strains in some pigs, a proportion of these animals become carriers and may develop chronic lesions.
- Countries maintain disease-free status by prohibiting importation of pigs and pig products. Waste food scraps from aircraft and ships must be boiled before inclusion in pig feed. In the face of an outbreak of ASF in countries free of infection, an eradication policy is implemented. The occurrence of low virulence strains renders eradication difficult.
- Restriction of pig movement, serological monitoring of carrier pigs and prevention of contact between domestic pigs and warthogs or ticks are important control measures in countries where the disease is endemic.
- Eradication of tick species which act as vectors of ASFV is an essential part of a control programme.

Parvoviridae

Key points

- ❖ Small, non-enveloped, single-stranded DNA viruses
- ❖ Icosahedral symmetry
- ❖ Replicate in the nucleus, form intranuclear inclusion bodies
- ❖ Require rapidly-dividing cells for replication
- ❖ Stable in the environment
- ❖ Resistant to heat, solvents, disinfectants and pH changes

Genus *Parvovirus:*

-Many have haemagglutinating activity

-Shed in large numbers in faeces

-Enteric and systemic diseases in dogs and cats

-Reproductive failure, SMEDI syndrome, in pigs

Clinical infections

Parvoviruses can infect many domestic and wild animals. The most important parvoviral diseases of domestic animals are feline panleukopenia, canine parvovirus infection and porcine parvovirus infection.

Canine parvovirus infection

Infection with canine parvovirus (CPV) emerged in the late 1970s as a worldwide disease in dogs with high morbidity and mortality. Acute or subacute heart failure in pups infected in ***utero*** or during the perinatal period, was a common manifestation of the disease. With the gradual development of immunity in the adult dog population, as a consequence of both natural exposure and vaccination, the clinical pattern of the disease changed. The most common clinical presentation now encountered is acute enteric disease in young dogs between weaning and 6 months of age.

Since the appearance of CPV in 1978, further mutations affecting the genome and antigenicity of the virus have occurred. Three subtypes of the virus are currently recognized. Infection or vaccination with one subtype confers immunity against the other subtypes. **Canine parvovirus is now considered to be a host-range mutant of feline panleukopenia virus or a closely related parvovirus.**

Epidemiology

Many canine species are susceptible to infection and transmission is predominantly by the faecal-oral route. Infected dogs shed large numbers of viruses in their faeces. The number of viruses in faeces may be 10^9/g from the fifth or sixth day after infection. The continued presence of the disease in dog populations depends mainly on the stability of the virus in the environment. The low dose of virus required to establish infection and the ease with which mechanical transfer can take place are important additional factors contributing to the spread of infection.

Pathogenesis and pathology

The virus replicates initially in pharyngeal lymphoid tissues and Peyer's patches. Viraemia develops and the main target tissues are those with rapidly multiplying cell populations. During the first two weeks of life there is active cardiac myocyte division allowing viral replication with resultant necrosis and myocarditis. In older pups, the virus invades the actively dividing epithelial cells of the crypts in the small intestine. Loss of cells from the intestinal crypts leads to blunting of villi, with resultant reduced absorptive and digestive capacity leading to diarrhoea. There may be extensive haemorrhage into the intestinal lumen in severely affected pups. Destruction of lymphoid tissues of the intestinal mucosa and mesenteric lymph nodes contributes to immunosuppression which allows proliferation of Gram-negative bacteria with secondary invasion of damaged intestinal tissues. Endotoxaemia, leading to endotoxic shock, may follow.

Clinical signs

- The age and the immune status of the animal largely determine the form and severity of the disease. After a short incubation period of four to seven days, animals with enteric disease show sudden onset of vomiting and anorexia. Depression and fever may also be observed. Diarrhoea, often blood-stained, develops within 48 hours and in severe cases, there may be frank haemorrhage. The faeces have a foetid smell.
- Concurrent intestinal parasitism and viral or bacterial infections may exacerbate the condition. Affected dogs deteriorate rapidly due to dehydration and weight loss. Prolonged illness is uncommon; severely affected animals die within three days. Animals which survive the disease develop a long-lasting immunity.
- In the myocardial form of the disease, which is now rare, affected pups usually show signs of acute heart failure before eight weeks of age. Some pups may develop congestive heart failure months after the initial infection, as a result of extensive fibrosis following myocardial necrosis.

Diagnosis

- Samples for laboratory examination should include faeces, blood and other tissues, particularly affected portions of intestines and myocardium.

- ❖ The nature and distribution of the gross and microscopic enteric lesions may point to a parvoviral infection.
- ❖ Immunocytochemical staining can confirm the presence of viral antigen in tissue sections.
- ❖ The presence of basophilic intranuclear inclusions in cardiac myocytes is confirmatory.
- ❖ A leukopenia may be detected, particularly in severely affected animals.
- ❖ Definitive diagnosis in clinically affected animals early in the course of the disease relies on the demonstration of virus or viral antigen in faeces:

  -Numerous virus particles can be demonstrated by electron microscopy.

  -ELISA or HA may be used to demonstrate viral antigen.

  -Virus can be isolated in a number of suitable canine and feline cell lines.

  -Serological tests, including HAI, VN and indirect immunofluorescence, may confirm the diagnosis.

Treatment

- ❖ No specific treatment is available.
- ❖ Intensive supportive therapy including anti-emetics and fluid administration is required for the treatment of parvoviral enteritis
- ❖ Broad-spectrum antibiotics, administered parenterally, reduce the risk of secondary bacterial infections.
- ❖ Dogs with subacute or chronic heart failure may improve with rest and diuretic therapy.

Control

- ➢ Vaccination alone usually cannot control the cycle of endemic parvovirus infection in kennels and therefore it is important to minimize exposure of pups to the virus until they reach 20 weeks of age.
- ➢ Non-protective, low levels of maternal antibody can interfere with the efficacy of some modified live vaccines. In many pups these critical levels last until eight to twelve weeks of age, but in some instances interference with vaccination continues until 18 weeks of age.
- ➢ Inactivated and modified live vaccines are commercially available: Inactivated vaccines usually provide protection for up to one year and are safe to use in pregnant bitches.
- ➢ Although modified live vaccines generally give good, long-lasting protection, annual boosters may be required. These vaccines should not be used for pregnant bitches. Modified live vaccines vary in their degree of viral attenuation. The less attenuated strains of vaccinal viruses can replicate in pups despite the presence of residual maternal antibodies and vaccination of pups at 12 weeks of age is often claimed by vaccine manufacturers to be protective.
- ➢ Thorough disinfection of premises must be carried out following a disease outbreak:

  -Effective disinfectants include 1% sodium hypochlorite and 2% formalin.

  -Fumigation with formaldehyde gas, where feasible, is the most efficient disinfection procedure.

Circoviridae

- Small, non-enveloped single-stranded DNA viruses with icosahedral symmetry
- Replicate in the nucleus of dividing cells
- Circoviruses cause infections in chickens, pigs and plants
- They are stable in the environment at pH 3 to pH 9 and are resistant to heating at 60°C for 30 minutes.
- The genome consists of a molecule of **circular single stranded DNA**.
- Replication occurs in the nuclei of dividing cells.

Chicken anaemia virus infection

Young birds infected with chicken anaemia virus (CAV) develop aplastic anaemia and generalized lymphoidatrophy. This virus, which infects only chickens, is present in poultry flocks worldwide. All of the field isolates of the virus appear to be equally pathogenic and belong to a single serotype.

Epidemiology

Both horizontal and vertical transmissions occur. Infectionis by the faecal-oral route. Vertical transmission through the egg occurs during the one to three week viraemic period following infection in laying hens. Once infectionis established in a breeder flock, most birds develop antibody before laying begins. Maternally-derived antibodies do not prevent chicks from becoming infected and shedding the virus. However, they prevent the development of clinical disease. An age resistance to disease but not to infection develops in chicks at about two weeks of age. In breeding flocks containing many serologically-positive adult birds, subclinical infection is common in chickens. However, age resistance and the protective effect of maternally-derived antibodies can be overcome if infections with other immunosuppressive viruses, such as infectious bursa1 disease virus or gallid herpesvirus 2, are present.

Pathogenesis and pathology

Viraemia develops following infection of susceptible day old chicks and virus can be recovered from most organsand from faeces for about three or four weeks. The principal target cells are precursor T cells in the thymus and haemocytoblasts in the bone marrow. Destruction ofthese cells results in immunosuppression and anaemia. Gross postmortem findings include atrophy of the thymus tissues and bursa, pale bone marrow and haemorrhages under the skin and in skeletal muscle.

Clinical signs

Chickens develop clinical signs at about two weeks of age.Affected birds are depressed, anorexic and pale. The mortality rate, which is usually about l0%, may be up to 50%. Birds which survive the acute phase of the disease recover slowly. Subclinical infection in broilers from breeder flocks can adversely affect weight gains.

Diagnosis

- A presumptive diagnosis is based on the clinical signs and gross lesions at postmortem. Laboratory confirmation relies on detection of viral antigen by immunocytochemical techniques. Viral DNA can be demonstrated in bone marrow and thymus by ***in situ*** hybridization, by dotblot hybridization or by PCR.
- Virus isolation, although possible, is exacting and expensive.
- Serum antibodies can be detected using virus neutralization, indirect immunofluorescence and ELISA. A commercial ELISA is available and can be used to identify seronegative breeder flocks before laying begins.

Control

- Because infection with CAV is common, it is difficult to maintain breeder flocks free of infection. Breeding birds should be exposed to infection before laying commences. Seronegative flocks can be deliberately exposed to virus by transfer of old litter from CAV-positive farms or by the addition of crude tissue homogenates from affected birds to drinking water but such exposure is unreliable and inherently unsafe.
- A live vaccine is available in some European countries. Vaccination does not prevent economic losses in broilers due to subclinical infection.

- Control of other immunosuppressive viruses in CAV-positive flocks is essential due to the additive effects of these infections.

PAPPILOMAVIRIDAE

Key points

- ➢ Non-enveloped, circular double-stranded DNA viruses, Icosahedral symmetry
- ➢ Replication takes place in the nucleus and release of new virions occurs by lysis of the infected cell.
- ➢ Members of the genus are resistant to lipid solvents, acid, and heating at 60°C f o r 30 minutes.
- ➢ Infections are often persistent and usually established early in life.
- ➢ Contains one genus, *Papillomavirus:*
- ➢ Have not been cultured *in vitro*
- ➢ Cause papillomas and fibropapillomas in domestic animals
- ➢ Malignant transformation of alimentary and urinary tract papillomas may occur in cattle ingesting bracken fern
- ➢ Bovine papillomas type 1 and 2 are aetiologically involved in equine sarcoids

Formerly papillomaviruses were grouped with polyomaviruses in the family *Papovaviridae.* The name *Papovaviridae* is derived from the first letters of the names of important members of the family: papillomaviruses, polyomaviruses and vacuolating agent.

Clinical infections

- ❖ The epitheliotropic, host-specific papillomaviruses (Latin *papilloma,* nipple) cause proliferative lesions (warts) in many mammalian and avian species.
- ❖ Although they have not been grown in cell culture, the DNA sequences of several papillomaviruses have been elucidated, allowing specific detection in lesions.
- ❖ In infected cells, the viral DNA is usually episomal. Papillomaviruses are used experimentally for inserting foreign DNA into cultured cells.
- ❖ Each papillomavirus tends to be host-specific and to produce proliferative lesions in specific anatomical sites. Lesions are most commonly observed in young animals and usually regress spontaneously after weeks or months. Regression is attributed to the development of cell-mediated immunity.
- ❖ Typical papillomas are composed of finger-like projections of proliferating epithelium supported by a thin core of mature fibrous tissue. In fibropapillomas, the fibrous tissue component predominates. In some host species, several types of papillomaviruses can cause neoplastic change.
- ❖ The six types of bovine papillomavirus (BPV), types 1, 2 and *5* (group A) and types 3, 4 and 6 (group B), are categorized on the basis of size of genome and the type and location of associated lesions.
- ❖ Progression of papillomas to malignant tumours has been documented in humans, cattle and rabbits.

Pathogenesis

Papillomaviruses usually infect the basal cells of squamous epithelium as a result of minute abrasions. They may also gain entry at vulnerable sites such as junctions between different types of epithelia. Infected cells proliferate and differentiation is delayed. Viral gene expression is restricted during this proliferative phase. Full gene expression results in the production of viral capsids only after cellular differentiation begins in the upper layers of the epithelium. New virions can be visualized by

electron microscopy in the nuclei of differentiated keratinized cells. The release of virus occurs during desquamation of infected cells from the surface of lesions.

Diagnosis

- ❖ The clinical appearance of papillomas (warts) is distinctive. Laboratory confirmation is not usually required for papillomatous lesions.
- ❖ Histopathological examination may be required to determine the nature of some lesions, especially equine sarcoids.
- ❖ Electron microscopic examination of specimens from the epidermis may reveal characteristic virus particles.
- ❖ Hybridization assays and PCR methods are available for the detection of papillomavirus DNA, but are not used routinely. Isolates can be typed by extraction of DNA and restriction endonuclease analysis or by Southern blotting.

Bovine cutaneous papillomatosis

- ✓ Bovine cutaneous papillomatosis is caused by several types of bovine papillomavirus. Cutaneous fibropapillomas are generally associated with BPV types 1, **2** or 5, whereas BPV types 3 or 6 are commonly linked to papillomas in which the fibrous tissue component is minimal.
- ✓ The different gross appearances of the two types of proliferative response are exemplified by lesions produced by types of papillomavirus on teats of cows. Teat fibropapillomas, associated with BPV-5 infection, have smooth surfaces and are described as 'rice grain' type. In contrast, 'frond' type teat papillomas arise from infection with BPV-6.
- ✓ The localization of the proliferative lesions caused by these two bovine papillomaviruses on the teats demonstrates the predilection of viruses in this group for particular anatomical sites.
- ✓ Fibropapillomas arising from infection with BPV types 1 or 2 are often found on the head and neck of cattle under two years of age. Spontaneous regression of the lesions generally occurs within 1 year.
- ✓ Cutaneous papillomas caused by BPV-3 tend to persist.
- ✓ Because infection with BPV is usually self-limiting, treatment is seldom required. Surgical removal of large lesions on teats may be necessary because of interference with milking.
- ✓ Although inactivated autogenous vaccines are used therapeutically, their efficacy is unproven. Inactivated vaccines can be used prophylactically.

Bovine alimentary papilloma-carcinoma complex

- ❖ Papillomas of the oesophagus, rumen and reticulum are associated with BPV-4 infection.
- ❖ The lesions, which are often solitary and relatively small, are found incidentally at postmortem examination.
- ❖ Epidemiological and experimental studies have demonstrated that there is an increased frequency in the occurrence of malignant transformation of virus-induced alimentary papillomas to squamous cell carcinomas when animals are ingesting bracken fern. Such malignant lesions, which are found in anatomical sites identical to those of the papillomas, may cause difficulty in swallowing, ruminal tympany and loss of condition.
- ❖ Nodular fibropapillomas caused by BPV-2, which are occasionally found in similar upper alimentary tract locations, do not appear to become malignant.

Enzootic haematuria

- Enzootic haematuria is encountered worldwide in cattle on poor pastures with abundant bracken fern growth.

- The haemorrhage originates from tumours in the bladder wall. Individual neoplastic lesions derive from either epithelial or mesenchymal tissues.
- Experimental studies suggest that BPV-2 and toxic compounds from bracken contribute to oncogenesis. It is probable that immunosuppression following ingestion of bracken may allow activation of latent BPV-2 in bladder tissues and this effect, together with the action of carcinogens also present in bracken, is responsible for the induction and progression of neoplastic lesions.

POXVIRIDAE

Key points

- Contains largest viruses
- Complex symmetry: The virions in this family are either brick-shaped (220 to 450 nm x 140 to 260 nm) with a surface membrane composed of tubular or globular proteins or ovoid (250 to 300 nm x 160 to 190 nm) with a surface membrane composed of a regular spiral filament
- Enveloped DNA viruses; A biconcave core or nucleoid contains linear double-stranded DNA and one or two lateral bodies within a surface membrane
- Replicate in cytoplasm of host cells takes place within defined areas (viral factories or viroplasm)
- Stable in the environment under dry conditions but sensitive to heat, detergents, formaldehyde and oxidizing agents. The genera differ in ether sensitivity.
- Skin lesions a prominent feature
- Individual poxviruses tend to infect particular host species; some poxviruses are not species-specific

Smallpox, caused by variola virus, was formerly a human disease of major international significance. The use of vaccinia virus for the prevention of smallpox, first introduced by Jenner in the late 18th century, eventually led to the eradication of this highly contagious disease at the close of the twentieth century.

Clinical infections

- ❖ Transmission of poxviruses can occur by aerosols, by direct contact, by mechanical transmission through arthropods and through fomites.
- ❖ Skin lesions are the principal feature of these infections. Several virus- encoded proteins are released from infected cells including a homologue of epidermal growth factor which stimulates cell proliferation.
- ❖ Typically, pox lesions begin as macules and progress through papules, vesicles and pustules to scabs which detach leaving a scar.
- ❖ In generalized infections there is a cell-associated viraemia and recovered animals have solid immunity. Some localized pox infections may induce transient immunity and reinfection can occur.
- ❖ Three closely related parapoxviruses, namely pseudocowpox virus, bovine papular stomatitis virus and orf virus, infect ruminants. These viruses are transmissible to humans producing lesions which are clinically similar. Moreover, the three viruses are morphologically indistinguishable and identification of the causal agent relies on nucleic acid analysis.
- ❖ Capripoxviruses are economically important viruses producing generalized infections with significant mortality in domestic ruminants. Sheeppox virus, goatpox virus and lumpy skin disease virus are closely related and share a group-specific structural protein (p32), which allows the same vaccine to be used against each virus.
- ❖ Many avian species are susceptible to infection with members of the genus ***Avipoxvirus.*** Although antigenic relationships exist among avian poxviruses, this relatedness is variable. Virus species

within the genus, named in accordance with their affinity for particular host species, include fowlpox virus, canarypox virus, pigeonpox virus and turkeypox virus. The type species is fowlpox virus.

Infections caused by vaccinia virus

1. Although the natural host species of vaccinia virus is unknown, mild infections have been described in a wide range of species including sheep, cattle, horses and humans. Formerly, the virus was used to vaccinate against smallpox.
2. Buffalopox virus and rabbitpox virus are considered to be subspecies of vaccinia virus.
3. Spread of vaccinia virus among cattle and transmission to humans occurs at milking. The lesions on the teats of cows resemble those caused by cowpox virus.
4. In recent years, vaccinia virus has been used as a recombinant virus vector for vaccination against several diseases including rabies, canine distemper and measles.

Cowpox

Etiology: Cowpox is caused by cowpox virus under genus *Orthopoxvirus*, subfamily *Chordovirinae* Although infection and disease have been described in cattle, cats, humans and a range of mammals in zoos, these species are considered to be incidental hosts. The reservoir hosts are probably wild rodents.

- ❖ In milking cows, lesions are usually confined to the teats.
- ❖ The domestic species in which the disease is most often recognized is the cat. Although cat-to-cat transfer occurs, this mode of transmission is rare.
- ❖ Small papules on the head or forelimbs, the first recognizable signs of the infection, eventually ulcerate. Scab formation is usually followed by complete resolution in about six weeks. Secondary skin lesions develop in some cats and a few animals may show signs of coryza or conjunctivitis. Rarely, pneumonia and pleural exudation have been described.
- ✓ Diagnosis can be confirmed by histopathology, electronmicroscopy or virus isolation.

No specific control measures are warranted. Human infections, which are usually contracted from infected cats, are uncommon.

Pseudocowpox / Milker's nodule

Etiology: Pseudocowpox is caused by pseudocowpox virus under genus *Parapoxvirus*, subfamily *Chordovirinae*

- It is a common mild condition affecting the teats of lactating cows. Infection spreads slowly through milking herds with variation in the number of affected animals at any particular time.
- Transmission is by direct or indirect contact. Transfer of infection can occur through teat cups and on milkers' hands. Transmission can also occur mechanically by flies or when calves are being suckled.
- Small red papules on teats or udder develop into ulcers with overlying scab formation. **Healing at the centre of the lesions produces characteristic ring- or horseshoe-shaped scabs.**
- Typical parapoxvirus particles can be demonstrated in scab material using electron microscopy.
- Control is based on appropriate hygienic measures at milking including the use of effective teat dips. In human infections, lesions are usually confined to the hand, forearm or face.

Orf / Contagious pustular dermatitis / Contagious ecthyma

Etiology: Orf is caused by orfvirus under genus *Parapoxvirus*, subfamily *Chordovirinae*

This important disease of sheep occurs world-wide and goats, camels and humans are also susceptible to infection.

Epidemiology

The virus is transmitted through direct or indirect contact. Under dry environmental conditions, it is stable and can survive in scab material for months. Infectivity is substantially reduced after exposure to adverse climatic conditions. In most flocks, infection is maintained by sheep with chronic lesions.

Pathogenesis

- The virus, which is epitheliotropic, produces proliferative wart-like lesions following entry through skin abrasions. The virus replicates in epidermal keratinocytes and infected cells release an endothelial growth factor which is implicated in epithelial cell proliferation.
- Papular lesions progress to vesicles, pustules and eventually to scab formation. Proliferation of cells underlying scabs produces verrucose masses. In the absence of secondary bacterial infection, lesions usually heal within four weeks.

Clinical signs

The disease primarily affects young sheep. The incubation period is up to seven days. Although lesions most often occur on the commissures of the lips and on the muzzle, they may also develop in the mouth and on the feet, genitalia and teats. Mild lesions may go unnoticed.

Severely affected lambs with lesions in the buccal cavity often fail to eat, lose condition and may die. Outbreaks last for some months and vary in severity from farm to farm and from year to year. The disease does not usually recur until the birth of new susceptible lambs in the flock.

Although isolates from individual flocks can differ in genotype, there is no evidence that the severity of the disease correlates with the strain of the virus involved. Environmental management factors may influence the outcome of infection. **Immunity following natural infection may not confer complete protection**. However, lesions in animals previously affected are usually less severe and heal more rapidly than those developing after first exposure. In chronically infected sheep, lesions may be either mild or proliferative. Neonatal lambs are susceptible to infection despite receipt of colostrum from previously infected ewes.

Diagnosis

Lesions of orf are readily recognized by their characteristic appearance and distribution. Virus present in lesion material can be identified by electron microscopy.

Treatment and control

- ✓ There is no specific treatment for infection with orf virus. Antibiotic therapy reduces the effect of secondary bacterial infection in young lambs.
- ✓ In endemically infected flocks, control is based on the use of a fully virulent live vaccine derived from scab material or cell culture. Ewes should be vaccinated by scarification in the axilla at least eight weeks before lambing. When close to lambing, they must be moved to a new grazing area in order to minimize exposure of lambs to infectious vaccinal scab material. Lambs should be vaccinated only if an outbreak occurs in a flock. If carried out effectively, thorough cleaning and disinfection of surfaces and equipment between periods of housing may reduce the amount of virus in the buildings.
- ✓ Humans are susceptible to infection with orf virus. Typically, a single lesion occurs on hands, forearm or face. Care should be exercised when handling affected sheep and when using live vaccines.

Sheeppox and goatpox

Etiology: The viruses of sheeppox and goatpox are members of the genus ***Capripoxvirus***, subfamily *Chordovirinae*

Both of these diseases are endemic diseases in south eastern Europe, the Middle East, Africa and Asia. A range of capripoxvirus strains have been isolated from sheep and goats and there is evidence of recombination between strains. Although some strains are extremely pathogenic in both sheep and goats, other strains produce severe disease in only one of these species.

Epidemiology

Virus particles are shed from skin lesions and in ocular and nasal discharges during the acute stages of the disease. Infection occurs through skin abrasions or by aerosol. Biting insects may also transmit the virus mechanically. Housing or stockading animals facilitates transmission of the virus. Following infection, capripox virus strains induce immunity. In endemic areas, where indigenous animals frequently have a high level of naturally acquired immunity, generalized disease and mortality are rare. In isolated flocks, outbreaks of severe disease may occur.

Pathogenesis and pathology

The virus replicates locally either in the skin or in the lungs. Spread to the regional lymph nodes is followed by viraemia and replication in various internal organs. Skin lesions, typical of poxvirus infection, appear about seven days post infection. Lung lesions often present as areas of consolidation and haemorrhage.

Clinical signs

- ➢ Following an incubation period of about one week, infected animals develop fever, oedema of the eyelids, conjunctivitis and nasal discharge. Within a few days, macules which rapidly develop into papules appear on the skin and external mucous membranes. Scabs form over necrotic papules.
- ➢ The severity of the clinical signs depends on the breed of host animal and the strain of capripoxvirus. Lesions in mild afebrile disease may be minimal and confined to the skin beneath the tail. Mortality rates with infections from some strains of capripoxvirus may be up to 50% even in indigenous breeds.
- ➢ The disease is most severe in young animals and in imported breeds. Some European breeds are extremely susceptible and mortality rates may approach 100%. Secondary bacterial infection or dissemination of the virus to other sites may result in a more severe form of the disease.

Diagnosis

- Diagnosis can often be made solely on clinical grounds. Skin biopsies or postmortem specimens may be used for laboratory confirmation. Eosinophilic intracytoplasmic inclusions may be demonstrable histologically in epidermal cells.
- Electron microscopy can be used for the rapid identification of poxvirus particles in material from lesions. Capripoxviruses can be readily distinguished from parapoxviruses.
- Virus may be isolated in lamb testis or kidney cell monolayers.
- An antigen-trapping ELISA has been developed for the detection of capripoxvirus antigen
- Several serological methods including virus neutralization, Western blot analysis and the indirect fluorescent antibody test are available.

Control

- In endemic areas, control is based on annual vaccination. Modified live and inactivated vaccines are available. Asubunit vaccine has also been developed. Inactivated vaccines are less effective than modified live vaccines because cell-mediated immunity is the predominant protective response.

- Capripoxviruses are being employed as vectors for important ruminant viral vaccines. These vector vaccines may provide protection against diseases caused by capripoxviruses as well as diseases such as rinderpest and peste des petits ruminants.

Fowlpox

Etiology: Fowlpox is caused by fowlpox virus under genus *Avipoxvirus*, subfamily *Chordovirinae*

This disease affects domestic poultry, including chickens and turkeys. The infection is slow-spreading and characterized by proliferative skin lesions and diphtheritic lesions in the upper digestive and respiratory tracts. Fowl pox has a worldwide distribution.

Epidemiology

Fowlpox, pigeonpox and turkeypox viruses are closely related and are not strictly host-specific. Several avian species are susceptible to infection with fowlpox virus.

Transmission occurs by contact and by mechanical transfer on the mouth parts of biting arthropods, particularly mosquitoes. Virus enters the body through abrasions on unfeathered skin, on oral mucosa or on respiratory mucosa. Aerosols generated from scab material may result in transmission by inhalation. There is some evidence that persistence as latent infection and reactivation by stress may occur in a few birds.

Pathogenesis

Virus multiplication occurs at the site of introduction and may be confined to that site when the strain of infecting virus is of low virulence. Infections caused by virulent strains result in viraemia with replication in internal organs. The route of introduction influences the distribution and severity of lesions. Factors such as malnutrition, debilitation and stress may contribute to the severity of the disease.

Clinical signs

The incubation period is up to 14 days. Two forms of fowlpox have been described, a cutaneous form (dry pox) and a diphtheritic form (wet pox).

Cutaneous form: Nodular lesions develop on the comb, wattles and other unfeathered areas of skin. Progression to vesicle formation is followed by ulceration and scab formation. Healing occurs within two weeks. In severely affected birds, lesions may involve both feathered and unfeathered areas of skin, and involvement of the eyelids may lead to complete closure.

Diphtheritic form: Yellowish necrotic lesions (cankers) develop on the mucous membranes of the mouth, oesophagus and trachea. Oral lesions may interfere with eating. Tracheal involvement may lead to laboured breathing and rales. The mortality rate, which is higher in birds with the diphtheritic form, may approach 50% in severe outbreaks, particularly when accompanied by secondary bacterial or fungal infection.

Economic losses are largely due to a transient drop in egg production in laying birds and reduced growth in young birds.

Diagnosis

- ➢ Large intracytoplasmic inclusions (Bollinger bodies) containing small elementary bodies (Borrel bodies) may be demonstrable in epithelial cells.
- ➢ Virus may be isolated on the chorioallantoic membrane of nine to 12 day old embryonated eggs.

- ➢ Typical poxvirus particles can be demonstrated by electron microscopy in material from lesions. Immunofluorescence and immunoperoxidase techniques can be used to identify viral antigen in intracytoplasmic inclusions.
- ➢ Nucleic acid probes can be used for diagnosis.
- ➢ Suitable serological tests include ELISA, virus neutralization, agar gel precipitation and passive haemagglutination.

Treatment and control

There is no specific treatment. Control of secondary bacterial infection is desirable.In endemic areas, improved management and hygiene along with regular vaccination have reduced the effect of the disease on commercial poultry production. Modified live fowlpox or pigeonpox virus vaccines, produced in tissue culture or chick embryo, are available commercially. Chickens are usually vaccinated at about one month of age. Recombinant vaccines employing fowlpox and canarypox viruses are being developed.

NEWCASTLE DISEASE

Newcastle disease (ND) is caused by avian paramyxovirus type I (APMV-1). The infection is considered as one of the very severe viral diseases of birds. The disease affects almost all poultry. The infection causes severe economic losses to farmers. The infection is also endemic in most of the countries in the world.

General characters:

Classification:

Baltimoregroup	-	**Group V** – Negative sense RNA viruses
Order	-	Mononegavirales
Family	-	Paramyxoviridae
Subfamily	-	Parmyxovirinae
Genus	-	Avulavirus
Species	-	Avian Paramyxovirus 1 (APMV-1) There are nine different species under the genus Avula virus and they have been numbered 1-9 indicated as APMV-1 to APMV-9. Strains of APMV-1 cause Newcastle disease. APMV-1 cross reacts with APMV-2 and APMV-3.

Synonyms: Pseudo fowlpest, Ranikhet disease, Vellaikalichal (vernacular)

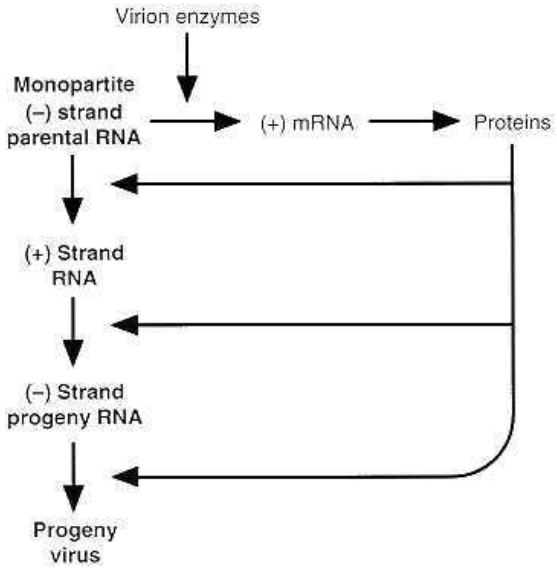
Morphology: APMV-1 are single stranded, non-segmented and negativesense RNA viruses. They are enveloped viruses and are ether sensitive. They possess helical symmetry. The surface has number of projections (HN Protein). They measure 150-300 nm. The virus contains six genes encoding six major structural proteins: nucleoprotein (NP), phosphoprotein (P), matrix protein (M), fusion protein (F), hemagglutinin-neuraminidase (HN), and RNA-dependent RNA polymerase (L). Apart from the six proteins APMV-1 also contains two proteins V and W. The RNA together with NP, P, and L proteins forms the ribonucleoprotein complex (RNP), which serves as a template for RNA synthesis. The hemagglutinin/neuraminidase (HN) is important for the attachment and release of the virus from the host cells in addition to its serologic identification. The other very important surface protein is the fusion (F) protein, which has a critical role in the pathogenesis of the disease.

OIE Listing: List A infection (*ND and highly pathogenic avian influenza are the only two poultry diseases specified as List A.*

Risk group(Animal Pathogen): Group II

Resistance: The virus is comparatively a stable virus and survives for long periods at ambient temperature, especially in faeces and can persist in houses (in faeces, dust etc.) for up to 12 months. However, it is sensitive to disinfectants, fumigants and sunlight. The virus is inactivated by temperatures of 56°C for 3 hours or 60°C for 30 min, acid pH, formalin and phenol, and is ether sensitive.

Replication:



Haemagglutination property : APMV-1 causes agglutination of chicken and human O erythrocytes. The agglutination reaction is not permanent and after 30 minutes as a result of loss of receptor the agglutination is lost and this phenomenon is referred as elution.

Virus strains : Though there are many strains, all strains are immunologically same. Some of the important virus strains are as follows

Pathotype	Strains
Velogenic	Hertz 33, Texas GB
Mesogenic	K, R2B
Lentogenic	F, LaSota, Ulster 2C
Avirulent	V4

Cultivation:

APMV-1 can be readily cultivated in embryonated eggs. The preferred route is allantoic cavity route. 9-11 days old embryos are preferred for cultivation. Virulent strains cause death of embryos in 24 hours with haemorrhages throughout the body particularly in occipital region. Lentogenic strains do not cause death. Presence virus is usually confirmed by HA and HI test using specific serum. The virus also grows in primary cells like Chicken embryo fibroblast and pig kidney cells. The virus also grows in BHK21 cellline.

Pathogenesis

ND has been reported at least 241 species of birds representing 27 of the 50 Orders of the class. Almost all birds are susceptible to infection. ND viruses have also been reported to infect animals other than birds, ranging from reptiles to man. The pathogenesis and production of clinical symptoms depend on the virus, the host species, age of host, infection with other organisms, environmental stress

and immune status. Infection with the extremely virulent viruses may result in sudden, high mortality with comparatively few clinical signs. The virus is grouped into five "pathotypes" on the basis of the predominant signs in affected chickens. They are as follows;

- **Viscerotropic velogenic (Doyles form)**: viruses responsible for disease characterized by acute lethal infections, usually with haemorrhagic lesions in the intestines of dead birds. The infection caused by this pathotype is also called as **Asiatic** or **exotic ND.**
- **Neurotropic velogenic (Beach's form)**: viruses causing disease characterized by high mortality, which follows respiratory and neurological disease, but where gut lesions are usually absent.
- **Mesogenic (Beaudettes form)**: viruses causing clinical signs consisting of respiratory and neurological signs, with low mortality.
- **Lentogenic (Hitchner's form)**: viruses causing mild infections of the respiratory tract.
- **Asymptomatic enteric**: viruses causing avirulent infections in which replication appears to be primarily in the gut.

Some workers differentiate NDV into three pathotypes: lentogenic, mesogenic, and velogenic, reflecting increasing levels of virulence. The most virulent (velogenic) isolates are further subdivided into neurotropic and viscerotropic types.

The infection is transmitted via aerosols, direct contact with infected birds, fomites, contaminated meat and water, through visitors and imported psittacines. Village chicken and birds like crows and sparrow can also spread the infection to chickens. The infection is not usually vertical but chicks may become infected in hatcheries from contaminated shells. The incubation period varies from 2 to 15 days. For velogenic ND in chickens, an incubation period of just 2 to 6 days is common. The incubation period in other species of birds may be longer.

Symptoms: Signs are predominantly seen in nervous, respiratory or reproductive systems. Morbidity is usually high and mortality varies from 0–100%. Higher mortality is seen in velogenic disease in unvaccinated stock. The clinical signs are highly variable and depend on the nature of the infecting virus (pathotype), the infective dose and the degree of immunity birds possess from previous infection or vaccination. The first clinical sign in laying chickens is usually a marked drop in egg production followed within 24 to 43 hours by high death losses. At the onset, 10-15 percent of a flock may be lost in 24 hours. After 7 to 10 days, deaths usually subside, and birds surviving 12 to 14 days generally do not die but may display permanent paralysis and other neurologic signs. The reproductive system may be permanently impaired, and thus egg production does not return to previous levels. In vaccinated chickens, or chicks protected by parental antibodies, the clinical signs are less severe and are proportional to the level of protective antibodies.

With viscerotropic strains (VVND), edema of the head, especially around the eyes may become apparent after birds have been sick for 2 or 3 days. This edema usually does not involve the comb and wattle as in case of highly pathogenic avian influenza (HPAI). A dark ring sometimes forms around the eye, probably due to cyanosis and poor blood circulation in the edematous tissue. This "black eye" appearance is especially visible in white chickens. Bile- stained, greenish-dark diarrhea may be noted 2 to 3 days after onset of illness. Some birds in an affected flock usually have diarrhea throughout the course of the disease. The most noteworthy clinical sign in unvaccinated flocks is sudden death without prior indications of illness. Neurotropic strains cause respiratory signs followed by neurologic signs, including muscular tremors, paralysis of legs or wings, torticollis, and opisthotonos. The peracute onset often causes the owner to suspect poisoning. Fall in egg production leading to complete cessation of egg laying anddeaths.

Lesions: As with clinical signs, no gross or microscopic lesions can be considered pathognomonic for any form of ND. Carcasses of birds dying as a result of virulent ND usually have a fevered, dehydrated appearance. Gross lesions vary with the infecting virus. Virulent ND viruses cause haemorrhagic lesions of the intestinal tract. These are most easily seen if the intestine is opened and may vary considerably in size. Haemorrhages are seen in the proventriculus, duodenum, jejunum and ileum. Haemorrhages and necrotic lesions in proventriculus and ileo-caecal junction are considered as specific for ND. Even in birds showing neurological signs prior to death, minimal gross lesions are seen in the central nervous system. Changes in the central nervous system are nonpurulent encephalomyelitis. Lesions are usually present in the respiratory tract when respiratory signs are observed. These generally appear as haemorrhagic lesions and congestion in lungs, tracheitis and airsacculitis. Egg peritonitis is seen in laying hens infected with virulent NDV. Ovaries may be edematous, hemorrhagic, or degenerated. Microscopic lesions are not considered to have any diagnostic significance. In most tissues and organs where changes occur, they consist of hyperaemia, necrosis, cellular infiltration and oedema.

Diagnosis:

1. **Field diagnosis: T**entative diagnosis of ND made on the basis of history, clinical signs, and gross lesions
2. Virus isolation and identification:

- **Clinical materials:** Samples from dead birds should consist of oro- nasal swabs, as well as samples collected from lung, kidneys, intestine (including contents), spleen, brain, liver and heart tissues. These may be collected separately or as apool.
- **Virus isolation:** virus isolation is attempted by inoculating 9- to 11-day-old embryonating chicken eggs. Amnioallantoic fluid (AAF) is collected from all embryos dying after 24 hours post inoculation and tested for hemagglutination (HA) activity. If positive, the hemagglutination-inhibition (Hl) test is used with known NDV- positive serum to confirm the presence of NDV in the AAF. If NDV is found, it is characterized by inoculating 4- to 6-week-old chickens free of ND antibodies with the suspect AAF by swabbing the cloaca, instilling into the nares or conjuctival sac, or injecting into the thoracic air sac. If VVND virus is present, the inoculated chicks usually die in 3 to 7 days, revealing typical visceral lesions on postmortem examination. Neurotrophic viruses will cause severe neurologic and respiratory signs in inoculated chickens but no visceral lesions. If no bird dies in 10 days, the APMV-1 is not considered to be the velogenic, viscerotropic type but is either a lentogenic or mesogenic.
- **Nucleic acid identification methods:** The virus can also be identified from clinical materials and AAF by RT-PCR (Reverse transcriptase Polymerase chain reaction) using primers for F gene.
- Nucleic acid probes
- **ELISA techniques:** Antigen capture and Antigen competitive ELISA techniques can also be used to identify APMV-1 from clinical materials and AAF.

3. Pathotyping: Newcastle disease virus are classified into Velogenic, mesogenic and lentogenic based on following three tests

- ❖ **Intracerebral pathogenicity index (ICPI)** in day old chickens. This involves the inoculation of virus derived from fresh infective allantoic fluid into the brain of ten day-old chicks from specific pathogen-free parents. Each bird is examined at 24-hour intervals for eight days and graded zero if normal, one if sick and two if dead. The index is the mean score per bird per observation over the 8-day period. The most virulent viruses give ICPI values approaching the maximum score of 2.0, while lentogenic viruses give values of, or close to, 0.0
- ❖ **Intravenous pathogenicty index (IVPI)** in 6 weeks old specific pathogen free (SPF) chickens. 0.1 ml of the diluted virus is injected intravenously into each of ten 6-week-old SPF chickens. Birds are

examined at 24-hour intervals for 10 days and scored at each observation - 0 if normal, 1 if sick, 2 if paralysed or showing other nervous signs, and 3 if dead. The intravenous pathogenicity index (IVPI) is the mean score per bird per observation over the 10-day period. Lentogenic strains and some mesogenic strains will have IVPI values of 0, whereas the indices for virulent strains will approach 3.0.

- **Mean death time (MDT)** in embryonated eggs. The mean death time (MDT) in eggs means time in hours for the minimum lethal dose to kill all the inoculated embryos. The MDT has been used to classify ND virus strains into velogenic (taking under 60 hours to kill); mesogenic (taking 60 to 90 hours to kill); and lentogenic (taking more than 90 hours to kill).
- Molecular typing by studying the aminoacid between 112- 119th position in F gene.
- **Panels of MAbs** are also used to establish antigenic profiles of ND virus isolates based on whether or not they react with the viruses.

4. Serology: A number of tests are performed to identify antibodies against NDV in chickens. The two common tests are haemagglutination inhibition (HI) test and ELISA. HI test is preformed in most of the laboratories. The HI test is based on the principle that the haemagglutinin on the viral envelope can bring about the agglutination of chicken red blood cells and that this can be inhibited by specific antibodies. A titre of $\log_2 3$ is indicative of protection and a titre of $\log_2 6$ or more suggests a recent infection by the virus. The ELISA works on the principle of recognition of anti-NDV antibodies, attached to a plate coated with viral antigen, by antibodies produced in another species against chicken antibodies. This anti-chicken antibody is conjugated to an enzyme that catalyses a reaction, causing a change of colour which can then be read quantitatively on a photo spectrometer designed to read microtitration plates

5. Monoclonal antibody based HI: Mouse monoclonal antibodies (MAbs) directed against strains of ND virus have been used in HI tests to allow rapid identification of ND virus without the possible cross-reactions with other APMV serotypes that may occur with polyclonal sera. MAbs have been produced that give reactions in HI tests that are specific for particular strains or variant ND virus isolates.

Control

Vaccination: These different approaches to vaccination have been applied to ND. There are three types of vaccines used for ND: live lentogenic, live mesogenic and inactivated vaccines.

1. Live lentogenic vaccines are usually derived from field viruses that have been shown to have low pathogenicity for poultry but produce an adequate immune response. Typical vaccine strains are HB1, La Sota, F, V4 or Ulster 2C viruses. These vaccines are given to young chicks through eyes are nostrils. They are very safe and only La Sota will produce mild post- vaccinal reaction in certain flocks. This type immunization is also called as priming.
2. Mesogenic strains are used only in areas where ND is epidemic and in village chickens. Normally mesogenic vaccines, such as Komarov and Mukteswar are used as secondary vaccines after a primary vaccination with a lentogenic vaccine.
3. Inactivated vaccines are produced by growing a ND virus in eggs, and then treating the infective allantoic fluid with an inactivating agent, such as formalin or betapropiolactone. An adjuvant, such as mineral oil, is usually then added to make the inactivated virus more immunogenic. Inactivated vaccines are usually applied after an initial priming vaccination with a live vaccine.
4. Thermostable live vaccines: Some asymptomatic enteric viruses have greater heat resistance than more conventional lentogenic viruses. This property has been enhanced by selection and cloning in the laboratory to produce heat tolerant vaccines. These have a distinct advantage in the village situation because it is possible to transport the vaccine without a coldchain.

5. Recombinant vaccine: APMV-1 has two surface glycoproteins, fusion [F] and haemagglutinin/ neuraminidase [HN]. The genes coding for either of these can be inserted into a different virus like herpes virus of turkeys to produce a vaccine, which gave good protection against virulent NDV.
6. Marker vaccine: Recently an immunodominat epitope of NP gene has been identified and replaced with mouse herpes virus gene. The resultant vaccine helps to differentiates between vaccinated and infected birds.

Vaccine application methods:

1. Mass administration through water or aerosol spray
2. Feed based techniques
3. As eye and/or nasal drops
4. As injection

Prevention:

Biosecurity is the best method to prevent the occurrence of infection. Farms and flocks should be well separated, hatcheries should be isolated from poultry farms, different species should be reared on different sites, and there should be an adequate fresh water supply, preferably one that does not draw on surface water.

On the farms the following points should be observed:

1. Regular vaccination and periodical antibody monitoring
2. Houses, food stores and water tanks should be bird-proofed.
3. Movements on and off the farm should be kept to a minimum.
4. All equipment, especially vehicles, should be disinfected before access to the site is permitted.
5. Movements between different farms for egg collection, carcass collection, food delivery and the like should be confined to a specified collection and delivery point away from the poultry flocks.
6. Visits by personnel such as bleeding or vaccination teams, inseminators and veterinarians are the most likely method of introduction of ND and if such visits are unavoidable, regimens of clothing change, equipment disinfection and other basic hygiene controls must been forced.

Public health hazards: People may become infected with APMV-1, the resulting disease is typically limited to a conjunctivitis. Recovery is usually rapid, and the virus is no longer present in eye fluids after 4 to 7 days. Infections have occurred mostly in laboratory workers and vaccinating crews with rare cases in poultry handlers. No instance of transmission to humans through handling or consuming of poultry products is known. Individuals with conjunctivitis from APMV-1 virus should not enter poultry premises or come in contact with live avian species.

RINDERPEST

Rinderpest is an acute, viral disease of domestic cattle, buffaloes and yaks characterized by high morbidity and mortality rates. Sheep, goats, pigs and wild ungulates may also be affected. It is characterized by fever, oral erosions, diarrhea, lymphoid necrosis, and high mortality.

General characters

Classification:

Baltimore group - **Group V** – Negative sense RNA viruses
Order - Mononegavirales
Family - Paramyxoviridae

Subfamily - Parmyxovirinae
Genus - Morbillivirus
Species - Rinderpestvirus

Rinderpest virus shares similarities in structure and antigenicity with measles and canine distemper viruses. These three viruses together form the **MEDIPEST** group.

Synonyms: Cattle plague

Morphology: Rinderpest virus is pleomorphic. It is single stranded, negativesense RNA viruses. They are enveloped viruses and are ether sensitive. They possess helical symmetry. Virus measures approximately100 to 300 nm in diameter. The envelopes are covered with minute projections, which are the surface glycoproteins (H and F proteins) responsible for cell attachment and fusion. Rinderpest virus is differentiated from PPR virus (*PPR is a very important infection resembling rinderpest in small ruminants)* through nucleic acid probes for N gene. In total The virus has 6 important proteins and two non-structuralproteins.

They are

N - nucleocapsid protein which covers and protects the virion RNA
P - polymerase-associated or phosphoprotein
M - matrix protein, interacts with cytoplasmically located nucleocapsids and membrane associated glycoproteins in cell envelope to produce virus bud
F - fusion protein enables virus and host cell membrane to fuse
H - haemagglutinin or attachment protein, enables virus to attach to host cell membrane
L - large protein, acts as a virus transcriptase and replicase in association with P protein
C/V - two non-structural virus-encoded proteins produced in infected cells, may have function in virus reproduction

The H and F proteins are very important in establishing the infection. H is responsible for attachment and F is responsible for entry into the cell.

OIE Listing: List A infection

Risk group: Class II Animal Pathogen

Resistance: Rinderpest virus is not a stable virus. The virus is readily destroyed at relative humidity between 50 and 60 percent. It is sensitive to heat, light and ultrasonic waves. High and low pH also denatures the virus. Being enveloped, rinderpest virus is destroyed by lipid solvents and lipophilic disinfectants. In dung material 5 percent sodium hydroxide and 50 percent lysol destroy the virus effectively. Molar concentration of magnesium sulphate heptahydrate (25 g in 100 ml) will slow down inactivation of the virus in water.

Virus strains: There is only one serotype of rinderpest virus, but field strains vary widely in virulence, ease of transmission, and host affinity. Some of the important strains of virus are Nakumara and Kabate O.

Cultivation:

Rinderpest virus is readily cultivable in cell culture. Some of the common cell cultures used for cultivation of rinderpest virus are calf kidney cells, African green monkey cells (Vero cells), lymphoblastoid cell line, B95a. Specific cytopathic effects (CPE) develop after three days. The CPE are in the form of small foci of rounded retractile cells with long cytoplasmic processes, stellate cells and small multinucleate syncytia with irregular shapes. With further incubation, the CPE generalizes to

involve large areas of the cell sheet and, by the ninth or tenth day after inoculation cells peel off from the surface. Rarely the syncytia may become highly vacuolated. The laboratory animal of choice for virus growth is rabbits. Rinderpest virus also grows in the chorioallantoic membrane of embryonated eggs, but no lesions are produced.

Pathogenesis:

1. **Host affected:** The common host is the domestic ungulates cattle and buffalo. Sheep, goat and pigs are the other important domestic animals affected by rinderpest. The infection has also been reported in a variety of wild animals including bushbuck, bush pig, giraffe, gaur, nilgai and sambar.
2. **Transmission:** The secretions and excretions from infected animals, particularly nasal-ocular discharges and faeces, 1 to 2 days before clinical signs to 8 to 9 days after onset of clinical signs contain large quantities of virus. Rinderpest virus spreads from healthy to susceptible animals mostly through these infected droplets, either in the breath of a sick animal or in its virus-rich secretions or excretions. The important source of infection for a new herd is arrival of live animal from infected areas. Since the droplets are large and short-lived, the contact between sick and healthy animals must be close for transmission to occur. Transmission also occurs through indirect methods like contact with contaminated bedding, fodder or water. In pigs infection also occurs through eating uncooked infected meat scraps. There is no vertical transmission. Further the infection is also not transmitted by arthropod vectors. There is no carrier state of infection also.
3. **Incubation period:** The incubation period varies with the strain of virus, dosage, and route of exposure. Following natural exposure, the incubation period ranges from 3 to 15 days but it is usually 4 to 5 days.
4. **Symptoms:**

Buffaloes and cattle: Clinical reactions in buffalo and cattle are similar and the infection may be exhibited as peracute, acute, subacute or even inapparent.

i. Peracute infection: The onset of a peracute reaction is sudden and unexpected. It is manifested by inappetence, high fever, depression, deep congestion of visible mucous membranes, severe panting and racing pulse. Death occurs within two or three days, even before mucosal erosions develop. The peracute reactions are not common and occur most frequently in young calves and exotic animals.

ii. Acute infection: This form is divided into five phases 1) an incubation period, 2) a prodromal fever, 3) an erosive-mucosa phase, 4) a diarrhoeic phase and 5) a convalescent phase in surviving animals. The onset of the **prodromal fever** is sudden, it is frequently missed because other clinical signs are minimal, except in lactating cows where there is drop in milk yield. Typical illness is clearly evident 24 to 48 hours later, when the animal becomes restless and then stands depressed, apart and alone. Respirations are shallow and rapid. The coat hairs stand erect, the muzzle dries, tears are wept and the nose runs. Appetite is impaired, rumination is retarded and defecation stops. Visible mucous membranes are congested but intact. The **erosive mucosl phase** is observed two to three days after prodormal fever and is characterized by raised pinheads of necrotic epithelium emerging from the surfaces of the mucous membranes lining the mouth, nasal passages and urogenital tracts. These lesions erode exposing a haemorrhagic layer. At this stage salivation is profuse. The erosions enlarge and coalesce. Thick yellow patches of necrotic cells begin to coat the nasal passages and mix with the nasal secretions, producing a fetid mucopurulent discharge. Secretions from eye also become mucopurulent. This stage is also characterized by intense thirst, poor appetite and voiding of soft faeces. The **dirrhoeic phase** begins after the fever subsides The dark, fluid faeces contain excess mucus and shreds of epithelium and necrotic debris streaked with blood. The smell is fetid and offensive. Affected animals arch their backs and strain frequently, exposing congested and eroded rectal mucosae. The diarrhoea is also referred

as shooting diarrhoea. Respirations are laboured and painful, characterized by an audible grunt when exhaling. In fatal cases of rinderpest the diarrhoea worsens progressively, causing rapid dehydration. Affected animals waste visibly, have sunken eyes and stand with lowered heads and arched backs. Most animals collapse and die six to 12 days after the onset of the prodromal fever. In surviving cases, the diarrhoea stops within a week of its onset. Pregnant animalsl abort during convalescence.

iii. Subacute infection: Subacute infections are seen in immature and young adult animals. The incubation period is longer than that of the acute syndrome and may even last 15 days .The clinical signs are mild and often one or more of the cardinal features of the classic disease, such as fever, mucosal erosions, mucopurulent nasal and ocular discharges or diarrhoea, are absent. Most affected animals survive the infection.

In cattle the rinderpest virus selectively destroys T- and B- lymphocytes. Because of this the latent pathogens are commonly activated and supra-infection occurs.

- **Sheep and goats:** Acute, subacute and inapparent rinderpest reactions occur in goats and sheep. Many of the clinical signs resemble those seen in cattle, but the course of the disease is shorter and pneumonic symptoms are more prominent. Acute cases die six to seven days after the onset of illness, whereas survivors show signs of recovery within two weeks.
- **Pigs:** Peracute reactions are characterized by sudden sharp fevers and death before other premonitory signs develops. Acute cases have a similar sudden onset, manifested by fever, inappetence and depression. Two days after the onset of fever, affected pigs shiver, vomit and bleeding is seen from the nose. Shallow erosions emerge in the oral mucosa while vesicles erupt in the perineal skin. Diarrhoea follows erosions in oral mucosa. The fluid faeces is fetid and heavily streaked with blood. Pregnant sows abort.
- **Lesions:** Rinderpest virus has affinity for lymphoid tissues and a secondary affinity for the epithelium of the alimentary, upper respiratory and urogenital tracts.
- Oral lesions start as small grey foci that may coalesce. The grey (necrotic) epithelium then sloughs off and leaves red erosion. At mouth lesions occur on the gums, lips, hard and soft palate, cheeks, and base of the tongue. The early lesions are grey, necrotic, pinhead- sized areas that later coalesce and erode and leave red areas.
- Intestinal lesions are characterized by necrosis or erosion of Peyer's patches in the jejunum, necrosis or erosions over the lymphoid area in the ileum. Lesions are more severe in the upper colon (edema of the wall, erosions in the mucosa, and congestion). Further down the colon, the colonic ridges may be congested; this is referred to as "**tiger striping**". Severity of intestinal lesions varies between isolates.
- Lymph nodes are swollen and edematous.
- Lung lesions are emphysema, congestion, and areas of pneumonia.

Diagnosis:

1. Field diagnosis
2. **Specimens to be collected:** specimens should preferably be collected from animals with a high fever and oral lesions. The following samples should be collected from live animals:

a. Live animals:

- Blood in EDTA or heparin
- Blood for serum
- Swabs containing lacrimal fluid

- Tears
- Necrotic tissue from the oral cavity
- Aspiration biopsies of superficial lymphnodes

b. Dead animals: Sections of

- ❖ Spleen
- ❖ Lymph nodes
- ❖ Tonsil

3. Differential diagnosis: Rinderpest should be differentiated from bovine virus diarrhea (mucosal disease), infectious bovine rhinotracheitis, malignant catarrhal fever, foot-and-mouth disease, vesicular stomatitis, salmonellosis, paratuberculosis, and arsenic poisoning.

4. Isolation and identification of virus in calf kidney cells and vero cells using specimens specified above.

5. Detection of virus specific antigen by

a. Agar gel immunodiffusion test

i. Counterimmunoelectrophoresis

ii. Passive haemagglutination

iii. Immunofluorescense and immunoperoxidase techniques

iv. Antigen capture ELISA

v. RT-PCR

vi. DNA probes for N gene

6. Serological tests

i. Competitive ELISA (test prescribed for International Trade)

ii. Virus neutralization test (Gold Standard)

iii. Measles haemagglutination inhibition test (MHI test)

Treatment: There is no effective treatment against rinderpest.

Vaccines: Since India has been declared provisionally free from rinderpest no vaccination is carried out in cattle. However two types of vaccines have been used previously a goat tissue adapted vaccine (GTV) and tissue culture adapted rinderpest virus vaccine (TCRV). Though GTV produces good immunity in native zebu cattle it caused severe post vaccinal reaction in cross bred animals. Hence, it was discontinued and TCRV was used. Since 1996-97, vaccination of cattle and sheep against rinderpest has been discontinued. Apart from GTV and TCRV, lapininsed (rabbit passaged), lapinised and avianised rinderpest virus have also been used as vaccine is many Asian countries. Vaccinia-vectored vaccine containing the F and H genes of RPV was also used experimentally.

Control and Eradication:

Eradication:

a. Countries and areas free of RP should prohibit unrestricted movement of RP-susceptible animals and uncooked meat products from areas infected by RP or practicing RP vaccination.

b. Since recovered animals are not carriers, serological techniques like ELISA can be effectively used to check the status of animals imported.

c. Wild ruminants and swine imported should be properly quarantined and tested with serological tests.

d. During an outbreak occurs, the area should be quarantined, infected and exposed animals slaughtered and buried or burned, and ring vaccination considered. (*Ring vaccination is a technique, wherein the animals in surrounding areas of actual location of infection are vaccinated*)

e. High-risk countries (those trading with, or geographically close to, infected countries) should protect themselves by having all susceptible animals vaccinated before they enter the country or vaccinating the national herd, or both.

f. Endemic countries should vaccinate their national herd. The schedule for vaccination is annual vaccination for at least 4 years, followed by annual vaccination of calves.

g. Foci of infection should be quarantined and stamped out. Wildlife, sheep, and goats should be monitored serologically.

h. RP vaccine should not be used to protect against *peste des petits ruminants* since it may interfere with serological monitoring of sheep and goats for rinderpest.

i. The Global Rinderpest Eradication Programme (GREP) is a time-bound programme to eliminate rinderpest from the world by the year 2010. This programme is operated by the FAO. It forms part of EMPRES. Strategies have been devised and programmes implemented to reduce the clinical incidence of rinderpest to zero. Elimination of disease and infection will be confirmed by statistically valid active disease

j. OIE Pathway: The OIE has developed a set of Recommended Standards for Epidemiological Surveillance for Rinderpest (the 'OIE Pathway') that governs the actions of Member Countries wishing to demonstrate that they have achieved freedom from infection. To this end, both competition and indirect enzyme-linked immunosorbent assays are available that will determine the presence of rinderpest antibodies in animals that have been infected with field virus or with rinderpest vaccine.

Public Health

There is no report of rinderpest infection in a human.

PESTE DES PETITS RUMINANTS

Peste des petits ruminants (PPR), also known as goat plague,is caused by a paramyxovirus of the *Morbillivirus* genus. It was first described in 1942 in Cote d'Ivoire, West Africa and is closely related to rinderpest virus, canine distemper virus, and human measles virus. PPR is an acute or subacute viral disease of goats and sheep characterized by fever, erosive stomatitis, conjunctivitis, gastroenteritis, and pneumonia. Goats are usually more severely affected than sheep (*hence called goat plague).*

Generalcharacters

Classification:

Baltimoregroup - **Group V** – Negative sense RNA viruses

Order - Mononegavirales

Family – Paramyxoviridae

Subfamily - Parmyxovirinae

Genus - Morbillivirus

Species - PPRvirus

Synonyms: Goat plague

Morphology: PPR virus is pleomorphic. It is single stranded, negative sense RNA viruses.They are enveloped viruses and are ether sensitive.

They possess helical symmetry. Virus measures approximately 150 nm in diameter. The envelopes are covered with minute projections, which are the surface glycoproteins (H and F proteins) responsible for cell attachment and fusion. The nucleocapsid is hollow, coiled, rod-shaped with herring-bone pattern of sub-units. PPR virus is differentiated from rinderpest virus (*PPR is a very important infection resembling rinderpest in small ruminants)* through nucleic acid probes for N gene. In total the virus has 6 six structural proteins:

Nucleocapsid protein - **Np** covers and protects the virion RNA

Phosphoprotein **P** - polymerase-associated or phosphoprotein

Large protein- **L** - acts as a virus transcriptase and replicase in association with P protein

Matrix protein – **M** – acts as a link between nucleocapsid and virus external glycoprotein H

Fusion – **F** - enables virus and host cell membrane to fuse

Haemagglutinin **H** - haemagglutinin or attachment protein, enables virus to attach to host cell membrane

C / V - two non-structural virus-encoded proteins produced in infected cells. They have function in virus reproduction

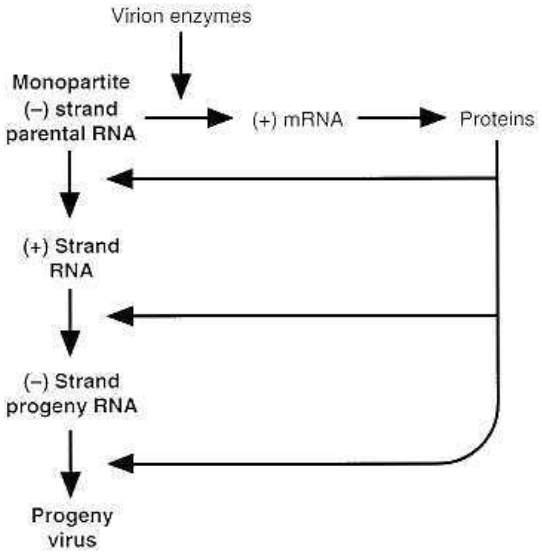
The matrix protein, intimately associated with the internal face of the viral envelope, makes a link between the nucleocapsid and the virus external glycoproteins H and F, which are responsible for the attachment and the penetration of the virus into the cell to be infected.

OIE Listing: List A infection

Risk group: Class II Animal Pathogen

Resistance: PPR virus is not a stable virus. PPRV is a heat sensitive virus and destroyed at 56^0C. It is stable at a pH between 4.0 to 10.0. However, the virus is killed by alcohol, ether, and detergents as well as by most disinfectants including common disinfectants like phenol, sodium hydroxide). PPRV also survives for a long time in chilled and frozen tissues.

Replication:



Haemagglutination: PPRV caused haemagglutination of chicken RBC. However, the pH of PBS used should be at 6.8.

Virus strains: PPR virus is closely related to rinderpest virus. But it is different from rinderpest. Different PPRV strains are grouped into four groups, of which 3 groups are found in Africa and 1 at Asia. One African group is also prevalent in Asia. The Arasur strain of PPRV (referred asAR-87) is used as vaccine in Southern India. Other South Indian PPRV isolates are Coimbatore (CBE) and Poondi (92).

Cultivation:

Tissue culture systems are ideal for the cultivation of PPRV. The virus grows well in lamb kidney cells and African green monkey cells (Vero cells). The CPE develop within 5 days and consists of cell rounding and aggregation and syncytia formation in lamb kidney cells. In Vero cells syncytia are rare and if present are very small. Syncytia are recognised by a circular arrangement of nuclei giving a "clock face" appearance. Other important CPE include intracytoplasmic and intranuclear inclusions and vacuolation of cells.

Pathogenesis:

1. Hosts affected: PPR is primarily a disease of sheep and goats. The infection also occurs in wild ungulates. Cattle, buffaloes, camels, and pigs are also susceptible to infection but do not exhibit clinical signs and are unable to transmit the disease to other animals.

2. Distribution: PPR is prevalent in Sub-Saharan Africa, Indian sub-continent and Middle East countries.

3. Transmission: PPR is not very contagious and transmission requires close contact between animals. Ocular, nasal, and oral secretions and faeces are the sources of virus. Infection occurs mainly through inhalation of aerosols produced by sneezing and coughing. Fomites such as bedding may also contribute to the onset of an outbreak. There is no known carrier state. The spread is not dependant on vectors. The appearance of clinical PPR may be associated with any of the following:

- ❖ History of recent movement or gathering together of sheep and/or goats of different ages
- ❖ Introduction of recently purchased animals; contact in a closed/village flock with sheep and/or goats that had been sent to market but returned unsold;
- ❖ Change in weather such as the onset of the rainy season (hot and humid) or dry, cold periods
- ❖ Contact with trade or nomadic animals
- ❖ A change in husbandry (e.g. towards increased intensification) and trading practices

4. Incubation period: The incubation period is from 4 to 5 days.

5. Morbidity and mortality: Up to 100 percent of the animals in a flock may be affected in a PPR outbreak with between 20 and 90 percent mortality.

6. Symptoms:

- ✓ Characterized by sudden onset of fever (40° to 41°C).
- ✓ Affected animals are markedly depressed and appear sleepy. Their hair stands erect giving them a bloated appearance.
- ✓ Soon after this stage, a clear watery discharge is noticed from the eyes, nose and mouth, which become thick and yellow as a result of secondary bacterial infection. The discharges wet the chin and the hair below the eye; they tend to dry, causing matting together of the eyelids, obstruction of the nose and difficulty in breathing.
- ✓ One to two days after fever has set in, the mucous membranes of the mouth and eyes become much reddened.
- ✓ This is followed by pin-point greyish areas on the gums, dental pad, palate, lips, inner aspects of the cheeks and upper surface of the tongue. These areas increase in number and size and join

together. In severe cases, the normal membrane may be completely covered by a thick cheesy material. Underneath the dead surface cells shallow erosions are noticed.

✓ Similar changes may also be seen in the mucous membranes of the nose, the vulva and the vagina. The lips tend to swell and crack and become covered with scabs.

✓ As the disease progresses, a characteristic foul smell exudes from the mouth. Affected animals resist attempts to open their mouths because of the pain.

✓ Diarrhoea commonly appears about two to three days after the onset of fever. The faeces are initially soft and then watery, foul- smelling and may contain blood streaks and pieces of dead gut tissue.

✓ Affected animals breathe fast exhibiting rocking movements of both the chest and abdominal walls. Severely affected cases show difficult and noisy breathing marked by extension of the head and neck, dilation of the nostrils, protrusion of the tongue and soft painful coughs.

✓ A common feature in later stages of the disease is the formation of small nodular lesions in the skin on the outside of the lips around the muzzle.

7. Lesions: The carcass of the affected animal is usually emaciated, the hindquarters soiled with soft/ watery faeces. The eyes and nose contain dried-up discharges. The following changes are also seen:

❖ **Mouth** - Dirty-white, false membranes and erosions are seen on the gums, soft and hard palates, tongue, cheeks and oesophagus.

❖ **Lips** - Appear Swollen with erosions and scabs or nodules

❖ Nasal cavity - Congested (reddened) lining with clear or creamy yellow exudates and erosions

❖ **Lungs** –Dark red or purple areas which are firm to the touch, are mainly seen in the anterior and cardiac lobes. Pleuritis and hydrothorax are also observed

❖ **Lymph nodes** (associated with the lungs and the intestines) - Appear soft and swollen.

❖ **Abomasum** - Congested (reddened) lining with haemorrhages

❖ **Small intestines** - Congested (reddened) lining with haemorrhages and erosions are noticed.

❖ **Large intestines** (caecum, colon and rectum) - Small red haemorrhages along the folds of the lining that join together and appear darker or green/black in stale carcasses (**Zebrastriping)**.

Diagnosis:

1. **Field diagnosis:** In the field, a presumptive diagnosis is made on the basis of clinical, pathological, and epizootiological findings.

2. **Clinical materials to be collected**

❖ Tears - Cotton buds or swabs of absorbent cotton wool are inserted into the conjunctival sac and swirled around to collect tears. The bud/swab is broken off into a container and about 150 microlitres of sterile phosphate-buffered saline (PBS pH 7.2 to7.6)

❖ Gum debris - This material can be collected by a spatula or finger rubbed across the gum and inside the upper and lower lips. The material collected is then scraped into a container and 150 microlitres of PBS are added

❖ Tissues – Tissue to be collected are from lymph nodes found around the lungs (mediastinal) and alimentary tract (mesenteric); portions of the spleen and the lungs. Two sets of each tissue are required; one set is chilled but not frozen, and the other is put in 10 percent formalin solution to preserve the samples

❖ Unclotted blood - This is needed for virus isolation and should be collected in bottles containing anticoagulants (heparin or ethylenediamine tetracetic acid [EDTA]).

❖ Clotted blood or serum – For identification of antiboies

3. Isolation and identification:

a. The tissue culture used is lamb kidney cells and Vero cells. The characteristic CPE appear around 5th day and are cell rounding and aggregation and syncytia formation

b. Identification of virus:

i. Immunocapture enzyme-linked immunosorbent assay (Sandwich ELISA)

ii. Counterimmunoelectrophoresis (CIEP): Most rapid test for detecting viral antigen

iii. Agar gel immunodiffusion (AGID): Very simple and inexpensive and gives results within 1 day, but not sensitive to mild forms of PPR

iv. Immunofluorescence and immunoperoxidase

v. Polymerase chain reaction (RT-PCR)

vi. Nucleic acid probes

4. Serology:

i. Competitive ELISA

ii. Virus neutralization test (Prescribed test for international trade)

5. Differential diagnosis:

Rinderpest	Clinically, RP and PPR are similar, but the former should be the prime suspect if the disease involves both cattle and small ruminants.
Pasteurellosis	Enzootic pneumonia or the septicemic form of pasteurellosis is characterized by obvious respiratory signs, infrequent diarrhea, and a fatality rate rarely exceeding 10 percent
Contagious caprine pleuropneumonia	There is no digestive system involvement, and the clinical signs and lesions are confined to the respiratory system and pericardium.
Bluetongue	Swelling of the lips, muzzle, and oral mucosa, together with edema of the head region, helps to differentiate bluetongue from PPR. Coronitis, common in bluetongue, is not a feature of PPR. Also, sheep are more affected than goats.
Heartwater	There is often central nervous system involvement, including convulsions. There is no diarrhea.
Contagious ecthyma (orf)	The orf virus causes proliferative, not necrotic lesions that involve the lips rather than the whole oral cavity. The absence of nasal discharges and diarrhea also distinguish orf from PPR.
Foot-and-mouth disease	This condition is comparatively mild, and the most characteristic clinical sign, lameness, is not a feature of PPR
Nairobi sheep disease	Sheep are more severely affected than goats
Coccidiosis	There is no upper digestive tract and respiratory system involvement.
Plant or mineral poisoning.	Several plants and minerals may cause severe intestinal lesions. Case history and absence of fevershould distinguish poisoning from PPR.

Control:

1. The tissue culture rinderpest vaccine at a dose of $10^{2.5}$ $TCID_{50}$ protects goats for at least 12 months against PPR. The vaccine is currently used in many African countries for vaccination against PPR. This vaccine interferes with rinderpest eradication programmes because it is impossible to determine if seropositive small ruminants have been vaccinated or naturally infected with RPV.
2. A homologous attenuated PPR vaccine is being tested and may soon be commercially available. In Southern India, a homologous PPR vaccine using AR-87 strain is used to control PPR in sheep and goat. This vaccine was developed at the Department of Veterinary Microbiology, Madras Veterinary College.

Eradication:

1. Methods that are applied for rinderpest eradication can be used for PPR eradication. These should include quarantine, slaughter, and proper disposal of carcasses and contact fomites, decontamination, and restrictions on importation of sheep and goats from affected areas.
2. Control of PPR outbreaks relies on movement control (quarantine) combined with the use of focused ("ring") vaccination and prophylactic immunization in high-risk populations.

Treatment:

There is no treatment for PPR. However, mortality rates may be decreased by the use of drugs that control the bacterial and parasitic complications. Specifically, oxytetracycline and chlortetracycline are recommended to prevent secondary pulmonary infections

Public Health

PPR does not cause infection in humans. Hence, no public health issues are concerned with this infection.

CANINE DISTEMPER

Canine distemper is a contagious, incurable, often fatal, multi systemic viral disease that affects the respiratory, gastrointestinal, and central nervous systems. CDV occurs among domestic dogs and many other carnivores, including raccoons, skunks, and foxes. CDV is fairly common in wildlife

General characters:

Classification:

Baltimoregroup - **Group V** – Negative sense RNA viruses
Order - Mononegavirales
Family - Paramyxoviridae
Subfamily - Parmyxovirinae
Genus - Morbillivirus
Species - Canine distemper virus (CDV)

Canine distemper virus (CDV) belongs to genus Morbillivirus within the *Paramyxoviridae* family, which includes other viruses such as: phocine distemper virus, dolphin morbillivirus, rinderpest virus, peste-des-petits- ruminants virus and measles virus.

Synonyms: Carre's disease, Hard paddisease

Morphology: CDV has a spherical shape and diameter ranging from 100 to 250 nm. It contains negative-stranded, non-segmented RNA. The virus contains six proteins: the nucleocapsid protein (NP), the phosphoprotein (P), the matrix protein (M), the fusion protein (F), the haemaglutinin attachment glycoprotein (H) and the large- polymerase protein (L).

Resistance: The virus is quickly killed by disinfectants and sunlight and heat. However, at normal temperatures the virus is very stable and can stay active in infected material for several weeks, provided the materials are not exposed to sunlight. At below zero temperatures the virus can stay active for many months, but at temperatures above 32°C it is rapidly inactivated. Routine disinfection and cleaning readily kills the distemper virus in the kennel setting.

Replication: Like other paramyxoviruses the CDV rapidly invades cells and uses the cell's reproductive mechanism to reproduce itself. Inside the cell the virus is protected and it is very difficult for the immune system to get at it to destroy it. Many thousands of new virus particles are released when the cell dies.

Virus strains: Rock born strain (Vaccine strain used in India)

Cultivation:

1. **Cell culture:** CDV grows well in dog kidney cells, ferret cells and chick fibroblasts. Formation of cytoplasm and acidophilic intracytoplasmic and intranuclear inclusion bodies are characteristic CPEs.
2. **Embryonated eggs:** The virus also grows in the Chorioallantoic membrane. However, no changes are produced upto 10th passage. Subsequent passages show increased thickening and whitish areas in the chorioallantoic membrane.
3. **Laboratory animals:** The distemper viruses are cultivated in live dogs or ferrets. Few strains also grown in suckling mouse.

Pathogenesis:

1. **Hosts affected:** Domestic dogs and many other carnivores, including raccoons, skunks, and foxes are affected by CDV. The infection is more severe in puppies.
2. **Transmission:** Infected dogs shed the virus through body secretions and excretions, especially respiratory secretions. The main mode of transmission is airborne. Normal animals get the infection by breathing the viral particles. Dogs in recovery may continue to shed the virus for several weeks after symptoms disappear and act as source for contamination. Once the virus enters the normal host, it is engulfed by macrophages. Viruses are not killed by the macrophages and the virus uses the macrophage as a means of transportation inside host's body. Within 24 hours, the virus reaches the lymph nodes of the lung. By the 6th day, the virus migrates to the spleen, stomach, small intestine, and liver and results in production of fever. After the host's immune system clears the virus from the different visceral organs, the virus still continues to remain in CNS and skin causing seizures for a long time and hardening of the skin. Though the virus spreads very rapidly inside the body their stability outside the body is questionable. The lipid envelope is easily disrupted in the environment, which makes it impossible for infectious virus to persist in the environment. Because an intact fatty envelope is required for infection, virus transmission must involve dog-to-dog contact or at least contact with extremely fresh (less than 30 minutes old) infected body secretions. As with other viruses, living virus happily freezes and can survive for years if kept frozen and protected from light. The initial disease in dogs appears quite, but the severity of the symptoms steadily increases despite treatment until the dog is often destroyed on humane grounds. CDV infection can take a month to show all of its worst characteristics, and things can still deteriorate from then on. There is no vector involvement in the transmission. In utero infection is rare.
3. **Incubation period –** 5-7 days
4. **Symptoms & lesions:** The initial symptom is fever (103^0F to 106^0F), which usually peaks 3 to 6 days after infection. The fever often goes unnoticed and may peak again a few days later. Eye and nose

discharge, depression, and loss of appetite (anorexia) are the associated symptoms during the period. After the fever, symptoms vary considerably, depending on the strain of the virus and the dog's immunity. At initial stages **gastrointestinal** and **respiratory symptoms** are more common, which include conjunctivitis, diarrhoea, pneumonia (cough, labored breathing), rhinitis (runny nose) and vomiting. These symptoms are often aggravated by secondary bacterial infections. Dogs develop encephalomyelitis (an inflammation of the brain and spinal cord), the symptoms of which are variable and progressive. Most dogs that die from distemper, die from neurological complications such as ataxia (muscle incoordination), depression, hyperesthesia (increased sensitivity to sensory stimuli, such as pain or touch), myoclonus (muscle twitching or spasm), which can become disabling, paralysis, Paresis (partial or incomplete paralysis), progressive deterioration of mental abilities, progressive deterioration of motor skills, and Seizures that can affect any part of the body (One type of seizure that affects the head, and is unique to distemper, is sometimes referred to as a "chewing gum fit" because the dog appears to be chewing gum). A variety of symptoms on the eye are also characteristic, which include keratoconjunctivitis, inflammation of the cornea and conjunctiva, or chorioretinitis, inflammation of the choroid and retina and optic neuritis (inflammation of the optic nerve which leads to blindness). Enamel hypoplasia (unenameled teeth that erode quickly in puppies whose permanent teeth haven't erupted yet) and hyperkeratosis (hardening of the foot pads and nose) are seen in chronic infection. In utero infection of fetuses is rare. When it happens, this can lead to spontaneous abortion, persistent infection in newborn puppies, or the birth of normal looking puppies that rapidly develop symptoms and die within 4 to 6 weeks.

Diagnosis:

1. **Field diagnosis:** Is based on gastrointestinal, respiratory and nervous symptoms. Biphasic fever curves and low white blood cell level are indicative of distemper infection.
2. **Microscopical identification of inclusion bodies:** Distemper inclusion bodies are actual clumps of virus that are visible under the microscrope within infected cells. Post-mortem inclusion bodies are readily visible in the urinary bladder tissues. In the living animals the inclusion bodies are readily visible in WBCs and conjunctival membranes. The inclusion bodies, which red stained (eosinophilic) oval structures are also found in the epithelial cells of the salivary glands, central nervous system, adrenal glands, bile duct, urinary tract, lymph nodes spleen and skin.

a. **Immunocytolgy:** To enhance the visibilty of inclusion bodies, "immunocytology" is used. In this technique, antibodies against distemper virus are tagged with fluorescent markers. The antibodies bind to virus if it is present effectively dying the inclusion body with glow-in-the- dark fluorescent color. The presence of inclusionbodiesconfirms distemper infection. The lack of dectectable inclusion bodies does not rule out distemper infection as inclusion bodies ultimately become coated with the host's own antibodies, which in turn block the fluorescent-tagged antibodies used in the test.

3. **Serology:** Distemper titers of either the "IgM" type (produced in early stages of infection) and the "IgG" type (produced in later phases of infection) can be checked. A high IgM titer indicates recent infection or recent vaccination. It is difficult to differentiate between antibodies initiated by vaccination and infection.

a. **Cerebrospinal fluid antibody levels –** In neurologic distemper cases, cerebrospinal fluid is often tapped and distemper antibody levels checked. Distemper antibodies in cerebrospinal fluid are highly indicative of distemper infection as vaccine-induced antibodies do not cross the blood-brain barrier into the CSF fluid.

4. **Isolation and identification:** The virus are identified by AGPT, Antigen capture ELISA and PCR techniques.

Treatment: There are no antiviral drugs to treat canine distemper virus. Because of this, only symptomatic treatment is attempte. Antibiotics coupled with antiallergic, antispasmodic and elctrolyte fluids are given. Once dogs develop nervous system signs treatment is of no use.

Prevention and control: Effective vaccines are available to protect dogs from canine distemper virus. These vaccines are manufactured from attenuated viruses to induce long-lasting immunity. These vaccines are produced in bird or dog cell cultures. Immunity lasts for many years. Vaccinations must not be given too early to puppies. Vaccination must be given to puppies at a time when the level of circulating antibodies that they have received from their mothers is in decline otherwise the vaccine's effects are neutralized (6-10 weeks). Twenty percent of this maternal immunity crosses the walls of the womb into the puppy while eighty percent is absorbed from colostrum milk across the intestine. Rarely dogs develop post vaccinal reaction referred as neurodistemper or vaccine distemper, which appear 10-21 days after vaccination. Primary vaccines are given at 6-8 weeks of age followed by booster at dose at 12-14 weeks of age with annual revaccination. CD vaccines are given in combination with *Leptospira* vaccine. Human measles virus can also be used to vaccinate dogs against CDV, but measles virus does not produce long lasting immunity. Older vaccines include virus adapted in ferrets and avianised virus.

Since the virus cannot live without fresh secretions and inactivated in minutes outside the living host's body only minimal disinfection is necessary.

Public health: There is no human hazard associated with canine distemper virus.

FAMILY - *ORTHOMYXOVIRIDAE*
AVIAN INFLUENZA

Avian influenza is caused by type A influenza virus. The symptoms can vary from amilddisease with littleornomortalitytoahighlyfatal,rapidlyspreadingepidemic (**highly pathogenic avian influenza - HPAI**) depending on the in fecting virus strain, hostfactors, andenvironmentalstressors. Highlypathogenic avian influenza (HPAI) is also known as fowl plague. The disease was first identified in Italy more than 100 years ago and now occurs worldwide.

General characters:

Classification:

Baltimoregroup - **Group V** – Negative sense RNA viruses
Order - Mononegavirales
Family – Orthomyxoviridae
Genus - Type A Influenzavirus
Species - Avian influenza virus(AIV)

Members of the family Orthomyxoviridae are classified into Influenza virus A, B and C. The type designation (A, B, or C) is based upon the antigenic character of the matrix (M) protein of the virus envelope and the nucleoprotein (NP) within the virus particle. All influenza viruses affecting domestic animals (equine, swine, avian) belong to Type A, and Type A influenza virus is the most common type producing serious epidemics in humans. Types B and C do not affect domestic animals.

Synonyms: Fowl plague

Morphology: Influenza virus particles are highly pleiomorphic (variable), mostly spherical/ovoid, 80-120nm diameter, but many forms occur, including long filamentous particles (up to 2000nm long x 80-120nm diameter). The outer surface of the particle consists of a lipid **envelope** from which project

prominent **glycoprotein spikes** of two types **Haemagglutinin** (**HA**) and **Neuraminidase** (**NA**). The inner side of the envelope is lined by the **matrix protein**. The genome is packed into the core along with nucleoprotein (NP). The genome (RNA) and nucleoprotein together are called RNP. The genome is of s/s (-) sense RNA arranged into 8 segments. The segments 1-3 code for various transcriptions related activities. Segments 4, 5 and 6 code for HA, NP and NA proteins. Segment 7 codes for matrix protein and 8 codes for non- structural proteins.

OIE Listing: List A disease

Risk group (Animal Pathogen): Group II

Resistance: The particles are not very stable and are killed within few hours on exposure to room temperature and are also not resistant to drying and disinfectants.

Replication: Entry of avian influenza virus into the cell is facilitated by binding of the HA spikes to mucoproteins on the surface of cells. After binding, the virus enters into the cell by endocytosis via coated pits into endocytotic vesicles as endosomes. Uncoating takes place after acidification of the endosomes. During the initial phase of infection (approximately 2h), active host cell DNA synthesis is required. Inside the nucleus the viral RNA uses the methylated cap of host cell mRNA as primers for RNA synthesis (**cap snatching)**. Two classes of (+) sense RNA are made inside the nucleus of infected cells.

a. Incomplete, 3' polyadenylated transcripts which are exported to the cytoplasm and serve as **mRNAs**

b. **cRNA** - complete, (+) sense copies of the (-) sense viral, which serve as template for the synthesis of progeny (-) sense vRNAs

Most of the proteins made (e.g. HA, NA) remain in the cytoplasm or become associated with the cell membrane. However, the NP protein migrates back into the nucleus, where it associates with newly-synthesized vRNA to form new nucleocapsids. These migrate back out into the cytoplasm and towards the cell membrane. The exact **packaging mechanism** by which eight distinct genome segments are packed into each progeny virion particle is not known.

2. **Haemagglutination property:** Avian influenza virus causes agglutination of chicken erythrocytes.

3. **Viral subtypes:** Influenza A viruses have antigenically related nucleocapsid and antigenically related matrix proteins, but are classified into subtypes on the basis of their haemagglutinin (HA *referred as H*) and neuraminidase (NA referred as N) antigens. At present, 15 H subtypes (H1–H15) and 9 neuraminidase subtypes (N1–N9) are recognised. There are three subtypes of avian influenza virus referred as H5, H7 and H9. H5 and H7 can produce low pathogenic and highly pathogenic avian influenza,whereas H9 produces only low pathogenic infection. Within these subtypes there are number of strains. These strains are formed by combination of any one of H5, H7 or H9 with 9 different NA proteins. The following table lists the details of subtypes and strains of avian influenza viruses

Subtype	Characters	Example
H5	1. Nine different strains. 2. Produce highly pathogenic or low pathogenicinfections 3. Human infection possible, sometimes causing severe illness and death	H5N1 (Hong Kong)

H7	1. Nine different strains 2. Produce highly pathogenic or low pathogenic infections. 3. Human infection is rare, but can occur among persons who have close contact with infected birds; symptoms may include conjunctivitis and/or upper respiratory symptom	H7N7 (Netherlands)
H9	1. Nine different strains 2. Produce only low pathogenic infection 3. Human infection possible	H9N2

Cultivation: The virus grows well in embryonated eggs by allantoic route (Nine to 11-day-old embryonated chicken eggs). The eggs used for isolation of AIV should be either from SPF birds or from SAN (Specific antibody negative birds). The eggs should be incubated at 35–37°C for 4–7 days. Avian influenza virus will usually kill embryos within 48-72 hours. Presence virus in allantoic fluid is confirmed by demonstration of HA activity or by AGID or by RT-PCR.

Pathogenesis:

1. **Host affected:** Many avian species are affected by AI. Certain subtypes of virus may infection predominantly in any one of the species of birds. AI has been reported in chickens, turkeys, ducks, waterfowls, gulls, shorebirds etc.
2. **Transmission:** Migratory waterfowl, sea birds, or shore birds are considered responsible for introducing the virus into poultry. Waterfowl serve as reservoir for AIV. Virus is found in large quantities in faeces and respiratory secretions of infected birds. The virus spreads to susceptible birds through inhalation of influenza particles in nasal and respiratory secretions and from contact with the faeces of infected birds. Once introduced into a flock, the virus spreads from flock to flock by the movement of infected birds, contaminated equipment, egg flats, feed trucks, and service crews etc. The infection spreads through shared and contaminated drinking water. Airborne transmission may occur if birds are in close proximity and with appropriate air movement. Vertical transmission is not completely established. There is also no vector involvement in the transmission of infection.
3. **Incubation period:** The incubation period is usually 3 to 7 days, depending upon the subtype of virus, the species, and age of the bird.
4. **Morbidity and mortality:** Morbidity and mortality rates are upto 100 percent within 2 to 12 days after the first signs of illness in HPAI.
5. **Symptoms (HPAI): Chickens (Layers)**

a. Marked depression with ruffled feathers, inappetence, excessive thirst, cessation of egg production, and watery diarrhea.

b. Mature chickens show swollen combs, wattle, and edema surrounding the eyes. The combs are cyanotic at the tips and may have plasma or blood vesicles on the surface with dark areas of ecchymotic hemorrhage and necrotic foci.

c. The last eggs laid, after the onset of illness may not have shells.

d. The diarrhea begins as watery bright green and progresses to almost totally white.

e. Edema of the head, if present, is often accompanied by edema of the neck.

f. The conjunctivae are congested and swollen with occasional hemorrhage.

g. The legs, between the hocks and feet, may have areas of diffuse hemorrhage.

h. Respiratory signs can be a significant feature of the disease, depending on the extent of tracheal involvement and mucus accumulation.

i. Death may occur within 24 hours of firstsignsofdisease,frequentlywithin 48 hours, or be delayed for as long as a week.Someseverely affected hens may occasionally recover.

Broilers: the signs of disease are frequently less obvious with severe depression, inappetence, and a marked increase in mortality being the first abnormalities observed. Edema of the face and neck and neurologic signs such as torticollis and ataxia may also beseen.

Turkeys: The disease in turkeys is similar to that seen in layers, but it lasts 2 or 3 days longer andisoccasionally accompanied by swollen sinuses.

Ducks: In domestic ducks and geese the signs of depression,

inappetence, and diarrhea are similar to those in layers, though frequently with swollen sinuses. Younger birds may exhibit neurologic signs.

6. Lesions:

- Birds that die with the peracute disease and young birds may not show significant gross lesions other than severe congestion of the musculature and dehydration.
- In the less acute form, and in mature birds, significantgrosslesions are frequently observed, which may consist of subcutaneous edema of the head and neck area, exudation of fluid from nares and oral cavity, congested conjuctiva with petechiation, tracheal lumen filled with excessive mucous exudates, hemorrhagic tracheitis, pinpoint petechial hemorrhages on the keel, very small petechia over the abdominal fat, serosal surfaces, and peritoneum and congestion of kidneys with urate deposition in the tubules.
- In layers, the ovary may be hemorrhagic or degenerated with darkened areas of necrosis and the peritoneal cavity is frequently filled with yolk from ruptured ova, causing severe airsacculitis and peritonitis.
- Hemorrhages may be present on the mucosal surface of the proventriculus — particularly at the juncture with the gizzard. The lining of the gizzard peels easily and frequently reveals hemorrhages and erosions underneath. The intestinal muscosa may have hemorrhagic areas — especially in the lymphoid foci such as the cecal tonsils.
- The lesions in turkeys and domestic ducks are similar to those in chickens but may not be as marked.

Diagnosis

1. **Field diagnosis:** The field diagnosis is based on sudden deaths followed with severe depression, inappetence, a drastic decline in egg production, presence of facial edema, swollen and cyanotic combs and wattles, and petechial hemorrhages on internal membrane are suggestive of HPAI.

2. Isolation and identification:

a. a. Samples from dead birds - Intestinal contents (faeces) or cloacal swabs, oro-nasal swabs, trachea, lungs, air sacs, intestine, spleen, kidney, brain, liver and heart are the ideal material

b. Samples from live birds - tracheal and cloacal swabs

c. The samples should be placed in isotonic phosphate buffered saline (PBS), pH 7.0–7.4, containing antibiotics, processed, centrifuged and the supernatant is inoculated in 9-11 day old embryonated eggs. The eggs should be incubated for 4 days at 35-37C. The presence of virus in allantoic fluid are confirmed by

- Detection of HA activity

- Agar gel immunodiffusion (AGID) test - by demonstrating the presence of the nucleocapsid or matrix antigens
- RT-PCR
- Antigen capture ELISA
- RT-PCR can also be used to identify viral RNA directly in the clinical samples without passaging the materials in embryonated eggs.

3. Serological tests:

i. AGID test

ii. HA / HI test

4. **Differential diagnosis:** Highly pathogenic avian influenza should be differentiated from VVND, infectious laryngotracheitis and acute bacterial diseases such as fowl cholera and *Escherichia coli.* However, in an area where AI is prevalent, such as during an outbreak, sound presumptive diagnosis can be made by flock history, signs, and gross lesions.

Treatment: Amantadine hydrochloride provides relief to birds when administered in drinking water. However, it results in emergency of strains that are resistant to Amantadine hydrochloride.

Control and eradication:

1. Vaccination:

a. Inactivated oil-emulsion vaccines are found to be effective in reducing mortality, preventing disease, or both, in chickens and turkeys.

b. Naturallyavirulentorattenuatedstrainscanalsobeusedasvaccines. The problem with live vaccine is emergence of new strains due to genetic reassortment. A recombinant fowl pox virus vaccine containing the gene that codes for the production of the H5 antigen

c. A recombinant fowl pox virus vaccine containing the gene that codes for the production of the H5 antigen is also used. A recombinant insect virus containing the gene for either the H5 or H7 antigen is also used to control AI.

2. Biosecurity procedures including control of human traffic, quarantine of birds of unknown disease status into the flock and proper cleaning and disinfection of fomites

3. Free range rearing should not be done in areas where, waterfowls frequent

Public Health: Severe human infection occurs as a result of close contact with live infected poultry was the source of human infection. Studies at the genetic level determined that the virus had jumped directly from birds to humans. Limited transmission to health care workers occurred, but did not cause severe disease. Some of the strains that caused human infection are H5N1, H7N7, H9N2 etc. Avian influenza viruses may be transmitted to humans in two main ways: - directly from birds or from avian virus-contaminated environments to people and through an intermediate host, such as a pig.

(Influenza viruses have eight separate gene segments. The segmented genome allows viruses from different species to mix and create a new influenza A virus if viruses from two different species infect the same person or animal. For example, if a pig were infected with a human influenza virus and an avian influenza virus at the same time, the viruses could reassort and produce a new virus that had most of the genes from the human virus, but a hemagglutinin and/or neuraminidase from the avian virus. The resulting new virus might then be able to infect humans and spread from person to person, but it would have surface proteins (hemagglutinin and/or neuraminidase) not previously seen in influenza viruses that infect humans. This type of major change in the influenza A viruses is known as antigenic shift. Antigenic shift results when a new influenza A

subtype to which most people have little or no immune protection infects humans. If this new virus causes illness in people and can be transmitted easily from person to person, an influenza pandemic can occur. It also is possible that the process of reassortment could occur in a human. For example, a person could be infected with avian influenza and a human strain of influenza at the same time. These viruses could reassort to create a new virus that had a hemagglutinin from the avian virus and other genes from the human virus. Theoretically, influenza A viruses with a hemagglutinin against which humans have little or no immunity that have reassorted with a human influenza virus are more likely to result in sustained human-to-human transmission and pandemic influenza. Thus, careful evaluation of influenza viruses recovered from humans who are infected with avian influenza is very important to identify reassortment if it occurs).

SWINE INFLUENZA

Swine influenza is a highly contagious respiratory viral infection of pigs characterized by coughing, sneezing, nasal discharge, elevated rectal temperatures, lethargy, difficult breathing, depressed appetite and rarely associated with reproductive disorders such as abortion.

General characters:

Classification:

Baltimoregroup - **Group V** – Negative sense RNA viruses

Order - Mononegavirales

Family - Orthomyxoviridae

Genus - Type A Influenzavirus

Species - Swine influenza virus(SIV)

Synonyms: Swine Flu, Flu, Influenza

Morphology: Influenza virus particles are highly pleiomorphic (variable). Mostly they appear as spherical/ovoid (80-120nm diameter). The outer surface of the particle consists of a lipid **envelope.** Hence these viruses are ether sensitive. **Glycoprotein spikes** project from these envelopes and these spikes are of two types **haemagglutinin** (**HA**) and **neuraminidase** (**NA**). The inner side of the envelope is lined by the **matrix protein.** The genome comprises of s/s (-) sense RNA in 8 segments. Each segment of RNA codes for one or two proteins. The fourth and sixth segment codes for haemagglutin (HA) and neuraminidase (NA) respectively. Variations observed in these two proteins are usedtoclassify the virus specied into many strains (subtypes). There are 13 H and 9 N types. Common strains of SIV belong to H1N1, H3N2 and H1N2.

Resistance: The particles are not very stable and are killed within few hours on exposure to room temperature and are also not resistant to drying and disinfectants.

OIE listing: Listed but not under A or B

Risk group: Human infection with SIV possible. It is an WHO surveyed infection. It is also a SPC List D disease. (*SPC – Southern Pacific Commission)*

Haemagglutination property: The virus causes agglutination of chicken and turkey RBCs.

Cultivation: The preferred cell culture systems for virus growth are Madin–Darby canine kidney (MDCK), swine kidney, swine testicle or swine lung epithelial cell lines. Specific cytopathic effects appear 7 days of incubation. The virus also grows well in embryonated eggs (9-11 days old). The eggs are incubated for four days. The presence of virus is confirmed by HA test using chicken or turkey RBCs.

Pathogenesis:

1. **Hosts affected:** Pig is the main host. The virus also affects avians and humans.
2. **Transmission:** This infection is transmitted by direct contact between pigs. The important source for spread is the nasal secretions. SIV can also be transmitted by aerosols over a short distance. The virus can be shed for 30 days after infection and has been recovered from clinically normal animals. The disease is spread to new areas and farms by the movement of infected pigs or carrier people. The virus is easily carried and spread by avian species, particularly water fowl and turkeys. In endemic areas virus is present throughout the year. However in most other areas, outbreaks are seasonal occurring in late autumn and early winter. The epidemics are very severe, with outbreaks occurring on most pig farms in a locality over a short period. The stress factors, which speed up the infection are transport, mixing of pigs, poor isolation facilities, marked diurnal (day and night) temperature fluctuations and overstocking.
3. **Incubation period:** The incubation period is very short 1 to 3 days. In very severe epidemics it can be as short as 4 hours.
4. **Morbidity and mortality:** Morbidity – upto 100%; mortality – very minimal 1%.
5. **Symptoms:** The first clinical signs are fever (40.5-41.5^{o}C), puffy eyes, anorexia leading to loss of weight, depression, prostration and huddling leading to weakness. These signs are followed by sudden onset of acute respiratory signs, which include paroxysmal coughing, sneezing, irregular abdominal breathing and ocular andnasal discharges. Most pigs recover about six days after the onset of the disease. The mortality rate is generally about 1%, but may be higher in young piglets. SIV can precipitate outbreak of more serious respiratory disease causing mortality. One of the important problems associated with SIV infection is the high rectal temperature, which in breeding stock result in abortions, infertility, production of small weak litters and increased stillbirths.
6. **Lesions:** Only few lesions are observed in SIV infection. The lesions are firm lobular lung lesions with interlobular oedema with enlargement of associated lymph nodes. The trachea may be filled with froth with severe damage to the lining of the trachea destroying the mucociliary escalator.

Diagnosis:

1. **Field diagnosis:** It is based on high rectal temperature, respiratory symptoms with very high morbidity rate and occurrence of the infection in autumn and winter months.
2. **Isolation and identification:** Lung tissues are the ideal material for isolation of virus. The supernatant from the lung tissue homogenates are used for infecting the cell culture system or embryonated eggs. The preferred cell culture systems for virus growth are Madin–Darby canine kidney (MDCK), swine kidney, swine testicle or swine lung epithelial cell lines. Specific cytopathic effects appear 7 days of incubation. The virus can also be isolated in embryonated eggs (9-11 days old). The presence of the virus are confirmed by

a. Haemagglutination test: This test is performed using 0.5% chicken or male turkey red blood cells. The HA activity is confirmed by HI test using specific serum (8HA units of virus are used unlike HA test for NDV).
b. Neuraminidase inhibition test
c. Fluorescent antibody test
d. Immunoperoxidase test
e. Antigen capture ELISA
f. Polymerase chain reaction

3. Serologicaltests:

a. Haemagglutination inhibition test: The primary serological test for detection of SIV antibodies is the HI test and it is subtype specific. It should be conducted on paired sera collected 10–21 days apart. A four-fold or greater increase in titre between the first and second sample is suggestive of a recent SIV infection.
b. Virus neutralization test: Not commonly performed
c. ELISA: Not commonly performed

4. Differential diagnosis: The following diseases should be differentiated - Aujeszky's disease, atrophic rhinitis, enzootic (mycoplasmal) pneumonia and bacterial pneumonia due to *Pasteurella* or *Haemophilus* spp.

Treatment: There is no specific treatment for SIV infection. Only supportive treatment is suggested for infected animals.

Control:

1. The best way to control swine influenza is to prevent the occurrence and spread of the disease.
2. Good husbandry practices including All-in/All-out to limit the spread of the disease, provision of fresh clean drinking water, avoiding ducks and turkey contamination's/contact including staff and proper use of disinfectants to clean infected buildings.
3. Commercial vaccines are available in Europe and North America

Public health:

Swine Influenza may rarely affect human beings. Hence, all work with infectious tissues, swabs, embryonated eggs, and cell cultures should be done in a class II biological safety cabinet.

EQUINE INFLUENZA

Equine influenza is an acute, contagious respiratory disease caused by two distinct subtypes (H7N7 - formerly equi-1, and H3N8 - formerly equi-2) of influenza A viruses within the genus Influenzavirus A of the family Orthomyxoviridae. The infection in horses ischaracterized by pyrexia, and a harsh dry cough followed by a mucopurulent nasal discharge.

General characters:

Classification:

Baltimoregroup - **Group V** – Negative sense RNA viruses
Order - Mononegavirales
Family – Orthomyxoviridae
Genus - Type A Influenzavirus
Species - Equine influenza virus(EIV)

Synonyms: Horse flu, Epizootic Cellulitis, Pinkeye, Stable Pneumonia, Shipping Fever

Morphology: The morphology of EIV is typical of Influenza virus A. They are pleomorphic enveloped viruses containing 8 segments of single stranded, negative sense, RNA. The virion consists of the viral ribonucleoprotein (RNA + NP) and polymerase complex (PA, PB1 and PB2) surrounded by the matrix proteins (M1 and M2) and a cellular lipid envelope containing 2 viral glycoproteins, the hemagglutinin (HA) and the neuraminidase (NA). The HA and NA contain the major neutralizing antigenic determinants of influenza viruses.

OIE listing: List B disease

Risk group: Risk Group II

Resistance: The particles are not very stable and are killed within few hours on exposure to room temperature and are also not resistant to drying and disinfectants

Replication: The replication strategy of EIV is typical of other members of the family *orthomyxoviridae*

Haemagglutination property: EIV agglutinates chicken and guinea pig RBCs.

Virus subtypes: The H7N7 (Equine-1) and H3N8 (Equine-2 + new avian strain) subtypes have been isolated in horses

Cultivation: EIV are cultivated in embryonated eggs and cell culture systems. 10-11 days old embryonated eggs are used for cultivation of EIV. The routes are allantoic or amniotic routes and the infected eggs are incubated for 2-3 days. The eggs are chilled after 2-3 days and the presence of EIV is confirmed by demonstration HA property using chicken or guinea pig RBCs. The cell culture system commonly used is MDCK (Madin–Darby canine kidney cell line). The cells are incubated for 7 days. The presence of EIV is confirmed by demonstration of HA property in tissue culture fluid using chicken or guinea pig RBCs and haemadsorption.

Pathogenesis:

1. **Hosts affected:** Horses and other equines are the normal hosts. Humans are also infected by H7N7 and H3N8 viruses. The infection is common in young horses, especially at racetracks, horse shows, and during transport (planes, etc.). The infection is seasonally more prevalent in the summer and autumn.
2. **Distribution:** The infection is worldwide except in Pacific regions.
3. **Transmission:** The disease is highly contagious in nature and can occur as epidemics. Infection spreads by direct contact, aerosol from coughing (up to 30m), wind (up to 8km) and indirect contact with infected material. Infected or recovered animals do not act as chronic carriers. There is no vertical transmission or involvement of vectors in spreading the infection. The HA glycoprotein (hemagglutinin) of EIV attaches to respiratory epithelial cell receptors, and enters the cell via endocytosis. Viral replication occurs and new viral particles are released into the airway to infect other cells or become aerosolized. In 1-3 days this invasion causes necrosis and desquamation of respiratory epithelial cells, exudation of protein-rich fluid into airways, and clumping of cilia, impairing the mucociliary apparatus for up to 4 weeks. Exudate accumulates and predisposes to secondary bacterial infections. Affected animals shed the virus for 8 days and these animals are important source of infection for healthy stock.
4. **Morbidity and mortality:** Morbidity very high (90-100%). However, mortality is almost nil.
5. **Incubation period:** The incubation period is very short ranging from 1-5 days.
6. **Symptoms:** Influenza in horses is similar to "flu" in humans. The first symptoms are poor performance, letharginess, depression followed by a harsh cough, which usually lasts around 10 days. Other symptoms include a high temperature (38.9^0C) lack of appetite and nasal discharge, which later turns into mucopurulent due to secondary bacterial infection. Watery eyes, enlarged lymph nodes between the mandibles, oedema and stiffness in the legs and breathing difficulty are also observed in affected horses. Horses with relatively mild cases of equine influenza usually recover in a week or so, but severely affected horses may require weeks to months to recover fully, especially if they are not allowed complete rest. Affected animals under stress, or ones not allowed to rest, may develop secondary pneumonia, a chronic cough, or inflammation of the heart muscle, sometimes resulting in death.

7. **Lesions:** The important lesions are bronchopneumonia,bronchitis,pharyngitis, pleuritis, guttural pouch infections and viral myocarditis. Bronchiolitis with abundant bilateral serous discharge is the most important post-mortem lesion.

Diagnosis:

1. **Field diagnosis:** Based on sudden occurrence with characteristic respiratory symptoms and high fever.

2. **Isolation and identification:**

   a. Clinical materials: Nasopharyngeal swabs and nasal or tracheal washings are the ideal clinical materials for isolation of viruses.

   b. Isolation systems: Embryonated eggs (allantoic or amniotic routes) and MDCK cell line.

   c. Identification of antigen:

i. Demonstration of HA activity using chicken or guinea pig RBCs.

ii. Neuraminidase typing

iii. Antigen capture ELISA using monoclonal antibody against nucleoprotein

iv. RT-PCR

3. Serology: Infections are detected by performing serological tests on paired sera to show a rise in antibody titre. Three tests that are commonly used include

a. Haemagglutination inhibition test

b. Single radial haemolysis: In this test, viral antigens are coupled to the fixed RBCs that are suspended in agarose containing guinea-pig complement (C'). Wells are punched in the agarose and filled with test sera. Influenza antibodies and C' lyse the antigen-coated RBCs, resulting in a clear, haemolytic zone around the well; the size of this zone is directly proportional to the level of strain-specific antibody in the serum sample

3. Complement fixation test: Not commonly used.

4. Differential diagnosis: The infection has to be differentiated from

a. Equine rhinopneumonitis

b. Equine viral arteritis

c. Equine rhinovirus and adenovirus infection

d. Pasteurellosis

e. Strangles

Treatment:

Symptomatic treatment is given, as the disease usually subsides in 7-14 days on its own. Complete rest for 3-4 weeks (guideline 1 week of rest for every day of fever) and good ventilation are very important to prevent systemic disease. Antibiotics are given only if secondary bacterial infection is observed.

Control:

1. **Vaccines:** Influenza vaccines are used in competition horses in countries where the infection is reported. In free countries vaccination is prohibited except for sport horses participating to international races. Both live and inactivated equine influnza vaccines are available.

a. Three doses of the killed-virus vaccine are administered 3-6 weeks apart starting at 6 months of age for younger horses. Revaccination should be done at intervals of 3 to12 months depending on age and risk of exposure

b. A modified live equine AE-2 influenza vaccine, when administered intranasally offers protection for 6 months.

2. **Biosecurity measures:** Since the infection is highly contagious the infected horses should be immediately separated and isolated.

Public health: The equine influenza strains are not known to cause illness in humans. However, rare infection in humans is possible.

FAMILY - *RHABDOVIRIDAE*
RABIES

Rabies is caused by a neurotropic virus of the genus *Lyssavirus* of the family Rhabdoviridae, and is transmissible to all mammals. It is considered as one of the very serious diseases.

General characters:

Classification:

Baltimoregroup - **Group V** – Negative sense RNA viruses

Order - Mononegavirales

Family - Rhabdoviridae

Genus - Lyssavirus

Species - Classical Rabies virus (CRBV)

There are seven distinct genetic lineages under lyssa viruses. They are

Classical rabies virus itself (RABV)

Lagos bat virus (LBV)

Mokola virus (MOKV)

Duvenhage virus (DUUV)

European bat lyssaviruses (EBLV) – 2 biotypes

Australian bat lyssavirus (ABLV)

Synonyms: hydrophobia, mad dog disease

Morphology: Rabies virus is approximately180 nm long and 75 nm wide and is bullet shaped. It is an enveloped virus and hence is ether sensitive.The nucleic acid is single stranded and negative sense RNA. The genome encodes 5 proteins designated as N, P, M, G, and L. The RNA is arranged as ribonucleoprote in core (RNP) along with nuceleoprotein (N), polymerase (L) and phosphoprotein (P). Inside the core the RNA is tightly encased by the nucleoprotein. The surface proteins are matrix protein (M) and glycoprotein (G). The glycoprotein forms approximately 400 trimeric spikes, which are tightly arranged on the surface of the virus. The M protein is associated both with the envelope and the RNP.

OIE listing: List A disease

Risk group: Risk Group III

Resistance: The virus is killed on exposure to 70^0C for 15 minutes. Virus can be preserved for more than one year in freezing temperatures. UV rays and disinfectants like phenol, chloroform destroys the virus rapidly.

Replication: Rabies virus attaches to a cell by interaction between G protein and receptors on the surface of cells. After adsorption, the virus penetrates the host cell and enters the cytoplasm by pinocytosis (via clathrin-coated pits). The virions are present in the cytoplasm as large endosomes (cytoplasmic vesicles). The viral membranes fuse to the endosomal membranes, causing the release of viral RNP into the cytoplasm (uncoating). Since rabies virus have a linear single-negative- stranded ribonucleic acid (RNA) genome, messenger RNAs (mRNAs) must be transcribed to permit virus replication, which is facilitated by polymerase (L) present in the core. The polymerase (L) transcribes the genomic strand of rabies RNA into leader RNA and five capped and polyadenylated mRNAs, which are translated into proteins.

Haemagglutination property: Virus does not haemagglutinate.

Virus strains: Flury, Street-Alabama-Dufferin [SAD], Vnukovo and Kelev. Rabies viruses are also referred as **Street viruses and fixed viruses.** Street viruses refer to virulent viruses obtained from rabid animals and fixed viruses are those, which are adapted in laboratories by passaging them in animals such as rabbits. Fixed viruses are characterized by a shortened incubation period and multiply faster. Fixed viruses also lose their ability to form Negri bodies, less pathogenic, stable and are used for vaccine production.

Cultivation: Rabies viruses are normally cultivated in tissue culture lines, such as WI-38, BHK-21, or CER. Since rabiesvirus induce minimal CPE, presence of rabies virus in tissue culture system is confirmed by antigen detection systems. The other more commonly used method for virus isolation is by the inoculation of saliva, salivary gland tissue and brain tissue intracerebrally into infant mice. The mice should develop paralysis and death within 28 days. Upon death, the brains are examined for the presence of the virus by immunofluorescence. The virus can be cultivated in chicken embryos but the lesions are irregular and not characteristic.

Pathogenesis:

1. **1. Hosts affected:** All mammals are affected by rabies.
2. **Transmission:** Rabies virus is more commonly transmitted to a new host only through an open wound or, less likely, through the mouth, the eyes, or the mucous membranes of the nose. Since the virus is present in the saliva and brain material of infected individuals, most transmission events occur through bite wounds. The respiratory transmission has been reported in very rare circumstances. The incubation period vary from as little as a few days to many years in rare cases. However, in most cases, incubation occurs within one to three months. Once an individual is infected with the rabies virus, it replicates within the cytoplasm of muscle cells and pass from cell to cell. Finally, it reaches nerve receptors and enters the nervous system. The virus passes along the nerve network, traveling to the central nervous system, where it concentrates in the brain and upper spinal cord. As the disease progresses, the virus continues to multiply and spreads back through the peripheral nervous system to the salivary glands. Not all animals or humans exposed to the virus contract the disease. However, once symptoms become evident, the disease usually is fatal. Many animals like dogs, bats, skunks act as reservoirs. When dogs act as reservoirs for spread of rabies to other animals including dogs it is referred as canine rabies. In Asia, Africa and Europe canine rabies are common. However, in US bat rabies pose major problem.
3. **Incubation period:** The incubation period is both prolonged and variable. Most cases in dogs occur within 21-80 days after exposure, but the incubation period may be shorter or considerably longer.
4. **Mortality:** Rabies is a fatal infection and once symptoms are exhibited animals will certainly die.
5. **Symptoms:** Clinical signs of rabies are not very definitive. Rabid animals of all species exhibit typical signs of CNS disturbance, with minor variations among species. The most reliable signs, regardless

of species, are behavioral changes and unexplained paralysis. Behavioral changes include anorexia, signs of apprehension or nervousness, irritability, and hyperexcitability. The animal may prefer to be in isolation. Ataxia, altered phonation (sound), and changes in temperament are apparent. Uncharacteristic aggressiveness may develop—a normally docile animal may suddenly become vicious. Commonly, rabid wild animals lose their fear of man, and species that are normally nocturnal may be seen wandering about during the day time.

The clinical course is divided into three phases—prodromal, excitative, and paralytic. During the prodromal period, which lasts 1-3 days, animals show only vague CNS signs, which intensify rapidly. The disease progresses rapidly after the onset of paralysis, and death is virtually certain within 10 days after the initial onset of signs. Some animals die rapidly without marked clinical signs.

Normally rabies is exhibited as "furious rabies" refers to animals in which aggression (the excitative phase) is pronounced and "Dumb or paralytic rabies" in which the behavioral changes are minimal or absent, and the disease is manifest principally by paralysis

Furious form: This is referred as "mad-dog syndrome," although it is seen in all species. There is rarely any evidence of paralysis during this stage. The animals become irrational and, with the slightest provocation, may viciously and aggressively use its teeth, claws, horns, or hooves. The posture and expression is of alertness and anxiety, with pupils dilated. Noise invites attack. Such animals lose all caution and fear of natural enemies. Carnivores with this form of rabies frequently roam extensively, attacking other animals, including people, and any moving object. They commonly swallow foreign objects, eg, feces, straw, sticks, and stones. Rabid dogs chew the wire and frame of their cages, breaking their teeth, and will follow a hand moved in front of the cage, attempting to bite. Young pups apparently seek human companionship and are overly playful, but bite even when petted, usually becoming vicious in a few hours. Rabid skunks appear to seek out and attack litters of puppies or kittens. Rabid domestic cats and bobcats attack suddenly, biting and scratching viciously. As the disease progresses, muscular incoordination and seizures are common. Death is the result of progressive paralysis

Dumb form: This is first manifested by paralysis of the throat and masseter muscles, often with profuse salivation and inability to swallow. Dropping of the lower jaw is common in dogs. Owners frequently examine the mouth of dogs and livestock searching for a foreign body or administer medication with their bare hands, thereby exposing themselves to rabies. These animals are not vicious and rarely attempt to bite. The paralysis progresses rapidly to all parts of the body, and coma and death follow in a few hours.

Cattle with furious rabies are dangerous, attacking and pursuing man and other animals. Lactation ceases abruptly in dairy cattle. Instead of the usual placid expression, there is one of alertness. The eyes and ears follow sounds and movement. A common clinical sign is a characteristic abnormal bellowing, which may continue intermittently until shortly before death.

Horses and mules frequently show evidence of distress and extreme agitation. These signs, especially when accompanied by rolling, may be interpreted as evidence of colic. As with other species, horses may bite or strike viciously and, because of size and strength, become unmanageable in a few hours. Such animals frequently suffer self-inflicted wounds

6. **Lesions:** The most important lesions in rabies is encephalomyelitis (inflammation) in brain tissue and meninges with mononuclear infiltration, perivascular cuffing of lymphocytes or polymorphonuclear cells, Babes nodules consisting of glial cells and **Negri bodies.** Negri bodies appear as round or oval inclusions within the cytoplasm of nerve cells of animals infected with rabies. Negri bodies may

vary in size from 0.25 to 27 μm. They are found most frequently in the pyramidal cells of Ammon's horn, and the Purkinje cells of the cerebellum. They are also found in the cells of the medulla and various other ganglia. Negri bodies can also be found in the neurons of the salivary glands, tongue, or other organs. Staining with Mann's, Giemsa, or Sellers stains can permit differentiation of rabies inclusions from other intracellular inclusions. With these stains, Negri bodies appear magenta in color and have small (0.2 μm to 0.5 μm), dark-blue interior basophilic granules.

Diagnosis:

1. **Field diagnosis:** Is based on characteristic symptoms of rabies coupled with presence of Negri bodies in the brain.
2. **Direct detection of virus antigen in tissue materials:** The ideal clinical materials for rabies virus identification are brain, salivary gland and saliva. In the brain, rabies virus is particularly abundant in the thalamus, pons and medulla.
   a. **Histological identification of characteristic lesions –** Demonstration of Negri bodies by Sellers method (for unfixed sections) or Mann's method (for fixed sections).
   b. **Fluorescent antibody test –** This is the most widely used test for rabies diagnosis and is recommended by both WHO and OIE.
   c. Immunoperoxidase test
   d. **ELISA –** Referred as rapid rabies enzyme immunodiagnosis test (RREID)
   e. **PCR and DNA sequencing**
   f. **DNA probe**
   g. **Electronmicroscopy**
   h. **Monoclonal antibody based typing**
3. **Isolation and identification:** Is attempted FAT gives an uncertain result or when the FAT is negative in the case of known human exposure.
   a. **Mouse inoculation test:** The infected mice develop paralysis and death occurs within 28 days. Upon death, the brain should be examined for the presence of the virus by immunofluorescence.
   b. **Cell culture system:** Neuroblastoma cell lines, e.g. CCL-131 is commonly used for isolation of rabies virus. Presence of rabies virus in the cells is revealed by the FAT.
4. **Serological tests:** Serological tests are rarely used due to late seroconversion and the low percentage of animals surviving the disease having post-infection antibodies. The tests that are normally performed are
   a. **Virus neutralisation test** in cell culture: **fluorescent antibody virus neutralisation test – FAVN** (accepted for international trade). The principle of the fluorescent antibody virus neutralisation (FAVN) test is the neutralisation *in vitro* of a constant amount of rabies virus before inoculating cells susceptible to rabies virus in BHK-21 C13 cells
   b. **Rapid fluorescent focus inhibition test (RFFIT)** for determining rabies virus-neutralising antibody (accepted test for international trade)
   c. Virus neutralization in mice
   d. **ELISA -** Indirect ELISA is used to determine the level of rabies antibodies in individual dog and cat serum samples following vaccination

Control: Comprehensive guidelines for control in dogs have been prepared by the World Health Organization and include the following:

1. Notification of suspected cases, and destruction of dogs with clinical signs and dogs bitten by a suspected rabid animal;

2. Reduction of contact rates between susceptible dogs by leash laws, dog movement control, and quarantine;
3. Mass immunization of dogs by campaigns and by continuing vaccination of young dogs;
4. Stray dog control and destruction of unvaccinated dogs with low levels of dependency on, or restriction by man
5. Dog registration
6. **Vaccines:**
   a. **Pasteur type vaccine:** One of the oldest vaccines against rabies. Prepared by injecting the fixed virus into rabbits. The brain cords are collected and dried by dessication and treating with KOH. Not in use at present.
   b. **Nervous tissue preparation and phenolized (Fermi, Semple and Umeno-doi vaccines)** - this consisted of a 5% suspension of infected animal nervous tissue which had been inactivated (eg. the Semple vaccine was derived from phenol-inactivated infected rabbit brain). These preparations are now out of date as they were associated with the rare complication of demyelinating allergic encephalitis. This appears to be related to myelin basic protein in the vaccine. This complication was shown to occur in 4.6 case for 1000 persons vaccinated by the Semple vaccine. The case-fatality proportion was 3.13%. The Semple vaccine is still used in some developing countries. A suckling mouse brain vaccine is used in some Central and South American countries.
   c. **UV treated vaccines (Webster):** Inactivated vaccines prepared from brain of infected animals, which were exposed to UV rays. UV rays readily kill the rabies virus.
   d. **Duck Embryo Vaccine (Peck)** - this vaccine strain is grown in embryonated duck eggs and is inactivated with β-propriolactone. This vaccine has a lower risk of allergic encephalitis. However, it is considerably less immunogenic and does have minor side effects. Almost all vaccinees experience local reactions, 33% have constitutional symptoms such as fever, malaise, myalgia, and generalized lymphadenopathy.
   e. **Live vaccines – (Hoegyes)**: Live fixed viruses are given at a very low level (below the infective dose).
   f. **Avianized vaccines (Koprowsky and Cox)**: The Flurry strain of rabies virus grows in chicken embryos. It loses its pathogenicity after 40-50 passages in embryos for rabbits and dogs (*referred as low egg passage vaccines – **LEP**)* and loses the pathogenicity completely except for mice after 180 passages (*referred as high egg passage vaccines – **HEP**)*. HEP were used for humans and LEP were used only for dogs and not for any other animals.
   g. **Human Diploid Cell Vaccine (HDCV)** - HDCV was introduced in 1978. It is a grown on WI-38 (U.S.) or MRC-5 (Europe) cells. The vaccine is highly effective, in several studies, antibodies have been demonstrated in 100% of all recipients. Serious adverse reactions to HDCV are extremely rare. However, the vaccine is very expensive, as human cell cultures are more difficult to handle than other animal cell culture systems.

Efforts are being made to use other inexpensive cell culture systems such as VERO cells.

Public health: Rabies virus causes highly fatal disease in human beings. The incubation period is highly variable, ranging from 7 days to several years. It depends on several factors such as dose of inoculum, severity of the wound, length of the neural path from the wound to the brain e.g. wounds on the face have a shorter incubation period than wounds in the leg. The illness begins with a non-specific prodrome period, comprising of fever, malaise, anorexia, sore throat, myalgia and headache. The patient nay exhibit irritability and abnormal sensations around the wound. The prodrome is followed by one of two basic clinical patterns: the more common "furious" form characterized by hyperexcitability, spasms and hydrophobia; or "dumb" rabies featuring an ascending paralysis. Survival tends to be

longer for patients with "dumb" rabies than those with "furious" rabies. Complications involving the Cardiovacular System, CNS, andtherespiratory systems eventually develop and contribute to death. Cardiac dysrhythmias of all types occur and respiratory disturbances occur in all cases. Raised intracranial pressure contributes to the decreased level of consciousness and to focal convulsions. Other CNS complications include disturbances of thermoregulation, diabetes insipidus, autonomic dysfunction and convulsions. Once rabies is established, it is always fatal (except in one case in US).

Postexposure prophylaxis: In cases of animal bites, dogs and cats in a rabies endemic area should be held for 10 days for observation. If signs develop, they should be killed and their tissue examined in the laboratory. Wild animals are not observed but if captured, the animal should be killed and examined. The essential components of postexposure prophylaxis are the local treatment of wounds and active and passive immunization.

Wound treatment - surgical debridement should be carried out. The wound should not be sutured up.

Passive immunization - human rabies immunoglobulin should be applied around the area of the wound; to be supplemented with an i.m. dose to confer short term protection. There is convincing evidence that combined treatment with rabies immunoglobulin and active immunization is much more effective than active immunization alone. Equine rabies immunoglobulin (ERIG) is available in many countries and is considerably cheaper than HRIG.

Active immunization - the human diploid cell vaccine is the best preparation available. The vaccine is usually administered into the deltoid region, and 5 doses are usually given.

Pre-exposure prophylaxis: Persons who are regularly at high risk of exposure, such as vets, laboratory workers, animal handlers and wildlife officers should be considered for preexposure prophylaxis by active immunization with the cell culture vaccine. Immunization normally consists of 3 doses of vaccine. Antibody can be demonstrated in the sera of virtually 100% of those vaccinated if the diploid cell culture vaccine is used. Booster doses should be offered to persons at continuing risk every one to three years. Local treatment of wounds should always be carried out in exposed persons who have been vaccinated previously.

BOVINE EPHEMERAL FEVER

Bovine ephemeral fever (BEF or Bovine Epizootic Fever) is a non-contagious arthropod-borne disease (arbovirus from the Rhabdoviridae family) of cattle, characterized by an acute fever of short duration, with high morbidity and low mortality. Typically, affected animals are only sick for a few days, and they recover and hence the infection is also known as **three Day Sickness.**

General characters

Classification:

Baltimoregroup	-	**Group V** – Negative sense RNA viruses
Order	-	Mononegavirales
Family	–	Rhabdoviridae
Genus	-	Ephemerovirus
Species	-	Bovine ephemeral fever virus (BEFV)

Synonyms: Three-day sickness, Bovine epizootic fever, Three-day stiff sickness, Dragon boat disease

Morphology: Typical of other *Rhabdoviridae*. The BEF virus is an enveloped virus and hence ether sensitive. The genome is single-stranded, negative sense linear RNA. The genome has five genes coding for five proteins.

OIE listing: No listing; SPC List D disease

Risk group: Group I

Resistance: Typical of other rhabdoviruses. The virus is rapidly inactivated at pH levels below 5

Replication: Typical of other rhabdoviruses

HA property: The virus has no HA property

Virus strains: This virus is antigenically related to at least three other viruses nonpathogenic for cattle: Kimberley virus, Berrimah virus, and Adelaide River virus and two that produce an ephemeral fever-like disease in cattle, Kotonkan and Puchongviruses.

Cultivation: BEFV are cultivated in embryonated eggs, suckling mice, vero cells and cattle. BEFV are cultivated through intravenous inoculation into chicken embryos. Lesions are not characteristic and infected embryos die. BEFV are identified by immunofluoresence in heart, brain, lung and liver of chicken embryos at 1-5 days and in lung and liver of one-day-old chickens at 1-2 days.

Pathogenesis:

Hosts affected: Cattle and buffaloes are the main host.

Distribution: BEF occurs widely across Africa and Asia, and in areas of northern and eastern Australia. It does not occur in Europe or the Americas

Transmission: The BEF virus spreads in nature only by an insect bite. The disease will not spread from cow to cow by close contact, droplet infection, bodily excretions including semen, or by the transfer or injection of exudates. Strong winds can transport vectors long distances, over land and water. Epizootics of BEF are associated with recent rainfall. Mosquitoes are the major vectors in spreading BEF. The BEF virus has been isolated from *Culicine* and *Anopheline* mosquitoes and biting midges. However, establishment of the disease depends on suitable environmental conditions for the vector to increase and spread. There is also no carrier stage as BEFV is only present in the blood for short periods during the acute stage of the diseases.

Incubation period: The incubation period is between 2 and 4 days, and 9 days in rare cases.

Morbidity and mortality: Morbidity and mortality are highly variable. Morbidity is very high and mortality is very low.

Symptoms: The clinical signs are very characteristic and can be very severe. Fever is most important symptom. The fever of ephemeral fever is generally biphasic, sometimes triphasic, with peaks of 40-41.5° C (104- 107° F) spaced 12-18 hours apart. The physical signs during the first febrile phase tend to be mild except for the dramatic fall in milk production of lactating cows. The characteristic signs associated with BEF are those of the second febrile phase. These signs include accelerated heart and respiratory rates, anorexia, ruminal atony, depression, serous or mucoid nasal and ocular discharges, salivation, muscle twitching or waves of shivering, a generalized stiffness or a shifting lameness. There may be submandibular edema or patchy edema on the head. Nasal and ocular discharges, drooling of saliva and periorbital swelling and increased excitability and agitation are other signs. With the onset of fever, there is a rapid fall in circulating lymphocytes with eosinopenia. Many animals become recumbent for 12-24 hours. Animals may be unable to rise and remain in sternal recumbency for hours or days with the head turned to the flank, or in lateral recumbency with or without loss of most reflexes. Recovery begins 1-2 days after the clinical signs are first noticed and is usually complete and without sequelae in a further 1- 2 days after the overt clinical signs are first noticed. By day three the affected animal is usually standing again and will begin to eat. However, lameness and weakness may last for

another two or three day. Milk yield will return to normal after three weeks. Death can occur suddenly in the febrile or in the recovery phase.

Lesions: The most important gross lesions are accumulation of fibrin-rich fluid in the pleural, peritoneal, and pericardial cavities and in the joint capsules. The lungs may show patchy edema. Lymphadenitis with petechial hemorrhages and focal necrosis are also observed.

Diagnosis:

Field diagnosis: Field diagnosis is made from clinical observations (presence of lameness, muscular stiffness, pain, rapid spread of the disease through herds and short fever) and the history of the outbreak.

Isolation and identification: Blood is the ideal material and should be taken during the period of fever and another sample 1-2 weeks later. Samples should be taken from animals in various stages of the disease to facilitate a rapid laboratory confirmation

- Part of the first sample of blood is allowed to clot, and another portion is mixed with anticoagulant. From the uncoagulated blood, a smear is made on a glass slide and allowed to dry in air. The balance is used for virus isolation. When blood taken during illness is allowed to clot, it usually fails to contract on standing, even over several days. It may be streaked with fibrin.
- Cattle are used for isolation of the virus. Blood from infected cattle is used as inoculum for infecting cattle. Exhibition of characteristic clinical symptoms are used to identify BEFV.
- PCR

3. Serology

- Virus neutralization test is commonly done to identify antibodies against BEFV.
- ELISA is also done

Differential diagnosis: BEF confuse with infections like early Rift Valley fever, FMD, heartwater, bluetongue, botulism, babesiosis and blackleg.

Treatment: Ephemeral fever is one of the rare virus diseases for which treatment is effective. Since the infection is inflammatory in nature, anti-inflammatory drugs are useful in controlling the infection. During fever, the paresis or paralysis responds to injected calcium borogluconate in the same manner as parturient paresis (milk fever).

Control:

1. **Vaccines:** Once cattle have been infected with the disease, most are resistant to infection for many years or for life. However, live attenuated vaccines, killed vaccine and a sub-unit vaccine has been used in certain part of the world. The live vaccine gives at least 12 months protection after two doses. The inactivated vaccine gives protection for about six months protection. Animals can be vaccinated from six months of age and should then be revaccinated each year to ensure continued protection
2. **Prevention:**
   a. Vaccination in endaemic areas
   b. Vector control - Useful steps are to place the cattle in an insect- proof area, spray with insecticides, or suppress insects in the local environment

Public health: There is no evidence of human infection.

FAMILY - *CORONAVIRIDAE*
AVIAN INFECTIOUS BRONCHITIS

Avian Infectious Bronchitis (IB) is a contagious disease of birds due to a *Coronavirus* and causing important economic loss in chicken operations. The infection was first described in 1931 in the USA as a respiratory disease of chicks. The virus – IBV is a corona virus and is prevalent in all countries with an intensive poultry industry including India. The infection has a significant economic impact; in broilers, production losses are due to poor weight gains, condemnation at processing and mortality. In laying birds, losses are due to suboptimal egg production and downgrading of eggs

General characters:

Classification:

Baltimoregroup - **Group IV** – Positive sense RNA viruses

Order - Nidovirales

Family – Coronaviridae

Genus - coronavirus

Species - Avian infectious bronchitis virus (IBV)

Characters of the order Nidovirales: Virions have a complex construction and consist of an envelope and a nucleocapsid. Virions are enveloped; spherical, or kidney-shaped, or pleomorphic. The surface projections are distinctive club-shaped peplomers that evenly cover the surface. They are densely dispersed. Capsid/nucleocapsid is elongated and exhibits helical symmetry. The nucleocapsid is cylindrical; 2 nm in diameter. The genome is not segmented and consists of a single molecule of linear positive-sense single-stranded RNA. The viral genome encodes structural proteins and non-structural proteins.

Morphology: The virions are enveloped and slightly pleomorphic or spherical in shape with 60-220 nm in diameter. The surface of the virus has surface projections, which are distinct and club- shaped. These projections are spaced widely apart and dispersed evenly over all over the surface providing a crown like appearance and hence these viruses are known as corona viruses. The nucleocapsid is filamentous with helical symmetry. The virions contain one molecule of linear positive-sense single stranded RNA, which also act as mRNA. The total genome length is 20000- 33000 nt. The 5' end of the genome has a cap and 3' end has a poly (A) tail. The virus has five structural proteins – S, M, N, HE and E.

Surface glycoprotein(or spike, S), which S protein is responsible for attachment to cells, hemagglutination and membrane fusion.

Integral membrane protein (M) which spans the virus envelope three times

Nucleocapsid protein (N)

Hemagglutinine-esterase protein (HE), which forms short surface projections, and have receptor binding, hemagglutination and receptor destroying activities

Envelope protein (E), small, envelope-associated protein

Unlike (-)sense RNA viruses polymerase enzymes are not present in the nucleocapsid

OIE Listing: List **B** infection

Risk group: Group I pathogen. No risk to human beings.

Resistance: The viruses are not very stable and are destroyed by heat and common disinfectants used. Following good biosecurity measures at the poultry house will greatly minimise the loss of birds due to this infection

Replication: The replication of corona viruses is slow compared to other enveloped viruses and the viruses replicate in the cytoplasm. The (+) sense genome is first translated to produce a viral polymerase (RNA dep RNA Polymerase), which then produces a full-length (-) sense strand.This (-) sense stand is used as a template to produce mRNA as a 'nested set' of transcripts, all with anidentical 5' non-translated leader sequence and coincident 3' polyadenylated ends. Each mRNA produced is monocistronic, the genes at the 5' end being translated from the longest mRNA and soon. These unusual cytoplasmic structures are produced not by splicing (post-transcriptional modification) but by the polymerase during transcription. Between each of the genes there is a repeated **intergenic sequence** - UCUAAAC - which interacts with the transcriptase plus cellular factors to 'splice' the leader sequence onto the start of each ORF. **Assembly** occurs by budding into the golgi apparatus, particles being transported to the surface of the cell by the secretory nature of this organelle & released.

HA property: Many IBV cause agglutination of chicken RBC after treatment with an enzyme neuraminidase (*Previously it was thought as Phospholipase C).*

Virus strains: The virus elicits distinct antigen determinants on envelope and spikes. Variations within the virus are due to differences in the S and HE proteins. As a result of this, these are many serotypes and within each serotype there are many strains. These serotypes are immunologically distinct (seropleurality; antigenic pleurality) and antibodies against one serotype may not protect the birds against the infection caused by other serotypes. Some of the common serotypes are Massachusetts, Connecticut, Arkansas etc. Some of the strains are M5, M41, H151, Holte, Gray, Florida etc.

Cultivation: IBV are cultivated in embryoated eggs and tracheal organ culture (TOC) system. Cell cultures are not used cultivation as it is usually necessary to adapt IBV isolates to grow in chicken embryos before cytopathic effects (CPE) of virus infection are seen in cell culture. The virus is inoculated into the allantoic cavity of 9–11-day-old embryos. The allantoic fluids of all eggs are pooled after harvesting 3–7 days after infection; this pool is diluted 1/5 or 1/10 in antibiotic broth and further passaged into another set of eggs. This is repeated as desired. A field strain will induce teratological changes in the embryo at the second or third passage consisting of stunted and curled embryos with feather dystrophy and urate deposits in the embryonic mesonephros. The allantoic fluid should not agglutinate red blood cells. TOCs prepared from 20-day-old embryos can be used to isolate IBV directly from field material. Tracheal rings of 0.5–1.0 mm thick are prepared using automatic tissue chopper. Infection of tracheal organ cultures produce ciliostasis within 24–48 hours.

Pathogenesis:

1. **Hosts affected:** Chickens and commercially reared pheasants are the only natural hosts for IBV. Other species are not considered as reservoirs of IBV.. Young birds are more prone than adult birds.
2. **Distribution:** The infection is worldwide in distribution.
3. **Transmission:** The affected birds shed the virus in respiratory secretion and faeces. The virus spreads rapidly among chickens in a flock through inhalation of virus droplets (coughing or sneezing) produced by infected chickens. Infection is also transmitted by aerosols, contaminated feed and water, contact with animals or material. Movement of live birds is considered as a potential source for the introduction of IBV. Affected birds recover within fourteen days. However, a small number of chicks may become latently infected with erratic shedding of virus for a prolonged period of time via both faeces and aerosol. There is no carrier state and vertical transmission. No vectors are also involved in the spread of infection.

4. **Incubation period:** The incubation period is 18 to 36 hours
5. **Morbidity and mortality:** Morbidity goes upto 100% and mortality is variable (upto 50%).
6. **Symptoms:** The infection in young chickens is characterized by gasping, coughing and nasal discharge with wet eyes and swollen sinuses. The chicks appear depressed and may huddle near the heat source. Food consumption and weight gain are also reduced. In adult laying flocks, the respiratory symptoms of gasping and coughing are usually followed by a drop in production and the birds may molt. Pullets in good condition may suffer only a slight drop in production and regain normal production within few weeks. Following the decline, production of misshapen soft-shelled eggs with inferior internal quality is often observed. This change in shell and internal quality may be permanent. Secondary bacterial infection due to *E.coli* complicates the disease scenario and lead increased condemnation of birds.
7. **Lesions:** The infection is exhibited as respiratory, renal and reproductive forms.
   a. **Respiratory forms** of the disease are usually observed in poorly vaccinated flocks. Secondary bacteria such as *E. coli,* as explained, can complicate the production parameters minimizing profit returns. Lesions include inflammation and accumulation of mucous in the trachea, nasal passages and sinuses. Air sacs may be cloudy and thickened.
   b. **Renal form** of the infections is characterized by swollen and inflamed kidneys, distension of the ureters with build up of urate deposits. Holte and Gray are considered as nephrogenic strains and they have a high affinity (attraction) for the kidneys and the ureters. Mortality in renal form may go upto 60%.
   c. **Reproductive form** characterized by damage to ovaries and shortened reproductive tract

Diagnosis

1. **Field diagnosis:** Sudden appearance of respiratory infection with high morbidity and characteristic respiratory symptoms help in field diagnosis. In adult birds, the causes for malformed egg should be analysed in details before confirming it as IB.
2. **Isolation and identification:**

a. Clinical materials:

i. Live birds: Tracheal mucus from young chicks

ii. Dead birds: Tissue samples from the trachea, kidney, oviduct and caecel tonsils in sterile transport media with antibiotics and dry swabs from the respiratory tract or cloaca

b. Isolation systems: Embryonated eggs or tracheal organ culture (TOC). In eggs characteristic lesions include teratological changes in the embryo at the second or third passage consisting of stunted and curled embryos with feather dystrophy and urate deposits in the embryonic mesonephros. The infection in TOCs is characterized by ciliastasis. The lesions have to confirmed by any one of the following antigen identification system. In outbreaks from affected birds, the virus can be isolated using sentinel birds.

c. Antigen identification system:

i. Fluorescent antibody test (FAT)

ii. Immunohistochemistry methods like IPT

iii. Antigen competitive ELISA

iv. RT-PCR for S gene

v. cDNA probe

d. Serotype identification: Since IBV are antigenically distinct identification of correct serotype is essential. Serotype identification is carried out by

i. Virus neutralization test in eggs, TOCs

ii. HI test

iii. Monoclonal antibody (serotype) based ELISA

iv. Monoclonal antibody (serotype) based immunohistochemistry

v. RT-PCR using serotype specific primers

3. **Serological tests:** The serological tests are normally performed to identify antibodies produced after vaccination. During infection, paired sera sample with a time interval of three weeks will yield clues about infection. Some of the commonly performed serological tests are

i. Virus neutralization test

ii. HI test

iii. AGID

iv. ELISA

4. **Differential diagnosis:** IB should be differentiated from Newcastle disease, Infectious Laryngeal Tracheitis and Infectious Coryza

Treatment: Not effective and only supportive treatment is provided.

Control

1. **Vaccines:** At present both live and inactivated vaccines are used. Live vaccines based on the Massachusetts serotype alone are used now. Live vaccines confer a better local immunity on the respiratory tract and may protect against a wider antigenic spectrum of field strains. While single dose of live vaccine may protect broilers, it may not protect for the life of the layer flock as variant serotype challenge is very high on farms. Further, the use of some live vaccines carries the risk of residual pathogenicity associated with vaccine back-passage in flocks. Inactivated oil emulsion vaccines preceded by a live virus vaccine produce persistent antibody response.
2. Strict biosecurity measures and good husbandry practices will minimize the spread of IBV.
3. During outbreaks, the virus should be serotyped and vaccine should be altered.
4. In areas where there is no IB, use of live vaccine should be minimized.

Public health: No human infection is reported.

TRANSMISSIBLE GASTROENTERITIS

Transmissible gastro-enteritis (TGE) is an acute highly contagious disease of pigs characterized by profuse diarrhoea and vomiting with high morbidity and mortality in piglets. Of late, a respiratory variant of the TGE virus (TGEV), known as the porcine respiratory coronavirus (PRCV) has emerged in Europe and is causing interference in the diagnosis of TGE. PRCV is probably a deletion mutant of TGEV. PRCV does not appear to be an important primary pathogen, but it has greatly complicated the diagnosis of TGE, particularly by serological means.

General characters:

Classification:

Baltimoregroup - **Group IV** – Positive sense RNA viruses

Order - Nidovirales

Family – Coronaviridae

Genus - Coronavirus

Species - Transmissible gastroenteritis virus (**TGEV**)

PRCV Porcine Respiratory Coronavirus is a mutant of TGE

Morphology: Morphological features are typical of other members of the family *Coronaviridae*. The virions are enveloped and slightly pleomorphic or spherical in shape. The surface of the virus has surface projections, which are distinct and club-shaped. The nucleocapsid is filamentous with helical symmetry. The virions contain one molecule of linear positive-sense single stranded RNA, which also act as mRNA. The virus has five structural proteins – S, M, N, HE and E

OIE Listing: List **B** infection

Risk group: Risk group I

Resistance: The viruses are not very stable, highly heat sensitive and common disinfectants used. The virus has some resistance for cold temperature. Following good biosecurity measures at the pig stall will greatly minimize the loss of piglets due to this infection

Replication: Typical of members of the family *Coronaviridae*

Cultivation: TGEV does not grow well in cell culture. However, they grow in 3–4- day-old primary or secondary pig kidney monolayers and low passage porcine cultures (such as thyroid or testis). The cytopathic effect (CPE) may be observed after 3–7 days, characterized by cell rounding, enlargement and formation of syncytia. Wild-type TGEV (*virus obtained from an outbreak in field is referred as wild type*) does not grow readily in tissue culture, so several subpassages may be necessary before distinctive CPE become apparent. The cytopathic isolates must be confirmed as TGEV by FAT or by *in-vitro* neutralisation tests or by nucleic acid identification methods. Young piglets are more susceptible to TEGV. Hence, they are commonly used to cultivate TEGV in laboratories.

Pathogenesis:

1. **Hosts affected:** The **c**linical disease only occurs in pigs. However, cats, dogs and foxes are potential carriers of the virus without clinical signs of the disease.
2. **Distribution:** TGE is present in Europe, Americas (North, South and Central), Asia (including China, Japan and Korea), and parts of West Africa.
3. **Transmission:** Outbreaks of TGE are seen commonly following the introduction of infected pigs into susceptible herd. Sub-clinically infected animals can also be a source of infection and recovered pigs often become carriers and can shed the virus for 2-3 weeks in their faeces. Large amounts of virus are excreted in the faeces of infected animals. The exact mechanisms of transmission of the disease are still unclear. However, the proposed means are, faeco-oral contamination, through milk from infected sows, contaminated litter, through carriers like dogs cats, foxes, infected swills and through infected husbandry material. The virus is relatively fragile and susceptible to disinfectants and drying. However, the virus can survive for few days in the cold temperature, hence the disease is more commonly seen during the winter months. There is no vertical transmission and no vectors are involved in the transmission.
4. **Incubation period:** Incubation period is relatively short – just 18 hours to 3 days
5. **Morbidity and mortality:** Morbidity and mortality are nearly 100% in piglets
6. **Symtoms:** The disease spreads rapidly around the farm. Piglets less than 21 days of age are all affected and generally die. Watery diarrhea (foul smelling yellowish-green often containing flecks of undigested milk particles in the piglet), vomiting, weight loss, dehydration and loss of appetite

are the common symptoms. The weaned piglets become unthrifty. The growers, finishers and adults are generally mildly affected and will survive if their water supplies are adequate (*prevention of dehydration)*. In adult pigs agalactia is a common symptom.

7. **Lesions:** Gross lesions are confined to the gastrointestinal tract, except for the dehydration. The common lesions are distended stomach with curdled milk, congestion with small area of haemorrhage on diaphragmatic surface, distended small intestine with yellow foamy fluid containing curdled milk, enteritis and thinning of intestinal due to severe atrophy of intestinal villi.

Diagnosis:

1. **Field diagnosis:** Is based on occurrence of infection more severely in piglets in winter months with characteristic symptoms.

2. **Isolation and identification:**

a. Clinical materials:

   i. Faeces from affected animals or from dead animals are the material of choice.

   ii. Loops of affected small intestine, ligated at each end to retain the contents are also the preferred specimens for isolation of virus.

b. Isolation systems: TGEV does not grow well in cell culture. However, they grow in 3–4-day-old primary or secondary pig kidney monolayers and low passage porcine cultures (such as thyroid or testis).

   i. The sample material is homogenised in cell culture medium or phosphate buffered saline (PBS), pH 7.2, containing antibiotics.

   ii. This homogenised sample is allowed to stand out off direct sunlight for 30 minutes at room temperature.

   iii. The suspension is then sonicated and clarified by low-speed centrifugation.

   iv. The supernatant fluid is mixed with an equal volume of heat-inactivated bovine serum in order to reduce the cytotoxic effect of the material and it is then used to inoculate susceptible tissue cultures.

   v. The cytopathic effect (CPE) may be observed after 3–7 days, characterized by cell rounding, enlargement and formation of syncytia. Wild-type TGEV (*virus obtained from an outbreak in field is referred as wild type*) does not grow readily in tissue culture, so several subpassages may be necessary before distinctive CPE become apparent.

c. Antigen identification tests: Since CPEs are not produced immediately and require number of blind passages, following antigen identification tests are performed.

   i. Immunofluorescence

   ii. Reversed passive hemagglutination

   iii. Enzyme-linked immunosorbent assays (ELISAs)

   iv. Radioimmunoassay (RIA)

   v. DNA probes

   vi. Electron microscopy

   vii. RT-PCR

3. **Identification of antigen directly in tissue sections/clinical materials:**

   a. Fluorescent antibody test: Used to identify viral antigens in sections of small intestine

   b. Double antibody sandwich ELISA: This test is based inthe capture of the viral antigen from

the faecal sample by three MAb, two specific for the S protein (site A and D) and one for the nucleoprotein N.

c. **in situ hybridisation (ISH) and RT-PCR:** This technique is used for the direct detection of TGEV in clinical samples and for differentiation from PRCV.

4. **Serological tests:** Serological tests are performed on paired serum samples. A rise in antibody titre is an indication of infection. The tests that are commonly performed include
   a. Virus neutralization test
   b. Indirect ELISA
   c. Competitive ELISA
   d. Blocking ELISA using monoclonal antibodies also allows the differentiation of TGEV and PCRV

5. **TEGV specific tests (test accepted for international trade):** TGEV- specific tests are ELISAs that use an MAb that recognises TGEV but not PRCV and are the tests of choice for qualifying animals for export.
   i. Haemagglutinating encephalomyelitis (vomiting and wasting disease)
   ii. Classical swine fever (Hog cholera)
   iii. Porcine rotavirus infection
   iv. Swine dysentery
   v. Colibacillosis,
   vi. Arsenic poisoning

Treatment:

No effective treatment is available. Treatment of TGE should be aimed towards treating dehydration and starvation

Control:

1. **Vaccines:** Vaccination of pregnant sows is being tried but commercial vaccines are not yet available.
2. To reduce risk of spread, animals from serologically negative herds alone should be introduced and the swill should always be sterilized.
3. If an outbreak occurs, good hygiene practices can help reducing the impact of the disease. The spread can be prevented by isolating the newborn piglets.
4. The other approach aims to minimize the duration of the disease by exposing all the pregnant sows to the disease to develop transplacental immunity of farrows.

FAMILY - *FLAVIVIRIDAE*

CLASSICAL SWINE FEVER

Classical swine fever (CSF) is a contagious viral disease of pigs. CSF is caused by a virus belonging to the family *Flaviviridae*. The infection can occur as acute, subacute, chronic or in a persistent form. The infection is characterized by high fever, severe depression, multiple superficial and internal hemorrhages, and high morbidity and mortality.

General characters:

Classification

Baltimoregroup - **Group IV** – Positive sense RNA viruses

Order - Nidovirales

Family – Flaviviridae

Genus - Pestivirus

Species - Classical swine fever virus (CSFV)

The family *Flaviviridae* is classified into three genus – flavivirus, pestivirus and hepacivirus. Most of the members under the genus flavivirus are arthropod transmiteed and cause very severe infection in human beings. The important infections are dengue fever, yellow fever, Kyasanur forest disease etc. Members under the genus pestivirus are generally not transmitted by arthropods. The important infections caused by the pestivirus include CSF and bovine viral diarrhoea (BVD)

Synonyms: English swine fever, hog cholera, *peste du porc, colera porcina*

Morphology: The virions are enveloped; spherical to pleomorphic with 40-60 nm in diameter. The surface projections are prominent with distinctive spikes surrounded by a prominent fringe. Surface projections also form ring-like subunits. The virions consist of structural protein(s) - nucleocapsid (C), matrix (M) glycoprotein (E) and a non-structural (NS) protein. The nucleocapsid is round and exhibits polyhedral symmetry. The genetic material comprises of a single molecule non-segmented linear positive-sense single-stranded RNA. The complete genome is 12,500 nucleotides long. The 5'-end of the genome has a cap, and the 3'-terminus has no poly (A) tail.

OIE Listing: List A infection

Risk group: No infection to human beings

Resistance: CSFV are resistant to temperature upto 56^0C and survive at wide range of pH between 3 and 11. They are susceptible to ether, chloroform and ß-propiolactone and are inactivated by cresol, sodium hydroxide (2%), formalin (1%), sodium carbonate (4% anhydrous or 10% crystalline, with 0.1% detergent), ionic and non-ionic detergents and strong iodophors (1%) in phosphoric acid. The virus is also sensitive to drying (desiccation). The virus also survives well in cold conditions and can survive some forms of meat processing (curing and smoking). In a protein-rich environment, the virus is very stable and can survive for months in refrigerated meat and for years in frozen meat.

Replication: The glycoprotein spikes are responsible for attaching to receptors on the cell. The entire replication of the virus takes place in the cytoplasm. The (+) sense genomic RNA acts directly as mRNA. The entire genome is translated as a single polyprotein, which is then cleaved into the mature proteins. The NS protein synthesizes complementary (-) strand, which used as a template for genomic progeny RNA synthesis. Assembly occurs during budding into cytoplasmic vacuoles than at the cell surface and release of progeny virion occurs when cell lyses.

HA property: The virus do not agglutinateerythrocytes.

Strains: The organism has a close antigenic relationship with the bovine viral diarrhea virus (BVDV) and the border disease virus (BDV). There are many strains – some are very virulent and some less pathogenic. Some of the important virus strains are Alfort/187, Alfort-Tübingen, Brescia and C strain.

Cultivation: The virus grows slowly in cell culture. The viruses are cultivated in PK-15 cell lines. Presence of the virus in cell lines is confirmed by FAT or RT- PCR.

Pathogenesis:

1. **Hosts affected:** Pigs and wild boar are the only natural host for CSF.
2. **Distribution:** The disease is present in Asia, Central and South America, and parts of Europe and Africa.
3. **Transmission:** Introduction of infected pigs is the principal source of infection in CSF free herds.

Blood, tissues, secretions and excretions from an infected animal contain viruses. Transmission occurs mostly by direct contact between animals through secretions, excretions, semen and blood, through oral route, through the conjunctiva, mucous membrane, skin abrasion, insemination, and percutaneous blood transfer (e.g., common needle, contaminated instruments). Infection also spreads by farm visitors, veterinarians and pig traders. Insufficiently cooked waste food fed to pigs are also a source of spread for the infection. Airborne transmission can occur in mechanically ventilated units when animal within close proximity to each other. During the warm season, CSF may be carried mechanically by insect vectors that are common to the farm environment. However, CSF doesn't replicate in invertebrate vectors. Transplacental infection also occurs in CSF. Congenitally infected piglets may shed the virus for months.

4. **Incubation period:** The incubation period is usually 3 to 4 days but can range from 2 to 14 days.
5. **Morbidity and mortality:** In acute CSF, the morbidity and mortality are high and go upto 100%.
6. **Symptoms:** The infection occurs as acute, subacute, chronic or congenital forms. The infection also appears in a milder form in sows.

a. **Peracute infection:** Observed in piglets, where young pigs may be found dead without any prior sign of illness.

b. **Acute infection:** In acute form the pigs appear sick, inactive and drowsy with arched back. Some pigs stand with droopy head and straight tail. They may huddle to a corner for warmth. Some pigs also vomit a yellow fluid containing bile. The most important symptom in acute infection is the high fever that may reach 108° F (42.2° C) with an average of 106° F (41.1° C) accompanied by anorexia and constipation. Conjunctivitis with encrustation of the eyelids and the presence of dirty streaks below the eyes is the other important symptom. Sick pigs will have a staggering gait and posterior weakness. In last stage of the infection, pigs will become recumbent, and convulsions may occur shortly before death. Sever diarrhoea will also occur during last stages. There may be purple discoloration of abdominal skin, or necrosis of the tips of extremities (ears, tail, vulva). Neurological signs may occur, which include incoordination, tremors, convulsions and circling. Death usually occurs 5-15 days after the onset of illness.

c. **Chronic infection:** The chronic form is characterized by dullness, capricious appetite, pyrexia and diarrhoea for up to 1 month. Weight loss, hair loss, dermatitis and discoloration of abdomen or ears are the other symptoms. A chronically infected pig may have a disproportionately large head relative to the small trunk. The affected pigs may stand with arched backs and their hind legs placed under the body. The animals may recover followed by relapse and death. Chronic form occurs in vaccinated herds or can also be caused by less virulent strains of CSFV.

d. **Congenital infection:** Congenital CSF infection by virulent strains results in abortions or in the birth of diseased pigs, which will die shortly after birth. Transplacental transmission with low-virulent CSF strains result in mummification, stillbirth, or the birth of weak and "shaker" pigs. Malformation of the visceral organs and of the central nervous system occurs frequently. Some pigs may be born virtually healthy but persistently infected with CSF. Such infection usually follows exposure of fetuses to CSF of low virulence in the first trimester of fetal life. Pigs thus infected do not produce neutralizing antibodies to CSF and have a lifelong viremia. The pigs may be free of disease for several months before developing mild anorexia, depression, conjunctivitis, dermatitis, diarrhoea, runting, and locomotive disturbance leading to paresis and death.

7. **Lesions:**

a. Peracute infection: No gross lesions.

b. Acute infection: Acute infection is characterized by following lesions;

i. Haemorrhages and purplish discoloration of the skin

ii. Necrotic foci in the tonsils

iii. Swollen and oedematous submandibular and pharyngeal lymphnodes with peripheral haemorrhages.

iv. Spleen with raised edges and dark wedge shaped areas

v. Pinpoint to ecchymotic hemorrhages on the surface of the kidneys (turkey egg appearance). The haemorrhages are most marked under the kidney capsule**.**

vi. Hemorrhages on the surface of the small and large intestine, larynx, heart, epiglottis, and the fascia lata of the back muscles

vii. Accumulation of straw-colored fluids in the peritoneal and thoracic cavities and in the pericardial sac

viii. Congestion of the lung and bronchopneumonia

ix. In brain the lesions are perivascular cuffing, endothelial proliferation, and microgliosis

c. Chronic infection:

i. Lesions are less severe and are often complicated by secondary bacterial infections

ii. Button ulcers are an expression of such a secondary bacterial infection

d. Congenital infection:

i. Hypoplasia of the cerebellum, thymus atrophy, ascites, and deformities of the head and of the limbs.

ii. Edema and petechial hemorrhages of the skin and of the internal organs

Diagnosis:

1. **Field diagnosis:** Based on high morbidity and mortality, high fever, diarrhoea kidney and lymph node lesions will help in field diagnosis.

2. **Clinical materials:** Tonsils, lymph nodes (pharyngeal, mesenteric), spleen, kidney, distal part of ileum and blood in EDTA (live cases) are the ideal materials for isolation and identification of viruses. These materials should be kept under refrigeration and shipped to laboratory as quickly as possible

3. **Direct identification of CSF from tissues:**

a. Immunological methods:

i. Fluorescent antibody test (FAT) – Also known as fluorescent antibody tissue section test (FATST). It a rapid test that can be used to detect CSFV antigen in cryostat sections of tonsils, spleen, kidney, lymph nodes or distal portions of the ileum. Cryostat sections are stained directly with anti-CSF immunoglobulin conjugated to fluorescein isothiocyanate (FITC) or indirectly using a secondary FITC conjugate and examined by fluorescence microscopy.

ii. Immunoperoxidase test (IPT) – This test is used to differentiate between field strains of CSFV and vaccine strain of CSFV. A panel of three monoclonal antibodies (Mabs) are used to detect all field strains of CSFV, vaccine strains of CSFV and ruminant pestiviruses.

iii. Antigen capture ELISA (Double antibody sandwich ELISA) - Blood leukocyte fraction or anticoagulated whole blood or clarified tissue homogenate are used as antigen for ELISA. Monoclonal and polyclonal antibodies are used as detector and capture antibodies respectively. The technique is relatively simple to perform, does not require tissue culture facilities, is suitable for automation and can provide results within half a day. The disadvantage is less sensitive than virus isolation, especially in adult pigs and in mild or subclinical cases.

iv. RT-PCR - This method is rapid and more sensitive than antigen-capture ELISAs or virus isolation.

4. **Isolation and identification:** Isolation of virus in cell cultures is a more sensitive but slower method for diagnosis of CSF. Isolation is best performed in rapidly dividing PK-15 cells. The cultures are examined for fluorescent foci by FAT after 24–72 hours.
5. **Serological tests:** Detection of virus-specific antibodies is particularly useful on premises suspected of having infections with CSF strains of low virulence. Due to the immunosuppressive effect of CSFV, antibodies cannot be detected with certainty until 21 days post-infection. Commonly performed serological tests are as below. All the three tests are specified tests for international trade
   a. Fluorescent antibody virus neutralisation test (FAVN)
   b. Neutralising peroxidase-linked assay (NPLA)
   c. ELISA
6. **Differential diagnosis:** CSF should be differentiated from following infections;
   a. African swine fever (indistinguishable clinico- pathologically. It is essential to send samples for laboratory examination)
   b. Infection with bovine viral diarrhoea virus
   c. Salmonellosis
   d. Erysipelas
   e. Acute pasteurellosis
   f. Other viral encephalomyelitis
   g. Streptococcosis
   h. Leptospirosis
   i. Salt poisoning

Treatment: Not effective. Affected pigs must be slaughtered and the carcasses buried or incinerated

Control:

Vaccination: Modified live vaccines (MLV) are used to control CSF. The lapinized Chinese (C) strain, the Japanese guinea pig cell culture-adapted strain, and the French Thiverval strain have been widely used. All three strains are considered safe for pregnant sows and piglets over 2 weeks old.

Eradication: Eradication is possible with strict vaccination of pigs, garbage cooking laws and serological surveys targeted primarily to breeding sows to detect subclinical infections. In an outbreak, slaughter of all pigs in affected farms, proper disposal of carcasses, bedding, etc., thorough disinfection, designation of infected zone, with control of pig movements, detailed epidemiological investigation, with tracing of possible sources (up-stream) and possible spread (down-stream) of infection and surveillance of infected zone, and surrounding area will prevent the further spread. Following good husbandry practices in the farm will also prevent the spread of infection.

BOVINE VIRAL DIARRHOEA / MUCOSAL DISEASE

Bovine virus diarrhoea / mucosal disease is observed as variety of clinical syndromes in cattle. This disease complex can cause significant economic losses in susceptible herds. Losses are due to abortions, congenital defects, poor growth, impaired reproductive performance and death due to mucosal disease and acute BVD infection.

General characters:

Classification:

Baltimoregroup	-	**Group IV** – Positive sense RNA viruses
Order	-	Nidovirales
Family	–	Flaviviridae
Genus	-	Pestivirus
Species	-	Bovine viral diarrhoea virus (BVDV)

Morphology: Is typical of other pestivirues. he virions are enveloped; spherical to pleomorphic with 40-60 nm in diameter. The genetic material comprises of a single molecule non-segmented linear positive-sense single-stranded RNA, which is about 12,300 nucleotides long. The BVDV genome encodes for both structural and nonstructural proteins. The structural proteins include the capsid protein C and three glycoproteins Erns, E1, and E2. The capsid protein functions to package the genomic RNA and to provide structure for the formation of the virion envelope. The three glycoproteins are associated with the lipid envelope.

OIE Listing: Listed infection but not under List A or B

Risk group: No infection to human beings

Resistance: As other pestiviruses. They are mildly thermo tolerant. They can survive under wide range of pH. They are sensitive to lipid soluble detergents and common disinfectants

Replication: Replication of BVDV begins with receptor-mediated endocytosis into a cell. The E2 glycoprotein mediates this step. Once inside the cell, the viral RNA is released and RNA translation begins immediately. Viral proteins can be obseved as early as three hours after cell infection. Following gene translation, the large polyprotein product is processed by both cellular and viral enzymes into mature proteins. One of the protein is the RNA-dependent RNA polymerase. New genomic RNA is produced to be packaged into virus packages. Viral packaging occurs in either the Golgi apparatus or endoplasmic reticulum where they acquire their lipid envelope through budding into the vesicle lumen. Mature virus packages are then released from the cell exocytosis. New virus can be released as early as 10 hours post cell infection.

HA property: The virus has no HA property.

Antigenicty and Strains: BVDV is closely related to classical swine fever and ovine Border disease viruses. BVDV occurs in two forms: noncytopathogenic and cytopathogenic. There are two antigenically distinct genotypes (types 1 and 2), and virus isolates within these groups exhibit considerable biological and antigenic diversity. BVD-I (genotype 1) comprises the classic BVD virus isolates including commonly used vaccine strains. BVD-II (genotype 2) comprises newly described BVD viruses strains but have been isolated from as early as 1981. Each genotype contains cytopathic and non-cytopathic biotypes and both genotypes are capable of producing clinically similar syndromes in cattle. However, it is the noncytopathogenic biotype that circulates in cattle populations. The two genotypes may be differentiated from each other, and from other pestiviruses, by monoclonal antibodies (MAbs) directed against the E2 and E^{RNS} major glycoproteins, or by genetic analysis.

Cultivation: The virus is cultivated in a number of bovine monolayer cell cultures (e.g. kidney, lung, testis or turbinate). Both biotypes grow well in cell cultures. Noncytopathogenic BVDV is a common contaminant of fresh bovine tissue. Hence, the cell cultures from bovine origin used for BVDV cultivation should be checked for residual contamination. Now, such problems are overcome by the

use of continuous cell lines, which can be obtained as BVD-free cell lines. The foetal bovine serum used in cell culture should also BVDV and antibodies against BVDV. The presence of virus in the cell culture is confirmed by immunoperoxidase technique, ELISA and nucleic acid identification methods.

Pathogenesis:

1. **Hosts affected:** Cattle is the main host for the infection. The infection is also observed in even-toed ungulates. All ages of cattle are susceptible, but infection usually occurs between 6 and 24 mo of age. Colostral antibody appears to protect most calves for 3-6 mo after birth.
2. **Distribution:** The infection is worldwide in distribution.
3. **Transmission:** The natural reservoir for BVDV is persistently infected cattle. Noncytopathic viruses cause the majority of acute BVDV infections.
   a. **Acute infection:** The virus spreads mainly by contact between cattle. Large numbers of BVDV are shed in the secretions and excretions of persistently infected cattle. Infected animals shed virus in nasal and oral secretions, feces and urine. The primary virus entry route is through oral and nasal. Other less important routes of entry may include infected semen, biting insects, and contaminated instruments. Vertical transmission plays an important role in its epidemiology and pathogenesis. Noncytopathic BVDV is transmitted transplacentally during the first 4 months of foetal development; therefore, infection is present at birth and lasts for life. Clinical disease and reproductive failure are reported in healthy cattle after contact with a persistently infected animal. Following entry and contact with the mucosal lining of the mouth or nose, initial replication occurs in epithelial cells with a predilection for the palatine tonsils. From here, the virus is able to spread systemically through the blood stream. Spread can occur through both free virus in the serum and virus infected leucocytes, particularly lymphocytes and monocytes. During systemic spread, the virus is able to gain entry to most tissues with a preference for lymphoid tissues. However, the tissues infected may vary between different virus strains.
   i. **Intrauterine infection:** BVDV can cross the placenta and infect foetuses of all ages. The outcome of these infections is largely dependent on the stage of gestation when infection occurs. Fetal calves develop immunocompetency at about 180 days ofgestationIf the cow is infected in the first month of pregnancy, the pregnancy is terminated either by abortion or resorption of the foetus by the dam.
   ii. If the foetus is infected in the second to the sixth month of pregnancy, a variety of different syndromes are observed. The foetus may still be aborted or the foetus survives full term with the resultant offspring born malformed, weak, "dwarfed", stillborn, or clinically healthy but "persistently infected" with pestivirus. Persistently infected calves may grow well, but generally are unthrifty compared to others the same age and they often suffer from chronic scours or pneumonia.
   iii. If infection occurs around three to five months of pregnancy, the virus affects the developing nervous system of the foetus. Calves may not be able to stand or suck or may develop convulsions after birth because they are missing parts of their brain. These calves may also have eye abnormalities such as blindness and cataracts or bent up front legs. Some calves will survive but have a wobbly gait and may have a permanent head tremor.
   iv. If exposure to a noncytopathic virus occurs between 42 and 125 days gestation, the fetus accepts the invading virus, becomes immunotolerant to that particular virus and will never mount an immune response to eliminate it. These calves are born persistently infected (PI).
4. **Incubation period:** Highly variable (7-14 days).
5. **Morbidity and mortality:** Variable (High morbidity and low mortality)

6. **Symptoms:** There are several clinical manifestations: BVD, acute and chronic mucosal disease, unthrifty persistently viremic calves, calves with congenital defects, and abortions.
   a. **Bovine viral diarrhoea:** It is most often a subclinical to mild infection in immunocompetent but naïve cattle. Occasionally, in naïve animals that are infected by virulent strains of BVD, death can occur. Biphasic fever (~104° F [40°C]), depression, decreased milk production, and inappetence are typical signs of acute BVD. An increased respiratory rate, diarrhoea, and excessive lacrimation may be seen. Disease lasts for 1-3 days, and is followed by rapid recovery with production of viral neutralizing antibody. There may be some loss of productivity. BVDV can be immunosuppressive and subclinical disease can predispose cattle to other infections such as pneumonia. Some noncytopathic BVDV induce clinically severe disease with a high fever (~107° F [41-42°C]), oral ulcerations, eruptive lesions of the coronary band and interdigital cleft, diarrhoea, dehydration, leukopenia, and thrombocytopenia.
   b. **Mucosal disease:** Acute and chronic mucosal disease are highly fatal forms of BVD seen in persistently infected cattle. These diseases occur when persistently infected cattle become superinfected with cytopathic BVDV. The origin of the cytopathic BVDV is usually internal, resulting from mutation of the persistent noncytopathic BVDV. Clinical signs of chronic mucosal disease may last several weeks to months. This form is characterized by fever, depression, anorexia and excessive salivation followed by profuse, watery diarrhoea 2-4 days later. Affected animals have erosive lesions in their mouths, mainly on the front of the hard palate, the corners of the mouth and the dental pad (gums). Other signs include a mucopurulent discharge from the nostrils, tearing of the eyes, corneal edema and lameness. Affected animals have a low white blood cell count, up to 50% below normal. Affected animals progressively become weaker and dehydrated and die in 5-7 days. Some animals do not die after developing mucosal disease but become chronically ill. They become progressively more emaciated with bouts of diarrhoea and bloat. Hoof deformities and erosive lesions on the skin that do not heal.
   c. **Unthrifty persistently viraemic calves**: These calves are stunted and have very poor growth rates. They do not have detectable mucosal disease lesions and are seronegative to BVDV. The life span of persistently infected cattle often is ≤ 2yr.
   d. **Congenital defects in calves:** These calves have defects relating to lesions in the brain (cerebellum) and the eyes. Signs range from inability to get up and incoordination to blindness.
   e. **Abortions**: Abortions occur in dams exposed during the first trimester of gestation. Early embryonic death can occur if the virus is present at breeding in either the dam or semen.

7. **Lesions:**
   a. **Bovine viral diarrhoe:** In acute infection petechial hemorrhages on conjunctiva, sclera, nictitating membrane and mouth are seen. Swollen lymph nodes, erosions and ulcerations of the GI tract, petechial and ecchymotic hemorrhages on the serosal surfaces of the viscera, and lymphoid depletion are other important lesions. These lesions are not observed in persistently infected cattle.
   b. **Mucosal disease:** Is characterized by erosions and ulcerations throughout the GI tract. The mucosa over Peyer's patches may be hemorrhagic and necrotic. Extensive necrosis of lymphoid tissues, especially those associated with the intestines, is seen on microscopical examination.

Diagnosis:

1. **Field diagnosis:** Is based on history, clinical signs, and gross and microscopic lesions. Diagnostic laboratory support is required when clinical signs and gross lesions are minimal and is essential in some outbreaks of mucosal disease or clinically severe acute bovine viral diarrhoea.
2. **Isolation and identification (*Accepted for International trade)*:**

a. **Clinical materials:** Whole blood, swabs from mucosal or nasal surfaces, semen and lymphnodes

b. **Isolation systems:** Bovine monolayer cell cultures from kidney, lung, testis or turbinate

c. **Antigen identification:**

i. Microplate immunoperoxidase method

ii. Double antibody sandwich ELISA

iii. RT-PCR

3. **Direct identification of BVDV from tissue sections**

a. **Clinical materials:** Tissues from lymph nodes, thyroid gland, skin, brain, abomasum and placenta are ideal for direct identification of BVDV.

b. **Techniques**

i. Immunoperoxidase method using monoclonal antibodies

ii. RT-PCR

iii. in situ hybridisation with enzyme-linked riboprobes

4. **Serology:** To properly assess serology, paired acute and convalescent samples should be collected 30 days apart to identify four fold increases in serum antibody titers following convalescence. Serological techniques that are commonly used are

a. Virus neutralization test

b. ELISA - Both indirect and blocking types of test can be used.

5. **Herd screening:** infection status of a herd are established by methods such as testing bulk milk for anti-BVDV antibody or the testing of the bulk milk sample for BVDV by RT-PCR amplification**.** Other screening methods such as serology on groups of nonvaccinated calves at 6 months of age also are useful.

Treatment: Not effective. Supportive therapy and administration of antiserum may be of benefit in outbreaks of acute disease

Control and eradication:

1. **Vaccination**: The use of killed or modified-live vaccines can provide protection by decreasing the consequences of acute infections. However, it is questionable whether killed or modified-live vaccines provide complete foetal protection from the development of in utero fetal infections. In a breeding herd all breeding females 3-6 weeks prior to breeding should be vaccinated. Both modified-live (MLV) and killed vaccines areavailable commercially, generally as a component in multivalent vaccines. Modified live virus (MLV) vaccines can potentially infect the fetus so should not be given to pregnant cows. The inactivated vaccines are safe but must be given twice, two to four weeks apart.

2. **Eradication**: In herds that are free of BVD, control is aimed at appropriate quarantine to keep replacement stock away from pregnant females by adopting strict biosecurity measures. Control is based on sound management practices, elimination of persistently infected cattle, and vaccination. Cattle should be tested for persistent infection and antibody against BVDV before entry into a herd, as should embryo donor and embryo recipients. Cattle selected as embryo recipients or herd replacements should be vaccinated if they are negative for antibody against BVDV. Cattle that are persistently infected should be sold for slaughter. Bull semen may transmit BVDV and should be tested for virus before use. All breeding bulls should be blood tested and shown to be free of persistent infection because the semen of persistently infected bulls contains high levels of virus.

Public Heath: No human infection is reported.

FAMILY - *TOGAVIRIDAE*

EQUINE ENCEPHALOMYELITIS (EASTERN, WESTERN AND VENEZUELAN)

The equine encephalitides are clinically similar and are characterized by signs of CNS dysfunction and moderate to high mortality. Arboviruses (Arthropod borne viruses) are the most common cause of equine encephalitis. Arboviruses are transmitted by mosquitoes or other hematophagous insects and infect a variety of vertebrate hosts, sometimes including man, and may cause serious disease. Four different species of the genus Alphavirus of the family *Togaviridae* cause equine encephalitides, which include Eastern (*Eastern equine encephalomyelitis) (EEE)* Western (*Western equine encephalomyelitis),* Highlands J and Venezuelan (*Venezuelan equine encephalomyelitis).*

General characters

Classification:

Baltimoregroup - **Group IV** – Positive sense RNA viruses

Order - Nidovirales

Family – Togaviridae

Genus - Alphavirus

Species - Eastern Equine Encephalomyelitis virus (EEEV)

Western Equine Encephalomyelitis virus (WEEV)

Venezuelan Equine Encephalomyelitis virus (VEEV)

Morphology: The virions are enveloped; spherical to pleomorphic with 65-70 nm in diameter. The surface projections are prominent with distinctive spikes. The genetic material comprises of a single molecule non-segmented linear positive-sense single-stranded RNA. The complete genome is 11,700 nucleotides long. The 5'-end of the genome has a cap, and the 3'-terminus has poly (A) tail. The viral RNA resembles cellular mRNA in structure.

OIE Listing: All three infections EEE, WEE and VEE are listed as List B infections.

Risk group: All three viruses can affect human beings and hence classified under Risk Group III.

Replication:

Replication occurs in the cytoplasm and is rapid. The glycoprotein spikes over the envelope of the virus are responsible for binding of the virus to cellular receptors. Upon entry into the cell the virus gets uncoated and (+) sense genomic RNA acts as mRNA and is partially translated (5' end) to produce NS proteins. These NS proteins are responsible for replication, forming a complementary (-) strand (replicative intermediate), which acts as the template for further (+) strand synthesis. Two species of (+) RNA are synthesized, full length genomic RNA and sub-genomic mRNAs. Translation of the newly synthesized sub-genomic RNA results in the production of structural proteins. Assembly occurs at the cell surface, and the envelope is acquired as the virus buds from the cell.

HA Property: These viruses agglutinate RBCs from geese. Male geese are preferred over female.

Virus strains:

Cultivation: Newborn mouse, chicken embryos and cell lines like African green monkey kidney (Vero), rabbit kidney (RK-13) and baby hamster kidney (BHK-21) cell lines are used for cultivation of viruses.

a. In **newborn mice** the virus are injected intracranially. The inoculation site is just lateral to the midline into the midportion of one lateral hemisphere. Mice are observed for 10 days; dead mice are collected daily and frozen at –70°C. Mouse brains are harvested for virus identification.

b. The **chicken embryo** is less sensitive than newborn mice. Tissue suspensions are inoculated by the yolk-sac route into 6–8-day-old embryonated chicken eggs. There are no diagnostic signs or lesions in the embryos infected with these viruses. Inoculated embryos are incubated for 7 days, but deaths usually occur between 2 and 4 days post- inoculation.

c. The most commonly used **cell cultures** are primary chicken or duck embryo fibroblasts, continuous cell lines of African green monkey kidney (Vero), rabbit kidney (RK-13), or baby hamster kidney (BHK-21). Cultures are incubated for 7 days. EEE and WEE viruses produce a cytopathic change in cell culture. The fluid from the thawed cultures is used for virus identification.

Pathogenesis

1. **Hosts affected:** . These viruses cycle between birds and mosquitoes. Horses and humans are dead-end hosts.
2. **Distribution:** Canada, States east of Mississippi river in US, Caribbean Islands, Mexico, and Central and South America.
3. **Transmission:**
   a. **EEE**: The main mode of transmission and amplification of EEE is a mosquito-vertebrate-mosquito cycle. The primary mosquito vector for the EEE is the *Culiseta melanura*. Large numbers of these insects are found in swamp areas. During the late summer and early fall, mosquitoes leave the swamp breeding sites and move to drier, upland forested habitats. *Aedes vexans* and *A canadensis* mosquitoes (which breed in containers) are also believed to be responsible for bird to mammal transmission. The identification of vectors in epidemics is difficult because no single species is consistently associated with the transmission of the virus to horses and people.
   b. **WEE:** WEE is transmitted by mosquito vectors (primarily *C tarsalis*) that breed in sunlit marshes and in pools of irrigation water in pastures and by *Dermacentor andersoni* . Epizootics of WEE are associated with increased rainfall in early spring followed by warmer than normal temperatures.
   c. **VEE:** VEE occurs as an epizootics in horses. VEE virus variants I-D to I-F and subtypes II through VI of VEE virus are associated with a rodent-mosquito transmission, in which human beings and horses are only incidentally get affected. Many species of mosquitoes under the genera *Aedes*, *Anopheles*, *Culex*, *Deinocerites*, *Mansonia*, and *Psorophora* and other hematophagous insects are involved in the explosive outbreaks. Horses are the most important amplifiers of VEE virus during epizootics due to the extremely high viraemias that they develop and the large numbers of hematophagous insects that can feed on an animal of such size. Human beings do not have a significant role in the maintenance and amplification of VEE virus.

      After inoculation by the vector, the virus travels via the lymphatics to lymph nodes and replicates in macrophages and neutrophils, resulting in lymphopenia, leukopenia, and fever. Subsequent replication occurs in other organs and is associated with viremia
4. **Incubation period:** The incubation period from the inoculation of the virus until the febrile response generally is 0.5 to 2 days but may be as long as 5 days, depending on the virus strain or quantity of virus in the inoculum.
5. **Morbidity and mortality:**
   a. **EEE –** Both are highly variable and mortality is between 50-90%
   b. **WEE –** Morbidity highly variable and mortality upto 30%
   c. **VEE -** Morbidity rates vary from 50 to 100 percent and mortality rates vary from 50 to 90 percent
6. **Symptoms:**
   a. **EEE & WEE:** The clinical signs of EEE and WEE can be identical. Following an incubation period of 5–14 days, clinical signs include fever, anorexia, and depression. In severe cases, the disease in

horses progresses to hyperexcitability, blindness, ataxia, severe mental depression, recumbency, convulsions, and death. Horses infected with WEE do not have a significant viraemia, whereas significant viraemia is characteristic for EEE.

VEE: VEE virus infection may be expressed as subclinical with no signs; (b) moderate and characterized primarily by anorexia, high fever, and depression; (c) severe, but nonfatal, and characterized by anorexia, high fever, stupor, weakness, staggering, blindness, and, occasionally, permanent neurologic sequelae; or (d) fatal, with the same clinical signs. In general, two forms of the disease exist: (a) the fulminating form in which signs of generalized, acute, febrile disease predominate, and (b) the encephalitic form in which the more impressive signs of central nervous system (CNS) involvement usually dominate. An incubation period of 0.5 to 5 days precedes a rise in body temperature to 39-41°C (103-105°F), which is accompanied by a hard, rapid pulse and depression. The onset of VEE virus infection is characterized with fever, inappetence, and mild excitability, which progresses to depression, weakness, and ataxia followed by signs of encephalitis such as muscle spasms, chewing movements, incoordination, and convulsions. Early encephalitic signs include loss of both cutaneous neck reflexes and visual responsiveness; diarrhoea and colic may also develop. Some animals may stand quietly in their surroundings whereas others may wander aimlessly or press their heads against solid objects. A braced stance or circling may occur late in the disease. A characteristic paddling motion of the limbs may be observed with lateral recumbency. The course of the disease may be rapid with death ensuing within hours after the observation of the first clinical manifestations of encephalitis.

7. **Lesions:**

❖. **EEE & WEE:** The extent of the lesions depends on the severity of the infection and the duration of the neurological involvement**.** Gross pathological lesions are rarely observed in horses and, if present, consist only of the congestion of the brain and meninges. Ecchymotic haemorrhages of traumatic origin may also be observed. Microscopic lesions are usually found throughout the central nervous system, which include inflammatory response involving the grey matter, neuronal degeneration with infiltration by polymorphonuclear leukocytes, diffuse and focal gliosis, perivascular cuffing with lymphocytes and neutrophils, neuronophagia and liquefaction of the neuropil.

❖. **VEE:** The macroscopic appearance of the CNS of horses with VEE virus varies from no visible lesion to extensive necrosis and hemorrhages. Lesions reported in other tissues have been too variable to be of any diagnostic significance.

Diagnosis

1. **Field diagnosis:** Very difficult. Is possible only when the infection occurs as an epidemic. Seasonality of the disease and association with large populations of mosquitoes help in field diagnosis. The signs of EEE, WEE and VEE are difficult to differentiate. Histopathologic lesions from tissues of dead animals will also help in field diagnosis

2. **Isolation and identification:**

   a. **Clinical materials:** Specimens for diagnosis are heparinized blood, serum (paired [acute and convalescent] sera if animal survives), and half the brain and piece of pancreas unfixed and a completed set of tissues in 10 percent formalin.

   b. **Isolation systems:** Newborn mice, embryonated eggs and cell lines like Vero, BHK21, RK13.

   c. **Antigen identification:**

   i. Complement fixation test

   ii. RT-PCR

iii. Antigen capture ELISA

iv. Immunofluorescence

3. **Identification of viral nucleic acid in mosquitoes by RT-PCR**

4. **Serological tests:** Serological confirmation of EEE, WEE and VEE virus infection is confirmed by a four-fold or greater increase or decrease in antibody titre in paired serum samples collected 10–14 days apart. The common serological tests performed are as below;

   a. Complement fixation test

   b. Haemagglutination inhibition test using male geese RBCs

   c. ELISA

   d. Plaque reduction neutralization test (PRN test) - The PRN test is very specific and can be used to differentiate between EEE and WEE virus infections. The PRN test is performed in duck embryo fibroblast, Vero, or BHK-21 cell cultures.

Treatment: Not very effective and only supportive treatment is given

Control:

1. **Vaccines: F**ormalin-inactivated viral vaccines for EEE, WEE, andVEE.Vaccines in mono-, bi-, or trivalent forms are available. The viral strain of VEE in vaccines is TC-83, which was originally developed as a modified live inoculation to protect laboratory workers investigating VEE. The vaccination protocol consists of two injections 30 days apart, followed by an annual or biannual booster dependent on the geographic location ofthe horses. An attenuated VEE virus vaccine is also used in certain parts of Americas.

2. **Eradication:** Eradication is aimed at restriction of the movement of infected horses and insect control programmes coupled with large scale mass immunization programmes in horses.

   a. Restriction of horse movement between the epizootic zone and non-infected areas is important to control the spread.

   b. Mosquito control measures such asaerial spraying with ultralow volumes of insecticides

Public health: EEE, WEE and VEE can cause human infections. EEE virus causes severe disease in humans with a mortality rate of 30–70% and a high frequency of permanent sequelae in patients who survive. WEE is usually mild in adult humans, but can be a severe disease in children. Human infections occur from bites of infected mosquitoes. Transmission can also occur by exposure to aerosolized infective material. In human beings, a flu-like syndrome predominates accompanied by high fever and frontal headache. Human deaths may occur in the young or the aged.

FAMILY - *PICORNAVIRIDAE*
FOOT AND MOUTH DISEASE

Members of the family *Picornaviridae* are very small. The work pico in Greek means small. This family contains important viruses like foot and mouth disease virus (first animal viruses to be known), poliomyelitis, duck viral hepatitis etc. Foot and mouth disease (FMD) is the most contagious disease of mammals and cause severe economic loss in susceptible cloven-hoofed animals (cattle, pigs, sheep, goats, and water buffalo). The disease is characterized by the formation of vesicles (fluid-filled blisters) and erosions in the mouth, nose, teats and feet. The infection is not very lethal in adult animals. But it causes serious production losses and is considered as a major problem in international trade.

General characters

Classification:

Baltimoregroup - **Group IV** – Positive sense RNA viruses

Order - Nidovirales

Family – Picornaviridae

Genus - Aphthovirus

Species - Foot and mouth disease virus Type species - MDV O strain – FMDV O

Synonyms: Afta epizootica, Bek-en-klouseer, Fiebra aftosa, Fievre aphteuse, Maul-und-Klauenseuche, Komari (vernacular), Kal voi noi (vernacular)

Morphology: The virions are non-enveloped and hence are not ether sensitive. The virion has icosahedral symmetry. The virion comprises of four polypeptides, VP1, 2,3and 4 and all these proteins are derived from cleavage of a single polypeptide VP0. The virion contains one molecule of linear positive- sense single stranded RNA. The Genomic RNA is infectious. Both ends of the genome are modified, the 5' end by a covalently attached small, basic protein VPg and the 3' end by polyadenylation. The genome contains approximately 7000-8500 nucleotides. 2,500nm and it is tightly packed into the capsid.

OIE Listing: List A infection.

Risk group: The virus can affect human beings. Since the infection is uncommon, FMD is not considered as a public health problem.

Resistance: Under *in vitro* conditions virions relatively stable. The FMD virus is pH sensitive and is inactivated at pH below 6.5 or above 11. In milk and milk products, the virion can survive at 70° C for 15 seconds and pH 4.6. Thevirusincellculturemediumwillremainviableforayearat4^{o}C. The virus in serum or other organic material will survive drying and can be carried on inanimate objects. In meat, the virus can survive for long periods in chilled or frozen bone marrow and lymph nodes. The virus inactivated by sodium hydroxide (2%), sodium carbonate (4%), and citric acid (0.2%) and is Resistant to iodophores, quaternary ammonium compounds, hypoclorite and phenol, especially in the presence of organic matter.

Replication: Replication occurs in the cytoplasm and is rapid. The replication cycle occurs very fast and is completed between 5-10 hours. Protein synthesis in the virus infected cell declines sharply almost to nil level and this is called **Shut off**. Since the genome is positive sense, it is immediately translated into a polyprotein, which is cleaved into enzymes required for genome replication and into structural protein by protease enzymes. One such product is the RNA-dependent RNA polymerase enzyme, which copies the genomic RNA to produce a (-) sense strand. This (-) sense strand replicative intermediate and from which (+) sense RNAs produced. These (+) sense RNAs are packed into the capsid. RNA is packed into preformed capsids. Hence, empty capsids (defective) are common in all Picornavirus infections. Release (in most cases) of the virus from the cytoplasm occurs when the cell lyses. This is a 'preprogrammed' event, which occurs a set time after the cessation of 'housekeeping' macromolecular synthesis at shutoff. The sequence of events that take place in a cell infected with FMDV is as below

Time after Infection:	Event:
1-2h	Sharp decrease in host cell macromolecular synthesis
2.5-3h	Start of viral protein synthesis (*Positive sense RNA hence translated immediately)*

3-4h	Permeabilization of plasma membrane
4-6h	Virus assembly in cytoplasm
6-10h	Lysis of cell and release of virus particles

HA Property: The virus has not HA property.

Virus strains: There are seven serotypes of FMDV: A, O, C, Asia 1, and Southern African Territories (SAT) 1, 2 and 3. Within these serotypes, over 60 subtypes have been described, and new subtypes occasionally arise spontaneously. The distribution of serotypes and subtypes globally is given below

Geographic location	Serotypes and subtypes
Europe (historically)	A (5), O (1), C (1)
Asia	
Near East	A (22), O (1)
Middle East	A (22), O (1), C, Asia (1)
Far East	A, O (1), C, Asia (1),
Africa	
Central East to West	A, O
Northeast Central and South	SAT 1, and 2
South	SAT 3
	Serotype C is uncommon in Africa
South America	A (24), A (27), O (1), C (3)

Cultivation:

1. **Cell cultures:** The virus grows well in cell cultures. The cell cultures that are commonly used are primary bovine thyroid cells, primary pig, calf or lamb kidney cells and cell lines, such as BHK-21 (baby hamster kidney) and IB-RS-2 cells. Cytopathic effects are produced in 24-48 hours.
2. **Laboratory animals:** Unweaned mice are used for cultivation FMDV. Mice of 2-7 days old are commonly used

Pathogenesis:

1. **Hosts affected:** Cloven-footed domestic and wild animals are primarily affected. Bovines (cattle, zebus, domestic buffaloes, yaks), sheep, goats, swine and all wild ruminants are severely affected. Members of the family Camelidae have low susceptibility. Other susceptible species are hedgehogs, armadillos, nutrias, elephants, capybaras, rats, and mice.
2. **Distribution:** Foot-and-mouth disease is widely distributed throughout the world. Endemic areas were Asia, Africa, and parts of South America. Most European countries have are generally free from FMD barring few outbreaks. Countries belonging to the European Union have stopped FMD vaccination. North and Central America, Australia, New Zealand, Japan, and the British Isles are free from FMD.
3. **Transmission:** FMDV spread by following methods
   a. Direct or indirect contact with infected animals.
   b. Through aerosol from infected animals
   c. Feeding contaminated garbage (meat, milk, blood, glands, bones, cheese, etc.)
   d. Contact with contaminated objects (hands, footwear, clothing).
   e. Artificial insemination.

f. Contaminated biological such as hormones

g. Through animate objects (*A person in contact with infected animals can have sufficient FMDV in his or her respiratory tract for 24 hours to serve as a source of infection for susceptible animals*)

h. As airborne, especially in temperate zones for a distance of (up to 60 km over land and 300 km by sea

In an outbreak of FMD, the roles of the three primary hosts in transmission are as follows:

- Sheep act as maintenance hosts,
- Pigs act as amplifiers,
- Cattle act as indicators.

When sheep or goats become infected with FMDV, the disease may not be diagnosed for a long time because signs and lesions are very mild. But during this time, the animals will be producing infectious aerosols, contaminating fomites, and spreading the virus by contact. Foot-and- mouth disease in pigs spreads very rapidly, for they produce 30 to 100 times more virus in aerosols than sheep or cattle. (*An infected pig can produce a hundred million infectious doses per day*). When cattle are infected with FMDV, signs and lesions usually develop more rapidly and are more severe than in pigs, sheep, or goats. If cattle, sheep, and pigs are exposed together, cattle will usually get sick first. Some animals particulary recovered cattle can be carriers of FMDV. Most ruminant species harbour the virus in their pharyngeal tissues for a long period.

4. **Incubation period:** In experimentally infected cattle the incubation period is 24-48 hours. In other conditions the incubation period is 2-14 days. In pigs the incubation period is as short as 1-3 days.

5. **Morbidity and mortality:** The morbidity is 100 percent and mortality is usually less than 1 percent, but in young animals and with certain isolates mortality can be high.

6. **Symptoms:**

   a. **Cattle:** Initial signs are pyrexia (39.4-40.6^{o} C), dullness, anorexia, and fall in milk production. These signs are followed by excessive salivation; drooling, serous nasal discharge; shaking, kicking of the feet or lameness; and vesicle (blister) formation. Sites of predilection for vesicles are the tongue, dental pad, gums, soft palate, nostrils, muzzle, interdigital space, coronary band, and teats. After vesicle formation, drooling may be more marked, and nasal discharge, lameness or both may increase. Pregnant cows may abort, and young calves may die without developing any vesicle. The course of an FMD infection is 2 to 3 weeks. Secondary infection may delay recovery. A lactating animal may not recover to preinfection production because of damage to the secretory tissue. Complications include tongue erosions, superinfection of lesions, hoof deformation, mastitis and permanent impairment of milk production, myocarditis, abortion, death of young animals, permanent loss of weight, loss of heat control ('such animals are called as panters')

   b. **Swine:** Initial signs are fever (40-40.6^{o} C), anorexia, reluctance to move, and scream when forced to move. These signs are followed by vesicles on the coronary band, heals, interdigital space and on the snout. Mouth lesions are not too common and when they occur are smaller and of shorter duration than in cattle and tend to be a "dry"-type lesion. There is no drooling. Sows may abort. Piglets may die without showing any clinical sign.

   c. **Sheep and Goats:** Clinical signs, if they occur, tend to be very mild, and may include dullness; fever; and small vesicles or erosions on the dental pad, lips, gums, and tongue. Mild lameness may be the only sign. In lame animals there may be vesicles or erosion on the coronary band or in the interdigital space. Infected animals may abort. Nursing lambs may die without showing any clinical sign.

7. **Lesions:** The main lesions are the vesicles in buccal and foot area. Another important lesion is the tiger heart (*yellowish streaking due to degeneration andnecrosis)*

   a. **Cattle:** The diagnostic lesions are single or multiple vesicles ranging from 2 mm to 10 cm. These can occur at all sites of predilection.

Gross lesions on the tongue: The lesions usually progress in the following manner:

- A small blanched whitish area develops in the epithelium.
- Fluid fills the area, and a vesicle (blister) is formed.
- Vesicle enlarges and may coalesce with adjacent ones.
- Vesicle ruptures.
- Vesicular covering sloughs leaving an eroded (red) area
- Gray fibrinous coating forms over the eroded area.
- Coating becomes yellow, brown or green.
- Epithelium is restored, but line of demarcation remains; line then gradually fades.

Occasionally "dry" FMD lesions develop. Instead of forming a vesicle, the fluid is apparently lost as it forms and the upper layers of the epithelium become necrotic and discolored. The lesion therefore appears necrotic rather than vesicular.

Gross Lesions on the Feet:The vesicle in the interdigital space is usually large because of the stress on the epithelium caused by movement and weight. The lesion at the coronary band at first appears blanched; then there is separation of the skin and horn. When healing occurs, new horn is formed, but a line resulting from the coronitis is seen on the wall of the hoof.

Gross Cardiac and Skeletal Lesions:Animals that die have grayish or yellowish streaking in themyocardium - degeneration and necrosis. These findings are known as "tiger heart".

   b. **Swine:** Vesicles on the snout can be large and filled with clear or bloody fluid. Mouth lesions are usually the "dry" type and appear as necrotic epithelium. Feet lesions are usually severe, and the hoof can become detached. Animals that die may have grayish or yellowish streaking in the myocardium with degeneration and necrosis ("tiger heart").

   c. **Sheep:** Lesions in the mouth and vesicles on the coronary band may be few, small, and difficult to find. Animals that die may have grayish or yellowish streaking in the myocardium with degeneration and necrosis ("tiger heart").

Diagnosis

1. **Field diagnosis:** Field diagnosis is based on characteristic symptoms in cattle, sheep and pigs. FMD should be considered whenever salivation and lameness occur simultaneously and a vesicular lesion is seen or suspected. In pigs, sheep, and goats, FMD should be considered when animals have sore feet, vesicular lesion is suspected, or both.
2. **Isolation and identification:** The following are preferred clinical materials.
   a. Vesicular fluid
   b. Epithelium covering a vesicle
   c. Flaps of epithelial tissue still attached
   d. 5 ml of blood with anticoagulant during viraemic phase
   e. Oesophageal—pharyngeal (OP) fluid from convalescent cattle, sheep, or goats.
   f. Serum

g. From dead animals - samples of epithelial lesions, lymphnodes, thyroid, adrenal gland, kidney, and heart

Specimens placed in ice should be delivered to a laboratory within 24 hours. When delivery may take longer, samples may be quick-frozen and should not thaw during transit.

Isolation should be attempted in primary bovine thyroid cells, primary pig, calf or lamb kidney cells and cell lines, such as BHK-21 (baby hamster kidney) and IB-RS-2 cells. The CPE appears after 24-48 hours and the presence of virus should be confirmed by ELISA or PCR

3. Immunological methods of identification of virus

a. ELISA: The preferred procedure for the detection of FMD viral antigen and identification of viral serotype is the ELISA as per the World Referral Laboratory for FMD. It is an indirect sandwich test in which different rows in multiwell plates are coated with rabbit antisera to each of the seven serotypes of FMD virus. These are the 'capture' sera. Test sample suspensions are added to each of the rows, and appropriate controls are also included. Guinea-pig antisera to each of the serotypes of FMD virus are added next, followed by rabbit anti- guinea-pig serum conjugated to an enzyme. Extensive washing is carried out between each stage to remove unbound reagents. A colour reaction on the addition of enzyme substrate indicates a positive reaction. The ELISA is preferable to the complement fixation (CF) test because it is more sensitive and it is not affected by pro- or anti-complementary factors.

b. Complement fixation test (CFT): Antisera to each of the seven types of FMD virus are diluted in veronal buffer diluent (VBD) in 1.5-fold dilution steps from an initial 1/16 dilution to leave 25 µl of successive antiserum dilutions in U-shaped wells across a microtitre plate. To these are added 50 µl of 3 units of complement, followed by 25 µl of test sample suspension(s). The test system is incubated at 37°C for 1 hour prior to the addition of 25 µl of 1.4% standardised sheep red blood cells (SRBC) in VBD sensitised with 5 units of rabbit anti-SRBC. The reagents are incubated at 37°C for a further 30 minutes and the plates are subsequently centrifuged and read. Appropriate controls for the test suspension(s), antisera, cells and complement are included. CF titres are expressed as the reciprocal of the serum dilution producing 50% haemolysis. A CF titregreaterthanorequalto36isconsideredtobeapositivereaction. Titre values of 24 should be confirmed by retesting an antigen that has been amplified through tissue culture passage.

4. Nucleic acid recognition methods:

a. PCR: The polymerase chain reaction is commonly performed to identify virus in diagnostic materials. Specific primers have been designed to distinguish between each of the seven serotypes.

***b. in situ* hybridisation:** This technique is used to identify viral RNA in tissue samples.

5. Serological tests: FMD virus infection can be diagnosed by the detection of a specific antibody response. The tests generally used are virus neutralisation (VN) and ELISA. These are also the prescribed tests for international trade.

a. Virus neutralization test: The VN test is serotype specific, requires cell culture facilities and takes 2–3 days to provide results.

b. ELISA: ELISA techniques are quantitative, serotype specific, sensitive and have the advantage of being quicker to perform, less variable, and are not dependent on tissue culture systems. The following types of ELISA are performed.

i. Solid-phase competitive enzyme - linked immunosorbent assay (Test for International trade)

ii. Liquid-phase blocking enzyme-linked immunosorbent assay

iii. Indirect ELISA using recombinant non-structural proteins

iv. Enzyme linked immunosorbent blot assay

6. **Differential diagnosis:** FMD should be differentiated from vesicular stomatitis, swine vesicular disease, vesicular exanthema of swine, foot rot, and chemical and thermal burns.

   a. In cattle, oral lesions caused by rinderpest, infectious bovine rhinopneumonitis, bovine virus diarrhea, malignant catarrhal fever, and bluetongue can be similar to the later lesions in FMD.

   b. In sheep, lesions caused by bluetongue, contagious ecthyma, and lip and leg ulceration can be similar to the later lesions of FMD.

Treatment: There is no effective treatment and only supportive treatment is given.

Control and eradication:

1. **Vaccines:** Different vaccines have been used to control FMD for a long time.

   a. Frenkel method of vaccine production is a preferred method. Normal tongue epithelium was removed, minced, placed in a nutrient broth, and inoculated with FMDV. After replication of FMDV, the virus was inactivated with formalin, and aluminum hydroxide was added as an adjuvant.

   b. At present instead of tongue epithelium BHK21 cell line is used. These vaccines are prepared using binary-ethyleneimine (BEI) as inactivating agent and aluminum hydroxide-saponin or oil as an adjuvant. According to prevalence of serotypes in a geographical area vaccine containing corresponding serotype(s) is used. Aluminum hydroxide vaccine offers immunity for duration of 4-6 months after two initial vaccinations, 1-month apart and double oil emulsion vaccine offers protection for 1 year.

   c. Molecular-engineered vaccines are not as effective or as economical as the cell culture vaccines

   d. When vaccinating animals, it is important that the vaccine contain the same subtype of virus as is in the area. This necessitates frequent checking of the serotype and subtype during an outbreak because FMD virus frequently changes during natural passage through various species

   e. Further vaccinated animals that are not completely protected can be a source of infection. The virus may replicate and be shed, but the animals may not show any clinical sign of infection.

2. To eradicate the disease a stamping out policy can be best applied. This involves quarantine, movement restrictions and slaughter and disposal of all affected an in-contact livestock on affected premises followed by cleaning and disinfection.

3. The following are the essential features of a control and eradication program:

   a. Prevention of movement of animals and animal products in the area affected.

   b. Slaughter of infected animals and known contact animals

   c. Destroy carcasses

   d. Disinfect vehicles leaving the infected area.

   e. Perform vaccination.

3. Global eradication programmes have also been initiated by the FAO to control FMD under EMPRES. The following two programmes are aimed at controlling FMD at continental Europe and South East Asia respectively

   a. EUFMD – European Union FMD control programme

   b. SEAFMD – South East Asia FMD control progrmme

DUCK VIRAL HEPATITIS

Duck hepatitis is caused by at least three different viruses, namely duck hepatitis virus (DHV) types I, II and III. The most common is DHV type I is an enterovirus. DHV type II is an astrovirus, and DHV type III is considered to be a picornavirus.

1. The more prevalent and internationally mentioned duck viral hepatitis is the duck hepatitis virus (DHV) type I, which is an enterovirus that causes a highly lethal, acute, contagious infection in ducklings upto 6 weeks of age. This infection is not seen in older birds.
2. DHV type II is caused by an astrovirus. The infection occurs in ducklings from 10 days to 6 weeks of age, and caused pathological changes similar to those of DHV type I.
3. DHV type III is reported only in the United States of America. It causes similar liver lesions in young ducklings, but is less virulent than DHV type I. It is believed to be a picornavirus, serologically unrelated to type I virus. The viruses that cause hepatitis in ducklings should not be confused with duck hepatitis B virus, a hepadnavirus infection of older ducks.

General characters:

Classification:

For DHV I

Baltimoregroup	-	**Group IV** – Positive sense RNA viruses
Order	-	Nidovirales
Family	–	Picornaviridae
Genus	-	Enterovirus
Species	-	Duck viral hepatitis virusI

For DHV II

Baltimoregroup	-	**Group IV** – Positive sense RNA viruses
Order	-	Nidovirales
Family	–	Astroviridae
Genus	-	Avastrovirus
Species	-	Duck viral hepatitis virusII

Morphology

DHV I (Enterovirus of *Pircornaviridae*) – The virions not enveloped and hence are not ether sensitive. The capsids are 28-30 nm in diameter and the symmetry is icosahedral. Nucleocapsids appear to be round. Incomplete virus particles (empty capsids) are often observed. The virions contain one molecule of linear positive-sense single stranded RNA. Total genome contains 7400 nt. The 5' end of the genome has a protein (VPg). 3' end has a poly (A) tract.

DHV II (Astrovirus) – The virions are not enveloped and hence are not ether sensitive. The capsid is roundand exhibits polyhedral symmetry. The capsid surface structure reveals a regular pattern with distinctive features (star-like with six points). The surface projections appear as small spikes protruding from the 12 vertices. The genome is not segmented and consists of a single molecule of linear positive-sense single- stranded RNA. The complete genome is about 6800-7900 nucleotides long. There is no poly A tail.

OIE Listing: List B infection

Risk group: No known infection to human beings

Resistance: For DHV I as described for FMD

Replication: For DHV I and DHV III as described for FMD. The full details of the replication cycle for Astroviruses are not known, but it is suspected that replication occurs primarily in the cytoplasm, with a possible nuclear step

HA property: The virus has no HA property

Serotypes / strains: Under DHV Type I a serological variant named DHV Type Ia has been identified

Cultivation: DHV I grow well in ducklings, embryonated eggs and in cell cultures.

1. Ducklings: Ducklings between 1 and 7 days of age are highly susceptible to DHV type I. The suspected materials are injected subcutaneously or intramuscularly. The characteristic clinical signs are followed by death of ducklings within 18–48 hours of inoculation, often in under 24 hours.
2. Duck eggs (10–14 days) and chicken eggs (8–10 days) are used for culturing DHV I. The route of inoculation is allantoic sac route. Duck embryos die between 24 and 72 hours later, whereas chicken embryos are more variable and die between 5 and 8 days. Gross pathological changes in the embryos include stunting and subcutaneous haemorrhages over the whole body, with oedema particularly of the abdominal and hind limb regions. The embryo livers may be red and yellowish, swollen and may show some necrotic foci. In embryos that take longer to die, the greenish colour of the allantois is more pronounced, and both the liver lesions and stunting become more prominent.
3. Primary cell cultures like duck embryo liver (DEL) cells are commonly used. DHV type I cause a cytopathic effect (CPE), which is characterized by cell rounding and necrosis. When overlaid with a maintenance medium containing 1% agarose (w/v), the CPE gives rise to plaques approximately 1 mm in diameter.
4. DHV II (Astrovirus) are difficult to cultivate under laboratory conditions.
5. DHV III (Pirconavirus) can be cultivated in the laboratory. The virus will not grow in embryonated chicken eggs

Pathogenesis:

1. **Hosts affected:** Ducks and gooses (young birds) are more commonly affected. The infection is seen in ducklings.
2. **Distribution:** DHV I is present in Northern America, Europe and Asia, DHV II is restricted to the United Kingdom and III is restricted to the United States of America. It has not been reported in Pacific region.
3. **Transmission:** The disease is very contagious and the virus excreted by faeces is transmitted by direct contact between birds or through fomites such as brooders, waterer, feeder and such other equipment. Recovered birds can shed the virus for up to 8 weeks. DHV can also be through import live ducks from an infected country. Introduction of virus into susceptible population is also possible through duck meat or duck product. Rats have also been described as a reservoir for DHV viruses.
4. **Incubation period:** Incubation period is 18-48 hours.
5. **Morbidity and mortality:** Morbidity is 100% and mortality is 80-95%.
6. **Symptoms:** Clinical signs include letharginess, loss of balance, spasmodic paddling, anorexia and sudden death with opisthotonos within a few days. At death the head is usually drawn back in the opisthotonos position. The whole disease sequence is rapid and can take as little as 1–2 hours. The clinical course of DHV Type II infection is similar to that of Type I and occurs in ducklings immune to Type I infection. DHV Type III infections occur in ducklings despite immunity to Type I virus. The clinical course of Type III infection is less severe, and mortality is rarely > 30%.

7. **Lesions:**

   a. **DHV I infection:** The liver is enlarged and covered with hemorrhagic foci up to 1 cm in diameter. The spleen may be enlarged and mottled. Kidneys may be swollen with congested blood vessels. Microscopic changes in the liver are characterized by extensive hepatocyte necrosis and bile duct hyperplasia, together with varying degrees of inflammatory cell response and haemorrhage.

   b. **DHV II infection:** Lesions include multiple haemorrhages, both punctate and confluent bands in the liver, swollen pale kidneys with congested blood vessels, and enlarged spleens. The alimentary tract is often empty although the small intestine may contain mucus, and haemorrhagic areas are occasionally seen. Petechial haemorrhages are also occasionally seen on the heart. Histologically, changes in the liver are similar to those seen in DHV type I infections;

   c. **DHV III infection:** The gross pathology is also similar to type I infection. The liver surface is pale and mottled with many red bands and some petechial haemorrhages. The spleen is paler, but not noticeably enlarged, and the kidneys may show patchy ongestion

Diagnosis:

1. **Field diagnosis:** Field diagnosis is based on symptoms and lesions. Sudden onset, rapid spread, and typical short course, together with characteristic liver lesions are highly suggestive of duck viral hepatitis

2. **Isolation and identification**

   a. **Clinical materials:** Liver is ideal clinical material for isolation of virus.

   b. **Systems for isolation:** Ducklings, Duck or chicken embryonated eggs and primary cell culture like Duck embryo liver

   c. **Changes / CPE / Lesions:** *As mentioned under cultivation.*

3. **Immunological tests:** The virus can be identified by neutralization with specific antisera or by inoculation into both susceptible and immune ducklings. Types II and III viruses are not neutralized by classic Type I antiserum.

4. **Serological tests:** These tests are not applicable since the infection sets in rapidly and affected birds die within short time. However, virus neutralization assays have been developed.

5. **Differential diagnosis:** The infection has to be differentiated from Duck virus enteritis (*caused by Herpes virus), c*occidiosis, mycotoxicosis and New duck disease (*caused by Riemerella anatipestifer)*.

Treatment: Not effective. Antibody against Type I virus, prepared from the eggs of hyperimmunized chickens, administered SC in the neck, at the time of initial loss, is an effective flock treatment.

Control and eradication:

1. **Vaccines:**

   a. Immunization of breeder ducks with modified live virus vaccines, using Types I, II, and III viruses, provides parenteral immunity that effectively prevents high losses in young ducklings. The Type I virus vaccine is administered SC in the neck to breeder ducks at 16, 20, and 24 wk of age and every 12 wk thereafter throughout the laying period. The three immunizations are advisable for passive protection of ducklings.

   b. Inactivated vaccines against DHV I can also be used in primed ducks, which give high antibody titre

   c. The chick-embryo origin, modified live Type I virus vaccine also can be used for early vaccination of ducklings susceptible to Type I (progeny of non-immune breeders). This vaccine is administered SC or by foot web stab in a single dose to day-old ducklings. These ducklings rapidly develop an active immunity over 3-4 days.

2. **Control:** Prevention is focused on strict isolation, particularly during the first 5 wk of age and during this period contact with wild waterfowl should be avoided. Since rats are reservoir host for DHV, rats should be controlled

REOVIRIDAE
BLUETONGUE

The term Reo refers to **R**espiratory **E**nteric **O**rphan viruses, i.e. infect the human respiratory and intestinal tracts, *usually* without disease symptoms. Bluetongue (BT) is a non-contagious, athropod-borne viral disease of sheep and other domestic and wild ruminants, such as goats, cattle and deer. Camelids are also affected by BT.The infection in animals is transmitted through the bite of *Culicoides* insects. Bluetongue virus (BTV) is endemic in some areas with cattle and wild ruminants serving as reservoirs for the virus.

General characters

Classification

Baltimore group – Group III double stranded RNA viruses

Family – Reoviridae

Genus – Orbivirus

Species – Bluetongue virus - BTV

Morphology: The virus measures approximately 70-85 nm in diameter. The virions not enveloped and hence are not ether sensitive. The capsid shell of virion iscomposedof two layers. The virus has icosahedral symmetry. The nucleocapsids appear to be round without any surface projections or distinct spikes. The outer layer contains two proteins, VP2 and VP5. VP2 is the major neutralising antigen and determinant of serotype specificity. This is followed by a core comprising of VP7, VP3 and three minor proteins and RNA. VP7 is a major determinant of serogroup specificity. VP7 can also mediate attachment of BTV to insect cells. The virions contain 10 segments of double stranded RNA. The total genomic length is 19200 nucelotides and the largest segment L1 has got approximately 2800-3900 nucelotides. The viral RNA has methylated cap structure, but no poly A tail.

Synonyms: Sore muzzle, pseudo foot-and-mouth disease, muzzle disease

OIE listing: List A infection

Risk group: No known human infection is associated with BTV

Resistance: The viruses are resistant to lipid. However, the viruses are relatively acid-labile, and slow freezing at -10 to -20°C (14 - 40^{0}F) is harmful to the virus. The viruses are inactivated by 50°C/3 hours and 60°C/15 minutes. ß-propiolactone, iodophores and phenolic compounds also inactivate the viruses. The viruses are very stable in the presence of protein and can survive for years in blood stored at 20°C.

Replication: The virus replicates in the cytoplasm. The virus enters into the cell through endocytosis. This is followed by partial uncoating. BTV is inactivated if complete uncoating occurs. Early transcription of the d/s RNA genome by viral polymerase occurs inside this sub-viral particle. The various genome segments are transcribed/translated at different frequencies. Transcription occurs in two stages, primary and secondary. Only (-) sense strands are transcribed, resulting in synthesis of (+) sense mRNAs, which are capped inside the core. These mRNAs leave the core and are translated in the cytoplasm. The genome is replicated in the cytoplasm in a conservative fashion. An excess of (+) sense strands are produced, which serve as late mRNAs and as template for (-)sense strand synthesis

(i.e. each (-) strand leads to many (+) strands - not one-for-one as semi-conservative replication). The mechanism responsible for segregation of the various genome segments into developing particles is not known.

HA property: BTV agglutinates mammalian RBCs.

Serotypes / strains: There are 14 serogroups and these serogroups in total contain 24 serotypes. The serogroups are differentiated by immunological tests that identify viral proteins, which are unique for each serogroup. Most serogroups are immunologically distinct, but there is considerable cross-reaction between members of the BT and EHD serogroups. *(EHD - epizootic haemorrhagic disease is a viral infection caused by a orbivirus and has clinical symptoms same as BT)*

Cultivation: The virus grows well in embryonated chicken eggs and cell culture systems.

1. In embryonted chickens the preferred route of inoculation is yolk sac. The eggs should be incubated in a humid chamber at 33.5C. The embryos die between 2 and 7 days. Infected haemorrhages have cherry red appearance due to extensive haemorrhages. The presence of virus can also be confirmed by immunofluorescense, immunoperoxidase or antigen capture ELISA tests.
2. BTV also grow in cell culture systems like baby hamster kidney (BHK)-21, African green monkey kidney (Vero) and *Aedes albopictus* (AA) cells. The cytopathic effects should be confirmed by antigen-capture ELISA, immunofluorescence, immunoperoxidase, or virus neutralisation (VN) tests.

Pathogenesis:

1. **Hosts affected:** The host range of BTV is very broad and includes all ruminants. However, the infection is well pronounced in sheep followed by other domestic and wild ruminants, such as goats, cattle and deer. Cattle act as reservoir for BTV.
2. **Distribution:** The bluetongue virus is present in regions where the *Culicoides* vector is present (e.g. Africa, the Americas, Australia and some countries of southern Asia and Oceania).
3. **Transmission:** BTV infection is transmitted in both wild and domestic ruminants/camelids from the bite of the vector midge of the genus *Culicoides. Culicoides* act as biological vectors for BTV. Experimental studies have also demonstrated that ticks are also capable of mechanically or biologically transmitting BTV. The virus can also be transmitted sexually in infected semen, embryos (*in embryo technology)* and transplacentally from dam to offspring. Mechanical transmission also occurs in BT through use for contaminated needles and equipment contaminated with blood of infected animals.Transmission by *Culicodes* is the most common method in the spread of BT. The *Culicoides* vector infects animals mostly during mid-summer and rainy seasons, when it is most active.
4. **Incubation period:** The incubation period of BT in sheep is usually 7-10 days. However, viraemia can occur within 4 days in sheep and cattle. The incubation period of BTV infection in deer is 7 to 12 days.
5. **Mortality and morbidity:** In sheep morbidity is 100 percent, whereas mortality is between 0 and 50 percent. Many affected animals usually recover within a few days to 2 weeks.
6. **Symptoms:** Bluetongue is clinically manifested as two syndromes: 1) vascular affection of several organ systems particularly oral and buccal mucosa and 2) a reproductive syndrome. Sheep are commonly seen with clinical disease, but other domestic ruminants such as cattle and goats only rarely show clinical signs
   a. **Sheep:** The clinical signs of BT are variable and not all strains of BTV that infect sheep cause clinical disease. The first clinical sign is the rise in temperature to 41.6-41.7° C. Within 24 hours of the initial rise in temperature, excessive salivation and frothing at the mouth develop and are associated with hyperemia and swelling of the buccal and nasal mucosa followed by erosions and

ulcerations. By 4 to 7 days in severe cases, extensive ulcerations may be covered by gray necrotic tissue on the dental pad and dorsal surface of the tongue. Hyperemia is also observed around the coronary bands of the hooves. The hooves are tender and varying degrees of lameness are also observed. In more severe cases, the animals stand with an arched back. Sloughing of hooves is common at this stage. Animals that recover may have a dark line in the wall of the hoof. The lesions in the mouth, the reluctance to move, and the necrosis of striated musculature lead to weakness, depression, and rapid weight loss. These can result in prostration and eventual death in severely affected animals. Sheep that recover from severe infections may have a break in the wool 3 to 4 weeks after the fever has subsided. This can lead to partial or complete shedding of wool. The reproductive form of the disease varies greatly. Signs include abortions, stillbirths, and weak "dummy lamb" live births.

b. Cattle: Bluetongue virus infection in cattle usually does not cause any clinical sign of disease. Clinical signs consist of

i. mild hyperemia in the buccal cavity and around the coronary band;

ii. vesicular lesions, which lead to ulcerations, in the buccal mucosa;

ii. erect hair over the cervical and dorsal thoracic areas; and

iii. hyperesthesia.

iv. In addition, the dermis becomes thickened with prominent folds apparent in the cervical areas, and a dry crusty exudate leads to matting of hair in affected areas. These lesions may persist for 10 to 20 days. Similar lesions have been reported on teats of cattle with clinical BT. Hoof lesions may be associated with lameness.

v. BTV is both abortigenic and teratogenic in cattle experimentally, but not observed commonly in field conditions. Early embryonic loss and decreased reproductive efficiency are the more frequently seen manifestations of the disease in cattle and can severely affect calf/milk production.

c. Goats: Bluetongue infection of goats is an inapparent infection similar to that described forcattle.

7. Lesions: The lesions of BT in sheep vary greatly depending on (1) the strain of virus, (2) individual animal and breed susceptibility and(3) environmental (stress) factors. Prominent lesions include facial edema, edematous ears, and dry, crusty exudate over the nostrils.

a. Oral cavity and digestive system: Lesions in the oral cavity include focal petechial hemorrhages that progress to gray necrotized debris over erosions and ulcerations on the lips; on the dorsal, lateral, and ventral surfaces of the tongue; and on the dental pad. The buccal mucosa may be cyanotic. Hyperemia and occasional erosions can occur on papillae and laminae in the reticulum and omasum.

b. Respiratory System: Lesions include cyanosis and edema of the nasal mucosa and pharynx.

c. Vascular system: Lesions include hyperemia, edema, and hemorrhages. A characteristic lesion is hemorrhage at the base of the pulmonary artery. Petechial and ecchymotic (larger than pinhead-size) hemorrhages may be observed at times in the endocardium. Focal gray-white areas of necrosis are often found in the papillary muscles and less frequently in other areas of the myocardium

d. Skin: Dermal and subcutaneous edema of the head and ears and an irregular rash (exanthematous eruptions) with serous and crusty exudates on the skin are the common lesions.

e. Muscle: A yellow gelatinous exudate is present in the fascia (connective tissue) along and between skeletal muscles. On the cut surface of the heavy muscle there may be focal hemorrhages and areas that appear dry and gray-white.

f. Newborn lambs: Newborn lambs with congenital BT have hydranencephaly or porencephaly. Abnormal development of cerebellum and spinal cord are the other lesions.

Diagnosis:

1. **Field diagnosis:** Field diagnosis is based on appearance of clinical signs in susceptible population, coincidence of occurrence of symptoms with prevalence of insect vectors, characteristic gross lesions, and a flock history of recent wasting (loss of weight) and pododermatitis (foot rot).

2. **Isolation and identification:**

   a. Clinical materials: Sterile heparinized blood samples from animals with clinical signs or spleen or bone marrow, or both, from dead animals are the clinical materials of choice. Samples from aborted and congenitally infected newborn animals should include heparinized blood and, if possible, spleen, lung, brain, and serum. Specimens should be shipped refrigerated, not frozen. Freezing makes virus isolation difficult.

   b. Isolation systems: Chicken embryos and cell cultures like baby hamster kidney (BHK)-21, African green monkey kidney (Vero) and *Aedes albopictus* (AA) cells.

   c. Immunological tests:

   i. Serogrouping: BTV are serogrouped on the basis of their reactivity with specific standard antisera that detect proteins, such as VP7 that are specific within each serogroup**.** Monoclonal antibodies specific for VP7 are used in the assay. The immunological assays that are performed using monoclonal antibodies against VP7 protein are immunofluorescense, antigen capture ELISA and immunospot test (dot ELISA).

   ii. Serotyping: Serotyping of BTV is carried out using serum specific for each of 24 serotypes. Virus neutralization assays are performed to serotype BTV. Four different types neutralization assays are performed in BHK21, Vero and L929 cells. These tests are plaque reduction test, plaque inhibition test, microtitre neutralization and fluorescent inhibition test.

   d. Nucleic acid detection methods: Polymerase chain reaction (PCR) is an accepted test for international trade. Serotyping is also possible in PCR.

3. **Serological tests:** These tests are performed to identify antibodies in the serum. The tests that are performed include

   a. Complement fixation test

   b. Agar gel immunodiffusion test: It is an accepted test for international trade.

   c. Competitive ELISA: It is also an accepted test for international trade.

4. **Differential diagnosis:** Contagious ecthyma, foot and mouth disease, photosensitisation, pneumonia, polyarthritis, footrot, foot abscesses, plant poisonings, *peste des petits ruminants*, coenurosis and epizootic haemorrhagic disease of deer

Treatment: Not very effective and only supportive. The only applicable treatment available is to minimize animal stress and administer broad-spectrum antibiotics to combat secondary infection.

Control and eradication:

1. Control is mainly aimed at vector control. Certain measures that have potential effectiveness against *Culicoides* include water management (reduction of *Culicoides* breeding sites), use of insecticides and larvacides (spraying breeding areas), and insect repellents in which animals are dipped.
2. Vaccination should be practiced only in endaemic areas. In such areas serotyping of available BTV should be carried out followed by incorporation of all serotypes into the vaccine. Because of the multiplicity of BTV serotypes and variable cross-protection between serotypes, vaccination has resulted in varying degrees of success. No inactivated or subunit vaccines are currently available.

AFRICAN HORSE SICKNESS

African horse sickness (AHS) is a non-contagious, highly fatal, viscerotropic, insect-borne viral disease of horses, mules and donkeys. The infection is caused by an orbivirus under the family reoviridae. AHS occurs in *Equidae*s in areas endemic for certain *Culicoides* midges or gnats. The infection is often fatal and is characterized by fever and oedema in the lungs and subcutaneous tissues. Nine different serotypes have been described as on date.

General characters:

Classification:

Baltimore group – Group III double stranded RNA viruses

Family – Reoviridae

Genus – Orbivirus

Species – African horse sickness virus - AHSV

Morphology: The virus measures approximately 55-70nm in diameter. The virions not enveloped and hence are not ether sensitive. The capsid shell of virion is composed of three layers (outer, middle and inner capsids). The virus has icosahedral symmetry. The genome of AHS virus (AHSV) is composed of ten double-stranded RNA segments comprising 18000 nucleotides, which encode seven structural proteins (VP1-7) and four nonstructural proteins (NS1, NS2, NS3, NS3A). Proteins VP2 and VP5 form the outer capsid of the virion, and proteins VP3 and VP7 are the major inner capsid (middle) proteins. Proteins VP1, VP4 and VP6 constitute minor inner capsid proteins. VP2 protein is highly variable and is responsible for serotype variations.

Synonyms: *Peste equina africana, Peste equine, perdesiekte*

OIE listing: List A infection

Risk group: No known human infection is associated with AHSV. However, occasional intranasal exposure to some vaccine strains has been reported.

Resistance: AHSV is comparatively a stable virus. The virus can survive at room temperature (37°C for 37 days) for nearly a month. However the viruses are inactivated by temperatures >50°C for 3 hr, 60°C for 15 min, 70°C for 5 min. The viruses are also stable for up to 20 yr at 4°C; however, inactivated by repeated freezing and thawing. The viruses are stable at basic pH range of 6.3-12.0. The viruses are inactivated byether, Beta-propiolactone, phenol, iodophores, acetic acid, 2% formalin and chlorine dioxide.

Replication: Typical of other orbiviruses under *Reoviridae.* The virus replicates in the cytoplasm. The virus enters into the cell through endocytosis. This is followed by partial uncoating. AHSV is inactivated if complete uncoating occurs. Early transcription of the d/s RNA genome by viral polymerase occurs inside this sub-viral particle. The various genome segments are transcribed/translated at different frequencies. Transcription occurs in two stages, primary and secondary. Only (-) sense strands are transcribed, resulting in synthesis of (+) sense mRNAs, which are capped inside the core. These mRNAs leave the core and are translated in the cytoplasm. The genome is replicated in the cytoplasm in a conservative fashion. An excess of (+) sense strands are produced, which serve as late mRNAs and as template for (-)sense strand synthesis (i.e. each (-) strand leads to many (+) strands - not one-for-one as semi-conservative replication). The mechanism responsible for segregation of the various genome segments into developing particles is not known.

HA property: The virus has HA property

Serotypes / strains: There are 9 serotypes of AHSV and all these serotypes are immunologically distinct. These serotypes are indicated as Serotype 1, 2, 3, 4, 5, 6. 7, 8 and 9. **Infection with serotype 9 has been reported in India.**

Cultivation:

1. **Cell culture:** AHSV can be cultivated in baby hamster kidney-21 (BHK- 21), monkey stable (MS) and African green monkey kidney (Vero).
2. **Embryonated chicken eggs (ECE):** The virus grows in chicken embryos but no lesions or death of embryo take place. The virus can also be cultivated in chicken embryo fibroblast, but no CPE are produced.
3. **Lab animals:** The virus can be cultivated in newborn mice and guinea pigs. In newborn mice the viruses are inoculated through intracerebral route.

Pathogenesis:

1. **Hosts affected:** Members of the family *Equidae* are affected by AHS. The important hosts are horses, mules and donkeys. Several other species in wild population like zebra, elephant and camels harbor the virus without showing clinical disease. Dogs can be infected by eating contaminated meat or animal products, and show severe clinical disease
2. **Distribution:** AHS occurs in the tropical area of central Africa where it is endemic. The disease spreads from there to the Southern Africa and occasionally to Northern Africa. Infection has also been reported in Asia (from Middle East to India) and in Europe (Spain and Portugal). AHS has never occurred in the Pacific region.
3. **Transmission:** AHS is not transmitted directly between horses. The infection is transmitted by via the biting *Culicoides. C imicola, C obsoletus* and *C pulicaris* are the common biological vectors for AHSV. The virus is also transmitted biologically by midges, and these insects are most active just after sunset and at about sunrise. Other insects such as mosquitoes also act as biological vectors, and large biting flies (e.g., *Stomoxys*, *Tabanus*) transmit AHS virus mechanically. Wind-borne spread of midges may assist the short-distance spread of the disease but that long-distance jumps of the infection are invariably the result of movement of infected *Equidae*. The spread of disease is influenced by climatic conditions, which favour the spread of carrier insects (vectors) including warm, moist weather and high rainfall, as well as spread by wind dispersal. In order for AHSV to become infective, it must be taken up in a blood meal from an infected host. After the virus matures within the arthropod vector, it can be introduced into a susceptible host at the following blood meal. Once inside the new host, the virus replicates in the local lymph nodes and disseminated through viraemia.
4. **Morbidity and mortality:** Morbidity depends on the number of infected *Culicoides* insects as well as the duration of exposure. Mortality varies between 70% and 95% horses, 50% in mules and 5-10% in donkeys.
5. **Incubation period:** Incubation period is usually 7-14 days, but may be as short as 2 days.
6. **Symptoms:** AHSV fever is a seasonal, viscerotropic disease characterized by subcutaneous/pleural edema and fever. The virus is manifested in several forms: peracute, subacute, acute, and horse sickness. Often, the associated respiratory and circulatory distress results in severe disease and death. Severity of disease varies among species, with horses being the most affected, followed in order by mules, donkeys, and zebras.

Horses

a. The **per-acute pulmonary form** (*also referred as **dunkob**)* has a short incubation period of 3-5 days with 95 per cent mortality. The course of the infection is 3 days. The infection is characterized by fever (40-41°C) lasting 1-2 days, congested ocular, nasal, and oral mucous membranes,

increased respiratory rate, wide abnormal stance with head and neck extension, flared nostrils, forced expiration resulting in heave lines, profuse sweating, spasmadic cough and appearance of frothy and serofibrinous blood-tinged fluid from nostrils. The death of animals is due to anoxia.

b. The **sub-acute cardiac form** (*also referred as **dikkob**)* has an incubation period of from 7-14 days and with a mortality rate of around 60 per cent. This form of the disease is characterized by fever (39-41°C) lasting 3-6 days, oedema of the supraorbital fossa, oedematous swellings over the head and eyelids, lips, cheeks and under the jaw, petechial hemorrhage of conjunctiva and ventral surface of tongue, depression and appearance of colic signs. Death results from cardiac failure.

c. The **acute or mixed form** is a combination of the previous two types with an incubation period of from 5-7 days and the disease shows itself initially by mild pulmonary symptoms followed by the typical oedernatous swellings of the cardiac form.

d. Horse sickness fever. This is the mildest form, characterized by a febrile (very active and nervous) reaction with low temperatures in the morning rising to a high peak in the afternoon. Other symptoms include slight congestion of conjunctivae, increased pulse rate, mild anorexia and mild depression

Other animals: The clinical disease in mules resembles that in horses. Donkeys and zebras, usually manifest the horse sickness form of the disease and mortality is significantly lower than for horses. Affected dogs exhibit the pulmonary form of the disease

7. Lesions:

a. In **peracute form** the most characteristic changes are edema of the lungs or hydrothorax. In very peracute cases, extensive alveolar edema and mottled hyperemia of the lungs are seen. Other less commonly observed lesions are periaortic and peritracheal edematous infiltration, diffuse or patchy hyperemia of the glandular fundus of the stomach, hyperemia and petechial hemorrhages in the mucosa and serosa of the small and large intestines. Most of the lymph nodes are enlarged and edematous, especially those in the thoracic and abdominal cavities.

b. In **cardiac form** the prominent lesion is a yellow gelatinous infiltration in the subcutaneous and intermuscular fascia primarily of the head, neck, and shoulders. Extensive petechial and ecchymotic hemorrhages on the epicardium and endocardium, particularly of the left ventricle are also observed.

c. In **mixed form** the lesions represent a combination of those found in the pulmonary and cardiac forms3

Diagnosis:

1. Field diagnosis: In early febrile phase field diagnosis is very difficult. It is based on season, prevalence of *Culicoides* insects, movement of horses from affected areas and characteristic symptoms and lesions.

2. Isolation and identification:

a. Clinical materials: Blood collected from early febrile period, spleen, lung, and lymph nodes. Specimens for virus isolation should be shipped to the laboratory refrigerated, NOT FROZEN.

b. Isolation systems: The virus can be isolated in baby hamster kidney-21 (BHK-21), monkey stable (MS), African green monkey kidney (Vero), newborn mice (intracerebral inoculation) and ECE (intravenous route).

c. Lesions / CPE: Not very characteristic and dependant on confirmation of antigen. In mice inoculatin, in positive cases, animals develop nervous signs between 3 and 15 days post-inoculation.

d. **Identification of antigen:** The presence of antigen in isolation systems are confirmed by ELISA, virus neutralization test or by PCR.

3. **Serological tests:** The following serological tests are used to identify antibodies against AHSV. They are

   a. Indirect ELISA (Accepted for international trade) - VP7 protein isn used as antigen for AHSV antibody determination. This test has high degree of sensitivity and specificity.

   b. Complement fixation test (Accepted for international trade)

   c. Immunoblot test: The binding of antibodies to viral proteins separated by electrophoresis and transferred to nitrocellulose paper is used for the determination of anti - AHSV antibodies.

   d. NS3 ELISA: This test distinguish between infected horses and horses vaccinated with the inactivated purified AHSV serotype 4 vaccine.

   e. Haemagglutination inhibition test

   f. Virus neutralization test: Done to serotype AHSV

4. **Differential diagnosis:** Clinical signs associated with AHSV are clearly not pathognomic. Hence, the infection has to be differentiated from

   a. Viral

   i. Equine infectious anemia (Retroviridae)

   ii. Equine viral arteritis (Arteriviridae)

   iii. Equine encephalosis (Reoviridae) (often occurs concurrently with AHSV; differentiable by absence of edema and lower mortality rates)

   b. Bacterial

   i. Anthrax (*Bacillus anthracis*)

   c. Protozoan

   i. Trypanosomiasis (*Trypanosoma evansi*)

   ii. Piroplasmosis (*Babesia caballi and Babesia equi*)

   d. Other

   i. Purpura haemorrhagica (lesions on necropsy tend to be more widely distributed in this condition than in AHS)

Treatment: No specific treatment is available for AHS. Recovered animals demonstrate life-long immunity to that serotype and partial immunity to similar serotypes. Passive immunity from maternal antibodies lasts about 6 months.

Control and eradication:

1. **Vaccination:** Since the serotypes are immunologically distinct monovalent, quadrivalent and polyvalent vaccines are used to control AHSV.The problem with modified live attenuated vaccines are that they do not allow differentiation between vaccinated and infected animals and may aid in persistence of the virus

   a. A live attenuated vaccine was developed by adapting the virus to the brains of adult mice. This vaccine though offers protection, occasionally caused encephalitis in horses, mules and particularly in donkeys.

   b. An alternate and safe vaccine was developed by attenuation of the virus in Vero cell cultures. This vaccine is currently used in South Africa, which consists of two quadrivalent vaccines that are administered 3 weeks apart.

2. **Eradication:** Aimed at following aspects
   a. Vaccination of susceptible animals
   b. Vector control (use of insecticides, repellents and protective screens)
   c. Reduction of exposure to arthropod vector by keeping horses in stables or barns from dusk till dawn, because *Culicoides* species are most active at sunrise and sunset

Public health: There is no evidence that man can become infected with field strains of AHS virus, either through contact with infected animals or from working in laboratories. However, it has been shown that certain neurotropic vaccine strains may cause encephalitis and retinitis in humans following transnasal infection.

BIRNAVIRIDAE
INFECTIOUS BURSAL DISEASE

Infectious bursal disease (IBD, Gumboro disease) is an immuno-suppressive disease of domestic poultry caused by a birnavirus. The disease was discovered in 1962 in Gumboro, Delaware, USA. Hence, it is also known as Gumboro disease. Chickens between two and three weeks of age were affected and showed signs of disease and a transient immunosuppression. However, in the 1980s the disease changed character. And the birds infected within the first week of life, developed bursal atrophy leading to a subclinical immunosuppression with increasing incidence of diseases of the skin, gastrointestinal and respiratory systems. Later a new strain appeared, causing a more rapid bursal atrophy, immunosuppression, diarrhoea and weight loss. Even vaccinated birds were affected and the natural immunity declined faster than after infection with the old type of virus. The mortality in Leghorns reached up to 90% and the morbidity almost 100%. Three different syndromes occur and these are referred to as "classical", "very virulent" (vvIBD), and "variant" infectious bursal disease. This classification is based on pathogenicity.

General characters

Classification:

Baltimore group – Group III double stranded RNA viruses
Family – Birnaviridae (*Birnavirus means bisegmented dsRNA virus)*
Genus – Avibirnavirus
Species – Infectious bursal disease virus – IBDV

Morphology: The virions are non-enveloped and hence are not ether sensitive. The capsid is composed of a single layer and appears hexagonal in outline with icosahedra symmetry. The capsid consists of 132 capsomers and surface projections are not present. The virions measure 60 nm in diameter. Virions contain two segments of linear double stranded RNA. The genome contains 5850-6050 nucleotides. The largest segment is termed as A and is comprised of 3100-3200. The second largest segment is termed as B and is comprised of 2750-2850. The 5' end of the genome has on both segments a genome-linked protein (VPg) and the 3' end has no poly (A) tract. Segment A codes for four proteins VP2, VP3, VP4 and VP5. VP2 and VP3 are capsid proteins and VP4 and VP5 are non-structural proteins. The antigenic determinants are located in the VP2 protein. VP5 is cytotoxic in nature and responsible for lysis of cell during release of virus. Segment B contains only one gene, VP1, which codes for the virus polymerase

Synonyms: Gumboro disease

OIE listing: List B infection

Risk group: No known human infection

Resistance: The virions are stable in acid environment of pH 3-7; stable in alkaline environment of pH 7-9. Virions are not sensitive to treatment with heat (60^0C, 1 hour), or ether (and 1% SDS at 20^0C, pH 7.5 for 20 min.)

Replication: Virus replication occurs in the cytoplasm and does not affect the cells own synthesis of RNA and proteins (*There is no complete shut off)*. The two double stranded DNA-segments, A and B, code for different viral proteins (VP). The replication is characterized by presence of subgenomic RNAs - a mRNA, a transcription of RNA-A and RNA-B. The non-structural protein VP5 has been shown to be important since it causes lysis of the host cells and release of the virus. The protein accumulates in the plasma membrane and causes cell deformation prior to cell lysis.

HA property: The virus has no HA property

Serotypes: There are two distinct serotypes of the virus named as Serotype 1 and Serotype 2, which are both capable of infecting chicken as well as some other species of domestic poultry like turkey and ducks. However, only serotype 1 cause clinical disease in chickens younger than 10 weeks. Serotype 2 does not cause any clinical disease in chickens. Using molecular techniques, serotype 1 has been further subdivided into six distinct groups.

Cultivation:

The virus is easily cultivable in chickens, ECE and chicken embryo fibroblasts.

1. **Chickens:** This method is not recommended due to animal welfare concerns. The virus is inoculated as eyedrop. The virus kills the chickens 72–80 hours after inoculation. The bursae of chickens infected with virulent serotype 1 IBDV appear yellowish (sometimes haemorrhagic) and turgid, with prominent striations. Peribursal oedema is sometimes present, and plugs of caseous material are occasionally found. The plicae are petechiated. The bursae of chickens infected with serotype 2 IBDV do not exhibit any gross lesions.
2. **Embryonated chicken eggs:** The virus is inoculated via yolk sac of five 6–8-day-old specific antibody negative (SAN) chicken embryos and on to the chorioallantoic membrane of five 9–11-day-old SAN chicken embryos. Serotype 1 IBD produces dwarfing of the embryo, subcutaneous oedema, congestion and subcutaneous or intracranial haemorrhages. The liver is usually swollen, with patchy congestion producing a mottled effect. In later deaths, the liver may be swollen and greenish, with areas of necrosis. The spleen is enlarged and the kidneys are swollen and congested, with a mottled effect. Serotype 2 IBDV does not induce subcutaneous oedema or haemorrhages in the infected embryos, but embryos are of a smaller size with a pale yellowish discolouration.
3. **Cell culture:** Chicken embryo fibroblasts are normally used for cultivation: The CPE is characterized by small round refractive cells.

Pathogenesis:

1. **Hosts affected:** The domestic fowl is the natural host. Sub-clinical infection may occur in turkeys. Disease is most common in 3 to 6 weeks old birds, however severe infection occurs in Leghorn up to 18 weeks.
2. **Distribution:** Very virulent IBD have been reported in Europe, Americas and South East Asia.
3. **Transmission:** The main route of infection is through ingestion. Affected birds serve as source of infection. Chickens infected with the IBD virus shed the virus in their feces. Feed, water, and poultry house litter become contaminated with faecal materials and they serve as source of infection. Other

chickens in the house become infected by ingesting the virus. The lesser mealworm (*Alphitobus diaperinus*) has also been shown to carry the virus. Because of the resistant nature of the IBD virus, it is easily transmitted mechanically among the farms by people, equipment and vehicles.

4. **Morbidity and mortality:** Morbidity rates are high (upto 100%). Mortality rates are variable and may go up to 25% in broilers and 60% in layers.

5. **Incubation period:** Incubation period is 2-3 days.

6. **Symptoms:** IBD is a disease only affecting chickens, and mainly those around 3-4 weeks old when the bursa is well developed. In younger birds the disease usually has a subclinical course. Diseased chickens are pale, depressed, dehydrated, anorectic, shivering, have an insecure, atactic pace and ruffled feathers. They often develop transient diarrhoea that can be white with red or green staining. Further there is self-inflicted vent picking. When the infection occurs earlier in life the immunosuppression becomes permanent. In other cases, it usually is transient. Some birds may also show growth retardation after recovery. No disease can occur after 15-16 weeks of age when the bursa is regressed. The immunodeficiency caused by the virus depends on viral strains, age of the chicken and co-infection with other pathogens

7. **Lesions:** The following lesions are observed in dead birds
   a. Enlareged cloacal bursa which is swollen and haemorrhagic in birds dead of the disease and is atrophied in recovered birds
   b. Dehydrated carcass
   c. Dark skeletal muscles with haemorrhages (especially thigh and pectoral muscles)
   d. Opaque thymus with thickened gelatinous capsule
   e. Fatty and yellow or pink bone marrow
   f. Swollen liver
   g. Swollen kidneys
   h. Intestines with increased mucus
   i. Changes in bursa: Within three to four days post infection there is an inflammatory hypertrophy of the bursa and apoptosis of B- lymphocytes. Within two days of infection there is complete depletion of cells in the follicular cortex and after a week the follicular structure may not be visible at all. At four to six days post infection the bursa is swollen, hemorrhagic and covered by a gelatinous exudate. The bursa will reach five times its normal size during acute infection before atrophy begins. Ten days post infection the bursa will be one eighth of its original size and after two months a repopulation of the bursa occurs. At necropsy an atrophied, grey bursa is seen with no B lymphocytes in the bursal follicles or in other lymphoid tissues. (*Recently it has been shown that T-cells also are affected by IBDV infection. This might be explained by a macrophage-mediated inhibition of the T-cells ability to respond to mitogenic stimulation or an activation of T- suppressor cells)*

Diagnosis:

1. **Field diagnosis:** Based on clinical symptoms and lesions, vaccination history, prevalence of the infection in the area and age of the birds affected by the infection. A flock will show very high morbidity with severe depression in most birds lasting for 5–7 days. Mortality rises sharply for 2 days then declines rapidly over the next 2–3 days. Usually between 5% and 10% of birds die, but mortality can reach 30–40%.

2. **Isolation and identification:**
   a. **Clinical materials:** Samples should be collected from sick or freshly dead birds.
   i. Fresh bursa and spleen for virus isolation.

ii. Samples of bursa, spleen, intestines, caecal tonsil, liver and kidney should be collected in neutral buffered formalin for histopathology

iii. Blood samples for serology.

b. Isolation systems: Chickens, ECE and Chicken embryo fibroblast (*follow as given under cultivation).*

3. Direct identification of antigen from bursa:

a. Agar gel immunodiffusion test

b. Immunofluorescense test

c. Antigen capture ELISA

d. RT-PCR

4. Strain differentiation: IBDV serotypes are differentiated by cross neutralization virus neutralization test or by serological tests using serotype specific monoclonal antibodies.

5. Pathotype identification: The terms 'variant', 'classical' and 'very virulent' have been used to qualify IBDV strains that exhibit a different pathogenicity. Based on the signs and lesions observed in White Leghorn SPF chickens during acute experimental IBD

a. 'Variant' IBDVs induce little if any clinical signs and no mortality but marked bursal lesions,

b. 'Classical' IBDVs induce approximately 10–50% mortality with typical signs and lesions

c. 'Very virulent' IBDVs induce approximately 50–100% mortality with typical signs and lesions

6. Serological tests:

a. AGID: The AGID test is the most useful of the serological tests for the detection of specific antibodies in serum

b. Quantitative AGID test: The AGID test is used to measure antibody levels by using dilutions of serum in the test wells and taking the titre as the highest dilution to produce a precipitin line. This can be very useful for measuring maternal or vaccinal antibodies and for deciding on the best time for vaccination.

c. Virus neutralization test: VN tests are carried out in cell culture. The test is more laborious and expensive than the AGID test, but is more sensitive in detecting antibody. This test is also useful for evaluating vaccine responses or for differentiating between IBDV 1 and 2 serotypes

d. ELISA: This is the preferred test for identification and quantification of antibodies against IBD.

Differential diagnosis: IBD should be differentiated from Marek's disease Mycotoxicosis, Coccidiosis, Haemorrhagic syndrome, Avian adenovirus infection and Infectious bronchitis.

Treatment:

Treatment is not effective.

Control and eradication:

1. Vaccines: Two types of vaccine are used for the control of IBD. These are live attenuated vaccines, or inactivated oil-emulsion adjuvanted vaccines. Of late a live recombinant vaccine expressing IBDV antigens is also being used. To date, IBD vaccines are made with serotype 1 IBDVonly.

a. Live vaccines: Live IBD vaccines are produced from fully or partially attenuated strains of virus, known as 'mild', 'intermediate', or 'intermediate plus' ('hot'), respectively. Mild or intermediate vaccines are used in parent chickens to produce a primary response prior to vaccination near to point-of-lay using inactivated vaccine. They are susceptible to the effect of maternally derived antibodies so should be administered only after all maternally derived antibodies has waned. Application is by means of intramuscular injection, spray or in the drinking water, usually at 8 weeks of age. Intermediate or intermediate plus vaccines are used to protect broiler chickens

and commercial layer replacements. Some of these vaccines are also used in young parent chickens if there is a high risk of natural infection with virulent IBD. Recently, technology has been developed to deliver live vaccine into eggs during the incubation period. Live vaccine virus is blended with IBD antibody and the complex is injected *in ovo* at 18 days of incubation. The eggs go on to hatch and the vaccine virus is released when the chicks are about 7 days of age. In this way, the problem of maternally derived IBD antibody is overcome and the chicks are effectively immunized. Live IBD vaccines are generally regarded as compatible with other avian vaccines. However, it is possible that IBD vaccines that cause bursal damage could interfere with the response to other vaccines. Only healthy birds should be vaccinated.

b. **Inactivated vaccines:** Inactivated IBD vaccines are used to produce high, long-lasting and uniform levels of antibodies in breeding hens that have previously been primed by live vaccine or by natural exposure to field virus during rearing. Only healthy birds, known to be sensitised by previous exposure to IBDV, should be vaccinated. Used in this way the vaccine should produce such a good antibody response that chickens hatched from those parents will have passive protection against IBD for up to about 30 days of age

c. **Recombinant vaccines:** A live recombinant vaccine expressing IBDV VP2 antigen has been licensed recently in Europe. There is limited information available on the use of this vaccine.

2. **Eradication*:*** Recovered and vaccinated birds can carry and shed virus for long periods. Control of the disease implies depopulation and rigorous disinfection of contaminated premises. Because of the stability of the virus and its ability to spread readily with infected birds and contaminated fomites, strict quarantine measures and movement restrictions should be used.

Public health

No human infection has been reported with IBDV.

RETROVIRIDAE

EQUINE INFECTIOUS ANAEMIA

Equine infectious anemia (EIA) is an infectious and potentially fatal viral disease of members of the horse family. EIA is significant historically because it is the first disease of horses proven to be caused by a "filterable virus". EIAV is the first lentivirus–induced disease proven to be transmitted by insects. And EIA is the first persistent virus for which antigenic drift was defined. EIA is also the first lentivirus–induced disease for which a diagnostic test was approved.

General characters

Classification:

Baltimoregroup	–	Group VI Reverse transcribing RNA viruses
Family	-	Retroviridae
Sub-family	-	Orthoretrovirinae
Genus	-	Lentivirus
Species	-	Equine infectious anaemia virus(EIAV)

Synonyms: Swamp fever

Morphology: The virions have a complex construction and consist of an envelope, a nucleocapsid, a nucleoid, and a matrix protein. Virions are enveloped, spherical to pleomorphic in shape and measure nmin diameter. The surface has numerous glycoprotein spikes, which are 8 nm in length. The core is rod- shaped andthe nucleoid is concentric. The genome is dimeric (diploid) comprises of two monomers. The genome consists of non-segmented two molecules of linear positive-sense, single-

stranded RNA joined by hydrogen bonds. Minor species of non-genomic nucleic acid are also found in virions. The genome is composed of 9200 nucleotides. The 5'-end of the genome has a methylated cap and the 3'-terminus has a poly (A) tract. The genome of retroviruses is unique in the following aspects.

- ♣ They are the only viruses, which are truly diploid.
- ♣ They are the only RNA viruses whose genome is produced by cellular transcriptional machinery
- ♣ They are the only viruses whose genome requires a specific cellular RNA (tRNA) for replication.
- ♣ They are the only (+) sense RNA viruses whose genome does not serve directly as mRNA immediately after infection. Hence, the nucleic acid is not infectious
- ♣ The genome is comprised of four genes *gag, pro, pol and env.* Each gene codes for a single or more than one protein. The important proteins are listed below.

Name:	Protein:	Function:
MA	Matrix	matrix protein (*gag* gene); lines envelope
CA	Capsid	capsid protein (*gag* gene); protects the core; most abundant protein in virus particle
NC	Nucleocapsid	capsid protein (*gag* gene); protects the genome; forms the core
PR	Protease	Essential for gag protein cleavage during maturation
RT	Reverse transcriptase (*RNA dependent DNA polymerase)*	Reverse transcribes the RNA genome
IN	Integrase	Encoded by the *pol* gene; needed for integration of the provirus
SU	Surface glycoprotein	The outer envelope glycoprotein; major virus antigen
TM	Transmembrane protein	The inner component of the mature envelope glycoprotein

OIE Classification: List B infection

Risk group: No known human infection

Resistance: EIAV are fragile. They are sensitive to treatment with heat, detergents, and formaldehyde. Their infectivity is not affected by irradiation. Theviruslasts for several months at room temperature, in urine, faeces, and dried blood etc.

Replication: Replication occurs inside the cytoplasm and it is the only RNA virus that depends on host cell transcriptionary mechanisms.The virus enters the cell by attaching it with a host cell receptor. The SU and TM proteins help in this stage. The virus enters by fusion. Only partial uncoating occurs inside the cytoplasm. Though the RNA is positive sense they are not translated immediately inside the cytoplasm. A copy of the DNA is synthesized inside the cell with a virus coded enzyme RNA dependent RNA polymerase (reverse transcriptase). The d/s DNA product formed by this reaction is known as the **provirus.** Three forms of provirus DNA are found in all infected cells. This DNA later migrates to the nucleus and integrates with host cell DNA with the help of the enzyme integrase. Retroviruses use the cellular transcriptional machinery for expression. Therefore they are expressed like cellular genes. Assembly occurs at the cytoplasm and the mature virions later bud out through the cell membrane, acquiring the envelope during this process. As a result of complex replication steps, the replication processes of retroviruses are error prone. Further changes in the retro viral genome occur due to high rate of mutation, recombination and exchange of genes with host genome.

HA property: The virus has no HA property

Serotypes / Strains: No distinct serotypes are present.

Cultivation:

The virus is cultivated in susceptible horses or in leukocyte cultures. The presence of virus in these systems is confirmed by antigen confirmation tests.

Pathogenesis:

1. **Hosts affected:** The infection is limited to members of the family equidae. Donkeys are less severely affected than horses.
2. **Distribution:** The infection is worldwide. Few countries like Australia and New Zealand have claimed freedom.
3. **Transmission:** EIA is considered a classic bloodborne infection. The infection occurs in summer and autumn since the insects appear in large numbers in these seasons. The spread of EIA is usually slow although it can be quite fast where large groups of horses are gathered in a small area. EIA is very easily spread by the injection of small quantities of infected blood. The EIAV most frequently is transmitted between horses in close proximity by large biting insects, such as horse flies and deer flies. The bites from these flies stimulate defensive movement by the horse, which often results in an interruption of the blood–feeding. When interrupted, the fly is motivated to complete the feeding as soon as possible. It then attacks the same or a second host and feeds to repletion. In this manner, any infective material from the blood of the first host that is present on the mouth parts of the insect can be mechanically transmitted to the second host. The virus can survive only less than 4 hours in the insect. Insect transmission of EIAV is dependent on the number and habits of the insects, the density of the horse population, the number of times the insect bites the same and other horses, the amount of blood transferred between horses, and the level of virus in the blood of the infected horse from which the initial blood meal was obtained. Iatraogenic transmission is also possible during any manipulation that may involve the transfer of infected blood or saliva, for example through the use of unsterilised instruments or equipment. Vertical transmission (*in utero* or through the milk) is also possible
4. **Morbidity and mortality:** Highly variable.
5. **Incubation period:** Incubation period is 2-4 weeks
6. **Symptoms:** The first attack is usually the most **acute** with the horse suffering from depression, profound weakness, and loss of condition. Intermittent fever with large and rapid swings in temperature may also occur. Jaundice and bleeding around the mouth and nasal areas may also occur. This form of the disease is the most damaging and the most difficult to diagnose because the signs appear rapidly, and often only an elevated body temperature is noted. One–fifth of a teaspoon of blood from a horse with acute EIA contains enough virus to infect 1 million horses. Animals may show a temporary recovery and appear normal for 2 - 3 weeks and then relapse with similar, but less severe symptoms. It is also referred as **chronic** form. The symptoms include fever, petechial hemorrhages on the mucous membranes, depression, weight loss, dependent edema under the skin in the legs and under the chest and other under body surfaces and anemia. The animal may also have an irregular heart beat, and a jugular pulse. One fifth of a tea spoon of blood from a chronic case during a feverish episode contains enough virus to infect 10,000 horses. Most horses also remain as **inapparent** carriers. These horses show no obvious clinical signs as a result of infection. They survive as reservoirs of the infection for extended periods. Inapparent carriers have dramatically lower concentrations of EIAV in their blood than horses with active clinical signs of

the disease. Only 1 horse fly out of 6 million is likely to pick up and transmit EIAV from this horse. All horses infected with EIAV are thought to remain virus carriers for life. The inapparent form may become chronic or acute due to severe stress, hard work, or the presence of other diseases.

7. **Lesions:** In horses dying of the disease there is haemorrhages in the liver, spleen, kidneys, on serous membranes and in the mucosa of the intestines. The liver is enlarged and often yellowish-brown in colour. The spleen is also enlarged with soft pulp.The heart muscle appear pale, flabby and friable

Diagnosis:

1. **Field diagnosis:** Field diagnosis is based on the Coggins test. Since, no vaccines are available against EIAV, presence of antibodies against EIA in the serum of horses is an indication of infection. Clinical signs and lesions also provide clue about EIA.
2. **Isolation and identification:** Rarely attempted since it is of no practical value. Blood samples should be collected to confirm the diagnosis using the Coggins (agar gel diffusion) test or the more recent ELISA test. Isolation of the virus from suspect horses may be made by inoculating their blood on to leukocyte cultures prepared from horses free of infection. Virus production in cultures can be confirmed by detection of specific EIA antigen by ELISA, or imunofluorescence assay.
3. **Nucleic acid identification method:** Recently a RT-PCR method has been developed to detect EIA proviral DNA from the peripheral blood of horses
4. **Serology:** Due to the persistence of EIA virus in infected equids, detection of serum antibody to EIA virus confirms the diagnosis of EIA virus infection.
   a. **Coggins test – (Agar gel immunodiffusion test):** It is a test prescribed for International trade. Precipitating antibody is rapidly produced as a result of EIA infection, and can be detected by the AGID test. Specific reactions are indicated by precipitin lines between the EIA antigen and the test serum and confirmed by their identity with the reaction between the antigen and the positive standard serum.
   b. **ELISA:** Competitive and non-competitive ELISAs are also used to identify antibodies against EIAV.
5. **Differential diagnosis:** EIA should be differentiated from purpura haemorrhagica, babesiosis, leptospirosis, severe strongylosis or fasciolasis, phenothiazine toxicity and auto-immune haemolytic anaemia

Treatment: There is no effective treatment available against EIA. Supportive treatment, including blood transfusions can be considered. However, because recovered horses can become carriers, if the disease occurs in a free area it is generally better to destroy affected horses.

Control and eradication:

1. **Vaccines:** There are no vaccines against EIA
2. **Eradication:**
   a. Eradication is aimed in minimizing or eliminating contact of horses with the secretions, excretions, and blood of EIAV–infected horses. This has to be carried out by testing and segregating test–positive horses from those with negative test results. Once the reservoirs of EIAV are identified, separated, and maintained a safe distance from the other horses.
   b. As recovered horses continue to carry the virus, the most likely source of introduction of EIA is through importation of horses. OIE guidelines suggest that, when importing horses, as well as showing no clinical signs, they should be tested negative during the 30 days prior to shipment.
   c. In endemic areas, risk of infection can be reduced by protecting horses from insect vectorsby
   i. Keeping horses away from low lying areas

ii. Providing draining swamps

iii. Through use of insecticides and by providing insect-proof stabling

Additional Reading:

d. By following precautions, horse owners can get rid of the infection. These precautions are specified by the USDA, APHIS, Veterinary Services, National Center for Animal Health Programs, Certification and Control Team

i. Use disposable syringes and needles. Follow the rule of one horse—one needle.

ii. Clean and sterilize all instruments thoroughly after each use.

iii. Keep stables and immediate facilities clean and sanitary. Remove manure and debris promptly, and ensure that the area is well drained.

iv. Implement insect controls. The local veterinarian or animal health official can provide information about approved insecticides and other insect–control measures. Avoid habitats favorable to insect survival.

v. Do not intermingle infected and healthy animals. Do not breed EIAV–positive horses.

vi. Isolate all new horses, mules, and asses brought to the premises until they have been tested for EIA.

vii. Obtain the required certification of negative EIA test status for horse shows, county fairs, race tracks, and other places where many animals are brought together.

Abide by State laws that govern EIA.